

Intelligent Security API

Developer Guide

drackum@gmail.com
DETEKTOR

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1 Reading Guide

Chapter	Description
Overview	Includes the ISAPI overview, applicable products, terms and definitions, abbreviations, and update history.
ISAPI Framework	Read the chapter to take a quick look at the ISAPI framework and basic functions.
Quick Start Guide	Read the chapter to quickly understand the programming process of basic functions such as authentication, message parsing, real-time live view, playback, and event uploading.
API Reference	Start programming according to API definitions.
How-To Video Guidance	How-to videos demonstrate detailed steps of different integration tasks.

2 Overview

2.1 Introduction

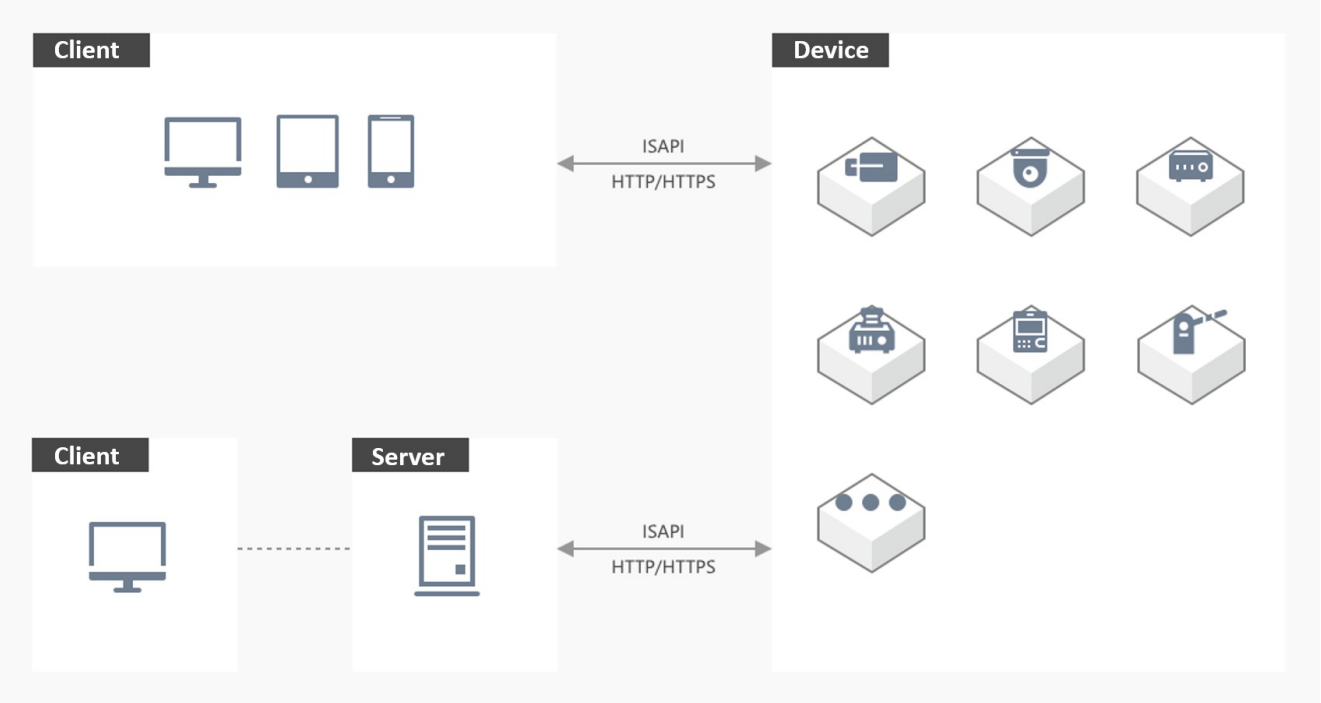
Intelligent Security API (hereinafter referred to as ISAPI) is an application layer protocol based on HTTP (Hypertext Transfer Protocol) and adopts the REST (Representational State Transfer) architecture for communication between security devices (cameras, DVRs, NVRs, etc.) and the platform or client software.

Since established in 2013, ISAPI has included more than 11,000 APIs for different functions, including device management, vehicle recognition, parking lot management, intelligent facial application, access control management, interrogation management, and recording management. It is applicable to industries such as traffic, fire protection, education, and security inspection.

dracrum@gmail.com
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2.1.1 Application Scenario

When you integrate devices via ISAPI, the device acts as the server to listen on the fixed port and the user's application acts as the client to actively log in to the device for communication. To achieve the above goals, the device should be configured with a fixed IP address and the requests from the client can reach the server.



2.1.2 Layers in the Network Model

ISAPI is an application layer protocol based on HTTP, thereby it inherits all specifications and properties from HTTP.

Protocols frequently used along with ISAPI include SADP (Search Active Device Protocol) based on multicast for discovering and activating devices, RTSP (Real-Time Streaming Protocol) based on TCP/UDP for live view and video playback of devices, etc.

Application Layer	Device Discovery SADP	Signaling Interaction ISAPI (HTTP)	Media Stream RTSP/RTP
Transport Layer	MCAST	TCP	TCP/UDP
Network Layer	IP		
Network Interface	Hardware Drive Interface		

2.2 Product Scope

- DVR
  - Pro Series With AcuSense

    DS-7204HQHI-S1/XT, DS-7208HQHI-S1/XT, DS-7208HQHI-S2/XT, DS-7216HQHI-S1/XT, DS-7216HQHI-S2/XT, iDS-7204HQHI-K1/2S, iDS-7204HQHI-K1/2S(C), iDS-7204HQHI-M1/FA, iDS-7204HQHI-M1/S, iDS-7204HQHI-M1/S-DIY, iDS-7204HTHI-M1/S, iDS-7204HTHI-M2/S, iDS-7204HUHI-K1/4S, iDS-7204HUHI-K1/4S(C), iDS-7204HUHI-K2/4S, iDS-7204HUHI-K2/4S(C), iDS-7204HUHI-M1/FA, iDS-7204HUHI-M1/P, iDS-7204HUHI-M1/S, iDS-7204HUHI-M1/S-DIY, iDS-7204HUHI-M1/X, iDS-7204HUHI-M2/FA, iDS-7204HUHI-M2/S, iDS-7204HUHI-M2/X, iDS-7208HQHI-K1/4S, iDS-7208HQHI-K1/4S(C), iDS-7208HQHI-K2/4S, iDS-7208HQHI-K2/4S(C), iDS-7208HQHI-M1/FA, iDS-7208HQHI-M1/S, iDS-7208HQHI-M1/S-DIY, iDS-7208HQHI-M2/FA, iDS-7208HQHI-M2/S, iDS-7208HTHI-M2/S, iDS-7208HUHI-K1/4S, iDS-7208HUHI-K1/4S(C), iDS-7208HUHI-K2/4S, iDS-7208HUHI-K2/4S(C), iDS-7208HUHI-M1/FA, iDS-7208HUHI-M1/S, iDS-7208HUHI-M1/S-DIY, iDS-7208HUHI-M1/X, iDS-7208HUHI-M2/FA, iDS-7208HUHI-M2/P, iDS-7208HUHI-M2/S, iDS-7208HUHI-M2/X, iDS-7216HQHI-K1/4S, iDS-7216HQHI-K1/4S(C), iDS-7216HQHI-K2/4S, iDS-7216HQHI-K2/4S(C), iDS-7216HQHI-M1/FA, iDS-7216HQHI-M1/S, iDS-7216HQHI-M2/FA, iDS-7216HQHI-M2/S, iDS-7216HUHI-M2/FA, iDS-7216HUHI-M2/P, iDS-7216HUHI-M2/S, iDS-7216HUHI-M2/X, iDS-7232HQHI-M2/S

2.3 Terms And Definitions

2.3.1 IoTChannel

IoTChannel refers to channels of IoT devices.

2.3.2 Event

Event refers to the information uploaded by the device actively. Event needs to be uploaded by the device in real time for the immediate response from the platform. If the device is offline, the event can be stored in the cache first and then be uploaded again when the connection is restored.

2.3.3 Arming

Arming means that the client establishes connection with the device so that events can be uploaded to the client via the connection. The client can subscribe to some event types, and the device will upload the specified events only, otherwise the device will upload all types of events to the client.

crackun@gmail.com DJIHKFDR

2.3.4 Motion Detection

It can realize image changes recognition and moving objects detection of a specified area by camera. For example, if a person passes by or a camera is moved, the motion will be recognized and then an alarm will be uploaded for notifications, and the event can trigger recording and alarm output.

2.4 Symbols And Acronyms

IOT: Internet of Things

OSD: On-screen display

ISAPI: Intelligent Security API.

ISUP: Intelligent Security Uplink Protocol.

HTTP: Hypertext Transfer Protocol.

NVR: Network Video Recorder

IPC: IP Camera

VMD: Motion Detection

PTZ: Pan-Tilt-Zoom

FDLib: Face Picture Library

FDID: Face Picture Library ID

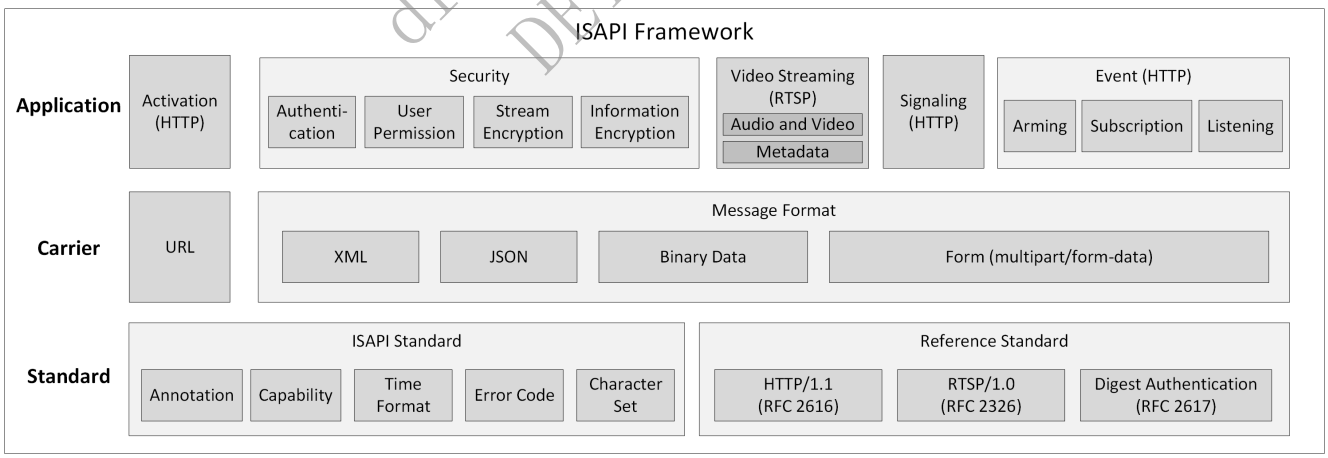
PID: Face Picture ID in the Face Picture Library

2.5 Update History

No update record

3 ISAPI Framework

3.1 Overview



Notes:

In general, ISAPI refers to the communication protocol based on the HTTP standard. As ISAPI is usually used along with RTSP (Real-Time Streaming Protocol), the RTSP standard is brought into the ISAPI system.

The metadata scheme for transmitting additional information of the stream is extended based on the RTSP standard to transmit the video stream and the structured intelligent information of the stream simultaneously. It is compatible with the RTSP standard.

3.2 Activation

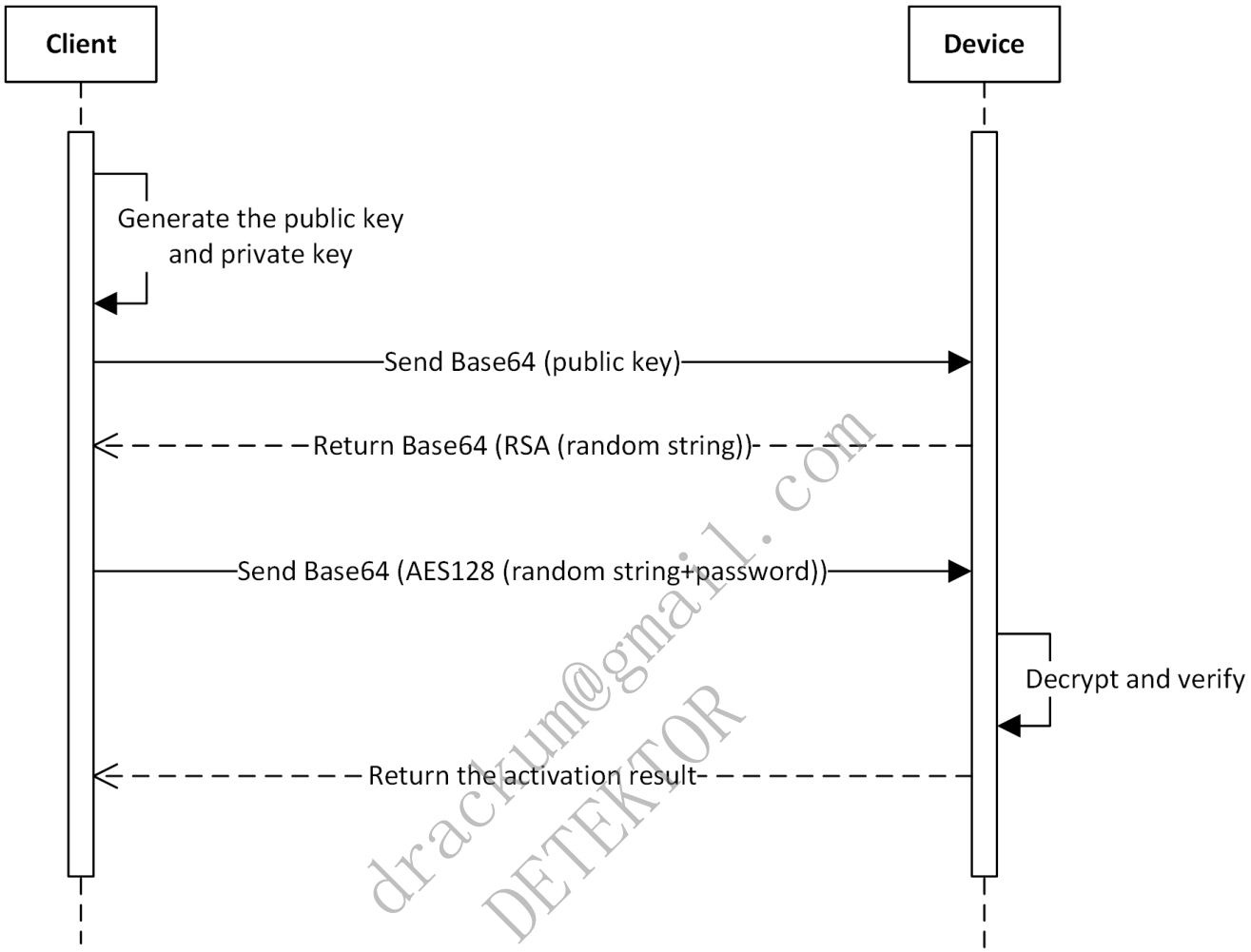
The purpose of activation is to ensure that the user can set the password for the device and the password meets the security requirement. After the device is activated, you can use the related functions.

drackum@gmail.com
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ISAPI is a communication protocol running on the application layer. When activating the device via ISAPI, you should know the device's IP address and make sure that the device is connected to the client.

The web application built in the device supports activating the device via ISAPI. When you enter the device's IP address in the address bar of the web browser on the PC, you can activate the device according to the activation guide.

If you want to activate the device on your own application, you need to integrate the activation function via ISAPI. The API calling flow and related APIs are shown below.



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Firstly, two operations are defined:

- bytesToHexstring: it is used to convert a byte array (the length is N) to a hexadecimal string (the length is 2N). For example, `127,10,23` -> `7f0a17`
- hexStringToBytes: it is used to convert a hexadecimal string (the length is 2N) to a byte array (the length is N). For example, `7f0a17` -> `127,10,23`

1. The client generates a public and private key pair (1024 bits), and gets the 128-byte modulus in the public key (hereinafter referred to as public key modulus). If the length is longer than 128, the leading 0 needs to be removed.
2. The client converts the public key modulus to a 256-byte public key string via bytesToHexstring and sends the public key string to the device in XML message (related URI: `POST /ISAPI/Security/challenge`) after being encoded by Base64.
3. The device parses the request to obtain a 256-byte public key string decoded by Base64 and converts it to a 128-byte public key modulus via hexStringToBytes. The complete public key is the combination of obtained public key modulus and public exponent (the default value is `'010001'`).
4. The device generates a 32-byte hexadecimal random string, calls the RSA API to encrypt the random string with the private key, converts the encrypted data to a string via bytesToHexstring, encodes the string by Base64, and then sends it to the client.
5. The client decodes the string from the device by Base64, converts it via hexStringToBytes to get the encrypted data, decrypts the encrypted data with the private key via RSA to obtain a 32-byte hexadecimal random string, converts the obtained string via hexStringToBytes to get a 16-byte AES key. Then the client uses the AES key to encrypt the `"string consisting of the first 16 characters of the random string and the real password"` by AES128

ECB mode (with zero-padding method) to get a ciphertext, and converts the ciphertext via bytesToHexstring, encodes it by Base64, and sends it to the device in XML message (related URI: `PUT /ISAPI/System/activate`). Note: If the first 16 characters of the random string are `aaaabbbbccccdddd` and the real password is `Abc12345`, the data before encryption is `aaaabbbbccccddddAbc12345`. This can ensure that the client uses the random string as the key for encryption.

6. The device decodes the string by Base64, converts it via hexStringToBytes to get the ciphertext, uses the AES key to decrypt the ciphertext by AES128 ECB mode, and gets the real password via removing the first 16 characters.
7. The device verifies the password and returns the activation result.

Notes:

- You can get the device's activation status by calling the URI `GET /SDK/activateStatus` which requires no authentication.
- Devices also support to be activated via SADP (Search Active Device Protocol) which is based on the communication protocol of the data link layer. With SADP, you do not have to know the IP address of the device but need to ensure that the device and the application running SADP are connected to the same router. SADP also supports discovering devices in the LAN, changing the password of the devices, and so on. The HCSadpSDK is provided for SADP integration, including the developer guide, plug-in, and sample demo which can be used as a simple SADP tool.

3.3 Security Mechanism

3.3.1 Authentication

When the client applications send requests to devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Currently, all mainstream request class libraries of HTTP have encapsulated digest authentication. See details in Authentication of Quick Start Guide.

crackum@gmail.com
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3.3.2 User Permission

There are three kinds of users with different permissions for access control and management.

Administrator: Has the permission to access all supported resources and should keep activated all the time. It is also known as "admin".

Operator: Has the permission to access general resources and a part of advanced resources.

Normal User: Only has the permission to access general resources.

3.3.3 Information Encryption

During ISAPI integration, the HTTPS service of devices is enabled by default. When the client applications communicate with devices via HTTPS, the information can be transmitted securely.

3.4 Video Streaming

3.4.1 Audio and Video Stream

ISAPI supports getting and setting stream media parameters of the device, such as video resolution, encoding format, and stream.

Cameras support standard RTSP (Real-Time Streaming Protocol, see details in RFC 7826). Client applications can get the stream from devices via RTSP.

For details about real-time streaming and video playback, refer to **Real-Time Live View** and **Playback** in Quick Start Guide.

3.4.2 Metadata

The metadata is the structured intelligent information generated by intelligent devices. When the client applications get the audio and/or video stream from devices via RTSP, the metadata will be returned by the device at the same time. For example, to display the face target frame, face information, vehicle target frame, license plate number, vehicle information, and other information on the video stream, the client applications can overlay the above information on the video image.

Before using the metadata, you need to enable the metadata function of the device and then get the stream from the device via RTSP. Some devices support subscribing to the metadata by type. For details about the process of integrating the metadata function, refer to **Metadata Management**.

4 Quick Start Guide

4.1 Authentication

When the client applications send requests to the devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Client applications only need to call APIs of the class library to implement the digest authentication. The sample code is shown below.

4.1.1 C/C++ (libcurl)

```
// #include <curl/curl.h>
// Callback Function
static size_t OnWriteData(void* buffer, size_t size, size_t nmemb, void* lpVoid)
{
    std::string* str = dynamic_cast<std::string*>((std::string *)lpVoid);
    if( NULL == str || NULL == buffer )
    {
        return -1;
    }

    char* pData = (char*)buffer;
    str->append(pData, size * nmemb);
    return nmemb;
}

std::string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
std::string strResponseData;
CURL *pCurlHandle = curl_easy_init();
curl_easy_setopt(pCurlHandle, CURLOPT_CUSTOMREQUEST, "GET");
curl_easy_setopt(pCurlHandle, CURLOPT_URL, strUrl.c_str());
// Set the user name and password
curl_easy_setopt(pCurlHandle, CURLOPT_USERPWD, "admin:admin12345");
// Set the authentication method to the digest authentication
curl_easy_setopt(pCurlHandle, CURLOPT_HTTPAUTH, CURLAUTH_DIGEST);
// Set the callback function
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEFUNCTION, OnWriteData);
// Set the parameters of the callback function to get the returned information
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEDATA, &strResponseData);
// Timeout settings for receiving the data. If receiving data is not completed within 5 seconds, the application will exit directly
curl_easy_setopt(pCurlHandle, CURLOPT_TIMEOUT, 5);
// Set the redirection times to avoid too many redirections
curl_easy_setopt(pCurlHandle, CURLOPT_MAXREDIRS, 1);
// Connection timeout duration. If the duration is too short, the client application will be disconnected before the data request sent by the application
reaches the device
curl_easy_setopt(pCurlHandle, CURLOPT_CONNECTTIMEOUT, 5);
CURLcode nRet = curl_easy_perform(pCurlHandle);
if (0 == nRet)
{
    // Output the received message
    std::cout << strResponseData << std::endl;
}
curl_easy_cleanup(pCurlHandle);
```

drackum@gmail.com
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4.1.2 C# (WebClient)

```
// using System.Net;
// using System.Net.Security;
try
{
    string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
    WebClient client = new WebClient();
    // Set the user name and password
    client.Credentials = new NetworkCredential("admin", "admin12345");
    byte[] responseData = client.DownloadData(strUrl);
    string strResponseData = Encoding.UTF8.GetString(responseData);
    // Output received information
    Console.WriteLine(strResponseData);
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
}
```

4.1.3 Java (HttpClient)

```
// import org.apache.commons.httpclient.HttpClient;
String url = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
HttpClient client = new HttpClient();
// Set the user name and password
UsernamePasswordCredentials creds = new UsernamePasswordCredentials("admin", "admin12345");
client.getState().setCredentials(AuthScope.ANY, creds);
GetMethod method = new GetMethod(url);
method.setDoAuthentication(true);
int statusCode = client.executeMethod(method);
byte[] responseData = method.getResponseBodyAsString().getBytes(method.getResponseCharSet());
String strResponseData = new String(responseData, "utf-8");
method.releaseConnection();
// Output received information
System.out.println(strResponseData);
```

4.1.4 Python (requests)

```
# - *- coding: utf-8 -*-

import requests
request_url = 'http://192.168.18.84:80/ISAPI/System/deviceInfo'
# Set the authentication information
auth = requests.auth.HTTPDigestAuth('admin', 'admin12345')
# Send the request and receive response
response = requests.get(request_url, auth=auth)
# Output response content
print(response.text)
```

drackum@gmail.com
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4.2 Message Parsing

4.2.1 Message Format

During the process of communication and interaction via ISAPI, the request and response messages are often text data in XML or JSON format. Besides that, the data of firmware packages and configuration files is in binary format. A request can also be in form format with multiple formats of data (multipart/form-data).

4.2.1.1 XML

Generally, the `Content-Type` in the headers of the HTTP request is `application/xml; charset="UTF-8"`.

Request and response messages in XML format are all encoded with UTF-8 standards in ISAPI.

The namespace `http://www.isapi.org/ver20/XMLSchema` and ISAPI version number `2.0` of XML messages are configured by default, see the example below.

```
<?xml version="1.0" encoding="UTF-8"?>
<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <Node>
    <id>1</id>
    <enabled>true</enabled>
    <nodeName>nodeName</nodeName>
    <level>level1</level>
  </Node>
</NodeList>
```

4.2.1.2 JSON

The `Content-Type` in the headers of the HTTP request is often `application/json`.

To distinguish between APIs with XML messages and those with JSON messages, ISAPI adds the query parameter format=json to all request URLs with JSON messages, e.g., `http://192.168.1.1:80/ISAPI/System/Sensor/thermometrySensor?format=json`. Messages of request URLs without the query parameter format=json are usually in XML format. However, there may be some exceptions, and the message format is subject to the API definition.

Request and response messages in JSON format are all encoded by UTF-8 in ISAPI.

4.2.1.3 Binary Data

For the firmware and configuration files, the `Content-Type` in the header of an HTTP request is often `application/octet-stream`.

4.2.1.4 Form (multipart/form-data)

When multiple pieces of data are submitted at the same time in an ISAPI request (e.g., the person information and face picture need to be submitted at the same time when a face record is added to the face picture library), the `Content-Type` in the header of the corresponding HTTP request is usually `multipart/form-data, boundary=AaB03x`, where the boundary is a variable used to separate the entire HTTP body into multiple units and each unit is a piece of data with its own headers and body. In `Content-Disposition` of form unit headers, the `name` property refers to the form unit name, which is required for all form units; the `filename` property refers to the file name of form unit body, which is required only when the form unit body is a file. In headers of form units, `Content-Length` refers to the body length, which starts after CRLF(`\r\n`) and ends before two hyphens (`--`) of next form. There should be a CRLF used as the delimiter of two form units before two hyphens (`--`), and the `Content-Length` of previous form unit does not include the CRLF length. For the detailed format description, refer to RFC 1867 (Form-Based File Upload in HTML). Pay attention to two hyphens (`--`) before and after the boundary.

Notes

- In RFC specifications, it is strongly recommended to contain the field `Content-Length` in the entity header, and there is no requirement that the field `Content-Length` should be contained in the header of each form element. The absence of field `Content-Length` in the header should be considered when the client and device programs parse the form data.
- To avoid the conflict between message content and boundary value, it is recommended to use a longer and more complex string as the boundary value.

The example of ISAPI form data submitted by a client to a device is as follows.

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```
POST /ISAPI/Intelligent/FDLib/pictureUpload
Content-Type: multipart/form-data; boundary=e5c2f8c5461142aea117791dade6414d
Content-Length: 56789

--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="PictureUploadData";
Content-Type: application/xml
Content-Length: 1234

<PictureUploadData/>
--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="face_picture"; filename="face_picture.jpg";
Content-Type: image/jpeg
Content-Length: 34567

Picture Data
--e5c2f8c5461142aea117791dade6414d--
```

The example of ISAPI form data responded from a device to a client is as follows.

In ISAPI messages, when there are multiple form units, three nodes (`pid`, `contentid`, and `filename`) are used for linking form units. The corresponding relations are as follows:

Node Name	Form Field	Description
pid	name	pid in XML/JSON messages corresponds to the name property of Content-Disposition in form headers.
contentid	Content-ID	contentid in XML/JSON messages corresponds to Content-ID in form headers.
filename	filename	filename in XML/JSON messages corresponds to filename property of Content-Disposition in form headers.

drackum@gmail.com
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```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=136a73438ecc4618834b999409d05bb9
Content-Length: 56789

--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="mixedTargetDetection";
Content-Type: application/json
Content-Length: 811

{
    "ipAddress": "172.6.64.7",
    "macAddress": "01:17:24:45:D9:F4",
    "channelID": 1,
    "dateTime": "2009-11-14T15:27+08:00",
    "eventType": "mixedTargetDetection",
    "eventDescription": "Mixed target detection",
    "deviceID": "123456789",
    "CaptureResult": [{
        "targetID": 1,
        "Human": {
            "Rect": {
                "height": 1.0,
                "width": 1.0,
                "x": 0.0,
                "y": 0.0
            },
            "contentID1": "humanImage", /*human body thumbnail*/
            "contentID2": "humanBackgroundImage", /*human body background picture*/
            "pId1": "9d48a26f7b8b4f2390c16808f93f3534", /*human body thumbnail ID */
            "pId2": "5EE7078E07BB47CF860DE8E4E9A85F28" /*ID of human body background picture*/
        }
    }]
}
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="9d48a26f7b8b4f2390c16808f93f3534"; filename="humanImage.jpg";
Content-Type: image/jpeg
Content-Length: 34567
Content-ID: humanImage

Picture Data
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="5EE7078E07BB47CF860DE8E4E9A85F28"; filename="humanBackgroundImage.jpg";
Content-Type: image/jpeg
Content-Length: 345678
Content-ID: humanBackgroundImage

Picture Data
--136a73438ecc4618834b999409d05bb9--
```

drackum@gmail.com
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4.2.2 Annotation

The field descriptions of ISAPI request and response messages are marked as annotations in the example messages as shown below.

```
<?xml version="1.0" encoding="UTF-8"?>

<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, node list, attr:version{req, string, version No., range:[,]}-->
  <Node>
    <!--ro, opt, object, node information-->
    <id>
      <!--ro, req, int, node No., range:[,], step:, unit:, unitType:-->1
    </id>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <nodeName>
      <!--ro, req, string, node name, range:[1,32]-->test
    </nodeName>
    <level>
      <!--ro, opt, enum, level, subType:string,
      [level1#Level 1,level2#Level 2,level3#Level 3]-->level1
    </level>
  </Node>
</NodeList>
```

```
{
    "name":  "test",
    /*ro, req, string, name, range:[1,32]*/
    "type":  "type1",
    /*ro, req, enum, type, subType:string, [type1#type 1,type2#type 2]*/
    "enabled":  true,
    /*ro, opt, bool, enable or not, desc:xxxxxxx*/
    "NodeList": {
    /*opt, object, node list, dep:and,{$.enabled,eq,true}*/
        "scene":  1,
        /*req, enum, scene, subType:int, [1#scene 1; 2#scene 2; 3#scene 3]*/
        "ID":  1
        /*req, int, No., range:[1,8], step:, unit:, unitType:*/
    }
}
```

Key annotations are shown in the table below.

Annotation	Description	Remark
ro	Attribute: Read-Only	This field can only be obtained and cannot be edited.
wo	Attribute: Write-Only	This field can only be edited and cannot be obtained.
req	Attribute: Required	This field is required for request messages sent to the device and response messages returned from the device.
opt	Attribute: Optional	This field is optional for request messages sent to the device and response messages returned from the device.
dep	Attribute: Dependent	This field is valid and required when specific conditions are satisfied.
object	Field Type: Object	The field of type object contains multiple sub-fields.
list	Field Type: List	The subType following it refers to the data type of sub-items in the list.
subType	Field Type: String	The range following it refers to the maximum and the minimum string size of the field.
int	Field Type: Int	The range following it refers to the maximum and the minimum value of the field.
float	Field Type: Float	The range following it refers to the maximum and the minimum value of the field.
bool	Field Type: Boolean	The value can be true or false.
enum	Field Type: Enumeration	The subType following it indicates that the enumerators are of type string or int. The [] following the subType contains all enumerators.
subType	Sub-Type of Field	When the type of field is list or enum, the value of subType is the data type of each sub-object.
desc	Field Description	The detailed description of the field.

crackum@gmail.com DETEKTOR

4.2.3 Capability Set

ISAPI has designed capability sets for almost all functions, APIs, and fields. URLs for getting the capability set end with `/capabilities`. Some URLs may contain query parameters in the format: `/capabilities?format=json&type=xxx`.

There are two types of fields in the capability message of ISAPI: whether the device supports a function and the value range of a field in an API.

Whether the device supports a function: it is often in the format `isSupportXxxxxxxx`, which indicates that whether the device supports a function and a set of APIs for implementing this function.

The capability message example in JSON format is shown below:

```
{
    "isSupportMap":  true,
    /*ro, opt, bool, whether it supports the e-map function, desc:/ISAPI/SDT/Management/map/capabilities?format=json*/
    "isSupportAlgTrainResourceInfo":  true,
    /*ro, opt, bool, whether it supports only getting the resource information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/ResourceInfo?format=json*/
    "isSupportAlgTrainAuthInfo":  true,
    /*ro, opt, bool, whether it supports ony getting the authorization information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/SoftLock/AuthInfo?format=json*/
    "isSupportAlgTrainNodeList":  true,
    /*ro, opt, bool, whether it supports only getting the node information of the algorithm training platform, desc:/ISAPI/SDT/algorithmTraining/NodeList?
format=json*/
    "isSupportNAS":  true
    /*ro, opt, bool, whether it supports mounting and unmounting NAS, desc:/ISAPI/SDT/Management/NAS/capabilities?format=json*/
}
```

The capability message example in XML format is shown below:

```
<isSupportNetworkStatus>
    <!--ro, opt, bool, whether it supports searching the network status, desc: related API (/ISAPI/System/Network/status?format=json)-->true
</isSupportNetworkStatus>
```

The value range of the field: the maximum value, minimum value, the maximum size, the minimum size, options, and so on of each field of the API.

The example of JSON format is shown below:

```
{
    "boolType": {
        /*req, object, example of the capability of type bool*/
        "@opt": [true, false]
        /*req, array, options, subType: bool*/
    },
    "integerType": {
        /*req, object, example of the capability of type integer*/
        "@min": 0,
        /*ro, req, int, the minimum value*/
        "@max": 100
        /*ro, req, int, the maximum value*/
    },
    "stringType": {
        /*req, object, example of the capability of type string*/
        "@min": 0,
        /*ro, req, int, the minimum string size*/
        "@max": 32
        /*ro, req, int, the maximum string size*/
    },
    "enumType": {
        /*req, object, capability example of type enum*/
        "@opt": ["enum1", "enum2", "enum3"]
        /*req, array, options, subType: string*/
    }
}
```

dr ackum@gmail.com
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The example of XML format is shown below:

```
<boolType opt="true,false" def="true">
    <!--ro, opt, bool, example of the capability of type bool-->true
</boolType>
<integerType min="0" max="100">
    <!--ro, opt, int, example of the capability of type int-->50
</integerType>
<stringType min="0" max="64">
    <!--ro, opt, string, example of the capability of type string-->test
</stringType>
<enumType opt="red,white,black" def="red">
    <!--ro, opt, string, example of the capability of type enum-->white
</enumType>
```

Note: For the same capability set, devices of different models and versions may return different results. The values shown in this document are only examples for reference. The capability set actually returned by the device takes precedence.

4.2.4 Time Format

ISAPI adopts ISO 8601 Standard Time Format, which is the same as W3C Standard Date and Time Formats.

Format: `YYYY-MM-DDThh:mm:ss.sTZD`

```
YYYY = the year consisting of four decimal digits
MM = the month consisting of two decimal digits (01-January, 02-February, and so forth)
DD = the day consisting of two decimal digits (01 to 31)
hh = the hour consisting of two decimal digits (00 to 23, a.m. and p.m. are not allowed)
mm = the minute consisting of two decimal digits (00 to 59)
ss = the second consisting of two decimal digits (00 to 59)
s = one or more digits representing the fractional part of a second
TZD = time zone identifier (Z or +hh:mm or -hh:mm)
```

Example: 2017-08-16T20:17:06.123+08:00 refers to 20:17:06.123 on August 16, 2017 (local time which is 8 hours ahead of UTC). The plus sign (+) indicates that the local time is ahead of UTC, and the minus sign (-) means that the local time is behind UTC.

After the DST is enabled, the local time and time difference will change compared with UTC, and the values of related fields also need to be changed. Disabling the DST will bring into the opposite effect.

Example: In 1986, the DST was in effect from May 4 at 2:00 a.m. (GMT+8). During the DST period, the clocks were moved one hour ahead, which means that there was one less hour on that day. When the DST ends at 2:00 a.m. on September 14, 1986, the clocks were moved one hour back and there was an extra hour on that day. The changes of the time are as follows:

- DST Starts: 1986-05-04T02:00:00+08:00 --> 1986-05-04T03:00:00+09:00
- DST Ends: 1986-09-14T02:00:00+09:00 --> 1986-09-14T01:00:00+08:00

Notes:

- The time difference cannot be simply used to determine the time zone. Because when the DST starts, the time difference will change and it cannot represent the actual time zone.
- Both TZ (UTC time, e.g., 1986-05-03T18:00:00Z) and TD (local time and time difference, e.g., 1986-05-04T02:00:00+08:00) meet the time format standards of ISO 8601. In ISAPI, the TD format is recommended to be used in messages sent from the user applications and the devices.
- A few old-version devices will return the time in TZ format. For representing the time difference information and forward compatibility, an extra field `timeDiff` is added as shown in the example below. User applications need to support both TD format and TZ format when parsing the time in the messages returned by devices.

```
{
    "dateTime": "1986-05-03T18:00:00Z", /*device time. The value in TZ format is the UTC time and the value in TD format is the time difference between the device's local time and UTC*/
    "timeDiff": "+08:00" /*optional, time difference between the local time and UTC time. If this field does not exist, the user application will convert the dateTime into the local time for use*/
}
```

dracktm@gmail.com DETHTOP

4.2.5 Character Set

To prevent characters not commonly used from resulting in exceptions in device programs and user applications, ISAPI limits the valid field values of type string to a specific range of characters. Character sets allowed to be used in the fields of type string in ISAPI are listed below.

- Single-byte character set: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and special characters (see details in the table below).
- Multi-byte character set: language characters based on Unicode and encoded by UTF-8 (UTF-8 encoding is a transformation format of Unicode character set. For details, refer to RFC 2044).

No.	Name	Special Character	No.	Name	Special Character
1	Open Parenthesis	(	18	Dollar Sign	$
2	Close Parenthesis	)	19	Percent Sign	%
3	Plus Sign	+	20	Ampersand	&
4	Comma	,	21	Close Single Quotation Mark	'
5	Minus Sign	-	22	Asterisk	*
6	Period	.	23	Slash	/
7	Semicolon	;	24	Smaller Than	<
8	Equal Sign	=	25	Greater Than	>
9	At Sign	@	26	Question Mark	?
10	Open Square Bracket	[	27	Caret	^
11	Close Square Bracket	]	28	Open Single Quotation Mark	'
12	Underscore	_	29	Vertical Bar	|
13	Open Brace	{	30	Tilde	~
14	Close Brace	}	31	Double Quotation Marks	"
15	Space		32	Colon	:
16	Exclamation Mark	!	33	Backslash	|
17	Octothorpe	#			

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The valid characters that can be used in some special fields are listed below.

- User name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 30 in the special character table.
- Password: User Name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 33 in the special character table.
- Names displayed on the UI (device name, person name, face picture library name, etc.): lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters.
- Normal fields of type string support lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters by default.

4.2.6 Error Processing

When requesting via ISAPI failed (the HTTP status code is not 200), the device will return the HTTP status code and ISAPI error code. For HTTP status codes, refer to 10 Status Code Definitions in RFC 2616. For ISAPI error codes, refer to Error Code Dictionary.

Message Example:

```
HTTP/1.1 403 Forbidden
Content-Type: application/json; charset="UTF-8"
Date: Thu, 15 Jul 2021 20:43:30 GMT
Content-Length: 229
Connection: Keep-Alive

{
    "requestURL": "/ISAPI/Event/triggers/notifications/channels/whiteLightAlarm",
    "statusCode": 4,
    "statusString": "Invalid Operation",
    "subStatusCode": "notSupport",
    "errorCode": 1073741825,
    "errorMsg": "notSupport"
}
```

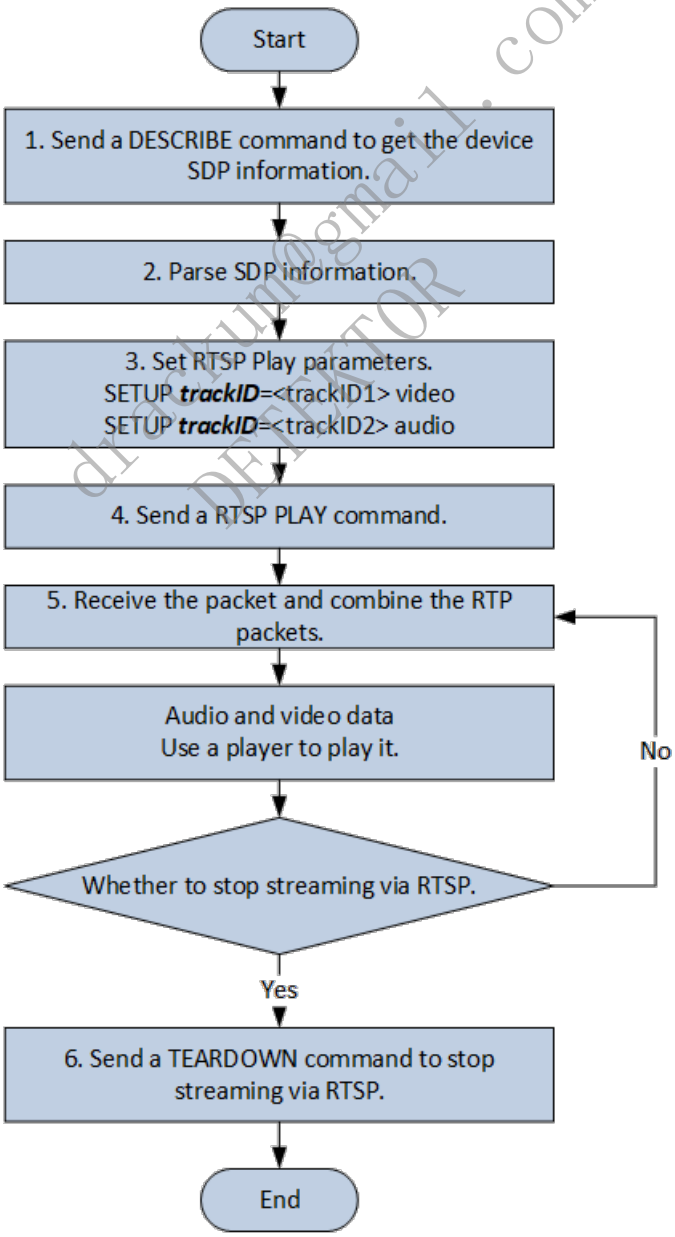
4.3 Real-Time Live View

4.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

4.3.2 API Calling Flow



1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`. Digest authentication with devices is required before this step. 2. The client parses the media SDP information returned by the device. 3. Set RTSP play parameters, that is to set the track ID parsed from SDP information via SETUP commands. For

example, trackID=1 indicates videos while trackID=2 indicates audios. 4. The client sends an RTSP PLAY command, and the device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`. 5. The client receives the RTP packet sent by the device. Divided RTP packets should be assembled on the client before being parsed. 6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address; `[:port]` is optional, and 554 by default; `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, the IP address of the target device is `172.7.203.11`, and the streaming address of main stream for channel 17 will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

4.3.3 Example

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

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```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.22.126
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000020000011171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

drackum@gmail.com
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```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

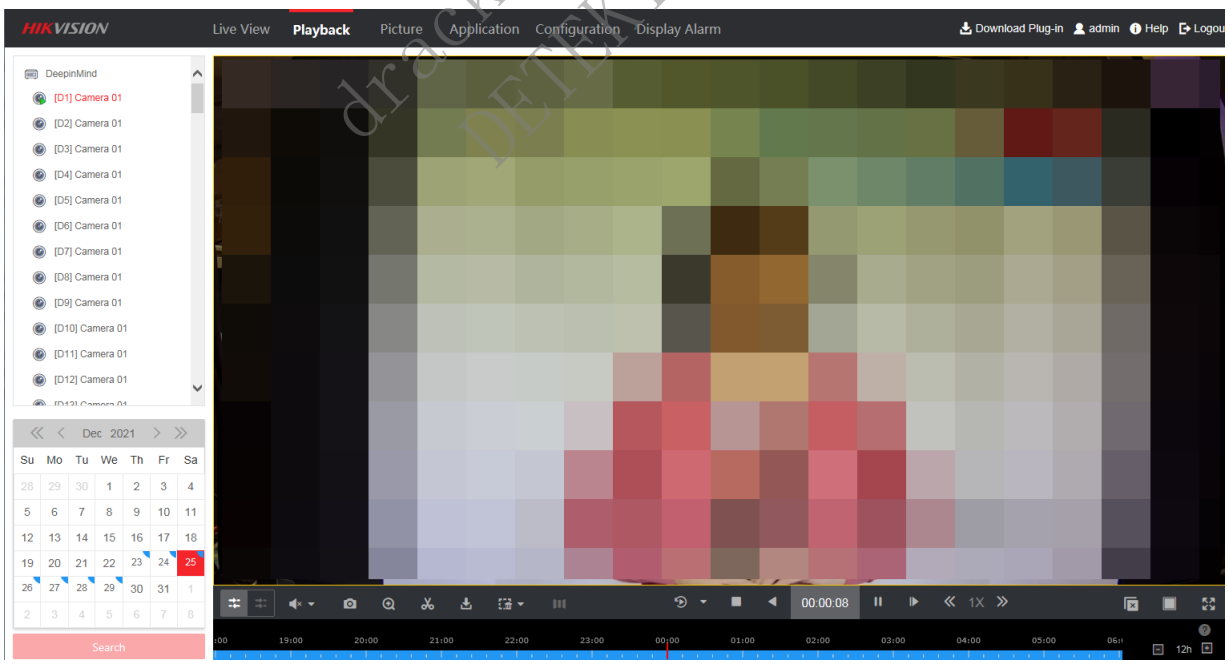
```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

4.4 Playback

4.4.1 Introduction to the Function

Devices such as cameras and NVRs can store videos. Storage mediums such as SD card, TF card and HDDs are needed before recording videos. The process of playback starts with searching for footages and then getting video stream via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

Below is an example of playback on a web client. The calendar in the bottom-left corner shows the results of searching for videos. Dates with videos are shown on the calendar and you can click a date to play back searched videos of the corresponding day.



4.4.2 API Calling Flow

1. (Optional) Check the calendar to find dates with stored videos: `POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution`. `<trackStreamID>` = channel number * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, The rackStreamID of the main stream for channel 17 is 1701.
2. Searching for videos by parsing the `playbackURI` of the device to get the RTSP address `POST /ISAPI/ContentMgmt/search`.

drackum@gmail.com DETEKTOR

3. Play videos back via RTSP. Steps of playback via RTSP is the same as that of real-time live view. Refer to streaming via RTSP in real-time streaming. See details in RFC 7826 (Real Time Streaming Protocol).

Notes:

- Playback via RTSP needs digest authentication. The method is the same as the one of ISAPI digest authentication.
- Send PAUSE command to pause playback. Send PLAY command to continue. If you want to perform fast forward and slow forward, you can send a PLAY command and set Scale in headers. See details in 10.6 PAUSE and 12.34 Scale in RFC 7826 (Real Time Streaming Protocol).

4.4.3 Example

4.4.3.1 Search for Videos According to Calendar

Request from Client

```
POST /ISAPI/ContentMgmt/record/tracks/101/dailyDistribution HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 119
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<trackDailyParam>
    <year>2021</year>
    <monthOfYear>08</monthOfYear>
</trackDailyParam>
```

Response from Device

drackum@gmail.com
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```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:47:43 GMT
Content-Length: 2915
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<trackDailyDistribution version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <dayList>
        <day><id>1</id><dayOfMonth>1</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>2</id><dayOfMonth>2</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>3</id><dayOfMonth>3</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>4</id><dayOfMonth>4</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>5</id><dayOfMonth>5</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>6</id><dayOfMonth>6</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>7</id><dayOfMonth>7</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>8</id><dayOfMonth>8</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>9</id><dayOfMonth>9</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>10</id><dayOfMonth>10</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>11</id><dayOfMonth>11</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>12</id><dayOfMonth>12</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>13</id><dayOfMonth>13</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>14</id><dayOfMonth>14</dayOfMonth><record>false</record></day>
        <day><id>15</id><dayOfMonth>15</dayOfMonth><record>false</record></day>
        <day><id>16</id><dayOfMonth>16</dayOfMonth><record>false</record></day>
        <day><id>17</id><dayOfMonth>17</dayOfMonth><record>false</record></day>
        <day><id>18</id><dayOfMonth>18</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>19</id><dayOfMonth>19</dayOfMonth><record>false</record></day>
        <day><id>20</id><dayOfMonth>20</dayOfMonth><record>false</record></day>
        <day><id>21</id><dayOfMonth>21</dayOfMonth><record>false</record></day>
        <day><id>22</id><dayOfMonth>22</dayOfMonth><record>false</record></day>
        <day><id>23</id><dayOfMonth>23</dayOfMonth><record>false</record></day>
        <day><id>24</id><dayOfMonth>24</dayOfMonth><record>false</record></day>
        <day><id>25</id><dayOfMonth>25</dayOfMonth><record>false</record></day>
        <day><id>26</id><dayOfMonth>26</dayOfMonth><record>false</record></day>
        <day><id>27</id><dayOfMonth>27</dayOfMonth><record>false</record></day>
        <day><id>28</id><dayOfMonth>28</dayOfMonth><record>false</record></day>
        <day><id>29</id><dayOfMonth>29</dayOfMonth><record>false</record></day>
        <day><id>30</id><dayOfMonth>30</dayOfMonth><record>false</record></day>
        <day><id>31</id><dayOfMonth>31</dayOfMonth><record>false</record></day>
    </dayList>
</trackDailyDistribution>
```

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4.4.3.2 Search for Videos

Request from Client

```
POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 486
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
    <searchID>88C2CD4D-D3FA-4AD4-BD80-555C18205DCC</searchID>
    <trackList>
        <trackID>101</trackID>
    </trackList>
    <timeSpanList>
        <timeSpan>
            <startTime>2021-08-16T00:00:00Z</startTime>
            <endTime>2021-08-18T23:59:59Z</endTime>
        </timeSpan>
    </timeSpanList>
    <maxResults>100</maxResults>
    <searchResultPostion>0</searchResultPostion>
    <metadataList>
        <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
    </metadataList>
</CMSearchDescription>
```

Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:19:13 GMT
Content-Length: 1021
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<CMSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <searchID>{88c2cd4d-d3fa-4ad4-bd80-555c18205dcc}</searchID>
    <responseStatus>true</responseStatus>
    <responseStatusStrg>OK</responseStatusStrg>
    <numOfMatches>1</numOfMatches>
    <matchList>
        <searchMatchItem>
            <sourceID>{00000000-0000-0000-0000-000000000000}</sourceID>
            <trackID>101</trackID>
            <timeSpan>
                <startTime>2021-08-18T15:18:15Z</startTime>
                <endTime>2021-08-18T15:19:08Z</endTime>
            </timeSpan>
            <mediaSegmentDescriptor>
            <contentType>video</contentType>
            <codecType>H.264-BP</codecType>
            <playbackURI>rtsp://10.14.97.40/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&amp;endtime=20210818T151908Z&amp;name=00000004667000100&amp;size=1400788</playbackURI>
            <lockStatus>unlock</lockStatus>
            <name>00000004667000100</name>
            </mediaSegmentDescriptor>
            <metadataMatches>
                <metadataDescriptor>recordType.meta..com/timing</metadataDescriptor>
            </metadataMatches>
        </searchMatchItem>
    </matchList>
</CMSearchResult>
```

drackum@gmail.com
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4.4.3.3 Playback via RTSP

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: application/sdp
```

4. The server responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 7
Content-Type: application/sdp
Content-Length: 546
Date: Wed, 18 Aug 2021 07:29:25 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.60.000
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:4106
a=range:clock=20210818T151815Z-20210818T151908Z
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H265/90000
a=control:trackID=video
b=AS:4096
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=audio
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000040005001171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP commands two times, and the server responds to them.

```
SETUP rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=video RTSP/1.0
CSeq: 8
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61048-61049
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61048-61049;server_port=62000-62001;ssrc=d6d3ed5
CSeq: 8
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

```
SETUP rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=audio RTSP/1.0
CSeq: 9
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61050-61051
Session: 225263317
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61050-61051;server_port=62002-62003;ssrc=d6d3ed6
CSeq: 9
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

6. The client sends an RTSP PLAY request, and the server starts to push streams for playback.

```
PLAY rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 10
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
Range: npt=0.000-
```

drackum@gmail.com
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```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

7. The client sends an RTSP TEARDOWN request to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

4.5 Event Uploading

When the rules configured on the device are triggered, the device will generate event messages (e.g., motion detection, etc.) and actively upload them to the client. ISAPI supports three methods to receive event messages uploaded by the device, that is, in arming mode, in listening mode, and via subscription.

dracklm@gmail.com
IJETER.COR

4.5.1 Arming

The client establishes a HTTP persistent connection with the device to receive event messages from the device.

There are two methods (arming with subscription and arming without subscription) to receive events from the device. The arming without subscription is to get all event messages from the device via HTTP GET method, while the arming with subscription is to get messages of subscribed events via HTTP POST method.

Notes

- ISAPI arming (with or without subscription) uses the HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send event messages continuously, while it's not supported for clients to send any message to the device via the connection.
- When the heartbeat timed out and no message is received from the device, you should terminate the arming connection and try establishing a new one.

4.5.1.1 Arming without Subscription

1. Establish the connection of arming without subscription: `GET /ISAPI/Event/notification/alertStream` and keep the connection alive via configuring `Connection: keep-alive` in HTTP headers on the client.
2. Receive events sent by the device. The event message will be separated and parsed by boundary. For parsing details, see **Event Message Parsing** below.
3. Terminate the arming connection when no event message needs to be received.

Event Message Parsing:

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

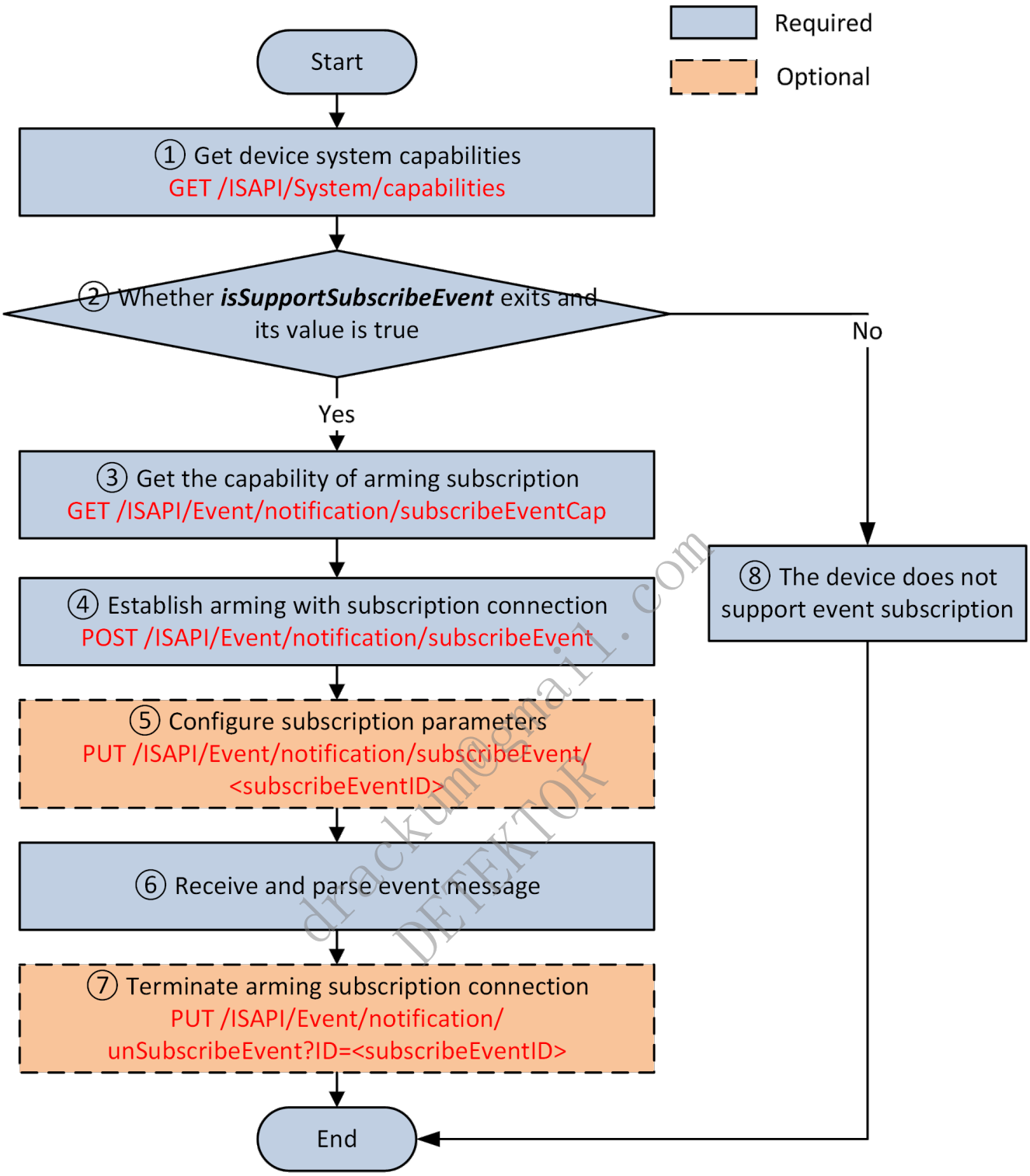
<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

Note: `<data_gateway_ip>` and `<frontier>` are variables, `[Picture Data]` indicates the raw data of a picture.

4.5.1.2 Arming with Subscription

drackum@gmail.com
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1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by the boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note:

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

Event Messages Parsing:

Client Creates Arming with Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

Server Responds to Request and Push Event Message

When the client sends a request to the device for establishing an arming connection, the device sends data in HTTP form format (multipart/form-data). In HTTP request headers, the `Content-Type` is `multipart/form-data, boundary=AaB03x`, of which the boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Please note two hyphens `--` before and after boundary. In normal conditions, devices will not close the arming link actively, and the end mark `--AaB03x--` will not be sent on the arming link.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Picture Data
--AaB03x--
```

drackum@gmail.com
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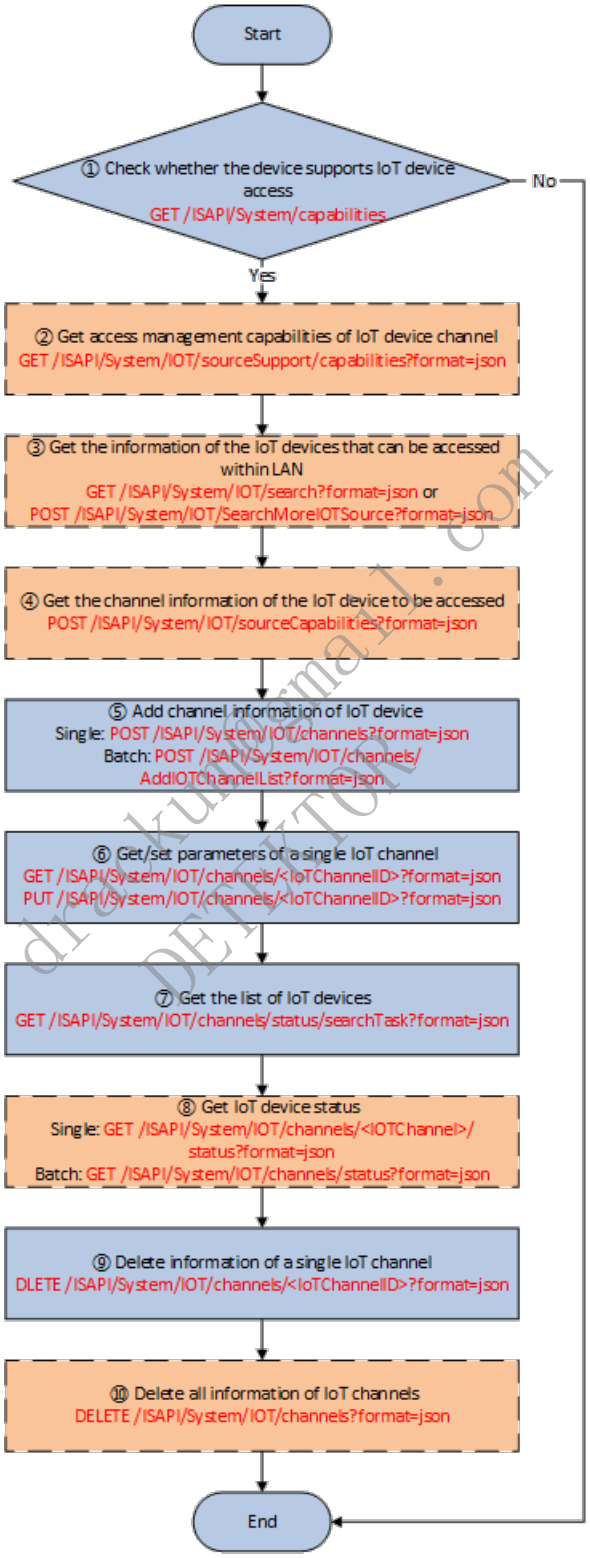
The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. A form message which starts with `--boundary` and ends with `--boundary--`.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description. `form-data` is a piece of form data.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next `\r\n` to the next `--boundary`.

5 Device Management (General)

5.1 4 IoT Channel Management

5.1.1 IoT Device



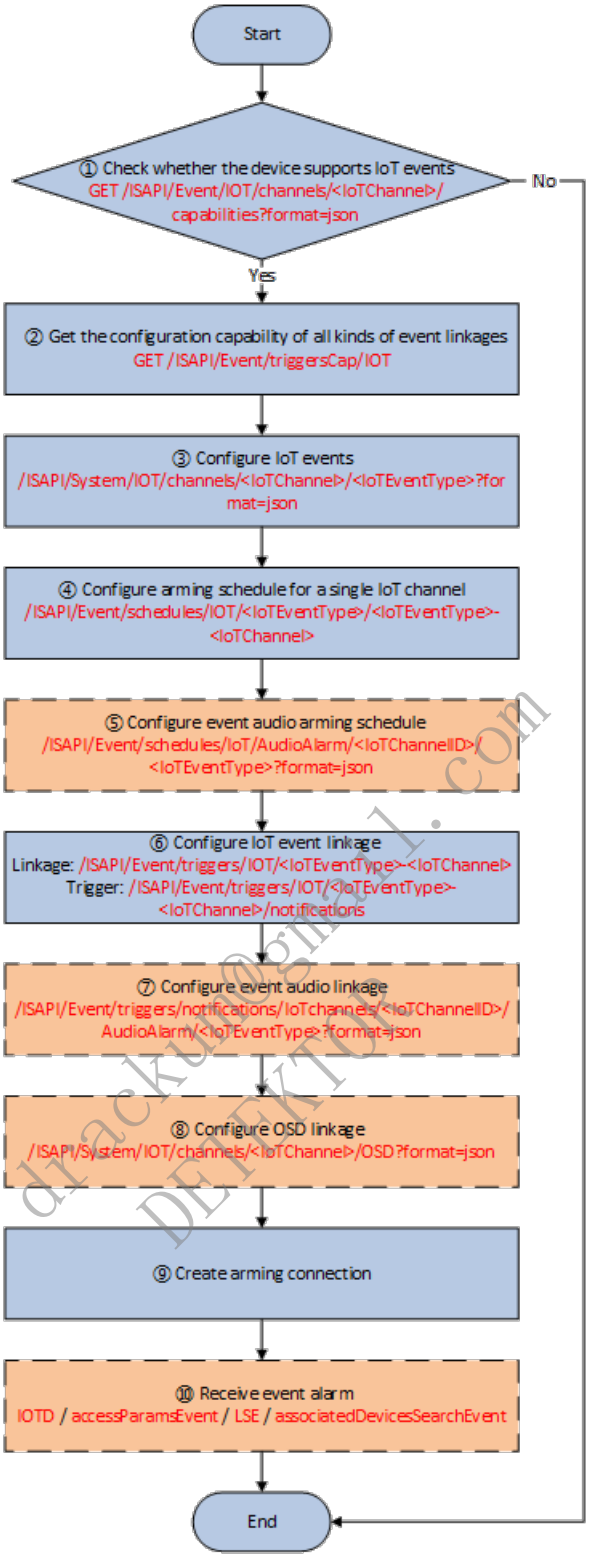
1. Check whether the device supports IoT device access: `GET /ISAPI/System/capabilities`. If the node `IOTCap` exists in the response message, it means that the device supports IoT device access. The content of `IOTCap` is the capability set of IoT device access.
2. (Optional) Get access management capabilities of IoT device channel: `GET /ISAPI/System/IOT/sourceSupport/capabilities?format=json`. Its function is similar to that of the previous step, so it is optional.
3. Get the information of the IoT devices that can be accessed within LAN.

- All results: `GET /ISAPI/System/IOT/search?format=json`
- Results with search conditions: `POST /ISAPI/System/IOT/SearchMoreIOTSource?format=json`

4. Get the channel information of the IoT device to be accessed: `POST /ISAPI/System/IOT/sourceCapabilities?format=json`
5. Add channel information of IoT device.
   - Single: `POST /ISAPI/System/IOT/channels?format=json`
   - Batch: `POST /ISAPI/System/IOT/channels/AddIOTChannelList?format=json`
6. Get/set parameters of a single IoT channel.
   - Get: `GET /ISAPI/System/IOT/channels/<IoTChannelID>?format=json`
   - Set: `PUT /ISAPI/System/IOT/channels/<IoTChannelID>?format=json`
7. Get the list of IoT devices: `GET /ISAPI/System/IOT/channels/status/searchTask?format=json`
8. (Optional) Get IoT device status. Its function is similar to that of the previous step, so it is optional. It is recommended to get device status via the device list in the previous step.
   - Single: `GET /ISAPI/System/IOT/channels/<IOTChannel>/status?format=json`
   - Batch: `GET /ISAPI/System/IOT/channels/status?format=json`
9. Delete information of a single IoT channel: `DLETE /ISAPI/System/IOT/channels/<IoTChannelID>?format=json`
10. Delete all information of IoT channels: `DELETE /ISAPI/System/IOT/channels?format=json`

5.1.2 IoT Event

drackum@gmail.com
DETEKTOR



1. Get the event capabilities supported by the IoT channel: `GET /ISAPI/Event/IOT/channels/<IoTChannel>/capabilities?format=json`
2. Get the configuration capability of all kinds of event linkages: `GET /ISAPI/Event/triggersCap/IOT`
3. Configure IoT events.
   - Get IoT event configuration capability: `GET /ISAPI/System/IOT/channels/<IoTChannel>/<IoTEventType>/capabilities?format=json`
   - Set IoT event parameters: `PUT /ISAPI/System/IOT/channels/<IoTChannel>/<IoTEventType>?format=json`
   - Get IoT event parameters: `GET /ISAPI/System/IOT/channels/<IoTChannel>/<IoTEventType>?format=json`
4. Configure arming schedule for a single IoT channel.
   - Get arming schedule of a single IoT channel: `GET /ISAPI/Event/schedules/IOT/<IoTEventType>/<IoTEventType>-<IoTChannel>`
   - Set arming schedule of a single IoT channel: `PUT`

`/ISAPI/Event/schedules/IOT/<IoTEventType>/<IoTEventType>-<IoTChannel>`

5. (Optional) Configure event audio arming schedule. If it is not configured, the device will play audio alarm whenever a linkage alarm is triggered. If it is configured, the device will only play the audio alarm within the arming schedule, and will upload the alarm without playing audio for the rest of the time.
   - Get event audio linkage configuration capability: `GET /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>/capabilities?format=json`
   - Set event audio arming schedule parameters: `PUT /ISAPI/Event/schedules/IoT/AudioAlarm/<IoTChannelID>/<IoTEventType>?format=json`
   - Get event audio arming schedule parameters: `GET /ISAPI/Event/schedules/IoT/AudioAlarm/<IoTChannelID>/<IoTEventType>?format=json`
6. Configure IoT event linkage, which consists of two parts: linkage method configuration and trigger method configuration.
   - Set IoT event linkage method: `PUT /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>`
   - Get IoT event linkage method: `GET /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>`
   - Set IoT event trigger method: `PUT /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>/notifications`
   - Get IoT event trigger method: `GET /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>/notifications`
7. (Optional) Configure event audio linkage.
   - Get event audio linkage configuration capability: `GET /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>/capabilities?format=json`
   - Set event audio linkage parameters: `PUT /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>?format=json`
   - Get event audio linkage parameters: `GET /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>?format=json`
8. Configure IoT channel OSD linkage.
   - Get OSD linkage configuration capability: `GET /ISAPI/System/IOT/channels/<IoTChannel>/OSD/capabilities?format=json`
   - Set OSD linkage parameters: `PUT /ISAPI/System/IOT/channels/<IoTChannel>/OSD?format=json`
   - Get OSD linkage parameters: `GET /ISAPI/System/IOT/channels/<IoTChannel>/OSD?format=json`
9. Upload alarm events. There are four types of IoT events:
   - IoT device events: `IOTD`
   - Getting access parameters: `accessParamsEvent`
   - Logistics scanning event: `LSE`
   - Getting the list of linked devices: `associatedDevicesSearchEvent`

drackun@gmail.com DETEKTOR

5.2 Arming and Subscription

5.2.1 Introduction to the Function

With arming and subscription, the client can establish HTTP persistent connection with the device, and continuously receive the event messages from the device.

For ISAPI event arming, the client can receive all types of events by GET method, or receive the subscribed events only by POST method.

5.2.2 API Calling Flow

5.2.2.1 Without Subscription

1. Establish a connection for arming: `GET /ISAPI/Event/notification/alertStream`. You need to set `Connection: keep-alive` in HTTP Headers.
2. When receiving events sent by the device, the event messages can be separated and parsed by boundary. See "Parsing Event Messages" below for details.
3. Disable the arming connection when you do not need to receive event messages.

5.2.2.1.2 Syntax

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/form-data; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!-- Some alarms are in JSON format, so the upper layer should parse based on the Content-Type field -->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length

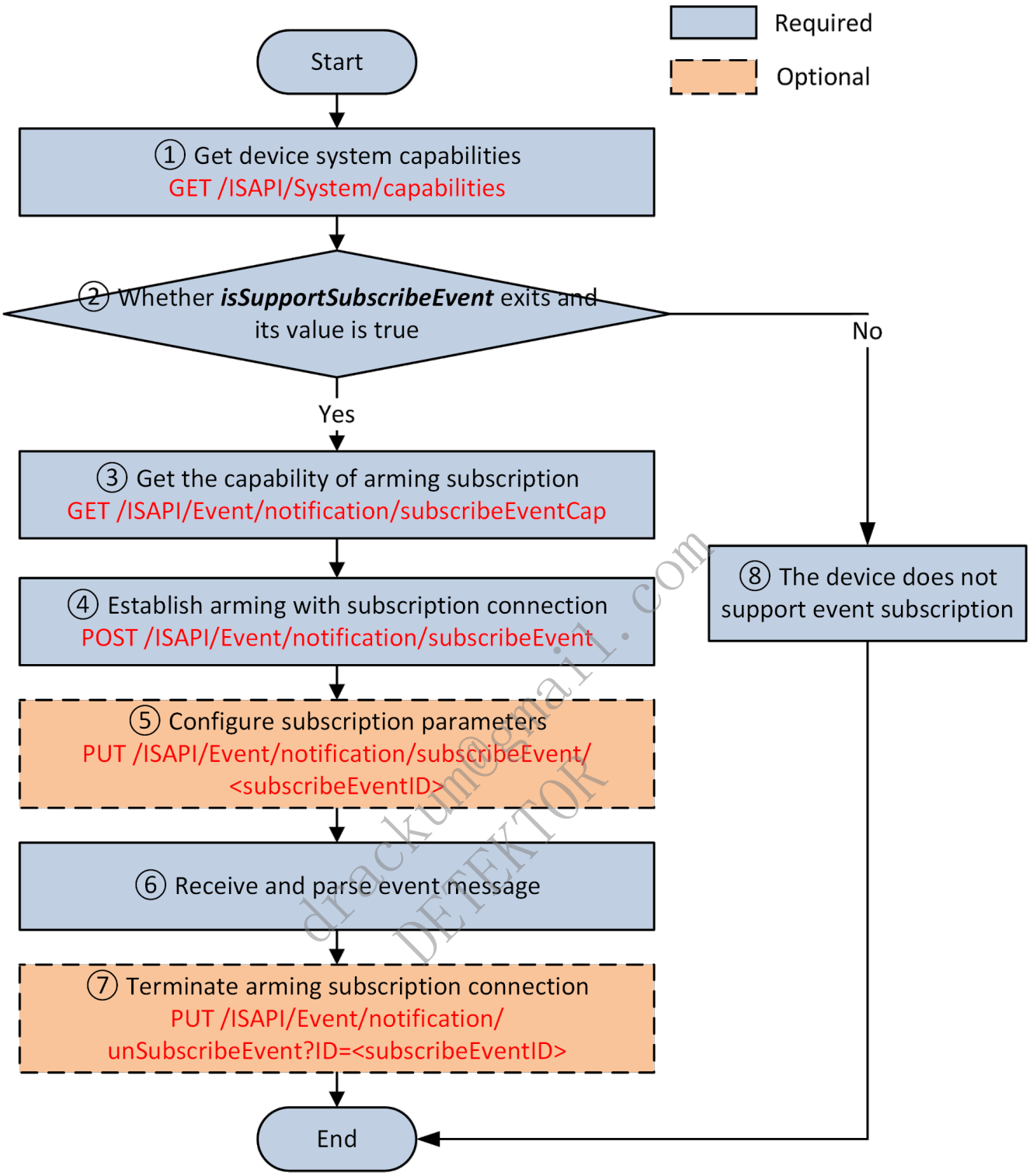
[Image Data]
--<frontier>
```

drackum@gmail.com
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Note: `<data_gateway_ip>` and `<frontier>` are variables, and `[Image Data]` is an abbreviated representation indicating the raw data of an image at this location.

5.2.2.2 Subscription

5.2.2.2.1 API Calling Flow



1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP Headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note:

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

5.2.2.2.2 Example

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Host: device_ip
Accept-Language: zh-cn
Date: YourDate
Content-Type: application/xml;
Content-Length: text_length
Connection: Keep-Alive

<SubscribeEvent/>
HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

drackum@gmail.com
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5.2.2.3 Event Messages Parsing

After the arming connection with the device is established, the data sent by the device is in HTTP form format (multipart/form-data). In an HTTP request, `Content-Type` in Headers is usually `multipart/form-data, boundary=AaB03x`, and boundary is a variable used to divide HTTP Body into multiple units, each being a set of data and has its own Headers and Body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Note the dash `--` before and after boundary. Under normal circumstances, the device will not actively close the arming connection, so the device will not send the form format end symbol `--AaB03x--` on the arming connection.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Image Data
--AaB03x--
```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. --boundary is the start of a form. --boundary-- is the end of the whole HTTP form message.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next \r\n to the next --boundary.

crackun@gmail.com DETEKTOR

5.2.3 Restriction Description

Note that ISAPI arming (with or without subscription) uses HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send out event messages continuously, while you cannot send any message to the device via the connection.

After the heartbeat time, if you do not receive any message from the device, you should disable the arming connection and try establishing a new one.

5.2.4 Sample Messages

5.2.4.1 Establish Arming Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

5.2.4.2 The Device Responses and Uploads an Event Message

```
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <ipAddress>10.17.133.46</ipAddress>
    <portNo>80</portNo>
    <protocol>HTTP</protocol>
    <macAddress>44:19:b6:6d:24:85</macAddress>
    <channelID>1</channelID>
    <dateTime>2017-05-04T11:20:02+08:00</dateTime>
    <activePostCount>0</activePostCount>
    <eventType>heartBeat</eventType>
    <eventState>active</eventState>
    <eventDescription>heartBeat</eventDescription>
</EventNotificationAlert>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length
Content-ID: image_ID

[Picture Data]
--<frontier>
```

5.3 Calling Flow of Device Packet Capture

5.3.1 Function Introduction

When problems arise after a device is deployed on site, interaction message between the device and the external network is necessary to help developers for troubleshooting. Packet capture can be stored on the local device, and packet capture files can be exported after capture is complete. Also, packet capture files can be uploaded to cloud storage, and packet capture data can be obtained in real-time even if the device does not have the storage space.

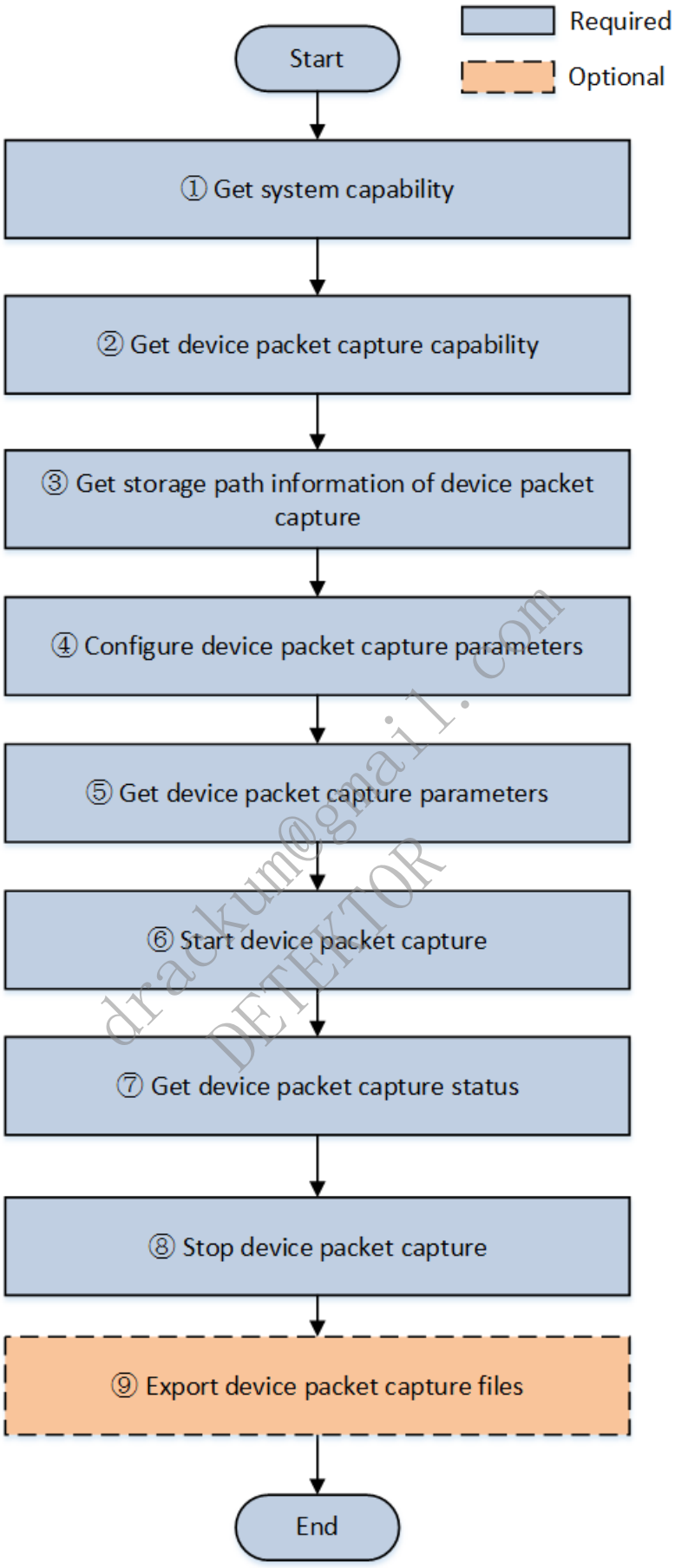
1. Device packet capture: save packet capture files on the local device, and export the files after capture is complete. Also, uploading packet capture files to cloud storage after capture is complete is supported. Then the client can obtain the storage URL and download packet capture files from the cloud storage.

2. Device real-time packet capture: after it is enabled, the device returns an URI for downloading packet capture data. The client can submit this URI to the browser to download the packet data. The device transmits packet data via HTTP Chunked, and users can store the packet data through the browser.

5.3.2 API Calling Flow

5.3.2.1 Device Packet Capture

dracknm@gmail.com
DEFEKTOR



drackum@gmail.com
DETEKTOR

1. Get device system capabilities: `GET /ISAPI/System/capabilities`. Get to know if the device supports packet capture by the field: `<isSupportNetworkCapture>true</isSupportNetworkCapture>`.

2. Check if the device supports packet captures: `GET /ISAPI/System/networkCapture/capabilities?format=json`. If `isSupportManualControl` is true, the device supports packet capture. If `isSupportManualControlAsyn` is true, the device supports asynchronous packet capture.

3. Get storage path information of device packet capture: `GET /ISAPI/System/networkCapture/StoragePathInfo?format=json`.

4. Configure device packet capture parameters such as capture duration, storage path, port, and address: `PUT /ISAPI/System/networkCapture/captureParams?format=json`.

5. Get device packet capture parameters such as capture duration, storage path, port, and address: `GET /ISAPI/System/networkCapture/captureParams?format=json`.

6. Start device packet capture: depending on the parameters, packet capture files can be saved on the local device for export after capture is complete, or packet capture files can be uploaded to cloud storage and downloaded via the storage URL, or the packet capture data can be returned in real-time.

Start device packet capture: `PUT /ISAPI/System/networkCapture/manualStart?format=json&asyn=<asyn>&realTime=<realTime>`.

Note:

- If the API does not contain URL parameters, the packet capture file is saved on the local device.
- If the API contains `asyn=true`, the packet capture file is automatically uploaded to cloud storage after capture is complete.

7. After starting capture, you can repeatedly get capture status, including whether the capture is ongoing, the size of the packet capture data, and the progress and storage URL for uploading the data to cloud storage.

Get status of device packet capture: `GET /ISAPI/System/networkCapture/manualStatus?format=json`.

8. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

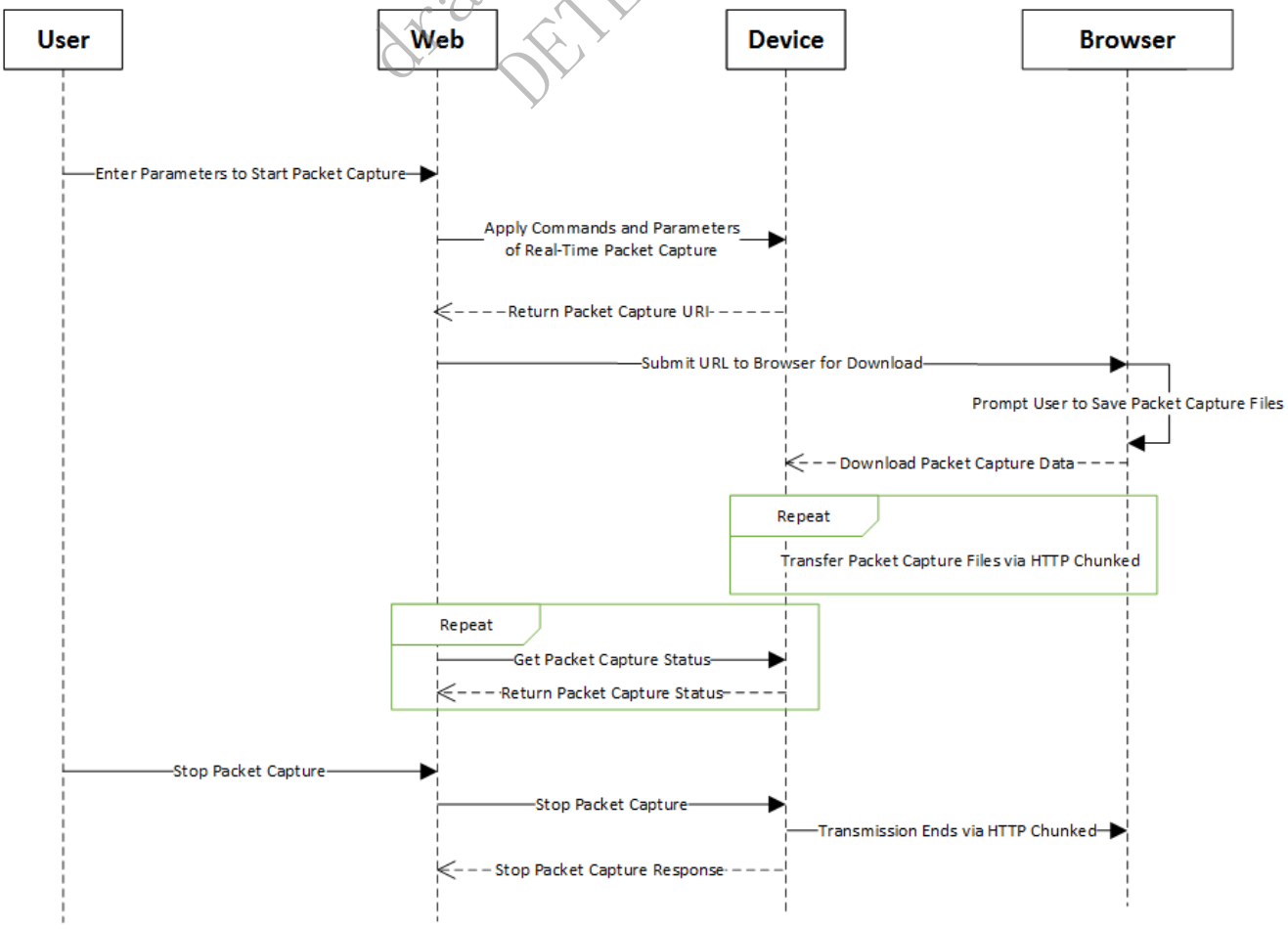
Stop device packet capture: `PUT /ISAPI/System/networkCapture/manualStop?format=json`.

9. (Optional) If packet capture data is stored on the local device, packet capture files need to be exported. If packet capture files are saved to cloud storage or captured in real-time, there is no need to export packet capture files.

Export device packet capture files: `GET /ISAPI/System/networkCapture/exportFile?format=json`.

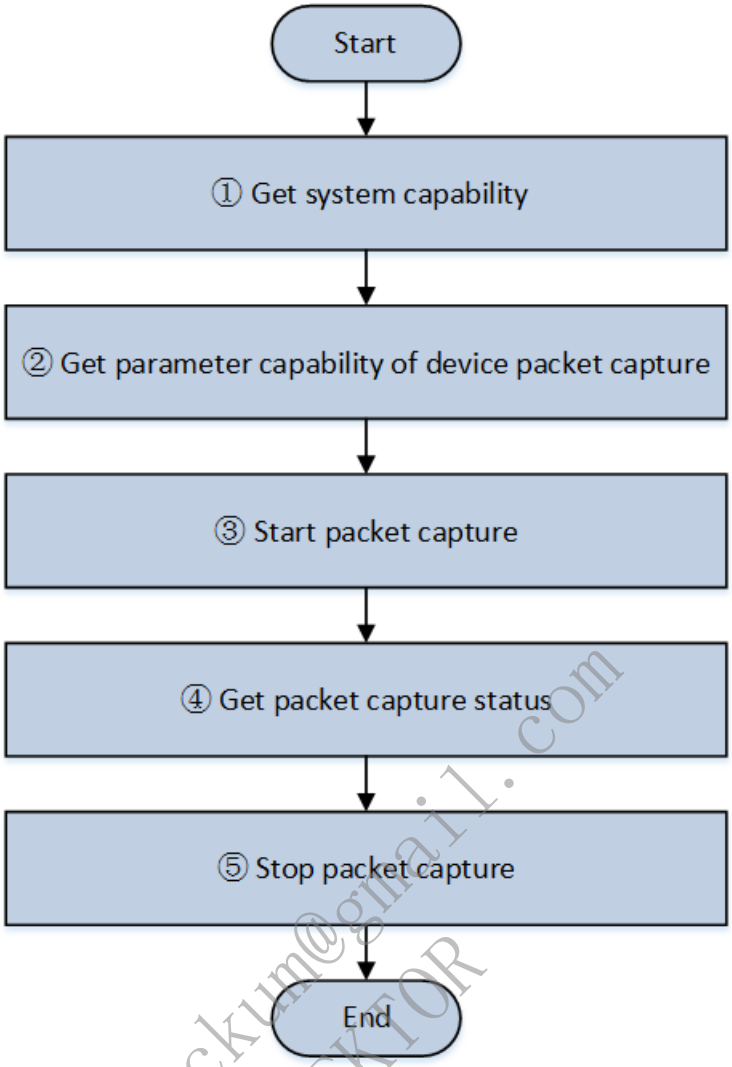
5.3.2.2 Device Real-Time Packet Capture

The process of real-time packet capture is shown as the following:



dracknn@gmail.com DETEKTOR

See the following figure for the calling flow:



crackun@gmail.com
DETECTOR

1. Get device system capabilities: `GET /ISAPI/System/capabilities`. `<isSupportStartNetworkCapture>true</isSupportStartNetworkCapture>` indicates the device supports starting packet capture. `<isSupportStopNetworkCapture>true</isSupportStopNetworkCapture>` indicates the device supports stopping packet capture. `<isSupportGetNetworkCaptureStatus>true</isSupportGetNetworkCaptureStatus>` indicates the device supports getting packet capture status.

2. Get capabilities of packet capture parameters: `GET /ISAPI/System/NetworkCaptureParams/capabilities?format=json`. The `realTimeEnabled` field indicates whether the device supports real-time packet capture.

3. Set the field realTimeEnabled as true in the parameters applied to the device to start device packet capture. The device returns an URI, and the client can download the real-time packet capture data from the device through a browser.

Start packet capture: `POST /ISAPI/System/StartNetworkCapture?format=json&security=<security>&iv=<iv>`.

Note:

- If `realTimeEnabled=true` is contained when starting device packet capture, it indicates packet capture data is uploaded in real-time by HTTP Chunked. The URL for downloading the returned packet capture data is valid for 30 seconds by default. If the download is attempted after this time, the device should return an HTTP 404 status code.

4. After starting packet capture, you can repeatedly get packet capture status, including whether packet capture is ongoing and the size of the packet capture data.

Get packet capture status: `GET /ISAPI/System/GetNetworkCaptureStatus?format=json`.

5. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

Stop device packet capture: `POST /ISAPI/System/StopNetworkCapture?format=json`.

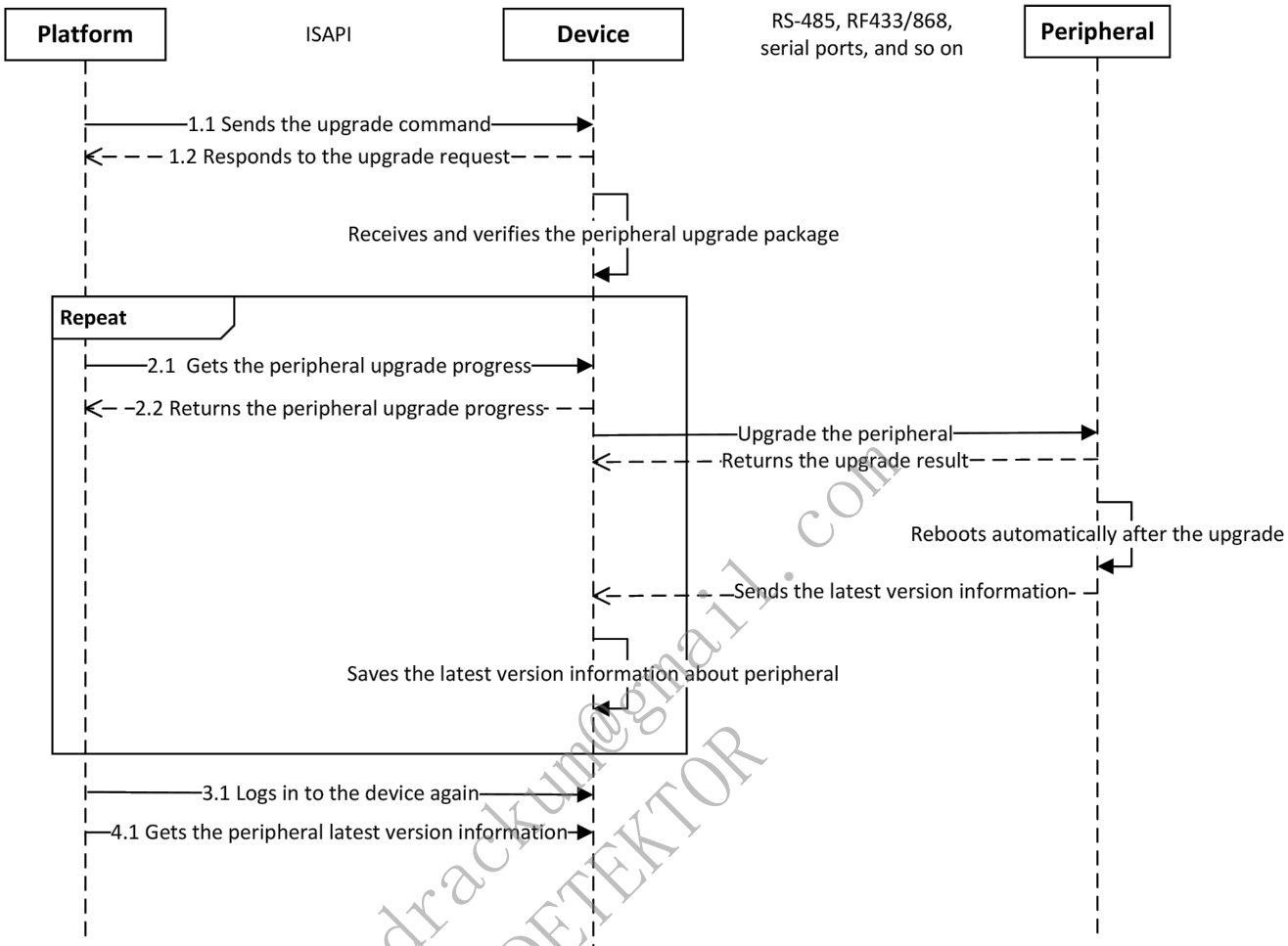
5.4 Device Peripherals Upgrade

5.4.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades device peripherals via ISAPI.

5.4.2 API Calling Flow

The sequence diagram of upgrading device peripherals by the platform is shown below.



drackum@gmail.com DETEKTOR

1. Get the device system capability `GET /ISAPI/System/capabilities` and check whether the device supports upgrading peripherals. If the field `isSupportAcsUpdate` is returned and its value is true, it indicates that the device supports this function, otherwise, the device does not support this function.
2. Get the capability of upgrading the peripherals module `GET /ISAPI/System/AcsUpdate/capabilities`, and get the types and IDs of peripherals that support upgrading.
3. The platform sends the upgrade command `POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>`. In the URL `type` refers to the peripheral type, `moduleAddress` refers to the peripheral module address, and `indexID` refers to the ID of peripheral to be upgraded. The platform will apply the upgrade peripheral package to the device.
4. Get the peripheral upgrade progress `GET /ISAPI/System/upgradeStatus?type=<Type>`.
5. Log in to the device again.
6. Get the peripheral latest version information.

5.5 Device Time Sync

5.5.1 Introduction to the Function

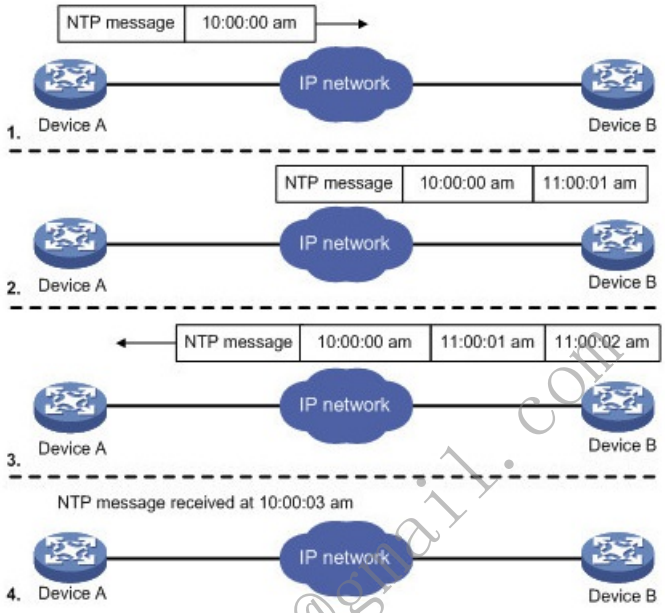
Time sync is a method to synchronize the time of all devices connecting to the NTP server, so that all devices can share the same clock time for providing related functions based on time. Supported time sync types: NTP time sync, manual sync, satellite time sync, platform time synchronization, etc. The following describes the method of NTP time sync.

5.5.1.1 NTP Time Sync

The local system of running NTP can receive sync from other clock sources (self as client), other clocks can sync from the local system (self as server), and sync with other devices.

The basic working principle of NTP is shown in the picture. Device A and Device B are connected via the network, and their systems follow their own independent system time. To auto sync their time, you can set device time auto sync via NTP. For example:

Before time sync between Device A and Device B, the time of Device A is 10:00:00 am, and that of Device B is 11:00:00 am. Device B is set as the server of NTP server, so that the time of Device A should be synchronized with that of Device B. The time of NTP message transmitted between Device A and Device B is 1 second.



The working process of system clock synchronization is as follows:

Device A sends an NTP message to Device B with a timestamp of 10:00:00 am (T1) that is when it leaves Device A.

When the NTP message reaches Device B, Device B will add its own timestamp, which is 11:00:01 am (T2).

Then the NTP message leaves Device B with Device B's timestamp, which is 11:00:02 am (T3).

Device A receives the response message, and the local time of Device A is 10:00:03 am (T4).

Above all, Device A can calculate two important parameters:

Round-trip delay of NTP message: Delay = (T4-T1) - (T3-T2) = 2 seconds.

Time difference between Device A and Device B: offset = ((T2-T1)+(T3-T4))/2=1 h.

Device A can sync its own time with that of Device B according to calculation results.

dracktum@gmail.com
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5.5.2 API Calling Flow

5.5.2.1 Time Sync Configuration

1. Get the Capability of Device Time synchronization Management

You can call this API to get the time sync types currently supported by the device, such as NTP time sync, manual time sync, satellite time sync, EZ platform time sync.

Get the capability: `GET /ISAPI/System/time/capabilities`.

2. Set device time synchronization management parameters

You can configure the time synchronization mode as follows：

Get device time synchronization management parameters: `GET /ISAPI/System/time`;

Set device time synchronization management parameters: `PUT /ISAPI/System/time`;

NTP time synchronization: See 4.2.2 NTP Time Sync (Client).

Manual time synchronization: Set the value of `timeMode` to `manual`, and set the device local time in nodes `localTime`、`timeZone`.

Satellite time synchronization: Set the value of `timeMode` to `satellite`, and set the device local time in nodes `satelliteInterval`.

Platform time synchronization: Set the value of `timeMode` to `platform`.

Note： For manual time synchronization (time offset including time zone offset): Set manual time synchronization: localTime refers to the local time on device (time offset excluded, in format like 2019-02-28T10:50:44); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00); Get manual time synchronization: localTime refers to the local time on device (time offset included, in format like 2019-02-28T10:50:44+8:30); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00).

5.5.2.2 Time Zone Configuration

1. Get device time zone configuration capability

Call `GET /ISAPI/System/capabilities` to get the system capability. When `isSupportTimeZone` is returned, the time zone configuration is supported by the device.

2. Configure time zone parameters

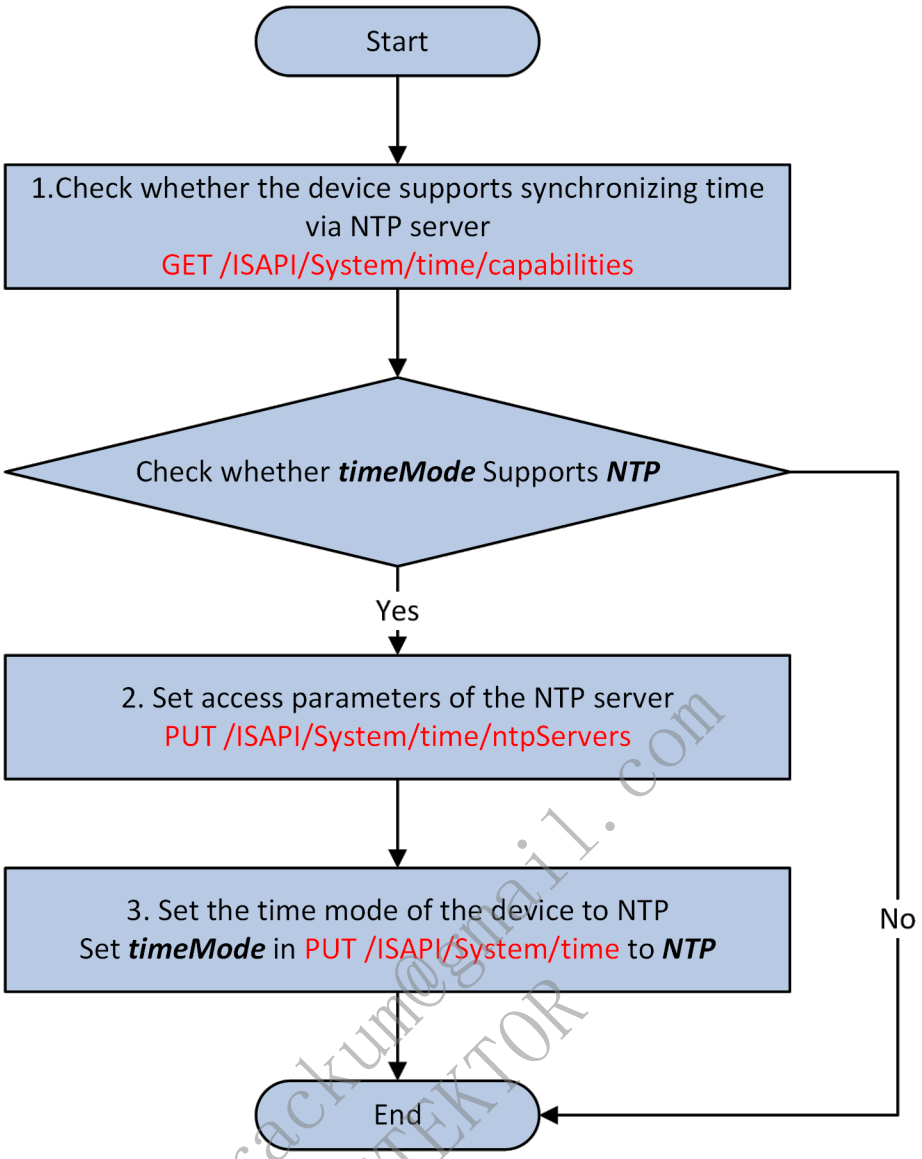
Get the device's time zone parameters: `GET /ISAPI/System/time/timeZone`. Set the device's time zone parameters: `PUT /ISAPI/System/time/timeZone`.

If DST (Daylight Saving Time) is disabled, the example of returned time zone parameters is: CST-8:00:00. It refers to UTC+8, and -8:00:00 is the UTC local time. If DST (Daylight Saving Time) is enabled, the example of returned time zone parameters is: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.2/02:00:00. It refers to UTC+8, the DST time is 30 minutes ahead of local time, the DST starts at 02:00:00 on the first Sunday of April and ends at 02:00:00 on the fifth Tuesday of October. MX.Y.Z: X is the month, Y is the week number in the month, Z is the day of a week (0-Sunday, 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday).

5.5.2.3 NTP Time Sync (Client)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as client):

drackum@gmail.com
DETEKTOR



drackum@gmail.com
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1. Check whether the device supports synchronizing time via NTP server Get the capability of the device: `GET /ISAPI/System/time/capabilities`, and check whether `timeMode` supports `NTP`.

2. Set access parameters of the NTP server

Supports accessing the NTP server by IP address to synchronize the device time.

Get the access parameter capability of the NTP server: `GET /ISAPI/System/time/ntpServers/capabilities`

Set access parameters of the NTP server: `PUT /ISAPI/System/time/ntpServers`

Get access parameters of the NTP server: `GET /ISAPI/System/time/ntpServers`

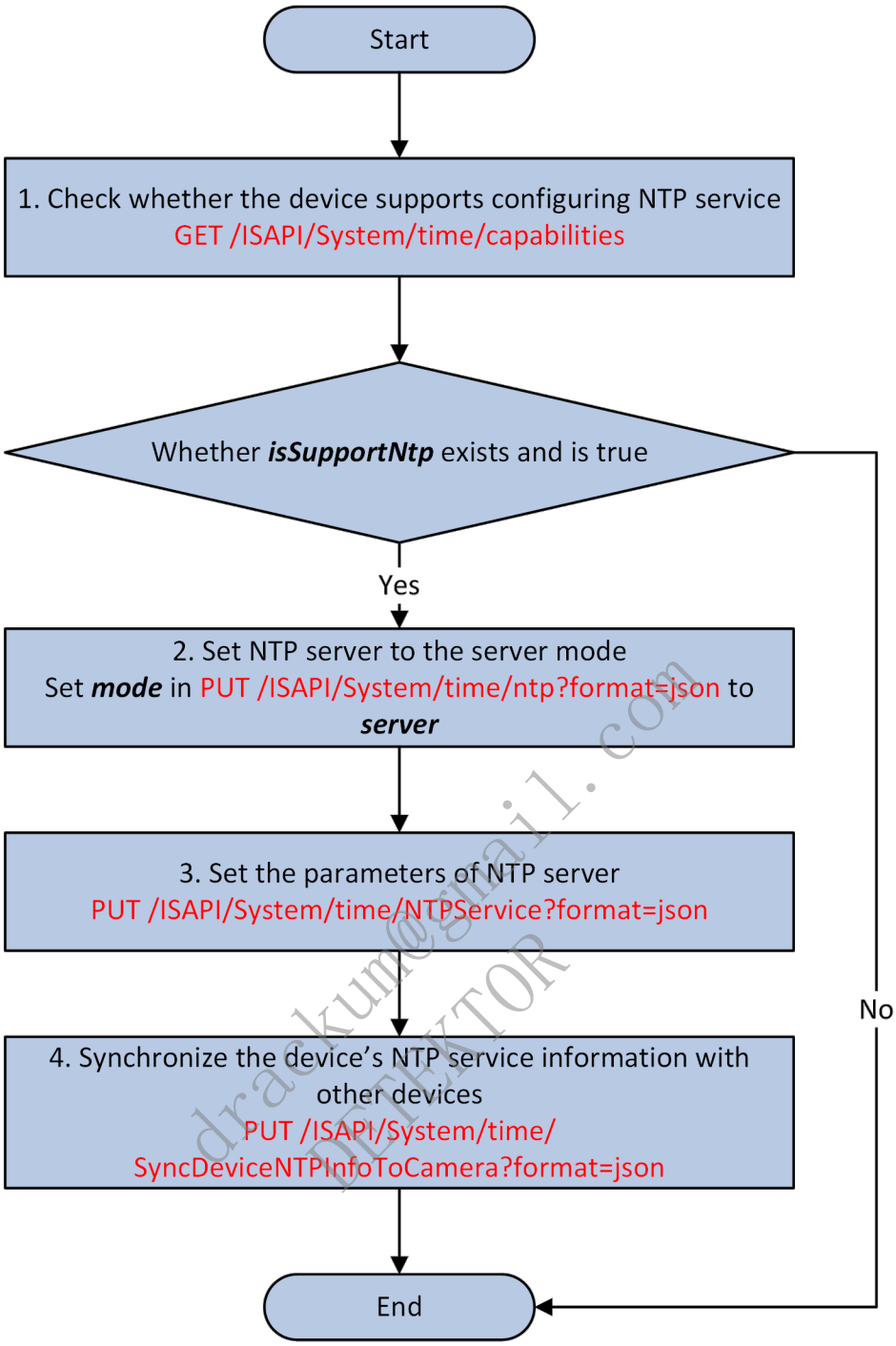
3. Set the time mode of the device to NTP

Supports setting the value of `timeMode` to `NTP`.

Get device time synchronization management parameters: `GET /ISAPI/System/time` Set device time synchronization management parameters: `PUT /ISAPI/System/time`

5.5.2.4 NTP Time Sync (Server Mode)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as server):



1. Check whether the device supports configuring NTP service Get the capability of device time synchronization management: `GET /ISAPI/System/time/capabilities`; If `isSupportNtp` is returned, it indicates that the device supports time synchronization management.

2. Set NTP server to the server mode

Supports setting the value of `mode` to `server`.

Get the capability of server mode: `GET /ISAPI/System/time/ntp/capabilities?format=json`

Set NTP to server mode: `PUT /ISAPI/System/time/ntp?format=json`

Get parameters of NTP server mode: `GET /ISAPI/System/time/ntp?format=json`

3. Set the parameters of NTP server

Supports setting the IP address of the NTP server.

Get the capability of NTP server: `GET /ISAPI/System/time/NTPService/capabilitis?format=json`

Set the NTP server parameters: `PUT /ISAPI/System/time/NTPService?format=json`

Get the parameters of the NTP server: `GET /ISAPI/System/time/NTPService?format=json`

4. Synchronize the device's NTP service information with other devices

Supports synchronizing the time information to the camera.

Get the capability set of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json`

Synchronize device's NTP service information with the camera: `PUT /ISAPI/System/time/SyncDeviceNTPInfoToCamera?format=json`

Get the progress of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/Progress?format=json`

Search for the results of synchronizing device's NTP service information with the camera: `POST /ISAPI/System/time/SyncDeviceNTPInfoToCamera/SearchResult?format=json`

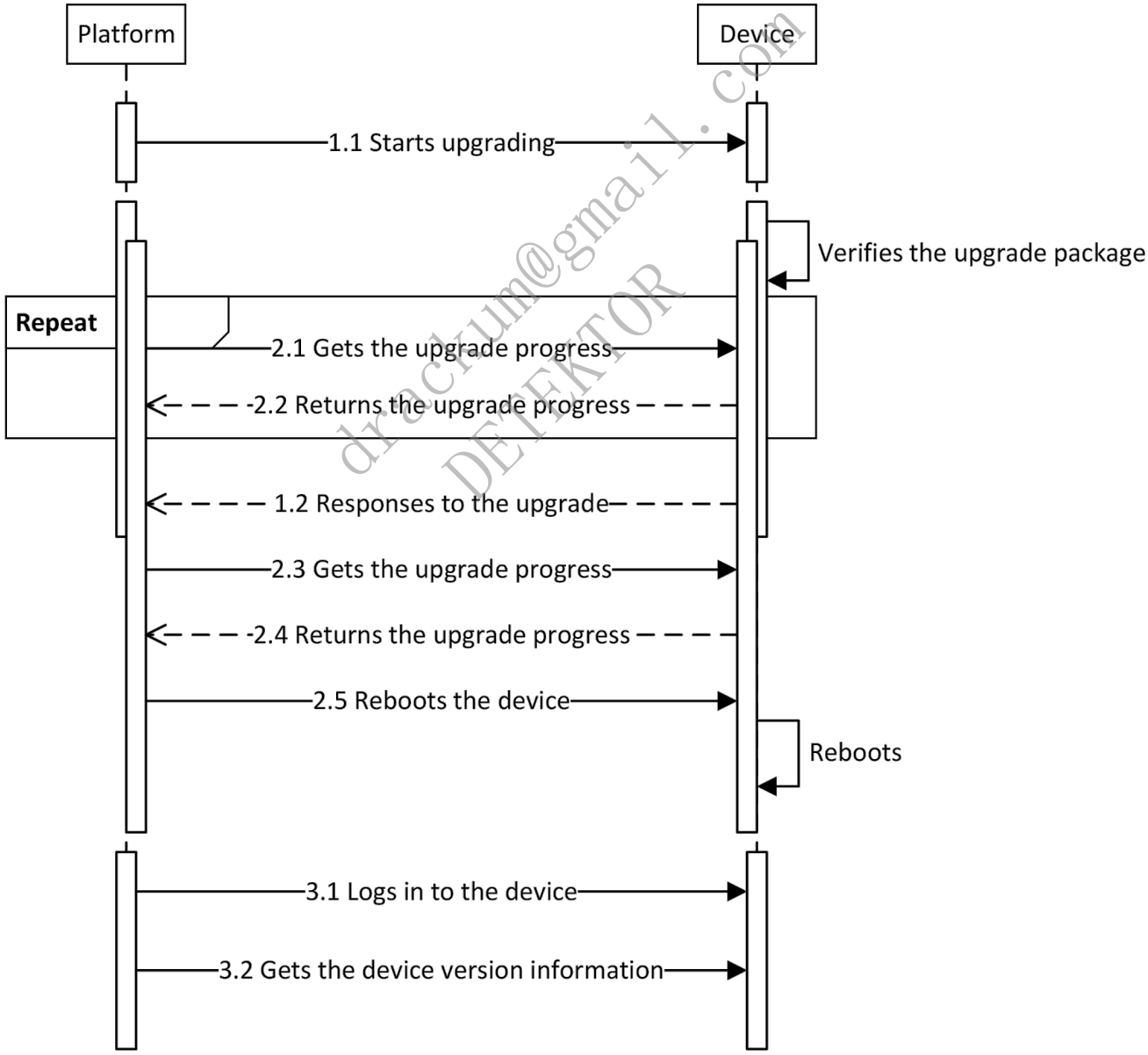
5.6 Device Upgrade

5.6.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades devices via ISAPI.

5.6.2 API Calling Flow

The sequence diagram of upgrading devices by the platform is shown below.



dracium@gmail.com
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1. **Upgrade devices.**

   Upgrade the device firmware: `POST /ISAPI/System/updateFirmware`.

2. **Get the device upgrade progress.**

Get the device upgrade progress: `GET /ISAPI/System/upgradeStatus`.

3. **Reboot devices.**

   Reboot devices: `PUT /ISAPI/System/reboot`.

5.7 Mutually Exclusive Functions

5.7.1 Introduction to the Function

Some functions are mutually exclusive due to the device performance (for example, function A and function B cannot run at the same time, i.e, only one of them is allowed at one time).

5.7.2 API Calling Flow

The following three APIs are available for the integration of mutually exclusive functions:

1. Get the information of mutually exclusive functions: `GET /ISAPI/System/mutexFunction/capabilities?format=json`. Call this URL to get the list of existing mutually exclusive functions supported by the device. Note: NVR devices only support setting exlusive function "perimeter" (perimeter protection), and do not support "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), or "regionExiting" (region exiting).
2. Search for the functions that are mutually exclusive with a specified function: `POST /ISAPI/System/mutexFunction?format=json`. Based on the list of mutually exclusive functions returned by `GET /ISAPI/System/mutexFunction/capabilities?format=json`, you can search for the mutual exclusion status of a specified function and see whether to change the settings and disbale the mutually exclusive function.
3. Get the mutual exclusion information when device function exception occurs: `GET /ISAPI/System/mutexFunctionErrorMsg`. After getting the error code, you can call this API to get the current mutually exclusive functions.

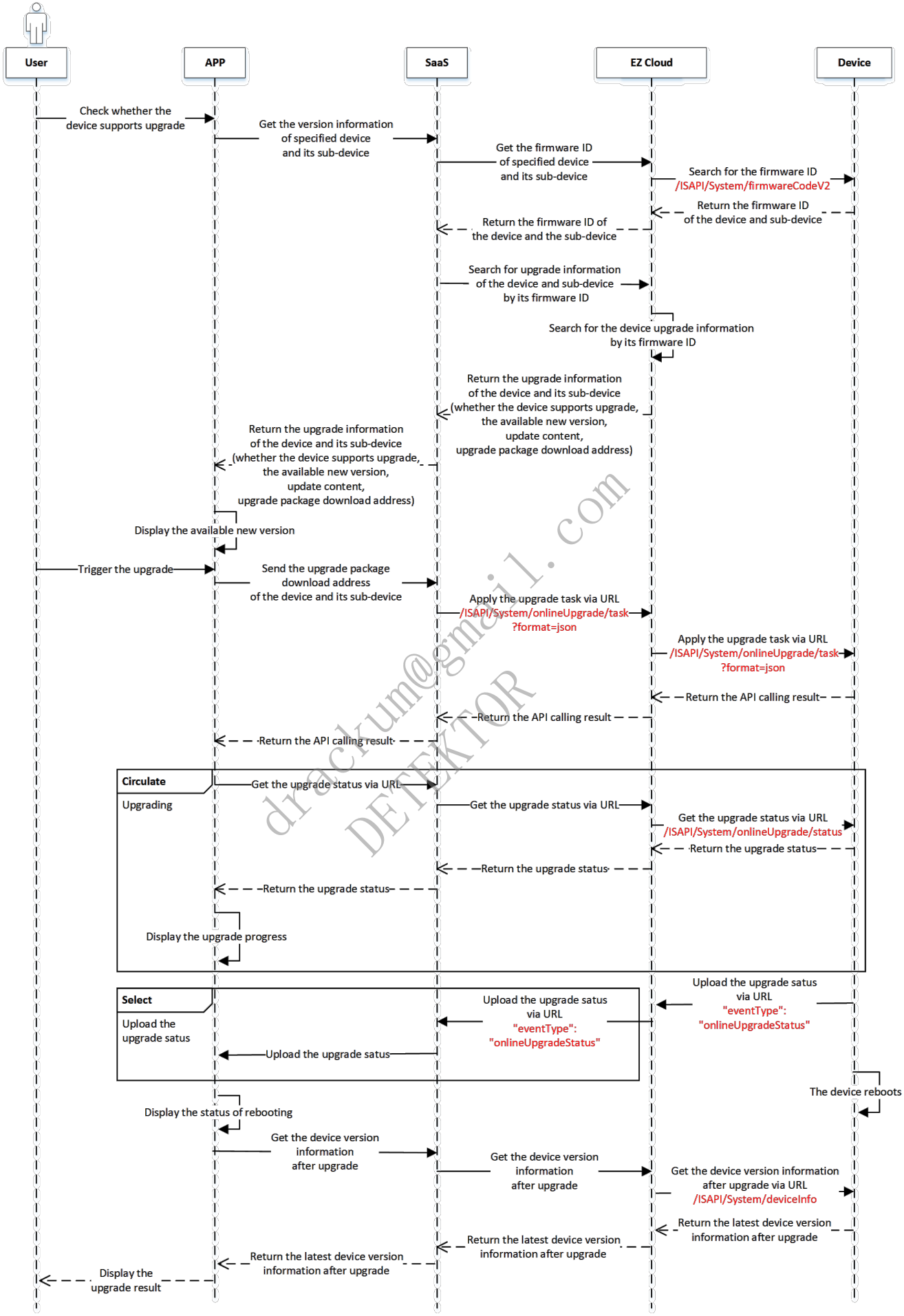
5.8 Remote Online Upgrade Based on The EZ2.0

5.8.1 Introduction to the Function

Considering that the device is connected to EZ via EZ 2.0 Protocol, EZ OTA upgrade solution is not supported. This solution applies to specified version upgrade.

5.8.2 API Calling Flow

dr.ackum@gmail.com DEFEKTOR



1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by its firmware ID. If there is an available new version, apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade devices (e.g., NVR) or sub-devices (e.g., IPC) via the node

device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.

3. During the device upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` and reboot automatically.
5. After both upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

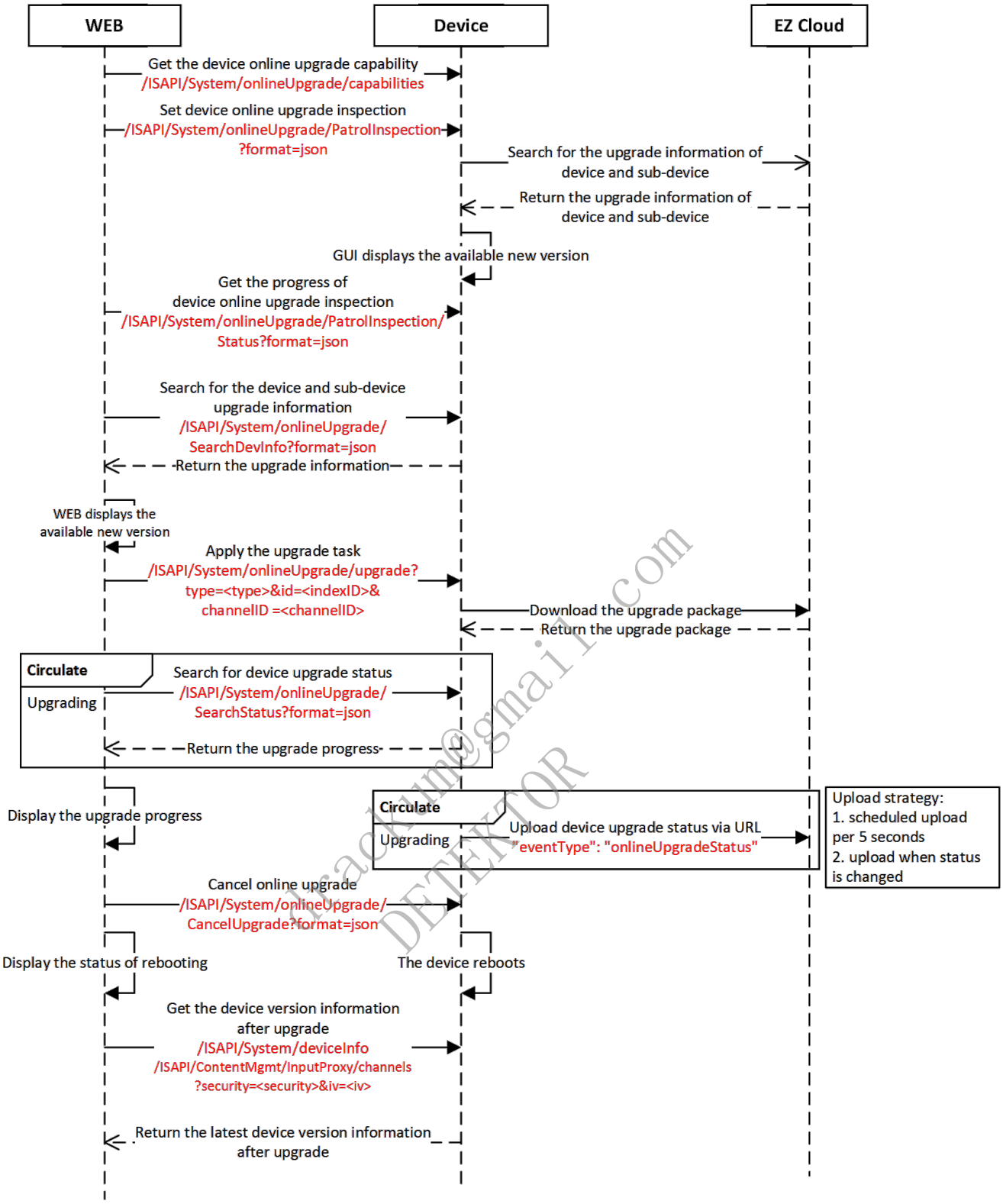
5.9 Remote Online Upgrade Based on the WEB Client

5.9.1 Introduction to the Function

This solution applies to remote online upgrade of the devices and sub-devices connected to WEB client.

5.9.2 API Calling Flow

drackum@gmail.com
DETEKTOR



1. Get device online upgrade capability: `GET /ISAPI/System/onlineUpgrade/capabilities`.
2. Manually trigger the device online upgrade inspection: `POST /ISAPI/System/onlineUpgrade/PatrolInspection?format=json`. After it is enabled, the device will search for the available new version.
3. During the inspection, the WEB client can get the real-time inspection progress: `GET /ISAPI/System/onlineUpgrade/PatrolInspection/Status?format=json`.
4. After the device inspection is finished, search for the device and sub-device upgrade information: `GET /ISAPI/System/onlineUpgrade/SearchDevInfo?format=json`.
5. Apply the upgrade task: `PUT /ISAPI/System/onlineUpgrade/upgrade?type=<type>&id=<indexID>&channelID =<channelID>`. You can upgrade a specified device via the node `<channelID>` in the URL. Specify sub-device upgrade by configuring `<type>`.

6. You can search for online upgrade status during the upgrade: `GET /ISAPI/System/onlineUpgrade/SearchStatus?format=json`.
7. You can cancel the upgrade during the upgrade: `PUT /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json`.
8. After the upgrade and reboot are finished, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

5.10 Serial Port Accessed External Device Management

5.10.1 Introduction to the Function

Information management of device accessed the serial are as follows: 1. Configure manufacturer, type, and model information of the specific serial port access device. 2. Search for the device type or model supported by the specific serial port.

5.10.2 API Calling Flow

1. Check whether the device supports information management of serial port. Get the capability of the device serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportDeviceInfo>` is returned, the device supports information configuration of devices access the serial port.

2. Set the information of serial port:

- Get the capability of device information parameters of a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Get device information parameters access single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Set device information parameter of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.

3. Check whether the device supports linking information of devices access the serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportSearchDeviceInfoRelations>` is returned, it indicates that the device supports searching for linked information od devices access the serial port.

4. Search for linked information of devices access the serial port.

- Get the capability of searching for linked parameters of information of devices access a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations/capabilities?format=json`.
- Search for linked parameters of information of devices access a single serial port:`POST /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations?format=json`.

5.11 Serial Port Data Transparent Transmission

5.11.1 Introduction to the Function

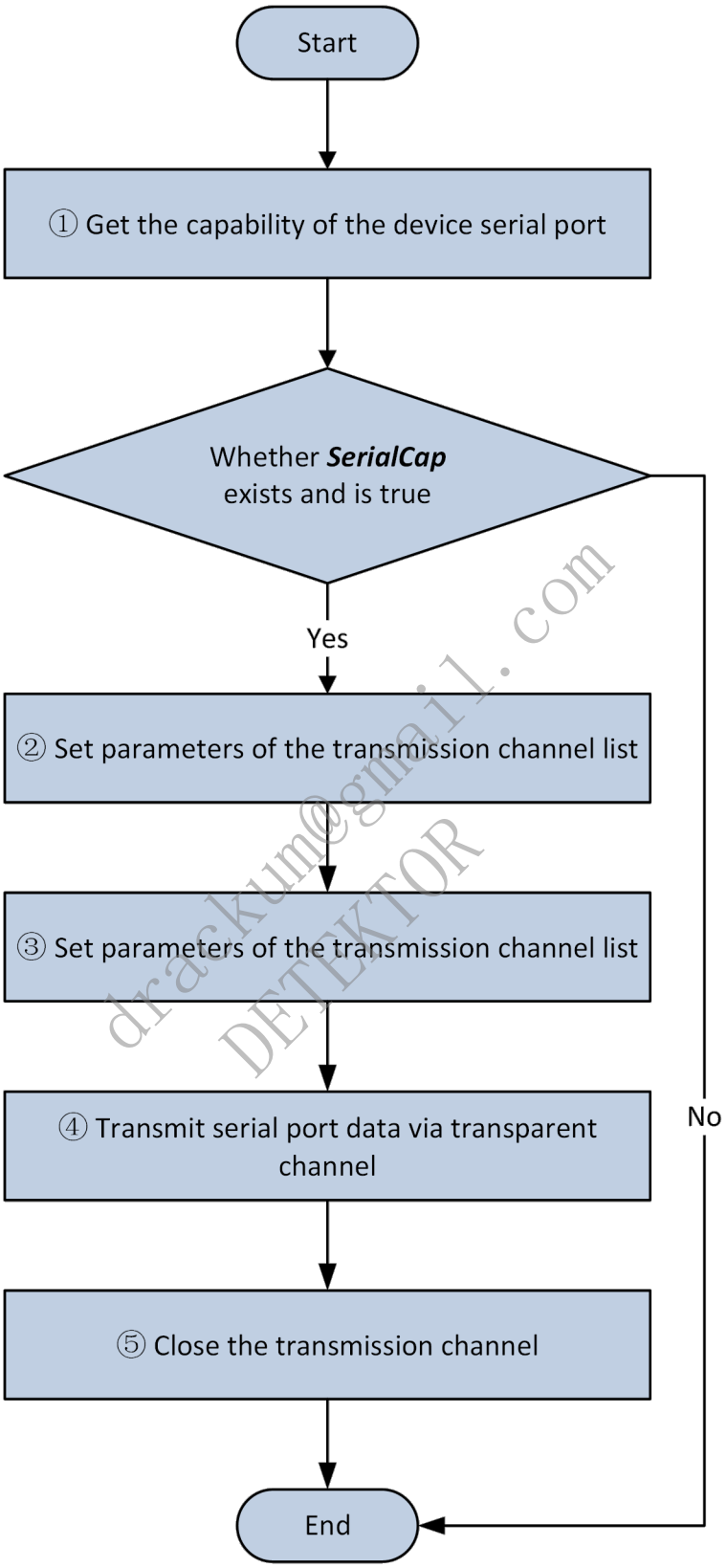
RS485, RS422 and RS232 serial ports external to the device are used as transparent channels to transmit serial port data. Supports the client sending serial data to the device, which then forwards the data to the serial port; conversely, when the serial port sends data to the device, which transparently transmits the data to the client. Note that the device's transparent transmission of serial data is half-duplex communication, meaning that bidirectional communication can be implemented between the device and the client, but the communication data is not necessarily in a question-and-answer format, and the client must match the request and response relationship on its own.

Since most standard HTTP request libraries do not support the persistent connection for receiving and sending data in real time, it is recommended that the client uses two TCP clients to implement the sending and receiving of

drackun@gmail.com DETEKTOR

transparently transmitted serial data.

5.11.2 API Calling Flow



drackum@gmail.com
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1. Check whether the device supports serial port data transmission.

Get the capability of the device serial port: `GET /ISAPI/System/capabilities`. If `SerialCap` is returned and the value is true, it indicates that the device supports the functions of the serial port.

2. Set parameters of the transmission channel list.

- Get parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.
- Configure parameters of the specific transmission channel: `GET`

`/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.

3. Open the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/open`;

4. Transmit serial port data via transparent channel.

- Receive data uploaded by device serial port through transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.
- Send data to device serial port through transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.

5. Close the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/close`.

5.12 Serial Port Parameter Configuration

5.12.1 Introduction to the Function

Serial port parameter configuration.

5.12.2 API Calling Flow

1. Check whether the device supports configuring serial port parameters.

Get the capability of device serial port: `GET /ISAPI/System/capabilities. If is returned and its value is true, the device supports functions of serial port.

3. Get parameters of all serial ports.

- Get the capability of all serial ports: `GET /ISAPI/System/Serial/capabilities`.
- Get control parameters of all serial ports: `GET /ISAPI/System/Serial/ports?permissionController=<indexID>`.

4. Set control parameters of a single serial port.

- Get control parameters of single serial port: `GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.
- Configure control parameters of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.

5. Get the status of single serial port:`GET /ISAPI/System/Serial/ports/<portID>/status`.

dracknn@gmail.com
DETEKTOR

5.13 Sub-device Batch Upgrade

5.13.1 Introduction to the Function

Sub-device Batch Upgrade (Single Task): applicable to situations of upgrading multiple sub-devices using one upgrade package. Application scenarios include: when the UWB positioning anchor connects to the web, upgrading multiple tags (sub-devices) through the positioning engine (gateway), with the same upgrade package.

Management of Sub-device Batch Upgrade Tasks: There are many types of sub-devices, such as LoRa nodes, which have a slow upgrade process and different models of LoRa nodes in the field also require different upgrade packages. Therefore, it is necessary to support the creation of multiple upgrade tasks at once, entrusting the upgrade to the devices, to enhance the practicality of batch upgrading sub-devices.

5.13.2 API Calling Flow

Sub-device Batch Upgrade (Single Task)

Get the device capability: `/ISAPI/System/capabilities`.

If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that the device support the function.

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that the device support getting the progress of batch upgraading sub-devices.

Sub-device batch upgrade: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.

Batch sub-device upgrade progress search: `POST /ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json`.

Sub-device Batch Upgrade Task Management

Get device system capabilities `/ISAPI/System/capabilities`, If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that batch upgrade of sub devices is supported,

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that progress of batch upgrading sub devices can be searched,

If the node `isSupportSearchBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading devices can be searched,

If the node `isSupportModifyBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading sub devices can be edited,

If the node `isSupportDeleteBulkUpgradeChildDeviceTask` exists and is true, it indicates that task of batch upgrading sub devices can be deleted.

Get the capability of batch upgrading sub-device: `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`.

Batch upgrade sub devices: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.****

Search for tasks of batch sub-device upgrade: `POST /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`.

Edit tasks of batch sub-device upgrade: **`PUT /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`**.****

Delete tasks of batch sub-device upgrade: `PUT /ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json`.

Note: 1. The two methods for batch upgrading child devices are the same API: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`. If the `BulkUpgradeChildDeviceListCap.taskID` is returned in `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`, it indicates that the task method is supported in the sub device upgrade. 2. To determine which sub devices can be upgraded, you can use the API `POST /ISAPI/IoTGateway/Childmanage/SearchChild?format=json`. Sub devices with the `tag` `isUpgradable` and `value` `true` indicate that the sub devices support upgrades.

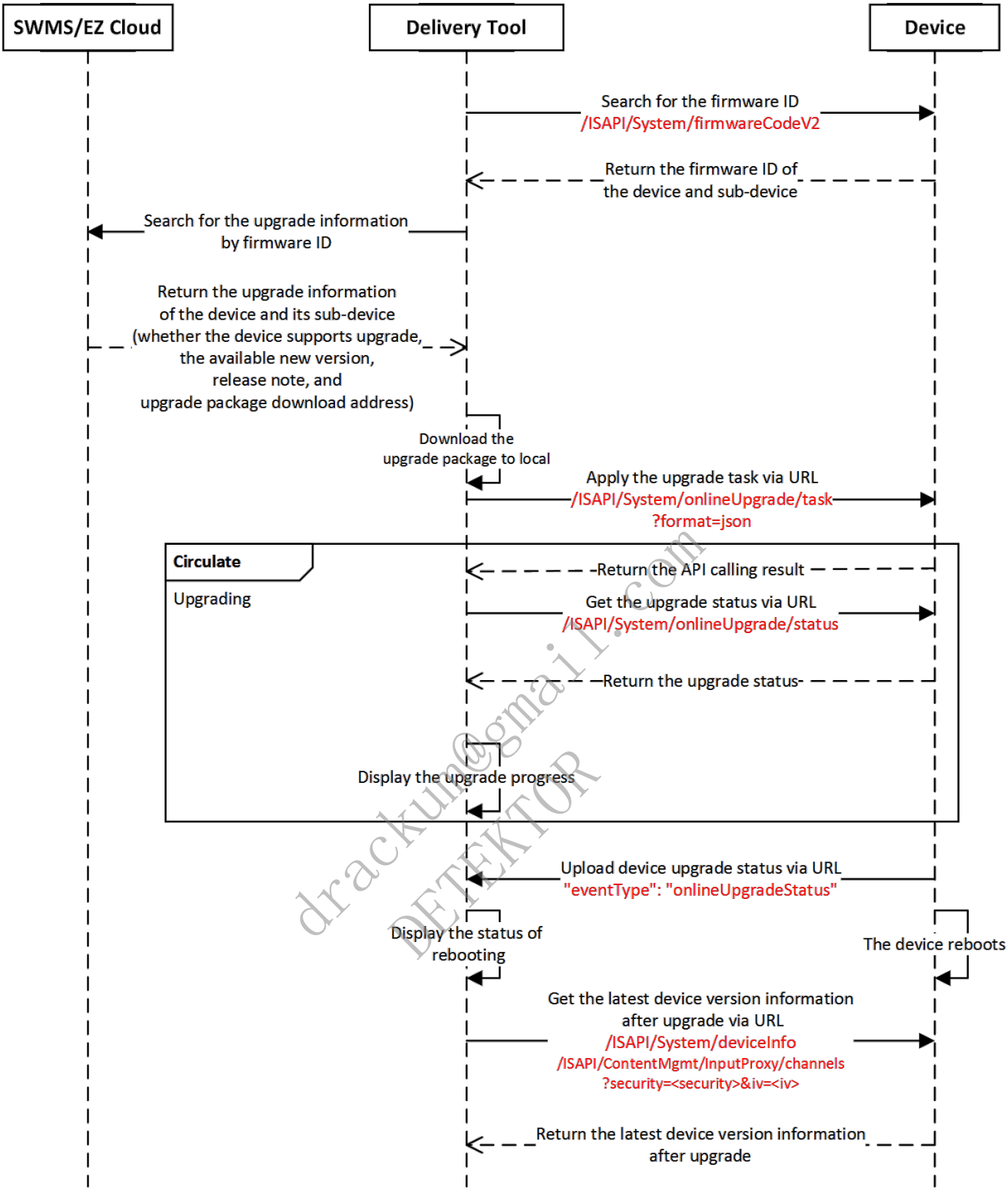
dracjun@gmail.com DEFKON

5.14 Upgrade Based on Interactive Tools Local Deployment

5.14.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI. If the device is connected via ISAPI protocols, it is required to set up HTTP(s) service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK, you can implement the device upgrade via transmitting ISAPI.

5.14.2 API Calling Flow



1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by firmware ID on the SWMS/EZ Cloud, download upgrade package to local storage, and apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., camera) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress: `GET /ISAPI/System/onlineUpgrade/status`.
4. After upgrade is completed, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and reboot automatically.
5. After the device rebooting, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=`

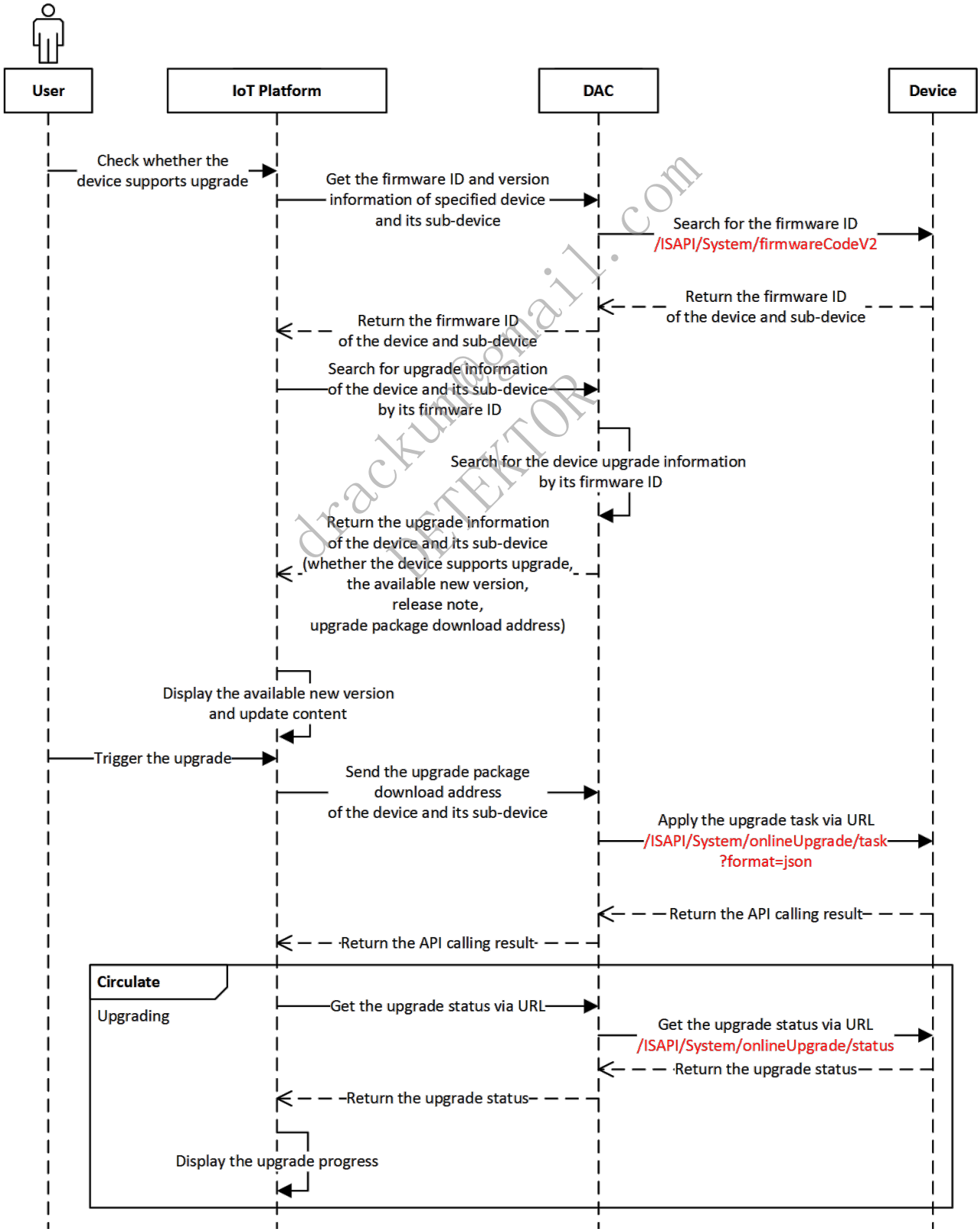
`<security>&iv=<iv>`.

5.15 Upgrade Based on Platform Local Deployment

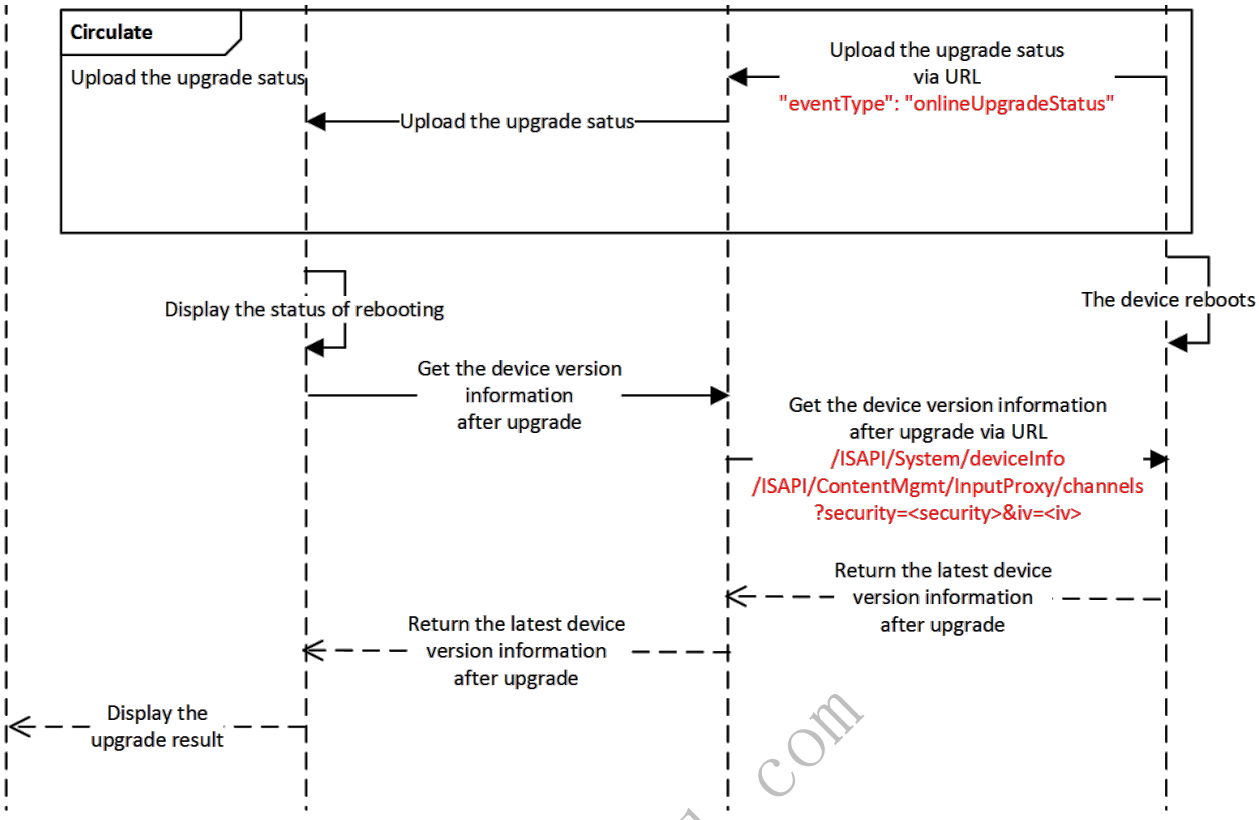
5.15.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI and ISUP. If via ISAPI, it is required to set up HTTP(s) service or transfer binary upgrade package files so as to store device upgrade package and generate URL for downloading upgrade package. If via ISUP, it is required to set up FTP service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK protocols, you can realize device upgrade via transmitting ISAPI.

5.15.2 API Calling Flow



dracku@gmail.com
HIKVISION



1. Search for device firmware ID on the platform: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information in DAC by device firmware ID on the platform. If there is an available new version, DAC will trigger an upgrade task and apply it to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and will reboot automatically.
5. After the upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

dracku@gmail.com
DETEKTOR

6 Video (General)

6.1 . OSD

6.1.1 Introduction to the Function

On-screen display is applied in the display interface of cameras, and is used to display some characters or graphics on the video to provide some information to the audience. After OSD configurations on the client software, the configured information text will be overlaid on the video stream of cameras and displayed on the screen.

The typical application scenarios of dynamic OSD: elevator floor number overlay for elevator monitoring, characters overlay for toll stations on highway/road, notification information overlay during live video programs. The typical application scenarios of static OSD: fixed information (such as camera position, date, time) overlay on the video signal.

6.1.2 API Calling Flow

6.1.2.1 OSD

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`.
2. The client software configures OSD parameters, including whether to enable OSD, custom text overlay, time overlay, and overlay format, for a specified channel.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

   Set: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

   Remarks:

   - The configuration URL supports various parameters, including custom text overlay, channel name overlay, time overlay, overlay format, and so on. Some of these parameters can also be configured by other URLs, as shown below: `TextOverlayList` can also be configured via URL of custom text overlay.

     `channelNameOverlay` can also be configured via URL of channel name overlay.

     `DateTimeOverlay` can also be configured via URL of channel time overlay.

   - The overlaid information alignment mode (defined by `alignment`) refers to the alignment mode of custom text:

     `customize`: custom mode. The client software sends the overlay position coordinates of custom text to the device, and the device will overlay the characters on the stream for display according to the position coordinates.

     `alignRight`: right align. The device will display custom text on the right boundary of screen.

     `alignLeft`: left align. The device will display custom text on the left boundary of screen.

     `allRight`: all right align. The device will display all overlay contents, including custom text, the channel name, the time on the right boundary of screen.

     `allLeft`： all left align. The device will display all overlay contents, including custom text, the channel name, the time on the left boundary of screen.

dracknn@gmail.com DETEKTOR

6.1.2.2 Custom Text Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `TextOverlayList` is returned, it indicates that custom text overlay is supported and `size` defines the maximum number of custom texts.
2. The client software configures custom text OSD. `textID` refers to the custom text No., and the value range is: [1, `size`]:

   Create a custom text for a specified channel: `POST /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

   Get overlay parameters of a custom text for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Set overlay parameters of a custom text for a specified channel: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Delete a custom text: `DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Get overlay parameters of all custom texts for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

Set overlay parameters of all custom texts for a specified channel: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

Delete all custom texts for a specified channel: `DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

6.1.2.3 Channel Name Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `channelNameOverlay` is returned, it indicates that channel name overlay is supported.
2. The client software configures the channel name overlay, including whether to enable channel name overlay, overlay position, and so on.

   Get channel name overlay parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay`.

   Set channel name overlay parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay`.

6.1.2.4 Channel Time Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `DateTimeOverlay` is returned, it indicates that channel time overlay is supported.
2. The client software configures the channel time overlay, including whether to enable channel time overlay, overlay position, date format, and so on.

   Get channel time overlay parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay`.

   Set channel time overlay parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay`.

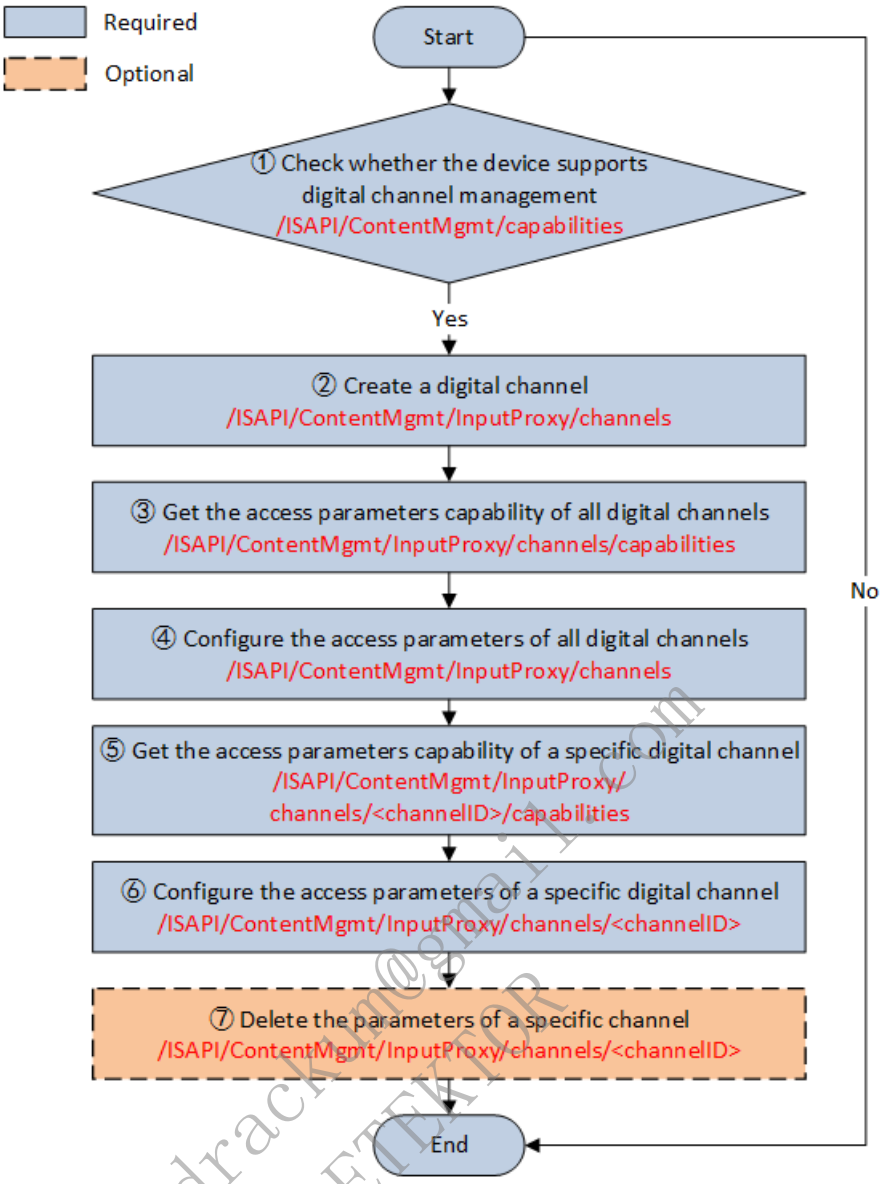
crackum@gmail.com DETEKTOR

6.2 Digital Channel Access Management

6.2.1 Introduction to the Function

- Video channel includes analog channel and digital channel. The service involves channel access protocols, streaming protocols, address information, version No. of channel access firmware, and so on.
- Analog channel refers to the local video channels of the device.
- Digital channel refers to the video channels accessed by the device by IP address or POE port (e.g., NVR accessing the video channel of network camera).
- Analog channel No. and digital channel No. are unique. Devices that do not support digital channels will only return analog channel No. For example, analog channel No. ranges from 1 to 8 (determined by the number of analog channel ports of the device), digital channel No. ranges from 9 to 32 (within the maximum number of digital channels, determined by the number of added network cameras).

6.2.2 API Calling Flow



1. Get the capability node to check whether the device supports digital channel management: `GET /ISAPI/ContentMgmt/capabilities`. If the value of the node `iSptInputProxyChanCap` is true, it indicates that the device supports digital channel management.
2. Add digital channels: `POST /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.
3. Get the access parameters capability of all digital channels: `GET /ISAPI/ContentMgmt/InputProxy/channels/capabilities?security=<security>&iv=<iv>`.
4. Configure the access parameters of all digital channels:

   Get: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

   Configure: `PUT /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.
5. Get the capability of configuring the access parameters of a specific digital channel: `GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/capabilities`.
6. Configure the access parameters of a specific digital channel:

   Get: `GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>?security=<security>&iv=<iv>`.

   Configure: `PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>?security=<security>&iv=<iv>`.
7. Delete the parameters of a specific digital channel: `DELETE /ISAPI/ContentMgmt/InputProxy/channels/<channelID>`.

6.2.3 Video Channel Application

1.Get the list of video channels, which is the sum of digital channels and analog channels. Digital channels and analog channels need to be obtained separately:

- 1) Call "Get all video input channel parameters" to get all analog channels via GET /ISAPI/System/Video/inputs/channels and VideoInputChannelList parameter is returned. The id represents the analog channel No.
- 2) Call the "Get all digital channel access parameters" to get all digital channels via GET /ISAPI/ContentMgmt/InputProxy/channels?security=&iv= and InputProxyChannelList parameter is returned. The id represents the digital channel No.

2. Video channel function configuration supports unified configuration of analog and digital channels. Both the analog channel interface starting with `/ISAPI/System/Video/inputs/channels/<channelID>/` and the digital channel interface starting with `/ISAPI/ContentMgmt/InputProxy/channels/<channelID>/` are supported. **It is recommended to use the channel configuration interface starting with `/ISAPI/System/Video/inputs/channels/<channelID>/`, where <channelID> represents all channel No.s supported by the device, including analog channels and digital channels.**

> For example, configure text overlay parameters for a channel: Both `GET/PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays` and `GET/PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/overlays` are supported, and it is recommended to use `/ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

6.3 Live View

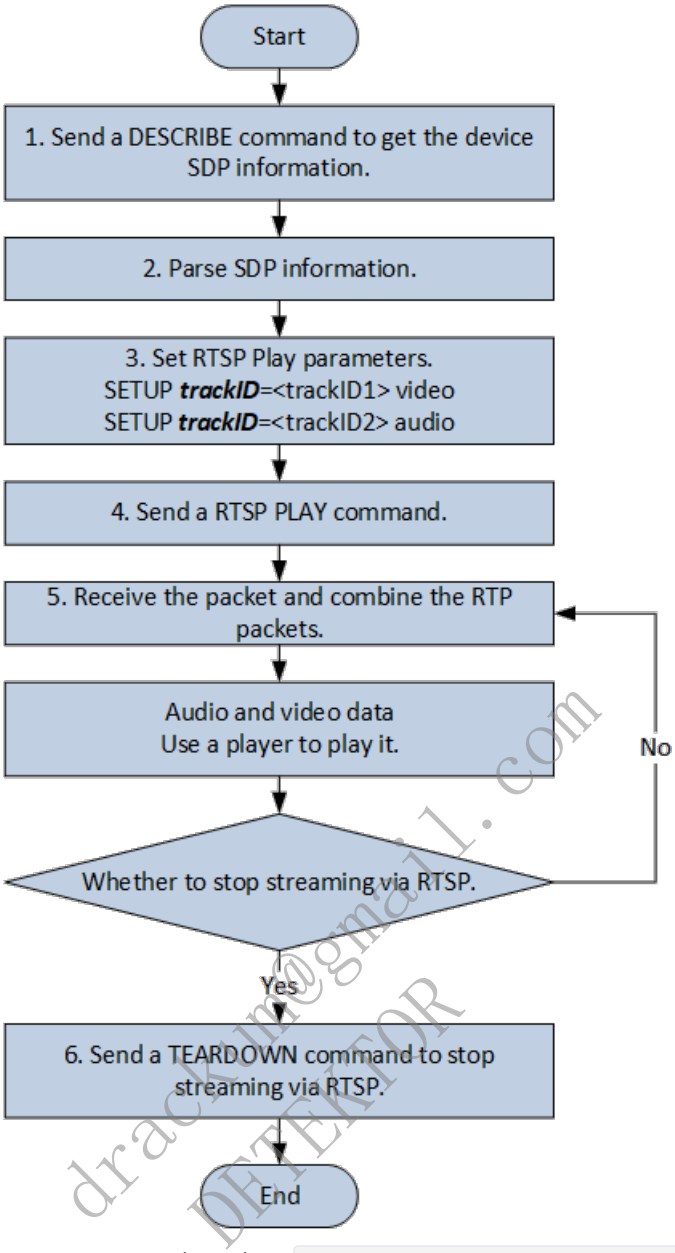
6.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

6.3.2 API Calling Flow

dracku@gmail.com
DETEKTOR



1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`; Digest authentication with devices is required.
2. The client parses the media SDP information returned by the device.
3. Set RTSP play parameters, set the track ID parsed from SDP information via SETUP commands. For example, trackID=1 indicates videos while trackID=2 indicates audios.
4. The client sends an RTSP PLAY command, and device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`.
5. The client receives the RTP package sent by device. Divided RTP packets should be assembled on the client before being parsed.
6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address, `[:port]` is optional, and 554 by default, `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream); for example, the IP address of the target device is `172.7.203.11`, and if the channel number is 172, the streaming address will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

6.3.3. Example

1. A client sends an RTSP DESCRIBE request.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s=Media Server V4.22.126
i=Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000020000011171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

drackum@gmail.com
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5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J.............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

dracknm@gmail.com
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6.4 Motion Detection

6.4.1 Introduction to the Function

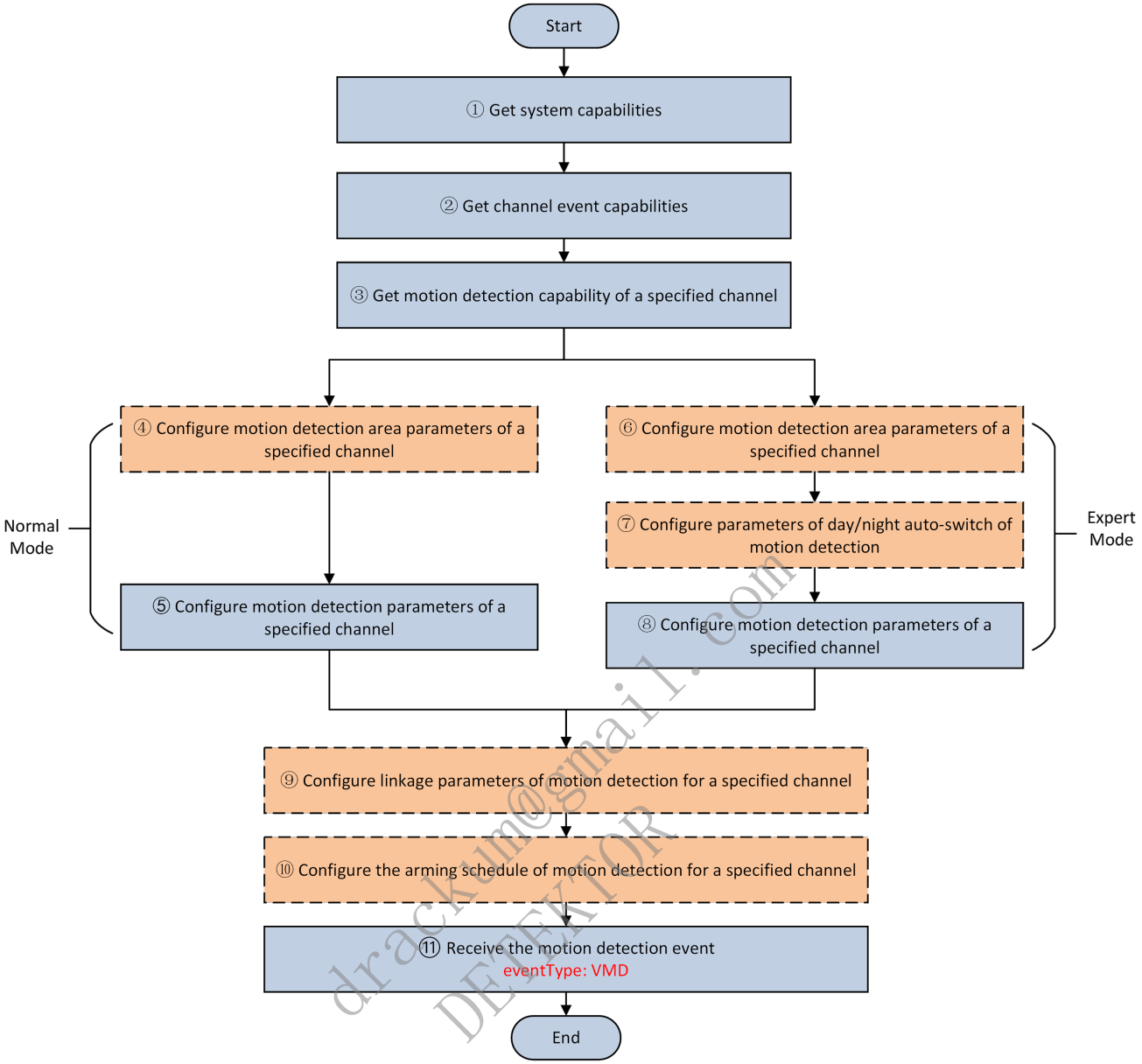
Commonly used in unattended camera recording and automatic alarm, motion detection can help to reduce the cost of manual monitoring for public institutions and companies. Besides, it can greatly improve monitoring efficiency and accuracy by avoiding the errors of manual monitoring due to fatigue after long-time work. For example, it can be applied in home and company theft proof, as well as monitoring and alarm in institutions like medical organizations and nursing homes.

Motion detection has two modes: normal mode and expert mode.

Normal mode: the default mode of motion detection. In this mode, all detected areas only support a same sensitivity.

Expert mode: in this mode, every detected area can have its own detection sensitivity and percentage of objects in area. It supports configuring different sensitivities according to area types. For example, for the motion detection in the company hall, you can set higher detection sensitivity for the gate area to focus on people's entering and exiting.

6.4.2 API Calling Flow



1. Get the system capability: `GET /ISAPI/System/capabilities`. If the node `isSupportMotionDetection` is returned and its value is `true`, it indicates that the device supports motion detection.
2. Get the device channel event capability. If the returned `eventType` contains `VMD`, it indicates that the corresponding channel supports motion detection.

   Get the event capability of all channels: `GET /ISAPI/Event/channels/capabilities`.

   Get the event capability of a specified channel: `GET /ISAPI/Event/channels/<channelID>/capabilities`.
3. Get the motion detection capability of a specified channel (normal mode): `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/capabilities`.

   Check the motion detection parameters (such as function enabling, detection sensitivity) supported by the device. You can check whether the device supports multiple areas via checking whether the node `regionType` contains `region`.
4. (Optional) Configure parameters of the motion detection area (normal mode).

   Capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout/capabilities`.

Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout?regionType=<regionType>`. **Note:** If the device supports multiple areas of motion detection, for compatibility, both `<gridMap>` and `<RegionList>` will be returned regardless the `regionType` value in URL.

Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout?regionType=<regionType>`. **Note:** If the device supports multiple areas of motion detection, to set multiple areas, you need to input `regionType=<region>` in URL.

5. Configure the motion detection parameters (such as motion detection enabling, detection sensitivity) (normal mode).

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection`. **Note:** If the device supports multiple areas of motion detection, for compatibility, both `<gridMap>` and `<RegionList>` will be returned regardless the `regionType` value in URL.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection`. **Note:** If the device supports multiple areas of motion detection, to set multiple areas, you need to input `regionType=<region>` in URL.

6. (Optional) Configure the parameters of motion detection area (expert mode). You can configure different detection conditions such as sensitivity for different areas.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/regions`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/regions`.

7. (Optional) Configure day/night auto switch of motion detection (expert mode). You can configure the time switch schedule of `day` and `night`.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/switch`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/switch`.

8. Configure motion detection conditions (expert mode).

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt`.

9. (Optional) Configure linkage parameters of the motion detection.

   Get: `GET /ISAPI/Event/triggers/VMD-<channelID>`.

   Configure: `PUT /ISAPI/Event/triggers/VMD-<channelID>`.

   Delete: `DELETE /ISAPI/Event/triggers/VMD-<channelID>`.

10. (Optional) Configure the arming schedule of motion detection.

    Get: `GET /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>`.

    Configure: `PUT /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>`.

11. Receive events reported by device in arming or listening mode: `eventType:VMD`.

Remarks:

1. For motion detection (normal mode), the uploaded `VMD` event includes device basic information only, such as channel No. and event time. For details, see 4.3.1 Motion Detection Event (Normal Mode).

2. For motion detection (expert mode), in addition to the device basic information, the uploaded `VMD` event includes the detection area ID, area coordinates and area detection sensitivity. For details, see 4.3.2 Motion Detection Event (Expert Mode).

drackum@gmail.com
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3. In practical employment, you can select one mode as needed. There are no logical relations between the two modes.

6.4.3 Message Example

6.4.3.1 Motion Detection Event (Normal Mode)

```
<?xml version="1.0" encoding="UTF-8"?>
<EventNotificationAlert version="1.0" xmlns="urn:psialliance-org">
<ipAddress>10.17.114.252</ipAddress>
<protocolType>HTTP</protocolType>
<macAddress>64:db:8b:6e:4f:08</macAddress>
<channelID>1</channelID>
<dateTime>2018-03-13T19:42:27+08:00</dateTime>
<activePostCount>4</activePostCount>
<eventType>VMD</eventType>
<eventState>active</eventState>
<eventDescription>Motion alarm</eventDescription>
<channelName>Camera 01</channelName>
<Extensions version="1.0" xmlns="urn:psialliance-org">
<serialNumber xmlns="urn:selfextension:psiaext-ver10-xsd">DS-2DF8236IX-AELW20170718CCWR799562898X</serialNumber>
<eventPush xmlns="urn:selfextension:psiaext-ver10-xsd">VMD&amp;amp;amp;DS-2DF8236IX-AELW20170718CCWR799562898X,2018-01-21T12:50:39+08:00,1,1.0</eventPush>
</Extensions>
</EventNotificationAlert>
```

6.4.3.2 Motion Detection Event (Expert Mode)

```
<?xml version="1.0" encoding="UTF-8"?>
<EventNotificationAlert version="1.0" xmlns="urn:psialliance-org">
<ipAddress>10.17.114.252</ipAddress>
<protocolType>HTTP</protocolType>
<macAddress>64:db:8b:6e:4f:08</macAddress>
<channelID>1</channelID>
<dateTime>2018-03-13T20:36:34+08:00</dateTime>
<activePostCount>11</activePostCount>
<eventType>VMD</eventType>
<eventState>active</eventState>
<eventDescription>Motion alarm</eventDescription>
<DetectionRegionList>
<DetectionRegionEntry>
<regionID>1</regionID>
<sensitivityLevel>50</sensitivityLevel>
<RegionCoordinatesList>
<RegionCoordinates>
<positionX>216</positionX>
<positionY>216</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>756</positionX>
<positionY>756</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>756</positionX>
<positionY>756</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>216</positionX>
<positionY>216</positionY>
</RegionCoordinates>
</RegionCoordinatesList>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName>Camera 01</channelName>
<Extensions version="1.0" xmlns="urn:psialliance-org">
<serialNumber xmlns="urn:selfextension:psiaext-ver10-xsd">DS-2DF8236IX-AELW20170718CCWR799562898X</serialNumber>
<eventPush xmlns="urn:selfextension:psiaext-ver10-xsd">VMD&amp;amp;amp;DS-2DF8236IX-AELW20170718CCWR799562898X,2018-01-21T12:50:39+08:00,1,1.0</eventPush>
</Extensions>
</EventNotificationAlert>
```

drackum@gmail.com
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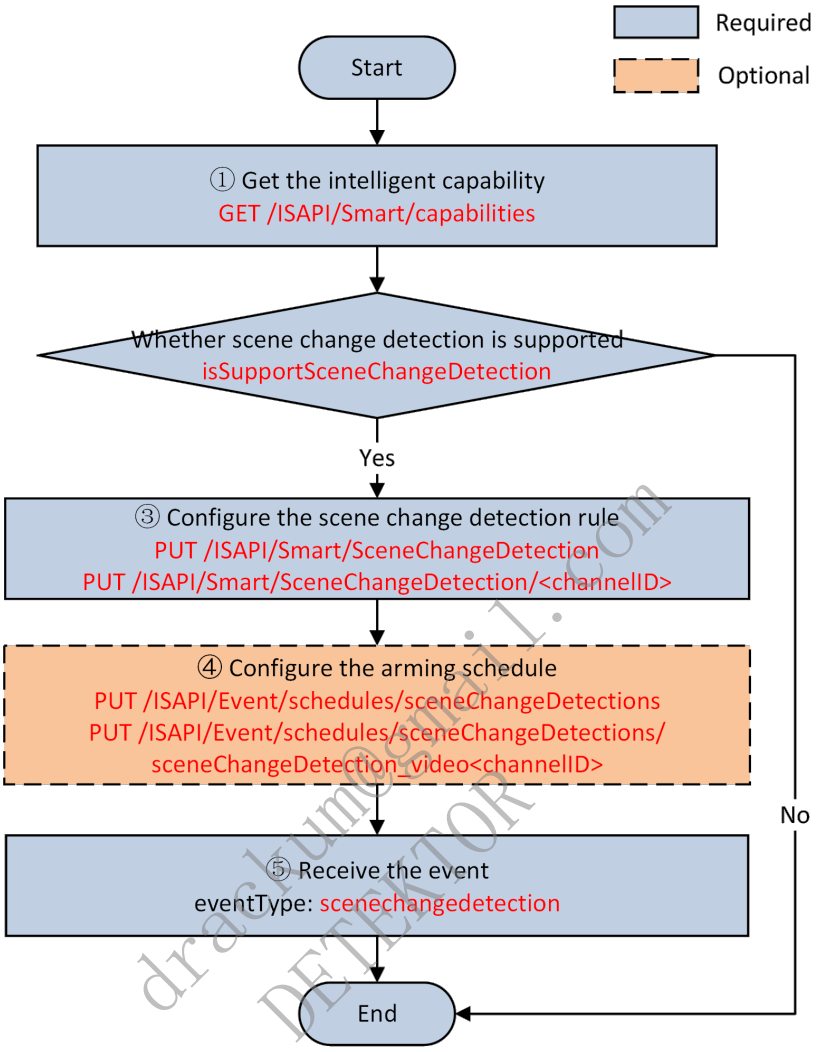
6.5 Scene Change Detection

6.5.1 Function Introduction

The scene change detection function identifies whether a scene has changed based on configured rules. Once a scene change is detected, an alarm is triggered.

This function is commonly used to monitor whether external forces have caused changes in camera's viewing angle. When the viewing angle is changed by external interference, or the monitoring scene is tampered, affecting normal monitoring, relevant personnel can be notified promptly to take actions.

6.5.2 API Calling Flow



1. Get the device's intelligent capability.

   `GET /ISAPI/Smart/capabilities`. If the response returns `isSupportSceneChangeDetection` as `true`, it indicates that the device supports the scene change detection.

2. Configure the detection rules.

   - Get the scene change detection capability: `GET /ISAPI/Smart/SceneChangeDetection/<channelID>/capabilities`.
   - Get scene change detection parameters: `GET /ISAPI/Smart/SceneChangeDetection`.
   - Set scene change detection parameters: `PUT /ISAPI/Smart/SceneChangeDetection`.
   - Get scene change detection parameters for a specific channel: `GET /ISAPI/Smart/SceneChangeDetection/<channelID>`.
   - Set scene change detection parameters for a specific channel: `PUT /ISAPI/Smart/SceneChangeDetection/<channelID>`.

   Note: A higher **sensitivityLevel** makes the camera more sensitive to scene changes. Adjust it based on actual environment.

3. (Optional) Configure the arming schedule.

- Get schedule parameters of scene change detection for all channels: `GET /ISAPI/Event/schedules/sceneChangeDetections`.
- Set schedule parameters of scene change detection for all channels: `PUT /ISAPI/Event/schedules/sceneChangeDetections`.
- Get schedule parameters of scene change detection for a specific channel: `GET /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>`.
- Set schedule parameters of scene change detection for a specific channel: `PUT /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>`.

Note:

- Up to 8 time periods can be configured per day.
- Time periods must not overlap.

4. Receive the event (`"eventType":"scenechangedetection"`).

6.6 Sub-device's Transparent Transmission

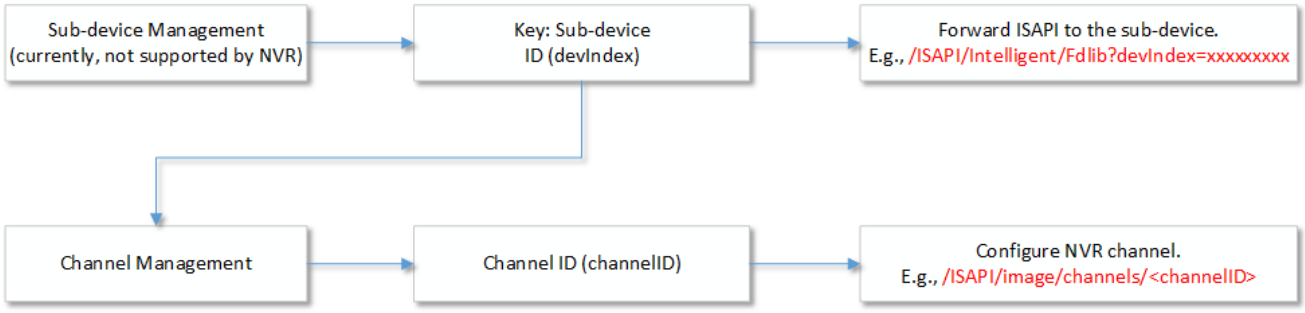
6.6.1 Introduction to the Function

In actual application scenarios, the NVR being used is generally not the latest version, if an IPC with latest functions is added to the system, the NVR will fail to recognize the IPC's new functions due to version incompatibility, so the platform (including device access platforms, client softwares, device's web plug-ins, etc, hereinafter referred to as the platform) will fail to operate new functions of the IPC.

Besides, in video channel resource management, the platform can only get information about digital channels (also known as IP channels) added to the NVR, but not the device that the channels belong to. For example, two video channels (channel 13 and channel 18) of a dual-lens people counting camera are added to the NVR. The platform can only get that digital channel 13 and 18 are added to the NVR, but not that the two channels belong to one camera. Such information loss may lead to repeated operations, which will further cause problems in actual practices. For example, users may fail to upgrade IPC via digital channel No. if not knowing the relationship between camera and digital channel.

To settle above problems, the sub-device's transmission function is introduced to the video channel resources management. In this way, users can directly access to the device of the digital channels via the devIndex (sub-device index) allocated to the device by NVR.

Notably, despite the introduction of the sub-device's transparent transmission function, the NVR does not function as an access gateway, and so it does not function with sub-device management on gateway. Refer to the Sub-device Management of Device Management for details.



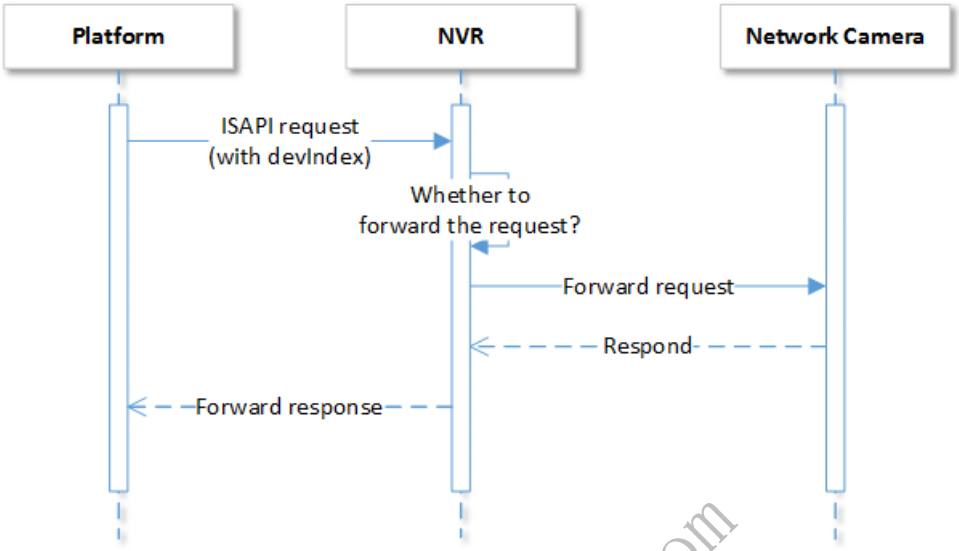
Steps are as follows:

1. First, the platform will check whether the NVR supports sub-device's transparent transmission. If it is supported, follow the next step. Second, add devIndex to the request URL.
2. For the ISAPI request sent to the NVR by the platform, if it contains devIndex, it will be forwarded to the specific sub-device; otherwise, the NVR will respond to the request.
3. When the request is forwarded to the sub-device by the NVR, delete the devIndex in the request URL so as to realize ISAPI interaction.

dr.ackun@gmail.com
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For example:

1. Send the request URL (`GET /ISAPI/System/deviceInfo`) to get the information of the platform's NVR device.
2. Send the request URL (`GET /ISAPI/System/deviceInfo?devIndex=<devIndex>`) to get the information of a specific sub-device of the NVR.

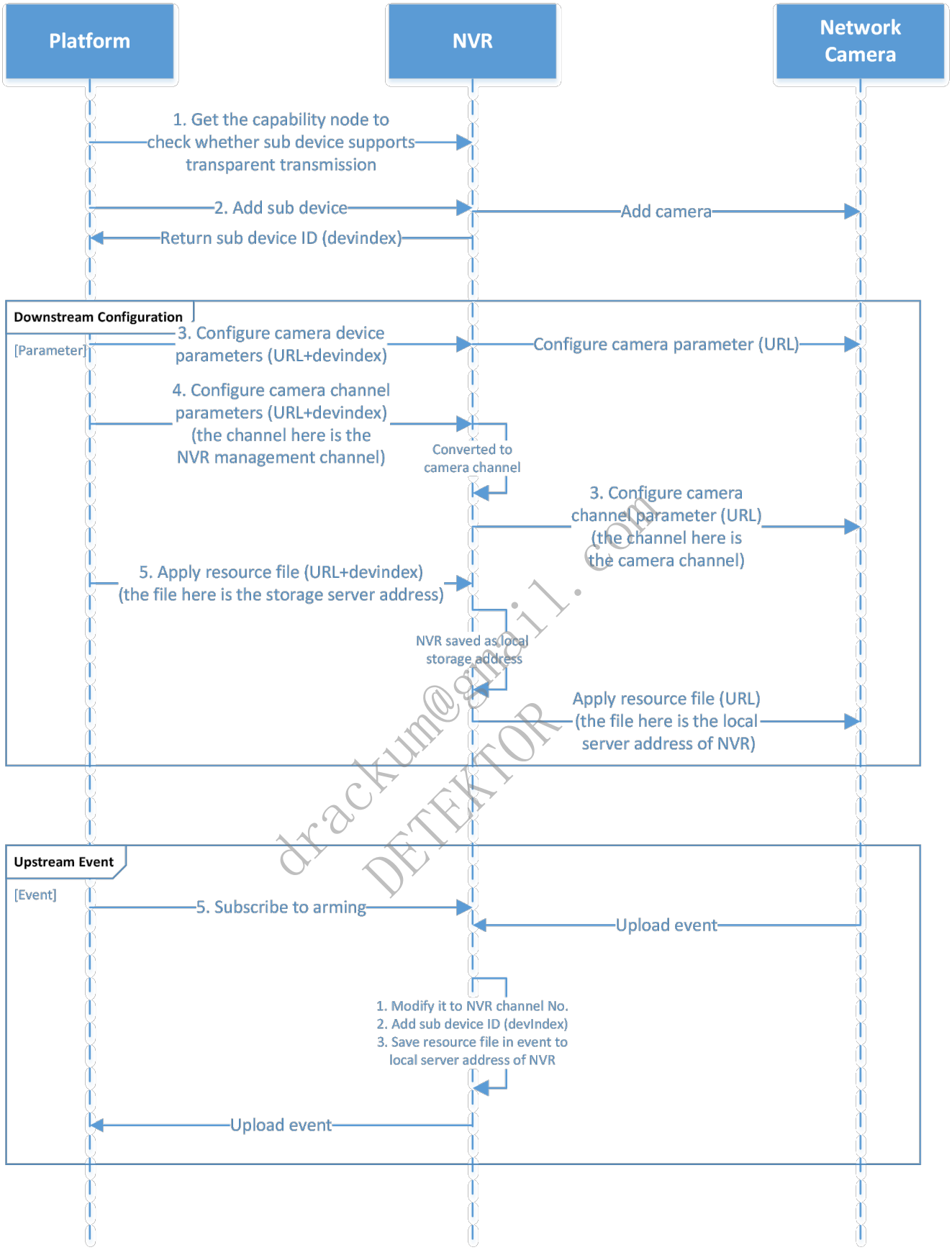


Note:

1. This function is applicable to transitory connections, while inapplicable to persistent connections (e.g., arming subscription, two-way audio).
2. It is required to delete the devIndex in the URL before NVR forwards the ISAPI request.
3. For the VCA function of NVR, the transmission solution of sub-devices is not adopted. You can switch the analysis unit by the following URI: /ISAPI/Smart/analysisUnitSwitch/channels/`<channelID>`/event/`<EventType>`.

6.6.2 API Calling Flow

drackun@gmail.com
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1. Get the capability node to check whether the device supports transparent transmission. If the value of the node `isSupportDevIndex` is true, it indicates that the device supports transparent transmission.
2. Create digital channels: `POST /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`. The sub device ID `devindex` will be returned.
3. Get access parameters of all digital channels: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.
   Each digital channel information contains one parameter `devIndex`, which is the same for digital channels of a same physical device.

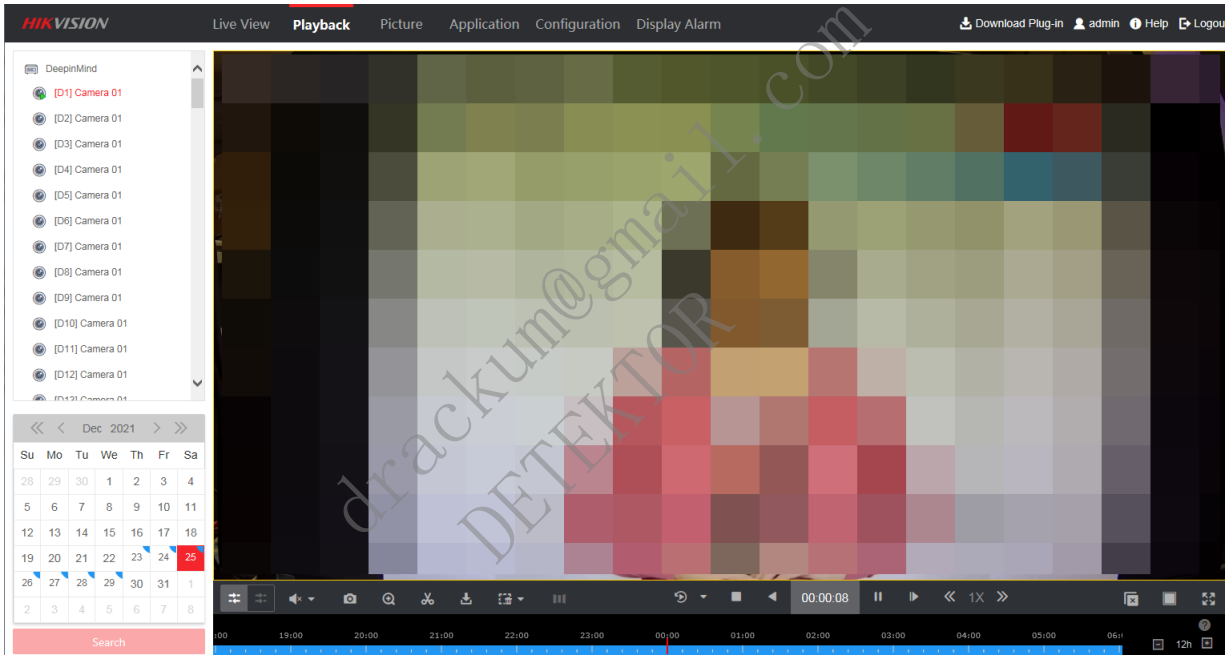
4. When splicing ISAPI request, if the URL contains `devIndex=<devIndex>`, it represents a request for the sub-device. For example, you can send `12345678` (the request URL) to get the information of the sub-device `GET /ISAPI/System/deviceInfo?devIndex=12345678` of the target device. Note: 1) For the downstream configuration of a video channel, the platform will apply the channel managed by NVR, and before applying, the NVR will transform channel into the video channel. 2) When the platform applies the storage address of resource files, the camera cannot access the storage service to get resource files because of the network isolation between the camera and platform, so the storage address must be saved as a local storage address by the NVR before being applied.

6.7 Video Playback

6.7.1 Introduction to the Function

Devices such as cameras and NVRs can store videos. Storage mediums such as SD card, TF card and HDDs are needed before recording videos. The process of playback starts with searching for footages and then getting video stream via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

The below is an example of playback on a Web Client. The calendar in the bottom-left corner shows the results of searching for videos. Dates with videos can be shown on the calendar and you can click a date to view videos.



6.7.2 API Calling Flow

1. (Optional) Check the calendar to find dates with stored videos: `POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution`. `<trackStreamID>` = channel number * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, The rackStreamID of the main stream for channel 17 is 1701.
2. Searching for videos by parsing the `playbackURI` of the device to get the RTSP address `POST /ISAPI/ContentMgmt/search`.
3. Play videos back via RTSP. Steps of playback via RTSP is the same as that of real-time live view. See details in RFC 7826 (Real Time Streaming Protocol).

Remarks:

- Playback via RTSP needs digest authentication. The method is the same as that of ISAPI digest authentication.
- Send PAUSE command to pause playback. Send PLAY command to continue. If you want to perform fast forward and slow forward, you can send a PLAY command and set the Scale in headers. See details in 10.6 PAUSE and 12.34 Scale in RFC 7826 (Real Time Streaming Protocol).

6.7.3 Example

6.7.3.1 Search for Video on Calendar

Request from Client

```
POST /ISAPI/ContentMgmt/record/tracks/101/dailyDistribution HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 119
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<trackDailyParam>
    <year>2021</year>
    <monthOfYear>08</monthOfYear>
</trackDailyParam>
```

Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:47:43 GMT
Content-Length: 2915
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<trackDailyDistribution version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <dayList>
        <day><id>1</id><dayOfMonth>1</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>2</id><dayOfMonth>2</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>3</id><dayOfMonth>3</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>4</id><dayOfMonth>4</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>5</id><dayOfMonth>5</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>6</id><dayOfMonth>6</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>7</id><dayOfMonth>7</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>8</id><dayOfMonth>8</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>9</id><dayOfMonth>9</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>10</id><dayOfMonth>10</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>11</id><dayOfMonth>11</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>12</id><dayOfMonth>12</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>13</id><dayOfMonth>13</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>14</id><dayOfMonth>14</dayOfMonth><record>false</record></day>
        <day><id>15</id><dayOfMonth>15</dayOfMonth><record>false</record></day>
        <day><id>16</id><dayOfMonth>16</dayOfMonth><record>false</record></day>
        <day><id>17</id><dayOfMonth>17</dayOfMonth><record>false</record></day>
        <day><id>18</id><dayOfMonth>18</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>19</id><dayOfMonth>19</dayOfMonth><record>false</record></day>
        <day><id>20</id><dayOfMonth>20</dayOfMonth><record>false</record></day>
        <day><id>21</id><dayOfMonth>21</dayOfMonth><record>false</record></day>
        <day><id>22</id><dayOfMonth>22</dayOfMonth><record>false</record></day>
        <day><id>23</id><dayOfMonth>23</dayOfMonth><record>false</record></day>
        <day><id>24</id><dayOfMonth>24</dayOfMonth><record>false</record></day>
        <day><id>25</id><dayOfMonth>25</dayOfMonth><record>false</record></day>
        <day><id>26</id><dayOfMonth>26</dayOfMonth><record>false</record></day>
        <day><id>27</id><dayOfMonth>27</dayOfMonth><record>false</record></day>
        <day><id>28</id><dayOfMonth>28</dayOfMonth><record>false</record></day>
        <day><id>29</id><dayOfMonth>29</dayOfMonth><record>false</record></day>
        <day><id>30</id><dayOfMonth>30</dayOfMonth><record>false</record></day>
        <day><id>31</id><dayOfMonth>31</dayOfMonth><record>false</record></day>
    </dayList>
</trackDailyDistribution>
```

liuxiaolin@gmail.com
VENDOR

6.7.3.2 Search for Video

Request from Client

```
POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 486
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
    <searchID>88C2CD4D-D3FA-4AD4-BD80-555C18205DCC</searchID>
    <trackList>
        <trackID>101</trackID>
    </trackList>
    <timeSpanList>
        <timeSpan>
            <startTime>2021-08-16T00:00:00Z</startTime>
            <endTime>2021-08-18T23:59:59Z</endTime>
        </timeSpan>
    </timeSpanList>
    <maxResults>100</maxResults>
    <searchResultPostion>0</searchResultPostion>
    <metadataList>
        <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
    </metadataList>
</CMSearchDescription>
```

Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:19:13 GMT
Content-Length: 1021
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<CMSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <searchID>{88c2cd4d-d3fa-4ad4-bd80-555c18205dcc}</searchID>
    <responseStatus>true</responseStatus>
    <responseStatusStrg>OK</responseStatusStrg>
    <numOfMatches>1</numOfMatches>
    <matchList>
        <searchMatchItem>
            <sourceID>{00000000-0000-0000-0000-000000000000}</sourceID>
            <trackID>101</trackID>
            <timeSpan>
                <startTime>2021-08-18T15:18:15Z</startTime>
                <endTime>2021-08-18T15:19:08Z</endTime>
            </timeSpan>
            <mediaSegmentDescriptor>
            <contentType>video</contentType>
            <codecType>H.264-BP</codecType>
            <playbackURI>rtsp://10.14.97.40/Streaming/tracks/101/?
starttime=20210818T151815Z&amp;endtime=20210818T151908Z&amp;name=00000004667000100&amp;size=1400788</playbackURI>
            <lockStatus>unlock</lockStatus>
            <name>00000004667000100</name>
            </mediaSegmentDescriptor>
            <metadataMatches>
                <metadataDescriptor>recordType.meta.com/timing</metadataDescriptor>
            </metadataMatches>
        </searchMatchItem>
    </matchList>
</CMSearchResult>
```

drackum@gmail.com
DETEKTOR

6.7.3.3 Playback via RTSP

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: application/sdp
```

4. The server responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 7
Content-Type: application/sdp
Content-Length: 546
Date: Wed, 18 Aug 2021 07:29:25 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.60.000
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:4106
a=range:clock=20210818T151815Z-20210818T151908Z
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H265/90000
a=control:trackID=video
b=AS:4096
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=audio
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000040005001171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

drackum@gmail.com
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5. The client sends RTSP SETUP commands two times, and the server responds to them.

```
SETUP rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=video
RTSP/1.0
CSeq: 8
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61048-61049
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61048-61049;server_port=62000-62001;ssrc=d6d3ed5
CSeq: 8
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

```
SETUP rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=audio
RTSP/1.0
CSeq: 9
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61050-61051
Session: 225263317
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61050-61051;server_port=62002-62003;ssrc=d6d3ed6
CSeq: 9
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

6. The client sends an RTSP PLAY request, and the server starts to push streams for playback.

```
PLAY rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 10
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
Range: npt=0.000-
```

```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

7. The client sends an RTSP TEARDOWN request to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

crackun@gmail.com
DEFEKTOR

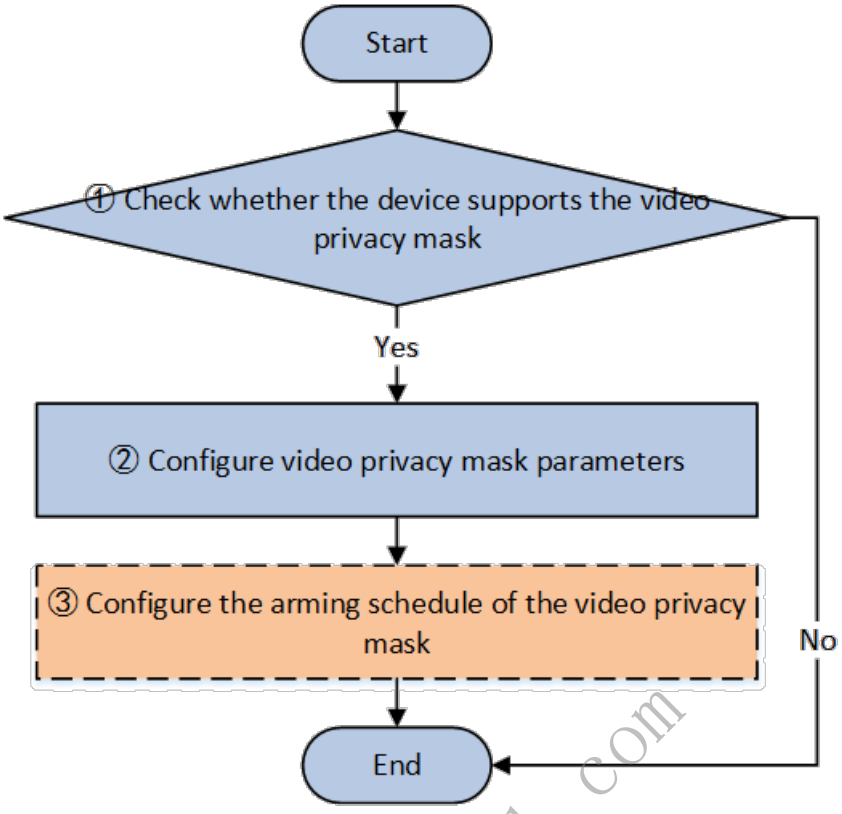
6.8 Video Privacy Mask

6.8.1 Introduction to the Function

Video privacy mask is to cover certain areas on the live image to protect personal privacy from being live viewed and recorded. You can use the masking method such as a mosaic or black frame as needed to make the area invisible. The device supporting privacy mask can specify the area(s) to be masked permanently on the live image, as a result of which the live view and recorded video during the live view will be influenced and the image cannot be restored. The following image shows the effect of masking an area with the mosaic.

6.8.2 API Calling Flow

6.8.2.1 Video Privacy Mask



1. Check whether the device supports the video privacy mask. Get the device's system capability: `GET /ISAPI/System/capabilities`. If `isSupportPrivacyMask` is returned and its value is true, it indicates that the video privacy mask is supported by the device.

2. Configure the video privacy mask parameters. You can specify sensitive area(s) to be masked on the live image (related field: `PrivacyMaskRegionList`), and up to 8 areas are supported. You can also configure the masking type, such as mosaic, black frame, and red frame (related field: `maskType`). Get the video privacy mask capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/privacyMaskCap`. Configure the video privacy mask parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask`. Get the video privacy mask parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask`.

3. (Optional) Configure the arming schedule of the video privacy mask. You can configure the privacy mask schedule as needed. If not configured, the privacy mask is performed 24 hours a day by default. Get the video privacy mask schedule: `GET /ISAPI/Event/schedules/privacyMask/privacyMask-<channelID>`. Set the video privacy mask schedule:`PUT /ISAPI/Event/schedules/privacyMask/privacyMask-<channelID>`.

dr.chun@gmail.com DETEKTOR

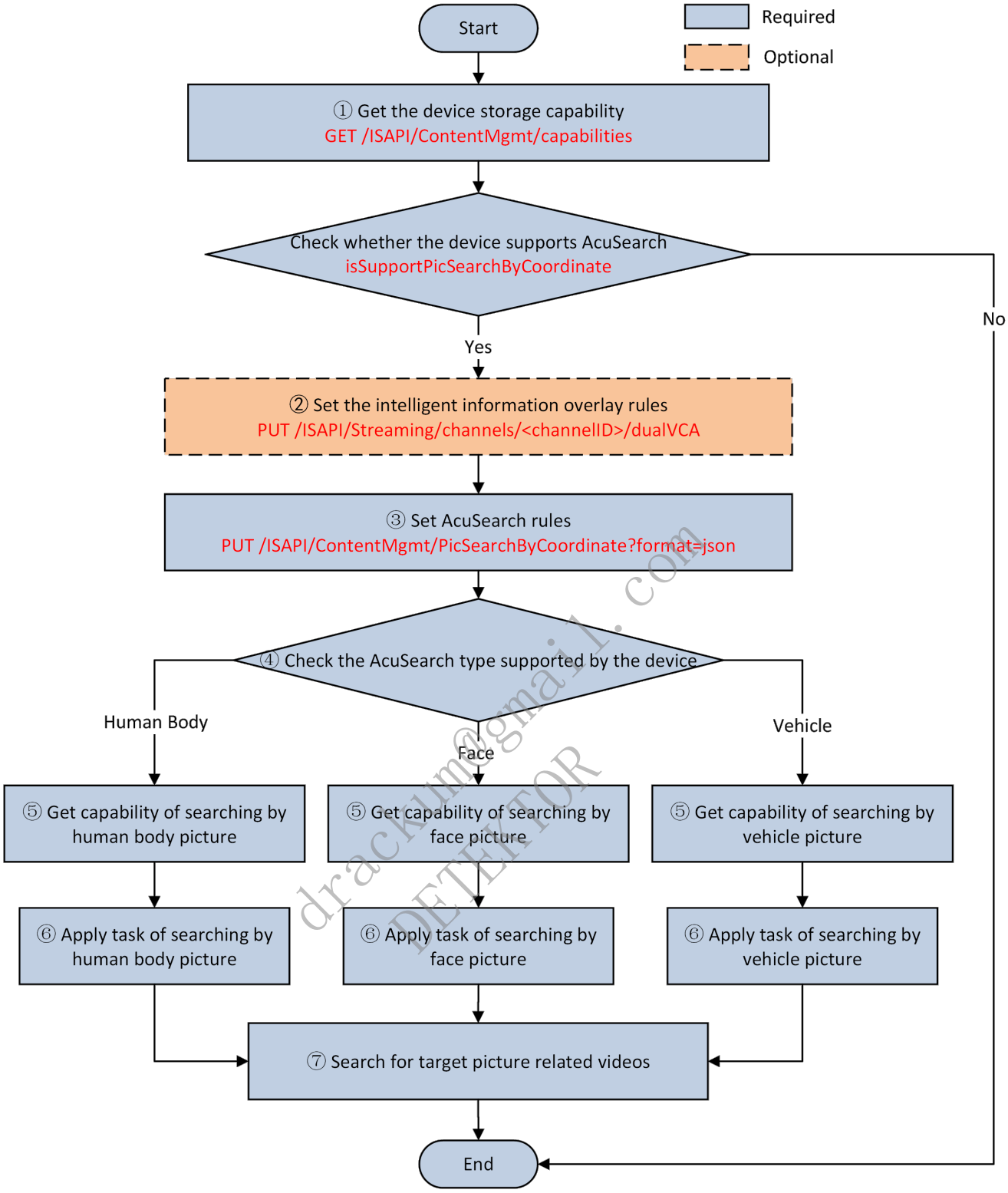
7 Video Management

7.1 AcuSearch Image Search

7.1.1 Function Introduction

The AcuSearch function enables extracting images of human faces or bodies during video playback, compares them with recorded videos, and ultimately identifies videos containing the target.

7.1.2 API Calling Flow



1. Check whether the device supports AcuSearch.

- Get the device storage capability: `GET /ISAPI/ContentMgmt/capabilities`.
- Verify: `<isSupportPicSearchByCoordinate>` = true. Explanation of sub nodes in node `<AcuSearchCap>`

Node	Description
`<targetType>`	Supported search type: face, vehicle, human body pictures.
`<dataType>`	Supported image format.
`<maxPicSize>`	Maximum size of image used for search (default: 1 MB if this node is absent).
`<maxPicWidthSize>` and `<maxPicHeightSize>`	Maximum resolution of image used for search (default: 2560 x 1440 if these two nodes are absent).

2. (Optional) Configure intelligent information overlay rules of channels.

- Get configurations of all channels: `GET /ISAPI/Streaming/channels/dualVCA?format=json`. If `dualVCAType` returns `remote`, it indicates the channel connected camera's intelligent information overlay (suitable for channels connected to intelligent cameras); if `dualVCAType` returns `local`, it indicates the NVR local intelligent information (suitable for channels connected to ordinary cameras without intelligent analysis capability).
- Get configurations of a single channel: `GET /ISAPI/Streaming/channels/<channelID>/dualVCA`. If `enabled` returns `false`, the current channel does not support intelligent information overlay.
- Configure parameters of a single channel (enable the intelligent information overlay): `PUT /ISAPI/Streaming/channels/<channelID>/dualVCA`.

3. Configure AcuSearch rules.

- Get the AcuSearch capability: `GET /ISAPI/ContentMgmt/PicSearchByCoordinate/capabilities?format=json`.
- Get the AcuSearch rule: `GET /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json`.
- Configure the AcuSearch rule: `PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json`.

4. (Optional) Search by human body picture.

Refer to the API calling flow of domain "Search by Human Body Picture".

5. (Optional) Use picture to search picture in face capture library.

Refer to the API calling flow of domain "Use Picture to Search Picture in Face Capture Library".

6. (Optional) Search by vehicle picture.

Refer to the API calling flow of domain "Search by Vehicle Picture".

7. Search for target related videos. Search for videos around the target's appearance time: `POST /ISAPI/ContentMgmt/search`, and play back the video segment using the returned `playbackUrl`.

dracknum@gmail.com DETEKTOR

7.2 Search for Event Video

7.2.1 Introduction to the Function

This function supports searching for the videos of linked cameras when an event occurs.

7.2.2 API Calling Flow

1. Check whether the device supports searching for event videos.

Get the device sysgem capability: `GET /ISAPI/System/capabilities`. If the node `<SerialCap>` is returned and its value is true, the device supports searching for event videos.

2. Search for event videos: `POST /ISAPI/ContentMgmt/eventRecordSearch?format=json`.

7.3 Search for Videos and Pictures

7.3.1 Introduction to the Function

This function supports searching for the video/picture data stored in the device. Video data source includes scheduled recording, manual recording and event recording, and picture data source includes scheduled capture and event linkage capture.

7.3.2 API Calling Flow

1. Check the device's supported time format. Get device's system capabilities: `GET /ISAPI/System/capabilities`.
    - If node `isSupportTimeTypeSTD` is returned with value `true`, it indicates that the device supports the ISO 8601 standard time, and URLs of functions related to ISO 8601 time parameters support additional query parameter `timeType=STD`.
    - If node `isSupportTimeTypeSTD` is not returned. Call `GET /ISAPI/System/time/timeType?format=json` to get the time type. If the `type` field returns `local`, it indicates that the device video search and playback use local

time. If the `type` field returns `UTC`, it indicates that the device video search and playback use ISO 8601 time format.

- If the device does not support API `GET /ISAPI/System/time/timeType?format=json`. Call `GET /ISAPI/System/deviceInfo` and check the time type of device via field `deviceType`. If `deviceType` (device type) is CVR, NVR, DVR, or ITCCAM, the device video search and playback use local time. Otherwise, follow the protocol specifications and use the TZ format, for example, `2024-05-13T16:00:00Z`.

2. Check whether the device supports video/picture search. Get the device's video search capability via `GET /ISAPI/ContentMgmt/capabilities`. If `<isSupportRecordSearch>` is returned, it indicates that the device supports video/picture search.

3. Search for videos and pictures:

   i. Get the capability of searching for videos and pictures: `GET /ISAPI/ContentMgmt/capabilities`. Get the picture search types that the device supports searching via the node `<pictureSearchType>` returned in the message, including pictures of event capture, scheduled capture and manual capture. Get the video types that the device supports searching for via the node `<recordSearchType>` returned in the message.

   ii. Get the parameters capability of searching for videos and pictures: `GET /ISAPI/ContentMgmt/search/capabilities`. Get the periods when the device supports searching for videos and pictures via the node `<timeSpanList>` returned in the message. Get the file tags that the device supports searching via the node `<labelSearch>` returned in the message.

   iii. Search for videos and pictures: `POST /ISAPI/ContentMgmt/search?security=<security>&iv=<iv>`.

   - The URL containing `timeType=STD` indicates that the time in the input and output messages are according to the ISO 8601 time format. The URL without `timeType=STD` indicates to use device's original time.
   - Check whether it is picture or video to be searched via the node `<trackID>` in the request message `<CMSearchDescription>`. "101" (search for the main stream videos of channel 1), "103" (search for the pictures of channel 1), and it is channel 1 by default .

Note:

- The value of `CMSearchDescription.trackIDList.trackID` is obtained from the `id` value returned in the response message of `GET /ISAPI/ContentMgmt/record/tracks`. The meaning of `id`: `101` is the main stream of channel 1, `102` is the sub-stream of channel 1, `103` is the capture schedule of channel 1, `201` is the main stream of channel 2, and so on. When the device saves only one copy of channel videos, it does not distinguish between the main and sub stream, and only supports setting `trackID` to `xx1`.
- If `<trackIDList>` is returned in response message of `GET /ISAPI/ContentMgmt/search/capabilities`, it indicates that the device supports accurate search, and the returned value in `<trackIDList>` indicates the channels that support independent video search.

7.4 Search for Videos by Target Type

7.4.1 Introduction to the Function

This function realizes persons, vehicles or mixed targets distinguishing when searching for motion detection event videos. Searching for videos by target type is newly added. The search conditions include target type information, and the returned result of the search contains the start and end time as well as the file name of the video.

7.4.2 API Calling Flow

1. Check whether the device supports searching by target type:

Get the device capability: `GET /ISAPI/System/capabilities`. If the node is returned and its value is true, it indicates that the device supports searching by target type. Currently, the device supports searching by person and by vehicle.

2. Get the capability of searching by target type: `GET /ISAPI/ContentMgmt/SearchByTargetType/capabilities?format=json`.

3. Search for videos by target type: `POST /ISAPI/ContentMgmt/SearchByTargetType?format=json`.

dlcrackum@gmail.com DLCRACKUM

7.5 Transfer Device Local File (Image) to Cloud

7.5.1 Function Introduction

The client can control the device to upload internally stored files (images) to a specified storage server. The interaction between the device and the storage server follows the server protocol. Here we only introduce the communication interaction between the client and the device.

There are two implementation methods for communication between the client and the device:

- Synchronous Use synchronous transfer for stable networks where devices can return results within 15 seconds.
- Asynchronous Use asynchronous transfer for unstable networks where responses may exceed 15 seconds.

7.5.2 API Calling Flow

7.5.2.1 Synchronous

1. The client obtains the device's local file transfer capability via `GET /ISAPI/ContentMgmt/RedirectStorageUrl/capabilities?format=json`. If `HTTP 200` is returned, it indicates that the device supports transferring local files to the server.
2. The client sends the source address information to the device via `POST /ISAPI/ContentMgmt/RedirectStorageUrl?format=json`. In the request, parameter `sourceUrl` carries the source address information. The response directly returns the file transfer result.

7.5.2.2 Asynchronous

1. The client checks the device's file asynchronous transfer capability. a. Call `GET /ISAPI/System/capabilities`. If `isSupportRedirectStorageUrlEvent` returns `true`, the device supports the asynchronous transfer event upload.

   b. Call `GET /ISAPI/ContentMgmt/RedirectStorageUrl/capabilities?format=json`. If `HTTP 200` is returned, the device supports transferring local files to the server.

   Both capabilities must be supported for asynchronous transfer.
2. The client sends the source address information to the device: `POST /ISAPI/ContentMgmt/RedirectStorageUrl?format=json`. In the request, `sourceUrl` carries the source address information. In the response, if `isAsyncResult` returns `true`, the client needs to wait for the event uploading.
3. Receive transfer results.

- After the device uploads the file to the server and gets the result, it uploads an event (`eventType`: `RedirectStorageUrlEvent`) to the client .
- This event does not depend on alarm subscription.

Notes:

- For multiple files to be transferred, devices may formulate custom strategies, e.g., upload events after partial or complete transfers.
- Each event includes `sourceUrl` (source address of each file) and `redirectUrl` (corresponding file address on server) . If transfer fails, `filePath` in the `redirectUrl` returns empty.

8 PTZ Control

8.1 . PTZ Lock

8.1.1 Introduction to the Function

The client can control a network camera's PTZ lock. During the locked period, the PTZ linked with alarm events will not work, but manual rotation is available, until the client unlocks the PTZ. This function can be used for long-term

dracknn@gmail.com DETEKTOR

monitoring of key locations to prevent the lens from being moved. When configuring or modifying intelligent applications that involve patrol, the PTZ should be locked first and then unlocked after the configuration is completed to achieve internal control of the intelligent application's operation, while the actual PTZ operation remains functional.

8.1.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `ManualPTZLockCap` indicates the support for manual `PTZ` control locking;
2. (Client) Control PTZ manually:

   Manually lock the PTZ: `PUT /ISAPI/PTZCtrl/channels/<channelID>/manualPTZLock?format=json`;

   Manually unlock the PTZ: `PUT /ISAPI/PTZCtrl/channels/<channelID>/manualPTZUnlock?format=json`.
3. (Optional) The client configures the PTZ lock parameters to specify the lock duration:

   Capability: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `lockTime` indicates the maximum lock duration.

   Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/lockPTZ`.

   Get: `GET /ISAPI/PTZCtrl/channels/<channelID>/lockPTZ`.
4. The client gets the remaining time for PTZ unlock: `GET /ISAPI/PTZCtrl/channels/<channelID>/manualPTZLockCfg?format=json`.

8.2 Auxiliary Device Control

8.2.1 Introduction to the Function

The PTZ auxiliary devices help the device better display video content. The auxiliary devices includes lights, wipers, fans, and heaters. This function is used to control PTZ auxiliary devices.

8.2.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `PTZAuxType` indicates the support for PTZ auxiliary device control;
2. The client configures the PTZ auxiliary device parameters to control the auxiliary devices. It is recommended to first get all auxiliary device parameters before modifying parameter configurations:

   (Optional) Get all auxiliary device parameters: `GET /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols`;

   (Optional) Get a single auxiliary device parameter: `GET /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols/<auxCtrlID>`;

   (Optional) Configure auxiliary device parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols`;

   Configure a single auxiliary device parameter: `PUT /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols/<auxCtrlID>`.

Notes:

The `id` in the parameter configuration message corresponds to the `auxCtrlID` in the `url`, which is the auxiliary device index and starts from 1 by default.

8.3 PTZ 3D Positioning Control

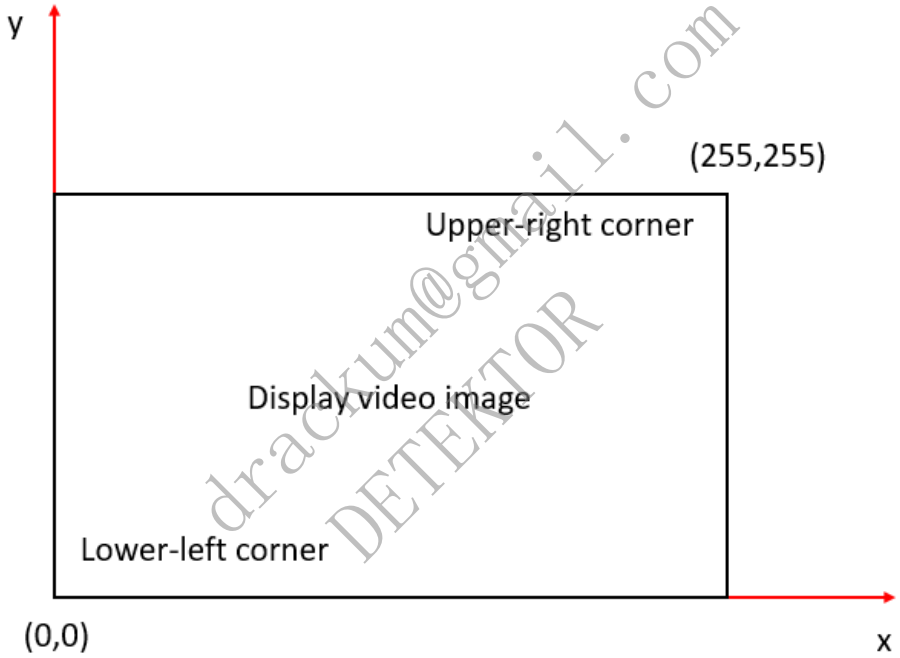
8.3.1 Introduction to the Function

drackum@gmail.com DETEKTOR

Supports centering a point in the video image, and supports centering and zooming a rectangular area in the video image.

8.3.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportPosition3D` indicates the support for 3D positioning control;
2. (Client) Send a control command to the device: `PUT /ISAPI/PTZCtrl/channels/<channelID>/position3D`:
    - The point in the video image is centered by the device, and the coordinates of `StartPoint` and `EndPoint` are the same;
    - The device zooms in and centers a rectangular area in the video image, and the X coordinate value of `StartPoint` is smaller than that of `EndPoint`;
    - The device zooms out and centers a rectangular area in the video image, and the X coordinate value of `StartPoint` is greater than that of `EndPoint`.



drackum@gmail.com
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8.4 PTZ Absolute Position

8.4.1 Introduction to the Function

The PTZ absolute position is unique and remains the same. The client can record certain PTZ absolute positions in actual application scenarios to repeatedly inspect at specified positions. For example, in a supermarket, record the PTZ absolute position of the cash register and inspect the position periodically.

8.4.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. the returned `isSupportAbsoluteEx` indicates the support for high-precision absolute `PTZ` movement control;
2. (Client) Get the high-precision absolute `PTZ` movement control capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/RailwayRobot/capabilities?format=json`. The value range of control parameters will be returned;
3. The client specifies the PTZ to move to an absolute position: `PUT /ISAPI/PTZCtrl/channels/<channelID>/absoluteEx`. The `elevation` and `azimuth` represent the angles in the vertical and horizontal directions, `absoluteZoom` represents the zoom range, `focus` represents the focus range, and

`absoluteRotate` represents the absolute rotation angle.

4. (Client) Get the PTZ absolute position: `GET /ISAPI/PTZCtrl/channels/<channelID>/absoluteEx`.

Notes:

- The zero coordinate position of `elevation` and `azimuth`, i.e., PT(0,0), can be specified by `PUT /ISAPI/PTZCtrl/channels/<channelID>/homeposition`.

8.5 PTZ Manual Control

8.5.1 Introduction to the Function

Adjust the horizontal and vertical positions of the PTZ, and control the zoom and focus of the lens at specified rotation/changing speeds. The client adjusts PTZ continuously through applying control commands, which is usually used for manual direction adjustment in remote live view.

8.5.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `ContinuousPanTiltSpace` in the message indicates the support for PTZ manual control;
2. The client sends a control command to the device: `PUT /ISAPI/PTZCtrl/channels/<channelID>/continuous`. The values of `pan`, `tilt`, `zoom`, and `rotate` represent the speed of rotation/change, and the characteristics (positive and negative) of the values represent the direction of rotation/change. If the control needs to be stopped, another command should be sent and values should be set as 0.

8.6 PTZ One-Touch Focus

8.6.1 Introduction to the Function

Help PTZ devices to focus. When defocus (blurry image) occurs, one-touch focus can make the image clear.

8.6.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportOnepushfoucsStart` indicates the support for one-touch focus;
2. (Client) Configure PTZ one-touch focus parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/onepushfoucs/start`;
3. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportOnepushfoucsReset` indicates the support for resetting focus parameters;
4. (Client) Reset the PTZ one-touch focus parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/onepushfoucs/reset`. The PTZ focus parameter setting will be restored.

8.7 Save PTZ Position

8.7.1 Introduction to the Function

The client controls the device to save the current PTZ position information. After the PTZ leaves the current position, the client can control the PTZ to return to the saved position, but only one position can be saved for each PTZ.

8.7.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportPTZSave` and `isSupportPTZSaveGet` indicates the support for keeping and getting `PTZ` parameters configurations;
2. (Client) Configure the device to save `PTZ` information: `PUT /ISAPI/PTZCtrl/channels/<channelID>/save?format=json`;
3. The client controls the PTZ to move to the saved position: `PUT /ISAPI/PTZCtrl/channels/<channelID>/gotoPosition?format=json`. The PTZ will rotates to the saved position;

drackum@gmail.com DEFEKTOR

4. The client gets the saved PTZ position information: `GET /ISAPI/PTZCtrl/channels/<channelID>/save`. The PTZ position information will be returned and PTZ does not move.

9 Abnormal Event Detection

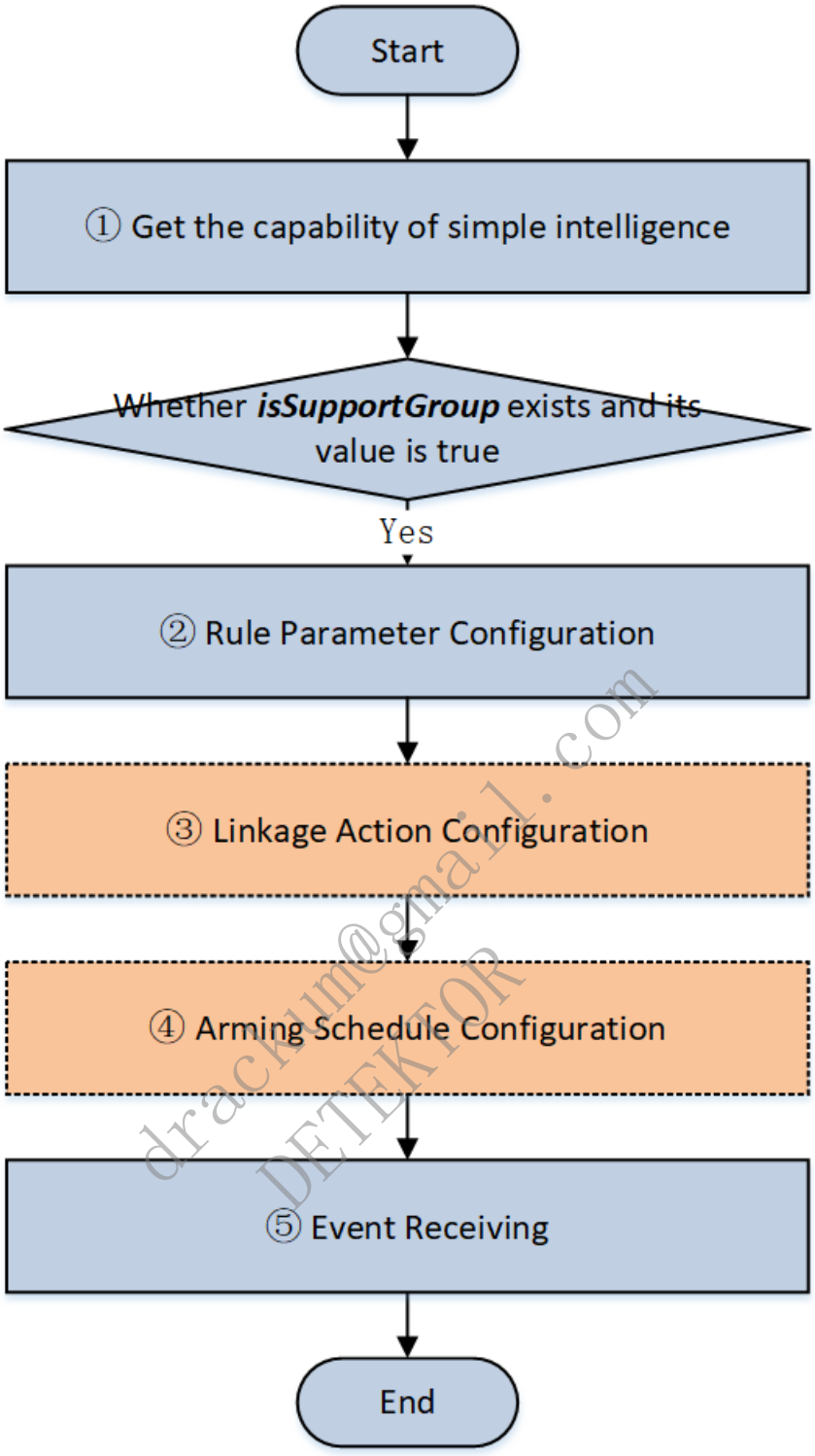
9.1 People Gathering

9.1.1 Introduction to the Function

The people gathering function is used for triggering the alarm in the fastest and best way when the people density in the set area exceeds the set threshold. The people gathering function is often used for performing the real-time monitoring of the abnormal behavior of people in the monitoring area, effectively preventing accidents caused by negligence, so as to avoid potential safety accidents and improve the supervision efficiency.

9.1.2 API Calling Flow

drackum@gmail.com
DETEKTOR



dracknm@gmail.com
DETEKTOR

1. Get the apability of AI intelligent search.

   Get the `isSupportGroup` in the capability of simple intelligence (related URI: `GET /ISAPI/Smart/capabilities`).

2. Configure rule parameters.

   Configure All Channels

- Get people gathering parameters (related URI: `GET /ISAPI/Smart/group`).
- Configure people gathering parameters (related URI: `PUT /ISAPI/Smart/group`).

  Configure Specified Channel

- Get the capability of people gathering parameters (related URI: `GET /ISAPI/Smart/group/<channelID>/capabilities`).

- Get the people gathering parameters of a specified channel (related URI: `GET /ISAPI/Smart/group/<channelID>`).
- Configure the people gathering parameters of a specified channel (related URI: `PUT /ISAPI/Smart/group/<channelID>`).

Configure All Areas of Specified Channel

- Get the capability of the parameters of all areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/capabilities`).
- Get the parameters of all areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions`).
- Configure the parameters of all areas for people gathering (related URI: `PUT /ISAPI/Smart/group/<channelID>/regions`).

Configure Specified Channel in Specified Channel

- Get the capability of the parameters of specified areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/<regionID>/capabilities`).
- Get the parameters of specified areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/<regionID>`).
- Configure the parameters of specified areas for people gathering (related URI: `PUT /ISAPI/Smart/group/<channelID>/regions/<regionID>`). Note: The percentage refers to the percentage of people number accounts for the alarming area. An alarm will be triggered in the system when the percentage is higher than the set percentage. Otherwise, no alarm will be triggered.

3. (Optional) Configure arming linkage parameters.

- Get the capability of event linkage configuration (related URI: `GET /ISAPI/Event/triggersCap`), whether the message `EventTriggersCap` contains the node `GroupDetectionTriggerCap`.
- Get the linkage parameters of people gathering (related URI: `GET /ISAPI/Event/triggers/group-<channelID>`).
- Configure linkage parameters of people gathering (related URI: `PUT /ISAPI/Event/triggers/group-<channelID>`).

4. (Optional) Configure arming schedule.

- Get the arming schedule of alarm outputs for all channels (related URI: `GET /ISAPI/Event/schedules/groups`).
- Configure the arming schedule of people gathering for all channels (related URI: `PUT /ISAPI/Event/schedules/groups`).
- Get the arming schedule of people gathering for a specified channel (related URI: `GET /ISAPI/Event/schedules/groups/group-<channelID>`).
- Configure the arming schedule of people gathering for a specified channel (related URI: `PUT /ISAPI/Event/schedules/groups/group-<channelID>`). Note: 1. Supports setting up to 8 time periods in one day on the timeline. 2. Overlapped time between any two time periods is not allowed when setting the arming schedule.

5. Receive event: `"eventType":"group"`, including the following. `detectionPictureTransType` refers to the transmission method of the pictures. The field value is bkgUrl for the picture URL or the picture encoding data via base64 when the value of detectionPictureTransType is "url" or "base64". The field value is the picture Content-ID when the value of detectionPictureTransType is "binary".

10 Facial Recognition

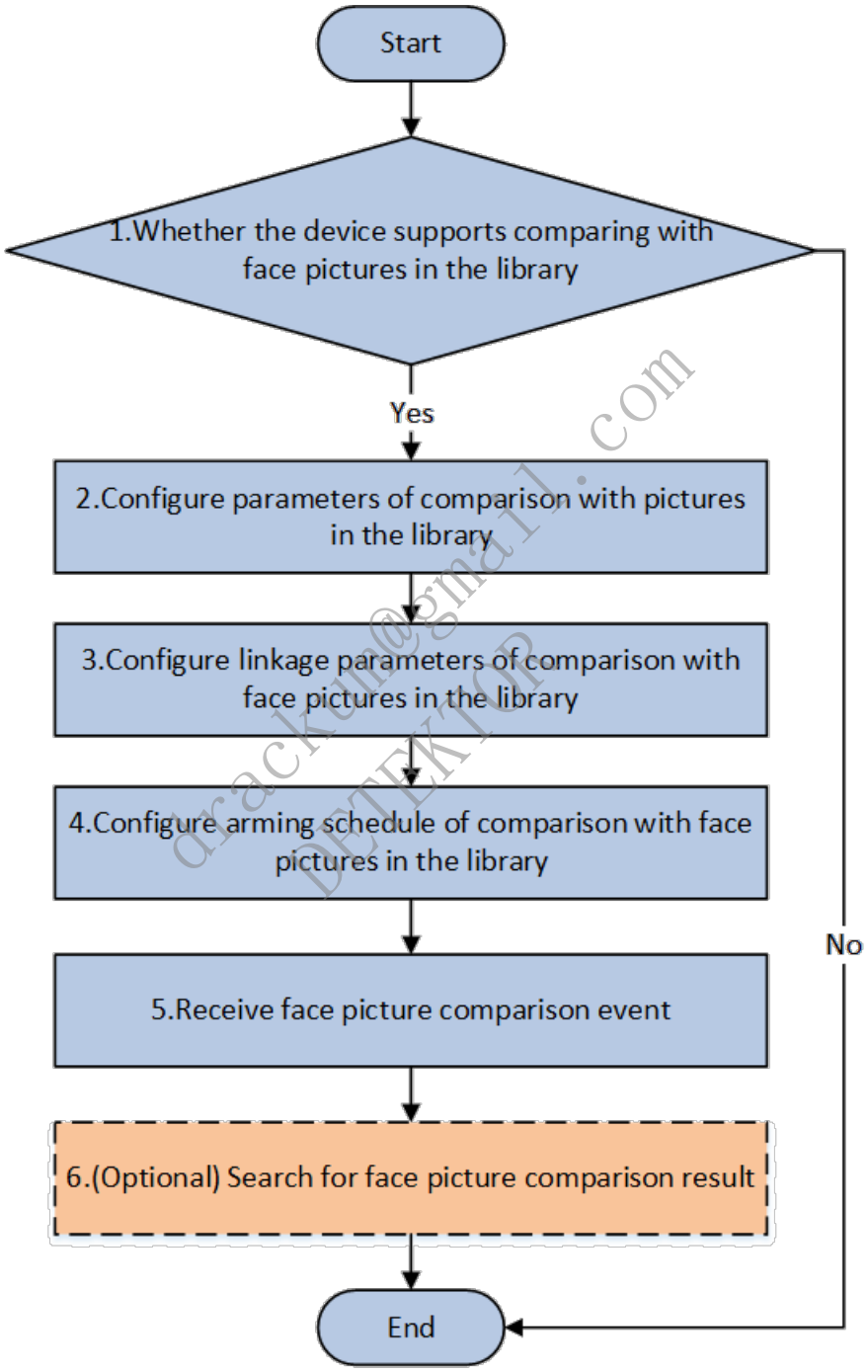
dracknn@gmail.com DETEKTOR

10.1 . Comparison with Pictures in the Library

10.1.1 Introduction to the Function

Note: This feature depends on the face picture library management (refer to Face Picture Library Management Field for details). Analyze and model the face captured in the camera, then compare the face picture with pictures in the library, and lastly return the comparison similarity.

10.1.2 API Calling Flow



dracku@gmail.com
DFTEK10P

1. **Whether the device supports comparing with face pictures in the library:**

   Get the capability node to evaluate whether the device supports comparing pictures with face pictures in the library, `GET /ISAPI/System/capabilities`; If the node `<isSupportFaceContrast>` is returned and its value is `true`, it indicates that the device supports comparing pictures with face pictures in the library.

2. **The parameters of comparison with pictures in the library including:**

   Get the capability of face picture comparison, `GET /ISAPI/Intelligent/channels/<channelID>/faceContrast/capabilities`. Configure the parameters of face

picture comparison, `PUT /ISAPI/Intelligent/channels/<channelID>/faceContrast`; Get the configuration of face picture comparison, `GET /ISAPI/Intelligent/channels/<channelID>/faceContrast`; **Note: The rule parameters support configuring the triggering threshold of face similarity event and specifying information including face picture library ID.**

3. **The linkage parameters of comparison with face pictures in the library:**

   Two sets of configurations linked to business systems are provided due to the front-end and back-end device differences:

   Linkage action of face picture library: for front-end devices (e.g., Deepinview), a business system of single channel and multiple face picture libraries. Therefore it supports configuring different arming linkage parameters via each compared face picture library. Call `GET /ISAPI/Event/triggersCap` to evaluate whether it supports the event linkage configuration capability, it does if `<FaceLibTriggerCap>` is returned, and the relevant configuration APIs are as follows: Configure the linkage action parameters, `PUT /ISAPI/Event/triggers/faceLib-<channelID>/<FDID>`; if users have added a customized face picture library ID when creating one, they can also call `PUT /ISAPI/Event/triggers/faceLib-<channelID>/<customFaceLibID>?FDType=custom` to configure via `customFaceLibID`; get the linkage action parameters, `GET /ISAPI/Event/triggers/faceLib-<channelID>/<FDID>`; Customizing the face picture library ID follows the same flow.

   Linkage action of channel: for back-end devices (e.g., DeepinMind), a business system of multiple channels and single face picture library. Therefore it supports configuring different arming linkage parameters via each video channel. Call `GET /ISAPI/Event/triggersCap` to evaluate whether it supports the event linkage configuration capability, it does if `<FaceContrastTriggerCap>` is returned, and the relevant configuration APIs are as follows: Configure the linkage action parameters of face picture comparison, `PUT /ISAPI/Event/triggers/faceContrast-<channelID>`; get the linkage action parameters of face picture comparison, `GET /ISAPI/Event/triggers/faceContrast-<channelID>`;

4. **Arming schedule of comparison with face pictures in the library**:

   Two sets of arming schedule configurations linked to business systems are provided due to the front-end and back-end device differences: Arming schedule of face picture library: Configure the single-channel arming schedule of face picture library, `PUT /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>`; if users have added a customized face picture library ID when creating one, they can also call `PUT /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>/<customFaceLibID>?FDType=custom` to configure via `customFaceLibID`; get the single-channel arming schedule of face picture library, `GET /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>`; Customizing the face picture library ID follows the same flow. Arming schedule of channel: Configure the arming schedule of face picture comparison, `PUT /ISAPI/Event/schedules/faceContrast-<channelID>`; get the arming schedule of face picture comparison, `GET /ISAPI/Event/schedules/faceContrast-<channelID>`;

5. **Face picture comparison event**:

   When the face captured on the camera is compared with pictures in the library, and the similarity has reached the rule parameters including the threshold, the device will upload the face picture comparison event; the event information mainly includes comparison similarity, captured face and its attributes, and the corresponding registered face picture and its attributes, `"eventType": "alarmResult"`.

6. **Search for face picture comparison result:**

   If supported by device, the face picture comparison result will be saved to the device local storage, and you can search for comparison history via the following API: Search for face picture comparison information, `GET /ISAPI/Intelligent/FDLib/FCSearch`.

10.2 . Search Via Picture Comparison In Face Capture Library

10.2.1 Introduction to the Function

dracktum@gmail.com DEFEKTOR

You can search for matched face pictures in face capture library via comparison with the uploaded local face picture. The general application scenario: compare the target face picture with face pictures in the face capture library to track the target according to the comparison result.

10.2.2 Search via Picture Comparison Synchronously

10.2.1.1 Introduction to the Function

After the request of searching via picture comparison synchronously is applied, the search result will be returned directly in the response message.

10.2.1.2 API Calling Flow

1. **Check whether the device supports searching via picture comparison synchronously.**

   Get the capability of searching via picture comparison: `GET /ISAPI/SDT/Face/searchByPic/capabilities`. If the node `isSupportSyncSearch` is returned and its value is `true`, it indicates that searching via picture comparison synchronously is supported.

   Note: For some devices supporting searching via picture comparison synchronously, the API for getting the capability is invalid, you can call `POST /ISAPI/SDT/Face/searchByPic?supportSync=true` directly.

2. **Submit the task of synchronously searching via picture comparison.**

   Synchronously search for pictures via picture comparison: `POST /ISAPI/SDT/Face/searchByPic?supportSync=true`.

   Note: Searching via picture comparison synchronously supports three types of pictures (`dataType`): picture URL (`faceURL`), picture model data (`targetModelData`, which should be generated in advance via `POST /ISAPI/SDT/Face/pictureAnalysis`, and binary picture data, which is in form format and follows the JSON message.

   The Request Message Example of Binary Picture Data

```
POST /ISAPI/SDT/Face/searchByPic?supportSync=true
Host: device_ip:port
Accept-Language: zh-cn
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: application/json

[JSON Message]
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture Data]
--<frontier>--
```

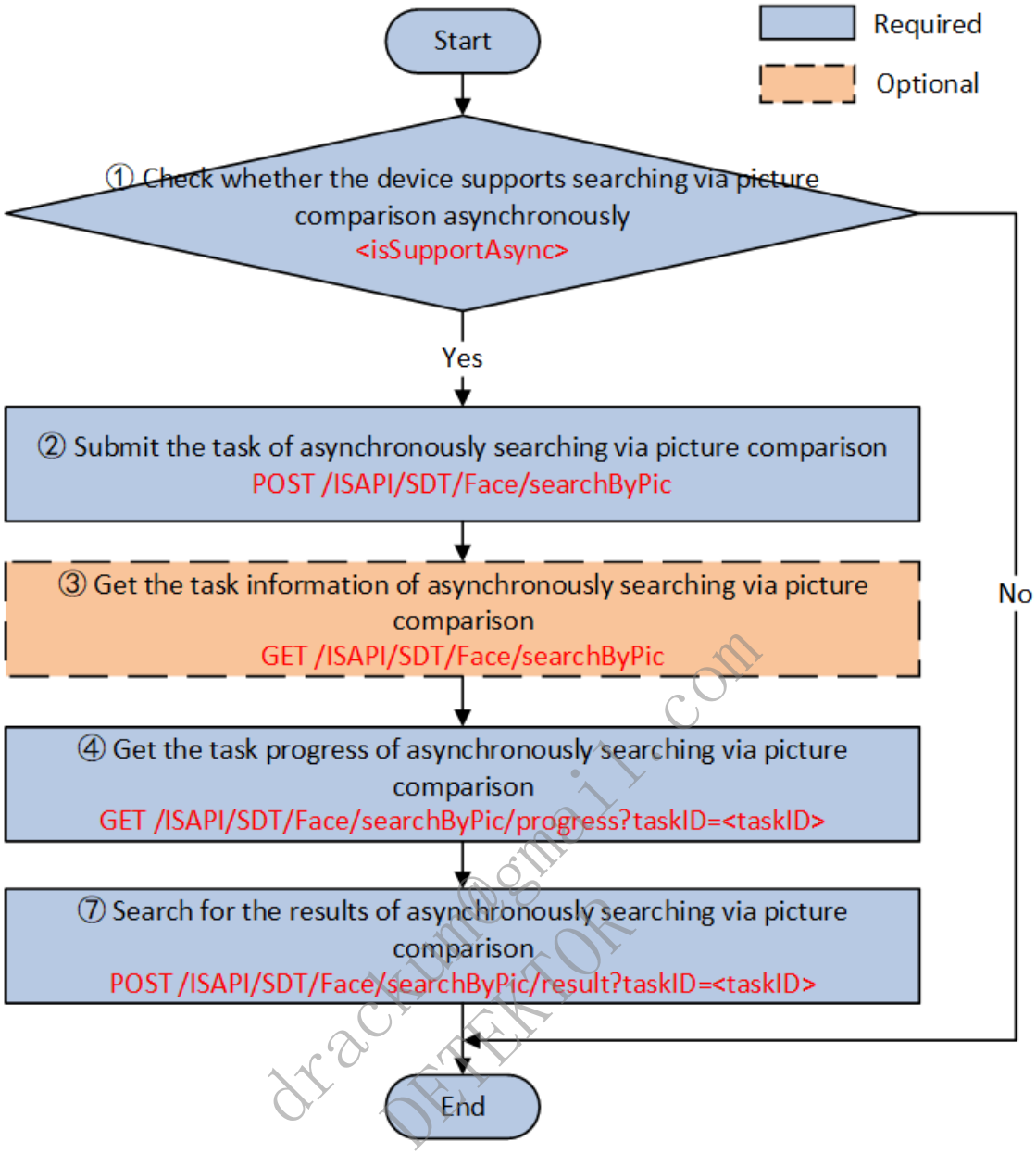
10.2.2 Search via Picture Comparison Asynchronously

10.2.2.1 Introduction to the Function

Submit the task of asynchronously searching via picture comparison and search for results by the returned task ID.

10.2.2.2 API Calling Flow

drackun@mail.com DETEKTOP



1. **Check whether the device supports searching via picture comparison asynchronously.**

   Get capabilities of searching via picture comparison: `GET /ISAPI/SDT/Face/searchByPic/capabilities`. If the node `isSupportAsync` is returned and its value is `true`, it indicates that searching via picture comparison asynchronously is supported.

2. **Submit the task of asynchronously searching via picture comparison.**

   Asynchronously search via picture comparison: `POST /ISAPI/SDT/Face/searchByPic`. The task ID (`taskID`) is returned and will be used in the APIs below.

   Delete the task of asynchronously searching via picture comparison: `DELETE /ISAPI/SDT/Face/searchByPic?format=json&taskID=<taskID>`.

3. **(Optional) Get the task information of asynchronously searching via picture comparison.**

   `GET /ISAPI/SDT/Face/searchByPic`.

4. **Get the task progress of asynchronously searching via picture comparison.**

   Get the task progress: `GET /ISAPI/SDT/Face/searchByPic/progress?taskID=<taskID>`. When the progress (`progress`) is 100%, you can proceed to the following step.

5. **Search for the results of asynchronously searching via picture comparison.**

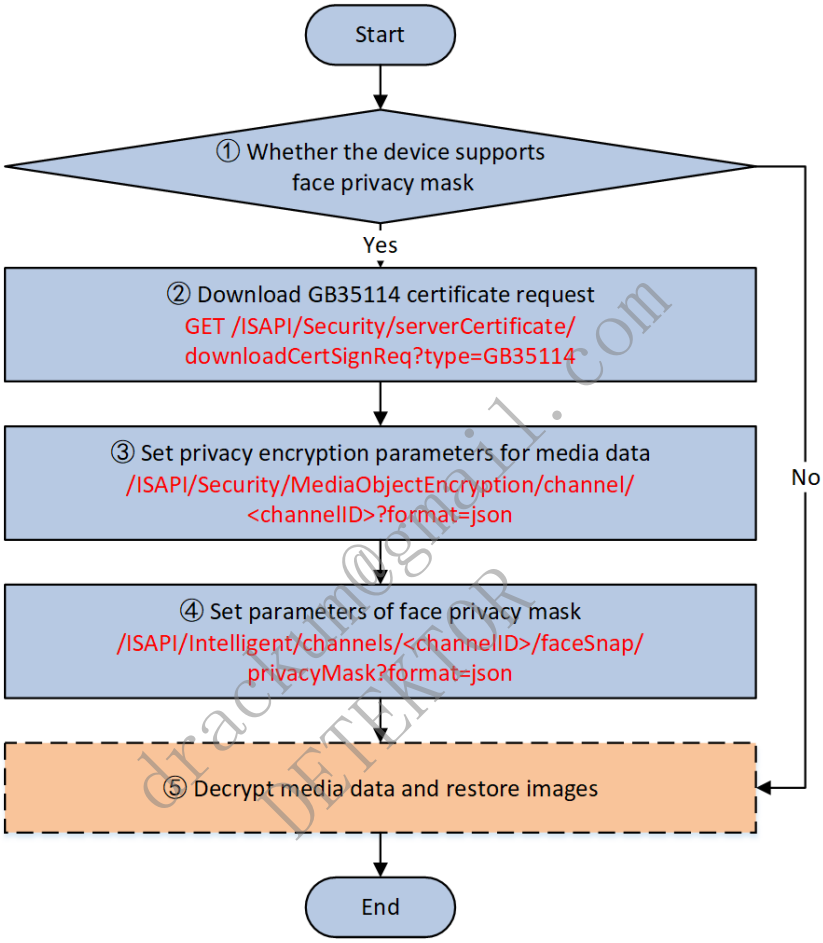
Search for results of asynchronously searching via picture comparison: `POST /ISAPI/SDT/Face/searchByPic/result?taskID=<taskID>`.

10.3 Face Capture

10.3.1 Introduction to the Function

Face capture is to detect whether face pictures show up in images. If set conditions are met and there are faces showing up, the face pictures will be captured and uploaded as alarms.

10.3.2 API Calling Flow



1. **Whether the device supports face capture:**

   Get the system capabilities: `GET /ISAPI/System/capabilities`. If the node (its parent node is) is returned as true, it indicates that the device supports face capture.

2. **Whether the device supports multi-scene face capture:**

   Get the intelligent capabilities of device: `GET /ISAPI/Intelligent/channels/<channelID>/capabilities`. If the node is returned and its value is true, it indicates that the device supports face capture.

3. **Set face picture capturing rules of multiple scenes:**

   If multi-scene capture of a channel is required, you can set the face picture capturing rules for multiple scenes.

   Set the face capture rules of a specified scene: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>`;

   Get the face capture rules of a specified scene: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>`;

Set the face picture capturing rules of multiple scenes for a specified channel: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene`;

Get the face picture capturing rules of multiple scenes for a specified channel: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene`.

4. **Set face picture capturing rules of a specific scene.**

   If specific-scene capture of a channel is required, you can set the face picture capturing rules.

   Get the capabilities of captured face picture rules `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/capabilities`;

   Set the face capture rule parameters: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule`;

   Get the face capture rule parameters: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule`;

   Get the default parameters of face capture rule: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule?defualt=<defualt>`.

5. **Set linkage mode of face picture capture.**

   Set the linkage parameters of face capture: `PUT /ISAPI/Event/triggers/faceSnap-<channelID>`;

   Get the linkage parameters of face capture: `GET /ISAPI/Event/triggers/faceSnap-<channelID>`;

   Delete the linkage of face capture: `DELETE /ISAPI/Event/triggers/faceSnap-<channelID>`.

6. **Configure the arming schedule of face capture.**

   Set the arming schedule of face capture for a specified channel: `PUT /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>`;

   Get the arming schedule of face capture for a specified channel: `GET /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>`;

   Set the arming schedule of face capture: `PUT /ISAPI/Event/schedules/faceSnap`;

   Get the arming schedule of face capture: `GET /ISAPI/Event/schedules/faceSnap`.

7. **Receive face capture and recognition events.**

   If the wanted face shows up in the set area, face recognition event will be reported. The event information includes face pictures and attribute information, etc. `"eventType": "faceCapture"`.

dtackun@gmail.com
DETEKTOR

10.4 Face Modeling and Face Picture Comparison

10.4.1 Face Picture Analysis

10.4.1.1 Introduction to the Function

Face picture analysis includes face detection and face modeling. After analyzing facial features (including age, gender, glasses, smiling, face angle, face frame, etc) of specified pictures, the platform performs modeling and returns model data. To perform face picture comparison in the name list, users need to submit the task of face picture comparison. For details, refer to the process of synchronous/asynchronous face picture comparison.

10.4.1.2 API Calling Flow

1. Get the capabilities of analyzing face pictures: `GET /ISAPI/SDT/Face/pictureAnalysis/capabilities`. If calling succeeded, it indicates that the device supports face picture analysis function.

2. Analyze face pictures (modeling): `POST /ISAPI/SDT/Face/pictureAnalysis`. The API will return the analysis result to the caller synchronously.

to the caller synchronously.

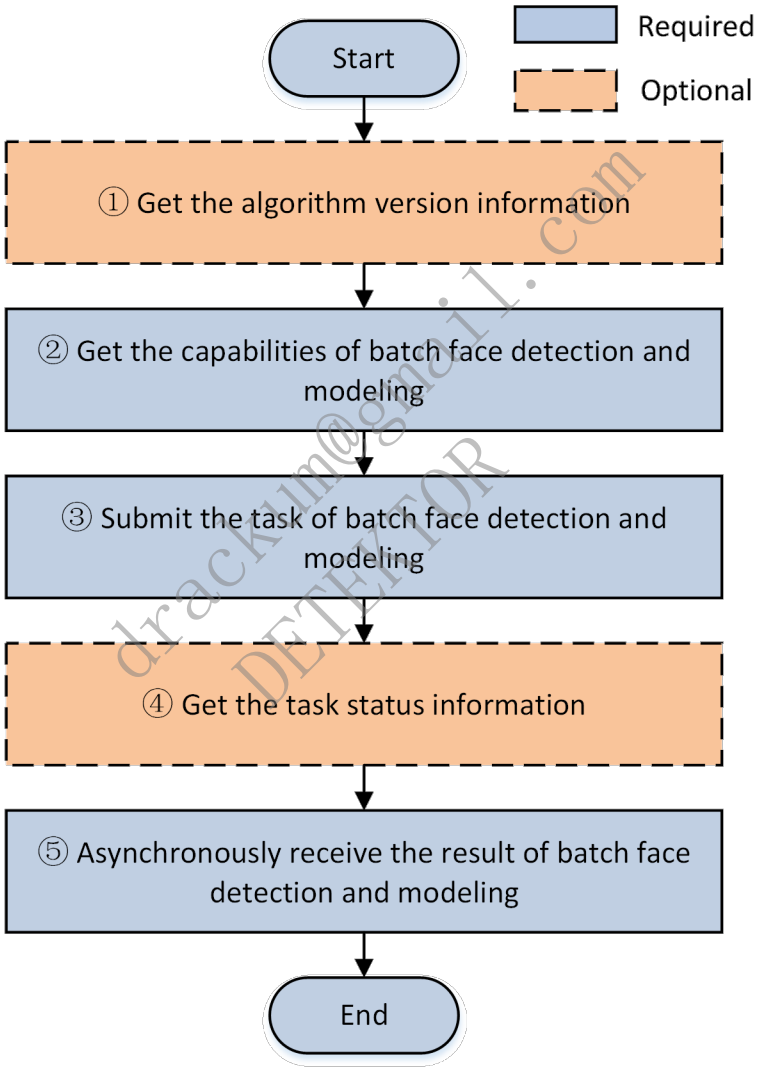
10.4.2 Batch Face Picture Analysis

10.4.2.1 Introduction to the Function

Batch face picture analysis supports detecting faces in multiple pictures, modeling and analyzing face thumbnails, uploading the analysis result via event type, and linking to corresponding tasks via task ID. Make sure the algorithm resources allocation is completed before performing batch face picture modeling detection. To perform face picture comparison in the name list, users need to submit the task of face picture comparison. For details, refer to the process of synchronous/asynchronous face picture comparison.

Batch picture analysis allows users to specify whether to import pictures into the face picture library. If importing pictures into library is supported, users can search for the status of the pictures imported to the library in a batch.

10.4.2.2 API Calling Flow



dracklm@gmail.com
DETECTOR

1. Get the version information of intelligent algorithm scheduling supported by the device: `GET /ISAPI/Intelligent/algorithmInfos?format=json`.
2. Get the capabilities of batch face detection and modeling: `GET /ISAPI/SDT/Face/pictureAnalysis/batch/capabilities`. Check whether the device supports batch face detection and modeling.
3. Submit the task of batch face detection and modeling: `POST /ISAPI/SDT/Face/pictureAnalysis/batch`. It is required to upload the pictures to server or cloud storage considering only importing the face pictures by URL is allowed.
4. Get the task status information: `GET /ISAPI/SDT/Management/Task/status`.

5. Asynchronously receive the result of batch face picture analysis mode: `eventType: analysisResult`.

10.4.3 Synchronous Face Picture Comparison

10.4.3.1 Introduction to the Function

Synchronous face picture comparison analyzes the facial features in a single picture first, and then compares the picture with pictures in the face picture library, and returns matched face records. It is commonly used for the analysis of a single picture. Users can obtain the face modeling data via face picture modeling function. For details, refer to the process of (batch) face picture analysis.

10.4.3.2 API Calling Flow

1. Get the capabilities of synchronous face picture comparison task: `GET /ISAPI/SDT/Management/Task/facePicture/sync/capabilities?format=json`.
2. Submit the task of synchronous face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/sync?format=json`. The API will return the analysis result to the caller synchronously.

10.4.3.3 Exception Handling

10.4.3.3.1 Error Code

statusCode	statusString	subStatusCode	Error Code	Description	Remarks
4	Invalid Operation	noMoreTasksCanBeAdded	0x4000118A	No more tasks can be added.	No more tasks can be added. Delete the completed or unneeded tasks and try again.
4	Invalid Operation	userScheOffline	0x40001040	Task scheduler server is offline.	Please contact the technical support.
4	Invalid Operation	clusterNodeOffline	0x40001075	Cluster node is offline.	Cluster node is offline. Check the cluster status.
4	Invalid Operation	destinationaddressinvalid	0x400011c6	Invalid third-party address.	Invalid third-party address. Check again.
4	Invalid Operation	cloudAnalysisRequestInformationBuildFailed	0x4000110C	Creating request message of cloud analysis failed	Please provide related log and packet capture information and contact the

drackum@gmail.com DETEKTOR

				failed.	the technical support.
4	Invalid Operation	cloudAnalysisResponseInformationParseFailed	0x4000110D	Parsing response message of cloud analysis failed.	Please provide related log and packet capture information and contact the technical support.
4	Invalid Operation	cloudAnalysisServerReturnFailed	0x400010FC	No response is returned by cloud analytic service.	Please provide related log and packet capture information and contact the technical support.
4	Invalid Operation	numberReachedLimit	0x40001119	The number reached limit.	No more configured items can be added. Delete unneeded items and try again.
4	Invalid Operation	tsModelError	0x40001254	Forwarding service module error.	Incorrect task receiving address. Check again.
4	Invalid Operation	anModelError	0x40001255	Computing node error.	Please contact the technical support.
4	Invalid Operation	schedulingdatabaseisfull	0x400011c4	Database disk is full.	Database disk is full. Try again later or cancel related tasks and try again.
					Scheduling module

drackum@gmail.com
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4	Invalid Operation	ssModelError	0x4000124c	Scheduling module error.	error. Please contact the technical support.
6	Invalid Content	requestMemoryNULL	0x6000003F	Request memory is null.	Please contact the technical support.
2	Device Busy	noMemory	0x20000001	Insufficient memory.	Confirm the remaining memory by checking the value of the free field via the directory path of the linux system. Please provide related log information and contact the technical support.

drackum@gmail.com
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10.4.4 Asynchronous Face Picture Comparison

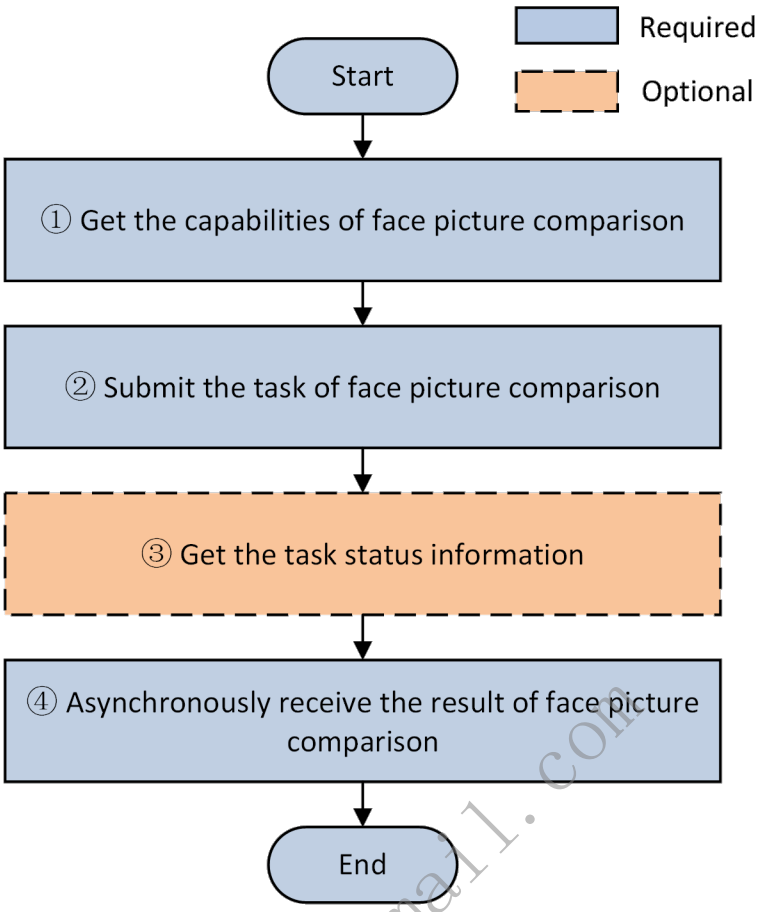
10.4.4.1 Introduction to the Function

Asynchronous face picture comparison analyzes face information in multiple pictures and compares with the faces in the face picture library, and then returns the matched face records. It is commonly used for analyzing multiple pictures. Users can obtain the face modeling data via face picture modeling function. For details, refer to the process of (batch) face picture analysis.

Asynchronous face picture comparison supports single and batch task.

10.4.4.2 API Calling Flow

10.4.4.2.1 Submit the Task of Face Picture Comparison



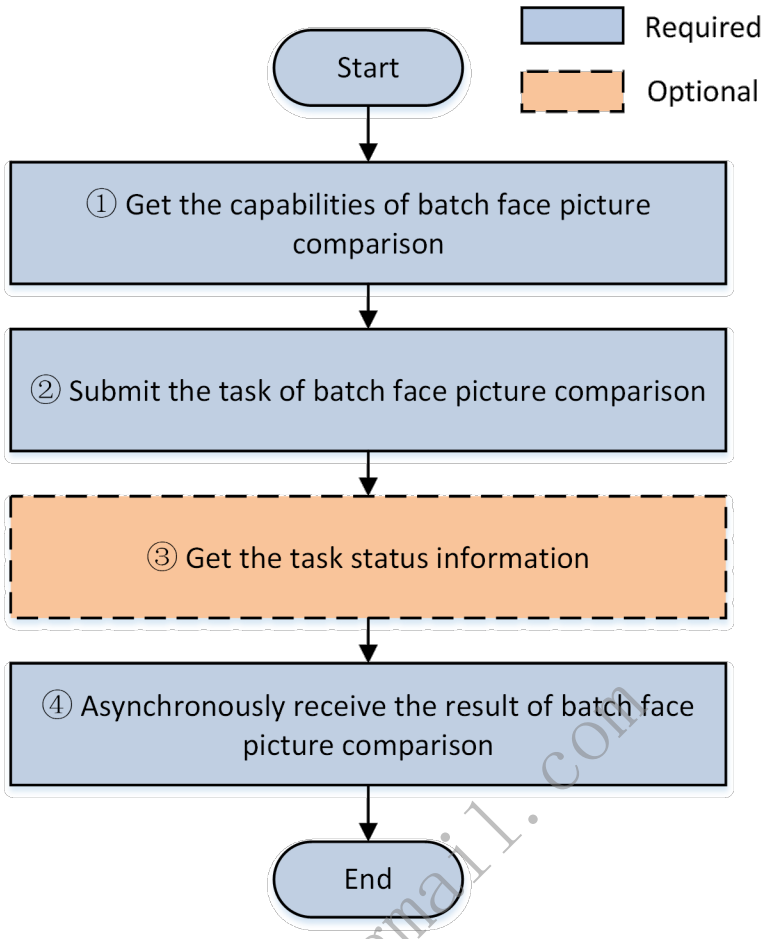
dracku@gmail.com
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1. Get the capability set of the face picture comparison task: `GET /ISAPI/SDT/Management/Task/facePicture/capabilities`.
2. Submit the face picture comparison task: `POST /ISAPI/SDT/Management/Task/facePicture`.

   Notes: It can be applied in scenes including blocklist alarm, stranger detection alarm, frequently appeared person alarm, and one person one archive. Users can distinguish various scenes according to the node `type` of the imported parameters `relateGroup`.
3. Get the task status information: `GET /ISAPI/SDT/Management/Task/status`.
4. Asynchronously receive the result of face picture comparison: `eventType: alarmResult`. The corresponding task is linked via the task ID.

Notes: Users can get the supported maximum number of pictures to be analyzed via the `maxPictureNum` of `GET /ISAPI/SDT/Management/Task/facePicture/capabilities`.

10.4.4.2.2 Submit the Task of Batch Face Picture Comparison



crackum@gmail.com
DETEKTOR

1. Get the capabilities of the task of batch face picture comparison: `GET /ISAPI/SDT/Management/Task/facePicture/batch/capabilities`.
2. Submit the task of batch face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/batch`.

   Notes: It can be applied in scenes including blocklist alarm, stranger detection alarm, frequently appeared person alarm, and one person one archive. Users can distinguish various scenes according to the node `type` of the imported parameters `relateGroup`.
3. Get the task status information: `GET /ISAPI/SDT/Management/Task/status`.
4. Asynchronously receive the result of face picture comparison: `eventType: alarmResult`. The corresponding task is linked via the task ID.

Notes: Users can get the supported maximum number of pictures to be analyzed via the `maxPictureNum` of `GET /ISAPI/SDT/Management/Task/facePicture/batch/capabilities`.

10.4.5 Transmitted Parameters

10.4.5.1 Introduction to the Function

For data center products, transmitted parameters (targetAttrs) are imported when the platform applies tasks. The transmitted parameters will not be parsed and will be returned when the analysis result is uploaded to link to corresponding alarms. Transmitted content is agreed between the platform and device.

Notes: The message length of the transmitted parameters is restricted, which depends on the corresponding capability set of the task API returned by the device.

10.4.5.2 Sample Message

10.4.5.2.1 The Task of Batch Face Picture Comparison

The task of batch face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/batch`. The sample message of the transmitted parameters targetAttrs are as follows:

```
"targetAttrs": {
/*opt, object, the analysis device will not parse the transmitted content*/
    "deviceId": "1",
    /*opt, string, device ID, range:[,]*/
    "deviceChannel": 1,
    /*opt, int, device channel No., range:[,], step:, unit:, unitType:*/
    "deviceName": "shebei",
    /*opt, string, device name, range:[,]*/
    "faceTime": "2016-09-10T01:01:00.000Z",
    /*opt, string, capture time, range:[,]*/
    "rect": {
        /*opt, object, background picture frame information of face*/
        "height": 1.0,
        /*opt, float, height, range:[,], step:, unit:, unitType:*/
        "width": 1.0,
        /*opt, float, width, range:[,], step:, unit:, unitType:*/
        "x": 0.0,
        /*opt, float, X coordinate, range:[,], step:, unit:, unitType:*/
        "y": 0.0
        /*opt, float, Y coordinate, range:[,], step:, unit:, unitType:*/
    },
    "bkgUrl": "http://absddsdfsdfsdk.jpg",
    /*opt, string, background picture, range:[,]*/
    "cameraIp": "10.11.41.26",
    /*req, string, camera IP address, range:[,]*/
    "cameraIndexCode": "3301023820382193",
    /*req, string, camera No., range:[,]*/
    "longitude": 116.39737,
    /*opt, float, latitude, range:[,], step:, unit:, unitType:*/
    "latitude": 116.39737,
    /*opt, float, latitude, range:[,], step:, unit:, unitType:*/
    "name": "Jack",
    /*opt, string, name, range:[,]*/
    "registerGender": "male",
    /*opt, string, gender, range:[,]*/
    "nation": 1,
    /*opt, int, nation, range:[,], step:, unit:, unitType:*/
    "bornTime ": "2016-01-30",
    /*opt, string, birthday, range:[,]*/
    "address": "New York, the United States",
    /*opt, string, address, range:[,]*/
    "certificateNumber": "111111111111111111",
    /*opt, string, ID card, range:[,]*/
    "credImgUrl": "http://XXX",
    /*opt, string, picture URL, range:[,]*/
    "matchResult": 0,
    /*opt, int, the number of matched results, range:[,], step:, unit:, unitType:*/
    "linkFaceBodyId": "XXX",
    /*opt, string, linkage ID between face and human body picture, range:[,]*/
    "linkFaceVehicleId": "XXX",
    /*opt, string, linkage ID between face and vehicle, range:[,]*/
    "linkFaceMacId": "XXX",
    /*opt, string, linkage ID between face and MAC, range:[,]*/
    "Coordinates": [
        /*req, array, coordinate, subType:object, range:[,]*/
        {
            "timeStamp": "test",
            /*req, string, timestamp, range:[,]*/
            "VT": {
                /*req, object, VT*/
                "h": 1.0,
                /*opt, float, height, range:[,], step:, unit:, unitType:*/
                "w": 1.0,
                /*opt, float, width, range:[,], step:, unit:, unitType:*/
                "x": 0.0,
                /*opt, float, X coordinate, range:[,], step:, unit:, unitType:*/
                "y": 0.0
                /*opt, float, Y coordinate, range:[,], step:, unit:, unitType:*/
            },
            "isPTValid": true,
            /*req, bool, whether the PT is valid*/
            "PT": {
                /*req, object, PT*/
                "x": 0.0,
                /*opt, float, X coordinate, range:[,], step:, unit:, unitType:*/
                "y": 0.0,
                /*opt, float, Y coordinate, range:[,], step:, unit:, unitType:*/
                "z": 0.0
                /*opt, float, Z coordinate, range:[,], step:, unit:, unitType:*/
            },
            "humanBodyClarityType": 1,
            /*ro, opt, enum, type of body completeness, subType:int, (1#totally covered,2#upper body not covered,3#whole body not covered)*/
            "posture": "sitting",
            /*ro, opt, enum, gesture, subType:string, ("sitting", "standingUp")*/
            "motion": "yes",
            /*ro, opt, enum, whether the person is in motion, subType:string, ("yes", "no")*/
            "height": 1,
            /*ro, opt, int, height, range:[0,1000], step:1, unit:cm, unitType:length*/
            "humanRect": {
            /*ro, opt, object, coordinates of the human body in the video, desc:coordinates in the video (x,y,width,height); float (accurate to 3 decimal
places). The origin is the upper-left corner of the screen */
                "h": 1.000,
```

drackum@gmail.com
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```
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "w": 1.000,
                /*ro, req, float, width, range:[0.000,1.000], step:, unit:, unitType:*/
                "x": 0.000,
                /*ro, req, float, X coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
                "y": 0.000
                /*ro, req, float, Y coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
            },
            "faceRect": {
            /*ro, opt, object, coordinates of the human face in the video, desc: coordinates in the video (x,y,width,height); float (accurate to 3 decimal
places). The origin is the upper-left corner of the screen */
                "h": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "w": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "x": 0.000,
                /*ro, req, float, X coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
                "y": 0.000
                /*ro, req, float, Y coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
            },
            "hsRect": {
            /*ro, opt, object, coordinates of the head and shoulder frame in the video, desc: coordinates in the video (x,y,width,height); float (accurate
to 3 decimal places). The origin is the upper-left corner of the screen */
                "h": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "w": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "x": 0.000,
                /*ro, req, float, X coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
                "y": 0.000
                /*ro, req, float, Y coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
            }
        }
    ],
    "targetID": "1",
    /*req, string, the tracking ID of a single camera, range:[,]*/
    "traComplete": true
    /*req, bool, boll*/
}
```

Notes: The length limit of the transmittance node `targetAttrs` depends on the `targetAttrsMaxLen` of the `GET /ISAPI/SDT/Management/Task/facePicture/batch/capabilities` (API of getting the capabilities of batch face picture comparison).

dr ackum@gmail.com DETEKTOR

10.4.6 Result Recipient List

10.4.6.1 Introduction to the Function

Users can specify address of the analysis result recipient by configuring the `destinationType` and `destinationUrl` of the `destination` node. Related APIs:

1. The task of face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture`.
2. The task of batch face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/batch`.

10.4.6.2 Corresponding Relationship List

The relationship between `destinationType` and `destinationUrl` is as follows:

destinationType	destinationType Description	destinationUrl Format	Rema
ServerKafka	Kafka address.	{"topic":"test","url":"10.17.139.184:2181"}	
faceCaptureLib	Captured face picture library of big data.	{"topic":"SNAP_IMAGE_INFO_TOPIC","url":"172.10.201.156:2181"}	
faceAlarmUpload	Upload list alarms by ISAPI.	http://IP:PORT/****(****represents custom string description and empty is allowed)	
AlarmRMQ	AlarmRMQ.	{"topic":"test_face_alarm:test_face","url":"10.15.117.65:6005"}	
KafkaFaceLabel	Fusion big data tagging.	" {"topic":"LABEL_SNAP_IMAGE_INFO_TOPIC","url":"10.66.106.149:2181"}"	
faceLabelUpload	Face picture tagging.	{"topic":"SNAP_IMAGE_INFO_TOPIC","url":"172.10.201.156:2181"}	Its UR same of the faceC

10.5 Face Picture Library Management

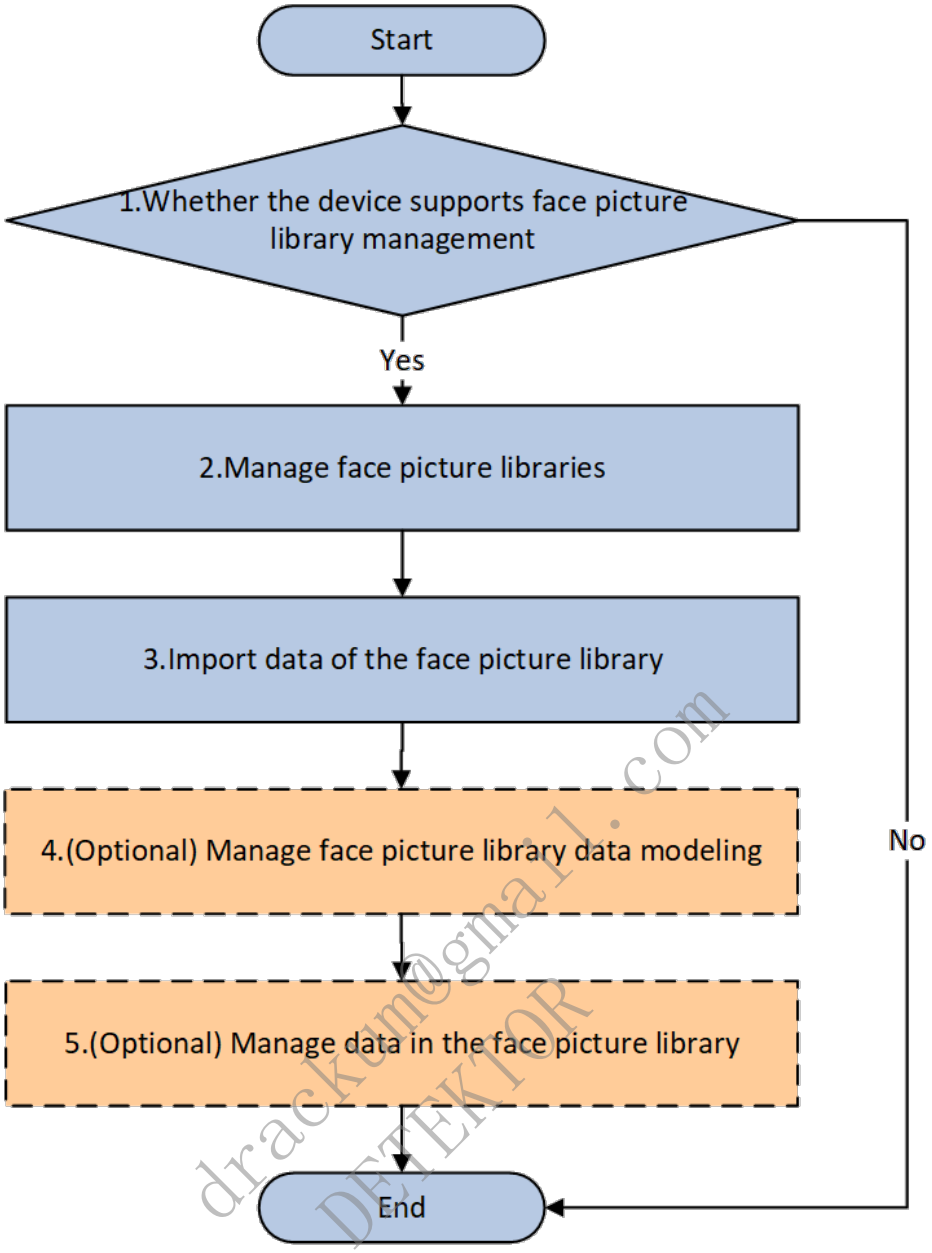
10.5.1 Face Picture Library Management

10.5.1.1 Introduction to the Function

This function is mainly used to manage operations on face picture libraries on the device, such as adding, deleting, editing, or searching for the face picture libraries, importing face pictures and additional person information to the libraries, editing, deleting, or searching for face pictures and additional person information in the libraries, and face picture modeling. This function is also the prerequisite of other face picture library functions such as face picture library comparison.

10.5.1.2 API Calling Flow

dr4ckun@gmail.com
DETERRED



drackum@gmail.com
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1. Get the capability of managing face picture libraries for checking functions supported by the device: `GET /ISAPI/Intelligent/FDLib/capabilities`.
2. Manage face picture libraries.
    i. Create a face picture library: `POST /ISAPI/Intelligent/FDLib`. After the face picture library is created, the device will automatically generate the face picture library ID (that is, `FDID` is returned in the message `<FDLibInfoList>`). If you need to manage the face picture library ID by yourself, you can configure the custom face picture library ID (`customFaceLibID`) in the request message `<CreateFDLibList>` as needed. Afterwards, you can perform operations on the face picture library by the custom ID instead of `FDID`.
    ii. Edit parameters of the face picture library: `PUT /ISAPI/Intelligent/FDLib/<FDID>`. If you have added the custom face picture library ID in the above steps, you can also call `PUT /ISAPI/Intelligent/FDLib/<customFaceLibID>?FDType=custom` to configure parameters using `customFaceLibID`.
    iii. Get parameters of the face picture library: `GET /ISAPI/Intelligent/FDLib/<FDID>`. Operations involved with the custom face picture library ID are the same as those mentioned above.
    iv. Delete the face picture library: `DELETE /ISAPI/Intelligent/FDLib/<FDID>`. Operations involved with the custom face picture library ID are the same as those mentioned above.
3. Import data of the face picture library.

i. Import face picture data and automatically perform modeling, `POST /ISAPI/Intelligent/FDLib/pictureUpload`. By repeatedly applying the previous API, you can import the face picture data (including the face picture and additional face picture information `FaceAppendData`) to the corresponding face picture library; after it is successfully imported, the device will automatically model on the picture which could take some time, then the a face picture ID will be automatically generated (i.e., `PID` is returned in response parameters `MaskInfo`) by device after modeling succeeds; if you want to manually manage the face picture ID, you can add customized face picture ID (`customHumanID`) as needed when applying the request parameter `PictureUploadData`. `PID` can be replaced with customized face picture ID for subsequent face-picture-related operations.

ii. Import face picture library data in concurrent operation mode, `POST /ISAPI/Intelligent/FDLib/pictureUpload?type=concurrent`. If you want to improve the import efficiency, you can apply concurrent operation that the device will only store the picture not model on the picture. You will need to manually model on the picture (refer to Face Picture Library Data Modeling Management Flow for details). Note: Back-end DeepinMind devices will automatically perform modeling after data is imported even though under concurrent operation circumstances; if you demand to manually perform modeling, you can still apply modeling management.

4. (Optional) Manage face picture library data modeling. If you imported face picture library data in concurrent operation mode, you can select a time for device data modeling management after importing completed:
   i. Modeling on registered face pictures, `GET /ISAPI/Intelligent/FDLib/manualModeling`;
   ii. Search for modeling control results, `GET /ISAPI/Intelligent/FDLib/manualModelingRangeTask`;
   iii. Get modeling progress status, `GET /ISAPI/Intelligent/FDLib/manualModeling/progressState`.

5. (Optional) Manage data in the face picture library. You can search, edit, and delete data in the face picture library.
   i. Search for face picture data: `POST /ISAPI/Intelligent/FDLib/FDSearch`;
   ii. Edit face picture data: `PUT /ISAPI/Intelligent/FDLib/FDModify?format=json`, it is for editing data of imported pictures in binary format;
   iii. Edit face picture data in the library: `PUT /ISAPI/Intelligent/FDLib/FDSearch?format=json&FDID=<FDID>&FPID=<FPID>&faceLibType=<faceLibType>`, it is for editing data of imported pictures in URL format;
   iv. Delete face picture data in the library: `PUT /ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&FDID=<FDID>&faceLibType=<FDType>`.

track.un@gmail.com DETEKTOR

10.5.1.3 Exception Handling

10.5.1.3.1 Error Code

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Rem
1	OK	ok	0x00000001			
2	Device Busy	noMemory	0x20000001		Not enough memory.	
4	Invalid Operation	notSupport	0x40000001		Not supported.	
4	Invalid Operation	faceLibDiskFull	0x400020F4		Face picture library storage media is full.	
4	Invalid Operation	faceLibraryIDNotExis	0x400020FD		The face picture library does not exist.	
					The face	

4	Invalid Operation	faceLibInitializing	0x40002136		picture library is initializing. Please wait.	
4	Invalid Operation	notMatchFacelib	0x4000a004		Face picture library is not matched.	
4	Invalid Operation	existSameNameFacelib	0x4000a00a		Face picture library with duplicate name exists.	
4	Invalid Operation	delFacelibFailed	0x4000a00b		Deleting the face picture library failed.	
4	Invalid Operation	modifyFacelibFailed	0x4000a00c		Editing the face picture library failed.	
4	Invalid Operation	createFacelibFailed	0x4000a00d		Creating a face picture library failed.	
4	Invalid Operation	facelibAlreadyUsed	0x4000a010		The face picture library has been used.	
6	Invalid Content	faceLibraryDatabaseError	0x60000032		Database version of face picture library mismatch.	
6	Invalid Content	faceLibraryError	0x60000034		Face picture library error.	
6	Invalid Content	multipleFaceObjectError	0x60000035		Multiple faces are recognized in the pictures to be imported into the face picture library.	
6	Invalid Content	faceLibPicImportingError	0x6000003A		Importing picture into library...	
6	Invalid Content	faceLibraryNoData	0x60000069		No data found in face picture	

dracrum@gmail.com
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					library.	
6	Invalid Content	NoDataForModelingError	0x6000006B		There is no data to be modeled in the face picture library.	
6	Invalid Content	customFaceLibIDRepeat	0x6000006C		Duplicate ID of custom face picture library.	
6	Invalid Content	customFaceDataDeleteing	0x600000A1		Deleting original customized face picture library data... The upper level cannot duplicate the customized face picture library of this ID.	
6	Invalid Content	analysisEnginesFacelibImporting	0x60001005		Importing face picture library to analysis engine...	
6	Invalid Content	facelibDataProcessing	0x6000100b		Processing face picture library data.	
6	Invalid Content	personInfoExtendValueIsTooLong	0x60001083		Customized tag for face picture library exceeded limit.	
6	Invalid Content	personInfoExtendValueContainsInvalidChar	0x60001084		Customized tag for face picture library contains invalid character(s).	

dracskum@gmail.com
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10.5.1.4 Message Example

Import face pictures to face picture library in binary format and automatically perform modeling, `POST /ISAPI/Intelligent/FDLib/pictureUpload`:

```
POST http://10.16.96.29/ISAPI/Intelligent/FDLib/pictureUpload HTTP/1.1
If-Modified-Since: 0
Accept: */*
X-Requested-With: XMLHttpRequest
Content-Type: multipart/form-data; boundary=----------------------------7e27010d1dba
Referer: http://10.16.96.29/doc/page/config.asp
Accept-Language: zh-CN
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/5.0 (compatible; MSIE 10.0; Windows NT 6.1; WOW64; Trident/6.0)
Connection: Keep-Alive
Content-Length: 903330
DNT: 1
Host: 10.16.96.29
Pragma: no-cache
Cookie: language=zh; WebSession=48b693ca2b7301e3119d; sdMarkMenu=14%3AfaceSnapLib; sdMarkTab_1_0=0%3AsettingBasic

----------------------------7e27010d1dba
Content-Disposition: form-data; name="PictureUploadData"

<?xml version='1.0' encoding='UTF-8'?
><PictureUploadData><FDID>55869AFEA1E840958120C97054720372</FDID><FaceAppendData><name>20171222103007395_2</name></FaceAppendData></PictureUploadData>
----------------------------7e27010d1dba
Content-Disposition: form-data; name="importImage"; filename="20171222103007395_2.jpeg"
Content-Type: image/jpeg

【Picture data in binary format】
----------------------------7e27010d1dba--
```

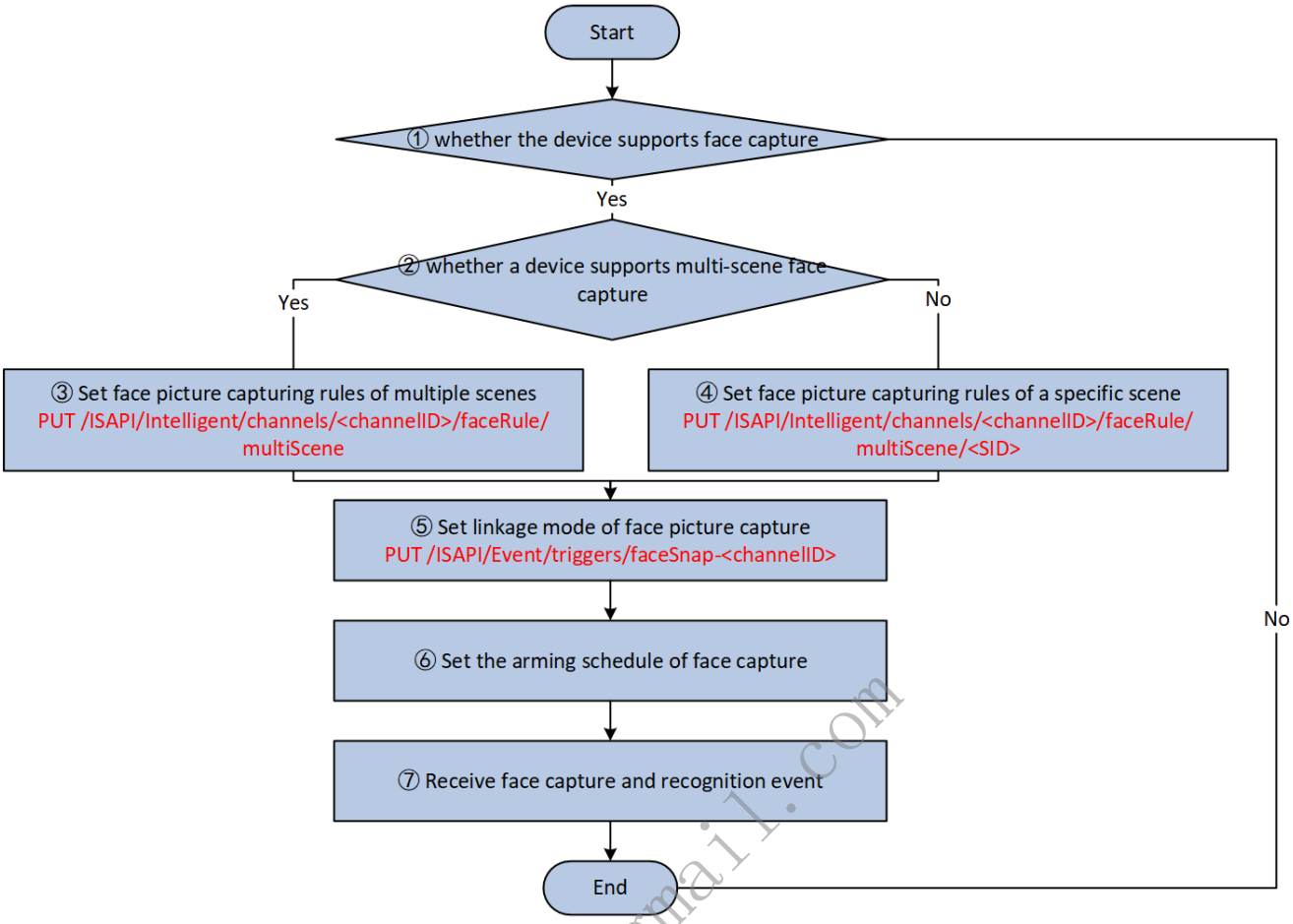
10.6 Privacy Mask on Faces

10.6.1 Introduction to the Function

Privacy mask on faces is to encrypt privacy contents for face capture and media data (see "Video Privacy Mask" API calling flow); it is the process of privacy masking for all faces of cameras and the encrypted images can be restored. First of all, all faces need to be identified, the original media data of faces in the images and masking block data will be replaced, the original media data of faces will be added to the private data of the live view stream and encrypted, then reported to the user synchronously; and the user can decrypt the private data to get the original media data and restore the face image. **Note: The current image restoration only supports private streams, which needs to be restored by integrating the Playback Library.** The following figure shows the effect of face privacy mask by mosaic processing.

dracklin@gmail.com
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10.6.2 API Calling Flow



1. **Whether the video channel supports face privacy mask.** Get the channel intelligence capabilities: `GET /ISAPI/Intelligent/channels/<channelID>/capabilities`, if the node `<isSupportFacePrivacyMask>` is returned as true, it indicates that the device supports face privacy masking.

2. **Set privacy encryption parameters in media data.** Users apply the encryption parameters including secondary key `secretKey` and the update time of the primary key `secretKeyUpdateFrequency` to devices, etc. Get the encryption capabilities of media data privacy: `GET /ISAPI/Security/MediaObjectEncryption/channel/<channelID>/capabilities?format=json`. Set the encryption parameters of media data privacy `PUT /ISAPI/Security/MediaObjectEncryption/channel/<channelID>?format=json`.Get the encryption parameters of media data privacy: `GET /ISAPI/Security/MediaObjectEncryption/channel/<channelID>?format=json`.

3. **Set parameters of face privacy mask.** Users can enable face privacy mask for a channel `enabled`. Get the capability of face privacy masking: `GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json`;
Set the parameters of face privacy mask: `PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json`; Get the parameters of face privacy mask: `GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json`;

4. **(Optional) Decrypt media data and restore images**. The device generates the primary key and updates the primary key periodically according to the primary key update time in the media data privacy encryption parameters, then replaces the original media data with the masking block in the specified sensitive area in the live view screen, after which the images in the sensitive area are encrypted by the primary key and added to the private data of the live view stream, and at the same time the primary key is encrypted by the secondary key and added to the private data of the preview stream. The key is also encrypted by the secondary key and added to the private data of the live view stream. The user can parse the private data in the live view stream and decrypt it with the secondary key to get the primary key, then decrypt the private data with the primary key to get the original media data and restore the image.

dracklm@gmail.com
HIKVISION

Note: At present, image restoration is only supported by private streams. You need to get the private data in the live view stream by integrating Playback Library, and get the primary key by decrypting the secondary key, then decrypt the private data by the primary key to get the original media data; and finally call the Playback Library API to restore the image. You can be refer to Playback library interface integration manual for playback library related APIs.

11 Human Body Recognition

11.1 Search by Human Body Picture

11.1.1 Function Introduction

Search by human body picture is the feature-based image search. It allows users to submit a human body picture as an input, and the system retrieves the most relevant images from a large-scale image records based on the input.

11.1.2 API Calling Flow

Capability Check

- Get device system capabilities: `GET /ISAPI/System/capabilities`. If `isSupportHumanRecognition` returns true, this function is supported.
- Get human body detection capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`.

Human Detection and Modeling

- Human detection: `POST /ISAPI/Intelligent/humanRecognition?format=json`.
- Human modeling: `POST /ISAPI/Intelligent/humanModel?format=json`.

Synchronous Search

- Get synchronous search capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If `isSupportSync` returns true, the synchronous search is supported.
- Submit a synchronous search request: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/sync?format=json`.

Asynchronous Search

- Get asynchronous search capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If `isSupportAsync` returns true, the asynchronous search is supported.
- Submit an asynchronous search task: `POST /ISAPI/Intelligent/humanRecognition/searchByPic?format=json`.
- Get the asynchronous search task progress: `GET /ISAPI/Intelligent/humanRecognition/searchByPic/progress?format=json&taskID=<taskID>`.
- Get the search task result: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/result?format=json`.
- Delete the search task: `DELETE /ISAPI/Intelligent/humanRecognition/searchByPic?format=json&taskID=<taskID>`.
- Search for task information.
  - Get human detection capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If `isSupportTaskInfo` returns true, searching for task information is supported.
  - Get task search capabilities: `GET /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo/capabilities?format=json`.
  - Search for task information: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo?format=json`.

Export Image Search Results

- Get human detection capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If the node `isSupportExport` returns true, exporting image search results is supported.
- Submit the exporting task: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/result/export?format=json`.
- Get the progress of exporting task: `GET`

drackun@gmail.com DETEKTOR

`/ISAPI/Intelligent/humanRecognition/searchByPic/result/export/progress?format=json&taskID=<taskID>`.

- Stop the exporting task: `PUT /ISAPI/Intelligent/humanRecognition/searchByPic/result/export/stop?format=json`.

12 Video Recognition

12.1 Face Picture Library Management

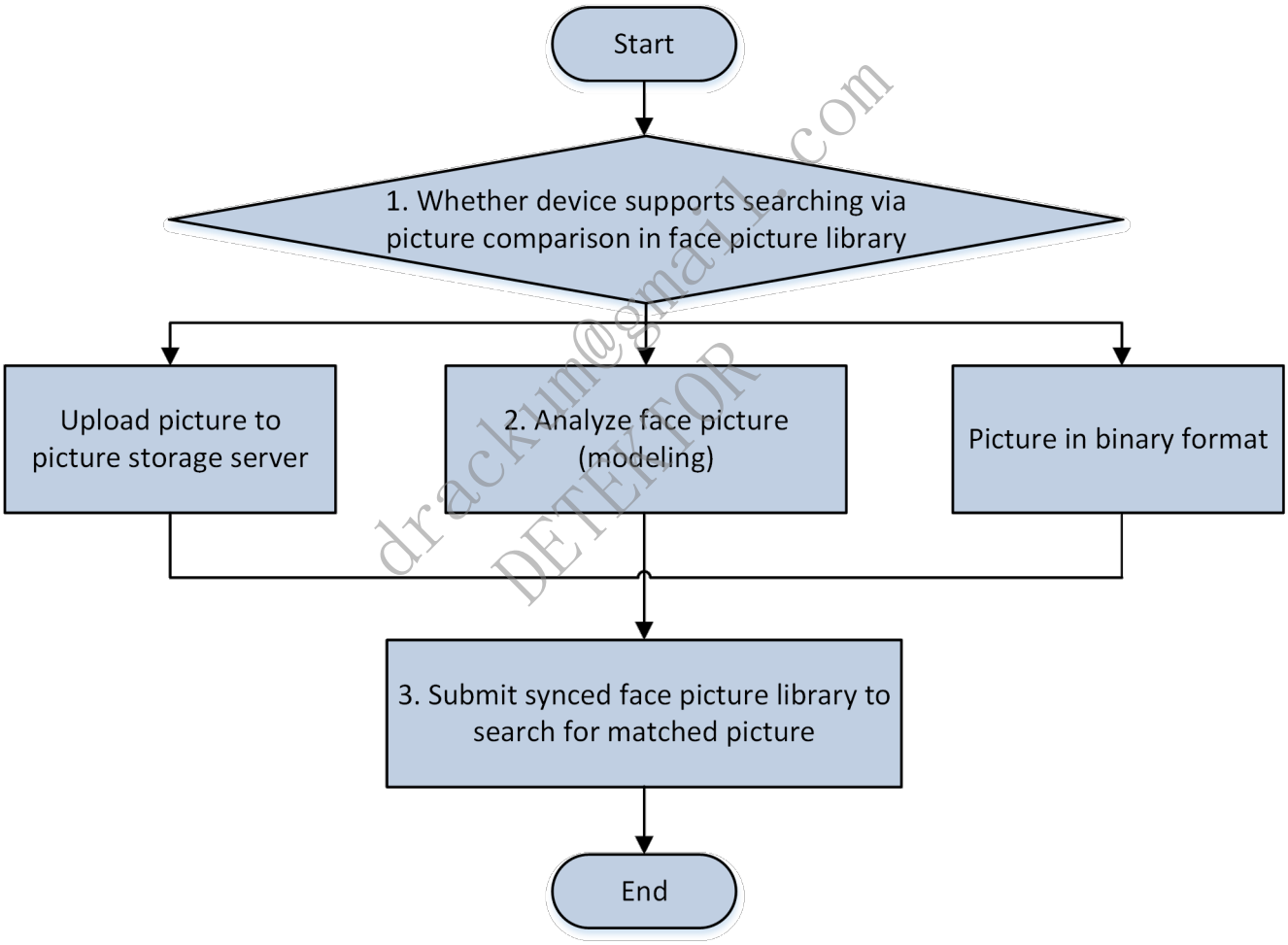
12.1.1. Search via Picture Comparison in Face Capture Library

12.1.1.1 Introduction to the Function

Picture comparison with captured picture library

Scene application: Search for similar person in face picture library, which can be used for identity confirmation.

12.1.1.2 API Calling Flow



dracknm@gmail.com
DETECTOR

1. **Check whether the device supports searching by picture synchronously.**

   Get the capability of searching via picture comparison: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`. If the node `isSuportFSSearchByPic` is returned and its value is `true`, it indicates that searching via picture comparison synchronously is supported.

   Note: For some devices supporting searching via picture comparison synchronously, the API for getting the capability is invalid, you can call `POST /ISAPI/Intelligent/FDLib/searchByPic?format=json` directly.

2. **Three methods are supported when searching picture.**

   By URL:

   Based on the picture storage server, picture can be uploaded and you can get the picture URL to download the

picture file via the URL.

By model data:

Get face picture capability: `GET /ISAPI/SDT/Face/pictureAnalysis/capabilities`;

Analyze (model) face pictures: `POSTT /ISAPI/SDT/Face/pictureAnalysis`; Return picture modeling data: `targetModelData`.

By binary picture:

Read binary picture files.

3. **Submit synced face picture library to use picture to search picture**

   Search via picture comparison in face capture library: `POST /ISAPI/Intelligent/FDLib/searchByPic?format=json`;

   Note: Input parameter node `dataType` include three modes. Picture URL corresponds to `faceURL`, model data corresponds to `targetModelData`, binary data mode is in form of form, and its JSON data is followed by binary picture data.

   The request message example of binary picture data is as follows:

```
POST /ISAPI/Intelligent/FDLib/searchByPic?format=json
Host: device_ip:port
Accept-Language: zh-cn
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive

--<frontier>
Content-Disposition: form-data; name=""
Content-Type: application/json

[JSON message]
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture data]
--<frontier>--
```

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12.2 Professional Smart Abnormal Event Detection

12.2.1 Introduction to the Function

Professional Smart Abnormal Event Detection includes the following features:

- Line Crossing Detection
- Intrusion Detection
- Region Entrance Detection
- Region Exiting Detection
- Loitering Detection
- People Gathering Detection
- Rapid Movement Detection
- Parking Detection
- Unattended Baggage Detection
- Object Removal Detection
- Running Detection
- Abnormal Number of People Detection
- Person Spacing Change Detection

- Violent Motion Detection
- On/Off Duty Detection
- Falling Down Detection
- Lingering Detection
- Person Density Detection
- Climbing Detection
- Trend Analysis
- In-Toilet Overtime Detection
- Sudden Change of Sound Intensity Detection
- Student Standing Up Detection
- Getting Up from Bed Detection

Professional smart abnormal event detection is designed for specific scenarios: workplace activity detection and pedestrian activity detection.

Workplace activity detection includes but not limited to:

- Using a Mobile Phone
- Number of People Changes
- On/Off Duty
- Person Lingering

Pedestrian activity detection includes but not limited to:

- Running Detection
- People Gathering Detection
- Violent Motion Detection
- Person Falling Detection

drackum@gmail.com
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You can configure workplace activity detection and pedestrian activity detection rules and alarm linkage actions to suit your needs.

Abnormal event detection can be categorized into two types: Basic smart abnormal event detection and professional smart abnormal event detection. Their differences are as follows:

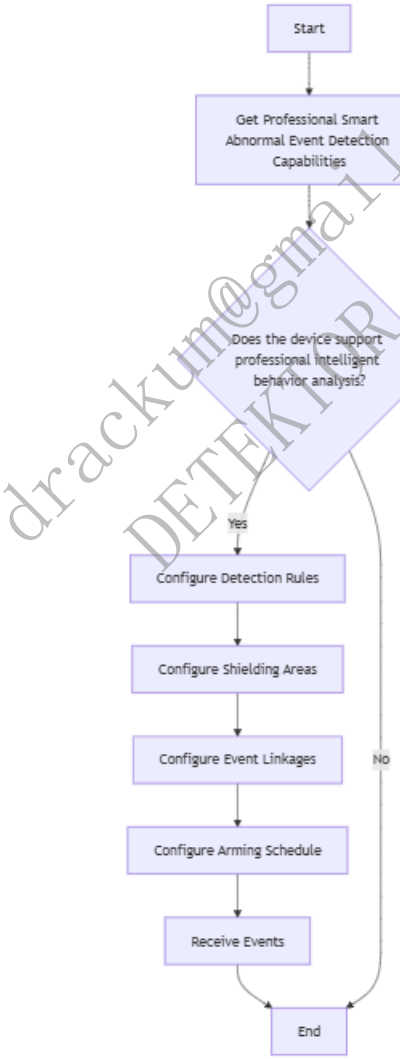
1. **Algorithms**: Traditional algorithms are used for basic smart abnormal event detection, while deep learning algorithms are used for professional smart abnormal event detection.
2. **Detection Effect**: Basic smart abnormal event detection can only analyze targets as people or vehicles. Professional smart abnormal event detection can not only distinguish between targets as people or vehicles but also identify specific behaviors, such as the number of people and human behaviors such as using a mobile phone, violent motion, and running.
3. **Supported Events**: Smart Basic Intelligent Abnormal Behavior Detection supports events such as:
    - Line Crossing Detection
    - Intrusion Detection
    - Region Entrance Detection
    - Region Exiting Detection
    - Loitering Detection
    - People Gathering Detection
    - Rapid Movement Detection
    - Parking Detection
    - Unattended Baggage Detection
    - Object Removal Detection

Professional Smart Abnormal Event Detection supports all the events mentioned above and additional events including:

- Running Detection

- Abnormal Number of People Detection
- Person Spacing Change Detection
- Violent Motion Detection
- On/Off Duty Detection
- Falling Down Detection
- Lingering Detection
- Person Density Detection
- Climbing Detection
- Trend Analysis
- In-Toilet Overtime Detection
- Sudden Change of Sound Intensity Detection
- Student Standing Up Detection
- Getting Up from Bed Detection

12.2.2 API Calling Flow



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Before taking the following steps, ensure that this smart detection has been enabled for devices.

1. **Get Professional Smart Abnormal Event Detection Capabilities**
   - Call `GET /ISAPI/Intelligent/capabilities` to get professional smart abnormal event detetection capabilities and the field `isBehaviorSupport` is returned as `true` in the response message.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/capabilities` to get professional smart abnormal event detetection capabilities for a specified channel and the field `isBehaviorSupport` is returned as `true` in the response message.

Note: `isBehaviorSupportas` `true` indicates support for professional smart abnormal event detetection. The fields of supported events are as follows:

- `isLineDetectionSupport` indicates support for **Line Crossing Detection**.
- `isFieldDetectionSupport` indicates support for **Intrusion Detection**.
- `isRegionEntranceSupport` indicates support for **Region Entrance Detection**.
- `isRegionExitingSupport` indicates support for **Region Exiting Detection**.
- `isLoiteringSupport` indicates support for **Loitering Detection**.
- `isGroupSupport` indicates support for **People Gathering Detection**.
- `isRapidMoveSupport` indicates support for **Rapid Movement Detection**.
- `isParkingSupport` indicates support for **Parking Detection**.
- `isUnattendedBaggageSupport` indicates support for **Unattended Baggage Detection**.
- `isAttendedBaggageSupport` indicates support for **Object Removal Detection**.
- `isRunningSupport` indicates support for **Running Detection**.
- `isPeopleNumChange` indicates support for **Abnormal Number of People Detection**.
- `isSpacingChangeSupport` indicates support for **Person Spacing Change Detection**.
- `isViolentMotionSupport` indicates support for **Violent Movement Detection**.
- `isLeavePositionSupport` indicates support for **On/Off Duty Detection**.
- `isFailDownSupport` indicates support for ** Falling Down Detection**.
- `isRetentionParamSupport` indicates support for **Lingering Detection**.
- `isSupportPersonDensityDetection` indicates support for **Person Density Detection**.
- `isVcaEventAdvReachHeightSupport` indicates support for **Climbing Detection**.
- `isSupportSituationAnalysis` indicates support for **Trend Analysis**.
- `isSupportToiletTarry` indicates support for **In-Toilet Overtime Detection**.
- `isSupportAudioAbnormal` indicates support for **Sudden Change of Sound Intensity Detection**.
- `isSupportStandUp` indicates support for **Student Standing Up Detection**.
- `isSupportGetUp` indicates support for **Getting Up from Bed Detection**.

2. **Configure Detection Rules**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/capabilities` to get professional smart abnormal event detection capabilities.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to get detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to configure detection rules.
   - Call `DELETE /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to delete detection rules.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>` to get all detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>` to configure all detection rules.

   Note:
   - The `eventType` field in the rules indicates the corresponding event type.
   - The `ruleType` field indicates the event support drawing a detection line or a polygon detection area. Some events (sudden change of sound intensity) do not support drawing a box or line, some events (line crossing, climbing) support drawing a line, and some events (except sudden change of sound intensity, line crossing, and climbing) support for polygon detection areas.

3. **(Optional) Configure Shielding Areas**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/Shield/capabilities` to get professional smart area shielding capabilities.

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- Call `GET /ISAPI/Intelligent/channels/<channelID>/Shield` to get shielding areas for all events.
- Call `PUT /ISAPI/Intelligent/channels/<channelID>/Shield` to configure shielding areas for all events.

4. **(Optional) Configure Event Linkages**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/notifications` to get event linkages.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/notifications` to configure event linkages.
5. **(Optional) Configure Arming Schedule**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/schedules` to get the schedule to disable the professional smart abnormal event detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/schedules` to configure the schedule to disable the professional smart abnormal event detection rules.

   Note:
   - Up to eight time periods can be configured per day.
   - Each two time periods cannot overlap.
6. **Receive Events**
   - Line Crossing Detection: `"eventType":"linedetection"`
   - Intrusion Detection: `"eventType":"fielddetection"`
   - Region Entrance Detection: `"eventType":"regionEntrance"`
   - Region Exiting Detection: `"eventType":"regionExiting"`
   - Loitering Detection: `"eventType":"loitering"`
   - People Gathering Detection: `"eventType":"group"`
   - Rapid Movement Detection: `"eventType":"rapidMove"`
   - Parking Detection: `"eventType":"parking"`
   - Unattended Baggage Detection: `"eventType":"unattendedBaggage"`
   - Object Removal Detection: `"eventType":"attendedBaggage"`
   - Running Detection: `"eventType":"running"`
   - Abnormal Number of People Detection: `"eventType":"peopleNumChange"`
   - Person Spacing Change Detection: `"eventType":"spacingChange"`
   - Violent Movement Detection: `"eventType":"violentMotion"`
   - On/Off Duty Detection: `"eventType":"leavePosition"`
   - Falling Down Detection: `"eventType":"failDown"`
   - Lingering Detection: `"eventType":"retention"`
   - Person Density Event: `"eventType":"personDensityDetection"`
   - Climbing Detection: `"eventType":"advReachHeight"`
   - Trend Analysis: `"eventType":"situationAnalysis"`
   - In-Toilet Overtime Detection: `"eventType":"toiletTarry"`
   - Sudden Change of Sound Intensity Detection: `"eventType":"audioAbnormal"`
   - Student Standing Up Detection: `"eventType":"standUp"`
   - Getting Up from Bed Detection: `"eventType":"getUp"`
   - Sitting Still Detection: `"eventType":"sitQuietly"`
   - Key Target Standing Up Detection: `"eventType":"keyPersonGetUp"`
   - Using Mobile Phone Detection: `"eventType":"playCellphone"`
   - Yard Lingering: `"eventType":"yardTarry"`
   - Key Target Standing Up Detection: `"eventType":"keyPersonGetUp"`

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- Sitting Still Detection: `"eventType":"sitQuietly"`
- Yard Lingering: `"eventType":"yardTarry"`
- Using Mobile Phone Detection: `"eventType":"playCellphone"`

Note:

- The content of `targetAttrs` is in JSON format. When the event format is XML, use CDATA (Character Data) to include this data in the event and transmit it to the platform.
- CDATA (Character Data) is used to include large amounts of text data in XML documents. This text may contain special characters (such as `<`, `>`, `&`), which have special meanings in XML and typically need to be escaped. CDATA blocks allow these special characters to be included directly in the document without escaping, such as including JavaScript scripts or CSS styles in XML documents.

12.3 Upgrading the Deep Learning Model for Perimeter Intrusion Detection

12.3.1 Introduction to the Function

You can upgrade the deep learning model for perimeter intrusion detection. You can also upgrade both the local training model for NVRs and the training model for network camera channels.

12.3.2 API Calling Flow

1. Determine if the device supports upgrading the deep learning model for perimeter intrusion detection: Call `GET /ISAPI/Intelligent/capabilities`. If the field `isSupportUpgradeDepthPerimeterModel` is returned as `true` in the response message, this function is supported.
2. Get the deep learning model upgrade capabilities: `GET /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/capabilities?format=json`
3. Set the model version: `PUT /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/VersionOperation?format=json`
   - You cannot specify a version to upgrade. Only rolling back to the previous version and restoring to the factory model version are currently supported.
4. Get the model upgrade progress: `GET /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/process?format=json`
5. Set the model version for a channel: `PUT /ISAPI/Intelligent/depthPerimeter/channels/<channelID>/upgradeTrainModel/VersionOperation?format=json`
6. Get the model upgrade progress for a channel: `GET /ISAPI/Intelligent/depthPerimeter/channels/<channelID>/upgradeTrainModel/process?format=json`

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13 Two-Way Audio

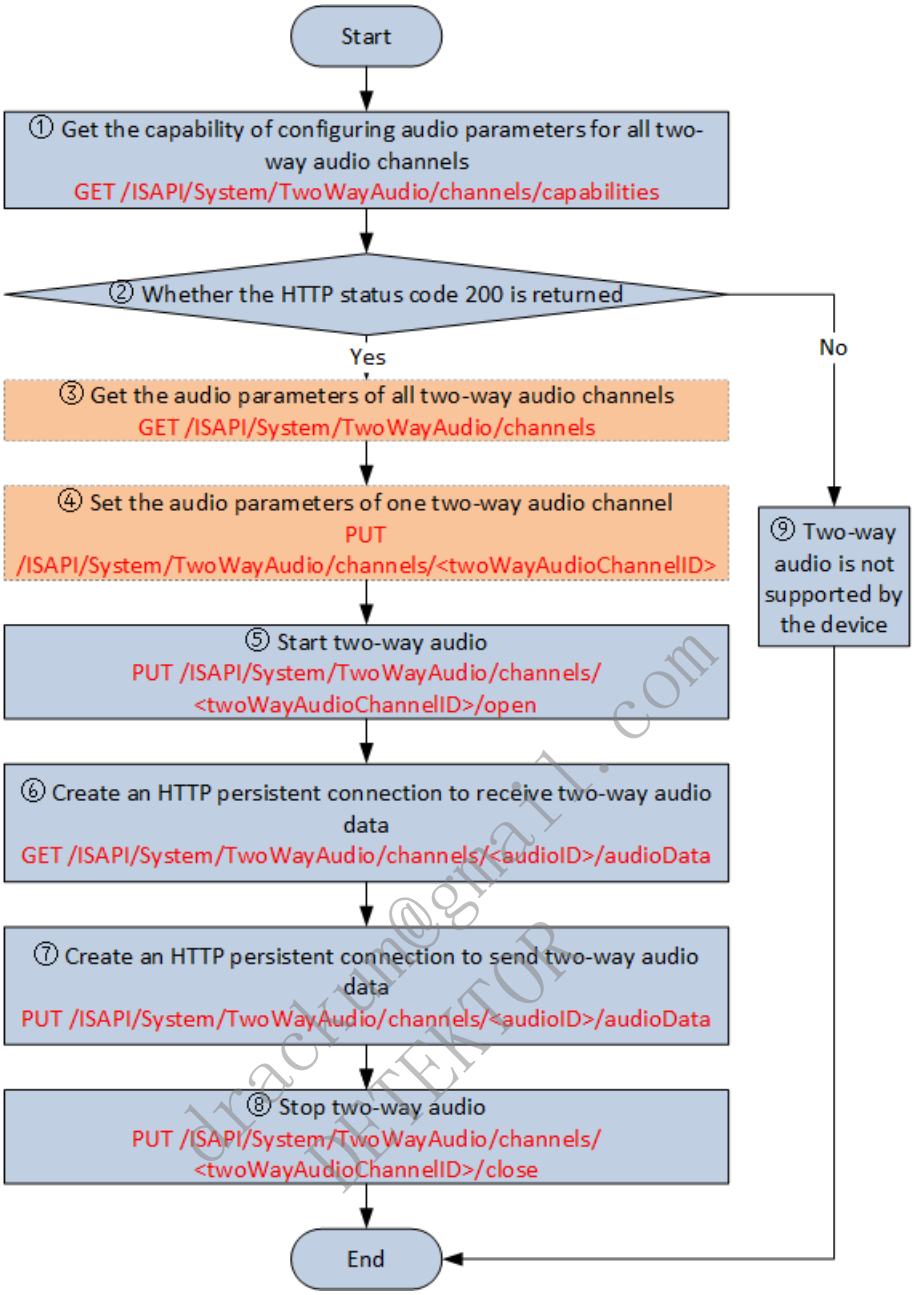
13.1 Two-Way Audio

13.1.1 Introduction to the Function

This function is for the two-way audio between the client and device. When you call the API to implement the two-way audio, it requires the client to collect and encode the local audios and to decode the device audio data.

Note: Before starting two-way audio, check whether the way of collecting device audio data, audio encoding format, volume, etc., is correct or not. The audio encoding format of the client should be consistent with the device audio encoding format.

13.1.2 API Calling Flow



1. Get the capability of configuring audio parameters for all two-way audio channels: `GET /ISAPI/System/TwoWayAudio/channels/capabilities`. Check whether the device supports two-way audio. If the HTTP status code `200 OK` is returned, then the device supports two-way audio.
2. Get the audio parameters of all two-way audio channels: `GET /ISAPI/System/TwoWayAudio/channels`. Get the No. of two-way audio channel for response and parsing `<id>`, audio type `<audioCompressionType>`, frame rate `<audioBitRate>`, sampling rate `<audioSamplingRate>` and so on, to ensure the consistent of audio format during the two-way audio process.
3. Set the audio parameters of one two-way audio channel: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>`. This step is optional, and the `<audioID>` is the channel No. `<id>` responded and parsed in step 2.
4. Start two-way audio: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/open`.
5. Create an HTTP persistent connection to receive the two-way audio data: `GET /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData`.
6. Create an HTTP persistent connection to send the two-way audio data: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData`.
7. When the two-way audio ends, the client disables the persistent connections in steps 6 and 7 and sends the message of stopping the two-way audio: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/close`.

Note:

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- Two-way audio over ISAPI supports digest authentication.
- It supports the two-way audio mode or sending only mode (the platform only call `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData` to send the audio data to devices).
- The `<audioID>` in step 3, 4, 5, 6, and 7 is the two-way audio channel No. which starts from 1. It can be parsed from the `<id>` in the device response message of the step 2.
- The audio type `<audioCompressionType>`, frame rate `<audioBitRate>`, and sampling rate `<audioSamplingRate>` can be parsed from the device response message of the step 2 for audio playing and collection in the step 5 and 6.
- The step 6 and 7 use persistent connections. In the HTTP request headers, you don't need to set `Content-Legth`, but need to set the `Connection: keep-alive` and `Content-Type: application/octet-stream`.
- When the encoding format of audio data in the step 5 and 6 is AAC/MP2L2/MP3, the audio data length is variable, so the frame header should be lengthened by 4 bytes. When the encoding format is G.722.1/G.711alaw/G.711ulaw/G.726/G.729/G.729a/G.729b/PCM/ADPCM/Opus, the audio data length is fixed, there is no need to lengthen the frame header. For example, if the content length of a frame of MP3 audio data is 576 bytes and the header length is 4 bytes, the big end data is 0x00000240, the total length is 580 bytes.
- The damaged audio file in the AC3 format is usually incompatible when using. Currently, the audio algorithm library does not support this format.

Statistics of audio types with fixed lengths:

Audio Type	Fixed Length	Frame Interval
G.722.1	80	40ms
G.711alaw	160/320	20ms/40ms
G.711ulaw	160/320	20ms/40ms
G.726	80	40ms
G.729	10	10ms
G.729a	10	10ms
G.729b	10	10ms
PCM	1920	None
ADPCM	80	20ms
Opus	32/64	20ms/40ms

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13.1.3 Exception Handling

13.1.3.1 Error Codes

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Re
4	Invalid Operation	twoWayAudioInProgressPleaseWait	0x40002068	Two-way audio in progress...Please wait.	Two-way audio in progress...Please wait.	Th op is all Tw au in pr Pl ag af tw au st

13.1.4 Message Format and Example

13.1.4.1 Receive Two-Way Audio Data

```
GET /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
Host: 10.17.115.128
Connection: keep-alive
Content-Type: application/octet-stream

HTTP/1.1 200 OK
Connection: keep-alive
Content-Type: application/octet-stream

//The following are the two-way audio data sent by the device.
```

13.1.4.2 Send Two-Way Audio Data

```
PUT /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
Host: 10.17.115.128
Connection: keep-alive
Content-Type: application/octet-stream

//The following are the two-way audio data sent by the client.
```

dr.ackum@gmail.com
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14 API Reference

14.1 设备通用

14.1.1 事件订阅

14.1.1.1 Establish arming connection

Request URL

GET /ISAPI/Event/notification/alertStream?deployID=<deployID>

Query Parameter

Parameter Name	Parameter Type	Description
deployID	string	0 (client-side arming), 1 (real-time arming). Defaults to the client-side arming when delpoyID does not exist.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, event massage, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPV4 address of the device that triggers an alarm-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPV6 address of the device that triggers an alarm-->test
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm, range:[0,65535]-->0
  </portNo>
  <protocol>
    <!--ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address, range:[0,17]-->00-16-EA-AE-3C-40
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, range:[0,65535]-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm triggering time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, the uploading times of the same alarm, range:[0,65535]-->0
  </activePostCount>
  <eventType>
    <!--ro, req, string, heartbeat type-->test
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:"active" (triggered), "inactive" (not triggered, the heartbeat data)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, range:[0,128]-->test
  </eventDescription>
</EventNotificationAlert>
```

drackum@gmail.com
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14.1.1.2 Set alarm/event subscription parameters

Request URL

PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all"(upload all alarms/events), "list"(upload specified alarm/event)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, "all"-upload all alarms/events,"list"-upload specified alarm/event, subType:string, desc:"all"(upload all alarms/events), "list"(upload
specified alarm/event)-->list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--opt, object, uploading mode of specified alarm/event-->
      <type>
        <!--req, enum, alarm/event types, subType:string, desc:see details in event types: "ADAS"(advanced driving assistance system), "ADASAlarm"(advanced
driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent", "CDsStatus", "DBD"
(driving behavior detection) "GPSUpload", "HFPD"(frequently appeared person), "IO"(I/O Alarm), "IOTD", "LES", "LFPD"(low frequency person detection),
"PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-
alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage",
"audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch",
"changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk
```

```
unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling",
"facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection",
"flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"
(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature
alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person
getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection), "listSyncException"(list synchronization exception),
"loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event), "mixedTargetDetection"(multi-target-type detection),
"modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"
(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm),
"personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection),
"personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception),
"policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection",
"radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient
recording period), "recordException", "regionEntrance", "regionExiting", "retention"(people overstay detection), "rollover", "running"(people running),
"safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"(angular acceleration alarm), "severeHDFailure"(HDD major fault
detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO",
"spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection),
"targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime
detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list
alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion
detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code
event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent",
"ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals),
"ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, operation sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, event sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is
"AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)--
>cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:event linked channel information,and multiple channels can be linked,each channel is
separated by comma,e.g.,"1,2,3,4..."-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get
all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--opt, array, subType:object-->
        <childDevID>
          <!--opt, string, range:[1,128]-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string-->1,2,3,4
      </audioChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4...", desc:event
linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4..."-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--opt, enum, alarm picture format, subType:string, desc:"binary"(binary), "localURL"(device local URL), "cloudStorageURL"(cloud storage URL)--
>cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:5 minute (default). If GUID is not reconnected in the internal, a new GUID
will be generated since the device starts a new arm period-->300
    </interval>
    <StatusSub>
```

drackum@gmail.com
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```
      <!--opt, object, status description-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:reporting is not required if all is true-->true
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <level>
    <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--opt, array, subType:object-->
    <childDevID>
      <!--opt, string, range:[1,128]-->test
    </childDevID>
  </childDevIDList>
  <radarChannels>
    <!--opt, string-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
      <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, subType:object-->
      <event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, string, range:[0,64]-->test
        </eventType>
        <level>
          <!--ro, req, enum, subType:int-->1
        </level>
        <channels>
          <!--ro, opt, string, range:[0,256]-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <audioChannels>
    <!--opt, string-->1,2,3,4
  </audioChannels>
</SubscribeEvent>
```

dracku@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

14.1.1.3 Get alarm/event subscription parameters

Request URL

GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), the (subscription ID) will be returned in message SubscribeEventResponse. When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), and if subscribeEventID is applied, it can be used to replace the in SubscribeEventResponse.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--ro, opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--ro, opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel
list)-->list
  </channelMode>
  <eventMode>
    <!--ro, req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)--
>list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:alarm/event types, which are obtained from the capability, refer to Alarm/Event Types
(eventType)-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
```

dracrun@gmail.com
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```
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--ro, opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--ro, opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--ro, opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--ro, opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--ro, opt, array, subType:object-->
        <childDevID>
          <!--ro, opt, string, range:[1,128]-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--ro, opt, string-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--ro, opt, string-->1,2,3,4
      </audioChannels>
      <childDeviceFormat>
        <!--ro, opt, bool-->true
      </childDeviceFormat>
    </Event>
  </EventList>
  <channels>
    <!--ro, opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is all, this node should not be applied-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--ro, opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, the <pictureURLType> of the <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--ro, opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--ro, opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--ro, opt, object, status description-->
      <all>
        <!--ro, opt, bool, whether to subscribe to all channels, HDDs, and capability set changes-->false
      </all>
      <channel>
        <!--ro, opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--ro, opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--ro, opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the value of <all> is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
```

dreckum@gmail.com
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```
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--ro, opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--ro, opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--ro, opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <level>
    <!--ro, opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--ro, opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--ro, opt, array, subType:object-->
    <childDevID>
      <!--ro, opt, string, range:[1,128]-->test
    </childDevID>
  </childDevIDList>
  <radarChannels>
    <!--ro, opt, string-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
      <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, subType:object-->
      <event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, string, range:[0,64]-->test
        </eventType>
        <level>
          <!--ro, req, enum, subType:int-->1
        </level>
        <channels>
          <!--ro, opt, string, range:[0,256]-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <audioChannels>
    <!--ro, opt, string-->1,2,3,4
  </audioChannels>
  <retransmissionID>
    <!--ro, opt, string, range:[1,64]-->123456
  </retransmissionID>
  <retransmissionTimeoutDuration>
    <!--ro, opt, int, range:[1,168], unit:h-->1
  </retransmissionTimeoutDuration>
  <SubscribeISAPIMessage>
    <!--ro, opt, object-->
    <EventTypeList>
      <!--ro, req, array, subType:object-->
      <eventType>
        <!--ro, req, enum, subType:string-->AIOP_Video
      </eventType>
      <uploadPicEnabled>
        <!--ro, opt, bool-->false
      </uploadPicEnabled>
    </EventTypeList>
    <EventList>
      <!--ro, req, array, subType:object, range:[0,200]-->
      <Event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, enum, subType:string-->AIOP_Video
        </eventType>
        <uploadPicEnabled>
          <!--ro, opt, bool-->false
        </uploadPicEnabled>
      </Event>
    </EventList>
  </SubscribeISAPIMessage>
</SubscribeEvent>
```

drackum@gmail.com
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14.1.1.4 Subscribe to events/alarms in arming mode

Request URL

POST /ISAPI/Event/notification/subscribeEvent?deployID=<deployID>

Query Parameter

Parameter Name	Parameter Type	Description
deployID	enum	0 (device local arming), 1 (real-time arming by platform). If the deployID is absent, it defaults to device local arming.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel
list)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)--
>list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--opt, object, subscription of a specified alarm/event-->
      <type>
        <!--req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection), "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioException", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fieldDetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "CertificateCaptureEvent"(person ID
capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
"TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status),
"certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)--
>mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, minor operation type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, minor event type, desc:this node is required when type is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)--
>cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
```

```
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get
all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel
check-in.
3. When this node is not applied, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
        <childDevID>
          <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string, arm by radar channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </audioChannels>
      <childDeviceFormat>
        <!--opt, bool, upload by sub device, desc:upload by sub device-->true
      </childDeviceFormat>
      <unitChannels>
        <!--opt, string, arm by optical fiber channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </unitChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is
all, this node should not be applied-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud
storage URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is
not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end
devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client
software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--opt, object, status information-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:it is not required if the value of <all> is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When
this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform
focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-
->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will
be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm--
>true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
```

dinckum@gmail.com
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```
<identityKey>
  <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
</identityKey>
<ConferenceIDList>
  <!--opt, array, video conference ID list, subType:object-->
  <ConferenceID>
    <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
  </ConferenceID>
</ConferenceIDList>
<subscribeEventID>
  <!--opt, string, custom subscription ID. The platform should guarantee the uniqueness of ID when applying it. The automatic network replenishment of events will be enabled by default after the node is applied, range:[1,128], desc:when an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), if this node is applied, it can be used to replace <id> (subscription ID) returned in message SubscribeEventResponse-->test
</subscribeEventID>
<level>
  <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
</level>
<middleLevelPictureEnabeld>
  <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
</middleLevelPictureEnabeld>
<childDevIDList>
  <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
  <childDevID>
    <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
  </childDevID>
</childDevIDList>
<SubscribeISAPIMessage>
  <!--opt, object-->
  <EventTypeList>
    <!--opt, array, event list, subType:object, desc:event list-->
    <eventType>
      <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
    </eventType>
    <uploadPicEnabled>
      <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
    </uploadPicEnabled>
  </EventTypeList>
  <EventList>
    <!--req, array, event list, subType:object, range:[0,200]-->
    <Event>
      <!--opt, object, a single event-->
      <eventType>
        <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
      </eventType>
      <uploadPicEnabled>
        <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
      </uploadPicEnabled>
    </Event>
  </EventList>
</SubscribeISAPIMessage>
<radarChannels>
  <!--opt, string, batch arming by radar channel, desc:if this field is applied, radarChannels in Event will not take effect. To arm all radar channels, this field should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
</radarChannels>
<qosLevel>
  <!--ro, opt, object, subscription events qos level, desc:subscription events qos level-->
  <defaultLevel>
    <!--ro, req, enum, default qos level, subType:int, desc:default qos level-->0
  </defaultLevel>
  <eventList>
    <!--ro, opt, array, custom event qos level list, subType:object-->
    <event>
      <!--ro, opt, object, custom event qos level-->
      <eventType>
        <!--ro, req, string, event type, range:[0,64], desc:its values refer to event types of different domains-->test
      </eventType>
      <level>
        <!--ro, req, enum, qos level of this event type, subType:int, desc:qos level of this event type-->1
      </level>
      <channels>
        <!--ro, opt, string, video resource channel No., range:[0,256], desc:no such node means it applies to all channels; otherwise it applies to the channels listed in the array, and channels not listed use default qos level-->0,1,2
      </channels>
    </event>
  </eventList>
</qosLevel>
<audioChannels>
  <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
</audioChannels>
<unitChannels>
  <!--opt, string, batch arming by optical fiber channel, desc:if this field is applied, unitChannels in Event will not take effect. To arm all fiber channels, this node should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
</unitChannels>
<retransmissionID>
  <!--opt, string, ANR arming ID, range:[1,64], desc:UUID (an unique ID generated by the client). If this field is applied during establishing the event arming/subscription connection, the ANR of subscribed event will be enabled by default; when the network reconnected, the platform can apply the same UUID to tell the device to re-upload saved event information during network disconnection period-->123456
</retransmissionID>
<retransmissionTimeoutDuration>
  <!--opt, int, data transmission timeout for ANR, range:[1,168], unit:h, desc:data transmission timeout for ANR-->1
</retransmissionTimeoutDuration>
<subscribeWithoutPic>
  <!--opt, bool-->false
```

crackum@gmail.com
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```
  </subscribeWithoutPic>
</SubscribeEvent>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEventResponse xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, subscription result, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, subscription ID, which is returned by the device. The device should guarantee the uniqueness of ID, range:[1,64]-->1
  </id>
  <FailedEventList>
    <!--ro, opt, array, list of events, of which subscription failed, subType:object, desc:when subscription of one or multiple events failed, this node
should be returned. The client software can check if all events are subscribed to by checking whether the node <FailedEventList> is returned-->
    <Event>
      <!--ro, opt, object, event, of which the subscription failed. The event/alarm type name can be obtained via the capability set-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:event type-->ADAS
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL).-->binary
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel No., desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <subStatusCode>
        <!--ro, req, enum, error code, subType:string, desc:refer to the error code dictionary-->activeNumMax
      </subStatusCode>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel
check-in.
3. When this node is not returned, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
    </Event>
  </FailedEventList>
</SubscribeEventResponse>
```

dfackum@gmail.com
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14.1.1.5 Get the alarm/event subscription capability

Request URL

GET /ISAPI/Event/notification/subscribeEventCap

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SubscribeEventCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <channelMode opt="all,list">
    <!--ro, opt, enum, channel subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel list)-->list
  </channelMode>
  <eventMode opt="all,list,allAndChild">
    <!--ro, req, enum, event subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)-->list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm" (advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpLoad" (GPS information upload), "HFPD"(frequently appeared person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch" (video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection), "listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention" (people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm" (angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "target(capture)", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event), "TransactionRecordEvent", "HealthInfoSyncQuery" (health information search event), "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery" (searching the consumption status), "certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
      </type>
      <childDeviceFormat opt="true,false">
        <!--ro, opt, bool, upload event by sub device, attr:opt{req, string}-->true
      </childDeviceFormat>
    </Event>
  </EventList>
  <ChangedUploadSub>
    <!--ro, opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--ro, opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--ro, opt, bool, status information-->true
      <all>
        <!--ro, opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--ro, opt, bool, status of subscribed channels, desc:reporting is not required if all is true-->true
      </channel>
      <hd>
        <!--ro, opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
      </hd>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey max="64">
    <!--ro, opt, string, subscription connection interaction key, range:[1,64], attr:max{req, int}, desc:N/A-->test
  </identityKey>
</SubscribeEventCap>
```

14.1.1.6 Cancel subscribing alarm/event

Request URL

PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	Subscription event ID, which should match the id field in the SubscribeEventResponse of the response message from the POST /ISAPI/Event/notification/subscribeEvent?deployID=.This field is optional. When the protocol is /ISAPI/Event/notification/unSubscribeEvent, it indicates that all subscription links are to be unsubscribed. When the protocol is /ISAPI/Event/notification/unSubscribeEvent?ID=XXX, it indicates that a specific subscription link is to be unsubscribed.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

drackum@gmail.com
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14.1.2 异常检测

14.1.2.1 Get the linkage parameters of HDD error

Request URL

GET /ISAPI/Event/triggers/diskerror

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->diskerror
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->diskerror
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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14.1.2.2 Set the linkage parameters of HDD error

Request URL

PUT /ISAPI/Event/triggers/diskerror

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->diskerror
  </id>
  <eventType>
    <!--req, enum, subType:string-->diskerror
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.2.3 Delete the linkage of HDD error

Request URL

DELETE /ISAPI/Event/triggers/diskerror

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
DETEKTOR

14.1.2.4 Get the linkage parameters of full HDD

Request URL

GET /ISAPI/Event/triggers/diskfull

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->diskfull
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->diskfull
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
DETEKTOR

14.1.2.5 Set the linkage parameters of full HDD

Request URL

PUT /ISAPI/Event/triggers/diskfull

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->diskfull
  </id>
  <eventType>
    <!--req, enum, subType:string-->diskfull
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.2.6 Delete the linkage of full HDD

Request URL

DELETE /ISAPI/Event/triggers/diskfull

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
DETEKTOR

14.1.2.7 Set the linkage parameters of illegal login

Request URL

PUT /ISAPI/Event/triggers/illaccess

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->illaccess
  </id>
  <eventType>
    <!--req, enum, subType:string-->illaccess
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.2.8 Get the linkage parameters of illegal login

Request URL

GET /ISAPI/Event/triggers/illaccess

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->illaccess
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->illaccess
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
DETEKTOR

14.1.2.9 Get the linkage parameters of IP address conflict event

Request URL

GET /ISAPI/Event/triggers/ipconflict

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->ipconflict
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->ipconflict
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
DETEKTOR

14.1.2.10 Delete the linkage of IP address conflict event

Request URL

DELETE /ISAPI/Event/triggers/ipconflict

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.2.11 Set the linkage parameters of IP address conflict event

Request URL

PUT /ISAPI/Event/triggers/ipconflict

Query Parameter

None

Request Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->ipconflict
  </id>
  <eventType>
    <!--req, enum, subType:string-->ipconflict
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.2.12 Get the linkage parameters of network disconnection

Request URL

GET /ISAPI/Event/triggers/nicbroken

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->nicbroken
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->nicbroken
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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14.1.2.13 Set the linkage parameters of network disconnection

Request URL

PUT /ISAPI/Event/triggers/nicbroken

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->nicbroken
  </id>
  <eventType>
    <!--req, enum, subType:string-->nicbroken
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.2.14 Delete the linkage of network disconnection

Request URL

DELETE /ISAPI/Event/triggers/nicbroken

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.1.2.15 HDD error

EventType:diskerror

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"diskerror" (HDD Error)-->diskerror
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:HDD Error-->diskerror
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <diskNo>
    <!--ro, opt, int, HDD number-->1
  </diskNo>
  <HDDList>
    <!--ro, opt, array, HDD list, subType:object-->
    <HDD>
      <!--ro, opt, object, HDD-->
      <id>
        <!--ro, req, int, index, desc:it starts from 1-->1
      </id>
      <diskNumber>
        <!--ro, req, int, HDD No.-->1
      </diskNumber>
    </HDD>
  </HDDList>
</EventNotificationAlert>
```

drackum@gmail.com
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14.1.2.16 Invalid access

EventType:illaccess

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event parameters, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the alarm device-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the alarm device-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the alarm device-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, transfer protocol type, range:[0,64], desc:transfer protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI
protocol is transmitted via EZVIZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, No, of the channel that triggers an alarm, desc:No. of the video channel that triggers an alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, the uploading times of the same alarm, desc:event frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"illaccess" (invalid access)-->illaccess
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for persistent events: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:illaccess-invalid access-->illaccess
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <username>
    <!--ro, opt, string, locked user name, range:[0,32], desc:the number of attempts by current user name has reached limit. The user name and IPv4 address
may be locked at the same time-->test
  </username>
  <ipV4Address>
    <!--ro, opt, string, locked IPv4 address, desc:the number of attempts by current IPv4 address has reached limit. The user name and Ipv4 address may be
locked at the same time-->test
  </ipV4Address>
  <lockStatus>
    <!--ro, opt, enum, locking status, subType:string, desc:; “unlocked”, “locked”-->unlocked
  </lockStatus>
  <retryLoginTime>
    <!--ro, opt, int, remaining attempts, range:[0,65535], desc:it is valid when the value of lockStatus is “unlocked”-->0
  </retryLoginTime>
  <lockedTime>
    <!--ro, opt, int, locking duration (minutes), range:[0,65535], unit:秒, desc:it is valid when the value of lockStatus is “locked”-->0
  </lockedTime>
  <accessInfoList>
    <!--ro, opt, array, invalid access information list, subType:object-->
    <accessInfo>
      <!--ro, opt, object, invalid access information-->
      <ipV4Address>
        <!--ro, opt, string, Invalid access IP address-->test
      </ipV4Address>
      <dateTime>
        <!--ro, opt, datetime, invalid access time-->2004-05-03T17:30:08+08:00
      </dateTime>
    </accessInfo>
  </accessInfoList>
</EventNotificationAlert>
```

drackum@gmail.com
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14.1.2.17 Network disconnected

EventType:nicbroken

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers an alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers an alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, communication protocol types, range:[0,64], desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome" (the value
should be "HTTP" when ISAPI protocol is transmitted via EZ protocol, and the value should be "EHome" when ISAPI protocol is transmitted via ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, the uploading times of the same alarm, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, string, event type, desc:"nicbroken" (network disconnected)-->nicbroken
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:"nicbroken" (network disconnected)-->nicbroken
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DevicePositionInfo>
    <!--ro, opt, object, device location, desc:the current location of the device when triggering the event, which is mainly for on-board device-->
    <divisionEW>
      <!--ro, opt, enum, eastern/western hemisphere, subType:string, desc:"E" (eastern hemisphere), "W" (western hemisphere)-->E
    </divisionEW>
    <longitude>
      <!--ro, opt, int, longitude, desc:degree*3600*100 + minute*60*100 + second*100-->100
    </longitude>
    <divisionNS>
      <!--ro, opt, enum, southern/northern hemisphere, subType:string, desc:"S" (southern hemisphere), "N" (northern hemisphere)-->N
    </divisionNS>
    <latitude>
      <!--ro, opt, int, latitude, desc:degree*3600*100 + minute*60*100 + second*100-->100
    </latitude>
    <direction>
      <!--ro, opt, int, vehicle direction, desc:actual direction (degree) (due north as 0 degree, calculated clockwise)-->100
    </direction>
    <speed>
      <!--ro, opt, int, speed, desc:cm/h-->100
    </speed>
    <height>
      <!--ro, opt, int, height, unit:cm, desc:cm-->1
    </height>
  </DevicePositionInfo>
</EventNotificationAlert>
```

drackum@gmail.com
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14.1.3 非视频通道管理(S)

14.1.3.1 Get the event search capability

Request URL

GET /ISAPI/SDT/Management/EventSearch/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "startTime":  "1970-01-01T00:00:00+08:00",
    /*ro, req, datetime, start time, desc:start time,ISO8601_time*/
    "endTime":  "1970-01-01T00:00:00+08:00",
    /*ro, req, datetime, end time, desc:end time,ISO8601_time*/
    "resultMaxNum":  100,
    /*ro, req, int, supported maximum number of searching*/
    "channelIDLen": {
    /*ro, opt, object, channel ID length*/
        "min":  0,
        /*ro, req, int, minimum value, step:1*/
        "max":  32
        /*ro, req, int, maximum value, step:1*/
    },
    "eventType": {
    /*ro, opt, object, event type, desc:studentStoodUp-student stands up, accessController-access controller, videoIntercom-video intercom, Luminite-
Luminite security control panel, GJD-GJD security control panel, cameraDetector-detector, securityControlPanel-security control panel. Multiple selections
are separated by comma*/
        "@opt":  "studentStoodUp, accessController,videoIntercom, OPTEX,Luminite,GJD, cameraDetector, securityControlPanel",
        /*ro, req, string, range*/
        "#text":  "studentStoodUp"
        /*ro, opt, string*/
    },
    "type": {
    /*ro, opt, object, channel type, desc:video-video channel, IOT-IOT channel; if this node does not exist, it indicates video channel*/
        "@opt":  "video,IOT",
        /*ro, req, string*/
        "#text":  "video"
        /*ro, opt, string*/
    },
    "AccessController": {
    /*ro, opt, object, access control event*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        },
        "nameLen": {
        /*ro, opt, object, name length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        },
        "cardNoLen": {
        /*ro, opt, object, card No. length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        }
    },
    "VideoIntercom": {
    /*ro, opt, object, Video Intercom Event*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        },
        "nameLen": {
        /*ro, opt, object, name length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        },
        "cardNoLen": {
        /*ro, opt, object, card No. length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        }
    },
    "OPTEX": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
```

drackum@gmail.com
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```
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "Luminite": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "GJD": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "CameraDetector": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "SecurityControlPanel": {
    /*ro, opt, object, security control panel event*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test"
            /*ro, req, string, range*/
        },
        "eventMainType": {
        /*ro, opt, object*/
            "@opt": ["alarm", "fault", "operation", "others"]
            /*ro, req, array, subType:string*/
        },
        "zoneNo": {
        /*ro, req, object, zone No.*/
            "@size":  1,
            /*ro, req, int, step:1*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  16,
            /*ro, req, int, maximum value, step:1*/
            "#text":  1
            /*ro, opt, int*/
        },
        "ipAddress": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, minimum value, range:[0,128], step:1*/
            "@max":  128
            /*ro, req, int, maximum value, range:[0,128], step:1*/
        },
        "devIndex": {
        /*ro, opt, object, maximum value*/
            "@min":  0,
            /*ro, req, int, minimum value, range:[0,256], step:1*/
            "@max":  256
            /*ro, req, int, maximum value, range:[0,256], step:1*/
        },
        "code": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, minimum value, range:[0,5000], step:1*/
            "@max":  5000
            /*ro, req, int, maximum value, range:[0,5000], step:1*/
        },
        "devName": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, range:[1,48], step:1*/
            "@max":  48
```


dracktm@gmail.com
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```
            /*ro, req, int, range:[1,48], step:1*/
        }
    },
    "isSupportEventExport":  true
    /*ro, opt, bool*/
}
```

14.1.3.2 Search for event information

Request URL

POST /ISAPI/SDT/Management/EventSearch?format=json

Query Parameter

None

Request Message

```
{
    "searchID":  "test",
    /*req, string, search ID, desc:this node is used to check whether the search is the same one. Due to device performance limitation, you can search
asynchronously by applying the search conditions multiple times and getting the searching progress (the searchID should be the same)*/
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  100,
    /*req, int, the maximum number of search results this time*/
    "startTime":  "1970-01-01T08:00:00+08:00",
    /*req, datetime, start time, desc:start time,ISO8601_time*/
    "endTime":  "1970-01-01T09:00:00+08:00",
    /*req, datetime, end time, desc:end time,ISO8601_time*/
    "choiceChannel": [
    /*opt, array, camera information, subType:object*/
        {
            "channelID":  "26c92893d2a94bc1840451b78444e0f9"
            /*opt, string, camera No.*/
        }
    ],
    "eventType":  "accessController",
    /*req, enum, event type, subType:string, desc:"studentStoodUp" (student stands up), "accessController" (access controller), "videoIntercom" (video
intercom), "OPTEX" (OPTEX security control panel), "Luminite" (Luminite security control panel), "GJD" (GJD security control panel), "cameraDetector"
(detector), "securityControlPanel" (security control panel), multiple selections are separated with commas*/
    "type":  "video",
    /*opt, enum, channel type, subType:string, desc:"video" (video channel), "IOT" (IOT channel), if this node does not exist, it indicates video channel*/
    "AccessController": {
    /*opt, object*/
        "eventType":  "test",
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
        "name":  "test",
        /*opt, string, name*/
        "cardNo":  "test"
        /*opt, string, card No.*/
    },
    "VideoIntercom": {
    /*opt, object*/
        "eventType":  "test",
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
        "name":  "test",
        /*opt, string, name*/
        "cardNo":  "test"
        /*opt, string, card No.*/
    },
    "OPTEX": {
    /*opt, object, OPTEX security control panel*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "Luminite": {
    /*opt, object, Luminite security control panel*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "GJD": {
    /*opt, object, GJD security control panel*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "CameraDetector": {
```

dracken@gmail.com
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```
    /*opt, object*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "SecurityControlPanel": {
    /*opt, object, security control panel*/
        "eventType":  "test",
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
        "eventMainType":  "alarm",
        /*opt, enum, subType:string*/
        "zoneNo": [1, 2],
        /*opt, array, zone No., subType:int*/
        "ipAddress":  "10.1.200.1",
        /*opt, string, device's IPv4 address, range:[0,128]*/
        "devIndex":  "test",
        /*opt, string, device No., range:[0,256]*/
        "code":  0,
        /*opt, int, range:[0,5000]*/
        "devName":  "test"
        /*opt, string, range:[1,48]*/
    }
}
```

Response Message

```
{
    "responseStatusStrg":  "MORE",
    /*ro, req, enum, searching status, subType:string, desc:"OK" (searching ended), "MORE" (search again for more results), "NO MATCH" (no matched data
found)*/
    "numOfMatches":  1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results*/
    "eventInfo": [
    /*ro, opt, array, event information, subType:object*/
        {
            "captureTime":  "1970-01-01T08:00:00+08:00",
            /*ro, req, datetime, capture time*/
            "channelID":  "test",
            /*ro, opt, string, camera No.*/
            "channelName":  "test",
            /*ro, opt, string, camera name*/
            "picUrl":  "test",
            /*ro, opt, string, picture URL*/
            "subPicUrl":  "test",
            /*ro, opt, string, thumbnail URL*/
            "eventType":  "accessController",
            /*ro, req, enum, event type, subType:string, desc:"studentStoodUp" (student stands up), "accessController" (access controller), "videoIntercom"
(video intercom), "OPTEX" (OPTEX security control panel), "Luminite" (Luminite security control panel), "GJD" (GJD security control panel), "cameraDetector"
(detector), "securityControlPanel" (security control panel), multiple selections are separated with commas*/
            "studentStoodUp": {
            /*ro, opt, object, student stand up event information*/
                "studentsStoodUp_number":  4
                /*ro, opt, int, the number of students who are standing up*/
            },
            "AccessController": {
            /*ro, req, object, event type*/
                "eventType":  "test",
                /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "name":  "test",
                /*ro, opt, string, name*/
                "cardNo":  "test",
                /*ro, opt, string, card No.*/
                "cardType":  "invalid",
                /*ro, opt, enum, card type, subType:string, desc:"invalid" (invalid card), "ordinary" (normal card), "disabled" (card for the disabled),
"blocklist" (card in blocklist), "patrol" (patrol card), "super" (super card), "guest" (visitor card), "remove" (removed card)*/
                "videoChannelInfo": [
                /*ro, opt, array, video channel information, subType:object*/
                    {
                        "videoChannel":  1,
                        /*ro, req, int, video channel No.*/
                        "picUrl":  "test"
                        /*ro, opt, string, picture URL*/
                    }
                ]
            },
            "VideoIntercom": {
            /*ro, opt, object, video intercom*/
                "eventType":  "test",
                /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "name":  "test",
                /*ro, opt, string, name*/
                "cardNo":  "test",
                /*ro, opt, string, card No.*/
                "cardType":  "invalid",
                /*ro, opt, enum, card type, subType:string, desc:"invalid" (invalid card), "ordinary" (normal card), "disabled" (card for the disabled),
"blocklist" (card in blocklist), "patrol" (patrol card), "super" (super card), "guest" (visitor card), "remove" (removed card)*/
```

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```
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ]
        },
        "OPTEX": {
        /*ro, opt, object, OPTEX security control panel*/
            "eventType":  "test",
            /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ]
        },
        "Luminite": {
        /*ro, opt, object, Luminite security control panel*/
            "eventType":  "test",
            /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ]
        },
        "GJD": {
        /*ro, opt, object, GJD security control panel*/
            "eventType":  "test",
            /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ]
        },
        "CameraDetector": {
        /*ro, opt, object*/
            "eventType":  "test",
            /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ]
        },
        "SecurityControlPanel": {
        /*ro, opt, object*/
            "eventType":  "test",
            /*ro, opt, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
            "eventMainType":  "alarm",
            /*ro, opt, enum, subType:string*/
            "videoChannelInfo": [
            /*ro, opt, array, video channel information, subType:object*/
                {
                    "videoChannel":  1,
                    /*ro, req, int, video channel No.*/
                    "picUrl":  "test"
                    /*ro, opt, string, picture URL*/
                }
            ],
            "zoneNo":  1,
            /*ro, opt, int, zone No., desc:zone No.,it is valid when eventType values "zone"*/
            "code":  1103,
            /*ro, opt, int, CID event No., desc:CID event No.,it is valid when eventType values "zone"*/
            "zoneName":  "test",
            /*ro, opt, string, zone name*/
            "ipAddress":  "test",
            /*ro, opt, string, device's IPv4 address, range:[0,128]*/
            "devIndex":  "test",
            /*ro, opt, string, device No., range:[0,256]*/
            "subSysNo":  1,
            /*ro, opt, int*/
```

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```
                "recordID":  "test",
                /*ro, opt, string*/
                "deviceName":  "test"
                /*ro, opt, string, range:[1,48]*/
            }
        }
    ]
}
```

14.1.4 在线升级管理

14.1.4.1 Get the device online upgrade capability

Request URL

GET /ISAPI/System/onlineUpgrade/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade, attr:version{req, string, protocolVersion}-->
  <firmwareNum max="10">
    <!--ro, req, int, number of supported online upgrade packages, attr:max{req, int}-->1
  </firmwareNum>
  <firmwareCode max="256">
    <!--ro, req, int, maximum length of firmware ID, attr:max{req, int}-->1
  </firmwareCode>
  <firmwareVersion max="64">
    <!--ro, req, int, maximum length of version, attr:max{req, int}-->1
  </firmwareVersion>
  <firmwareCodeNumOnce max="10">
    <!--ro, req, int, maximum number of obtained firmware IDs, attr:max{req, int}-->1
  </firmwareCodeNumOnce>
  <upgradePercent min="0" max="100">
    <!--ro, req, int, upgrade progress capability in percentage, attr:min{req, int},max{req, int}-->1
  </upgradePercent>
  <Version>
    <!--ro, opt, object, upgrade package version information-->
    <newVersion max="64">
      <!--ro, req, string, version information of the new upgrade package, attr:max{req, int}-->test
    </newVersion>
    <changeLog max="64">
      <!--ro, req, string, log of the new upgrade package, attr:max{req, int}-->test
    </changeLog>
  </Version>
  <rebootAfterUpgrade>
    <!--ro, opt, enum, upgrade options after device reboot, subType:string, desc:"auto" (auto upgrade), "manual" (manual upgrade)-->auto
  </rebootAfterUpgrade>
  <DeviceParameter>
    <!--ro, opt, object, online upgrade parameters-->
    <isSupportAutoDownloadPackage>
      <!--ro, opt, bool, whether it supports automatic download of upgrade package-->true
    </isSupportAutoDownloadPackage>
    <notSupportAutoUpgrade>
      <!--ro, opt, bool, whether it no longer supports automatic download of upgrade package and automatic upgrade, desc:corresponding URL: PUT
/ISAPI/System/onlineUpgrade/upgrade-->true
    </notSupportAutoUpgrade>
    <isSupportTimingUpgrade>
      <!--ro, opt, bool, whether it supports scheduled upgrade-->true
    </isSupportTimingUpgrade>
  </DeviceParameter>
  <ManualDownloadPackage>
    <!--ro, opt, object, manual download of upgrade package-->
    <supportOperation opt="start,cancel,pause,resume">
      <!--ro, opt, enum, supported operations, subType:string, attr:opt{req, string}, desc:"start", "cancel", "pause", "resume"-->start
    </supportOperation>
  </ManualDownloadPackage>
  <isSupportIgnoreCurrentVersion>
    <!--ro, opt, bool, whether it supports ignoring the current version-->true
  </isSupportIgnoreCurrentVersion>
  <UpgradePackageTask>
    <!--ro, opt, object, whether the device supports applying upgrade tasks, desc:if this node is returned, it indicates that the following URLs are also
supported: /ISAPI/System/onlineUpgrade/task?format=json, /ISAPI/System/onlineUpgrade/status/<ID>-->
    <isSupportUpgradeModules>
      <!--ro, opt, bool, whether the device supports upgrading sub modules-->true
    </isSupportUpgradeModules>
    <isSupportUpgradeChannels>
      <!--ro, opt, bool, whether the device supports upgrading sub channels-->true
```

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```
    </isSupportUpgradeChannels>
    <isSupportDownloadStrategy>
      <!--ro, opt, bool, whether the device supports strategy parameters of downloading upgrade package, desc:corresponding filed is "downloadStrategy" of
URL (/ISAPI/System/onlineUpgrade/task?format=json)-->true
    </isSupportDownloadStrategy>
    <isSupportDelayedUpgradeByDateTime>
      <!--ro, opt, bool, whether the device supports delaying the upgrade by time, desc:corresponding filed is "delayedUpgradeDateTime" of URL
(/ISAPI/System/onlineUpgrade/task?format=json)-->true
    </isSupportDelayedUpgradeByDateTime>
    <upgradePackageAddressType opt="URL, ftpIPAddress, ftpDomainAddress">
      <!--ro, opt, string, attr:opt{req, string}-->URL
    </upgradePackageAddressType>
  </UpgradePackageTask>
  <isSupportPatrolInspection>
    <!--ro, opt, bool, whether the device supports patrol inspection-->true
  </isSupportPatrolInspection>
  <isSupportBatchUpgrade>
    <!--ro, opt, bool, whether the device supports batch online upgrades-->true
  </isSupportBatchUpgrade>
  <isSupportCancelUpgrade>
    <!--ro, opt, bool, whether the device supports canceling online upgrade-->true
  </isSupportCancelUpgrade>
  <isSupportPackagePause>
    <!--ro, opt, bool, whether the device supports pausing upgrade package download, desc:/ISAPI/System/onlineUpgrade/PackagePause?format=json-->true
  </isSupportPackagePause>
  <isSupportPackageResume>
    <!--ro, opt, bool, whether the device supports resuming upgrade package download, desc:/ISAPI/System/onlineUpgrade/PackageResume?format=json-->true
  </isSupportPackageResume>
  <isSupportCancelUpgradeByTaskID>
    <!--ro, opt, bool, whether the device supports canceling online upgrade by task ID, desc:corresponding to the field assignByTaskID of operationType in
URL PUT  /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json-->true
  </isSupportCancelUpgradeByTaskID>
</OnlineUpgradeCap>
```

14.1.4.2 Set the online device upgrade parameters

Request URL

PUT /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Query Parameter

None

Request Message

```
{
    "OnlineUpgradeParameter": {
    /*req, object, online upgrade parameters*/
        "autoDownloadPackage":  true,
        /*req, bool, whether to enable automatic download of upgrade package*/
        "timingUpgrade":  true,
        /*opt, bool, whether to enable scheduled upgrade*/
        "upgradeTime":  "00:00:00"
        /*opt, time, upgrade time, dep:and,{$.OnlineUpgradeParameter.timingUpgrade,eq,true}*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode when statusCode is not 1*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

14.1.4.3 Get the device online upgrading parameters

Request URL

GET /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Query Parameter

None

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Request Message

None

Response Message

```
{
    "OnlineUpgradeParameter": {
    /*ro, req, object, online upgrade parameters*/
        "autoDownloadPackage":  true,
        /*ro, req, bool, whether to enable automatic downloading of upgrade package*/
        "timingUpgrade":  true,
        /*ro, opt, bool, whether to enable scheduled upgrade*/
        "upgradeTime":  "00:00:00"
        /*ro, opt, time, upgrade time, dep:and,{$.OnlineUpgradeParameter.timingUpgrade,eq,trur}*/
    }
}
```

14.1.4.4 Get the online upgrade package downloading progress

Request URL

GET /ISAPI/System/onlineUpgrade/downloadPackage/status?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DownloadPackageStatus": {
    /*ro, req, object, upgrade package downloading status*/
        "status":  "notDownload",
        /*ro, req, enum, download status, subType:string, desc:"notDownload", "downloading", "downloadFailed", "pause", "finish", "incorrectPackage"
(incorrect upgrade package format), "hdOperationFailed" (HD operation failed)*/
        "total":  0.0,
        /*ro, opt, float, upgrade package total size, desc:unit: MB, corrects to one decimal place*/
        "remain":  0.0,
        /*ro, opt, float, remaining space, desc:unit: MB, corrects to one decimal place*/
        "speed":  0.0,
        /*ro, opt, float, download speed, desc:unit: MB, corrects to one decimal place*/
        "remainTime":  0,
        /*ro, opt, int, estimated remaining time, unit:s*/
        "progress":  0
        /*ro, req, int, progress, range:[0,100]*/
    }
}
```

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14.1.4.5 Start to download upgrade package to device

Request URL

PUT /ISAPI/System/onlineUpgrade/downloadPackage?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.1.4.6 Get the online upgrade progress of device

Request URL

GET /ISAPI/System/onlineUpgrade/status

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade status, attr:version{req, string, protocolVersion}-->
  <status>
    <!--ro, req, enum, online upgrade status, subType:string, desc:"notUpgrade" (not upgraded), "upgrading", "successful" (upgraded), "languageMismatch" (language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error), "engineVersionMismatch" (mismatch of upgrade package engine version)-->notUpgrade
  </status>
  <percent>
    <!--ro, req, int, current upgrade progress in percentage, range:[0,100]-->1
  </percent>
  <onlineUpdateDevInfoList>
    <!--ro, opt, object, online upgrade device information, desc:containing the online upgrade status of sub devices, sub modules, or sub channels-->
    <onlineUpdateDevInfo>
      <!--ro, opt, object, online upgrade device information-->
      <moduleType>
        <!--ro, req, enum, module type, subType:string, desc:"keypad", "detector", "cardReader" (card reader), "siren" (sounder), "relay", "transmitter" (transmitter), "remoteCtrl" (remote control),"repeater" (repeater), "SSD", "extensionModule" (extension module)-->keypad
      </moduleType>
      <moduleID>
        <!--ro, req, int, module ID, range:[1,128], step:1-->1
      </moduleID>
      <model>
        <!--ro, opt, string, model, range:[1,32]-->test
      </model>
      <status>
        <!--ro, opt, enum, upgrade status, subType:string, desc:"allowance" (upgradeable, the upgrade command is not triggered), "ready" (waiting for upgrade, the upgrade command has been triggered), "upgriding" (upgrading), "success" (upgraded), "failed" (upgrade failed)-->allowance
      </status>
      <percent>
        <!--ro, opt, int, current upgrade percentage, range:[0,100]-->1
      </percent>
      <updateTime>
        <!--ro, opt, datetime, upgrade completion time-->2004-05-03T17:30:08+08:00
      </updateTime>
    </onlineUpdateDevInfo>
  </onlineUpdateDevInfoList>
</OnlineUpgradeStatus>
```

14.1.4.7 Get the progress of a specified online upgrade task

Request URL

GET /ISAPI/System/onlineUpgrade/status/task/<taskID>

Query Parameter

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Parameter Name	Parameter Type	Description
taskID	string	in the URL refers to the task ID, which is returned by the URL (POST /ISAPI/System/onlineUpgrade/task?format=json). It is to get the upgrade status of internal sub modules or sub channels of devices at the same time.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, online upgrade status, attr:version{req, string, protocolVersion}-->
  <status>
    <!--ro, req, enum, online upgrade status, subType:string, desc:"notUpgrade" (not upgraded), "upgrading", "successful" (upgraded), "languageMismatch"
(language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade
version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error), "waitManualTrigger" (waiting for manual trigger; this status
will be returned if delayed upgrade is set when the task is applied and not triggered manually)-->notUpgrade
  </status>
  <percent>
    <!--ro, req, int, current upgrade progress in percentage, range:[0,100]-->1
  </percent>
  <taskID>
    <!--ro, opt, string, task ID, range:[1,64], desc:generated when creating upgrade tasks, related URI: /ISAPI/System/onlineUpgrade/task?format=json-->test
  </taskID>
  <moudleType>
    <!--ro, opt, enum, component type, subType:string, desc:"stationHostAndScreen" (dock station upgrade package, including host and screen), "stationHost"
(dock station host), "bodyCamera" (body camera)-->stationHostAndScreen
  </moudleType>
  <packageName>
    <!--ro, opt, string, upgrade package name, range:[1,128]-->ZCS-1_target_V1.0.1_build201123.zip
  </packageName>
  <taskTime>
    <!--ro, opt, datetime, time of applying the upgrade task-->1970-01-01T00:00:00+08:00
  </taskTime>
</OnlineUpgradeStatus>
```

drackum@gmail.com
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14.1.4.8 Get all of the online upgrade task progress

Request URL

GET /ISAPI/System/onlineUpgrade/status/task?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "onlineUpgradeStatusList": [
    /*ro, req, array, online upgrade status list, subType:object*/
        {
            "status":  "notUpgrade",
            /*ro, req, enum, upgrade status, subType:string, desc:“notUpgrade”, “upgrading”, “successful” (upgrading succeeded), “languageMismatch”,
“writeFlashError”, “packageTypeMismatch”, “packageVersionMismatch”, “netUnreachable” (network disconnected), “unknownError”, “waitManualTrigger”*/
            "percent":  0,
            /*ro, req, int, upgrade progress percentage, range:[0,100]*/
            "taskID":  "test",
            /*ro, opt, string, task ID, range:[1,64], desc:generated when creating upgrade tasks, related URI: /ISAPI/System/onlineUpgrade/task?
format=json*/
            "moudleType":  "stationHostAndScreen",
            /*ro, opt, enum, component type, subType:string, desc:"stationHostAndScreen" (dock station upgrade package, including host and screen),
"stationHost" (dock station host), "spellingControllerMainAndSubBoard" (main board and sub board of splicing controller), "spellingControllerMainBoard"
(main board of splicing controller), "spellingControllerSubBoard" (sub board of splicing controller), "bodyCamera" (body camera)*/
            "packageName":  "ZCS-H1_target_V1.0.1_build201123.zip",
            /*ro, opt, string, upgrade package name, range:[1,128]*/
            "taskTime":  "1970-01-01T00:00:00+08:00",
            /*ro, opt, datetime, applying upgrade task time*/
            "taskParam": {
            /*ro, opt, object, task parameters, desc:corresponding URL: POST /ISAPI/System/onlineUpgrade/task?format=json*/
                "URL":  "test",
                /*ro, req, string, upgrade package address, range:[1,1024]*/
                "delayedUpgrade":  true,
                /*ro, opt, bool, whether to delay the upgrade, desc:if this node is not applied, it indicates that the upgrade will be started immediately;
if this node is applied and the value is false, it indicates that the upgrade will be delayed*/
                "delayedUpgradeDateTime":  "1970-01-01T00:00:00+08:00"
                /*ro, opt, datetime, time of starting the delayed upgrade, dep:or,{$.PackageDescription.delayedUpgrade,eq,false}*/
            }
        }
    ]
}
```

14.1.4.9 Apply the task of online upgrade

Request URL

POST /ISAPI/System/onlineUpgrade/task?format=json

Query Parameter

None

Request Message

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```
{
    "PackageDescription": {
    /*req, object, description of the upgrade package*/
        "name":  "DSJ_N3_GML_DEBUG_STD_1.2.0_190524.zip",
        /*opt, string, upgrade package name, range:[1,256]*/
        "URL":  "test",
        /*req, string, URL of the upgrade package, range:[1,1024], desc:the task ID is contained in the URL. The uploading and downloading of the upgrade package is based on the object storage protocol (version 2.0). If this node is null, the FTP address will take effect. If both the URL and FTP address are configured, the URL has priority over FTP address*/
        "MD5":  "test",
        /*opt, string, range:[0,256]*/
        "moduleType":  "cardReader",
        /*opt, enum, module type, subType:string, desc:“cardReader” (card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader), "dockStationHost" (dock station host), "dockStationScreen" (dock station screen), "stationHostAndScreen" (dock station host and screen)*/
        "moduleID":  1,
        /*opt, int, module ID*/
        "spellingControllerSubBoardList": [
        /*opt, array, splicing controller sub board list, subType:object, range:[1,1024], dep:or, {$.PackageDescription.moduleType,eq,spellingControllerMainAndSubBoard},{$.PackageDescription.moduleType,eq,spellingControllerSubBoard}, desc:it is valid when the value of the node moduleType is "spellingControllerMainAndSubBoard" or "spellingControllerSubBoard". If it is not configured, it indicates that all the sub boards will be updated*/
            {
                "subBoardID":  1,
                /*opt, int, sub board ID, range:[1,1024]*/
                "ipV4Address":  "test"
                /*ro, opt, string, IPv4, range:[1,16], desc:when an exception occurs on the main board, the sub board can be updated through the IP address. The default sub board interface will be used*/
            }
        ],
        "channel":  1,
        /*opt, int, channel No., desc:this node is used for upgrading the IP channel of the NVR*/
        "childDevID":  "3434645611",
        /*opt, string, range:[9,32]*/
        "customTaskID":  "test",
        /*opt, string, user-defined task ID, range:[1,128], desc:if this node exists, it will be returned in the result*/
        "delayedUpgrade":  true,
        /*opt, bool, whether to delay upgrading the package, desc:if this node is not applied, it indicates that the package will be upgraded immediately; if this node is applied and the value is false, it indicates that the upgrade will be delayed. User can decide the time to upgrade the package of the dock station or body camera if the upgrade is delayed*/
        "delayedUpgradeDateTime":  "1970-01-01T00:00:00+08:00",
        /*opt, datetime, time of starting the delayed upgrade, dep:or,{$.PackageDescription.delayedUpgrade,eq,false}, desc:the related capability is "isSupportDelayedUpgradeByDateTime" in GET  /ISAPI/System/onlineUpgrade/capabilities*/
        "ftpAddress":  "test",
        /*opt, string, FTP address of the upgrade package, desc:this node will take effect when URL is null and the format is: ftp://[username]:[password]@[IP address]:[port No.]*/
        "downloadStrategy": {
        /*opt, object, device's upgrade package downloading strategy parameters*/
            "sequelEnabled":  true,
            /*opt, bool, whether to enable ANR (Automatic Network Replenishment), desc:if enabled, when downloading is interrupted, the device will try downloading the package again; if disabled, the default strategy of the device will be adopted*/
            "sequelTimeout":  3000,
            /*opt, int, timeout of a single ANR attempt, step:1, unit:ms, desc:when a single ANR attempt exceeds the timeout period but the download does not continue, the ANR attempt will be recognized as failed*/
            "maxRetryTimes":  5
            /*opt, int, ANR maximum attempt times, desc:when the device ANR exceeds the maximum attempt times but the download does not continue, the download and the upgrade will be recognized as failed*/
        },
        "upgradePackageURLType":  "full"
        /*opt, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1-succeeded, other value-error occurred. This field is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"OK" (succeeded), other value (error occurred). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded), other value (error occurred). This field is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error message, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "test"
    /*ro, opt, string, ID of the upgrade task, desc:it can be used to get the status and progress of the task*/
}
```

14.1.4.10 Get the new version information

Request URL

GET /ISAPI/System/onlineUpgrade/version?check=<check>

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Query Parameter

Parameter Name	Parameter Type	Description
check	string	When check is false (default), the device does not need to interact with the server and only needs to return the results of periodic checks. When check is true, the device needs to interact with the server and return the latest results.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeVersion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, new version information, attr:version{req, string, protocolVersion}-->
  <newVersionAvailable>
    <!--ro, req, bool, whether the new version already is available-->true
  </newVersionAvailable>
  <newVersion>
    <!--ro, opt, string, version information of the new upgrade package-->test
  </newVersion>
  <changeLog>
    <!--ro, opt, string, updating logs of the new upgrade package version-->test
  </changeLog>
  <changeLevel>
    <!--ro, opt, enum, subType:int-->2
  </changeLevel>
</OnlineUpgradeVersion>
```

14.1.5 串口配置

14.1.5.1 Get serial port authentication service capability

Request URL

GET /ISAPI/System/Serial/authentication/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AuthenticationCap": {
    /*ro, opt, object, the serial port authentication service*/
        "enabled": {
        /*ro, req, object, whether to enable serial port authentication service*/
            "@opt": [true, false]
            /*ro, opt, array, whether to enable serial port authentication service, subType:bool*/
        },
        "serviceClosurePeriod": {
        /*ro, opt, object, service life, desc:unit: day it is required when enabled is "false"*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  30,
            /*ro, opt, int, the maximum value*/
            "@def":  7
            /*ro, opt, int, the default value*/
        }
    }
}
```

14.1.5.2 Set parameters of serial authentication service

Request URL

PUT /ISAPI/System/Serial/authentication?format=json

Query Parameter

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None

Request Message

```
{
    "Authentication": {
    /*opt, object*/
        "enabled":  false,
        /*req, bool, whether to enable serial authentication service*/
        "serviceClosurePeriod":  7
        /*opt, int, service life, unit:d, desc:it is required when enabled is "false"*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

14.1.5.3 Get parameters of serial authentication service

Request URL

GET /ISAPI/System/Serial/authentication?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "Authentication": {
    /*ro, opt, object*/
        "enabled":  false,
        /*ro, req, bool, whether to enable serial authentication service*/
        "serviceClosurePeriod":  7
        /*ro, opt, int, service life, unit:d, desc:it is required when enabled is "false”*/
    }
}
```

14.1.5.4 Get the serial port capability of the device

Request URL

GET /ISAPI/System/Serial/capabilities

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<SerialCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, range of RS-485 serial port numbers supported by the device, attr:version{req, string, protocolVersion}-->
  <rs485PortNums>
    <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->0
  </rs485PortNums>
  <supportRS232Config>
    <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
  </supportRS232Config>
  <rs422PortNums>
    <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->0
  </rs422PortNums>
  <rs232PortNums>
    <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->0
  </rs232PortNums>
  <rs485WorkMode opt="Led, CaptureTrigger,transparent">
    <!--ro, opt, string, range of RS-485 serial port No. supported by the device, attr:opt{req, string}-->test
  </rs485WorkMode>
  <rs232SerialNumber opt="1,2,3">
    <!--ro, opt, int, range of RS-232 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs232SerialNumber>
  <rs485SerialNumber opt="4,5,6">
    <!--ro, opt, int, range of RS-485 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs485SerialNumber>
  <isSupportAuthenticationService>
    <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
  </isSupportAuthenticationService>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
  </isSupportDeviceInfo>
  <isSupportSearchDeviceInfoRelations>
    <!--ro, opt, bool, whether the device supports searching the linkage information of the serial port device-->true
  </isSupportSearchDeviceInfoRelations>
</SerialCap>
```

14.1.5.5 Get the capability of a specific serial port

Request URL

GET /ISAPI/System/Serial/ports/<portID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port No., attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->0
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <encryptKey min="0" max="10">
    <!--ro, opt, string, key, attr:min{req, int},max{req, int}, desc:key-->test
  </encryptKey>
  <serialNumber opt="1,2,3,4,5">
    <!--ro, opt, int, serial No., attr:opt{req, string}-->0
  </serialNumber>
  <serialPortType opt="RS485,RS422,RS232">
    <!--ro, opt, string, serial port type, attr:opt{req, string}-->RS485
  </serialPortType>
  <serialAddress min="0" max="10">
    <!--ro, opt, int, serial port address, attr:min{req, int},max{req, int}, desc:serial port address-->1
  </serialAddress>
  <duplexMode opt="half,full">
    <!--ro, opt, string, duplex mode of the serial port, attr:opt{req, string}-->half
  </duplexMode>
  <direction opt="monodirectional,bdirectional">
    <!--ro, opt, string, attr:opt{req, string}-->monodirectional
  </direction>
  <baudRate opt="1200,2400,4800,9600,19200,38400,57600,115200">
    <!--ro, opt, int, attr:opt{req, string}-->1200
  </baudRate>
  <dataBits opt="6,7,8">
```

drackun@gmail.com
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```
    <!--ro, opt, int, attr:opt{req, string}-->6
  </dataBits>
  <parityType opt="none,even,odd,mark,space">
    <!--ro, opt, string, attr:opt{req, string}-->none
  </parityType>
  <stopBits opt="1,1.5,2">
    <!--ro, opt, string, stop bit, attr:opt{req, string}-->1
  </stopBits>
  <workMode opt="console,transparent,audiomixe,screenCtrl,ptzCtrl,keyboard,matrix,audioMixers">
    <!--ro, opt, string, working mode, attr:opt{req, string}-->console
  </workMode>
  <flowCtrl opt="none,software,hardware">
    <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
  </flowCtrl>
  <deviceName min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </deviceName>
  <deviceProtocol min="0" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
  </deviceProtocol>
  <mode opt="readerMode,clientMode,externMode,stairsControl,accessControlHost,disabled,custom,cardReceiver,QRCodeReader">
    <!--ro, opt, string, working mode, attr:opt{req, string}-->readerMode
  </mode>
  <modeChangeReboot>
    <!--ro, opt, bool-->true
  </modeChangeReboot>
  <outputDataType opt="cardNo,employeeNo,auto">
    <!--ro, opt, string, output data type, attr:opt{req, string}, desc:data type output from the door station to the elevator controller: floorNumber (floor
No.,default),cardNo (card No.)-->cardNo
  </outputDataType>
  <copyToAll opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </copyToAll>
  <isVariable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isVariable>
  <PTZInfo>
    <!--ro, opt, object-->
    <bindChannelNo min="0" max="10">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->0
    </bindChannelNo>
    <PTZProtocolList>
      <!--ro, req, object-->
      <PTZProtocol>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, protocol index-->0
        </id>
        <protoDesc>
          <!--ro, req, string, protocol description-->test
        </protoDesc>
      </PTZProtocol>
    </PTZProtocolList>
  </PTZInfo>
  <ScreenCtrlInfo>
    <!--ro, opt, object, ScreenCtrlInfo, desc:this field is valid when the value of workMode is "screenCtrl"-->
    <ScreenCtrlProtoList>
      <!--ro, req, object, ScreenCtrlProtoList-->
      <ScreenCtrlProto>
        <!--ro, opt, object, ScreenCtrlProto-->
        <id>
          <!--ro, req, int, protocol index-->0
        </id>
        <protoDesc>
          <!--ro, req, string, protocol description-->test
        </protoDesc>
      </ScreenCtrlProto>
    </ScreenCtrlProtoList>
  </ScreenCtrlInfo>
  <MatrixCtrlInfo>
    <!--ro, opt, object, MatrixCtrlInfo-->
    <MatrixCtrlProtoList>
      <!--ro, req, object, MatrixCtrlProtoList-->
      <MatrixCtrlProto>
        <!--ro, opt, object, MatrixCtrlProto-->
        <id>
          <!--ro, req, int, protocol index-->0
        </id>
        <protoDesc>
          <!--ro, req, string, protocol description-->test
        </protoDesc>
      </MatrixCtrlProto>
    </MatrixCtrlProtoList>
  </MatrixCtrlInfo>
  <KeyBoardCtrlInfo>
    <!--ro, opt, object, KeyBoardCtrlInfo-->
    <KeyBoardCtrlProtoList>
      <!--ro, req, object, KeyBoardCtrlProtoList-->
      <KeyBoardCtrlProto>
        <!--ro, opt, object, KeyBoardCtrlProto-->
        <id>
          <!--ro, req, int, protocol index-->0
        </id>
        <protoDesc>
```

drackum@gmail.com
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```
          <!--ro, req, string, protocol description-->test
        </protoDesc>
      </KeyBoardCtrlProto>
    </KeyBoardCtrlProtoList>
  </KeyBoardCtrlInfo>
  <ControlAddress>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <address min="0" max="255">
      <!--ro, req, int, IP address, attr:min{req, int},max{req, int}-->0
    </address>
  </ControlAddress>
  <rs485WorkMode
opt="Led,sensor,CaptureTrigger,transparent,GPS,ModbusRTU,supplementLight,audibleVisualAlarm,columnSpeaker,commModeUsing433MHz,signalLampControl">
    <!--ro, opt, string, working mode of RS-485 serial port, dep:and,{$.SerialPortList.SerialPort.serialPortType,eq,RS485}, attr:opt{req, string},
desc:working mode of RS-485 serial port,which is used for LED display or triggering transmission of captured pictures. This node is valid only when
<serialPortType> is "RS485"-->sensor
  </rs485WorkMode>
  <SensorCtrlList>
    <!--ro, opt, object-->
    <Sensor>
      <!--ro, opt, object-->
      <id min="1" max="3">
        <!--ro, req, int, index, attr:min{req, int},max{req, int}-->1
      </id>
      <name min="1" max="32">
        <!--ro, req, string, attr:min{req, int},max{req, int}-->visibilitySensor
      </name>
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable, attr:opt{req, string}, desc:whether to enable the serial port: "true,false"-->false
      </enabled>
    </Sensor>
  </SensorCtrlList>
  <stairsControl>
    <!--ro, opt, object, elevator control parameters-->
    <outputDataType opt="floorNumber,cardNo">
      <!--ro, opt, enum, data type output from the door station to the elevator controller, subType:string, attr:opt{req, string}, desc:floorNumber (floor
No.,default),cardNo (card No.)-->floorNumber
    </outputDataType>
  </stairsControl>
  <comMode opt="active,passive">
    <!--ro, opt, enum, communication mode, subType:string, attr:opt{req, string}, desc:communication mode-->active
  </comMode>
  <ModbusRTU>
    <!--ro, opt, object-->
    <ModbusRTUBaudRate opt="1200,2400,4800,9600,19200" def="9600">
      <!--ro, opt, int, attr:opt{req, string},def{req, string}-->9600
    </ModbusRTUBaudRate>
    <ModbusRTUDataBits opt="5,6,7,8" def="8">
      <!--ro, opt, int, attr:opt{req, string},def{req, string}-->6
    </ModbusRTUDataBits>
    <ModbusRTUParityType opt="none,even,odd" def="none">
      <!--ro, opt, string, attr:opt{req, string},def{req, string}-->none
    </ModbusRTUParityType>
    <ModbusRTUStopBits opt="1,2" def="1">
      <!--ro, opt, string, stop bit, attr:opt{req, string},def{req, string}-->1
    </ModbusRTUStopBits>
    <ModbusRTUFlowCtrl opt="none,software,hardware">
      <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
    </ModbusRTUFlowCtrl>
    <decoderAddress min="0" max="255" def="0">
      <!--ro, opt, int, decoder IP address, attr:min{req, int},max{req, int},def{req, int}-->1
    </decoderAddress>
    <online opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </online>
  </ModbusRTU>
  <accessName min="1" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </accessName>
  <serialPortUse opt="acquAndCtrl,swipe,strongElectric">
    <!--ro, opt, string, attr:opt{req, string}-->acquisition
  </serialPortUse>
  <acquisitionProtocol opt="modbus,custom">
    <!--ro, opt, string, dep:and,{$.SerialPort.serialPortUse,eq,acquisition}, attr:opt{req, string}-->modbus
  </acquisitionProtocol>
  <serialProtocol opt="FECbus,GB26875">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->GB26875
  </serialProtocol>
  <SupplementLightList size="5">
    <!--ro, opt, array, subType:object, dep:and,{$.SerialPort.rs485WorkMode,eq,supplementLight}, attr:size{req, int}-->
    <SupplementLight>
      <!--ro, opt, object-->
      <SupplementLightID min="1" max="5">
        <!--ro, req, int, range:[1,5], attr:min{req, int},max{req, int}-->1
      </SupplementLightID>
    </SupplementLight>
  </SupplementLightList>
  <permissionController opt="0,1">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
  </permissionController>
```

dracium@gmail.com
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```
    <deviceModel min="0" max="32">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </deviceModel>
    <deviceSoftwareVersion min="0" max="32">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </deviceSoftwareVersion>
    <audibleVisualAlarmType opt="CIRCLE,AFE80,AFE82,XY">
      <!--ro, opt, enum, CIRCLE; AFE80; AFE82; XY, subType:string, attr:opt{req, string}, desc:CIRCLE; AFE80; AFE82; XY-->CIRCLE
    </audibleVisualAlarmType>
    <cardReaderType opt="K1108AM">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </cardReaderType>
    <strongElectricType opt="19M04-RO/T,19M08-RO/T">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </strongElectricType>
    <signalLampControl>
      <!--ro, opt, object, dep:and,{$.SerialPort.rs485WorkMode,eq,signalLampControl}-->
      <protocolType opt="GB920,Hisense,LESER,JERE">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->GB920
      </protocolType>
    </signalLampControl>
  </SerialPort>
```

14.1.6 系统维护

14.1.6.1 Get the storage capability of the device

Request URL

GET /ISAPI/ContentMgmt/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<RacmCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device storage capability, attr:version{req, string, protocolVersion}-->
  <isSupportZeroChan>
    <!--ro, opt, bool, whether the device supports channel-zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
  </isSupportZeroChan>
  <inputProxyNums>
    <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->0
  </inputProxyNums>
  <eSATANums>
    <!--ro, opt, int, supported number of eSATA interfaces-->0
  </eSATANums>
  <miniSASNums>
    <!--ro, opt, int, supported number of miniSAS interfaces-->0
  </miniSASNums>
  <nasNums>
    <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->0
  </nasNums>
  <ipSanNums>
    <!--ro, opt, int, supported number of IPSANs-->0
  </ipSanNums>
  <isSupportRaid>
    <!--ro, opt, bool, whether the device supports RAID-->true
  </isSupportRaid>
  <isSupportExtHdCfg>
    <!--ro, opt, bool, whether the device supports advanced HDD management-->true
  </isSupportExtHdCfg>
  <isSupportTransCode>
    <!--ro, opt, bool, whether the device supports live view transcoding-->true
  </isSupportTransCode>
  <isSupportIpcImport>
    <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfig-->true
  </isSupportIpcImport>
  <isSupportIpcStreamType>
    <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
  </isSupportIpcStreamType>
  <isSupportIOInputProxy>
    <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
  </isSupportIOInputProxy>
  <isSupportIOOutputProxy>
    <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
  </isSupportIOOutputProxy>
  <isSupportPTZRs485Proxy>
    <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
  </isSupportPTZRs485Proxy>
  <isSupportSMARTTest>
```

drackum@gmail.com
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```
    <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
  </isSupportSMARTTest>
  <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,ge
tUp,toiletTarry,
playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePh
one,fatigueDriving,
notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringW
arning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideS
hop,stallOccupyingRoad,     illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,temperatureIntervalMeasurement,AIOP,tail,abnormalFace,normalFace,multiFace,callFace,sunglasses
Face,humanEnter,operateOverTime,stickUp,installScanner,objectPlacedOrTaken,fakeCard,securityChecking,pothole,crack,roadMaint,imageCapture,methaneConcentrati
onException,methaneLightIntensityException,bannerDetection,variableDataEvent,shipChannelAbnormal,radarVideoDetection,smokeAndFireDetection,smokeDetection,se
nsorAlarm,alarmResult,AIOP_Video,AIOP_Polling_Snap,swipePatrol,BatteryCarInfoEvent,indoorPasswayBlock,detectorTemperatureAlarm,fireNoRegulation,detectorMoti
onAlarm,detectorShelterAlarm,detectorRunningStatus,faceCapture,UVPhotonDetection,DevPowerStatusReport,HFPD,audioAnalysis,IndoorPasswayRecovery,BatteryCarRec
overy,DetectorTemperatureRecovery,RadarFallenObjectsOnTracks,VegetationSpectrumDetection,personDensityDetection,ChannelClearEvent,channelOccupy,ChannelPassi
ngEvent,FallenObjectsOnTracksEvent,audioexception,CarWindow,shipsDetection,InsulatorAbnormalAlarm,PackingSpaceRecognition,smokeAlarm,SmokeAlarmRecovery,fire
SmartFireDetect,FirePointAlarmRecovery,offDuty,backDuty,sleepDuty,batteryCarDisobey,SmokingDetectionRecoverEvent,noFireExtinguisherRecog,FireExtinguisherRec
overyEvent,TelephoningBehaviorAlarmEvent,TelephoningBehaviorRecoveryEvent,GasTankMonitoringEvent,TemperatureMeasurementPreviewRecovery,TMPA,TMA,TemperatureM
easurementPreviewAlarmRecovery,TemperatureMeasurementAlarmRecovery,DetectorShelterAlarmRecovery,TemperatureUp,TemperatureDown,TemperatureEvent,TemperatureUp
Recover,TemperatureDownRecover,TemperatureEventRecover,ElectricFenceAlarmEvent,cityManagement,flowVelocity,volumeFlowRate,rainfall,partialDischarge,gasLeaka
ge,manualSnapshot,AlarmTriggeredByPanicButton,detectorSmokeAlarm,smokeAndFireSample,synthesis,FireStatusSwitch,FlameCombustionCounting,linkedTracking,smartS
hipDetection,targetModeling,SafetyWorkwearDetectionEvent,SafetyWorkwearRecoveryEvent,DeskMatOffset">
    <!--ro, opt, enum, picture search type, subType:string, attr:opt{req, string}, desc:"AllEvent", "CMR", "MOTION", "ALARM", "EDR", "ALARMANDMOTION",
"pir:PIR", "wlsensor",  "callhelp", "facedetection", "FieldDetection", "scenechangedetection", "LineDetection", "regionEntrance", "regionExiting",
"loitering", "group", "rapidMove", "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual", "manualSnapShot",  "playSnapShot",
"safetyHelmet", "personQueueCounting", "personQueueTime", "violentMotion", "LeavePosition", "advReachHeight", "peopleNumChange", "mixedTargetDetection",
"illegalParking", "pedestrian", "construction", "roadblock", "evidence", "crosslane", "vehicleexist", "lanechange", "wrongdirection", "congestion",
"turnround", "parallelParking", "abandonedObject", "trafficAccident", "fogDetection", "smokeDetection", "personDensity", "failDown", "accessController",
"videoInterCom", "GJD", "Luminite", "OPTEX", "securityControlPanel", "getUp", "toiletTarry", "playCellphone", "vehicleMonitor", "running", "retention",
"spacingChange", "yardTarry", "keyPersonGetUp", "sitQuietly", "standup", "plateRecognition", "smoking", "onThePhone", "fatigueDriving",
"notLookingStraightAhead", "leaveThePost", "seatbelt", "sunglasses", "identityAbnormal", "ForwardCollisionWarning", "LaneDepartureWarning",
"BlindSpotDetection", "HeadwayMonitoringWarning", "PedestrianCollisionWarning", "SuddenSpeedUp", "SuddenSpeedDown", "SuddenLeftTurn", "SuddenRightTurn",
"Turnover", "NotYieldToPedestrian;allPic", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin", "smokeDetectAlarm", "temperatureIntervalMeasurement",
"AIOP", "tail", "abnormalFace", "normalFace;callFace", "sunglassesFace", "humanEnter", "operateOverTime", "stickup", "installScanner",
"objectPlacedOrTaken", "fakeCard", "securityChecking", "pothole", "crack", "roadMaint", "imageCapture", "ReID", "thermalVehicleDetection",
"methaneConcentrationException", "methaneLightIntensityException", "bannerDetection", "waterQualityDetection", "waterLevelDetection", "parallelParking",
"methaneConcentrationException", "variableDataEvent", "VOCsGasDetection", "shipChannelAbnormal", "radarVideoDetection", "AIDfogDetection:AID",
"smokeAndFireDetection", "smokeDetection", "sensorAlarm", "GasTankDetection", "alarmResult", "AIOP_Video", "AIOP_Polling_Snap", "swipePatrol",
"BatteryCarInfoEvent", "indoorPasswayBlock", "detectorTemperatureAlarm", "fireNoRegulation", "detectorMotionAlarm", "detectorShelterAlarm",
"detectorRunningStatus", "faceCapture", "UVPhotonDetection", "DevPowerStatusReport", 'HFPD"-->AllEvent
  </pictureSearchType>
  <recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
,temperatureIntervalMeasurement,AIOP,securityChecking,scheduledRecordUpload,variableDataEvent,smokeAndFireDetection,smokeDetection,AIOP_Video,AIOP_Polling_S
nap,BatteryCarInfoEvent,indoorPasswayBlock,detectorTemperatureAlarm,fireNoRegulation,detectorMotionAlarm,detectorShelterAlarm,detectorRunningStatus,faceCapt
ure,UVPhotonDetection,DevPowerStatusReport,ChannelClearEvent,channelOccupy,ChannelPassingEvent,personQueueCounting,personQueueTime,leavePosition,localButton
Record,callRecord,partialDischarge,gasLeakage,smokeAndFireSample,passback,linkedTracking,CallRecordAnswered,CallRecordNotAnswered,lensMotorLocking">
    <!--ro, opt, enum, video search type, subType:string, attr:opt{req, string}, desc:"AllEvent" (all events), "CMR" (capture by schedule), "MOTION" (motion
detection), "ALARM", "EDR", "ALARMANDMOTION", "pir", "wlsensor", "callhelp", "facedetection", "FieldDetection", "scenechangedetection", "LineDetection",
"regionEntrance", "regionExiting", "loitering", "group", "rapidMove", "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual",
"manualSnapShot", "playSnapShot", "safetyHelmet",  "personDensityDetection", "smokeDetection", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin",
"temperatureIntervalMeasurement", "AIOP", "securityChecking", "scheduledRecordUpload", "thermalVehicleDetection", "waterQualityDetection",
"waterLevelDetection", "methaneConcentrationException", "variableDataEvent"-->AllEvent
  </recordSearchType>
  <isSupportActivateIpc>
    <!--ro, opt, bool, whether the device supports activating the network camera-->true
  </isSupportActivateIpc>
  <isSupportCheckIpcSecurity>
    <!--ro, opt, object, whether the device supports IPC security verification-->
    <isSupportCheckPassword>
      <!--ro, opt, bool, whether the device supports password detection-->true
    </isSupportCheckPassword>
  </isSupportCheckIpcSecurity>
  <isSupportMainAndSubRecord>
    <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
  </isSupportMainAndSubRecord>
  <isSupportSyncIPCPassword>
    <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
  </isSupportSyncIPCPassword>
  <isSupportTransferIPC>
    <!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
  </isSupportTransferIPC>
  <isSupportSmartSearch>
    <!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
  </isSupportSmartSearch>
  <isSupportPOS>
    <!--ro, opt, bool, whether the device supports POS-->true
  </isSupportPOS>
  <isSupportLinkNum>
    <!--ro, opt, bool, whether the device supports getting the number of links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/linkNum-->true
```

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```
  </isSupportLinkNum>
  <isSupportStorageExtraInfo>
    <!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
  </isSupportStorageExtraInfo>
  <ExternalDevice>
    <!--ro, opt, object, external device storage-->
    <USB>
      <!--ro, opt, object, USB information-->
      <debugLogOutput>
        <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
      </debugLogOutput>
      <isSupportCapturePackage>
        <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
      </isSupportCapturePackage>
      <isSupportStreamStorage>
        <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
      </isSupportStreamStorage>
    </USB>
  </ExternalDevice>
  <customProtocolNums>
    <!--ro, opt, int, number of custom protocols for accessing network cameras, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols-->0
  </customProtocolNums>
  <isSupportIPCTiming>
    <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:relate to enableTiming on URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>-->true
  </isSupportIPCTiming>
  <certificateValidationEnabled>
    <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <certificateValidationEnabled> node supports-->true
  </certificateValidationEnabled>
  <defaultAdminPortEnabled>
    <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <defaultAdminPortEnabled> node supports-->true
  </defaultAdminPortEnabled>
  <isSupportLogDataPackage>
    <!--ro, opt, bool, whether the device supports exporting the log files, desc:related URI: /ISAPI/ContentMgmt/logSearch/dataPackage-->true
  </isSupportLogDataPackage>
  <isSupportWebPrivatePlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback (via private protocol) according to UTC-->true
  </isSupportWebPrivatePlaybackByUTC>
  <isSupportFindCommonFileByUTC>
    <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
  </isSupportFindCommonFileByUTC>
  <isSupportFindEventFileByUTC>
    <!--ro, opt, bool, whether it supports searching for event related videos (via private protocol) according to UTC-->true
  </isSupportFindEventFileByUTC>
  <isSupportSmartSearchRecordByUTC>
    <!--ro, opt, bool, whether it supports VCA search by UTC time-->true
  </isSupportSmartSearchRecordByUTC>
  <isSupportMRDSearchByTimeZone>
    <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
  </isSupportMRDSearchByTimeZone>
  <isSupportSearchRecordLabelByUTC>
    <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
  </isSupportSearchRecordLabelByUTC>
  <isSupportSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
  </isSupportSearchPictureByUTC>
  <isSupportSmartSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
  </isSupportSmartSearchPictureByUTC>
  <isSupportFindLogByUTC>
    <!--ro, opt, bool, whether it supports searching log by UTC time-->true
  </isSupportFindLogByUTC>
  <isSupportUploadRecordByUTC>
    <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
  </isSupportUploadRecordByUTC>
  <isSupportPlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
  </isSupportPlaybackByUTC>
  <isSupportFaceDetecttionAlarmByTimeZone>
    <!--ro, opt, bool, whether it supports reporting face detection by UTC time, desc:the time should be before isSupportCountingSearchByUTC and after
isSupportPlaybackByUTC-->true
  </isSupportFaceDetecttionAlarmByTimeZone>
  <isSupportRecordSearchByTargetType>
    <!--ro, opt, bool, whether it supports searching for event recordings by target types (pedestrians and vehicles)-->true
  </isSupportRecordSearchByTargetType>
  <isSupportLinkToIPCWeb>
    <!--ro, opt, bool, whether it supports going to the network camera web page (the digital channel management page), desc:N/A-->true
  </isSupportLinkToIPCWeb>
  <isSupportPicSearchByCoordinate>
    <!--ro, opt, bool, whether the device supports AcuSearch configurations, desc:GET or PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json-->true
  </isSupportPicSearchByCoordinate>
  <AcuSearchCap>
    <!--ro, opt, object, AcuSearch capability-->
    <targetType opt="face,human,vehicle">
      <!--ro, opt, enum, AcuSearch supported target type, subType:string, attr:opt{req, string}, desc:"face", "human" (human body), "vehicle"-->face
    </targetType>
    <dataType opt="url,binary,modelData">
      <!--ro, opt, enum, supported picture transmission format, subType:string, attr:opt{req, string}, desc:"URL", (picture URL), "modelData" (model data),
"binary" (picture binary form)-->url
```

dr4ckun@gmail.com
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```
      </dataType>
      <maxPicSize>
        <!--ro, opt, int, maximum size of image can be searched, range:[0,5], desc:unit: MB-->1
      </maxPicSize>
      <maxPicWidthSize>
        <!--ro, opt, int, maximum width of image can be searched, range:[0,4096]-->4096
      </maxPicWidthSize>
      <maxPicHeightSize>
        <!--ro, opt, int, maximum height of image can be searched, range:[0,4096]-->4096
      </maxPicHeightSize>
    </AcuSearchCap>
    <isSupportRedirectStorageUrl>
      <!--ro, opt, bool-->true
    </isSupportRedirectStorageUrl>
    <isSupportSSDTemperatureAbnormalTrigger>
      <!--ro, opt, bool-->true
    </isSupportSSDTemperatureAbnormalTrigger>
    <isSupportSSDLifeTimeAbnormalTrigger>
      <!--ro, opt, bool-->true
    </isSupportSSDLifeTimeAbnormalTrigger>
  </RacmCap>
```

14.1.6.2 Get the system security capability

Request URL

GET /ISAPI/Security/capabilities?username=<userName>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
userName	string	user name
deviceIndex	string	For centralized splicing controllers connecting to LED sending cards, the URL needs to include the parameter deviceIndex= to specify LED sending card The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SecurityCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, system security capability, attr:version{req, string, protocolVersion}-->
  <supportUserNums>
    <!--ro, opt, int, supported maximum number of users-->0
  </supportUserNums>
  <userBondIpNums>
    <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->0
  </userBondIpNums>
  <userBondMacNums>
    <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->0
  </userBondMacNums>
  <isSupportOnlineUser>
    <!--ro, opt, bool, whether the device supports the online user configuration-->true
  </isSupportOnlineUser>
  <securityVersion opt="1,2,3,4">
    <!--ro, opt, string, encryption capability set, attr:opt{req, string}, desc:the encryption capability of each version consists of two parts: encryption algorithm and the range of encrypted nodes currently 1 refers to AES128 encryption and 2 refers to AES256 encryption, the range of encrypted nodes is described in each protocol-->1
  </securityVersion>
  <keyIterateNum>
    <!--ro, opt, int, secret key iteration times, dep:or,{$.SecurityCap.securityVersion,eq,1},{$.SecurityCap.securityVersion,eq,2}, desc:this node depends on the node securityVersion, the range is between 100 and 1000-->100
  </keyIterateNum>
  <isSupportUserCheck>
    <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, dep:or,{$.SecurityCap.securityVersion,eq,0},{$.SecurityCap.securityVersion,eq,1}, desc:it is an added capability, which indicates that whether supporting the login password verification for editing/adding/deleting user parameters, this node depends on the node securityVersion, which means that it is only valid for the versions that support encrypting the sensitive information-->true
  </isSupportUserCheck>
  <isSupportGUIDFileDataExport>
    <!--ro, opt, bool, whether the device supports exporting the device's GUID file, desc:related URI: /ISAPI/Security/GUIDFileData-->true
  </isSupportGUIDFileDataExport>
  <isSupportSecurityQuestionConfig>
    <!--ro, opt, bool, whether the device supports answering security questions, desc:related URI: /ISAPI/Security/questionConfiguration-->true
  </isSupportSecurityQuestionConfig>
  <isSupportGetOnlineUserListSC>
```

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```
    <!--ro, opt, bool, whether the device supports searching the online user list, desc:related URI: /ISAPI/Security/onlineUser-->true
  </isSupportGetOnlineUserListSC>
  <SecurityLimits>
    <!--ro, opt, object, capability of configuring security limit parameters-->
    <LoginPasswordLenLimit min="1" max="16">
      <!--ro, opt, string, length limit of the user's login password, attr:min{req, int},max{req, int}-->1
    </LoginPasswordLenLimit>
    <SecurityAnswerLenLimit min="1" max="128">
      <!--ro, opt, string, length limit of the security question's answer, attr:min{req, int},max{req, int}-->1
    </SecurityAnswerLenLimit>
  </SecurityLimits>
  <isSupportONVIFUserManagement>
    <!--ro, opt, bool, whether the device supports user management of Open Network Video Interface Protocol-->true
  </isSupportONVIFUserManagement>
  <WebCertificateCap>
    <!--ro, opt, object, HTTP authentication capability, desc:if this node is not returned, it indicates that device supports basic and digest
authentication-->
    <CertificateType opt="basic,digest,digest/basic">
      <!--ro, req, enum, certificate type, subType:string, attr:opt{req, string}, desc:“basic” (basic authentication), “digest” (digest authentication),
“digest/basic” (digest/basic authentication)-->basic
    </CertificateType>
    <SecurityAlgorithm>
      <!--ro, opt, object, security algorithm, dep:or,{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest},
{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
      <algorithmType opt="MD5,SHA256,MD5/SHA256">
        <!--ro, opt, enum, algorithm type, subType:string, attr:opt{req, string}, desc:“MD5”, “SHA256”, “MD5/SHA256”-->MD5
      </algorithmType>
    </SecurityAlgorithm>
  </WebCertificateCap>
  <isSupportConfigFileImport>
    <!--ro, opt, bool, whether the device supports importing the configuration file, desc:true (support), this node is not returned (not support)-->true
  </isSupportConfigFileImport>
  <isSupportConfigFileExport>
    <!--ro, opt, bool, whether the device supports exporting the configuration file, desc:true (support), this node is not returned (not support)-->true
  </isSupportConfigFileExport>
  <cfgFileSecretKeyLenLimit min="0" max="16">
    <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{req, int},max{req, int}-->1
  </cfgFileSecretKeyLenLimit>
  <supportIPCActivatePassword>
    <!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
  </supportIPCActivatePassword>
  <isIrreversible>
    <!--ro, opt, bool, whether the device supports irreversible password storage, desc:If this function is not supported, the plaintext password of the user
information will be stored in the device; otherwise, the password will be hashed for storage in the device.-->true
  </isIrreversible>
  <salt>
    <!--ro, opt, string, the specific salt used by the user to log in-->test
  </salt>
  <isSupportPictureURlCertificate opt="true,false">
    <!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt{req, string}, desc:if this node is not
returned, it indicates the digest authentication for picture URL by default-->true
  </isSupportPictureURlCertificate>
  <isSupportUnloggedUserPermissionConfig opt="true,false">
    <!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt{req, string}-->true
  </isSupportUnloggedUserPermissionConfig>
  <isSupportDeviceCertificatesManagement>
    <!--ro, opt, bool, whether the device supports device certificate management, desc:if this node is not returned, it indicates that device does not
support this function-->true
  </isSupportDeviceCertificatesManagement>
  <isSupportDeviceSelfSignCertExport>
    <!--ro, opt, bool, whether the device supports exporting self-signed certificate-->true
  </isSupportDeviceSelfSignCertExport>
  <isSupportSecurityEmail>
    <!--ro, opt, bool, whether the device supports configuring the security email, desc:if this node is not returned, it indicates that device does not
support this function-->true
  </isSupportSecurityEmail>
  <isSupportEncryptCertificate>
    <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
  </isSupportEncryptCertificate>
  <isSupportPasswordProtection>
    <!--ro, opt, bool, whether it supports configuring SDK protocol authentication mode parameters, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
  </isSupportPasswordProtection>
</SecurityCap>
```

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14.1.6.3 Get device system capability

Request URL

GET /ISAPI/System/capabilities?type=<all>

Query Parameter

Parameter Name	Parameter Type	Description
all	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device system capability, attr:version{opt, string, protocolVersion}-->
  <SysCap>
    <!--ro, opt, object, system capability-->
    <isSupportDst>
      <!--ro, opt, bool, whether the device supports DST (Daylight Saving Time)-->true
    </isSupportDst>
    <NetworkCap>
      <!--ro, opt, object, network capability, desc:related URI: /ISAPI/System/Network/capabilities-->
      <isSupportWireless>
        <!--ro, req, bool, whether the device supports wireless network-->true
      </isSupportWireless>
      <isSupportWAN>
        <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
      </isSupportWAN>
      <isSupportPPPoE>
        <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
      </isSupportPPPoE>
      <isSupportBond>
        <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
      </isSupportBond>
      <isSupport802_1x>
        <!--ro, req, bool, whether the device supports 802.1x protocol-->true
      </isSupport802_1x>
      <isSupportNtp>
        <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
      </isSupportNtp>
      <isSupportFtp>
        <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
      </isSupportFtp>
      <isSupportUpnp>
        <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play ) protocol-->true
      </isSupportUpnp>
      <isSupportPNP>
        <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
      </isSupportPNP>
      <isSupportDdns>
        <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
      </isSupportDdns>
      <isSupportHttps>
        <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
      </isSupportHttps>
      <SnmpCap>
        <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
        <isSupport>
          <!--ro, req, bool, whether the device supports SNMP-->true
        </isSupport>
      </SnmpCap>
      <isSupportExtNetCfg>
        <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
      </isSupportExtNetCfg>
      <isSupportIPFilter>
        <!--ro, opt, bool, whether the device supports IP filtering-->true
      </isSupportIPFilter>
      <isSupportNetPreviewStrategy>
        <!--ro, opt, bool, whether the device supports network preview strategy-->true
      </isSupportNetPreviewStrategy>
      <isSupportEZVIZ>
        <!--ro, opt, bool, whether the device supports EZ protocol-->true
      </isSupportEZVIZ>
      <isSupportWPS>
        <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
      </isSupportWPS>
      <isWirelessMutexWithWirelessServer>
        <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
      </isWirelessMutexWithWirelessServer>
      <isSupportMACFilter>
        <!--ro, opt, bool, whether the device supports physical address filtering-->true
      </isSupportMACFilter>
      <GB28181Cap>
        <!--ro, opt, object, N/A-->
        <isSupportGB28181Service>
          <!--ro, opt, bool, N/A-->true
        </isSupportGB28181Service>
      </GB28181Cap>
      <VerificationCodeModification>
        <!--ro, opt, object, device verification code configuration-->
        <verificationCodeType opt="normal,empty">
          <!--ro, opt, string, verification code type, attr:opt{opt, string}-->empty
        </verificationCodeType>
        <isSupportDeclarationURL>
          <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
        </isSupportDeclarationURL>
        <isSupportPrivacyPolicyURL>
          <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
        </isSupportPrivacyPolicyURL>
```

dracrum@gmail.com
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```
        <verificationCodeModify opt="true,false">
          <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{opt, string}-->true
        </verificationCodeModify>
        <Hyperlinks>
          <!--ro, opt, object, hyperlink configuration-->
          <declarationURL>
            <!--ro, opt, string, hyperlink to the Terms of Service-->test
          </declarationURL>
          <privacyPolicyURL>
            <!--ro, opt, string, hyperlink to the Privacy Policy-->test
          </privacyPolicyURL>
        </Hyperlinks>
        <isSupportVerificationCodeCheck>
          <!--ro, opt, bool, whether the device supports verifying the verification code-->true
        </isSupportVerificationCodeCheck>
      </VerificationCodeModification>
      <isSupportIntegrate>
        <!--ro, opt, bool, whether the device supports access protocol configuration-->true
      </isSupportIntegrate>
      <isSupportResourceStatistics>
        <!--ro, opt, bool, whether the device supports getting the network resource information-->true
      </isSupportResourceStatistics>
      <isSupportBandwidthLimit>
        <!--ro, opt, bool, whether the device supports bandwidth limit-->true
      </isSupportBandwidthLimit>
      <isSupportGetLinkSocketIP>
        <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
      </isSupportGetLinkSocketIP>
      <isSupportEZVIZTiming>
        <!--ro, opt, bool, whether the device supports synchronizing time with EZ platform, desc:related URI: enabledTiming in the message of
/ISAPI/System/Network/EZVIZ-->true
      </isSupportEZVIZTiming>
      <isSupportRequestOutlookAuth>
        <!--ro, opt, bool, whether the device supports applying authentication request of Outlook email, desc:POST
/ISAPI/System/Network/Email/RequestOutlookAuth?format=json-->true
      </isSupportRequestOutlookAuth>
      <isSupportOutlookEmailList>
        <!--ro, opt, bool, whether the device supports Outlook email information, desc:GET or PUT /ISAPI/System/Network/Email/OutlookEmailList?format=json--
>true
      </isSupportOutlookEmailList>
      <isSupportDSTPServerCfg>
        <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET or PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
      </isSupportDSTPServerCfg>
      <isSupportDSTPProtocolsCap>
        <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
      </isSupportDSTPProtocolsCap>
      <isSupportSearchOutlookAuthResult>
        <!--ro, opt, bool-->true
      </isSupportSearchOutlookAuthResult>
    </NetworkCap>
    <IOCap>
      <!--ro, opt, object, I/O capability, desc:related URI: /ISAPI/System/IO/capabilities-->
      <IOInputPortNums>
        <!--ro, opt, int, number of local input ports of the device-->true
      </IOInputPortNums>
      <IOOutputPortNums>
        <!--ro, opt, int, number of local output ports of the device-->true
      </IOOutputPortNums>
      <isSupportIOOutputAdvanceParameter>
        <!--ro, opt, bool, whether the device supports configuring advanced parameters of the output port, desc:this node determines whether the URI
/ISAPI/System/IO/outputs/advanceParameter/capabilities?format=json is supported-->true
      </isSupportIOOutputAdvanceParameter>
      <isSupportSetAllIOOutput>
        <!--ro, opt, bool, whether the device supports configuring all output ports, desc:this node determines whether the PUT method of the URI
/ISAPI/System/IO/outputs is supported-->true
      </isSupportSetAllIOOutput>
      <enabledIOOutputPortNums>
        <!--ro, opt, int, number of enabled alarm outputs-->0
      </enabledIOOutputPortNums>
    </IOCap>
    <SerialCap>
      <!--ro, opt, object, range of RS-485 serial port numbers supported by the device-->
      <rs485PortNums>
        <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->true
      </rs485PortNums>
      <supportRS232Config>
        <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
      </supportRS232Config>
      <rs422PortNums>
        <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->true
      </rs422PortNums>
      <rs232PortNums>
        <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->true
      </rs232PortNums>
      <isSupportAuthenticationService>
        <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
      </isSupportAuthenticationService>
    </SerialCap>
    <VideoCap>
      <!--ro, opt, object, video encoding capability, desc:related URI: /ISAPI/System/Video/capabilities-->
      <videoInputPortNums>
        <!--ro, opt, int, number of video input ports-->true
      </videoInputPortNums>
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      <videoOutputPortNums>
        <!--ro, opt, int, number of video output ports-->true
      </videoOutputPortNums>
      <menuNums>
        <!--ro, opt, int, number of local menus that can be displayed-->true
      </menuNums>
    </VideoCap>
    <AudioCap>
      <!--ro, opt, object, audio encoding capability, desc:related URL: /ISAPI/System/Audio/capabilities-->
      <audioInputNums>
        <!--ro, req, int, number of audio inputs-->1
      </audioInputNums>
      <audioOutputNums>
        <!--ro, req, int, number of audio outputs-->1
      </audioOutputNums>
    </AudioCap>
    <isSupportHolidy>
      <!--ro, opt, bool, this node is located after the node <AudioCap>-->true
    </isSupportHolidy>
    <RebootConfigurationCap>
      <!--ro, opt, object, capability of configuring parameters for rebooting the device, desc:related URI: /ISAPI/System/reboot/capabilities. This node is
located before the node <isSupportSubscribeEvent> and after the node <isSupportHolidy>-->
      <Genetec>
        <!--ro, opt, bool, whether to enable the Genetec protocol-->true
      </Genetec>
      <ONVIF>
        <!--ro, opt, bool, whether to enable the ONVIF protocol-->true
      </ONVIF>
      <RTSP>
        <!--ro, opt, bool, whether to enable the RTSP protocol-->true
      </RTSP>
      <HTTP>
        <!--ro, opt, bool, whether to enable the HTTP protocol-->true
      </HTTP>
      <SADP>
        <!--ro, opt, object, SADP (Search Active Devices Protocol, including the link layer and multicast discovery)-->
        <ISDiscoveryMode>
          <!--ro, opt, bool, mode of the SADP multicast discovery protocol-->true
        </ISDiscoveryMode>
        <PcapMode>
          <!--ro, opt, bool, mode of the SADP link layer discovery protocol-->true
        </PcapMode>
      </SADP>
      <IPCAddStatus>
        <!--ro, opt, bool, IP address status-->true
      </IPCAddStatus>
    </RebootConfigurationCap>
    <isSupportSubscribeEvent>
      <!--ro, opt, bool, whether the device supports subscribing to events, desc:related URI: /ISAPI/Event/notification/subscribeEventCap-->true
    </isSupportSubscribeEvent>
    <isSupportDiagnosedData>
      <!--ro, opt, bool, whether the device supports exporting the device diagnosis data, desc:true (support), this node is not returned (not support).
Related URI: /ISAPI/System/diagnosedData-->true
    </isSupportDiagnosedData>
    <isSupportDiagnosedDataParameter>
      <!--ro, opt, bool, whether the device supports enabling or disabling exporting the diagnosis information-->true
    </isSupportDiagnosedDataParameter>
    <isSupportSimpleDevStatus>
      <!--ro, opt, bool, whether the device supports getting the device's simple status, desc:related URI: /ISAPI/System/workingstatus?format=json-->true
    </isSupportSimpleDevStatus>
    <isSupportFlexible>
      <!--ro, opt, bool, whether the device supports the flexible combination configuration. Related URI: /ISAPI/System/Video/inputs/channels?format=json--
>true
    </isSupportFlexible>
    <isSupportTimeCap>
      <!--ro, opt, bool, whether the device supports the time configuration, desc:related URI: /ISAPI/System/time/capabilities-->true
    </isSupportTimeCap>
  </SysCap>
  <voicetalkNums>
    <!--ro, opt, int, number of two-way audio channels-->2
  </voicetalkNums>
  <isSupportSnapshot>
    <!--ro, opt, bool, whether the device supports capturing pictures-->true
  </isSupportSnapshot>
  <SecurityCap>
    <!--ro, opt, object, encryption capability set-->
    <supportUserNums>
      <!--ro, opt, int, supported maximum number of users-->1
    </supportUserNums>
    <userBondIpNums>
      <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->1
    </userBondIpNums>
    <userBondMacNums>
      <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->1
    </userBondMacNums>
    <isSupportOnlineUser>
      <!--ro, opt, bool, whether the device supports the online user configuration-->true
    </isSupportOnlineUser>
    <securityVersion opt="1,2,3,4,7">
      <!--ro, opt, string, encryption capability, attr:opt{opt, string}, desc:the encryption capability of each version consists of two parts: encryption
algorithm and the range of encrypted nodes. Currently 1 refers to AES128 encryption and 2 refers to AES256 encryption. The range of encrypted nodes is
described in each protocol-->test
    </securityVersion>
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    <keyIterateNum>
      <!--ro, opt, int, iteration times, desc:the value is usually between 100 and 1000-->100
    </keyIterateNum>
    <isSupportUserCheck>
      <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, desc:this node
depends on the node <securityVersion>, which means that it is only valid for the versions that support encrypting the sensitive information-->true
    </isSupportUserCheck>
    <isSupportGUIDFileDataExport>
      <!--ro, opt, bool, whether the device supports exporting the device's GUID file-->true
    </isSupportGUIDFileDataExport>
    <isSupportSecurityQuestionConfig>
      <!--ro, opt, bool, whether the device supports answering security questions-->true
    </isSupportSecurityQuestionConfig>
    <isSupportGetOnlineUserListSC>
      <!--ro, opt, bool, whether the device supports searching the online user list-->true
    </isSupportGetOnlineUserListSC>
    <SecurityLimits>
      <!--ro, opt, object, capability of configuring security limit parameters-->
      <LoginPasswordLenLimit min="1" max="16">
        <!--ro, opt, string, length limit of the user's login password, attr:min{opt, string},max{opt, string}-->test
      </LoginPasswordLenLimit>
      <SecurityAnswerLenLimit min="1" max="128">
        <!--ro, opt, string, length limit of the security question's answer, attr:min{opt, string},max{opt, string}-->test
      </SecurityAnswerLenLimit>
    </SecurityLimits>
    <isSupportONVIFUserManagement>
      <!--ro, opt, bool, whether the device supports ONVIF user management-->true
    </isSupportONVIFUserManagement>
    <WebCertificateCap>
      <!--ro, opt, object, HTTP authentication capability-->
      <CertificateType opt="basic,digest,digest/basic">
        <!--ro, req, string, certificate type: basic (authentication), attr:opt{opt, string}-->test
      </CertificateType>
      <SecurityAlgorithm>
        <!--ro, opt, object, encryption algorithm, dep:or,{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,degist},
{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
        <algorithmType opt="MD5,SHA256,MD5/SHA256">
          <!--ro, opt, string, algorithm type, attr:opt{req, string}-->test
        </algorithmType>
      </SecurityAlgorithm>
    </WebCertificateCap>
    <isSupportConfigFileImport>
      <!--ro, opt, bool, whether the device supports importing the configuration file-->true
    </isSupportConfigFileImport>
    <isSupportConfigFileExport>
      <!--ro, opt, bool, whether the device supports exporting the configuration file-->true
    </isSupportConfigFileExport>
    <cfgFileSecretKeyLenLimit min="0" max="16">
      <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{opt, string},max{opt, string}-->0
    </cfgFileSecretKeyLenLimit>
    <supportIPCActivatePassword>
      <!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
    </supportIPCActivatePassword>
    <isSupportDeviceCertificatesManagement>
      <!--ro, opt, bool, whether the device supports certificate management-->true
    </isSupportDeviceCertificatesManagement>
    <isSupportDeviceSelfSignCertExport>
      <!--ro, opt, bool, whether the device supports exporting the self-signed certificate-->true
    </isSupportDeviceSelfSignCertExport>
    <isSupportSecurityEmail>
      <!--ro, opt, bool, whether the device supports configuring the security email-->true
    </isSupportSecurityEmail>
    <isSupportPictureURlCertificate opt="true,false">
      <!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt{opt, string}-->true
    </isSupportPictureURlCertificate>
    <isSupportUnloggedUserPermissionConfig opt="true,false">
      <!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt{opt, string}-->true
    </isSupportUnloggedUserPermissionConfig>
    <isSupportEncryptCertificate>
      <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
    </isSupportEncryptCertificate>
    <isSupportPasswordProtection>
      <!--ro, opt, bool, whether it supports parameter configuration of SDK protocol authentication mode, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
    </isSupportPasswordProtection>
  </SecurityCap>
  <EventCap>
    <!--ro, opt, object, event capability-->
    <isSupportHDFull>
      <!--ro, opt, bool, whether the device supports HDD full alarm-->true
    </isSupportHDFull>
    <isSupportHDError>
      <!--ro, opt, bool, whether the device supports HDD error alarm-->true
    </isSupportHDError>
    <isSupportNicBroken>
      <!--ro, opt, bool, whether the device supports the network disconnected event-->true
    </isSupportNicBroken>
    <isSupportIpConflict>
      <!--ro, opt, bool, whether the device supports the IP address conflict event-->true
    </isSupportIpConflict>
    <isSupportIllAccess>
      <!--ro, opt, bool, whether the device supports the illegal login event-->true
    </isSupportIllAccess>
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drackum@gmail.com
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    <isSupportViException>
      <!--ro, opt, bool, whether the device supports video input exception (only for analog channels)-->true
    </isSupportViException>
    <isSupportViMismatch>
      <!--ro, opt, bool, whether the device supports the event of video standard mismatch-->true
    </isSupportViMismatch>
    <isSupportRecordException>
      <!--ro, opt, bool, whether the device supports the recording exception event-->true
    </isSupportRecordException>
    <isSupportRaidException>
      <!--ro, opt, bool, whether the device supports the array exception event-->true
    </isSupportRaidException>
    <isSupportViResMismatch>
      <!--ro, opt, bool, whether the device supports the event of video resolution mismatch-->true
    </isSupportViResMismatch>
    <isSupportPOCException>
      <!--ro, opt, bool, whether the device supports the event of powering over coaxial exception-->true
    </isSupportPOCException>
    <isSupportSpareException>
      <!--ro, opt, bool, whether the device supports hot spare exception-->true
    </isSupportSpareException>
    <isSupportHumanRecognition>
      <!--ro, opt, bool, whether the device supports target capture events-->true
    </isSupportHumanRecognition>
    <isSupportFaceContrast>
      <!--ro, opt, bool, whether the device supports face picture comparison events-->true
    </isSupportFaceContrast>
    <isSupportFaceSnap>
      <!--ro, opt, bool, whether the device supports face capture events-->true
    </isSupportFaceSnap>
    <isSupportTriggerCapCheck>
      <!--ro, opt, bool, whether the device supports verifying the capability of triggering linkage actions, desc:this node indicates that the capability
message returned by calling the URI GET /ISAPI/Event/triggersCap has been verified, and the event linkage actions can be displayed exactly according to the
capability-->true
    </isSupportTriggerCapCheck>
  </EventCap>
  <RacmCap>
    <!--ro, opt, object, UI before picture search-->
    <isSupportZeroChan>
      <!--ro, opt, bool, whether the device supports channel-zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
    </isSupportZeroChan>
    <inputProxyNums>
      <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->1
    </inputProxyNums>
    <eSATANums>
      <!--ro, opt, int, supported number of eSATA interfaces-->1
    </eSATANums>
    <miniSASNums>
      <!--ro, opt, int, supported number of miniSAS interfaces-->1
    </miniSASNums>
    <nasNums>
      <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->1
    </nasNums>
    <ipSanNums>
      <!--ro, opt, int, supported number of IPSANs-->1
    </ipSanNums>
    <isSupportRaid>
      <!--ro, opt, bool, whether the device supports RAID-->true
    </isSupportRaid>
    <isSupportExtHdCfg>
      <!--ro, opt, bool, whether the device supports advanced HDD management-->true
    </isSupportExtHdCfg>
    <isSupportTransCode>
      <!--ro, opt, bool, whether the device supports live view transcoding-->true
    </isSupportTransCode>
    <isSupportIpcImport>
      <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfig-->true
    </isSupportIpcImport>
    <isSupportIpcStreamType>
      <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
    </isSupportIpcStreamType>
    <isSupportIOInputProxy>
      <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
    </isSupportIOInputProxy>
    <isSupportIOOutputProxy>
      <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
    </isSupportIOOutputProxy>
    <isSupportPTZRs485Proxy>
      <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
    </isSupportPTZRs485Proxy>
    <isSupportSMARTTest>
      <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
    </isSupportSMARTTest>
    <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,ge
tUp,toiletTarry,
playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePh
one,fatigueDriving,
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drackum@gmail.com
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notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringW
arning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideS
hop,stallOccupyingRoad,     illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,alarmResult">
      <!--ro, opt, string, picture search type, attr:opt{opt, string}-->test
    </pictureSearchType>
    <recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
">
      <!--ro, opt, string, video search type, attr:opt{opt, string}, desc:"llEvent" (all events), "CMR" (capture by schedule), "MOTION" (capture motion
detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures
and motion detection pictures), "Command", "pir" (pir alarm), "wlsensor" (wireless alarm), "callhelp" (panic alarm), "facedetection" (face detection),
"FieldDetection" (intrusion detection), "scenechangedetection" (scene change detection), "LineDetection" (line crossing detection)-->test
    </recordSearchType>
    <isSupportActivateIpc>
      <!--ro, opt, bool, whether the device supports activating the network camera-->true
    </isSupportActivateIpc>
    <isSupportCheckIpcSecurity>
      <!--ro, opt, object, device password detection-->
      <isSupportCheckPassword>
        <!--ro, opt, bool, whether the device supports password detection-->true
      </isSupportCheckPassword>
    </isSupportCheckIpcSecurity>
    <isSupportMainAndSubRecord>
      <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
    </isSupportMainAndSubRecord>
    <isSupportSyncIPCPassword>
      <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
    </isSupportSyncIPCPassword>
    <isSupportTransferIPC>
      <!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
    </isSupportTransferIPC>
    <isSupportSmartSearch>
      <!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
    </isSupportSmartSearch>
    <isSupportPOS>
      <!--ro, opt, bool, whether the device supports POS-->true
    </isSupportPOS>
    <isSupportLinkNum>
      <!--ro, opt, bool, whether the device supports getting the number of links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/linkNum--
>true
    </isSupportLinkNum>
    <isSupportStorageExtraInfo>
      <!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
    </isSupportStorageExtraInfo>
    <ExternalDevice>
      <!--ro, opt, object, external device storage-->
      <USB>
        <!--ro, opt, object, USB information-->
        <debugLogOutput>
          <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
        </debugLogOutput>
        <isSupportCapturePackage>
          <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
        </isSupportCapturePackage>
        <isSupportStreamStorage>
          <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
        </isSupportStreamStorage>
      </USB>
    </ExternalDevice>
    <customProtocolNums>
      <!--ro, opt, int, number of custom protocols connected via IPC, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols, supported number of
custom protocols-->1
    </customProtocolNums>
    <isSupportIPCTiming>
      <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <enableTiming> node supports-->true
    </isSupportIPCTiming>
    <certificateValidationEnabled>
      <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <certificateValidationEnabled> node supports-->true
    </certificateValidationEnabled>
    <defaultAdminPortEnabled>
      <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <defaultAdminPortEnabled> node supports-->true
    </defaultAdminPortEnabled>
    <isSupportLogDataPackage>
      <!--ro, opt, bool, whether the device supports exporting the log files, desc:related URI: /ISAPI/ContentMgmt/logSearch/dataPackage-->true
    </isSupportLogDataPackage>
    <isSupportWebPrivatePlaybackByUTC>
      <!--ro, opt, bool, whether it supports playback (via private protocol) by UTC-->true
    </isSupportWebPrivatePlaybackByUTC>
    <isSupportFindCommonFileByUTC>
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drackum@gmail.com
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    <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
  </isSupportFindCommonFileByUTC>
  <isSupportFindEventFileByUTC>
    <!--ro, opt, bool, whether it supports searching for event files (private protocols) by UTC time-->true
  </isSupportFindEventFileByUTC>
  <isSupportSmartSearchRecordByUTC>
    <!--ro, opt, bool, whether it supports smart video search by UTC time-->true
  </isSupportSmartSearchRecordByUTC>
  <isSupportMRDSearchByTimeZone>
    <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
  </isSupportMRDSearchByTimeZone>
  <isSupportSearchRecordLabelByUTC>
    <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
  </isSupportSearchRecordLabelByUTC>
  <isSupportSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
  </isSupportSearchPictureByUTC>
  <isSupportSmartSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
  </isSupportSmartSearchPictureByUTC>
  <isSupportFindLogByUTC>
    <!--ro, opt, bool, whether it supports searching log by UTC time-->true
  </isSupportFindLogByUTC>
  <isSupportUploadRecordByUTC>
    <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
  </isSupportUploadRecordByUTC>
  <isSupportPlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
  </isSupportPlaybackByUTC>
  <isSupportFaceDetecttionAlarmByTimeZone>
    <!--ro, opt, bool, no file is found-->true
  </isSupportFaceDetecttionAlarmByTimeZone>
</RacmCap>
<PTZCtrlCap>
  <!--ro, opt, object, device's PTZ capability-->
  <isSupportPatrols>
    <!--ro, opt, bool, whether the device supports patrol-->true
  </isSupportPatrols>
</PTZCtrlCap>
<SmartCap>
  <!--ro, opt, object, compact and smart capability, desc:/ISAPI/Smart/capabilities-->
  <isSupportROI>
    <!--ro, opt, bool, whether the device supports ROI encoding configuration-->true
  </isSupportROI>
  <isSupportFaceDetect>
    <!--ro, opt, bool, whether the device supports face detection-->true
  </isSupportFaceDetect>
  <isSupportIntelliTrace>
    <!--ro, opt, bool, whether the device supports smart tracking-->true
  </isSupportIntelliTrace>
  <isSupportFieldDetection>
    <!--ro, opt, bool, whether the device supports intrusion detection-->true
  </isSupportFieldDetection>
  <isSupportDefocusDetection>
    <!--ro, opt, bool, whether the device supports defocus detection-->true
  </isSupportDefocusDetection>
  <isSupportAudioDetection>
    <!--ro, opt, bool, whether the device supports audio detection-->true
  </isSupportAudioDetection>
  <isSupportSceneChangeDetection>
    <!--ro, opt, bool, whether the device supports scene change detection-->true
  </isSupportSceneChangeDetection>
  <isSupportLineDetection>
    <!--ro, opt, bool, whether the device supports line crossing detection-->true
  </isSupportLineDetection>
  <isSupportRegionEntrance>
    <!--ro, opt, bool, whether the device supports region entrance detection-->true
  </isSupportRegionEntrance>
  <isSupportRegionExiting>
    <!--ro, opt, bool, whether the device supports region exiting detection-->true
  </isSupportRegionExiting>
  <isSupportLoitering>
    <!--ro, opt, bool, whether the device supports loitering detection-->true
  </isSupportLoitering>
  <isSupportGroup>
    <!--ro, opt, bool, whether the device supports people gathering detection-->true
  </isSupportGroup>
  <isSupportRapidMove>
    <!--ro, opt, bool, whether the device supports fast moving detection-->true
  </isSupportRapidMove>
  <isSupportParking>
    <!--ro, opt, bool, whether the device supports parking detection-->true
  </isSupportParking>
  <isSupportUnattendedBaggage>
    <!--ro, opt, bool, whether the device supports unattended baggage detection-->true
  </isSupportUnattendedBaggage>
  <isSupportAttendedBaggage>
    <!--ro, opt, bool, whether the device supports object removal detection-->true
  </isSupportAttendedBaggage>
  <isSupportAnalysisUnitSwitch opt="true,false">
    <!--ro, opt, bool, whether the device supports switching analysis unit, attr:opt{opt, string}-->true
  </isSupportAnalysisUnitSwitch>
</SmartCap>
<isSupportEhome>
```

drackun@gmail.com
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```
    <!--ro, opt, bool, whether the device supports ISUP functions-->true
  </isSupportEhome>
  <isSupportStreamingEncrypt>
    <!--ro, opt, bool, whether the device supports stream encryption-->true
  </isSupportStreamingEncrypt>
  <TestCap>
    <!--ro, opt, object, test capability-->
    <isSupportEmailTest>
      <!--ro, opt, bool, whether the device supports email test-->true
    </isSupportEmailTest>
  </TestCap>
  <ThermalCap>
    <!--ro, opt, object, thermal capability, desc:related URI: /ISAPI/Thermal/capabilities-->
    <isSupportFireDetection>
      <!--ro, opt, bool, whether the device supports fire detection, desc:related URI: /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities-->true
    </isSupportFireDetection>
    <isSupportThermometry>
      <!--ro, opt, bool, whether the device supports thermography, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities-->true
    </isSupportThermometry>
    <isSupportNVR>
      <!--ro, opt, bool, it is used for client to check-->true
    </isSupportNVR>
    <isSupportThermometrySchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of temperature measurement, desc:related URI: /ISAPI/Event/schedules/thermometry--
>true
    </isSupportThermometrySchedule>
    <isSupportTemperatureSchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of temperature difference, desc:related URI: /ISAPI/Event/schedules/temperature--
>true
    </isSupportTemperatureSchedule>
    <isSupportFireDetectionSchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of fire detection, desc:related URI: /ISAPI/Event/schedules/fireDetections-->true
    </isSupportFireDetectionSchedule>
  </ThermalCap>
  <isSupportCloud>
    <!--ro, opt, bool, whether the device supports cloud storage-->true
  </isSupportCloud>
  <isSupportGetmutexFuncErrMsg>
    <!--ro, opt, bool, whether the device supports the mutual exclusion information-->true
  </isSupportGetmutexFuncErrMsg>
  <isSupportTokenAuthenticate>
    <!--ro, opt, bool, whether the device supports token authentication-->true
  </isSupportTokenAuthenticate>
  <isSupportStreamDualVCA>
    <!--ro, opt, bool, whether the device supports overlaying intelligent private data on the stream, desc:related URI: in the
/ISAPI/Intelligent/channels/<ID>/intelliResource; corresponding node: <isOverlapIntelli>-->true
  </isSupportStreamDualVCA>
  <isSupportIntelligentAnalysisEngines>
    <!--ro, opt, bool, whether the device supports configuring the intelligent engine, desc:related URI: /ISAPI/Intelligent/analysisEngines/capabilities--
>true
  </isSupportIntelligentAnalysisEngines>
  <isSupportChannelEventCap>
    <!--ro, opt, bool, whether the device supports getting the event capability by channel, desc:related URI: /ISAPI/Event/channels/<ID>/capabilities-->true
  </isSupportChannelEventCap>
  <isSupportPanoramaCalibration>
    <!--ro, opt, bool, whether the device supports configuring the calibration of the panoramic camera-->true
  </isSupportPanoramaCalibration>
  <isSupportAutoMaintenance>
    <!--ro, opt, bool, whether the device supports automatic maintenance, desc:related URI: /ISAPI/System/autoMaintenance/capabilities?format=json-->true
  </isSupportAutoMaintenance>
  <isSupportRtspOverHTTPS>
    <!--ro, opt, bool, supports tspOverHTTPS-->true
  </isSupportRtspOverHTTPS>
  <isSupportPictureServer>
    <!--ro, opt, bool, whether the device supports configuring the picture storage server, desc:related URI: /ISAPI/System/PictureServer/capabilities?
format=json-->true
  </isSupportPictureServer>
  <isSupportISUPHttpPassthrough>
    <!--ro, opt, bool, whether the device supports HTTP transparent transmission via ISUP V5.0-->true
  </isSupportISUPHttpPassthrough>
  <isSupportInputProxyDeviceInfo>
    <!--ro, opt, bool, whether the device supports getting the information of the devices added to the proxy device channel, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities-->true
  </isSupportInputProxyDeviceInfo>
  <isSupportOnlineUpgradeTask>
    <!--ro, opt, bool, whether device online upgrading is supported-->true
  </isSupportOnlineUpgradeTask>
  <isSupportChannelsTimeSync>
    <!--ro, opt, bool, whether the device supports batch device time synchronization of the IPC/IOT channel, desc:related URL:
/ISAPI/System/time/channelsTimeSync/capabilities?format=json-->true
  </isSupportChannelsTimeSync>
  <isSupportSeachMutexFunction>
    <!--ro, opt, bool, whether the device supports searching for disabling the mutual exclusion function, desc:related URL: GET
/ISAPI/System/mutexFunction/capabilities?format=json; function description: whether the device supports searching for mutual exclusion function-->true
  </isSupportSeachMutexFunction>
  <isSupportPowerUpPicture>
    <!--ro, opt, bool, whether the device supports importing and exporting pictures displayed during startup,
desc:/ISAPI/System/powerUpPicture/capabilities?format=json-->true
  </isSupportPowerUpPicture>
  <isSupportSerialLogUpload>
    <!--ro, opt, bool, whether the device supports uploading serial port log, desc:true (support), false (not support)-->true
  </isSupportSerialLogUpload>
  <isSupportTimeTypeSTD>
```

dracku@gmail.com
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```
    <!--ro, opt, bool, whether the device supports the time type of ISO8601, desc:if "true" is returned, it indicates that all the devices containing
ISO8601 parameters support adding the timeType=STD, such as /ISAPI/ContentMgmt/search?timeType=STD; the time will be implemented according to the standard-->
>true
  </isSupportTimeTypeSTD>
  <isSupportResources>
    <!--ro, opt, bool, whether it supports resources, desc:GET /ISAPI/System/DynamicResource/Resources?format=json, Resources-->true
  </isSupportResources>
  <isSupportChildmanage>
    <!--ro, opt, bool, whether the device supports gateway sub-device topology management, desc:GET /ISAPI/IoTGateway/Childmanage/capabilities?format=json-->
>true
  </isSupportChildmanage>
  <isSupportSearchResources>
    <!--ro, opt, bool, whether the device supports searching the resource list, desc:POST /ISAPI/System/DynamicResource/SearchResources?format=json-->true
  </isSupportSearchResources>
  <isSupportSearchChildDevResourcesStatus>
    <!--ro, opt, bool, whether the device supports searching for sub-device resource status, desc:POST
/ISAPI/System/DynamicResource/SearchChildDevResourcesStatus?format=json-->true
  </isSupportSearchChildDevResourcesStatus>
  <isSupportDeleteResources>
    <!--ro, opt, bool, whether the device supports deleting resources, desc:PUT /ISAPI/System/DynamicResource/DeleteResources?format=json-->true
  </isSupportDeleteResources>
  <isSupportOTAPService>
    <!--ro, opt, bool, whether the device supports OTAP service configuration, desc:/ISAPI/System/Network/OTAPService/capabilities?format=json-->true
  </isSupportOTAPService>
  <isSupportAuthorizeMicAcquisitionParam>
    <!--ro, opt, bool, whether the device supports audio authorization information, desc:GET/PUT
/ISAPI/System/InformedConsentMgr/AuthorizeMicAcquisitionParam?format=json-->true
  </isSupportAuthorizeMicAcquisitionParam>
  <isSupportMutexFunctionClose>
    <!--ro, opt, bool, whether the device supports quickly disabling exclusive functions, desc:POST /ISAPI/System/mutexFunctionClose?format=json-->true
  </isSupportMutexFunctionClose>
  <isSupportAddChildDevResource>
    <!--ro, opt, bool, whether the device supports adding gateway resource information, desc:POST /ISAPI/System/DynamicResource/AddChildDevResource?
format=json-->true
  </isSupportAddChildDevResource>
  <isSupportModifyChildDevResource>
    <!--ro, opt, bool, whether the device supports editing gateway resource information, desc:POST /ISAPI/System/DynamicResource/ModifyChildDevResource?
format=json-->true
  </isSupportModifyChildDevResource>
  <isSupportUserLoginAbnormal>
    <!--ro, opt, bool, whether the device supports user login exception information, desc:GET/PUT /ISAPI/Security/users/UserLoginAbnormal?format=json-->true
  </isSupportUserLoginAbnormal>
  <isSupportRTCTimeTrigger>
    <!--ro, opt, bool, whether the device supports RTC time exception linkage, desc:GET/PUT /ISAPI/Event/RTCTimeTrigger?format=json-->true
  </isSupportRTCTimeTrigger>
  <isSupportVideoChannelEnhance>
    <!--ro, opt, bool, whether it supports enhancement configuration of adding video channel, desc:GET/PUT
/ISAPI/ContentMgmt/InputProxy/VideoChannelEnhance?format=json-->true
  </isSupportVideoChannelEnhance>
  <isSupportGetChannelEnhancePreOperationTips>
    <!--ro, opt, bool, whether it supports getting the prompt information before the enhancement configuration of adding the channel, desc:POST
/ISAPI/ContentMgmt/InputProxy/GetChannelEnhancePreOperationTips?format=json-->true
  </isSupportGetChannelEnhancePreOperationTips>
  <isSupportEventRuleAndTriggerParams>
    <!--ro, opt, bool, whether it supports configuring the rule linkage parameters of a specified event, desc:POST
/ISAPI/Event/EventBatchRulesMgr/EventRuleAndTriggerParams?format=json-->true
  </isSupportEventRuleAndTriggerParams>
  <isSupportSearchEventRuleAndTriggerParams>
    <!--ro, opt, bool, whether it supports searching for the rule linkage parameters of a specified event, desc:POST
/ISAPI/Event/EventBatchRulesMgr/SearchEventRuleAndTriggerParams?format=json-->true
  </isSupportSearchEventRuleAndTriggerParams>
  <isSupportDeviceRegionParams>
    <!--ro, opt, bool, whether it supports device area parameters, desc:GET/PUT /ISAPI/System/DeviceRegionParams?format=json-->true
  </isSupportDeviceRegionParams>
  <isSupportSearchRegionDefaultInfo>
    <!--ro, opt, bool, whether it supports searching for the default parameters for the region, desc:POST /ISAPI/System/SearchRegionDefaultInfo?format=json-
->true
  </isSupportSearchRegionDefaultInfo>
  <isSupportPicSearchByCoordinate>
    <!--ro, opt, bool, whether the device supports configuring AcuSearch parameters, desc:GET/PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json-->
>true
  </isSupportPicSearchByCoordinate>
  <AcuSearchCap>
    <!--ro, opt, object, AcuSearch capability-->
    <targetType opt="face,human,vehicle">
      <!--ro, opt, enum, supported target types of AcuSearch, subType:string, attr:opt{req, string}, desc:"face", "human", "vehicle"-->face
    </targetType>
    <dataType opt="url,binary,modelData">
      <!--ro, opt, enum, supported image transmission format types, subType:string, attr:opt{req, string}, desc:"url", "binary", "modelData"-->url
    </dataType>
  </AcuSearchCap>
  <isSupportSearchEventResources>
    <!--ro, opt, bool, whether it supports searching for event resource information, desc:POST /ISAPI/Event/EventBatchRulesMgr/SearchEventResources?
format=json-->true
  </isSupportSearchEventResources>
  <isSupportSearchRealStreamStatus>
    <!--ro, opt, bool, whether it supports searching for the real-time streaming status, desc:POST /ISAPI/Streaming/SearchRealStreamStatus?format=json-->
>true
  </isSupportSearchRealStreamStatus>
  <isSupportSearchStreamBandwidthInfo>
    <!--ro, opt, bool, whether it supports searching for stream bandwidth status, desc:POST /ISAPI/Streaming/SearchStreamBandwidthInfo?format=json-->true
  </isSupportSearchStreamBandwidthInfo>
  <isSupportMultipleIntelligentParams>
    <!--ro, opt, bool, whether it supports multi-intelligent mode parameters, desc:GET/PUT /ISAPI/System/VCAResource/MultipleIntelligentParams?format=json-->
```

dr3ckum@gmail.com
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```
>true
  </isSupportMultipleIntelligentParams>
</DeviceCap>
```

14.1.6.4 Export device configuration file

Request URL

GET /ISAPI/System/configurationData?secretkey=<secretkey>

Query Parameter

Parameter Name	Parameter Type	Description
secretkey	string	The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.

Request Message

None

Response Message

Binary Data

14.1.6.5 Get the languages supported by the device

Request URL

GET /ISAPI/System/DeviceLanguage

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--ro, req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian",
"Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish",
"Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian"-Slovenian,
"Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian", "Uzbek", "Kazak",
"Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
</DeviceLanguage>
```

14.1.6.6 Set device language parameters

Request URL

PUT /ISAPI/System/DeviceLanguage

Query Parameter

None

Request Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian", "Bulgarian",
"Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese",
"Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian" (Slovenian), "Croatian",
"Lithuanian", "Serbian", "BrazilianPortuguese" (Brazilian Portuguese), "Indonesian", "Ukrainian", "EURSpanish"-->SimChinese
  </language>
</DeviceLanguage>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.1.6.7 Get the capability of configuring the device language

Request URL

GET /ISAPI/System/DeviceLanguage/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device language configuration, attr:version{req, string, protocolVersion}-->
  <language
opt="SimChinese,TraChinese,English,Russian,Bulgarian,Hungarian,Greek,German,Italian,Czech,Slovakia,French,Polish,Dutch,Portuguese,Spanish,Romanian,Turkish,J
apanese,Danish,Swedish,Norwegian,Finnish,Korean,Thai,Estonia,Vietnamese,Hebrew,Latvian,Arabic,Sovenian,Croatian,Lithuanian,Serbian,BrazilianPortuguese,Indon
esian,Ukrainian,EURSpanish,Uzbek,Kazak,Kirghiz,Farsi,Azerbaidzhan,Burmese,Mongolian,Anglicism,Estonian">
    <!--ro, req, enum, language, subType:string, attr:opt{req, string}, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese),
"English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish",
"Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic",
"Sovenian"-Slovenian, "Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian",
"Uzbek", "Kazak", "Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
  <upgradeFirmWareEnabled>
    <!--ro, opt, bool, whether to enable upgrading the firmware-->true
  </upgradeFirmWareEnabled>
</DeviceLanguage>
```

14.1.6.8 Set restoring devices' default parameters

Request URL

PUT /ISAPI/System/factoryReset?mode=<mode>&childDevID=<devIndex>&loginPassword=<loginPassword>&module=<module>

Query Parameter

Parameter Name	Parameter Type	Description
mode	enum	--
devIndex	string	--
loginPassword	string	--
module	enum	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

drackun@gmail.com
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14.1.6.9 Get the capability of exporting files from the device

Request URL

GET /ISAPI/System/fileExport/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FileExportCap": {
    /*ro, opt, object, the capability of exporting files from the device*/
        "fileType": {
        /*ro, req, object, file type*/
            "@opt": ["serialLog", "coreDump", "debug", "selfDefine", "errorInfo", "playService", "communicationService", "attendanceService", "faceService",
"mainApp", "produceApp", "mainDsp", "daemon", "subAPP", "subDsp", "TVIDsp", "TVIApp", "SSDSerialLog"]
            /*ro, req, array, options, subType:string, desc:"serialLog"-serial port log, "coreDump"-core dump log, "debug"-debug log, "selfDefine"-custom
file type to be exported, "errorInfo"-error information*/
        },
        "selfDefineCmd": {
        /*ro, opt, object, custom file type to be exported, desc:custom file type to be exported,it is valid when fileType is set to "selfDefine"*/
            "@min":  0,
            /*ro, req, int, the minimum length of the custom file type to be exported*/
            "@max":  0
            /*ro, req, int, the maximum length of the custom file type to be exported*/
        },
        "devType": {
        /*ro, opt, object, device type*/
            "@opt": ["mainDev", "subDev", "D04"]
            /*ro, req, array, options, subType:string, desc:file type: "serialLog"-serial port log,"coreDump"-core dump log,"debug"-debug log,"selfDefine"-
custom file type to be exported,"errorInfo"-error information*/
        },
        "subSys": {
        /*ro, opt, object, partition No.*/
            "@min":  0,
            /*ro, req, int, the minimum length of the custom file type to be exported*/
            "@max":  0
            /*ro, req, int, the maximum length of the custom file type to be exported*/
        },
        "encryptionKey": {
        /*ro, opt, object, file encryption key*/
            "@min":  0,
            /*ro, req, int, the minimum length of the custom file type to be exported*/
            "@max":  0
            /*ro, req, int, the maximum length of the custom file type to be exported*/
        },
        "slotNo": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, the minimum length of the custom file type to be exported*/
            "@max":  0
            /*ro, req, int, the maximum length of the custom file type to be exported*/
        },
        "channelID": {
        /*ro, opt, object, channel No.*/
            "@min":  1,
            /*ro, req, int, the minimum length of the custom file type to be exported*/
            "@max":  1024
            /*ro, req, int, the maximum length of the custom file type to be exported*/
        },
        "fileList": {
        /*ro, opt, object*/
            "fileName": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  256
                /*ro, req, int*/
            },
            "time": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  25
                /*ro, req, int*/
            },
            "fileSize": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  100000
                /*ro, req, int*/
            }
        }
    }
}
```

drackum@gmail.com
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14.1.6.10 Get the capability of mutual exclusion information

Request URL

GET /ISAPI/System/mutexFunction/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "function": {
    /*ro, req, object, mutually exclusive functions, desc:"VMDHumanVehicle" (motion detection), "linedetection" (line crossing detection), "fielddetection"
(intrusion detection), "regionEntrance" (region entrance), "regionExiting" (region exiting), "faceCapture" (face capture), "faceRecognition" (face
recognition), "IPEnhance" (IP enhancement), "SVCEnhance" (SVC enhancement), "4KOutput" (4K output), "ZCR" (zero channel), "misinfoFilter" (alarm filtering),
"transcodeCfg" (stream transcoding), "AIOP_Video" (AI video task analysis), "mixedTargetDetection" (mixed targets detection), "RAID"*/
        "@opt": ["VMDHumanVehicle", "faceCapture", "faceRecognition", "IPEnhance", "SVCEnhance", "4KOutput", "linedetection", "fielddetection",
"regionEntrance", "regionExiting", "ZCR", "misinfoFilter", "transcodeCfg", "mixedTargetDetection", "RAID", "perimeter"]
        /*ro, opt, array, options, subType:string,
        desc:"misinfoFilter" (filter false alarms. Conduct secondary recognition of human targets in perimeter alarm, e.g., line crossing detection and
region entrance, so as to reduce false alarms caused by other factors such as swaying leaves, shadows, light, vehicles, animals, etc.)
            "transcodeCfg"  (transcoded stream compression configuration, including preview and playback transcoded stream. This transcoding configuration
is exclusive with 4K output), "perimeter" (perimeter protection, corresponding to image_recognition_human in /ISAPI/Intelligent/analysisEngines. When the
current engine mode is image_recognition_human, it indicates that perimeter protection algorithm mode is running)*/
    },
    "channelID": {
    /*ro, req, object, channel No.*/
        "@min":  1,
        /*ro, opt, int, the minimum value*/
        "@max":  16
        /*ro, opt, int, the maximum value*/
    }
}
```

14.1.6.11 Search for the mutual exclusion information of a specified function

Request URL

POST /ISAPI/System/mutexFunction?format=json

Query Parameter

None

Request Message

```
{
    "function":  "faceCapture",
    /*req, enum, function, subType:string, desc:the value of functions can be obtained via calling URL (/ISAPI/System/mutexFunction/capabilities?
format=json) "VMDHumanVehicle" (motion detection), "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance"
(region entrance), "regionExiting" (region exiting), "faceCapture" (face capture), "faceRecognition" (face recognition), "IPEnhance" (IP enhancement),
"SVCEnhance" (SVC enhancement), "4KOutput" (4K output), "ZCR" (zero channel), "misinfoFilter" (alarm filtering), "transcodeCfg" (stream transcoding),
"AIOP_Video" (AI video task analysis)*/
    "channelID":  1
    /*opt, int, channel No., range:[1,16]*/
}
```

Response Message

```
{
    "MutexFunctionList": [
    /*ro, req, array, mutually exclusive function list, subType:object*/
        {
            "mutexFunction":  "linedetection",
            /*ro, opt, enum, mutually exclusive functions, subType:string, desc:"VMDHumanVehicle" (motion detection), "linedetection" (line crossing
detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), "regionExiting" (region exiting), "faceCapture" (face capture),
"faceRecognition" (face recognition), "IPEnhance" (IP enhancement), "SVCEnhance" (SVC enhancement), "4KOutput" (4K output), "ZCR" (zero channel),
"misinfoFilter" (alarm filtering), "transcodeCfg" (stream transcoding), "AIOP_Video" (AI video task analysis)*/
            "channelID": [1, 2, 3]
            /*ro, opt, array, channel No. corresponding to the mutually exclusive function, subType:int*/
        }
    ]
}
```

14.1.6.12 Get the mutual exclusion information when device function exception occurred

Request URL

GET /ISAPI/System/mutexFunctionErrorMsg

Query Parameter

None

crackum@gmail.com
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Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MutexFunctionErrorMsg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, mutual exclusion information, attr:version{req, string, protocolVersion}-->
  <MutexFunctionList size="10">
    <!--ro, opt, array, mutual exclusion information list, subType:object, attr:size{req, int}, desc:get the functions that are mutually exclusive to the enabled functions-->
    <MutexFunction>
      <!--ro, opt, object, mutual exclusion message-->
      <id>
        <!--ro, req, int, No.-->1
      </id>
      <describe>
        <!--ro, req, enum, description, subType:string, desc:"SMD", "vehicleDetection" (vehicle detection), "counting" (people counting), "heatMap" (heat map statistics), "fieldDetection" (intrusion detection), "lineDetection" (line crossing detection), "sceneChangeDetection" (scene change detection), "VMDHumanVehicle" (motion detection), "regionEntrance" (region entrance), "regionExiting" (region exiting), "faceCapture" (face capture), "faceRecognition" (face recognition), "IPEnhance" (IP enhancement), "SVCEnhance" (SVC enhancement), "4KOutput" (4K output), "ZCR" (zero channel), "misinfoFilter" (alarm filtering), "transcodeCfg" (stream transcoding), "AIOP_Video" (AI video task analysis)-->SMD
      </describe>
      <channelNo>
        <!--ro, req, string, channel No. corresponding to the mutually exclusive function, desc:multiple channel numbers are supported and should be separated by commas-->1
      </channelNo>
    </MutexFunction>
  </MutexFunctionList>
</MutexFunctionErrorMsg>
```

14.1.6.13 Get the network service capability

Request URL

GET /ISAPI/System/Network/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<NetworkCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, get the network service capability, attr:version{opt, string, protocolVersion}-->
  <isSupportWireless>
    <!--ro, req, bool, whether the device supports wireless network-->true
  </isSupportWireless>
  <isSupportWAN>
    <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
  </isSupportWAN>
  <isSupportPPPoE>
    <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
  </isSupportPPPoE>
  <isSupportBond>
    <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
  </isSupportBond>
  <isSupport802_1x>
    <!--ro, req, bool, whether the device supports 802.1x protocol-->true
  </isSupport802_1x>
  <isSupportNtp>
    <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
  </isSupportNtp>
  <isSupportFtp>
    <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
  </isSupportFtp>
  <isSupportUpnp>
    <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play) protocol-->true
  </isSupportUpnp>
  <isSupportPNP>
    <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
  </isSupportPNP>
  <isSupportDdns>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
  </isSupportDdns>
  <isSupportHttps>
    <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
  </isSupportHttps>
  <SnmpCap>
    <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
```

drackum@gmail.com
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```
    <isSupport>
      <!--ro, req, bool, whether the device supports SNMP-->true
    </isSupport>
  </SnmpCap>
  <isSupportExtNetCfg>
    <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
  </isSupportExtNetCfg>
  <isSupportIPFilter>
    <!--ro, opt, bool, whether the device supports IP filtering-->true
  </isSupportIPFilter>
  <isSupportNetPreviewStrategy>
    <!--ro, opt, bool, whether the device supports network live view strategy-->true
  </isSupportNetPreviewStrategy>
  <isSupportEZVIZ>
    <!--ro, opt, bool, whether the device supports EZ protocol-->true
  </isSupportEZVIZ>
  <isSupportWPS>
    <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
  </isSupportWPS>
  <isWirelessMutexWithWirelessServer>
    <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
  </isWirelessMutexWithWirelessServer>
  <isSupportMACFilter>
    <!--ro, opt, bool, whether the device supports physical address filtering-->true
  </isSupportMACFilter>
  <GB28181Cap>
    <!--ro, opt, object, N/A-->
    <isSupportGB28181Service>
      <!--ro, opt, bool, N/A-->true
    </isSupportGB28181Service>
  </GB28181Cap>
  <VerificationCodeModification>
    <!--ro, opt, object, verification code editing-->
    <verificationCodeType opt="normal,empty">
      <!--ro, opt, string, verification code type, attr:opt{req, string}-->test
    </verificationCodeType>
    <isSupportDeclarationURL>
      <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
    </isSupportDeclarationURL>
    <isSupportPrivacyPolicyURL>
      <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
    </isSupportPrivacyPolicyURL>
    <verificationCodeModify opt="true,false">
      <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{req, string}-->true
    </verificationCodeModify>
    <Hyperlinks>
      <!--ro, opt, object, hyperlink-->
      <declarationURL>
        <!--ro, opt, string, hyperlink to the Terms of Service-->test
      </declarationURL>
      <privacyPolicyURL>
        <!--ro, opt, string, hyperlink to the Privacy Policy-->test
      </privacyPolicyURL>
    </Hyperlinks>
    <isSupportVerificationCodeCheck>
      <!--ro, opt, bool, whether the device supports verifying the verification code-->true
    </isSupportVerificationCodeCheck>
  </VerificationCodeModification>
  <isSupportIntegrate>
    <!--ro, opt, bool, whether the device supports access protocol configuration-->true
  </isSupportIntegrate>
  <isSupportResourceStatistics>
    <!--ro, opt, bool, whether the device supports getting the network resource information-->true
  </isSupportResourceStatistics>
  <isSupportBandwidthLimit>
    <!--ro, opt, bool, whether the device supports bandwidth limit-->true
  </isSupportBandwidthLimit>
  <isSupportGetLinkSocketIP>
    <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
  </isSupportGetLinkSocketIP>
  <isSupportEZVIZTiming>
    <!--ro, opt, bool, whether the device supports EZ platform synchronization (enabledTiming of message returned by/ISAPI/System/Network/EZVIZ)-->true
  </isSupportEZVIZTiming>
  <isSupportDSTPServerCfg>
    <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET/PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
  </isSupportDSTPServerCfg>
  <isSupportDSTPProtocolsCap>
    <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
  </isSupportDSTPProtocolsCap>
  <isSupportIPAdaptiveParams>
    <!--ro, opt, bool, whether the device supports self-adaptive IP address parameters, desc:GET/PUT /ISAPI/System/Network/IPAdaptiveParams?format=json--
>true
  </isSupportIPAdaptiveParams>
  <isSupportUSBNetworkCard>
    <!--ro, opt, bool, whether the device supports USB NIC-->true
  </isSupportUSBNetworkCard>
  <isSupportSearchOutlookAuthResult>
    <!--ro, opt, bool-->true
  </isSupportSearchOutlookAuthResult>
  <isSupportRequestOutlookAuth>
    <!--ro, opt, bool, whether the device supports applying authentication request of Outlook email, desc:POST
/ISAPI/System/Network/Email/RequestOutlookAuth?format=json-->true
```

drackum@gmail.com
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```
    </isSupportRequestOutlookAuth>
    <isSupportOutlookEmailList>
      <!--ro, opt, bool, whether the device supports Outlook email information, desc:GET or PUT /ISAPI/System/Network/Email/OutlookEmailList?format=json--
>true
    </isSupportOutlookEmailList>
  </NetworkCap>
```

14.1.6.14 Set SSH parameters

Request URL

PUT /ISAPI/System/Network/ssh?readerID=<readerID>&type=<type>&SOCChipID=<SOCChipID>&controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	--
type	enum	--
SOCChipID	string	--
controllerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SSH xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, opt, object, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--wo, req, bool, whether to enable the function-->true
  </enabled>
  <port>
    <!--wo, opt, int-->22
  </port>
</SSH>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, read-only,status description: OK,Device Busy,Device Error,Invalid Operation,Invalid XML Format,Invalid XML Content,Reboot,Additional
Error, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot"
(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.6.15 Get the capability of importing and exporting parameters of pictures displayed during startup

Request URL

GET /ISAPI/System/powerUpPicture/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
{
    "pictureFormat": {
    /*ro, opt, object, picture format*/
        "@opt": ["GIF", "BMP", "JPG", "PNG"]
        /*ro, req, array, options, subType:string*/
    },
    "pictureSize": {
    /*ro, opt, object, picture size, desc:unit: KB*/
        "@min":  1,
        /*ro, req, int, options*/
        "@max":  2048
        /*ro, req, int, options*/
    },
    "pictureURL": {
    /*ro, opt, object, picture URL*/
        "@min":  1,
        /*ro, req, int, options*/
        "@max":  256
        /*ro, req, int, options*/
    },
    "pictureResolution": {
    /*ro, opt, object, picture resolution*/
        "@opt": ["600*1024", "800*1280", "800*480", "1024*768"]
        /*ro, req, array, options, subType:string*/
    },
    "ImportCap": {
    /*ro, opt, object, importing capability*/
        "filePathType": {
        /*ro, opt, object, file path type*/
            "@opt": ["binary", "URL"]
            /*ro, req, array, options, subType:string*/
        },
        "applyType": {
        /*ro, opt, object, application range of the imported pictures*/
            "@opt": ["all", "powerUpPicture", "noVideoPicture"],
            /*ro, req, array, options, subType:string*/
            "@def":  "powerUpPicture"
            /*ro, req, string, the default value*/
        }
    },
    "ExportCap": {
    /*ro, opt, object, exporting capability*/
        "filePathType": {
        /*ro, opt, object, file path type*/
            "@opt": ["localPath", "URL"]
            /*ro, req, array, options, subType:string*/
        }
    }
}
```

drackum@gmail.com
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14.1.6.16 Import startup pictures

Request URL

POST /ISAPI/System/powerUpPicture?format=json

Query Parameter

None

Request Message

```
{
    "filePathType":  "binary",
    /*req, enum, file path type, subType:string, desc:“binary”, "URL", when the value of filePathType is "URL", the message is in application/json format.
When the value of filePathType is "binary", the the message is in multipart/form-data*/
    "pictureURL":  "test",
    /*opt, string, picture URL, range:[1,256], dep:and,{$.filePathType,eq,URL}*/
    "applyType":  "powerUpPicture"
    /*opt, enum, imported picture application range, subType:string, desc:"all" (all the pictures), "powerUpPicture" (pictures displayed during startup),
"noVideoPicture" (no captured pictures of the video). If this node does not exist, it indicates that the default imported picture is used for the startup*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
powerUpPicture	[Message content]	application/json	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:this field is required when the status code is not 1 (success)*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status description is not "ok" (success)*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status description is not "ok" (success)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not "1"*/
}
```

14.1.6.17 Delete startup picture

Request URL

DELETE /ISAPI/System/powerUpPicture?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:this field is required when the status code is not 1 (success)*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status description is not "ok" (success)*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status description is not "ok" (success)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not "1"*/
}
```

14.1.6.18 Get capability of startup picture parameters

Request URL

GET /ISAPI/System/powerUpPictureCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
{
    "enabled": {
    /*ro, opt, object, whether to enable startup picture*/
        "@opt": [true, false]
        /*ro, req, array, range, subType:bool*/
    }
}
```

14.1.6.19 Get startup picture parameters

Request URL

GET /ISAPI/System/powerUpPictureCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "enabled":  true,
    /*ro, opt, bool, whether to enable startup picture*/
    "importPowerUpPictureStatus":  "on"
    /*ro, opt, enum, string, desc:"on" (using), "off" (uploaded but not be used), "noUpload" (not uploaded)*/
}
```

14.1.6.20 Configure startup picture parameters

Request URL

PUT /ISAPI/System/powerUpPictureCfg?format=json

Query Parameter

None

Request Message

```
{
    "enabled":  true
    /*opt, bool, whether to enable startup picture*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:this field is required when the status code is not 1 (success)*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status description is not "ok"(success)*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status description is not "ok"(success)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not "1"*/
}
```

14.1.6.21 Reboot device

Request URL

PUT /ISAPI/System/reboot?childDevID=<devIndex>&module=<module>&loginPassword=<loginPassword>

Query Parameter

drackun@gmail.com
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Parameter Name	Parameter Type	Description
devIndex	string	Reboot a sub-device via specifying sub-device ID. The ID of the sub-device connected to the current device can be searched via /ISAPI/IoTGateway/Childmanage/SearchChild?format=json.
module	enum	The "module" specifies a module to restart, without requiring a device reboot, corresponding to the capability (/ISAPI/System/capabilities/deviceReboot->module). "algorithmProgram (traffic terminal radar-assisted algorithm program, supports independent algorithm program startup), "radar" (restart radar module of the radar-assisted camera).
loginPassword	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

dreckum@gmail.com
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14.1.7 时间管理

14.1.7.1 Get device time synchronization management parameters

Request URL

GET /ISAPI/System/time

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--ro, req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--ro, opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:30
  </localTime>
  <timeZone>
    <!--ro, opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:daylight saving time configuration in time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the device time returned currently is in DST (Daylight Saving Time) system, desc:whether the device time returned currently is in DST (Daylight Saving Time) system-->true
  </isSummerTime>
  <platformType>
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform-->EZVIZ
  </platformType>
  <platformNo>
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--ro, opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--ro, opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--ro, opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

drackum@gmail.com
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14.1.7.2 Set device time synchronization management parameters

Request URL

PUT /ISAPI/System/time

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}, desc:the time difference between local time and coordinated universal time (UTC) is configured via the timeZone field-->2019-02-28T10:50:44
  </localTime>
  <timeZone>
    <!--opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the time returned by the current device is that in the DST (daylight saving time), desc:whether the time returned by the current device is that in the DST (daylight saving time)-->true
  </isSummerTime>
  <platformType>
    <!--opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform, related URI: /ISAPI/System/Network/EZVIZ-->EZVIZ
  </platformType>
  <platformNo>
    <!--opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <FailedNodeInfoList>
    <!--ro, opt, object, information list of failed nodes, desc:for the manual time synchronization of central analysis cluster, this field is returned if time synchronization failed-->
    <FailedNodeInfo>
      <!--ro, opt, object, information of failed nodes-->
      <nodeID>
        <!--ro, req, string, node ID, range:[0,64]-->test
      </nodeID>
      <nodeIP>
        <!--ro, req, string, node IP, range:[0,20]-->test
      </nodeIP>
      <reason>
        <!--ro, opt, string, reason why the node failed to synchronize time, range:[0,128]-->test
      </reason>
    </FailedNodeInfo>
  </FailedNodeInfoList>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

drackum@gmail.com DETEKTOR

14.1.7.3 Get the capability of device time synchronization management

Request URL

GET /ISAPI/System/time/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management capability set, attr:version{opt, string, protocolVersion}-->
  <timeMode opt="NTP,manual,satellite,SDK,28181,ONVIF,ALL(任何支持的校时方式都允许校时),NONE(不允校时或无校时源),platform,PTP">
    <!--ro, req, enum, time synchronization mode, subType:string, attr:opt{opt, string}, desc:“NTP” (NTP time synchronization), “manual” (manual time
synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or
no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime min="0" max="256">
    <!--ro, opt, string, local time, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTime>
  <timeZone min="0" max="256">
    <!--ro, opt, string, time zone, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </timeZone>
  <satelliteInterval min="0" max="3600">
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, attr:min{opt, string},max{opt, string}, desc:unit: minute-->60
  </satelliteInterval>
  <timeType opt="local,UTC">
    <!--ro, opt, enum, time type, subType:string, attr:opt{opt, string}, desc:“local” (local time), “UTC” (UTC time)-->local
  </timeType>
  <platformType opt="EZVIZ">
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, attr:opt{opt, string}, desc:platform type-->EZVIZ
  </platformType>
  <platformNo min="1" max="2">
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, attr:min{req, int},max{req, int}, desc:it is the only ID, which is
configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <isSupportHistoryTime>
    <!--ro, opt, bool, supported capability of the historical time synchronization list, desc:related URI: /ISAPI/System/time/historyInfo?format=json-->true
  </isSupportHistoryTime>
  <isSupportTimeFilter>
    <!--ro, opt, bool, supported capability of filtering time synchronization, desc:related URI: /ISAPI/System/time/filter/capabilities?format=json-->true
  </isSupportTimeFilter>
  <displayFormat opt="MM/dd/yyyy hh:mm,mm,dd-MM-yyyy hh,MM-dd-yyyy hh:mm,yyyy-MM-dd hh:mm,MM/dd/yyyy hh:mm:ss,dd-MM-yyyy hh:mm:ss,MM-dd-yyyy hh:mm:ss,yyyy-
MM-dd hh:mm:ss,yyyy-MM-dd,dd-MM-yyyy,MM-dd-yyyy">
    <!--ro, opt, enum, time display format, subType:string, attr:opt{req, string}, desc:if this node is returned, it indicates that the device supports
configuring time display format, related URI: /ISAPI/System/time/timeType?format=json-->MM/dd/yyyy hh:mm
  </displayFormat>
  <isSupportSyncDeviceNTPInfoToCamera>
    <!--ro, opt, bool, the capability of synchronizing device’s NTP service information with the camera, desc:related URI:
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json-->true
  </isSupportSyncDeviceNTPInfoToCamera>
  <isSupportNTPService>
    <!--ro, opt, bool-->true
  </isSupportNTPService>
  <isSupportTimeZoneListInfo>
    <!--ro, opt, bool-->true
  </isSupportTimeZoneListInfo>
  <isSupportAutoEnterSummerTime>
    <!--ro, opt, bool-->true
  </isSupportAutoEnterSummerTime>
  <ethernetPort min="0" max="255">
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}, attr:min{req, int},max{req, int}-->1
  </ethernetPort>
  <IANA
opt="Africa/Abidjan,Africa/Accra,Africa/Addis_Ababa,Africa/Algiers,Africa/Blantyre,Africa/Brazzaville,Africa/Cairo,Africa/Casablanca,Africa/Conakry,Africa/D
akar,Africa/Dar_es_Salaam,Africa/Djibouti,Africa/Freetown,Africa/Gaborone,Africa/Harare,Africa/Johannesburg,Africa/Kampala,Africa/Khartoum,Africa/Kigali,Afr
ica/Kinshasa,Africa/Lagos,Africa/Maseru,Africa/Mogadishu,Africa/Nairobi,Africa/Sao_Tome,Africa/Timbuktu,Africa/Tripoli,Africa/Tunis,America/Anchorage,Americ
a/Aruba,America/Asuncion,America/Barbados,America/Belize,America/Bogota,America/Buenos_Aires,America/Cancun,America/Caracas,America/Cayman,America/Chicago,A
merica/Curacao,America/Dawson_Creek,America/Denver,America/Detroit,America/Dominica,America/Edmonton,America/El_Salvador,America/Fortaleza,America/Grand_Tur
k,America/Grenada,America/Guatemala,America/Guyana,America/Halifax,America/Havana,America/Indiana/Indianapolis,America/Indiana/Knox,America/Indiana/Marengo,
America/Indiana/Petersburg,America/Indiana/Tell_City,America/Indiana/Vevay,America/Indiana/Vincennes,America/Indiana/Winamac,America/Jamaica,America/La_Paz,
America/Lima,America/Los_Angeles,America/Louisville,America/Managua,America/Martinique,America/Mendoza,America/Metlakatla,America/Mexico_City,America/Monter
rey,America/Montevideo,America/Montreal,America/Nassau,America/New_York,America/North_Dakota/Beulah,America/North_Dakota/Center,America/North_Dakota/New_Sal
em,America/Panama,America/Phoenix,America/Port_of_Spain,America/Port-au-
Prince,America/Puerto_Rico,America/Santo_Domingo,America/Sao_Paulo,America/St_Johns,America/St_Kitts,America/St_Lucia,America/St_Thomas,America/Tijuana,Amer
ica/Toronto,America/Vancouver,America/Winnipeg,Antarctica/South_Pole,Arctic/Longyearbyen,Asia/Almaty,Asia/Amman,Asia/Anadyr,Asia/Aqtau,Asia/Baghdad,Asia/Bah
rain,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia/Dubai,Asia/Gaza,Asia/Hebron,Asia/Ho_Chi_Minh,Asia/Hong_Kong,Asia/Jakarta
,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kuala_Lumpur,Asia/Kuwait,Asia/Macau(China),Asia/Manila,Asia/Muscat,Asia/Phnom_Penh,Asia/Pyongyan
g,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tel_Aviv,Asia/Tokyo,Asia/Ulaanbaatar,Atlantic/Bermuda,Atlant
ic/Canary,Atlantic/Cape_Verde,Atlantic/Reykjavik,Atlantic/Stanley,Australia/Adelaide,Australia/Brisbane,Australia/Canberra,Australia/Darwin,Australia/Melbou
rne,Australia/NSW,Australia/Perth,Australia/Queensland,Australia/Sydney,Australia/Victoria,Canada/Newfoundland,Canada/Saskatchewan,Chile/EasterIsland,Europe
/Amsterdam,Europe/Andorra,Europe/Athens,Europe/Belfast,Europe/Belgrade,Europe/Berlin,Europe/Bratislava,Europe/Brussels,Europe/Bucharest,Europe/Budapest,Euro
pe/Chisinau,Europe/Copenhagen,Europe/Dublin,Europe/Gibraltar,Europe/Helsinki,Europe/Isle_of_Man,Europe/Istanbul,Europe/Kiev,Europe/Lisbon,Europe/London,Euro
```

drackun@gmail.com DETEKTOR

```
pe/Luxembourg,Europe/Madrid,Europe/Malta,Europe/Minsk,Europe/Monaco,Europe/Moscow,Europe/Oslo,Europe/Paris,Europe/Prague,Europe/Riga,Europe/Rome,Europe/San_
Marino,Europe/Sarajevo,Europe/Simferopol,Europe/Skopje,Europe/Sofia,Europe/Stockholm,Europe/Tallinn,Europe/Vatican,Europe/Vienna,Europe/Vilnius,Europe/Warsa
w,Europe/Zagreb,Europe/Zurich,Indian/Antananarivo,Indian/Maldives,Indian/Mauritius,Pacific/Auckland,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific/Kirit
imati,Pacific/Noumea,America/Antigua,MST,Asia/Kolkata,America/Santiago,Africa/Monrovia,Asia/Colombo,America/Chihuahua,America/Tegucigalpa,America/Paramaribo
,Europe/Ljubljana,Asia/Ashgabat,Africa/Asmara,Asia/Brunei,Africa/Bangui,Africa/Banjul,Asia/Bishkek,Africa/Bissau,Africa/Bujumbura,Asia/Dili,Asia/Dushanbe,Pa
cific/Funafuti,Pacific/Guadalcanal,Africa/Juba,America/St_Vincent,Africa/Libreville,Africa/Lome,Africa/Luanda,Africa/Lusaka,Pacific/Majuro,Africa/Malabo,Afr
ica/Maputo,Africa/Mbabane,Indian/Comoro,Africa/Ndjamena,Pacific/Palau,Africa/Niamey,Europe/Nicosia,Africa/Nouakchott,Pacific/Tongatapu,Africa/Ouagadougou,Am
erica/Lower_Princes,Europe/Podgorica,Pacific/Efate,Africa/Porto-
Novo,Asia/Tashkent,Asia/Tbilisi,Asia/Thimphu,Europe/Tirane,Europe/Vaduz,Asia/Vientiane,Africa/Windhoek,Africa/Douala,Pacific/Nauru,Asia/Yerevan,Etc/GMT+12,E
tc/GMT+11,America/Adak,Pacific/Marquesas,Etc/GMT+9,Etc/GMT+8,America/Whitehorse,Pacific/Easter,America/Regina,America/Indianapolis,America/Cuiaba,America/Ar
aguaina,America/Cayenne,America/Godthab,America/Punta_Arenas,America/Miquelon,America/Bahia,Etc/GMT+2,Atlantic/Azores,Etc/UTC,Europe/Kaliningrad,Europe/Astr
akhan,Europe/Samara,Europe/Saratov,Europe/Volgograd,Asia/Yekaterinburg,Asia/Qyzylorda,Asia/Katmandu,Asia/Omsk,Asia/Barnaul,Asia/Hovd,Asia/Krasnoyarsk,Asia/N
ovosibirsk,Asia/Tomsk,Asia/Irkutsk,Australia/Eucla,Asia/Chita,Asia/Yakutsk,Pacific/Port_Moresby,Australia/Hobart,Asia/Vladivostok,Australia/Lord_Howe,Pacifi
c/Bougainville,Asia/Srednekolymsk,Asia/Magadan,Pacific/Norfolk,Asia/Sakhalin,Asia/Kamchatka,Etc/GMT-12,Pacific/Chatham,Etc/GMT-13,Pacific/Apia">
    <!--ro, opt, enum, IANA synchronization, subType:string, attr:opt{opt, string}-->Africa/Abidjan
  </IANA>
  <isSupportSearchDeviceIANAList>
    <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
  </isSupportSearchDeviceIANAList>
  <windowsZone opt="Dateline Standard Time,UTC-11,Aleutian Standard Time,Hawaiian Standard Time,Marquesas Standard Time,Alaskan Standard Time,UTC-09,Pacific
Standard Time (Mexico),UTC-08,Pacific Standard Time,US Mountain Standard Time,Mountain Standard Time (Mexico),Mountain Standard Time,Yukon Standard
Time,Central America Standard Time,Central Standard Time,Easter Island Standard Time,Central Standard Time (Mexico),Canada Central Standard Time,SA Pacific
Standard Time,Eastern Standard Time (Mexico),Eastern Standard Time,Haiti Standard Time,Cuba Standard Time,US Eastern Standard Time,Turks And Caicos Standard
Time,Paraguay Standard Time,Atlantic Standard Time,Venezuela Standard Time,Central Brazilian Standard Time,SA Western Standard Time,Pacific SA Standard
Time,Newfoundland Standard Time,Tocantins Standard Time,E. South America Standard Time,SA Eastern Standard Time,Argentina Standard Time,Greenland Standard
Time,Montevideo Standard Time,Magallanes Standard Time,Saint Pierre Standard Time,Bahia Standard Time,UTC-02,Azores Standard Time,Cape Verde Standard
Time,UTC,GMT Standard Time,Greenwich Standard Time,Sao Tome Standard Time,Morocco Standard Time,W. Europe Standard Time,Central Europe Standard Time,Romance
Standard Time,Central European Standard Time,W. Central Africa Standard Time,Jordan Standard Time,GTB Standard Time,Middle East Standard Time,Egypt Standard
Time,E. Europe Standard Time,Syria Standard Time,West Bank Standard Time,South Africa Standard Time,FLE Standard Time,Israel Standard Time,South Sudan
Standard Time,Kaliningrad Standard Time,Sudan Standard Time,Libya Standard Time,Namibia Standard Time,Arabic Standard Time,Turkey Standard Time,Arab
Standard Time,Belarus Standard Time,Russian Standard Time,E. Africa Standard Time,Iran Standard Time,Arabian Standard Time,Astrakhan Standard
Time,Azerbaijan Standard Time,Russia Time Zone 3,Mauritius Standard Time,Saratov Standard Time,Georgian Standard Time,Volgograd Standard Time,Caucasus
Standard Time,Afghanistan Standard Time,West Asia Standard Time,Ekaterinburg Standard Time,Pakistan Standard Time,Qyzylorda Standard Time,India Standard
Time,Sri Lanka Standard Time,Nepal Standard Time,Central Asia Standard Time,Bangladesh Standard Time,Omsk Standard Time,Myanmar Standard Time,SE Asia
Standard Time,Altai Standard Time,W. Mongolia Standard Time,North Asia Standard Time,N. Central Asia Standard Time,Tomsk Standard Time,China Standard
Time,North Asia East Standard Time,Singapore Standard Time,W. Australia Standard Time,Taipei Standard Time,Ulaanbaatar Standard Time,Aus Central W. Standard
Time,Transbaikal Standard Time,Tokyo Standard Time,North Korea Standard Time,Korea Standard Time,Yakutsk Standard Time,Cen. Australia Standard Time,AUS
Central Standard Time,E. Australia Standard Time,AUS Eastern Standard Time,West Pacific Standard Time,Tasmania Standard Time,Vladivostok Standard Time,Lord
Howe Standard Time,Bougainville Standard Time,Russia Time Zone 10,Magadan Standard Time,Norfolk Standard Time,Sakhalin Standard Time,Central Pacific
Standard Time,Russia Time Zone 11,New Zealand Standard Time,UTC+12,Fiji Standard Time,Chatham Islands Standard Time,UTC+13,Tonga Standard Time,Samoa
Standard Time,Line Islands Standard Time">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}-->Dateline Standard Time
  </windowsZone>
  <isSupportTimeStateParam>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeStateParam?format=json-->true
  </isSupportTimeStateParam>
  <localTimeDST min="0" max="256">
    <!--ro, opt, string, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTimeDST>
  <isSupportStandard>
    <!--ro, opt, bool-->true
  </isSupportStandard>
  <isSupportPTPParams>
    <!--ro, opt, bool, desc:GET/PUT /ISAPI/System/time/PTPParams?format=json-->true
  </isSupportPTPParams>
  <isSupportTimeZoneLock>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeZoneLock?format=json-->true
  </isSupportTimeZoneLock>
</Time>
```


dracrum@gmail.com DETEKTOR

14.1.7.4 Set parameters of all NTP servers

Request URL

PUT /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--opt, object, NTP server information-->
    <id>
      <!--req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

dracium@gmail.com
DETEKTOR

14.1.7.5 Get the parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{req, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--ro, req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--ro, opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--ro, opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

14.1.7.6 Set the parameters of a NTP server

Request URL

PUT /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, NTP server domain name地址去掉, range:[1,64]-->12345
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,32], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--opt, bool-->false
  </enabled>
</NTPServer>
```

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

14.1.7.7 Get the parameters of a NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	NTP server No.

Request Message

None

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--ro, req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, NTP server domain name, range:[1,64], dep:and,{$.NTPServer.addressingFormatType,eq,hostname}-->xxx12345
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,32], dep:and,{$.NTPServer.addressingFormatType,eq,ipAddress}-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--ro, opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--ro, opt, bool-->false
  </enabled>
  <devicePortNoValid>
    <!--ro, opt, enum, subType:string-->yes
  </devicePortNoValid>
  <portType>
    <!--ro, opt, enum, subType:string-->auto
  </portType>
  <customPortNo>
    <!--ro, opt, int, range:[1,65535], step:1, dep:and,{$.NTPServer.portType,eq,custom}-->1
  </customPortNo>
  <hostNameExampleList>
    <!--ro, opt, array, subType:object-->
    <hostNameExample>
      <!--ro, opt, object-->
      <hostName>
        <!--ro, opt, string-->xxx12345
      </hostName>
    </hostNameExample>
  </hostNameExampleList>
</NTPServer>
```

drackum@gmail.com
DETEKTOR

14.1.7.8 Get the configuration capability of a specific NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id min="1" max="1">
    <!--ro, req, string, ID, range:[0,1], attr:min{req, int},max{req, int}-->1
  </id>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, IP address type of NTP server, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)--
>hostname
  </addressingFormatType>
  <hostName min="0" max="68">
    <!--ro, opt, string, NTP server domain name, range:[0,68], attr:min{req, int},max{req, int}-->time.windows.com
  </hostName>
  <ipAddress min="0" max="32">
    <!--ro, opt, string, IPv4 address, range:[0,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address min="0" max="128">
    <!--ro, opt, string, IPv6 address, range:[0,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo min="0" max="65535">
    <!--ro, opt, int, port No., range:[0,65535], attr:min{req, int},max{req, int}, desc:port No.-->123
  </portNo>
  <synchronizeInterval min="0" max="10800">
    <!--ro, opt, int, time synchronization interval, range:[0,10800], unit:min, attr:min{req, int},max{req, int}, desc:NTP time synchronization interval,
unit: minute-->1440
  </synchronizeInterval>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <portType opt="auto,custom" def="auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->auto
  </portType>
  <customPortNo min="1" max="65535">
    <!--ro, opt, int, range:[0,65535], dep:and,{$.NTPServer.portType,eq,custom}, attr:min{req, int},max{req, int}-->123
  </customPortNo>
</NTPServer>
```

14.1.7.9 Get the configuration capability of parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers/capabilities

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, NTP server address type, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName min="1" max="64">
      <!--ro, opt, string, NTP server domain name, range:[1,64], attr:min{req, int},max{req, int}-->12345
    </hostName>
    <ipAddress min="1" max="32">
      <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
    </ipAddress>
    <ipv6Address min="1" max="128">
      <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo min="1" max="65535">
      <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}, desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval min="1" max="10800">
      <!--ro, opt, int, time synchronization interval, unit:min, attr:min{req, int},max{req, int}-->1440
    </synchronizeInterval>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
  </NTPServer>
</NTPServerList>
```

14.1.7.10 Set the device time type

Request URL

PUT /ISAPI/System/time/timeType?format=json

Query Parameter

None

Request Message

```
{
    "TimeType": {
    /*req, object, time type information, desc:time type information*/
        "type":  "UTC",
        /*req, enum, time type, subType:string, desc:"UTC, local "*/
        "displayFormat":  "MM/dd/yyyy hh:mm"
        /*opt, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

14.1.7.11 Get the device time type

Request URL

GET /ISAPI/System/time/timeType?format=json

Query Parameter

None

Request Message

drackum@gmail.com
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None

Response Message

```
{
    "TimeType": {
    /*ro, req, object*/
        "type":  "UTC",
        /*ro, req, enum, value: local,UTC, subType:string, desc:"UTC,local"*/
        "displayFormat":  "MM/dd/yyyy hh:mm"
        /*ro, opt, enum, subType:string*/
    }
}
```

14.1.8 设备本地升级

14.1.8.1 Get the device upgrade progress

Request URL

GET /ISAPI/System/upgradeStatus?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL":  "/ISAPI/Streaming/channels/1",
    /*ro, opt, string, request URL*/
    "statusCode":  "test",
    /*ro, req, string, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, This field is required when the value of statusCode is not 1, and it corresponds to subStatusCode.*/
    "errorMsg":  "ok",
    /*ro, opt, string, This field is required when the value of statusCode is not 1. Detailed error description of a certain parameter can be provided*/
    "upgrading":  "TRUE",
    /*ro, opt, string, whether the device is upgrading: “TRUE" (upgrading), "FALSE" (not in upgrading)*/
    "percent":  22,
    /*ro, opt, int, upgrade progress (% complete)*/
    "idList": [
    /*ro, opt, array, ID list, subType:object*/
        {
            "id":  "test",
            /*ro, req, string, analysis unit ID*/
            "percent":  22,
            /*ro, opt, int, upgrade progress (% complete)*/
            "status":  "test"
            /*ro, opt, string, "backingUp" (backing up upgrade)*/
        }
    ]
}
```

dr4ckum@gmail.com
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14.1.8.2 Get the device upgrading status and progress

Request URL

GET /ISAPI/System/upgradeStatus?type=<type>&childDevID=<childDevID>&id=<id>

Query Parameter

Parameter Name	Parameter Type	Description
type	string	"usbmic" (array microphone module)
childDevID	string	--
id	string	The id indicates the peripheral No. to be upgraded.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<upgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, upgrade status and result, attr:version{req, string, protocolVersion}-->
  <upgrading>
    <!--ro, req, bool, upgrade status-->true
  </upgrading>
  <percent>
    <!--ro, req, int, upgrade progress (% complete), range:[0,100]-->1
  </percent>
  <reboot>
    <!--ro, opt, bool, dep:and,{$.upgradeStatus.percent,eq,100}-->false
  </reboot>
  <upgradeStage>
    <!--ro, opt, enum, subType:string-->software
  </upgradeStage>
</upgradeStatus>
```

14.2 视频通用

14.2.1 图像调节

14.2.1.1 Get the image processing parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

crackum@gmail.com
DEFEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <videoInputID>
    <!--ro, req, int-->1
  </videoInputID>
  <NoiseReduce>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->close
    </mode>
    <GeneralMode>
      <!--ro, opt, object-->
      <generalLevel>
        <!--ro, req, int, range:[0,100]-->1
      </generalLevel>
    </GeneralMode>
  </NoiseReduce>
  <Color>
    <!--ro, opt, object-->
    <brightnessLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </contrastLevel>
    <saturationLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </saturationLevel>
    <hueLevel>
      <!--ro, opt, int, range:[0,100]-->0
    </hueLevel>
    <sharpnessLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </sharpnessLevel>
  </Color>
  <enableImageLossDetection>
    <!--ro, opt, bool-->true
  </enableImageLossDetection>
</ImageChannel>
```

14.2.1.2 Get the image mode parameters of all channels

Request URL

GET /ISAPI/Image/channels/imageModes

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageModeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, image mode configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <ImageMode>
    <!--ro, opt, object, image mode-->
    <type>
      <!--ro, req, enum, type, subType:string, desc:"standard", "indoor", "outdoor", "dimLight", "bright", "soft", "vivid"-->standard
    </type>
    <recommendation>
      <!--ro, req, object, recommended configurations-->
      <brightnessLevel>
        <!--ro, opt, int, brightness, range:[0,100]-->1
      </brightnessLevel>
      <contrastLevel>
        <!--ro, opt, int, contrast, range:[0,100]-->1
      </contrastLevel>
      <sharpnessLevel>
        <!--ro, opt, int, sharpness, range:[0,100]-->1
      </sharpnessLevel>
      <saturationLevel>
        <!--ro, opt, int, saturation, range:[0,100]-->1
      </saturationLevel>
      <hueLevel>
        <!--ro, opt, int, hue, range:[0,100]-->1
      </hueLevel>
      <deNoiseLevel>
        <!--ro, opt, int, noise reduction, range:[0,100]-->1
      </deNoiseLevel>
    </recommendation>
  </ImageMode>
</ImageModeList>
```

14.2.1.3 Get the configuration parameters of all image modes of a specific channel

Request URL

GET /ISAPI/Image/channels/imageModes/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageMode xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <type>
    <!--ro, req, enum, "standard,indoor,outdoor,dimLight", subType:string, desc:"standard,indoor,outdoor,dimLight"-->standard
  </type>
  <recommendation>
    <!--ro, req, object, ro-->
    <brightnessLevel>
      <!--ro, opt, int, ranging from 0 to 100, range:[0,100]-->1
    </brightnessLevel>
    <contrastLevel>
      <!--ro, opt, int, ranging from 0 to 100, range:[0,100]-->1
    </contrastLevel>
    <sharpnessLevel>
      <!--ro, opt, int, ranging from 0 to 100, range:[0,100]-->1
    </sharpnessLevel>
    <saturationLevel>
      <!--ro, opt, int, ranging from 0 to 100, range:[0,100]-->1
    </saturationLevel>
    <hueLevel>
      <!--ro, opt, int, ranging from 0 to 100, range:[0,100]-->1
    </hueLevel>
    <deNoiseLevel>
      <!--ro, opt, int, ranging from 0 to 100, range:[0,100]-->1
    </deNoiseLevel>
  </recommendation>
</ImageMode>
```

14.2.2 实时取流

drackum@gmail.com DETEKTOR

14.2.2.1 Get multicast parameters

Request URL

GET /ISAPI/System/Network/extension

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<networkExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <multicastAddress>
    <!--ro, opt, object-->
    <ipVersion>
      <!--ro, req, enum, IP address version, subType:string, desc:IP address version-->v4
    </ipVersion>
    <ipAddress>
      <!--ro, opt, string, IPv4 address-->172.6.64.7
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address-->1080:0:0:0:8:800:200C:417A
    </ipv6Address>
  </multicastAddress>
  <enVirtualHost>
    <!--ro, opt, bool-->true
  </enVirtualHost>
</networkExtension>
```

14.2.2.2 Set the multicast address

Request URL

PUT /ISAPI/System/Network/extension

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<networkExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, req, object, advanced network parameters, attr:version{req, string, protocolVersion}-->
  <multicastAddress>
    <!--wo, opt, object, multicast address-->
    <ipVersion>
      <!--wo, req, enum, IP address version, subType:string, desc:"v4" (IPv4), "v6" (IPv6), "dual" (IPv4 and IPv6)-->v4
    </ipVersion>
    <ipAddress>
      <!--wo, opt, string, IPv4 address-->172.6.64.7
    </ipAddress>
    <ipv6Address>
      <!--wo, opt, string, IPv6 address-->1080:0:0:0:8:800:200C:417A
    </ipv6Address>
  </multicastAddress>
  <enVirtualHost>
    <!--wo, opt, bool, whether to enable the virtual host-->true
  </enVirtualHost>
</networkExtension>
```

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.3 移动侦测

14.2.3.1 Get motion detection parameters by channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/motionDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<MotionDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, motion detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable motion detection-->true
  </enabled>
  <enableHighlight>
    <!--ro, opt, bool, whether to enable highlight-->true
  </enableHighlight>
  <samplingInterval>
    <!--ro, opt, int, frame interval, range:[0,2]-->0
  </samplingInterval>
  <startTriggerTime>
    <!--ro, opt, int, unit: millisecond, range:[0,500], unit:ms-->0
  </startTriggerTime>
  <endTriggerTime>
    <!--ro, opt, int, unit: millisecond, range:[0,500], unit:ms-->0
  </endTriggerTime>
  <regionType>
    <!--ro, req, enum, region type, subType:string, desc:"grid", "roi", "none", "region"-->grid
  </regionType>
  <Grid>
    <!--ro, opt, object, grid-->
    <rowGranularity>
      <!--ro, req, int, total number of rows, range:[0,18]-->0
    </rowGranularity>
    <columnGranularity>
      <!--ro, req, int, total number of columns, range:[0,22]-->0
    </columnGranularity>
  </Grid>
  <ROI>
    <!--ro, opt, object, ROI-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[0,1000]-->0
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height, range:[0,1000]-->0
    </normalizedScreenHeight>
  </ROI>
  <MotionDetectionLayout>
    <!--ro, opt, object, rule region configuration of motion detection-->
    <sensitivityLevel>
      <!--ro, req, int, sensitivity, range:[0,100]-->0
    </sensitivityLevel>
```

drackum@gmail.com
DEFEKTOR

```
      <layout>
        <!--ro, opt, object, layout information-->
        <gridMap>
          <!--ro, opt, string, grid map, desc:the image is divided in to N × N grids, and motion detection can be enabled (value: 1) or disabled (value: 0)
for each grid. Each four horizontal grids consist of a hexadecimal number, e.g., the corresponding hexadecimal number of "0011" is 3-->test
        </gridMap>
        <roiMap>
          <!--ro, opt, string, ROI map-->test
        </roiMap>
        <RegionList>
          <!--ro, opt, array, this node is available when the regionType is set to "region", subType:object-->
          <Region>
            <!--ro, opt, object, region information-->
            <id>
              <!--ro, req, string, region No., range:[1,4]-->test
            </id>
            <RegionCoordinatesList>
              <!--ro, opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
              <RegionCoordinates>
                <!--ro, opt, object, region coordinate, desc:the origin is the lower-left corner of the screen-->
                <positionX>
                  <!--ro, req, int, x-coordinate, range:[0,1000]-->0
                </positionX>
                <positionY>
                  <!--ro, req, int, y-coordinate, range:[0,1000]-->0
                </positionY>
              </RegionCoordinates>
            </RegionCoordinatesList>
          </Region>
        </RegionList>
      </layout>
      <targetType>
        <!--ro, opt, enum, target type, subType:string, desc:"human"; "vehicle";-->human
      </targetType>
    </MotionDetectionLayout>
    <enableWithMoving>
      <!--ro, opt, bool, whether to enable motion detection-->true
    </enableWithMoving>
    <discardFalseAlarm>
      <!--ro, opt, bool, whether to filter false alarms, desc:a choice "Enable False Alarms Filtering" is added and it is disabled by default-->true
    </discardFalseAlarm>
    <maxMegaPixels>
      <!--ro, opt, int, maximum video resolution pixel supported for analysis, range:[4,6], desc:when SVC is disabled, the maximum supported channel
resolution is 4 MP; when SVC is enabled, the maximum supported channel resolution is 6 MP. When the channel streaming resolution exceeds the limit, an error
occurs and the prompt of changing channel resolution pops up-->4
    </maxMegaPixels>
  </MotionDetection>
```

14.2.3.2 Set motion detection parameters by channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/motionDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<MotionDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, motion detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable motion detection-->true
  </enabled>
  <enableHighlight>
    <!--opt, bool, whether to enable highlight-->true
  </enableHighlight>
  <samplingInterval>
    <!--opt, int, number of frames, range:[0,2]-->0
  </samplingInterval>
  <startTriggerTime>
    <!--opt, int, unit: millisecond, range:[0,500], unit:ms-->0
  </startTriggerTime>
  <endTriggerTime>
    <!--opt, int, unit: millisecond, range:[0,500], unit:ms-->0
  </endTriggerTime>
  <regionType>
    <!--req, enum, region type, subType:string, desc:“grid”, “roi”, “none”, “region”-->grid
  </regionType>
  <Grid>
    <!--opt, object, grid-->
    <rowGranularity>
      <!--req, int, range:[0,18]-->0
    </rowGranularity>
    <columnGranularity>
      <!--req, int, range:[0,22]-->0
    </columnGranularity>
  </Grid>
  <ROI>
    <!--opt, object, region of interest (ROI)-->
    <normalizedScreenWidth>
      <!--req, int, normalized width, range:[0,1000]-->0
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height, range:[0,1000]-->0
    </normalizedScreenHeight>
  </ROI>
  <MotionDetectionLayout>
    <!--opt, object, layout configuration-->
    <sensitivityLevel>
      <!--req, int, sensitivity, range:[0,100]-->0
    </sensitivityLevel>
    <layout>
      <!--opt, object, layout information-->
      <gridMap>
        <!--opt, string, grid map, desc:the image is divided in to N × N grids; each grid can be enabled (value: 1) or disabled (value 0) motion detection;
each four horizontal grids consist of a hexadecimal number, e.g., the corresponding hexadecimal number of “0011” is 3-->test
      </gridMap>
      <roiMap>
        <!--opt, string, ROI picture-->test
      </roiMap>
      <RegionList>
        <!--opt, array, this node is available when the regionType is set to "region", subType:object-->
        <Region>
          <!--opt, object, region information-->
          <id>
            <!--req, string, region ID, range:[1,4]-->test
          </id>
          <RegionCoordinatesList>
            <!--opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
            <RegionCoordinates>
              <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--req, int, X-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--req, int, Y-coordinate, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </Region>
      </RegionList>
    </layout>
    <targetType>
      <!--opt, enum, target type, subType:string, desc:“human”, “vehicle”-->human
    </targetType>
  </MotionDetectionLayout>
  <enableWithMoving>
    <!--opt, bool, whether to enable motion detection-->true
  </enableWithMoving>
  <discardFalseAlarm>
    <!--opt, bool, whether to filter false alarms, desc:whether to filter false alarms-->true
  </discardFalseAlarm>
</MotionDetection>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.3.3 Set rule region parameters of motion detection by channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/motionDetection/layout

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<MotionDetectionLayout xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, motion detection parameters, attr:version{req, string, protocolVersion}-->
  <sensitivityLevel>
    <!--req, int, sensitivity, range:[0,100]-->0
  </sensitivityLevel>
  <layout>
    <!--opt, object, layout information-->
    <gridMap>
      <!--opt, string, grid map, desc:the image is divided in to N × N grids; each grid can be enabled (value: 1) or disabled (value 0) motion detection;
each four horizontal grids consist of a hexadecimal number, e.g., the corresponding hexadecimal number of "0011" is 3-->test
    </gridMap>
    <roiMap>
      <!--opt, string, ROI picture-->test
    </roiMap>
    <RegionList>
      <!--opt, array, this node is available when the regionType is set to "region", subType:object-->
      <Region>
        <!--opt, object, region information-->
        <id>
          <!--req, string, region ID, range:[1,4]-->test
        </id>
        <RegionCoordinatesList>
          <!--opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType>
    <!--opt, enum, target type, subType:string, desc:"human", "vehicle”-->human
  </targetType>
</MotionDetectionLayout>
```

Response Message

dr.ackum@gmail.com DEFEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.3.4 Get the rule region parameters of the specified digital channel for motion detection

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/motionDetection/layout

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<MotionDetectionLayout xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, motion detection parameters, attr:version{req, string, protocolVersion}-->
  <sensitivityLevel>
    <!--ro, req, int, sensitivity, range:[0,100]-->0
  </sensitivityLevel>
  <layout>
    <!--ro, opt, object, layout information-->
    <gridMap>
      <!--ro, opt, string, grid map, desc:the image is divided in to N × N grids, each grid can be enabled (value: 1) or disabled (value 0) for motion
detection; each four horizontal grids consist of a hexadecimal number, e.g., the corresponding hexadecimal number of "0011" is 3-->test
    </gridMap>
    <roiMap>
      <!--ro, opt, string, region of interest (ROI) map-->test
    </roiMap>
    <RegionList>
      <!--ro, opt, array, list of multiple regions, subType:object-->
      <Region>
        <!--ro, opt, object, region information-->
        <id>
          <!--ro, req, string, region ID, range:[1,4]-->test
        </id>
        <RegionCoordinatesList>
          <!--ro, opt, array, list of region coordinates, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType>
    <!--ro, opt, enum, target classification, subType:string, desc:"human", "vehicle"-->human
  </targetType>
</MotionDetectionLayout>
```

14.2.3.5 Get rule region configuration capability of motion detection by channel

Request URL

dracknm@gmail.com DETEKTOR

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/motionDetection/layout/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<MotionDetectionLayout xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <sensitivityLevel min="0" max="100" step="20">
    <!--ro, req, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int},step{opt, int}-->0
  </sensitivityLevel>
  <layout>
    <!--ro, opt, object-->
    <gridMap>
      <!--ro, opt, string, the image is divided in to N × N grids,each grid can be enabled (value: 1) or disabled (value 0) motion detection,each four horizontal grids consist of a hexadecimal number,e.g.,the corresponding hexadecimal number of "0011" is 3, desc:the image is divided in to N × N grids, each grid can be enabled (value: 1) or disabled (value 0) motion detection, each four horizontal grids consist of a hexadecimal number, e.g., the corresponding hexadecimal number of "0011" is 3-->test
    </gridMap>
    <roiMap>
      <!--ro, opt, string-->test
    </roiMap>
    <RegionList size="4">
      <!--ro, opt, array, this node is available when the regionType is set to "region", subType:object, attr:size{req, int}-->
      <Region>
        <!--ro, opt, object, region information-->
        <id>
          <!--ro, req, string-->test
        </id>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinates list, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType opt="human,vehicle">
    <!--ro, opt, enum, target type, subType:string, attr:opt{req, string}, desc:"human", "vehicle"-->human
  </targetType>
  <regionEdgesNum min="4" max="10">
    <!--ro, opt, int, range:[4,10], attr:min{req, int},max{req, int}-->0
  </regionEdgesNum>
</MotionDetectionLayout>
```

dracium@gmail.com DETEKTOR

14.2.3.6 Get the linkage parameters of motion detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/VMD-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupporttriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, string-->VMD
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <triggerEnabled>
        <!--ro, opt, bool-->true
      </triggerEnabled>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->[email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string--> 1
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string--> 1
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[1,11]-->test
      </cellphoneNumber>
      <LightAudioByIPAction>
        <!--ro, opt, object-->
        <notificationType>
          <!--ro, req, enum, subType:string-->audio
        </notificationType>
        <notificationName>
          <!--ro, opt, string-->test
        </notificationName>
```

drackum@gmail.com
DETEKTOR

```
    </LightAudioByIPAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <preRecordTime>
      <!--ro, opt, int, range:[0,10], unit:s-->1
    </preRecordTime>
    <recordTime>
      <!--ro, opt, int, range:[0,180], unit:s-->5
    </recordTime>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
      <lightFlashFrequency>
        <!--ro, opt, enum, subType:string-->high
      </lightFlashFrequency>
      <brightness>
        <!--ro, opt, int, range:[0,100]-->1
      </brightness>
      <timeRangeList>
        <!--ro, opt, array, subType:object-->
        <timeRangeItem>
          <!--ro, opt, object-->
          <week>
            <!--ro, req, enum, subType:int-->1
          </week>
          <timeRange>
            <!--ro, req, array, subType:object-->
            <timeItem>
              <!--ro, req, object-->
              <timeRangeID>
                <!--ro, req, int-->1
              </timeRangeID>
              <beginTime>
                <!--ro, req, string-->08:00
              </beginTime>
              <endTime>
                <!--ro, req, string-->24:00
              </endTime>
            </timeItem>
          </timeRange>
        </timeRangeItem>
      </timeRangeList>
    </WhiteLightAction>
    <AudioAction>
      <!--ro, opt, object-->
      <audioID>
        <!--ro, req, int-->1
      </audioID>
      <alarmTimes>
        <!--ro, opt, int, range:[0,9]-->1
      </alarmTimes>
      <childDevID>
        <!--ro, opt, string, range:[9,32]-->3434645611
      </childDevID>
    </AudioAction>
    <audioAlarmTriggerParam>
      <!--ro, opt, object, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,audio}-->
      <audioIndexList>
        <!--ro, opt, array, subType:object, range:[1,3]-->
        <audioIndexItem>
          <!--ro, req, object-->
          <indexID>
            <!--ro, req, int, range:[1,3]-->1
          </indexID>
          <indexName>
            <!--ro, opt, string, range:[1,128]-->test
          </indexName>
          <alarmTimes>
            <!--ro, opt, int, range:[1,50]-->1
          </alarmTimes>
          <audioID>
            <!--ro, opt, enum, subType:int-->1
          </audioID>
        </audioIndexItem>
      </audioIndexList>
      <timeRangeList>
        <!--ro, opt, array, subType:object-->
        <timeRangeItem>
          <!--ro, opt, object-->
          <week>
            <!--ro, req, enum, subType:int-->1
          </week>
          <timeRange>
            <!--ro, req, array, subType:object-->
            <timeItem>
              <!--ro, req, object-->
              <timeRangeID>
```

drackum@gmail.com
DETEKTOR

```
                <!--ro, req, int-->1
              </timeRangeID>
              <audioID>
                <!--ro, opt, enum, subType:int-->1
              </audioID>
              <beginTime>
                <!--ro, req, string-->08:00
              </beginTime>
              <endTime>
                <!--ro, req, string-->24:00
              </endTime>
            </timeItem>
          </timeRange>
        </timeRangeItem>
      </timeRangeList>
    </audioAlarmTriggerParam>
  </EventTriggerNotification>
</EventTriggerNotificationList>
</EventTrigger>
```

14.2.3.7 Set the linkage parameters of motion detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/VMD-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->VMD
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->1
      </outputIOPortID>
      <dynOutputIOPortID>
```

drackun@gmail.com
DETEKTOR

```
      <dynOutputIOPortID>
        <!--opt, string-->1
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[1,11]-->test
      </cellphoneNumber>
      <LightAudioByIPAction>
        <!--opt, object-->
        <notificationType>
          <!--req, enum, subType:string-->audio
        </notificationType>
        <notificationName>
          <!--opt, string-->test
        </notificationName>
      </LightAudioByIPAction>
      <SupplementLightAlarm>
        <!--opt, object-->
        <durationTime>
          <!--opt, int, range:[0,90], unit:s-->5
        </durationTime>
      </SupplementLightAlarm>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

crackum@gmail.com
DETEKTOR

14.2.3.8 Motion detection

EventType:VMD

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, EventNotificationAlert, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPv4 address-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 Address-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, device's communication port No.-->1
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol, subType:string, desc:"HTTP", "HTTPS", "EHome"-->HTTP
  </protocol>
  <macAddress>
```

```
    <!--ro, opt, string, MAC address-->test
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->1
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No.-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, event occurred time-->2017-04-22T15:39:01+08:0
  </dateTime>
  <activePostCount>
    <!--ro, req, int, event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"VMD" (motion detection)-->VMD
  </eventType>
  <eventState>
    <!--ro, req, enum, event status (for persistent event), subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <DetectionRegionList>
    <!--ro, opt, array, detection area parameter list (for expert mode only), subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, parameters of a detection area (for expert mode only), desc:parameters of a detection area (for expert mode only)-->
      <regionID>
        <!--ro, req, string, detection area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, req, array, detection area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, detection area origin coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID-->1
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, req, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, req, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <targetType>
    <!--ro, opt, string, target classification, desc:"human", "vehicle", "human,vehicle" (human and vehicle). If detection target classification is not
supported, this field is not returned-->human
  </targetType>
  <bkgUrl>
    <!--ro, opt, string, background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (it is for the cloud storage protocol), "digest" (digest authentication,
it is for the device local storage)-->no
  </URLCertificationType>
</EventNotificationAlert>
```

drackum@gmail.com
DETEKTOR

14.2.4 视频遮盖

14.2.4.1 Get the video tampering parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/tamperDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TamperDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, req, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <tampersensitivityLevel>
    <!--ro, req, int, range:[0,100]-->1
  </tampersensitivityLevel>
  <TamperDetectionRegionList>
    <!--ro, req, array, subType:object-->
    <TamperDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <enabled>
        <!--ro, opt, bool-->true
      </enabled>
      <sensitivityLevel>
        <!--ro, req, int, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, req, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--ro, req, object-->
          <positionX>
            <!--ro, req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </TamperDetectionRegion>
  </TamperDetectionRegionList>
</TamperDetection>
```

drackum@gmail.com
DETEKTOR

14.2.4.2 Set the video tampering parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/tamperDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TamperDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <normalizedScreenSize>
    <!--req, object-->
    <normalizedScreenWidth>
      <!--req, int-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <tampersensitivityLevel>
    <!--req, int, range:[0,100]-->1
  </tampersensitivityLevel>
  <TamperDetectionRegionList>
    <!--req, array, subType:object-->
    <TamperDetectionRegion>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <enabled>
        <!--opt, bool-->true
      </enabled>
      <sensitivityLevel>
        <!--req, int, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--req, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </TamperDetectionRegion>
  </TamperDetectionRegionList>
</TamperDetection>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
DETEKTOR

14.2.4.3 Get the video tampering capabilities of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/tamperDetection/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TamperDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, req, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <tampersensitivityLevel min="1" max="3">
    <!--ro, req, int, range:[0,100], attr:min{req, int},max{req, int}-->1
  </tampersensitivityLevel>
  <TamperDetectionRegionList>
    <!--ro, req, array, subType:object-->
    <TamperDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <enabled>
        <!--ro, opt, bool-->true
      </enabled>
      <sensitivityLevel>
        <!--ro, req, int, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, req, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--ro, req, object-->
          <positionX>
            <!--ro, req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </TamperDetectionRegion>
  </TamperDetectionRegionList>
</TamperDetection>
```

drackum@gmail.com
DETEKTOR

14.2.4.4 Set parameters of all video tampering detection areas of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/tamperDetection/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TamperDetectionRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <TamperDetectionRegion>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <enabled>
      <!--opt, bool-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, range:[0,100]-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--req, array, subType:object, range:[0,4]-->
      <RegionCoordinates>
        <!--req, object-->
        <positionX>
          <!--req, int, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--req, int, range:[0,576]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </TamperDetectionRegion>
</TamperDetectionRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
DETEKTOR

14.2.4.5 Delete all video tampering detection areas of a specified digital channel

Request URL

DELETE /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/tamperDetection/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.4.6 Get parameters of all video tampering detection areas of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/tamperDetection/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TamperDetectionRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <TamperDetectionRegion>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, range:[0,100]-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, req, array, subType:object, range:[0,4]-->
      <RegionCoordinates>
        <!--ro, req, object-->
        <positionX>
          <!--ro, req, int, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, range:[0,576]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </TamperDetectionRegion>
</TamperDetectionRegionList>
```

14.2.4.7 Delete a specified area of tampering detection of a digital channel

Request URL

DELETE /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/tamperDetection/regions/<regionID>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

dracKum@gmail.com
DETEKTOR

14.2.4.8 Get the linkage parameters of video tampering detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/tamper-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->tamper-1
  </id>
  <eventType>
    <!--ro, req, string-->tamper
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s-->0
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <modbus>
        <!--ro, opt, object-->
        <outputPortID>
          <!--ro, opt, int, range:[0,15]-->1
        </outputPortID>
        <delayTime>
          <!--ro, opt, int, range:[1,10], unit:s-->1
        </delayTime>
        <alarmStatus>
          <!--ro, opt, enum, subType:string-->open
        </alarmStatus>
      </modbus>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
DETEKTOR

14.2.4.9 Delete the linkage of video tampering detection for a specified channel

Request URL

DELETE /ISAPI/Event/triggers/tamper-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

dr.ackum@gmail.com
DETEKTOR

14.2.4.10 Set the linkage parameters of video tampering detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/tamper-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->tamper-1
  </id>
  <eventType>
    <!--req, string-->tamper
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s-->0
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <modbus>
        <!--opt, object-->
        <outputPortID>
          <!--opt, int, range:[0,15]-->1
        </outputPortID>
        <delayTime>
          <!--opt, int, range:[1,10], unit:s-->1
        </delayTime>
        <alarmStatus>
          <!--opt, enum, subType:string-->open
        </alarmStatus>
      </modbus>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.4.11 Delete the linkage triggering parameters of video tampering detection for a specified channel

Request URL

DELETE /ISAPI/Event/triggers/tamper-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.4.12 Set the linkage triggering parameters of video tampering detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/tamper-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

dracksm@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <notificationMethod>
    <!--req, enum, subType:string-->FTP
  </notificationMethod>
  <notificationRecurrence>
    <!--opt, enum, subType:string-->beginning
  </notificationRecurrence>
  <notificationInterval>
    <!--opt, int, unit:ms-->1
  </notificationInterval>
  <outputIOPortID>
    <!--opt, string-->test
  </outputIOPortID>
  <dynOutputIOPortID>
    <!--opt, string-->test
  </dynOutputIOPortID>
  <videoInputID>
    <!--opt, string-->1
  </videoInputID>
  <dynVideoInputID>
    <!--opt, string-->1
  </dynVideoInputID>
  <ptzAction>
    <!--opt, object-->
    <ptzChannelID>
      <!--req, string-->1
    </ptzChannelID>
    <actionName>
      <!--req, enum, subType:string-->preset
    </actionName>
    <actionNum>
      <!--opt, int-->1
    </actionNum>
    <presetDurationTime>
      <!--opt, int, unit:s-->0
    </presetDurationTime>
  </ptzAction>
  <cellphoneNumber>
    <!--opt, string, range:[0,11]-->test
  </cellphoneNumber>
  <modbus>
    <!--opt, object-->
    <outputPortID>
      <!--opt, int, range:[0,15]-->1
    </outputPortID>
    <delayTime>
      <!--opt, int, range:[1,10], unit:s-->1
    </delayTime>
    <alarmStatus>
      <!--opt, enum, subType:string-->open
    </alarmStatus>
  </modbus>
</EventTriggerNotification>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.4.13 Get the linkage triggering parameters of video tampering detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/tamper-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <notificationMethod>
    <!--ro, req, enum, subType:string-->FTP
  </notificationMethod>
  <notificationRecurrence>
    <!--ro, opt, enum, subType:string-->beginning
  </notificationRecurrence>
  <notificationInterval>
    <!--ro, opt, int, unit:ms-->1
  </notificationInterval>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <dynOutputIOPortID>
    <!--ro, opt, string-->test
  </dynOutputIOPortID>
  <videoInputID>
    <!--ro, opt, string-->1
  </videoInputID>
  <dynVideoInputID>
    <!--ro, opt, string-->1
  </dynVideoInputID>
  <ptzAction>
    <!--ro, opt, object-->
    <ptzChannelID>
      <!--ro, req, string-->1
    </ptzChannelID>
    <actionName>
      <!--ro, req, enum, subType:string-->preset
    </actionName>
    <actionNum>
      <!--ro, opt, int-->1
    </actionNum>
    <presetDurationTime>
      <!--ro, opt, int, unit:s-->0
    </presetDurationTime>
  </ptzAction>
  <cellphoneNumber>
    <!--ro, opt, string, range:[0,11]-->test
  </cellphoneNumber>
  <modbus>
    <!--ro, opt, object-->
    <outputPortID>
      <!--ro, opt, int, range:[0,15]-->1
    </outputPortID>
    <delayTime>
      <!--ro, opt, int, range:[1,10], unit:s-->1
    </delayTime>
    <alarmStatus>
      <!--ro, opt, enum, subType:string-->open
    </alarmStatus>
  </modbus>
</EventTriggerNotification>
```

drackum@gmail.com
DETEKTOR

14.2.5 场景变更

14.2.5.1 Set the arming schedule of scene change detection for all channels

Request URL

PUT /ISAPI/Event/schedules/sceneChangeDetections

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SceneChangeDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->sceneChangeDetection_video1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->sceneChangeDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</SceneChangeDetectionScheduleList>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.5.2 Get the arming schedule of scene change detection for all channels

Request URL

GET /ISAPI/Event/schedules/sceneChangeDetections

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SceneChangeDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->sceneChangeDetection_video1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->sceneChangeDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</SceneChangeDetectionScheduleList>
```

drackum@gmail.com
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14.2.5.3 Sudden scene change detection

EventType:scenechangedetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"scenechangedetection" (sudden scene change detection)-->scenechangedetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
</EventNotificationAlert>
```

drackum@gmail.com
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14.2.6 视频通道资源管理

14.2.6.1 Set the image parameters of all channels

Request URL

PUT /ISAPI/ContentMgmt/ImageProxy/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageChannellist xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <ImageChannel>
    <!--opt, object-->
    <id>
      <!--req, int-->1
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <inputPort>
      <!--req, int-->1
    </inputPort>
    <Color>
      <!--opt, object-->
      <brightnessLevel>
        <!--opt, int, range:[0,100]-->1
      </brightnessLevel>
      <contrastLevel>
        <!--opt, int, range:[0,100]-->1
      </contrastLevel>
      <saturationLevel>
        <!--opt, int, range:[0,100]-->1
      </saturationLevel>
    </Color>
    <enableImageLossDetection>
      <!--opt, bool-->true
    </enableImageLossDetection>
  </ImageChannel>
</ImageChannellist>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.2.6.2 Get the image parameters of all channels

Request URL

GET /ISAPI/ContentMgmt/ImageProxy/channels

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageChannellist xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <ImageChannel>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->1
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <inputPort>
      <!--ro, req, int-->1
    </inputPort>
    <Color>
      <!--ro, opt, object-->
      <brightnessLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </brightnessLevel>
      <contrastLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </contrastLevel>
      <saturationLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </saturationLevel>
    </Color>
    <enableImageLossDetection>
      <!--ro, opt, bool-->true
    </enableImageLossDetection>
  </ImageChannel>
</ImageChannellist>
```

14.2.6.3 Get the image parameters of a specified channel

Request URL

GET /ISAPI/ContentMgmt/ImageProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

dracken@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <NoiseReduce>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->close
    </mode>
    <GeneralMode>
      <!--ro, opt, object-->
      <generalLevel>
        <!--ro, req, int, range:[0,100]-->1
      </generalLevel>
    </GeneralMode>
  </NoiseReduce>
  <Color>
    <!--ro, opt, object-->
    <brightnessLevel>
      <!--ro, opt, int, range:[0,100]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--ro, opt, int, range:[0,100]-->0
    </contrastLevel>
    <saturationLevel>
      <!--ro, opt, int, range:[0,100]-->0
    </saturationLevel>
    <hueLevel>
      <!--ro, opt, int, range:[0,100]-->0
    </hueLevel>
    <sharpnessLevel>
      <!--ro, opt, int, range:[0,100]-->0
    </sharpnessLevel>
  </Color>
  <enableImageLossDetection>
    <!--ro, opt, bool-->true
  </enableImageLossDetection>
</ImageChannel>
```

14.2.6.4 Set the image parameters of a specified channel

Request URL

PUT /ISAPI/ContentMgmt/ImageProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int-->1
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <inputPort>
    <!--req, int-->1
  </inputPort>
  <FocusConfiguration>
    <!--opt, object-->
    <focusStyle>
      <!--req, enum, subType:string-->AUTO
    </focusStyle>
    <focusLimited>
      <!--opt, enum, subType:string-->10
    </focusLimited>
    <focusPosition>
      <!--opt, int, range:[1,10]-->1
    </focusPosition>
    <focusSpeed>
      <!--opt, int, range:[1,3]-->1
    </focusSpeed>
    <focusSensitivity>
      <!--opt, int, range:[0,2]-->1
    </focusSensitivity>
    <temperatureChangeAdantEnabled>
```

dr4ckun@gmail.com
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```
    <!--opt, bool-->true
   </temperatureChangeAdaptEnabled>
   <relativeFocusPos>
    <!--opt, int, range:[0,4294967295]-->1
   </relativeFocusPos>
   <lensType>
    <!--opt, enum, subType:string-->1140
   </lensType>
   <highTemperaturePriority>
    <!--opt, bool-->true
   </highTemperaturePriority>
   <focusRange>
    <!--opt, enum, subType:string-->max
   </focusRange>
   <foucusDistanceMode>
    <!--opt, enum, subType:string-->compatible
   </foucusDistanceMode>
  </FocusConfiguration>
  <LensInitialization>
   <!--opt, object-->
   <enabled>
    <!--req, bool-->true
   </enabled>
  </LensInitialization>
  <ImageFlip>
   <!--opt, object-->
   <enabled>
    <!--req, bool-->true
   </enabled>
   <ImageFlipStyle>
    <!--opt, enum, subType:string-->LEFTRIGHT
   </ImageFlipStyle>
   <flipAngle>
    <!--opt, enum, subType:string-->90
   </flipAngle>
  </ImageFlip>
  <ImageFreeze>
   <!--opt, object-->
   <enabled>
    <!--req, bool-->true
   </enabled>
  </ImageFreeze>
  <proportionalpan>
   <!--opt, object-->
   <enabled>
    <!--req, bool-->true
   </enabled>
  </proportionalpan>
  <WDR>
   <!--opt, object-->
   <mode>
    <!--req, enum, subType:string-->[open
   </mode>
   <WDRLevel>
    <!--opt, int, range:[1,100]-->1
   </WDRLevel>
   <WDRContrastLevel>
    <!--opt, int, range:[1,100]-->1
   </WDRContrastLevel>
   <WDRLevel1>
    <!--opt, int, range:[1,100]-->1
   </WDRLevel1>
  </WDR>
  <BLC>
   <!--opt, object-->
   <enabled>
    <!--req, bool-->true
   </enabled>
   <BLCMode>
    <!--opt, enum, subType:string-->UP
   </BLCMode>
   <BLCLevel>
    <!--opt, int, range:[0,10]-->0
   </BLCLevel>
   <BLCRegionList>
    <!--opt, array, subType:object-->
    <BLCRegion>
     <!--opt, object-->
     <id>
      <!--req, int, range:[1,10]-->1
     </id>
     <RegionCoordinatesList>
      <!--req, array, subType:object, range:[0,4]-->
      <RegionCoordinates>
       <!--opt, object-->
       <positionX>
        <!--req, int, range:[0,704]-->1
       </positionX>
       <positionY>
        <!--req, int, range:[0,576]-->1
       </positionY>
      </RegionCoordinates>
     </RegionCoordinatesList>
```

drackum@gmail.com
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```
      </BLCRegion>
    </BLCRegionList>
  </BLC>
  <NoiseReduce>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->close
    </mode>
    <GeneralMode>
      <!--opt, object-->
      <generalLevel>
        <!--req, int, range:[0,100]-->1
      </generalLevel>
    </GeneralMode>
    <AdvancedMode>
      <!--opt, object-->
      <FrameNoiseReduceLevel>
        <!--req, int, range:[0,100]-->1
      </FrameNoiseReduceLevel>
      <InterFrameNoiseReduceLevel>
        <!--req, int, range:[0,100]-->1
      </InterFrameNoiseReduceLevel>
    </AdvancedMode>
  </NoiseReduce>
  <ImageEnhancement>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <ImageEnhancementLevel>
      <!--opt, int, range:[0,10]-->0
    </ImageEnhancementLevel>
  </ImageEnhancement>
  <DSS>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <DSSLevel>
      <!--opt, enum, subType:string-->*1.25
    </DSSLevel>
    <mode>
      <!--opt, enum, subType:string-->frame
    </mode>
    <slowShutterLevel>
      <!--opt, int, range:[1,6]-->1
    </slowShutterLevel>
  </DSS>
  <WhiteBalance>
    <!--opt, object-->
    <WhiteBalanceStyle>
      <!--opt, enum, subType:string-->auto
    </WhiteBalanceStyle>
    <WhiteBalanceRed>
      <!--opt, int, range:[0,100]-->1
    </WhiteBalanceRed>
    <WhiteBalanceBlue>
      <!--opt, int, range:[0,100]-->1
    </WhiteBalanceBlue>
    <whiteBalanceLevel>
      <!--opt, int, range:[0,100]-->1
    </whiteBalanceLevel>
  </WhiteBalance>
  <Exposure>
    <!--opt, object-->
    <ExposureType>
      <!--req, enum, subType:string-->auto
    </ExposureType>
    <autoIrisLevel>
      <!--opt, int, range:[0,100]-->1
    </autoIrisLevel>
    <OverexposeSuppress>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <Type>
        <!--opt, enum, subType:string-->AUTO
      </Type>
      <DistanceLevel>
        <!--opt, int, range:[1,100]-->1
      </DistanceLevel>
      <shortIRDistanceLevel>
        <!--opt, int, range:[1,100]-->1
      </shortIRDistanceLevel>
      <longIRDistanceLevel>
        <!--opt, int, range:[1,100]-->1
      </longIRDistanceLevel>
      <supplementLightIntensity1>
        <!--opt, int, range:[1,100]-->1
      </supplementLightIntensity1>
      <supplementLightIntensity2>
        <!--opt, int, range:[1,100]-->1
      </supplementLightIntensity2>
```

drackum@gmail.com
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```
      </supplementLightIntensity2>
      <supplementLightIntensity3>
        <!--opt, int, range:[1,100]-->1
      </supplementLightIntensity3>
      <supplementLightIntensity4>
        <!--opt, int, range:[1,100]-->1
      </supplementLightIntensity4>
    </OverexposeSuppress>
    <pIris>
      <!--opt, object-->
      <pIrisType>
        <!--opt, enum, subType:string-->MANUAL
      </pIrisType>
      <IrisLevel>
        <!--opt, int, range:[1,100]-->1
      </IrisLevel>
    </pIris>
    <faceExposure>
      <!--opt, object-->
      <enabled>
        <!--opt, bool-->true
      </enabled>
      <sensitivity>
        <!--opt, int, range:[0,100]-->0
      </sensitivity>
    </faceExposure>
    <PIrisGeneral>
      <!--opt, object-->
      <irisLevel>
        <!--opt, int, range:[1,100]-->1
      </irisLevel>
    </PIrisGeneral>
  </Exposure>
  <Sharpness>
    <!--opt, object-->
    <SharpnessMode>
      <!--opt, enum, subType:string-->manual
    </SharpnessMode>
    <SharpnessLevel>
      <!--req, int, range:[0,100]-->1
    </SharpnessLevel>
  </Sharpness>
  <gammaCorrection>
    <!--opt, object-->
    <gammaCorrectionEnabled>
      <!--opt, bool-->true
    </gammaCorrectionEnabled>
    <gammaCorrectionLevel>
      <!--opt, int, range:[0,100]-->1
    </gammaCorrectionLevel>
  </gammaCorrection>
  <powerLineFrequency>
    <!--opt, object-->
    <powerLineFrequencyMode>
      <!--opt, enum, subType:string-->50hz
    </powerLineFrequencyMode>
  </powerLineFrequency>
  <Color>
    <!--opt, object-->
    <brightnessLevel>
      <!--opt, int, range:[0,100]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--opt, int, range:[0,100]-->0
    </contrastLevel>
    <saturationLevel>
      <!--opt, int, range:[0,100]-->0
    </saturationLevel>
    <hueLevel>
      <!--opt, int, range:[0,100]-->0
    </hueLevel>
    <grayScale>
      <!--opt, object-->
      <grayScaleMode opt="indoor,outdoor">
        <!--opt, enum, subType:string, attr:opt{req, string}-->indoor
      </grayScaleMode>
    </grayScale>
    <nightMode>
      <!--opt, bool-->true
    </nightMode>
    <sharpnessLevel>
      <!--opt, int, range:[0,100]-->0
    </sharpnessLevel>
  </Color>
  <IrcutFilter>
    <!--opt, object-->
    <IrcutFilterType>
      <!--opt, enum, subType:string-->auto
    </IrcutFilterType>
    <dayToNightFilterLevel>
      <!--opt, enum, subType:string-->low
    </dayToNightFilterLevel>
    <dayToNightFilterTime>
      <!--opt, int, range:[0,10]-->1
```

drackum@gmail.com
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```
    </dayToNightFilterTime>
    <nightToDayFilterLevel>
      <!--opt, enum, subType:string-->low
    </nightToDayFilterLevel>
    <nightToDayFilterTime>
      <!--opt, int, range:[0,10]-->1
    </nightToDayFilterTime>
    <Schedule>
      <!--opt, object-->
      <scheduleType>
        <!--req, enum, subType:string-->day
      </scheduleType>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </Schedule>
    <EventTrigger>
      <!--opt, object-->
      <eventType>
        <!--req, enum, subType:string-->IO
      </eventType>
      <IrcutFilterAction>
        <!--req, enum, subType:string-->day
      </IrcutFilterAction>
    </EventTrigger>
  </IrcutFilter>
  <Scene>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->indoor
    </mode>
  </Scene>
  <EPTZ>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <mode>
      <!--opt, enum, subType:string-->cruise
    </mode>
  </EPTZ>
  <EIS>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <EISLevel>
      <!--opt, int, range:[0,10]-->0
    </EISLevel>
  </EIS>
  <HLC>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <HLCLevel>
      <!--opt, int, range:[0,10]-->0
    </HLCLevel>
  </HLC>
  <ZoomLimit>
    <!--opt, object-->
    <ZoomLimitRatio>
      <!--opt, enum, subType:int-->4
    </ZoomLimitRatio>
  </ZoomLimit>
  <corridor>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
  </corridor>
  <Dehaze>
    <!--opt, object-->
    <DehazeMode>
      <!--opt, enum, subType:string-->open
    </DehazeMode>
    <DehazeLevel>
      <!--opt, int, range:[0,10]-->0
    </DehazeLevel>
  </Dehaze>
  <ImageMode>
    <!--opt, enum, subType:string-->standard
  </ImageMode>
  <enableImageLossDetection>
    <!--opt, bool-->true
  </enableImageLossDetection>
  <CaptureMode>
```

drackum@gmail.com
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```
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->close
    </mode>
  </CaptureMode>
  <IrLight>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->auto
    </mode>
    <brightnessLevel>
      <!--opt, int, range:[0,100]-->1
    </brightnessLevel>
    <sensitivityLevel>
      <!--opt, int, range:[0,100]-->1
    </sensitivityLevel>
    <brightnessLimit>
      <!--opt, int, range:[0,100]-->1
    </brightnessLimit>
    <triggerMode>
      <!--opt, enum, subType:string-->camera
    </triggerMode>
  </IrLight>
  <LensDistortionCorrection>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <level>
      <!--opt, enum, subType:string-->level1
    </level>
    <accurateLevel>
      <!--opt, int, range:[0,100]-->0
    </accurateLevel>
    <zoomedInDistantViewLevel>
      <!--opt, int, range:[0,100]-->0
    </zoomedInDistantViewLevel>
    <horizontalFOV>
      <!--opt, int, range:[0,100]-->0
    </horizontalFOV>
    <verticalFOV>
      <!--opt, int, range:[0,100]-->0
    </verticalFOV>
  </LensDistortionCorrection>
  <SupplementLight>
    <!--opt, object-->
    <mode>
      <!--opt, enum, subType:string-->auto
    </mode>
    <Schedule>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:08:00+08:00
        </beginTime>
        <endTime>
          <!--req, time-->00:18:00+08:00
        </endTime>
      </TimeRange>
    </Schedule>
    <brightnessLimit>
      <!--opt, int, range:[0,100]-->80
    </brightnessLimit>
    <supplementLightMode>
      <!--opt, enum, subType:string-->mixed
    </supplementLightMode>
    <irLightBrightness>
      <!--opt, int, range:[0,100]-->80
    </irLightBrightness>
    <mixedLightBrightnessRegulatMode>
      <!--opt, enum, subType:string-->auto
    </mixedLightBrightnessRegulatMode>
    <highIrLightBrightness>
      <!--opt, int, range:[0,100]-->80
    </highIrLightBrightness>
    <highWhiteLightBrightness>
      <!--opt, int, range:[0,100]-->80
    </highWhiteLightBrightness>
    <lowIrLightBrightness>
      <!--opt, int, range:[0,100]-->80
    </lowIrLightBrightness>
    <lowWhiteLightBrightness>
      <!--opt, int, range:[0,100]-->80
    </lowWhiteLightBrightness>
    <whiteLightBrightness>
      <!--opt, int, range:[0,100]-->80
    </whiteLightBrightness>
    <WhiteLightModeCfg>
      <!--opt, object-->
      <highWhiteLightBrightness>
        <!--opt, int, range:[0,100]-->80
      </highWhiteLightBrightness>
      <lowWhiteLightBrightness>
```

drackum@gmail.com
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```
      <!--opt, int, range:[0,100]-->80
    </lowWhiteLightBrightness>
    <whiteLightBrightness>
      <!--opt, int, range:[0,100]-->80
    </whiteLightBrightness>
  </WhiteLightModeCfg>
  <isAutoModeBrightnessCfg>
    <!--opt, bool-->true
  </isAutoModeBrightnessCfg>
  <irLightbrightnessLimit>
    <!--opt, int, range:[0,100]-->80
  </irLightbrightnessLimit>
  <whiteLightbrightnessLimit>
    <!--opt, int, range:[0,100]-->80
  </whiteLightbrightnessLimit>
  <isNotSupportMixedWhiteLight>
    <!--opt, bool-->true
  </isNotSupportMixedWhiteLight>
  <EventIntelligenceModeCfg>
    <!--opt, object-->
    <brightnessRegulatMode>
      <!--opt, enum, subType:string-->auto
    </brightnessRegulatMode>
    <whiteLightBrightness>
      <!--opt, int, range:[0,100]-->80
    </whiteLightBrightness>
    <irLightBrightness>
      <!--opt, int, range:[0,100]-->80
    </irLightBrightness>
  </EventIntelligenceModeCfg>
  <MultiAzimuthMixedModeCfg>
    <!--opt, object-->
    <brightnessRegulatMode>
      <!--opt, enum, subType:string-->auto
    </brightnessRegulatMode>
    <LightCtrlList>
      <!--opt, array, subType:object-->
      <LightCtrl>
        <!--opt, object-->
        <azimuthType>
          <!--opt, enum, subType:string-->left
        </azimuthType>
        <lightType>
          <!--opt, enum, subType:string-->highWhite
        </lightType>
        <brightness>
          <!--opt, int, range:[0,100]-->80
        </brightness>
      </LightCtrl>
    </LightCtrlList>
  </MultiAzimuthMixedModeCfg>
  <MultiAzimuthWhiteLightCfg>
    <!--opt, object-->
    <brightnessRegulatMode>
      <!--opt, enum, subType:string-->auto
    </brightnessRegulatMode>
    <LightCtrlList>
      <!--opt, array, subType:object-->
      <LightCtrl>
        <!--opt, object-->
        <azimuthType>
          <!--opt, enum, subType:string-->left
        </azimuthType>
        <lightType>
          <!--opt, enum, subType:string-->highWhite
        </lightType>
        <brightness>
          <!--opt, int, range:[0,100]-->80
        </brightness>
      </LightCtrl>
    </LightCtrlList>
  </MultiAzimuthWhiteLightCfg>
  <MultiAzimuthIrLightCfg>
    <!--opt, object-->
    <brightnessRegulatMode>
      <!--opt, enum, subType:string-->auto
    </brightnessRegulatMode>
    <LightCtrlList>
      <!--opt, array, subType:object-->
      <LightCtrl>
        <!--opt, object-->
        <azimuthType>
          <!--opt, enum, subType:string-->left
        </azimuthType>
        <lightType>
          <!--opt, enum, subType:string-->highIrLight
        </lightType>
        <brightness>
          <!--opt, int, range:[0,100]-->80
        </brightness>
      </LightCtrl>
    </LightCtrlList>
  </MultiAzimuthIrLightCfg>
```

drackum@gmail.com
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```
    <PanoramicWhiteLightCfg>
      <!--opt, object-->
      <brightnessRegulatMode>
        <!--opt, enum, subType:string-->auto
      </brightnessRegulatMode>
      <LightCtrlList>
        <!--opt, array, subType:object-->
        <LightCtrl>
          <!--opt, object-->
          <lightID>
            <!--opt, int, range:[1,4]-->1
          </lightID>
          <brightness>
            <!--opt, int, range:[0,100]-->80
          </brightness>
        </LightCtrl>
      </LightCtrlList>
    </PanoramicWhiteLightCfg>
    <PanoramicIrLightCfg>
      <!--opt, object-->
      <brightnessRegulatMode>
        <!--opt, enum, subType:string-->auto
      </brightnessRegulatMode>
      <LightCtrlList>
        <!--opt, array, subType:object-->
        <LightCtrl>
          <!--opt, object-->
          <lightID>
            <!--opt, int, range:[1,4]-->1
          </lightID>
          <brightness>
            <!--opt, int, range:[0,100]-->80
          </brightness>
        </LightCtrl>
      </LightCtrlList>
    </PanoramicIrLightCfg>
    <whiteLightEnhanceEnabled>
      <!--opt, bool-->true
    </whiteLightEnhanceEnabled>
  </SupplementLight>
  <OpticalDehaze>
    <!--opt, object-->
    <enabled>
      <!--opt, bool-->true
    </enabled>
  </OpticalDehaze>
  <ManualRanging>
    <!--opt, object-->
    <enabled>
      <!--opt, bool-->true
    </enabled>
  </ManualRanging>
  <OIS>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->close
    </mode>
    <OISLevel>
      <!--opt, enum, subType:string-->high
    </OISLevel>
    <OISSensitivity>
      <!--opt, string-->test
    </OISSensitivity>
  </OIS>
  <isSupportLaserSpotManual>
    <!--opt, bool-->true
  </isSupportLaserSpotManual>
  <isSupportLaserSpotAdjustment>
    <!--opt, bool-->true
  </isSupportLaserSpotAdjustment>
  <DigitalZoom>
    <!--opt, object-->
    <ZoomRatio>
      <!--opt, enum, subType:string-->1x
    </ZoomRatio>
  </DigitalZoom>
  <Palettes>
    <!--opt, object-->
    <mode>
      <!--opt, enum, subType:string-->WhiteHot
    </mode>
    <ColorateTarget>
      <!--opt, object-->
      <ColorateTargetModeList>
        <!--opt, array, subType:object-->
        <ColorateTargetMode>
          <!--opt, object-->
          <id>
            <!--req, int-->0
          </id>
          <mode>
            <!--req, enum, subType:string-->colorateHotAreae
          </mode>
          <enabled>
```

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```
            <enabled>
              <!--req, bool-->true
            </enabled>
            <TemperatureLimit>
              <!--req, object-->
              <minTemperature>
                <!--opt, float, range:[-20,150]-->0.000
              </minTemperature>
              <maxTemperature>
                <!--opt, float, range:[-20,150]-->0.000
              </maxTemperature>
            </TemperatureLimit>
            <Color>
              <!--req, object-->
              <R>
                <!--req, int, range:[0,255]-->0
              </R>
              <G>
                <!--req, int, range:[0,255]-->0
              </G>
              <B>
                <!--req, int, range:[0,255]-->0
              </B>
            </Color>
          </ColorateTargetMode>
        </ColorateTargetModeList>
      </ColorateTarget>
    </Palettes>
    <ExposureSync>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
    </ExposureSync>
    <BrightnessSuddenChangeSuppression>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
    </BrightnessSuddenChangeSuppression>
    <TargetEnhancement>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
    </TargetEnhancement>
    <TempRange>
      <!--req, object-->
      <mode>
        <!--req, enum, subType:string-->automatic
      </mode>
      <temperatureUpperLimit>
        <!--opt, float, range:[-20,550], unit:℃-->0.000
      </temperatureUpperLimit>
      <temperatureLowerLimit>
        <!--opt, float, range:[-20,550], unit:℃-->0.000
      </temperatureLowerLimit>
    </TempRange>
    <Beauty>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <whiteningStrength>
        <!--opt, int, range:[0,10]-->1
      </whiteningStrength>
      <skinSmoothingStrength>
        <!--opt, int, range:[0,10]-->1
      </skinSmoothingStrength>
    </Beauty>
  </ImageChannel>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.5 Get the image parameters capability of a specified channel

Request URL

GET /ISAPI/ContentMgmt/ImageProxy/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <NoiseReduce>
    <!--ro, opt, object-->
    <mode opt="close,general,advanced">
      <!--ro, req, enum, subType:string, attr:opt{req, string}-->close
    </mode>
    <GeneralMode>
      <!--ro, opt, object-->
      <generalLevel min="0" max="10">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->1
      </generalLevel>
    </GeneralMode>
  </NoiseReduce>
  <Color>
    <!--ro, opt, object-->
    <brightnessLevel def="50" min="0" max="100">
      <!--ro, opt, int, attr:def{req, int},min{req, int},max{req, int}-->1
    </brightnessLevel>
    <contrastLevel def="50" min="0" max="100">
      <!--ro, opt, int, attr:def{req, int},min{req, int},max{req, int}-->1
    </contrastLevel>
    <saturationLevel def="50" min="0" max="100">
      <!--ro, opt, int, attr:def{req, int},min{req, int},max{req, int}-->1
    </saturationLevel>
    <hueLevel min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </hueLevel>
    <sharpnessLevel min="0" max="100" def="0">
      <!--ro, opt, int, attr:min{req, int},max{req, int},def{req, int}-->0
    </sharpnessLevel>
  </Color>
  <enableImageLossDetection opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enableImageLossDetection>
</ImageChannel>
```

dracku@gmail.com
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14.2.6.6 Delete a specific digital channel

Request URL

DELETE /ISAPI/ContentMgmt/InputProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

drackum@gmail.com
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14.2.6.7 Set the device parameters of a digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/deviceInfo

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName>
    <!--req, string, device name-->test
  </deviceName>
  <deviceID>
    <!--req, string, UUID device No.-->fb8d4000-8158-11b5-8450-b4a38245654c
  </deviceID>
  <deviceDescription>
    <!--opt, string, device description-->test
  </deviceDescription>
  <deviceLocation>
    <!--opt, string, device location-->hangzhou
  </deviceLocation>
  <deviceStatus>
    <!--opt, enum, device status, subType:string, desc:"normal", "abnormal"-->normal
  </deviceStatus>
  <DetailAbnormalStatus>
    <!--opt, object, error status details-->
    <hardDiskFull>
      <!--opt, bool, the disk is full-->true
    </hardDiskFull>
    <hardDiskError>
      <!--opt, bool, the disk error-->true
```

```
    </hardDiskError>
    <ethernetBroken>
      <!--opt, bool, network disconnected-->true
    </ethernetBroken>
    <ipaddrConflict>
      <!--opt, bool, IP address is conflicted-->true
    </ipaddrConflict>
    <illegalAccess>
      <!--opt, bool, illegal access-->true
    </illegalAccess>
    <recordError>
      <!--opt, bool, recording exception-->true
    </recordError>
    <raidLogicDiskError>
      <!--opt, bool, virtual disk exception in the array-->true
    </raidLogicDiskError>
    <spareWorkDeviceError>
      <!--opt, bool, backup device exception-->true
    </spareWorkDeviceError>
  </DetailAbnormalStatus>
  <systemContact>
    <!--opt, string, manufacturer-->STD-CGI
  </systemContact>
  <model>
    <!--req, string, device model-->iDS-9632NX-I8/X
  </model>
  <serialNumber>
    <!--req, string, serial No.-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
  </serialNumber>
  <macAddress>
    <!--req, string, MAC address-->44:47:cc:c8:d9:e4
  </macAddress>
  <firmwareVersion>
    <!--req, string, firmware version No.-->V4.1.40
  </firmwareVersion>
  <firmwareReleasedDate>
    <!--opt, string, firmware release date-->build 181129
  </firmwareReleasedDate>
  <encoderVersion>
    <!--opt, string, device encoder version No.-->V7.3
  </encoderVersion>
  <encoderReleasedDate>
    <!--opt, string, compile date of the device encoder version-->build 20161223
  </encoderReleasedDate>
  <bootVersion>
    <!--opt, string, boot version-->V1.3.4
  </bootVersion>
  <bootReleasedDate>
    <!--opt, string, release date of boot-->100316
  </bootReleasedDate>
  <panelVersion>
    <!--opt, string, panel version-->V1.0
  </panelVersion>
  <hardwareVersion>
    <!--opt, string, hardware version-->0x0
  </hardwareVersion>
  <decoderVersion>
    <!--opt, string, device decoder version-->V1.0
  </decoderVersion>
  <decoderReleasedDate>
    <!--opt, string, compile date of the device decoder version-->build 20161223
  </decoderReleasedDate>
  <softwareVersion>
    <!--opt, string, software version-->build 181129
  </softwareVersion>
  <capacity>
    <!--opt, int, device capacity, unit: MB-->10240
  </capacity>
  <usedCapacity>
    <!--opt, int, capacity usage, unit: MB-->5120
  </usedCapacity>
  <telecontrolID>
    <!--opt, int, remote control control ID, range:[1,255]-->1
  </telecontrolID>
  <supportBeep>
    <!--opt, bool, whether to support buzzer-->true
  </supportBeep>
  <supportVideoLoss>
    <!--opt, bool, whether to support video loss-->true
  </supportVideoLoss>
  <firmwareVersionInfo>
    <!--opt, string, firmware version information-->1
  </firmwareVersionInfo>
  <actualFloorNum>
    <!--opt, int, actual number of floors, range:[1,255]-->1
  </actualFloorNum>
  <localZoneNum>
    <!--opt, int, number of local zones-->1
  </localZoneNum>
  <alarmOutNum>
    <!--opt, int, number of alarm outputs-->1
  </alarmOutNum>
  <distanceResolution>
```

drackum@gmail.com
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```
    <!--opt, float, distance resolution-->0.000
  </distanceResolution>
  <angleResolution>
    <!--opt, float, angle resolution-->0.000
  </angleResolution>
  <speedResolution>
    <!--opt, float, speed resolution-->0.000
  </speedResolution>
  <detectDistance>
    <!--opt, float, detection distance-->0.000
  </detectDistance>
  <relayNum>
    <!--opt, int, number of local relays-->1
  </relayNum>
  <electroLockNum>
    <!--opt, int, number of local electronic locks-->1
  </electroLockNum>
  <RS485Num>
    <!--opt, int, number of local RS-485-->1
  </RS485Num>
  <radarVersion>
    <!--opt, string, radar version-->test
  </radarVersion>
  <cameraModuleVersion>
    <!--opt, string, camera module version-->test
  </cameraModuleVersion>
  <mainversion>
    <!--opt, int, main version No., range:[1,255]-->1
  </mainversion>
  <subversion>
    <!--opt, int, sub version No., range:[1,255]-->1
  </subversion>
  <upgradeversion>
    <!--opt, int, updated version No., range:[1,255]-->1
  </upgradeversion>
  <customizeversion>
    <!--opt, int, customized version No., range:[1,255]-->1
  </customizeversion>
  <companyName>
    <!--opt, string, manufacturer name-->test
  </companyName>
  <copyright>
    <!--opt, string, copyright Information-->test
  </copyright>
  <systemName>
    <!--opt, enum, storage system name, subType:string, desc:"storageManagement" (storage management system), "distributedStorageManagement" (distributed
storage management system)-->storageManagement
  </systemName>
  <systemStatus>
    <!--opt, enum, system status, subType:string, desc:"configured", "unConfigured" (not configured)-->configured
  </systemStatus>
  <isLeaderDevice>
    <!--opt, bool, whether it is the corresponding device of the resource IP address-->true
  </isLeaderDevice>
  <clusterVersion>
    <!--opt, string, system cluster version, desc:this node is valid when the value of isLeaderDevice is true-->test
  </clusterVersion>
  <centralStorageVersion>
    <!--opt, string, center storage version, range:[1,16]-->test
  </centralStorageVersion>
  <powerOnMode>
    <!--opt, enum, startup mode, subType:string, desc:"button" (press button to power on), "adapter" (connect adapter to power on)-->button
  </powerOnMode>
  <customizedInfo>
    <!--opt, string, customized project No., desc:order No. of the customization project. For baseline devices,this node is empty; for custom devices,the
order No. of the customization project will be returned by this node-->test
  </customizedInfo>
  <charEncodeFormat>
    <!--opt, enum, character encoding, subType:string, desc:GB2312:GB2312;GBK:GBK-->GB2312
  </charEncodeFormat>
  <languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech">
    <!--opt, enum, language type, subType:string, attr:opt{req, string}, desc:"Chinese", "English", "Spanish", "Portuguese", "Italian", "French", "Russian",
"Turkish", "Greek", "Czech"-->chinese
  </languageType>
  <DockStation>
    <!--opt, object, dock station information configuration-->
    <Platform>
      <!--opt, object, platform dock station information configuration-->
      <ip>
        <!--opt, string, IP address-->test
      </ip>
      <port>
        <!--opt, int, communication port-->1
      </port>
      <userName>
        <!--req, string, user name-->test
      </userName>
      <password>
        <!--req, string, password-->test
      </password>
    </Platform>
    <centralStorageBackupEnabled>
      <!--opt, bool, whether to enable center storage backup-->true
    </centralStorageBackupEnabled>
```

dracku m@gmail.com
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```
    </centralStorageBackupEnabled>
  </DockStation>
  <webVersion>
    <!--opt, string, Web version No.-->test
  </webVersion>
  <deviceRFProgramVersion>
    <!--opt, string, version No. of the device's RF (Radio Frequency) program-->test
  </deviceRFProgramVersion>
  <securityModuleSerialNo>
    <!--opt, string, security module serial No.-->test
  </securityModuleSerialNo>
  <securityModuleVersion>
    <!--opt, string, security module version No.-->test
  </securityModuleVersion>
  <securityChipVersion>
    <!--opt, string, security chip version No.-->test
  </securityChipVersion>
  <securityModuleKeyVersion>
    <!--opt, string, security module key version No.-->test
  </securityModuleKeyVersion>
  <UIDLampRecognition>
    <!--opt, object, information of the UID lamp recognition device-->
    <enabled>
      <!--opt, bool, whether to enable the function-->true
    </enabled>
  </UIDLampRecognition>
  <confDeviceIdPrefix>
    <!--opt, bool, whether the meeting uses the device name as its prefix-->true
  </confDeviceIdPrefix>
  <OEMCode>
    <!--opt, enum, manufacturer OME code, subType:int, desc:1 (Hikvision device), 0 (neutral device)-->1
  </OEMCode>
  <bootTime>
    <!--opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
  </bootTime>
  <intelligentAnalysisEngineModel>
    <!--opt, string, engine model-->test
  </intelligentAnalysisEngineModel>
</DeviceInfo>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

dracktm@gmail.com
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14.2.6.8 Get the device parameters of a digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/deviceInfo

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device information, attr:version{opt, string, protocolVersion}-->
```

```
<!--ro, opt, object, device information, attr:version[opt, string, protocolVersion]-->
<deviceName>
  <!--ro, req, string, device name-->test
</deviceName>
<deviceID>
  <!--ro, req, string, UUID device No.-->fb8d4000-8158-11b5-8450-b4a38245654c
</deviceID>
<deviceDescription>
  <!--ro, opt, string, device description-->test
</deviceDescription>
<deviceLocation>
  <!--ro, opt, string, actual location of the device-->hangzhou
</deviceLocation>
<deviceStatus>
  <!--ro, opt, enum, device status, subType:string, desc:"normal", "abnormal"-->normal
</deviceStatus>
<DetailAbnormalStatus>
  <!--ro, opt, object, error status details-->
  <hardDiskFull>
    <!--ro, opt, bool, the disk is full-->true
  </hardDiskFull>
  <hardDiskError>
    <!--ro, opt, bool, the disk error-->true
  </hardDiskError>
  <ethernetBroken>
    <!--ro, opt, bool, network disconnected-->true
  </ethernetBroken>
  <ipaddrConflict>
    <!--ro, opt, bool, IP address is conflicted-->true
  </ipaddrConflict>
  <illegalAccess>
    <!--ro, opt, bool, illegal access-->true
  </illegalAccess>
  <recordError>
    <!--ro, opt, bool, recording exception-->true
  </recordError>
  <raidLogicDiskError>
    <!--ro, opt, bool, virtual disk exception in the array-->true
  </raidLogicDiskError>
  <spareWorkDeviceError>
    <!--ro, opt, bool, backup device exception-->true
  </spareWorkDeviceError>
</DetailAbnormalStatus>
<systemContact>
  <!--ro, opt, string, manufacturer-->STD-CGI
</systemContact>
<model>
  <!--ro, req, string, device model-->iDS-9632NX-I8/X
</model>
<serialNumber>
  <!--ro, req, string, serial No.-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
</serialNumber>
<macAddress>
  <!--ro, req, string, MAC address-->44:47:cc:c8:d9:e4
</macAddress>
<firmwareVersion>
  <!--ro, req, string, firmware version No.-->V4.1.40
</firmwareVersion>
<firmwareReleasedDate>
  <!--ro, opt, string, firmware release date-->build 181129
</firmwareReleasedDate>
<encoderVersion>
  <!--ro, opt, string, device encoder version No.-->V7.3
</encoderVersion>
<encoderReleasedDate>
  <!--ro, opt, string, compile date of the device encoder version-->build 20161223
</encoderReleasedDate>
<bootVersion>
  <!--ro, opt, string, boot version-->V1.3.4
</bootVersion>
<bootReleasedDate>
  <!--ro, opt, string, release date of boot-->100316
</bootReleasedDate>
<panelVersion>
  <!--ro, opt, string, panel version-->V1.0
</panelVersion>
<hardwareVersion>
  <!--ro, opt, string, hardware version-->0x0
</hardwareVersion>
<decoderVersion>
  <!--ro, opt, string, device decoder version-->V1.0
</decoderVersion>
<decoderReleasedDate>
  <!--ro, opt, string, compile date of the device decoder version-->build 20161223
</decoderReleasedDate>
<softwareVersion>
  <!--ro, opt, string, software version-->build 181129
</softwareVersion>
<capacity>
  <!--ro, opt, int, device capacity, unit: MB-->10240
</capacity>
<usedCapacity>
  <!--ro, opt, int, capacity usage, unit: MB-->5120
</usedCapacity>
```

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```
<telecontrolID>
  <!--ro, opt, int, remote control ID, range:[1,255]-->1
</telecontrolID>
<supportBeep>
  <!--ro, opt, bool, whether to support buzzer-->true
</supportBeep>
<supportVideoLoss>
  <!--ro, opt, bool, whether it supports video loss-->true
</supportVideoLoss>
<firmwareVersionInfo>
  <!--ro, opt, string, firmware version information-->1
</firmwareVersionInfo>
<actualFloorNum>
  <!--ro, opt, int, actual number of floors, range:[1,255]-->1
</actualFloorNum>
<localZoneNum>
  <!--ro, opt, int, number of local zones-->1
</localZoneNum>
<alarmOutNum>
  <!--ro, opt, int, number of alarm outputs-->1
</alarmOutNum>
<distanceResolution>
  <!--ro, opt, float, distance resolution-->0.000
</distanceResolution>
<angleResolution>
  <!--ro, opt, float, angle resolution-->0.000
</angleResolution>
<speedResolution>
  <!--ro, opt, float, speed resolution-->0.000
</speedResolution>
<detectDistance>
  <!--ro, opt, float, detection distance-->0.000
</detectDistance>
<relayNum>
  <!--ro, opt, int, number of local relays-->1
</relayNum>
<electroLockNum>
  <!--ro, opt, int, number of local electronic locks-->1
</electroLockNum>
<RS485Num>
  <!--ro, opt, int, number of local RS-485-->1
</RS485Num>
<radarVersion>
  <!--ro, opt, string, radar version-->test
</radarVersion>
<cameraModuleVersion>
  <!--ro, opt, string, camera module version-->test
</cameraModuleVersion>
<mainversion>
  <!--ro, opt, int, main version No., range:[1,255]-->1
</mainversion>
<subversion>
  <!--ro, opt, int, sub version No., range:[1,255]-->1
</subversion>
<upgradeversion>
  <!--ro, opt, int, updated version No., range:[1,255]-->1
</upgradeversion>
<customizeversion>
  <!--ro, opt, int, customized version No., range:[1,255]-->1
</customizeversion>
<companyName>
  <!--ro, opt, string, manufacturer name-->test
</companyName>
<copyright>
  <!--ro, opt, string, copyright Information-->test
</copyright>
<systemName>
  <!--ro, opt, enum, storage system name, subType:string, desc:“storageManagement” (storage management system), “distributedStorageManagement”
(distributed storage management system)-->storageManagement
</systemName>
<systemStatus>
  <!--ro, opt, enum, system status, subType:string, desc:“configured”, “unConfigured” (not configured)-->configured
</systemStatus>
<isLeaderDevice>
  <!--ro, opt, bool, whether it is the corresponding device of the resource IP address-->true
</isLeaderDevice>
<clusterVersion>
  <!--ro, opt, string, system cluster version, desc:this node is valid when the value of isLeaderDevice is true-->test
</clusterVersion>
<centralStorageVersion>
  <!--ro, opt, string, center storage version, range:[1,16]-->test
</centralStorageVersion>
<powerOnMode>
  <!--ro, opt, enum, startup mode, subType:string, desc:device startup mode: "button"-press button to power on (default),"adapter"-connect adapter to
power on-->button
</powerOnMode>
<customizedInfo>
  <!--ro, opt, string, customized project No., desc:no value will be returned if it is baseline device. custom project No. will be returned if is custom
device-->test
</customizedInfo>
<charEncodeFormat>
  <!--ro, opt, enum, character encoding, subType:string, desc:GB2312:GB2312;GBK:GBK-->GB2312
</charEncodeFormat>
```

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```
    <languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech">
      <!--ro, opt, enum, language type, subType:string, attr:opt{req, string}, desc:"Chinese", "English", "Spanish", "Portuguese", "Italian", "French",
  "Russian", "Turkish", "Greek", "Czech"-->chinese
    </languageType>
    <DockStation>
      <!--ro, opt, object, dock station information configuration-->
      <Platform>
        <!--ro, opt, object, platform dock station information configuration-->
        <ip>
          <!--ro, opt, string, IP address-->test
        </ip>
        <port>
          <!--ro, opt, int, communication port-->1
        </port>
        <userName>
          <!--ro, req, string, user name-->test
        </userName>
        <password>
          <!--ro, req, string, password-->test
        </password>
      </Platform>
      <centralStorageBackupEnabled>
        <!--ro, opt, bool, whether to enable central storage backup-->true
      </centralStorageBackupEnabled>
    </DockStation>
    <webVersion>
      <!--ro, opt, string, Web version No.-->test
    </webVersion>
    <deviceRFProgramVersion>
      <!--ro, opt, string, version No. of the device's RF (Radio Frequency) program-->test
    </deviceRFProgramVersion>
    <securityModuleSerialNo>
      <!--ro, opt, string, security module serial No.-->test
    </securityModuleSerialNo>
    <securityModuleVersion>
      <!--ro, opt, string, version No. of the security module-->test
    </securityModuleVersion>
    <securityChipVersion>
      <!--ro, opt, string, security chip version-->test
    </securityChipVersion>
    <securityModuleKeyVersion>
      <!--ro, opt, string, security module key version-->test
    </securityModuleKeyVersion>
    <UIDLampRecognition>
      <!--ro, opt, object, information of the UID lamp recognition device-->
      <enabled>
        <!--ro, opt, bool, whether to enable the function-->true
      </enabled>
    </UIDLampRecognition>
    <confDeviceIdPrefix>
      <!--ro, opt, bool, whether the meeting uses the device name as its prefix-->true
    </confDeviceIdPrefix>
    <OEMCode>
      <!--ro, opt, enum, manufacturer OME code, subType:int, desc:1 (Hikvision device), 0 (neutral device)-->1
    </OEMCode>
    <bootTime>
      <!--ro, opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
    </bootTime>
    <intelligentAnalysisEngineModel>
      <!--ro, opt, string, engine model-->test
    </intelligentAnalysisEngineModel>
  </DeviceInfo>
```

dr'ackum@gmail.com
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14.2.6.9 Set parameters of a specific digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/password

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelPassword xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <oldPassword>
    <!--req, string, old password-->test
  </oldPassword>
  <newPassword>
    <!--req, string, new password-->test
  </newPassword>
</InputProxyChannelPassword>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.10 Set the iris control parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/iris

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IrisData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <iris>
    <!--opt, int, range:[-100,100]-->1
  </iris>
</IrisData>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.11 Get the capability of activating digital channels

Request URL

drackum@gmail.com
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GET /ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoSourceActivationCapability xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <ActivateByChannel>
    <!--ro, opt, object-->
  </ActivateByChannel>
  <ActivateByAddress>
    <!--ro, opt, object-->
  </ActivateByAddress>
</VideoSourceActivationCapability>
```

14.2.6.12 Activate the network devices

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/activate

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoSourceActivation xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <userName>
    <!--req, string, user name, range:[1,32]-->test
  </userName>
  <passwordType>
    <!--opt, enum, password types, subType:string, desc:"default" (NVR password), "selfDefine", "IPCActivatePasswd" (activation password)-->default
  </passwordType>
  <password>
    <!--req, string, password, range:[1,16]-->test
  </password>
  <ActivationDevice>
    <!--opt, object-->
    <mode>
      <!--req, enum, "channel,descriptor", subType:string, desc:"channel, descriptor"-->channel
    </mode>
    <channel>
      <!--opt, object, channel information, desc:channel information-->
      <dynVideoInputChannelID>
        <!--req, int-->1
      </dynVideoInputChannelID>
    </channel>
    <descriptor>
      <!--opt, object-->
      <macAddress>
        <!--req, string, MAC address, range:[1,48]-->test
      </macAddress>
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"hostname" (host name), "ipaddress" (ip address)-->hostname
      </addressingFormatType>
      <hostName>
        <!--opt, string, host name, range:[1,64]-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, range:[1,32]-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6 address, range:[1,128]-->test
      </ipv6Address>
      <portNo>
        <!--req, int, port No., range:[1,65535]-->1
      </portNo>
    </descriptor>
  </ActivationDevice>
</VideoSourceActivation>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.13 Get the digital channel access capability

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelListCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, digital channel access management capability, attr:version{req, string, protocolVersion}-->
  <InputProxyChannel>
    <!--ro, opt, object, input channel-->
    <id>
      <!--ro, req, string, index-->test
    </id>
    <name>
      <!--ro, opt, string, name, range:[1,32]-->test
    </name>
    <sourceInputPortDescriptor>
      <!--ro, req, object, input channel description-->
      <addressingFormatType opt="ipaddress,hostname">
        <!--ro, req, enum, address type, subType:string, attr:opt{req, string}, desc:"hostname", "ipaddress". It is invalid when the protocol type is
GB28181-->hostname
      </addressingFormatType>
      <hostName min="1" max="64">
        <!--ro, opt, string, domain name, range:[1,64], attr:min{req, int},max{req, int}, desc:it is invalid when the protocol type is GB28181-->test
      </hostName>
      <ipAddress min="1" max="32">
        <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:it is invalid when the protocol type is GB28181-->test
      </ipAddress>
      <ipv6Address min="1" max="128">
        <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:it is invalid when the protocol type is GB28181-->test
      </ipv6Address>
      <managePortNo min="1" max="65535">
        <!--ro, req, int, port No. under management, attr:min{req, int},max{req, int}-->1
      </managePortNo>
      <srcInputPort min="1" max="256">
        <!--ro, req, string, device channel No., range:[1,256], attr:min{req, int},max{req, int}-->test
      </srcInputPort>
      <userName min="1" max="32">
        <!--ro, req, string, user name, range:[1,32], attr:min{req, int},max{req, int}-->test
      </userName>
      <password min="1" max="16">
        <!--ro, req, string, password, range:[1,16], attr:min{req, int},max{req, int}-->test
      </password>
      <streamType opt="auto,tcp,udp">
        <!--ro, opt, enum, streaming protocol type, subType:string, attr:opt{req, string}, desc:"auto", "TCP", "UDP"-->auto
      </streamType>
      <deviceID min="1" max="32">
        <!--ro, opt, string, device ID, range:[1,32], attr:min{req, int},max{req, int}-->test
      </deviceID>
      <activateEnabled opt="true,false">
        <!--ro, opt, bool, whether the device needs to be activated, attr:opt{req, string}, desc:if this node does not exist, it indicates that the device
has already been activated. Application scenario: supports adding an unactivated device to the NVR channel. If this node is true, the device is unactivated
and needs to be activated by the NVR. The NVR will auto-activate the device (MAC address is needed because IP access is unavailable for unactivated
devices), and will add the device to NVR-->true
      </activateEnabled>
      <macAddress min="0" max="48">
        <!--ro, opt, string, MAC address, range:[0,48], attr:min{req, int},max{req, int}, desc:it is required when the value of activateEnabled is true--
```

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```
>test
      </macAddress>
      <cameraType>
        <!--ro, opt, enum, camera type, subType:string, desc:"", "", "teacherLocation" (teacher locating), "studentLocation" (student locating),
"teacherFull" (teacher full view), "studentFull" (student full view), "media", "blackboard",  "blackboardLocation", "IPCamera", "IPDome", "none"-->none
      </cameraType>
      <streamMode opt="main,sub">
        <!--ro, opt, enum, stream type, subType:string, attr:opt{req, string}, desc:"main", "sub"-->main
      </streamMode>
      <getSubStream opt="true,false">
        <!--ro, opt, bool, whether to get sub-stream, attr:opt{req, string}, desc:when this node is not configured, its value is true by default, which
means the sub-streaming will start after the device is connected-->true
      </getSubStream>
    </sourceInputPortDescriptor>
    <online opt="true,false">
      <!--ro, opt, bool, whether the channel is online, attr:opt{req, string}-->true
    </online>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the channel, attr:opt{req, string}, desc:the default value is true-->true
    </enabled>
    <anrInterval min="1" max="24">
      <!--ro, opt, int, time interval for automatic network replenishment, attr:min{req, int},max{req, int}-->1
    </anrInterval>
  </InputProxyChannel>
  <FilterchannelsByIp>
    <!--ro, opt, bool, whether filtering channel information by IP address is supported-->true
  </FilterchannelsByIp>
  <FilterchannelsByStreamID>
    <!--ro, opt, bool, whether filtering channel information by stream ID is supported-->true
  </FilterchannelsByStreamID>
  <isSupportSmartConfig>
    <!--ro, opt, bool, -ro,
desc:search for connected wireless network camera automatically and then activate and add the network camera. The automatic match period is 120 seconds. The
match process will end as soon as the number of channels reaches the upper limit.
    If its value is true, it indicates that the following URLs are supported:
PUT /ISAPI/ContentMgmt/InputProxy/channels/smartConfig?format=json;
GET ISAPI/ContentMgmt/InputProxy/channels/smartConfig/status?format=json;
PUT /ISAPI/ContentMgmt/InputProxy/channels/smartConfig/stop?format=json-->true
  </isSupportSmartConfig>
  <isSupportDevIndex>
    <!--ro, opt, bool, whether transparent transmission of sub-devices ID is supported, desc:related URLs: /ISAPI/ContentMgmt/InputProxy/channels. Each
channel corresponds to a devIndex node. The sub-devices do not include the gateway sub-devices in /ISAPI/ContentMgmt/DeviceMgmt/addDevice?format=json-->true
  </isSupportDevIndex>
  <displayGetStream>
    <!--ro, opt, bool, whether the node configuration is displayed on the web channel configuration page, desc:currently, the configuration is not displayed
on the NVR channel page, and is displayed on the AI DeepinMind channel page-->true
  </displayGetStream>
</InputProxyChannelListCap>
```

14.2.6.14 Get status of all digital channels

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/status

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelStatusList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, channel status list, subType:object, attr:version{req, string, protocolVersion}-->
  <InputProxyChannelStatus>
    <!--ro, opt, object, channel status-->
    <id>
      <!--ro, req, string, index-->test
    </id>
    <sourceInputPortDescriptor>
      <!--ro, req, object, input channel description-->
      <addressingFormatType>
        <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress". It is invalid when the protocol type is GB28181-->hostname
      </addressingFormatType>
      <ipAddress>
        <!--ro, opt, string, IPv4 address, range:[1,32], desc:it is invalid when the protocol type is GB28181-->test
      </ipAddress>
      <managePortNo>
        <!--ro, req, int, port No. under management, range:[0,65535]-->0
      </managePortNo>
      <srcInputPort>
        <!--ro, req, string, device channel No., range:[1,16]-->test
      </srcInputPort>
      <userName>
        <!--ro, req, string, user name, range:[0,32]-->test
      </userName>
      <streamType>
        <!--ro, opt, enum, streaming protocol type, subType:string, desc:“auto”, “tcp”, “udp”-->auto
      </streamType>
      <deviceID>
        <!--ro, opt, string, device ID, range:[1,20], desc:the node exists only when the protocol type is GB28181-->test
      </deviceID>
    </sourceInputPortDescriptor>
    <online>
      <!--ro, req, bool, whether the channel is online-->true
    </online>
    <streamingProxyChannelIdList>
      <!--ro, req, array, streaming channel list, subType:object-->
      <streamingProxyChannelId>
        <!--ro, opt, string, streaming channel-->test
      </streamingProxyChannelId>
    </streamingProxyChannelIdList>
    <relatedIOProxy>
      <!--ro, opt, object, linked IO port-->
      <inputProxyPortIdList>
        <!--ro, opt, array, input port No. list, subType:object-->
        <inputProxyPortId>
          <!--ro, opt, string, input port No.-->test
        </inputProxyPortId>
      </inputProxyPortIdList>
      <outputProxyPortIdList>
        <!--ro, opt, array, output port No. list, subType:object-->
        <outputProxyPortId>
          <!--ro, opt, string, output port No.-->test
        </outputProxyPortId>
      </outputProxyPortIdList>
    </relatedIOProxy>
    <chanDetectResult>
      <!--ro, opt, enum, channel detection status, subType:string, desc:"connect" (connected), "overSysBandwidth" (exceeded the system bandwidth),
"domainError" (domain error), "ipcStreamFail" (network camera streaming failed), "connecting" (connecting), "chanNoError" (incorrect channel number),
"ipAddrConflictWithDev" (IP address is conflicted with current device’s), "ipAddrConflicWithIpc" (IP address is conflicted with other device's.),
"errorUserNameOrPasswd" (incorrect user name or password), "netUnreachable" (network is unreachable), "unknownError" (unknown error), "notExist" (not
exist), "ipcStreamTypeNotSupport" (the network camera stream type is not supported), "ipcResolutionNotSupport" (the network camera resolution is not
supported), "userLocked" (the user is locked), "userNotExist" (the user does not exist), "ipcUnregistered" (network camera is not registered),
"ipcNotActivated" (network camera is inactivated), "poePortDetecting" (detecting POE port), "uploadingCloudFile" (uploading files to cloud storage),
"certificateValidationError" (certification validation failed), "ONVIFRegistrationError" (ONVIF registration failed), "streamingParameters" (streaming
parameter error), "sleep", "wakeUping" (waking up)-->connect
    </chanDetectResult>
    <SecurityStatus>
      <!--ro, opt, object, security status-->
      <PasswordStatus>
        <!--ro, opt, enum, password status, subType:string, desc:“ipcNotActivated” (IPC is not activated), “risk”, “medium”, “strong:”, “invalid”--
>ipcNotActivated
      </PasswordStatus>
    </SecurityStatus>
  </InputProxyChannelStatus>
</InputProxyChannelStatusList>
```

drackum@gmail.com
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14.2.6.15 Set parameters of all digital channels

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, digital channel input information list, subType:object, attr:version{req, string, protocolVersion}-->
  <InputProxyChannel>
    <!--opt, object, digital channel input information-->
    <id>
      <!--req, string, index-->test
    </id>
    <name>
      <!--opt, string, name-->test
    </name>
    <sourceInputPortDescriptor>
      <!--req, object, input channel source description-->
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"hostname", "ipaddress", it is invalid when the protocol type is GB28181-->hostname
      </addressingFormatType>
      <hostName>
        <!--opt, string, domain name, range:[1,64], desc:it is invalid when the protocol type is GB28181-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, range:[1,32], desc:it is invalid when the protocol type is GB28181-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6 address, range:[1,128], desc:it is invalid when the protocol type is GB28181-->test
      </ipv6Address>
      <managePortNo>
        <!--req, int, port No., range:[1,65535]-->1
      </managePortNo>
      <srcInputPort>
        <!--req, string, device channel No., range:[1,16]-->test
      </srcInputPort>
      <userName>
        <!--req, string, user name, range:[1,32]-->test
      </userName>
      <password>
        <!--req, string, password, range:[1,16], desc:this node is empty in the response message-->test
      </password>
      <connMode>
        <!--opt, enum, connect mode, subType:string, desc:“plugplay” (plug and play), “manual”-->plugplay
      </connMode>
      <streamType>
        <!--opt, enum, streaming protocol type, subType:string, desc:"auto", "tcp", "udp"-->auto
      </streamType>
      <deviceID>
        <!--opt, string, device ID, range:[1,20], desc:the node exists only when the protocol type is GB28181-->test
      </deviceID>
      <getStreamMode>
        <!--opt, enum, streaming mode, subType:string, desc:"active", "passive". The node exists only when the protocol type is GB28181-->active
      </getStreamMode>
      <cameraType>
        <!--opt, enum, camera type, subType:string, desc:"", "", "teacherLocation" (teacher locating), "studentLocation" (student locating), "teacherFull"
(teacher full view), "studentFull" (student full view), "media", "blackboard",  "blackboardLocation", "none"-->none
      </cameraType>
      <isDoubleAdd>
        <!--opt, bool, add two streams of a same camera, desc:it is used when DeepinView is added to education sharing server-->true
      </isDoubleAdd>
      <streamUrl>
        <!--opt, string, RTSP streaming URL, range:[1,256], desc:if this node exists, use this URL for streaming; if not, follow the original rule-->test
      </streamUrl>
      <getStream>
        <!--opt, bool, whether to stream, desc:its value is true by default, which means getting device information after the device is connected but not
recording video-->true
      </getStream>
      <RTSP>
        <!--opt, object, RTSP related nodes-->
        <userName>
          <!--req, string, user name, range:[1,64], desc:"admin" by default-->test
        </userName>
        <password>
          <!--req, string, password, range:[0,64], desc:empty by default-->test
        </password>
        <portNo>
          <!--ro, opt, int, RTSP streaming port, range:[1,65535], step:1, desc:default port is 554-->554
        </portNo>
      </RTSP>
      <customChanID>
        <!--opt, string, custom camera channel No., range:[1,64], desc:support configuring remotely. When adding sub devices, the platform applies a custom
ID to the accessed digital channel. The ID is the unique identifier with 63 bytes. After the platform applies, the digital channel will upload the ID when
an alarm is triggered, and the platform can find the corresponding device through the ID-->test
      </customChanID>
      <model>
        <!--ro, opt, string, device model, range:[1,64]-->iDS-9632NX-I8/X
      </model>
      <serialNumber>
        <!--ro, opt, string, serial No., range:[0,64]-->test
      </serialNumber>
      <firmwareVersion>
        <!--ro, opt, string, firmware version, range:[0,64]-->test
      </firmwareVersion>
      <getSubStream>
        <!--opt, bool, whether to get sub-stream, desc:when this node is not configured, its value is true by default, which means the sub-streaming will
start after the device is connected-->true
```

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```
      </getSubStream>
      <cloudStorageCfg>
        <!--opt, object, dep:or,{$.InputProxyChannel.sourceInputPortDescriptor.proxyProtocol,eq,cloudStrage}-->
        <accessKey>
          <!--req, string, range:[1,64]-->test
        </accessKey>
        <secretKey>
          <!--req, string, range:[1,64]-->test
        </secretKey>
        <encoderID>
          <!--opt, string, range:[1,64]-->test
        </encoderID>
      </cloudStorageCfg>
    </sourceInputPortDescriptor>
    <enableAnr>
      <!--opt, bool, GB23181 push-flow port, desc:return to the push-flow port after adding the GB23181-->true
    </enableAnr>
    <enableTiming>
      <!--opt, bool, whether to enable synchronizing the time of the device-->true
    </enableTiming>
    <streamID>
      <!--opt, string, stream ID, range:[1,32], desc:if this node is transmitted, the stream ID of the current channel can be specified-->test
    </streamID>
    <enabled>
      <!--opt, bool, whether to enable channel ID, desc:the default value is true-->true
    </enabled>
    <anrInterval>
      <!--ro, opt, int, time interval for automatic network replenishment, range:[1,24], unit:h, dep:and,{$.InputProxyChannel.enableAnr,eq,true}, desc:NVR
will auto-reload the video according to the time interval-->1
    </anrInterval>
  </InputProxyChannel>
</InputProxyChannelList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

drackun@gmail.com
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14.2.6.16 Get access parameters of all digital channels

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, channel list, subType:object, attr:version{req, string, protocolVersion}-->
  <InputProxyChannel>
    <!--ro, opt, object, channel information-->
    <id>
      <!--ro, req, string, index-->test
    </id>
    <name>
      <!--ro, opt, string, name-->test
    </name>
    <sourceInputPortDescriptor>
```

```
      <!--ro, req, object, input channel description-->
      <addressingFormatType>
        <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress". It is invalid when the protocol type is GB28181-->hostname
      </addressingFormatType>
      <hostName>
        <!--ro, opt, string, domain name, range:[1,64], desc:it is invalid when the protocol type is GB28181-->test
      </hostName>
      <ipAddress>
        <!--ro, opt, string, IPv4 address, range:[1,32], desc:it is invalid when the protocol type is GB28181-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address, range:[1,128], desc:it is invalid when the protocol type is GB28181-->test
      </ipv6Address>
      <managePortNo>
        <!--ro, req, int, port No. under management, range:[1,65535]-->1
      </managePortNo>
      <srcInputPort>
        <!--ro, req, string, device channel No., range:[1,16]-->test
      </srcInputPort>
      <userName>
        <!--ro, req, string, user name, range:[1,32]-->test
      </userName>
      <password>
        <!--ro, req, string, password, range:[0,16], desc:the node is empty in the returned results-->test
      </password>
      <connMode>
        <!--ro, opt, enum, connection mode, subType:string, desc:"plugplay" (plug and play), "manual"-->plugplay
      </connMode>
      <streamType>
        <!--ro, opt, enum, streaming protocol type, subType:string, desc:"auto", "tcp", "udp"-->auto
      </streamType>
      <deviceID>
        <!--ro, opt, string, device ID, range:[1,20], desc:the node exists only when the protocol type is GB28181-->test
      </deviceID>
      <getStreamMode>
        <!--ro, opt, enum, streaming mode, subType:string, desc:"active", "passive". The node exists only when the protocol type is GB28181-->active
      </getStreamMode>
      <cameraType>
        <!--ro, opt, enum, camera type, subType:string, desc:"", "", "teacherLocation" (teacher locating), "studentLocation" (student locating),
"teacherFull" (teacher full view), "studentFull" (student full view), "media", "blackboard",  "blackboardLocation", "IPCamera", "IPDome", "none"-->none
      </cameraType>
      <isDoubleAdd>
        <!--ro, opt, bool, add two streams of the same camera, desc:it is used when DeepinView is added to education sharing server-->true
      </isDoubleAdd>
      <streamUrl>
        <!--ro, opt, string, RTSP stream URL, range:[1,256], desc:if this node exists, use this URL for streaming; if not, follow the original rule-->test
      </streamUrl>
      <transMode>
        <!--ro, opt, enum, subType:int-->0
      </transMode>
      <getStream>
        <!--ro, opt, bool, whether to stream, desc:its value is true by default, which means getting device information after the device is connected and
not recording video-->true
      </getStream>
      <RTSP>
        <!--ro, opt, object, RTSP related nodes-->
        <userName>
          <!--ro, req, string, user name, range:[1,64], desc:"admin" by default-->test
        </userName>
        <password>
          <!--ro, req, string, password, range:[0,64], desc:empty by default-->test
        </password>
        <portNo>
          <!--ro, opt, int, RTSP streaming port, range:[1,65535], step:1, desc:default port is 554-->554
        </portNo>
      </RTSP>
      <customChanID>
        <!--ro, opt, string, custom camera channel ID, range:[1,64], desc:support configuring remotely. When adding sub devices, the platform applies a
custom ID to the accessed digital channel. The ID is the unique identifier with 63 bytes. After the platform applies, the digital channel will upload the ID
when an alarm is triggered, and the platform can find the corresponding device through the ID-->test
      </customChanID>
      <model>
        <!--ro, opt, string, device model, range:[1,64]-->iDS-9632NX-I8/X
      </model>
      <serialNumber>
        <!--ro, opt, string, serial No., range:[1,64]-->test
      </serialNumber>
      <firmwareVersion>
        <!--ro, opt, string, firmware version, range:[1,64]-->test
      </firmwareVersion>
      <streamMode>
        <!--ro, opt, enum, stream type, subType:string, desc:"main", "sub"-->main
      </streamMode>
      <getSubStream>
        <!--ro, opt, bool, whether to get sub-stream, desc:when this node is not configured, its value is true by default, which means the sub-streaming
will start after the device is connected-->true
      </getSubStream>
    </sourceInputPortDescriptor>
    <certificateValidationEnabled>
      <!--ro, opt, bool, whether to enable the certification validation-->true
    </certificateValidationEnabled>
    <defaultAdminPortEnabled>
      <!--ro, opt, bool, whether to enable the default communication port-->true
```

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```
    </defaultAdminPortEnabled>
    <enableAnr>
      <!--ro, opt, bool, whether to enable ANR (Automatic Network Replenishment), desc:after the camera goes offline, it will reload all the video files
during the offline period once it goes online again-->true
    </enableAnr>
    <enableTiming>
      <!--ro, opt, bool, whether to perform time sync on the device-->true
    </enableTiming>
    <streamID>
      <!--ro, opt, string, stream ID, range:[1,32], desc:if this node is transmitted, the stream ID of the current channel can be specified-->test
    </streamID>
    <online>
      <!--ro, opt, bool, whether the channel is online-->true
    </online>
    <NVRInfo>
      <!--ro, opt, object-->
      <ipAddressNVR>
        <!--ro, opt, string, range:[1,32]-->192.168.1.100
      </ipAddressNVR>
      <portNVR>
        <!--ro, opt, int, range:[0,65535]-->1
      </portNVR>
      <ipcChannelNo>
        <!--ro, opt, int, range:[0,64]-->1
      </ipcChannelNo>
    </NVRInfo>
    <enabled>
      <!--ro, opt, bool, whether to enable the channel, desc:the default value is true-->true
    </enabled>
    <devIndex>
      <!--ro, opt, string, sub-device ID, range:[0,64], desc:used to specify a sub-device to which ISAPI is transmitted-->test
    </devIndex>
    <boundAudioChannelID>
      <!--ro, opt, int, linked audio channel No., desc:if this node does not exist, no audio channel is linked to-->1
    </boundAudioChannelID>
    <anrInterval>
      <!--ro, opt, int, time interval for automatic network replenishment, range:[1,24], unit:h, dep:and,{$.InputProxyChannel.enableAnr,eq,true}, desc:NVR
will auto-reload the video according to the time interval-->1
    </anrInterval>
    <offlineReason>
      <!--ro, opt, enum, channel offline reason, subType:string, dep:and,{$.InputProxyChannelList[*].InputProxyChannel.online,eq,false}, desc:passwordError:
incorrect password-->passwordError
    </offlineReason>
    <streamTypeReflectList>
      <!--ro, opt, array, subType:object, range:[0,1]-->
      <streamTypeReflect>
        <!--ro, opt, object-->
        <streamType>
          <!--ro, req, enum, subType:string-->sub
        </streamType>
        <streamReflectType>
          <!--ro, req, enum, subType:string-->sub
        </streamReflectType>
      </streamTypeReflect>
    </streamTypeReflectList>
    <recordTranscodeEnabled>
      <!--ro, opt, bool-->true
    </recordTranscodeEnabled>
    <recordTranscodeConfig>
      <!--ro, opt, bool-->true
    </recordTranscodeConfig>
  </InputProxyChannel>
</InputProxyChannelList>
```

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14.2.6.17 Search for device information of all digital channels

Request URL

POST /ISAPI/ContentMgmt/InputProxy/deviceInfo

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ProxyDevicesInfoCondition xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time-->1
  </maxResults>
</ProxyDevicesInfoCondition>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ProxyDevicesInfoResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <responseStatus>
    <!--ro, req, bool, search status-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, search status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data),"OK" (searching
completed)-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <DeviceInfoList>
    <!--ro, opt, array, refer to XML_DeviceInfo, subType:object-->
    <DeviceInfo>
      <!--ro, opt, object-->
      <deviceName>
        <!--ro, req, string, device name-->test
      </deviceName>
      <deviceID>
        <!--ro, req, string, uuid-->fb8d4000-8158-11b5-8450-b4a38245654c
      </deviceID>
      <deviceDescription>
        <!--ro, opt, string, device description-->test
      </deviceDescription>
      <deviceLocation>
        <!--ro, opt, string, device location-->hangzhou
      </deviceLocation>
      <deviceStatus>
        <!--ro, opt, enum, device status, desc:"normal", "abnormal"-->normal
      </deviceStatus>
      <DetailAbnormalStatus>
        <!--ro, opt, object, abnormal status-->
        <hardDiskFull>
          <!--ro, opt, bool, whether disk is full-->true
        </hardDiskFull>
        <hardDiskError>
          <!--ro, opt, bool, whether disk has error-->true
        </hardDiskError>
        <ethernetBroken>
          <!--ro, opt, bool, whether it is disconnected-->true
        </ethernetBroken>
        <ipaddrConflict>
          <!--ro, opt, bool, whether IP address conflicts-->true
        </ipaddrConflict>
        <illegalAccess>
          <!--ro, opt, bool, whether it is illegal access-->true
        </illegalAccess>
        <recordError>
          <!--ro, opt, bool, whether video exception occurred-->true
        </recordError>
        <raidLogicDiskError>
          <!--ro, opt, bool, whether raid logic disk exception occurred-->true
        </raidLogicDiskError>
        <spareWorkDeviceError>
          <!--ro, opt, bool, whether spare device exception occurred-->true
        </spareWorkDeviceError>
      </DetailAbnormalStatus>
      <systemContact>
        <!--ro, opt, string, manufacturer-->STD-CGI
      </systemContact>
      <model>
        <!--ro, req, string, device model-->iDS-9632NX-I8/X
      </model>
      <serialNumber>
        <!--ro, req, string, serial No.-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
      </serialNumber>
      <macAddress>
        <!--ro, req, string, physical address-->44:47:cc:c8:d9:e4
```

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```
</macAddress>
<firmwareVersion>
  <!--ro, req, string, firmware version-->V4.1.40
</firmwareVersion>
<firmwareReleasedDate>
  <!--ro, opt, string-->build 181129
</firmwareReleasedDate>
<encoderVersion>
  <!--ro, opt, string, device encoder version No.-->V7.3
</encoderVersion>
<encoderReleasedDate>
  <!--ro, opt, string-->build 20161223
</encoderReleasedDate>
<bootVersion>
  <!--ro, opt, string, boot version-->V1.3.4
</bootVersion>
<bootReleasedDate>
  <!--ro, opt, string, release date of boot-->100316
</bootReleasedDate>
<panelVersion>
  <!--ro, opt, string, panel version-->V1.0
</panelVersion>
<hardwareVersion>
  <!--ro, opt, string, hardware version-->0x0
</hardwareVersion>
<decoderVersion>
  <!--ro, opt, string, device decoder version-->V1.0
</decoderVersion>
<decoderReleasedDate>
  <!--ro, opt, string, compile date of the device decoder version-->build 20161223
</decoderReleasedDate>
<softwareVersion>
  <!--ro, opt, string, software version-->build 181129
</softwareVersion>
<capacity>
  <!--ro, opt, int, device capacity (MB)-->10240
</capacity>
<usedCapacity>
  <!--ro, opt, int, used capacity (MB)-->5120
</usedCapacity>
<deviceType>
  <!--ro, req, enum, device type, subType:string, desc:read-only,"IPCamera,IPDome,DVR,HybirdNVR,NVR,DVS,IPZoom,CVR"-->Blade
</deviceType>
<telecontrolID>
  <!--ro, opt, int, "1-255", range:[1,255]-->1
</telecontrolID>
<supportBeep>
  <!--ro, opt, bool, whether to support buzzer-->true
</supportBeep>
<supportVideoLoss>
  <!--ro, opt, bool, whether it supports video loss-->true
</supportVideoLoss>
<firmwareVersionInfo>
  <!--ro, opt, string, firmware version information-->1
</firmwareVersionInfo>
<actualFloorNum>
  <!--ro, opt, int, "1-128", range:[1,255]-->1
</actualFloorNum>
<localZoneNum>
  <!--ro, opt, int, number of local zones-->1
</localZoneNum>
<alarmOutNum>
  <!--ro, opt, int, number of alarm outputs-->1
</alarmOutNum>
<distanceResolution>
  <!--ro, opt, float, distance resolution-->0.000
</distanceResolution>
<angleResolution>
  <!--ro, opt, float, angle resolution-->0.000
</angleResolution>
<speedResolution>
  <!--ro, opt, float, speed resolution-->0.000
</speedResolution>
<detectDistance>
  <!--ro, opt, float, detection distance-->0.000
</detectDistance>
<relayNum>
  <!--ro, opt, int, number of local relays-->1
</relayNum>
<electroLockNum>
  <!--ro, opt, int, number of local electronic locks-->1
</electroLockNum>
<RS485Num>
  <!--ro, opt, int, number of local RS-485-->1
</RS485Num>
<radarVersion>
  <!--ro, opt, string, radar version-->test
</radarVersion>
<cameraModuleVersion>
  <!--ro, opt, string, camera module version-->test
</cameraModuleVersion>
<mainversion>
  <!--ro, opt, int, main version No., range:[1,255]-->1
</mainversion>
```

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```
        </mainVersion>
        <subversion>
          <!--ro, opt, int, sub version No., range:[1,255]-->1
        </subversion>
        <upgradeversion>
          <!--ro, opt, int, update version No., range:[1,255]-->1
        </upgradeversion>
        <customizeversion>
          <!--ro, opt, int, custom version No., range:[1,255]-->1
        </customizeversion>
        <companyName>
          <!--ro, opt, string, manufacturer name-->test
        </companyName>
        <copyright>
          <!--ro, opt, string, copyright Information-->test
        </copyright>
        <systemName>
          <!--ro, opt, enum, storage system name, subType:string, desc:"storageManagement (storage management system), "distributedStorageManagement"
(distributed storage management system)-->storageManagement
        </systemName>
        <systemStatus>
          <!--ro, opt, enum, system status, subType:string, desc:"configured" (system has been configured), "unConfigured" (system has not been configured)--
>configured
        </systemStatus>
        <isLeaderDevice>
          <!--ro, opt, bool, whether it is the corresponding device of the resource IP-->true
        </isLeaderDevice>
        <clusterVersion>
          <!--ro, opt, string, system cluster version, desc:this field is valid only when isLeaderDevice is "true"-->test
        </clusterVersion>
        <manufacturer>
          <!--ro, opt, enum, manufacturer information, subType:string, desc:it is used to differentiate OEM devices-->test
        </manufacturer>
        <centralStorageVersion>
          <!--ro, opt, string, center storage version, range:[1,16]-->test
        </centralStorageVersion>
        <powerOnMode>
          <!--ro, opt, enum, startup mode, subType:string, desc:“button” (press button to power on), “adapter” (connect adapter to power on)-->button
        </powerOnMode>
        <customizedInfo>
          <!--ro, opt, string, custom project No., desc:custom project information-->test
        </customizedInfo>
        <charEncodeFormat>
          <!--ro, opt, enum, subType:string-->GB2312
        </charEncodeFormat>
        <languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->chinese
        </languageType>
        <DockStation>
          <!--ro, opt, object, dock station information configuration-->
          <Platform>
            <!--ro, opt, object, platform dock station information configuration-->
            <type>
              <!--ro, req, enum, platform which is connected to the dock station, subType:string, desc:“none”, “9533”, “8618”, “ISAPI”-->none
            </type>
            <ip>
              <!--ro, opt, string, IP address-->test
            </ip>
            <port>
              <!--ro, opt, int, communication port-->1
            </port>
            <userName>
              <!--ro, req, string, user name-->test
            </userName>
            <password>
              <!--ro, req, string, password-->test
            </password>
          </Platform>
          <centralStorageBackupEnabled>
            <!--ro, opt, bool-->true
          </centralStorageBackupEnabled>
        </DockStation>
        <webVersion>
          <!--ro, opt, string, Web version No.-->test
        </webVersion>
        <deviceRFProgramVersion>
          <!--ro, opt, string-->test
        </deviceRFProgramVersion>
        <securityModuleSerialNo>
          <!--ro, opt, string, security module serial No.-->test
        </securityModuleSerialNo>
        <securityModuleVersion>
          <!--ro, opt, string-->test
        </securityModuleVersion>
        <securityChipVersion>
          <!--ro, opt, string-->test
        </securityChipVersion>
        <securityModuleKeyVersion>
          <!--ro, opt, string-->test
        </securityModuleKeyVersion>
        <UIDLampRecognition>
          <!--ro, opt, object, recognize the device information by UID-->
          <enabled>
            <!--ro, opt, bool, whether to enable the function-->true
```

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```
        </enabled>
      </UIDLampRecognition>
      <confDeviceIdPrefix>
        <!--ro, opt, bool, whether the meeting uses the device name as its prefix-->true
      </confDeviceIdPrefix>
      <OEMCode>
        <!--ro, opt, enum, manufacturer OME code, subType:int, desc:1 (standard device), 0 (neutral device)-->1
      </OEMCode>
      <bootTime>
        <!--ro, opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
      </bootTime>
      <intelligentAnalysisEngineModel>
        <!--ro, opt, string, engine model-->test
      </intelligentAnalysisEngineModel>
      <channel>
        <!--ro, opt, int, channel No. of the NVR with the network camera-->1
      </channel>
    </DeviceInfo>
  </DeviceInfoList>
</ProxyDevicesInfoResult>
```

14.2.6.18 Set the digital channel configuration file

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/ipcConfig

Query Parameter

None

Request Message

Binary Data

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImportIpcCfgError xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <existError>
    <!--ro, opt, bool-->true
  </existError>
  <errorCode>
    <!--ro, opt, enum, subType:string-->chanNumReachLimit
  </errorCode>
  <errorRowNo>
    <!--ro, opt, string, range:[1,2]-->4
  </errorRowNo>
  <errorColNo>
    <!--ro, opt, string, range:[1,12]-->4
  </errorColNo>
  <IpcErrorList>
    <!--ro, opt, object-->
    <IpcError>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, int-->1
      </id>
      <errorRowNo>
        <!--ro, opt, string-->4
      </errorRowNo>
      <errorType>
        <!--ro, opt, enum, subType:string-->adminPortError
      </errorType>
    </IpcError>
  </IpcErrorList>
</ImportIpcCfgError>
```

dr4ckun@gmail.com
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14.2.6.19 Get the digital channel configuration file

Request URL

GET /ISAPI/ContentMgmt/InputProxy/ipcConfig

Query Parameter

None

Request Message

None

Response Message

Binary Data

14.2.6.20 Search for accessible network cameras on LAN (Local Area Network)

Request URL

GET /ISAPI/ContentMgmt/InputProxy/search?serialNumber=<serialNumber>

Query Parameter

Parameter Name	Parameter Type	Description
serialNumber	string	Device short serial No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoSourceList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, accessible network cameras on LAN, subType:object, attr:version{req, string, protocolVersion}-->
  <VideoSourceDescriptor>
    <!--ro, opt, object, video source description-->
    <id>
      <!--ro, req, string, index No.-->test
    </id>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:; "hostname" (domain name), "ipaddress" (IP address)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host server name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,32]-->test
    </ipAddress>
    <subnetMask>
      <!--ro, opt, string, subnet mask for IPv4 address, range:[0,32]-->test
    </subnetMask>
    <gateway>
      <!--ro, opt, string, IPv4 address of gateway, range:[0,32]-->test
    </gateway>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->test
    </ipv6Address>
    <bitMask>
      <!--ro, opt, string, subnet mask for IPv6 address, range:[0,128]-->test
    </bitMask>
    <serialNumber>
      <!--ro, opt, string, serial No., range:[0,64]-->test
    </serialNumber>
    <macAddress>
      <!--ro, opt, string, MAC address, range:[0,48]-->test
    </macAddress>
    <firmwareVersion>
      <!--ro, req, string, firmware version, range:[0,64]-->test
    </firmwareVersion>
    <managePortNo>
      <!--ro, opt, int, management port No., range:[0,65535]-->1
    </managePortNo>
    <userName>
      <!--ro, opt, string, user name, range:[0,32]-->test
    </userName>
    <password>
      <!--ro, opt, string, password, range:[0,16], desc:this is a default password from NVR, it may not be the actual password scenario: the device will use
the default password to batch add network cameras; if the default password is not the same as the camera, you can manual edit the camera password
thereafter-->test
    </password>
    <srcInputPortNums>
      <!--ro, req, int, the number of input ports-->1
    </srcInputPortNums>
    <deviceID>
      <!--ro, opt, string, device No., range:[1,64], desc:device No.-->test
    </deviceID>
    <deviceModel>
      <!--ro, opt, string, device model, range:[0,64]-->test
    </deviceModel>
    <softwareVersion>
      <!--ro, opt, string, software version, range:[0,64]-->test
    </softwareVersion>
    <activated>
      <!--ro, opt, bool, device activation status-->true
    </activated>
    <registerStatus>
      <!--ro, opt, enum, ISUP registration status, subType:string, desc:; "unknown" (unknown"; "unregister" (not registered); "registered" (registered)--
>unknown
```

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```
    </registerStatus>
    <isSupportGB35114>
      <!--ro, opt, bool,   -->true
    </isSupportGB35114>
    <readerID>
      <!--ro, opt, int, card reader No., range:[1,8], desc:supports being searched over SADP-->1
    </readerID>
    <POEID>
      <!--ro, opt, int, PoE port No., range:[1,8], step:1, desc:the No. of PoE port connected with the device-->1
    </POEID>
    <isIPConflict>
      <!--ro, opt, bool-->true
    </isIPConflict>
    <personnelChannelInfo>
      <!--ro, opt, object, dep:or,{$.VideoSourceList[*].VideoSourceDescriptor.deviceType,eq,PersonnelChannel}-->
      <isMainPermissionController>
        <!--ro, req, bool-->true
      </isMainPermissionController>
      <subPermissionControllerSerialNumber>
        <!--ro, opt, string, range:[9,9], dep:or,{$.VideoSourceList[*].VideoSourceDescriptor.personnelChannelInfo.isMainPermissionController,eq,true}-->test
      </subPermissionControllerSerialNumber>
    </personnelChannelInfo>
    <deviceName>
      <!--ro, opt, string, range:[0,64]-->test
    </deviceName>
    <addStatus>
      <!--ro, opt, enum, subType:string-->added
    </addStatus>
  </VideoSourceDescriptor>
</VideoSourceList>
```

14.2.6.21 Set the network camera's password to its activation password

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/syncIPCActivatePasswd?loginPassword=<loginPassword>

Query Parameter

Parameter Name	Parameter Type	Description
loginPassword	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
	[Message content]	application/json	--	--	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit

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name.

- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.22 Synchronize the device password to the digital channel password

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/syncIPCPasswd

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.23 Get PTZ control capability of digital channel

drackum@gmail.com
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Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZCtrlCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, PTZ control capability of digital channel, attr:version{req, string, protocolVersion}-->
  <maxPTZChannleNum>
    <!--ro, opt, int, maximum number of PTZ channels-->0
  </maxPTZChannleNum>
  <isSupportPTZWithPreview>
    <!--ro, opt, bool, whether the device supports live view, desc:;:;-->true
  </isSupportPTZWithPreview>
</PTZCtrlCap>
```

14.2.6.24 Get parameters of all digital PTZ channels

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>
<PTZChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZChannel>
    <!--ro, req, object-->
    <id>
      <!--ro, req, int, ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <serialNumber>
      <!--ro, req, int, serial port No. of channel, desc:if the value is 0, it indicates that no serial port is linked-->1
    </serialNumber>
    <videoInputID>
      <!--ro, req, int-->1
    </videoInputID>
    <panMaxSpeed>
      <!--ro, opt, int, panning speed-->1
    </panMaxSpeed>
    <tiltMaxSpeed>
      <!--ro, opt, int, tilting speed-->1
    </tiltMaxSpeed>
    <presetSpeed>
      <!--ro, opt, int, preset speed, range:[1,8]-->1
    </presetSpeed>
    <autoPatrolSpeed>
      <!--ro, opt, int, patrol speed, range:[0,100]-->1
    </autoPatrolSpeed>
    <keyBoardControlSpeed>
      <!--ro, opt, int, PTZ control speed, range:[0,100]-->1
    </keyBoardControlSpeed>
    <controlAddress>
      <!--ro, req, object-->
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <Address>
        <!--ro, opt, string, address, range:[1,255]-->1
      </Address>
    </controlAddress>
    <defaultPresetID>
      <!--ro, opt, string, default preset No.-->test
    </defaultPresetID>
    <PTZRs485Para>
      <!--ro, req, object-->
      <baudRate>
        <!--ro, req, int, baud rate-->9600
      </baudRate>
      <dataBits>
        <!--ro, req, int, data bit-->1
      </dataBits>
      <parityType>
        <!--ro, req, enum, parity type, subType:string, desc:“none”, “even”, “odd”, “mark”, “space”-->space
      </parityType>
      <stopBits>
        <!--ro, req, enum, stop bit, subType:string, desc:“1”, “1.5”, “2”-->1
      </stopBits>
      <flowCtrl>
        <!--ro, req, enum, encoding or decoding type, subType:string, desc:“none”, “software”, “hardware”-->hardware
      </flowCtrl>
    </PTZRs485Para>
    <manualControlSpeed>
      <!--ro, opt, enum, manual control speed type, subType:string, desc:“pedestrian”, “nonMotorVehicle”, “motorVehicle”, “selfadaptive”, “compatible”--
>pedestrian
    </manualControlSpeed>
  </PTZChannel>
</PTZChannelList>
```

drackum@gmail.com
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14.2.6.25 Set parameters of all digital PTZ channels

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, PTZ channels' configuration list, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZChannel>
    <!--req, object, PTZ channel configuration-->
    <id>
      <!--req, int, ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <serialNumber>
      <!--req, int, serial port No., desc:this field should be supported by both GET and PUT methods. If the value is 0, it indicates that no serial port is
linked-->1
    </serialNumber>
    <videoInputID>
      <!--req, int, video input-->1
    </videoInputID>
    <panMaxSpeed>
      <!--opt, int, panning speed, unit: degree/second-->1
    </panMaxSpeed>
    <tiltMaxSpeed>
      <!--opt, int, tilting speed, unit: degree/second-->1
    </tiltMaxSpeed>
    <presetSpeed>
      <!--opt, int, preset speed, range:[1,8]-->1
    </presetSpeed>
    <autoPatrolSpeed>
      <!--opt, int, auto patrol speed, range:[0,100]-->1
    </autoPatrolSpeed>
    <keyBoardControlSpeed>
      <!--opt, int, keypad control speed, range:[0,100]-->1
    </keyBoardControlSpeed>
    <controlAddress>
      <!--req, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <Address>
        <!--opt, string, range: [1,255], range:[1,255]-->1
      </Address>
    </controlAddress>
    <defaultPresetID>
      <!--opt, string, default preset No.-->test
    </defaultPresetID>
    <PTZRs485Para>
      <!--req, object-->
      <baudRate>
        <!--req, int-->9600
      </baudRate>
      <dataBits>
        <!--req, int, data bit-->1
      </dataBits>
      <parityType>
        <!--req, enum, parity type, subType:string, desc:"none", "even", "odd", "mark", "space"-->space
      </parityType>
      <stopBits>
        <!--req, enum, stop bit, subType:string, desc:1, 1.5, 2-->1
      </stopBits>
      <flowCtrl>
        <!--req, enum, encoding or decoding type, subType:string, desc:"none", "software", "hardware"-->hardware
      </flowCtrl>
    </PTZRs485Para>
    <manualControlSpeed>
      <!--opt, enum, manual control speed type, subType:string, desc:"pedestrian", "nonMotorVehicle" (non-motor vehicles), "motorVehicle" (vehicle),
"selfadaptive" (self-adaptive), "compatible"-->pedestrian
    </manualControlSpeed>
  </PTZChannel>
</PTZChannelList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.26 Get parameters of a digital PTZ channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>
<PTZChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, PTZ channel configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <serialNumber>
    <!--ro, req, int, serial No., desc:serial No.-->1
  </serialNumber>
  <videoInputID>
    <!--ro, req, int, video input-->1
  </videoInputID>
  <panMaxSpeed>
    <!--ro, opt, int, the maximum panning speed-->1
  </panMaxSpeed>
  <tiltMaxSpeed>
    <!--ro, opt, int, the maximum tilting speed-->1
  </tiltMaxSpeed>
  <presetSpeed>
    <!--ro, opt, int, predefined speed, range:[1,8]-->1
  </presetSpeed>
  <autoPatrolSpeed>
    <!--ro, opt, int, patrol speed, range:[0,100]-->1
  </autoPatrolSpeed>
  <keyBoardControlSpeed>
    <!--ro, opt, int, range:[0,100]-->1
  </keyBoardControlSpeed>
  <controlAddress>
    <!--ro, req, object, control address-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <Address>
      <!--ro, opt, string, address, range:[1,255]-->1
    </Address>
  </controlAddress>
  <defaultPresetID>
    <!--ro, opt, string, the default ID-->test
  </defaultPresetID>
  <PTZRs485Para>
    <!--ro, req, object-->
    <baudRate>
      <!--ro, req, int, baud rate-->9600
    </baudRate>
    <dataBits>
      <!--ro, req, int, data bit-->1
    </dataBits>
    <parityType>
      <!--ro, req, enum, verification type, subType:string, desc:verification type-->space
    </parityType>
    <stopBits>
      <!--ro, req, enum, stop bit, subType:string, desc:stop bit-->1
    </stopBits>
    <flowCtrl>
      <!--ro, req, enum, "none, software, hardware", subType:string, desc:"none, software, hardware"-->hardware
    </flowCtrl>
  </PTZRs485Para>
  <manualControlSpeed>
    <!--ro, opt, enum, "pedestrian, nonMotorVehicle, motorVehicle, selfadaptive, compatible", subType:string, desc:"pedestrian, nonMotorVehicle,
motorVehicle, selfadaptive, compatible"-->pedestrian
  </manualControlSpeed>
  <panSupport>
    <!--ro, opt, bool, whether it supports panning-->true
  </panSupport>
  <tiltSupport>
    <!--ro, opt, bool, whether it supports tilting-->true
  </tiltSupport>
  <zoomSupport>
    <!--ro, opt, bool, whether it supports zooming-->true
  </zoomSupport>
</PTZChannel>
```

drackum@gmail.com
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14.2.6.27 Set the PTZ parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int-->1
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <serialNumber>
    <!--req, int-->1
  </serialNumber>
  <videoInputID>
    <!--req, int-->1
  </videoInputID>
  <panMaxSpeed>
    <!--opt, int-->1
  </panMaxSpeed>
  <tiltMaxSpeed>
    <!--opt, int-->1
  </tiltMaxSpeed>
  <presetSpeed>
    <!--opt, int, range:[1,8]-->1
  </presetSpeed>
  <autoPatrolSpeed>
    <!--opt, int, range:[0,100]-->1
  </autoPatrolSpeed>
  <keyBoardControlSpeed>
    <!--opt, int, range:[0,100]-->1
  </keyBoardControlSpeed>
  <controlAddress>
    <!--req, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <Address>
      <!--opt, string, range:[1,255]-->1
    </Address>
  </controlAddress>
  <defaultPresetID>
    <!--opt, string-->test
  </defaultPresetID>
  <PTZRs485Para>
    <!--req, object-->
    <baudRate>
      <!--req, int-->9600
    </baudRate>
    <dataBits>
      <!--req, int-->1
    </dataBits>
    <parityType>
      <!--req, enum, subType:string-->space
    </parityType>
    <stopBits>
      <!--req, enum, subType:string-->1
    </stopBits>
    <flowCtrl>
      <!--req, enum, subType:string-->hardware
    </flowCtrl>
  </PTZRs485Para>
  <manualControlSpeed>
    <!--opt, enum, subType:string-->pedestrian
  </manualControlSpeed>
  <panSupport>
    <!--opt, bool-->true
  </panSupport>
  <tiltSupport>
    <!--opt, bool-->true
  </tiltSupport>
  <zoomSupport>
    <!--opt, bool-->true
  </zoomSupport>
</PTZChannel>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.28 Manually control PTZ

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/continuous

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <pan>
    <!--opt, int, pan range, range:[-100,100]-->60
  </pan>
  <tilt>
    <!--opt, int, tilt range, range:[-100,100]-->60
  </tilt>
  <zoom>
    <!--opt, int, zoom range, range:[-100,100]-->60
  </zoom>
</PTZData>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.29 Perform scheduled PTZ control via digital channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/momentary

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

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Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <pan>
    <!--opt, int, pan range, range:[-100,100]-->1
  </pan>
  <tilt>
    <!--opt, int, tilt range, range:[-100,100]-->1
  </tilt>
  <zoom>
    <!--opt, int, zoom range, range:[-100,100]-->1
  </zoom>
  <Momentary>
    <!--opt, object-->
    <duration>
      <!--opt, int, unit: millisecond, unit:ms-->1
    </duration>
  </Momentary>
</PTZData>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.2.6.30 Perform one-touch focus via a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/onepushfoucs/start

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.31 Set all patrols of a digital PTZ channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPatrol>
    <!--ro, opt, object, patrol parameters-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <patrolName>
      <!--ro, req, string, patrol name-->test
    </patrolName>
    <resumeType>
      <!--ro, opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
    </resumeType>
    <PatrolSequenceList>
      <!--ro, req, array, subType:object-->
      <PatrolSequence>
        <!--ro, req, object, list-->
        <presetID>
          <!--ro, req, int, preset ID-->1
        </presetID>
        <seqSpeed>
          <!--ro, req, int, patrol speed-->1
        </seqSpeed>
        <delay>
          <!--ro, req, int, delay time, unit:ms-->1
        </delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

drackum@gmail.com
DETEKTOR

14.2.6.32 Add a patrol of a digital channel

Request URL

POST /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string-->test
  </id>
  <patrolName>
    <!--req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--opt, enum, type, subType:string, desc:“relative”, “absolute”-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--req, array, subType:object-->
    <PatrolSequence>
      <!--req, object-->
      <presetID>
        <!--req, string, preset ID-->test
      </presetID>
      <seqSpeed>
        <!--req, string, patrol speed-->test
      </seqSpeed>
      <delay>
        <!--req, int, delay time, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

dracku m@gmail.com
DETEKTOR

14.2.6.33 Delete all patrols of a digital PTZ channel

Request URL

DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.6.34 Set all patrols' parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPatrol>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <id>
      <!--req, string-->test
    </id>
    <patrolName>
      <!--req, string-->test
    </patrolName>
    <resumeType>
      <!--opt, enum, subType:string-->relative
    </resumeType>
    <PatrolSequenceList>
      <!--req, array, subType:object-->
      <PatrolSequence>
        <!--req, object-->
        <presetID>
          <!--req, int-->1
        </presetID>
        <seqSpeed>
          <!--req, int-->1
        </seqSpeed>
        <delay>
          <!--req, int, unit:ms-->1
        </delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.35 Get the patrol parameters of a digital PTZ channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <patrolName>
    <!--ro, req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--ro, opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--ro, req, array, patrol result list, subType:object-->
    <PatrolSequence>
      <!--ro, req, object, patrol result-->
      <presetID>
        <!--ro, req, int, preset ID in the patrol-->1
      </presetID>
      <seqSpeed>
        <!--ro, req, int, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--ro, req, int, delay time, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

14.2.6.36 Set a patrol of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>

Query Parameter

dracium@gmail.com
DETEKTOR

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <id>
    <!--req, string, ID-->test
  </id>
  <patrolName>
    <!--req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--opt, enum, type: relative,absolute, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--req, array, subType:object-->
    <PatrolSequence>
      <!--req, object, list-->
      <presetID>
        <!--req, int, preset ID-->1
      </presetID>
      <seqSpeed>
        <!--req, int, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--req, int, delay time, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
DEFEKTOR

14.2.6.37 Delete a patrol of a digital PTZ channel

Request URL

DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.6.38 Pause calling a patrol of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>/pause

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.39 Start calling a patrol of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>/start

Query Parameter

drackum@gmail.com
DETEKTOR

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.40 Search a patrol status of a digital channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>/status

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrolStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, patrol status, attr:version{req, string, protocolVersion}-->
  <patrolID>
    <!--ro, req, string, patrol ID-->test
  </patrolID>
  <patrolStatus>
    <!--ro, req, enum, patrol status, subType:string, desc:"running", "stopped", "paused"-->running
  </patrolStatus>
</PTZPatrolStatus>
```

14.2.6.41 Stop calling a patrol of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>/stop

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

dr.actum@gmail.com
DETEKTOR

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.42 Perform 3D positioning

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/position3D

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<Position3D xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <StartPoint>
    <!--opt, object, start point, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, x-coordinate, range:[0,255]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,255]-->1
    </positionY>
  </StartPoint>
  <EndPoint>
    <!--opt, object, end point, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, x-coordinate, range:[0,255]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,255]-->1
    </positionY>
  </EndPoint>
</Position3D>
```

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.43 Set presets of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPresetList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in XML_PTZPreset, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPreset>
    <!--opt, object, PTZ preset-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <id>
      <!--req, string-->test
    </id>
    <presetName>
      <!--req, string, preset name-->test
    </presetName>
  </PTZPreset>
</PTZPresetList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.44 Get parameters of all presets of a digital PTZ channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets

Query Parameter

drackum@gmail.com
DETEKTOR

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPresetList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, presets list, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPreset>
    <!--ro, opt, object, PTZ preset-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <presetName>
      <!--ro, req, string, preset name-->test
    </presetName>
  </PTZPreset>
</PTZPresetList>
```

14.2.6.45 Delete presets of a digital PTZ channel.

Request URL

DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.6.46 Add a preset to a digital PTZ channel

Request URL

POST /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets

drackum@gmail.com
DETEKTOR

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPreset xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, PTZ preset, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string, listening host ID-->test
  </id>
  <presetName>
    <!--req, string, preset name-->test
  </presetName>
</PTZPreset>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.2.6.47 Delete a preset of a digital PTZ channel

Request URL

DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets/<presetID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.48 Set a preset of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets/<presetID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPreset xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, PTZ preset, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable preset configuration-->true
  </enabled>
  <id>
    <!--req, string, preset No.-->test
  </id>
  <presetName>
    <!--req, string, preset name-->test
  </presetName>
</PTZPreset>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.6.49 Get parameters of a preset of a digital PTZ channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets/<presetID>

Query Parameter

dracku@gmail.com
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Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPreset xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, PTZ preset, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <presetName>
    <!--ro, req, string, preset name-->test
  </presetName>
</PTZPreset>
```

14.2.6.50 Call a preset of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets/<presetID>/goto

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.7 OSD

14.2.7.1 Get text overlay parameters of a specific video digital input channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays

Query Parameter

drackum@gmail.com
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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, req, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[0,1000]-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height, range:[0,1000]-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent>
      <!--ro, req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--ro, req, bool, Flashing Status-->true
    </flashing>
  </attribute>
  <TextOverlayList>
    <!--ro, opt, object, list of overlay information-->
    <TextOverlay>
      <!--ro, opt, object, list-->
      <id>
        <!--ro, req, string, text No., range:[1,8]-->test
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
      </positionY>
      <displayText>
        <!--ro, req, string, overlaid text, range:[1,32]-->test
      </displayText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
    <dateStyle>
      <!--ro, opt, enum, date format, subType:string, desc:date format-->YYYY-MM-DD
    </dateStyle>
    <timeStyle>
      <!--ro, opt, enum, time format, subType:string, desc:time format-->12hour
    </timeStyle>
    <displayWeek>
      <!--ro, opt, bool, whether to display the day of the week-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
  </channelNameOverlay>
  <fontSize>
    <!--ro, opt, enum, font size, subType:string, desc:font size-->adaptive
  </fontSize>
  <frontColorMode>
    <!--ro, opt, enum, font color mode, subType:string, desc:font color mode-->auto
```

drackum@gmail.com
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```
    <!--ro, opt, enum, font color mode, subType:string, desc:font color mode-->auto
  </frontColorMode>
  <frontColor>
    <!--ro, opt, string, font color, dep:and,{$.VideoOverlay.frontColorMode,eq,customize}, desc:font color-->test
  </frontColor>
  <BatteryPowerOverlay>
    <!--ro, opt, object, battery power overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
  </BatteryPowerOverlay>
  <alignment>
    <!--ro, opt, enum, alignment mode, subType:string, desc:alignment mode-->customize
  </alignment>
  <publicSecurity>
    <!--ro, req, bool, public security-->true
  </publicSecurity>
</VideoOverlay>
```

14.2.7.2 Set text overlay parameters of a specific video digital input channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--req, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width, range:[0,1000]-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height, range:[0,1000]-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--opt, object-->
    <transparent>
      <!--req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--req, bool, whether it is flickering-->true
    </flashing>
  </attribute>
  <TextOverlayList>
    <!--opt, object-->
    <TextOverlay>
      <!--opt, object, list-->
      <id>
        <!--req, string, text No., range:[1,8]-->test
      </id>
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--req, int, x-coordinate, range:[0,704], desc:X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,576], desc:Y-coordinate-->1
      </positionY>
      <displayText>
        <!--req, string, overlaid text, range:[1,32]-->test
      </displayText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to display the date-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:X-coordinate-->1
```

dracknm@gmail.com
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```
      <!--req, int, x-coordinate, range:[0,704], desc:X-coordinate-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:Y-coordinate-->1
    </positionY>
    <dateStyle>
      <!--opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
    </dateStyle>
    <timeStyle>
      <!--opt, enum, time format, subType:string, desc:time format: "12hour","24hour"-->12hour
    </timeStyle>
    <displayWeek>
      <!--opt, bool, whether to display the day of the week-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:X-coordinate-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:Y-coordinate-->1
    </positionY>
  </channelNameOverlay>
  <fontSize>
    <!--opt, enum, font size, subType:string, desc:"adaptive", "16*16", "32*32", "48*48", "64*64"-->adaptive
  </fontSize>
  <frontColorMode>
    <!--opt, enum, font color mode, subType:string, desc:font color mode: "auto","customize"-->auto
  </frontColorMode>
  <frontColor>
    <!--opt, string, font color, dep:and,{$.VideoOverlay.frontColorMode,eq,customize}, desc:font color-->test
  </frontColor>
  <BatteryPowerOverlay>
    <!--opt, object, battery power overlay-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:X-coordinate-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:Y-coordinate-->1
    </positionY>
  </BatteryPowerOverlay>
  <alignment>
    <!--opt, enum, alignment mode, subType:string, desc:alignment mode: "customize"-custom,"alignRight"-align to right,"alignLeft"-align to left-->customize
  </alignment>
  <publicSecurity>
    <!--req, bool, public security-->true
  </publicSecurity>
</VideoOverlay>
```

dr4ckum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

14.2.7.3 Get the text overlay configuration capability of a specific video digital input channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, number of characters in the lower-right corner, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, req, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object-->
    <transparent opt="true,false">
      <!--ro, req, bool, whether it is transparent, attr:opt{req, string}-->true
    </transparent>
    <flashing opt="true,false">
      <!--ro, req, bool, Flashing Status, attr:opt{req, string}-->true
    </flashing>
  </attribute>
  <TextOverlayList size="8">
    <!--ro, opt, array, subType:object, range:[0,8], attr:size{req, int}-->
    <TextOverlay>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <positionX>
        <!--ro, req, int, range:[0,704]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,576]-->1
      </positionY>
      <displayText>
        <!--ro, req, string-->test
      </displayText>
      <isPersistentText>
        <!--ro, opt, bool-->true
      </isPersistentText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
    <dateStyle opt="YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY,CHR-YYYY-MM-DD,CHR-MM-DD-YYYY,CHR-DD-MM-YYYY,CHR-YYYY/MM/DD,CHR-MM/DD/YYYY,CHR-DD/MM/YYYY">
      <!--ro, opt, enum, date format, subType:string, attr:opt{req, string}, desc:date format-->YYYY-MM-DD
    </dateStyle>
    <timeStyle opt="12hour,24hour">
      <!--ro, opt, enum, time format, subType:string, attr:opt{req, string}, desc:time format-->12hour
    </timeStyle>
    <displayWeek opt="true,false">
      <!--ro, opt, bool, Display Day, attr:opt{req, string}-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, Show Channel Name-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
  </channelNameOverlay>
  <fontSize opt="16*16,32*32,48*48,64*64,adaptive">
```

drackum@gmail.com
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```
    <!--ro, opt, enum, font size, subType:string, attr:opt{req, string}, desc:font size,unit: pixel-->adaptive
  </fontSize>
  <frontColorMode opt="auto,customize">
    <!--ro, opt, enum, font color mode, subType:string, attr:opt{req, string}, desc:font color mode-->auto
  </frontColorMode>
  <frontColor>
    <!--ro, opt, string, font color, dep:and,{$.VideoOverlay.frontColorMode,eq,customize}-->test
  </frontColor>
  <BatteryPowerOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
  </BatteryPowerOverlay>
  <DeviceStatusDisplay>
    <!--ro, opt, object, device status display-->
    <batteryCapacityDisplay>
      <!--ro, opt, bool, whether to display battery capacity-->true
    </batteryCapacityDisplay>
    <chargingStatusDisplay>
      <!--ro, opt, bool, whether to display charging status-->true
    </chargingStatusDisplay>
    <bluetoothStatusDisplay>
      <!--ro, opt, bool, whether to display bluetooth status-->true
    </bluetoothStatusDisplay>
    <dialStatusDisplay>
      <!--ro, opt, bool, whether to display dial-up status-->true
    </dialStatusDisplay>
  </DeviceStatusDisplay>
  <alignment opt="customize,alignRight,alignLeft,GB,allRight,allLeft">
    <!--ro, opt, enum, alignment mode, subType:string, attr:opt{req, string}, desc:alignment mode: "customize"-custom,"alignRight"-align to
right,"alignLeft"-align to left,"allRight"-align all to right (align all OSD texts to the right,including characters,time,title,and so on),"allLeft"-align
all to left (align all OSD texts to the left,including characters,time,title,and so on)-->customize
  </alignment>
  <lowRightTextNumber min="1" max="6">
    <!--ro, opt, int, number of characters in the lower-left corner, attr:min{req, int},max{req, int}-->1
  </lowRightTextNumber>
  <lowLeftTextNumber min="7" max="8">
    <!--ro, opt, int, number of characters in the lower-right corner, attr:min{req, int},max{req, int}-->1
  </lowLeftTextNumber>
  <boundary min="0" max="2">
    <!--ro, opt, int, boundary, attr:min{req, int},max{req, int}-->1
  </boundary>
  <TextOverlaySize min="0" max="0">
    <!--ro, opt, int, number of overlaid texts, attr:min{req, int},max{req, int}-->1
  </TextOverlaySize>
  <FloorNumOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
  </FloorNumOverlay>
  <upDownboudary>
    <!--ro, opt, int, minimum size of the upper or lower boundary, range:[0,2], desc:unit: number of characters, range: [0,2] (the value can be 0,1,or 2)--
>1
  </upDownboudary>
  <leftRightboudary>
    <!--ro, opt, int, minimum size of the left or right boundary, range:[0,2], desc:unit: number of characters, range: [0,2] (the value can be 0,1,or 2)-->1
  </leftRightboudary>
  <Angle>
    <!--ro, opt, object, device angle-->
    <TiltAngle>
      <!--ro, opt, object, pitch angle-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
      </enabled>
      <positionX min="0" max="1000">
        <!--ro, req, int, X-coordinate, range:[0,1000], attr:min{req, int},max{req, int}, desc:the origin is the lower-left corner of the screen-->1
      </positionX>
      <positionY min="0" max="1000">
        <!--ro, req, int, Y-coordinate, range:[0,1000], attr:min{req, int},max{req, int}, desc:the origin is the lower-left corner of the screen-->1
      </positionY>
    </TiltAngle>
  </Angle>
</VideoOverlay>
```

drackun@gmail.com DETEKTOR

14.2.7.4 Set the channel name overlay parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--req, int, range:[0,576]-->1
  </positionY>
</channelNameOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.2.7.5 Get the channel name overlay parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <positionX>
    <!--ro, req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--ro, req, int, range:[0,576]-->1
  </positionY>
</channelNameOverlay>
```

14.2.7.6 Get the time information overlay parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <positionX>
    <!--ro, req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--ro, req, int, range:[0,576]-->1
  </positionY>
  <dateStyle>
    <!--ro, opt, enum, subType:string-->YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--ro, opt, enum, subType:string-->12hour
  </timeStyle>
  <displayWeek>
    <!--ro, opt, bool-->true
  </displayWeek>
</DateTimeOverlay>
```

14.2.7.7 Set the time information overlay parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--req, int, range:[0,576]-->1
  </positionY>
  <dateStyle>
    <!--opt, enum, subType:string-->YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--opt, enum, subType:string-->12hour
  </timeStyle>
  <displayWeek>
    <!--opt, bool-->true
  </displayWeek>
</DateTimeOverlay>
```

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.7.8 Get all text overlay parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <TextOverlay>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, range:[1,8]-->test
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <positionX>
      <!--ro, req, int, range:[0,704]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,576]-->1
    </positionY>
    <displayText>
      <!--ro, req, string, range:[1,32]-->test
    </displayText>
  </TextOverlay>
</TextOverlayList>
```

dracku@gmail.com
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14.2.7.9 Set all text overlay parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <TextOverlay>
    <!--opt, object-->
    <id>
      <!--req, string, range:[1,8]-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <positionX>
      <!--req, int, range:[0,704]-->1
    </positionX>
    <positionY>
      <!--req, int, range:[0,576]-->1
    </positionY>
    <displayText>
      <!--req, string, range:[1,32]-->test
    </displayText>
  </TextOverlay>
</TextOverlayList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.2.7.10 Get the text overlay parameters of the specific digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
textID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, text ID, range:[1,8]-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable text overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-
->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-
->1
  </positionY>
  <displayText>
    <!--ro, req, string, overlaid text, range:[1,32]-->test
  </displayText>
</TextOverlay>
```

14.2.7.11 Set parameters of a specific digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, text No., range:[1,8]-->test
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, the origin is the lower-left corner of the screen, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, the origin is the lower-left corner of the screen, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, overlaid text, range:[1,32]-->test
  </displayText>
</TextOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

dr.ackum@gmail.com
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14.2.7.12 Get the OSD parameters of a specified channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, req, object, screen size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent>
      <!--ro, req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--ro, req, bool, whether it is flickering-->true
    </flashing>
  </attribute>
  <TextOverlayList>
    <!--ro, opt, array, text content list, subType:object-->
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object, channel time information overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
    <dateStyle>
      <!--ro, opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY",
"CHR-YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
    </dateStyle>
    <timeStyle>
      <!--ro, opt, enum, time format, subType:string, desc:"12hour", "24hour"-->12hour
    </timeStyle>
    <displayWeek>
      <!--ro, opt, bool, whether to display week information-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object, channel name overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable channel name overlay-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </channelNameOverlay>
  <fontSize>
    <!--ro, opt, enum, font size, subType:string, desc:"adaptive", "16*16", "32*32", "48*48", "64*64", "96*96", "128*128"-->96*96
  </fontSize>
</VideoOverlay>
```

drackum@gmail.com
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14.2.7.13 Set OSD parameters of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, text overlay, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--opt, bool, whether to enable, desc:true (OSD is enabled), false (OSD is disabled). The default value is true-->true
  </enabled>
  <normalizedScreenSize>
    <!--req, object, screen size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--opt, object, attributes-->
    <transparent>
      <!--req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--req, bool, whether it is flickering-->true
    </flashing>
    <backgroundColor>
      <!--opt, object, background color-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <color>
        <!--opt, string, background color RGB, desc:hexadecimal digit-->000000
      </color>
      <transparency>
        <!--opt, int, background color transparency, range:[0,100]-->50
      </transparency>
    </backgroundColor>
  </attribute>
  <TextOverlayList>
    <!--opt, array, text content list, subType:object-->
    <TextOverlay>
      <!--opt, object, text content-->
      <id>
        <!--req, string, text No.-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
      <displayText>
        <!--req, string, overlay text-->test
      </displayText>
      <isPersistentText>
        <!--opt, bool, whether the overlaid text remains the previous settings after device rebooting, desc:if the value is true or the node is not
transmitted, it indicates to write OSD to flash, and it is not suitable for frequent or fast callings. If the value is false, it indicates not to writer OSD
to flash and the OSD settings will not restore-->true
      </isPersistentText>
      <OverlayRegion>
        <!--opt, object, OSD overlay frame of a single text will be returned when text overlay succeeded (the origin is the upper-left corner of the
screen)-->
        <width>
          <!--req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--req, int, height, range:[0,1000]-->0
        </height>
      </OverlayRegion>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--opt, object, channel time information overlay-->
```

drackum@gmail.com
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```
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
    <dateStyle>
      <!--opt, enum, date, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-
YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
    </dateStyle>
    <timeStyle>
      <!--opt, enum, time format, subType:string, desc:time format-->12hour
    </timeStyle>
    <displayWeek>
      <!--opt, bool, whether to display week information-->true
    </displayWeek>
    <displayDate>
      <!--opt, bool, whether to display the date (year-month-day)-->true
    </displayDate>
    <displayTime>
      <!--opt, bool, whether to display the time (hour-minute-second)-->true
    </displayTime>
    <OverlayRegion>
      <!--opt, object, OSD overlay frame of a single text will be returned when text overlay succeeded (the origin is the upper-left corner of the screen)--
>
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--opt, object, channel name overlay-->
    <enabled>
      <!--req, bool, whether to enable channel name overlay-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
    <OverlayRegion>
      <!--opt, object, OSD overlay frame of a single text will be returned when text overlay succeeded (the origin is the upper-left corner of the screen)--
>
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </channelNameOverlay>
  <HumitureOverlay>
    <!--opt, object, temperature and humidity-->
    <enabled>
      <!--req, bool, whether to display temperature and humidity-->true
    </enabled>
    <updateInterval>
      <!--opt, int, update interval-->1
    </updateInterval>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </HumitureOverlay>
  <fontSize>
    <!--opt, enum, font size, subType:string, desc:"adaptive", "16*16", "32*32", "48*48", "64*64", "96*96", "128*128"-->96*96
  </fontSize>
  <frontColorMode>
    <!--opt, enum, font color mode, subType:string, desc:"auto", "customize", "outline"-->auto
  </frontColorMode>
  <frontColor>
    <!--opt, string, font color, desc:hexadecimal digit-->000000
  </frontColor>
  <BatteryPowerOverlay>
    <!--opt, object, battery power information overlay-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <positionX>
```

drackun@gmail.com
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```
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </BatteryPowerOverlay>
  <alignment>
    <!--opt, enum, alignment mode, subType:string, desc:"customize" (custom), "alignRight" (right align), "alignLeft" (left align), "GB", "adsorption"
(adsorption mode)-->alignRight
  </alignment>
  <publicSecurity>
    <!--opt, bool-->true
  </publicSecurity>
  <DeviceStatusDisplay>
    <!--opt, object, whether to display the device status-->
    <batteryCapacityDisplay>
      <!--opt, bool, whether to display the battery capacity-->true
    </batteryCapacityDisplay>
    <chargingStatusDisplay>
      <!--opt, bool, whether to display the charging status-->true
    </chargingStatusDisplay>
    <bluetoothStatusDisplay>
      <!--opt, bool, whether to display the bluetooth status-->true
    </bluetoothStatusDisplay>
    <dialStatusDisplay>
      <!--opt, bool, whether to display the dial-up status-->true
    </dialStatusDisplay>
  </DeviceStatusDisplay>
  <boundary>
    <!--opt, int, range:[0,2], dep:and,{$.VideoOverlay.alignment,eq,GB}, desc:unit: number of characters-->000000
  </boundary>
  <FloorNumOverlay>
    <!--opt, object, elevator floor No.-->
    <enabled>
      <!--req, bool, whether to display the elevator floor No.-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </FloorNumOverlay>
  <upDownboundary>
    <!--opt, int, range:[0,2]-->1
  </upDownboundary>
  <leftRightboundary>
    <!--opt, int, range:[0,2]-->1
  </leftRightboundary>
  <Angle>
    <!--opt, object, device angle-->
    <TiltAngle>
      <!--opt, object, pitch angle-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
    </TiltAngle>
    <MultiAngleOverlay>
      <!--opt, object, multiple angles information overlay, desc:pitch angle, range, and rotation angle-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
    </MultiAngleOverlay>
  </Angle>
  <GPSOverlay>
    <!--opt, object, GPS information overlay-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
```

drackum@gmail.com
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    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </GPSOverlay>
  <PackingSpaceRecognitionOverlay>
    <!--opt, object, display of parking space detection result-->
    <isParkedEnabled>
      <!--opt, bool, whether to display the parking space status (occupied or free)-->true
    </isParkedEnabled>
  </PackingSpaceRecognitionOverlay>
  <WirelessDialOverlay>
    <!--opt, object, wireless dial-up status overlay-->
    <enabled>
      <!--req, bool, whether to enable. If it is enabled, the information such as wireless network type (5G/4G), upstream speed, and downstream speed will
be overlaid-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </WirelessDialOverlay>
  <MethaneDetectionOverlay>
    <!--opt, object, methane (CH4) detection informaiton overlay-->
    <enabled>
      <!--req, bool, whether to enable. If it is enabled, the information such as methane density and light intensity will be overlaid-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </MethaneDetectionOverlay>
  <WaterLevelDetectionOverlay>
    <!--opt, object, water level monitoring-->
    <enabled>
      <!--req, bool, whether to display the water level height-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </WaterLevelDetectionOverlay>
  <VisibilityDisplay>
    <!--opt, object, visibility information overlay-->
    <enabled>
      <!--req, bool, whether to enable. If it is enabled, the visibility information will be overlaid-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </VisibilityDisplay>
  <RoadSurfaceStatusDisplay>
    <!--opt, object, road status overlay-->
    <enabled>
      <!--req, bool, whether to enable. If it is enabled, the road status information will be overlaid-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </RoadSurfaceStatusDisplay>
  <FiveElementWeatherDisplay>
    <!--opt, object, 5-element weather overlay-->
    <enabled>
      <!--req, bool, whether to enable. If it is enabled, the 5-element weather information will be overlaid-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
```

dfsckum@gmail.com
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    </positionY>
  </FiveElementWeatherDisplay>
  <MEFStatusDisplay>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
  </MEFStatusDisplay>
  <RainfallOverlay>
    <!--opt, object, rainfall information overlay-->
    <enabled>
      <!--req, bool, whether to enable. If it is enabled, the rainfall information will be overlaid-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </RainfallOverlay>
  <WaterLevelThresholdOverlay>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </WaterLevelThresholdOverlay>
  <SolarPanelOverlay>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </SolarPanelOverlay>
  <BatteryInfoOverlay>
    <!--opt, object, desc:display the output voltage and output current of battery-->
    <enabled>
      <!--req, bool, whether to enable battery information overlay-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </BatteryInfoOverlay>
  <WirelessDataOverlay>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </WirelessDataOverlay>
  <AccelerationOverlay>
    <!--opt, object, display the acceleration-->
    <enabled>
      <!--req, bool, whether to enable acceleration overlay-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </AccelerationOverlay>
  <FirmwareVersionOverlay>
```

drackum@gmail.com
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    <!--opt, object, device firmware version information-->
    <enabled>
      <!--req, bool, whether to enable overlaying of device firmware version information-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </FirmwareVersionOverlay>
  <WifiOverlay>
    <!--opt, object, Wi-Fi status information, desc:enable/disable-->
    <enabled>
      <!--req, bool, whether to display the Wi-Fi status information-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </WifiOverlay>
  <StorageStatusOverlay>
    <!--opt, object, storage status information overlay, desc:whether there is SD card, the remaining storage space-->
    <enabled>
      <!--req, bool, whether to display the storage status information-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </StorageStatusOverlay>
  <RecordStatusOverlay>
    <!--opt, object, recording status overlay, desc:enable/disable-->
    <enabled>
      <!--req, bool, whether to enable the recording status overlay-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </RecordStatusOverlay>
  <laserRangingOverlay>
    <!--opt, object, laser distance measurement-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </laserRangingOverlay>
  <sensorDataOverlay>
    <!--opt, object, sensor data overlay-->
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </sensorDataOverlay>
  <frameCentreOverlayEnabled>
    <!--opt, bool-->true
  </frameCentreOverlayEnabled>
  <elevatorRunningOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
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dracknn@gmail.com
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      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </elevatorRunningOverlay>
  <elevatorPeopleNumOverlay>
    <!--ro, opt, object, overlay the number of passengers in an elevator-->
    <enabled>
      <!--ro, req, bool, whether to enable overlaying the number of passengers in an elevator-->false
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </elevatorPeopleNumOverlay>
</VideoOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, it is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

drackum@gmail.com
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14.2.7.14 Get the OSD capability of a specific video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, see details in the message of XML_BatteryPowerOverlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent opt="true,false">
      <!--ro, req, bool, whether it is transparent, attr:opt{req, string}-->true
    </transparent>
    <flashing opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </flashing>
  </attribute>
  <TextOverlayList size="8">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <positionX>
      <!--ro, req, int, range:[0,704]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,576]-->1
    </positionY>
    <dateStyle opt="YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY,CHR-YYYY-MM-DD,CHR-MM-DD-YYYY,CHR-DD-MM-YYYY,CHR-YYYY/MM/DD,CHR-MM/DD/YYYY,CHR-DD/MM/YYYY">
      <!--ro, opt, string, attr:opt{req, string}-->YYYY-MM-DD
    </dateStyle>
    <timeStyle opt="12hour,24hour">
      <!--ro, opt, string, time format, attr:opt{req, string}-->12hour
    </timeStyle>
    <displayWeek opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <positionX>
      <!--ro, opt, int, range:[0,1920]-->1
    </positionX>
    <positionY>
      <!--ro, opt, int, range:[0,1920]-->1
    </positionY>
  </channelNameOverlay>
  <fontSize opt="16*16,32*32,48*48,64*64,80*80,96*96,128*128,adaptive">
    <!--ro, opt, string, font size, attr:opt{req, string}-->adaptive
  </fontSize>
</VideoOverlay>
```

drackum@gmail.com
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14.2.7.15 Get the channel name overlay parameters of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, channel name overlay parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable channel name overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate-->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate-->1
  </positionY>
  <name>
    <!--ro, opt, string, channel name, range:[0,128]-->Camera 01
  </name>
  <OverlayRegion>
    <!--ro, opt, object, coordinates of overlaid string frame (the origin is the is the upper-left corner of the screen)-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</channelNameOverlay>
```

14.2.7.16 Set the channel name overlay parameters for a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable channel name overlay-->true
  </enabled>
  <positionX>
    <!--req, int, X-coordinate-->1
  </positionX>
  <positionY>
    <!--req, int, Y-coordinate-->1
  </positionY>
  <name>
    <!--opt, string, channel name, range:[0,32]-->Camera 01
  </name>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</channelNameOverlay>
```

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.7.17 Set the parameters of date and time information overlay for a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, configurations of date and time information overlay of video input channels, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable date and time information overlay-->true
  </enabled>
  <positionX>
    <!--req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <dateStyle>
    <!--opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-
YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--opt, enum, time format, subType:string, desc:"12hour" (12-hour system), "24hour" (24-hour system)-->12hour
  </timeStyle>
  <displayWeek>
    <!--opt, bool, whether to display week information-->true
  </displayWeek>
  <displayDate>
    <!--opt, bool, whether to display date-->true
  </displayDate>
  <displayTime>
    <!--opt, bool, whether to display time-->true
  </displayTime>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</DateTimeOverlay>
```

Response Message

dracku@gmail.com DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.7.18 Get the parameters of date and time information overlay of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable date and time overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlay area, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlay area, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <dateStyle>
    <!--ro, opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY"--
>YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--ro, opt, enum, time format, subType:string, desc:time format-->12hour
  </timeStyle>
  <displayWeek>
    <!--ro, opt, bool, whether to display week information-->true
  </displayWeek>
  <displayDate>
    <!--ro, opt, bool-->true
  </displayDate>
  <displayTime>
    <!--ro, opt, bool-->true
  </displayTime>
  <OverlayRegion>
    <!--ro, opt, object-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</DateTimeOverlay>
```

14.2.7.19 Delete the text overlay parameters of a specific video input channel

Request URL

drackum@gmail.com
DETEKTOR

DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.7.20 Set the text overlay parameters of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

dr.ackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in the message of XML_TextOverlay, subType:object, attr:version{opt, string, protocolVersion}-->
  <TextOverlay>
    <!--opt, object-->
    <id>
      <!--req, string, text No.-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <positionX>
      <!--req, int, X-coordinate-->1
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate-->1
    </positionY>
    <displayText>
      <!--req, string, overlay character-->test
    </displayText>
    <OverlayRegion>
      <!--opt, object-->
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </TextOverlay>
</TextOverlayList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

dracium@gmail.com
DETEKTOR

14.2.7.21 Get parameters of text information overlay of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, text information overlay, subType:object, attr:version{req, string, protocolVersion}-->
  <TextOverlay>
    <!--ro, opt, object, text overlay configuration-->
    <id>
      <!--ro, req, string, text No.-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the
screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the
screen-->1
    </positionY>
    <displayText>
      <!--ro, req, string, overlaid text-->test
    </displayText>
    <OverlayRegion>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </TextOverlay>
</TextOverlayList>
```

14.2.7.22 Add a text to overlay on the video for a specific video input channel

Request URL

POST /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, text overlay settings, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <positionX>
    <!--req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--req, int, the origin is the lower-left corner of the screen, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, text overlay-->test
  </displayText>
</TextOverlay>
```

Response Message

drackum@gmail.com DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.7.23 Set the parameters of a specific piece of text overlay information of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, text No.-->test
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, overlay character-->test
  </displayText>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</TextOverlay>
```

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.7.24 Get the text overlay parameters of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
textID	string	Text overlay ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay configuration, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, text No.-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-
->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-
->1
  </positionY>
  <displayText>
    <!--ro, req, string, overlaid text-->test
  </displayText>
  <OverlayRegion>
    <!--ro, opt, object-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</TextOverlay>
```

14.2.7.25 Delete a specified piece of text overlay of a video input channel

Request URL

DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.
textID	string	Text overlay ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.2.8 隐私遮蔽

14.2.8.1 Get the privacy mask parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/privacyMask

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, privacy mask, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--ro, opt, array, privacy mask area configurations, subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--ro, opt, object, configurations of a privacy mask area-->
      <id>
        <!--ro, req, int, area ID-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable the area-->true
      </enabled>
      <RegionCoordinatesList>
        <!--ro, req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--ro, opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
      </zoomdoorlimit>
      <mosaicThick>
        <!--ro, opt, enum, mosaic blur strength, subType:int, dep:and,{$.PrivacyMask.PrivacyMaskRegionList[*].PrivacyMaskRegion.maskType,eq,mosaic}, desc:it
is valid when the value of maskType is "mosaic". 1 (low, the block size is 8*8), 2 (medium, the block size is 16*16), 3 (high, the block size is 32*32)-->1
      </mosaicThick>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--ro, opt, enum, area type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

drackum@gmail.com
DEFEKTOR

14.2.8.2 Set the privacy mask parameters of the specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/privacyMask

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, privacy mask, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--opt, array, privacy mask area configurations, subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--opt, object, configurations of a privacy mask area-->
      <id>
        <!--req, int, area ID-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable the area-->true
      </enabled>
      <RegionCoordinatesList>
        <!--req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:The origin is the lower-left corner of the screen.-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
      </zoomdoorlimit>
      <mosaicThick>
        <!--opt, enum, mosaic blur strength, subType:int, dep:and,{$.PrivacyMask.PrivacyMaskRegionList[*].PrivacyMaskRegion.maskType,eq,mosaic}, desc:It is
valid when the value of maskType is "mosaic". 1 (low, the block size is 8*8), 2 (medium, the block size is 16*16), 3 (high, the block size is 32*32)-->1
      </mosaicThick>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--opt, enum, area type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.2.8.3 Get the privacy mask area parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/privacyMask/regions

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PrivacyMaskRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, privacy mask area configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <PrivacyMaskRegion>
    <!--ro, opt, object, configurations of a privacy mask area-->
    <id>
      <!--ro, req, int, area ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the area-->true
    </enabled>
    <RegionCoordinatesList>
      <!--ro, req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,576]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <privacymaskName>
      <!--ro, opt, string, privacy mask name-->test
    </privacymaskName>
    <maskType>
      <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
    </maskType>
    <zoomdoorlimit>
      <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
    </zoomdoorlimit>
    <mosaicThick>
      <!--ro, opt, enum, mosaic blur strength, subType:int, dep:and,{$.PrivacyMaskRegion.maskType,eq,mosaic}, desc:it is valid when the value of maskType is
"mosaic". 1 (low, the block size is 8*8), 2 (medium, the block size is 16*16), 3 (high, the block size is 32*32)-->1
    </mosaicThick>
  </PrivacyMaskRegion>
</PrivacyMaskRegionList>
```

drarkum@gmail.com
DETEKTOR

14.2.8.4 Set parameters of all privacy mask areas of the specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/privacyMask/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PrivacyMaskRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, privacy mask area configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <PrivacyMaskRegion>
    <!--opt, object, configurations of a privacy mask area-->
    <id>
      <!--req, int, area ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the area-->true
    </enabled>
    <RegionCoordinatesList>
      <!--req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,576]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <privacymaskName>
      <!--opt, string, privacy mask name-->test
    </privacymaskName>
    <maskType>
      <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
    </maskType>
    <zoomdoorlimit>
      <!--opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
    </zoomdoorlimit>
    <mosaicThick>
      <!--opt, enum, mosaic blur strength, subType:int, dep:and,{$.PrivacyMaskRegion.maskType,eq,mosaic}, desc:it is valid when the value of maskType is
"mosaic". 1 (low, the block size is 8*8), 2 (medium, the block size is 16*16), 3 (high, the block size is 32*32)-->1
    </mosaicThick>
  </PrivacyMaskRegion>
</PrivacyMaskRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0:OK;1:OK;2:Device Busy;3:Device Error;4:Invalid Operation;5:Invalid XML Format;6:Invalid XML
Content;7:Reboot Required;-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:OK, Device Busy, Device Error, Invalid Operation, Invalid XML Format, Invalid XML Content,
Reboot-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

dracium@gmail.com
DETEKTOR

14.2.8.5 Get the privacy mask parameters of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask?switch=<switchType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
switchType	string	Analysis unit switch: "local" (analysis by local device, e.g., executed locally by NVR), "remote" (the device transmits data to sub-device for analysis, e.g., from the NVR to the network camera).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, privacy mask configurations, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the privacy mask-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--ro, opt, array, list of video tampering region(s), subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--ro, opt, object, configurations of a privacy mask region-->
      <id>
        <!--ro, req, int, region ID-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable the region-->true
      </enabled>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--ro, opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private region will be displayed, range:[10,1000]-->1
      </zoomdoorlimit>
      <mediaObjectEncryption>
        <!--ro, opt, object, encryption of the private target in media data, desc:If it is enabled, the main steam of corresponding region will be covered,
but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?format=json). You can use the key to decrypt the
encrypted region.-->
        <enabled>
          <!--ro, req, bool, whether to enable-->true
        </enabled>
        <maskType>
          <!--ro, opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
        </maskType>
      </mediaObjectEncryption>
      <regionRangeType>
        <!--ro, opt, enum, subType:string-->region
      </regionRangeType>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--ro, opt, enum, region type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

drackum@gmail.com
DETEKTOR

14.2.8.6 Set the privacy mask parameters of a specified channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask?switch=<switchType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
switchType	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, privacy mask configurations, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the privacy mask-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--opt, array, list of video privacy mask area(s), subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--opt, object, configurations of a privacy mask area-->
      <id>
        <!--req, int, area ID-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable the area-->true
      </enabled>
      <RegionCoordinatesList>
        <!--req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--opt, int, zooming value limit range, range:[10,1000]-->1
      </zoomdoorlimit>
      <mediaObjectEncryption>
        <!--opt, object, encryption of the private target in media data, desc:it is disabled when this field is not configured. If it is enabled, the main
steam of corresponding area will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted area-->
        <enabled>
          <!--req, bool, whether to enable-->true
        </enabled>
        <maskType>
          <!--opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
        </maskType>
      </mediaObjectEncryption>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--opt, enum, area type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

dr'ackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.3 录像管理

14.3.1 云存储管理

14.3.1.1 Get the cloud storage access capability

Request URL

GET /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage/<indexID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No. The default value is 1.
indexID	string	Storage server index, which starts from 1.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, cloud storage service parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index-->test
  </id>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, address types, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <port min="1" max="65535">
    <!--ro, req, int, port No., range:[1,65535], attr:min{req, int},max{req, int}-->1
  </port>
  <userName min="1" max="128">
    <!--ro, opt, string, range:[1,128], attr:min{req, int},max{req, int}-->test
  </userName>
  <password min="1" max="32">
    <!--ro, opt, string, password, attr:min{req, int},max{req, int}, desc:password-->null
  </password>
  <postPoolID min="0" max="4294967295">
    <!--ro, opt, int, checkpoint picture pool ID, attr:min{req, int},max{req, int}-->1
  </postPoolID>
  <illegalPoolID min="0" max="4294967295">
    <!--ro, opt, int, violation picture pool ID, attr:min{req, int},max{req, int}-->1
  </illegalPoolID>
  <vehicleDetectionID min="0" max="4294967295">
```

drackum@gmail.com
DETEKTOR

```
    <!--ro, opt, int, ID of vehicle detection picture pool, attr:min{req, int},max{req, int}-->1
  </vehicleDetectionID>
  <cloudStorageType opt="0,1,2">
    <!--ro, opt, enum, storage type, subType:int, attr:opt{req, string}, desc:0-cloud storage (default), 1-CVR, 2-pcnvr2.0; it is the cloud storage by
default if this node is empty-->1
  </cloudStorageType>
  <V2.0>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, address types, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port min="0" max="65535">
      <!--ro, req, int, port No., attr:min{req, int},max{req, int}-->1
    </port>
    <isSupportAccessKey>
      <!--ro, opt, bool, whether it supports access key configuration-->true
    </isSupportAccessKey>
    <isSupportSecretKey>
      <!--ro, opt, bool, whether it supports secret key configuration-->true
    </isSupportSecretKey>
    <postPoolID min="0" max="4294967295">
      <!--ro, opt, int, checkpoint picture pool ID, attr:min{req, int},max{req, int}, desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID min="0" max="4294967295">
      <!--ro, opt, int, violation picture pool ID, attr:min{req, int},max{req, int}-->1
    </illegalPoolID>
    <vehicleDetectionID min="0" max="4294967295">
      <!--ro, opt, int, ID of vehicle detection picture pool, attr:min{req, int},max{req, int}-->1
    </vehicleDetectionID>
    <accessKeyLen min="1" max="64">
      <!--ro, opt, int, access key length, dep:and,{$.CloudStorage.V2.0.isSupportAccessKey,eq,true}, attr:min{req, int},max{req, int}-->1
    </accessKeyLen>
    <secretKeyLen min="1" max="64">
      <!--ro, opt, int, secret key length, dep:and,{$.CloudStorage.V2.0.isSupportSecretKey,eq,true}, attr:min{req, int},max{req, int}-->1
    </secretKeyLen>
    <resourceID min="0" max="32">
      <!--ro, opt, string, resource pool ID, attr:min{req, int},max{req, int}, desc:resource pool ID-->test
    </resourceID>
    <transportProtocolType opt="HTTP,HTTPS" def="HTTPS">
      <!--ro, opt, enum, "HTTP", "HTTPS", subType:string, attr:opt{req, string},def{req, string}, desc:"HTTP", "HTTPS"-->HTTP
    </transportProtocolType>
  </V2.0>
  <ImageAndVideoCloud>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, address types, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port min="0" max="65535">
      <!--ro, req, int, port No., attr:min{req, int},max{req, int}-->1
    </port>
    <isSupportAccessKey>
      <!--ro, opt, bool, whether it supports access key configuration-->true
    </isSupportAccessKey>
    <isSupportSecretKey>
      <!--ro, opt, bool, whether it supports secret key configuration-->true
    </isSupportSecretKey>
    <postPoolID min="0" max="4294967295">
      <!--ro, opt, int, checkpoint picture pool ID, attr:min{req, int},max{req, int}, desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID min="0" max="4294967295">
      <!--ro, opt, int, violation picture pool ID, attr:min{req, int},max{req, int}-->1
    </illegalPoolID>
    <vehicleDetectionID min="0" max="4294967295">
      <!--ro, opt, int, ID of vehicle detection picture pool, attr:min{req, int},max{req, int}-->1
    </vehicleDetectionID>
    <accessKeyLen min="1" max="64">
      <!--ro, opt, int, access key length, dep:and,{$.CloudStorage.ImageAndVideoCloud.isSupportAccessKey,eq,true}, attr:min{req, int},max{req, int}-->1
    </accessKeyLen>
    <secretKeyLen min="1" max="64">
      <!--ro, opt, int, secret key length, dep:and,{$.CloudStorage.ImageAndVideoCloud.isSupportSecretKey,eq,true}, attr:min{req, int},max{req, int}-->1
```

dr.ckum@gmail.com
DETEKTOR

```
      <!--ro, opt, int, secret key length, dep:and,{$.CloudStorage.ImageAndVideoCloud.isSupportSecretKey,eq,true}, attr:min{req, int},max{req, int}-->1
    </secretKeyLen>
  </ImageAndVideoCloud>
  <protocolType opt="V1.0,V2.0,Image and Video Cloud,objectStorage,videoStreamingOS">
    <!--ro, opt, enum, cloud storage protocol type, subType:string, attr:opt{req, string}, desc:"V1.0", "V2.0", "Image and Video Cloud"-->V1.0
  </protocolType>
  <isSupportCloudStorageServerParam>
    <!--ro, opt, bool, whether it supports getting the parameters of cloud storage server, desc:whether it supports getting the parameters of cloud storage
server-->true
  </isSupportCloudStorageServerParam>
  <bucketName>
    <!--ro, opt, string, bucket name of cloud storage-->test
  </bucketName>
  <associateToCenter>
    <!--ro, opt, object-->
    <centerType opt="all,ISUP,OTAP">
      <!--ro, req, enum, subType:string, attr:opt{req, string}-->ISUP
    </centerType>
    <centerID min="1" max="4">
      <!--ro, req, int, range:[1,4], attr:min{req, int},max{req, int}-->0
    </centerID>
  </associateToCenter>
</CloudStorage>
```

14.3.1.2 Get the access parameters of specified cloud storage

Request URL

GET /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No. The default value is 1.
indexID	string	Storage server index, which starts from 1.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, cloud storage service, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index-->test
  </id>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <addressingFormatType>
    <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <port>
    <!--ro, req, int, port No., range:[1,65535]-->1
  </port>
  <userName>
    <!--ro, req, string, range:[1,64]-->test
  </userName>
  <postPoolID>
    <!--ro, opt, int, checkpoint picture pool ID, range:[0,4294967295]-->1
  </postPoolID>
  <illegalPoolID>
    <!--ro, opt, int, violation picture pool ID, range:[0,4294967295]-->1
  </illegalPoolID>
  <vehicleDetectionID>
    <!--ro, opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
  </vehicleDetectionID>
  <protocolType>
    <!--ro, opt, enum, cloud storage protocol type, subType:string, desc:"V1.0", "V2.0", "Image and Video Cloud"-->V1.0
  </protocolType>
  <cloudStorageType>
    <!--ro, opt, enum, 0-cloud storage (default),1-CVR,2-pcnvr2.0; if this node is empty,it is the cloud storage by default, subType:int, desc:0-cloud
storage (default), 1-CVR, 2-pcnvr2.0; if this node is empty, it is the cloud storage by default-->0
```

drackum@gmail.com
HIKVISION

```
  </cloudStorageType>
  <V2.0>
    <!--ro, opt, object, this node in valid only when the vaule of protocolType is V2.0, dep:and,{$.CloudStorage.protocolType,eq,V2.0}, desc:this node in valid only when the vaule of protocolType is V2.0-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <addressingFormatType>
      <!--ro, req, enum, address type: "ipaddress","hostname", subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--ro, req, int, port No., range:[1,65535]-->1
    </port>
    <postPoolID>
      <!--ro, opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID>
      <!--ro, opt, int, violation picture pool ID, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--ro, opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
    </vehicleDetectionID>
    <resourceID>
      <!--ro, opt, string, resource pool ID, range:[1,32], desc:resource pool ID-->test
    </resourceID>
    <transportProtocolType>
      <!--ro, opt, enum, "HTTP", "HTTPS", subType:string, desc:"HTTP", "HTTPS"-->HTTP
    </transportProtocolType>
  </V2.0>
  <ImageAndVideoCloud>
    <!--ro, opt, object, dep:and,{$.CloudStorage.protocolType,eq,Image and Video Cloud}-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--ro, req, int, port No., range:[1,65535]-->1
    </port>
    <postPoolID>
      <!--ro, opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID>
      <!--ro, opt, int, violation picture pool ID, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--ro, opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
    </vehicleDetectionID>
  </ImageAndVideoCloud>
  <bucketName>
    <!--ro, opt, string, bucket name of cloud storage, desc:bucket name of cloud storage-->test
  </bucketName>
  <status>
    <!--ro, opt, enum, cloud storage status, subType:string, desc:only supports getting this node. “ok” (normal), “connectServerFail” (connecting to server failed), “passwordError” (incorrect password), “exchangeServerFail”, “createDirectoryFailed”, “noWritePermission” (no writing permission), “noBucket” (bucket does not exist)-->ok
  </status>
  <associateToCenter>
    <!--ro, opt, object-->
    <centerType>
      <!--ro, req, enum, subType:string-->ISUP
    </centerType>
    <centerID>
      <!--ro, req, int, range:[1,4]-->0
    </centerID>
  </associateToCenter>
</CloudStorage>
```

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14.3.1.3 Set the access parameters of specified cloud storage

Request URL

PUT /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
indexID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, cloud storage service, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index-->test
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:ipaddress: IP address; hostname: host server name;-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--opt, string, host server name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <port>
    <!--req, int, port No., range:[1,65535]-->1
  </port>
  <userName>
    <!--req, string, range:[1,64]-->test
  </userName>
  <password>
    <!--opt, string, password, range:[1,32]-->test
  </password>
  <postPoolID>
    <!--opt, int, checkpoint picture pool ID, range:[0,4294967295]-->1
  </postPoolID>
  <illegalPoolID>
    <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
  </illegalPoolID>
  <vehicleDetectionID>
    <!--opt, int, vehicle detection picture pool ID, range:[0,4294967295]-->1
  </vehicleDetectionID>
  <protocolType>
    <!--opt, enum, cloud storage protocol type, subType:string, desc:"V1.0"; "V2.0"; "Image and Video Cloud"-->V1.0
  </protocolType>
  <cloudStorageType>
    <!--opt, enum, storage type, subType:int, desc:0-cloud storage (default),1-CVR,2-pcnvr2.0; if this node is empty,it is the cloud storage by default-->1
  </cloudStorageType>
  <V2.0>
    <!--opt, object, this node in valid only when the vaule of protocolType is V2.0, dep:and,{$.CloudStorage.protocolType,eq,V2.0}-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:ipaddress: IP address; hostname: host server name;-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host server name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--req, int, port No., range:[1,65535]-->1
    </port>
    <accessKey>
      <!--opt, string, access key, range:[1,64]-->test
    </accessKey>
    <secretKey>
      <!--opt, string, secret key, range:[1,64]-->test
    </secretKey>
    <postPoolID>
      <!--opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID>
      <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
```

dracknum@gmail.com
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```
    </illegalPoolID>
    <vehicleDetectionID>
      <!--opt, int, vehicle detection picture pool ID, range:[0,4294967295]-->1
    </vehicleDetectionID>
    <resourceID>
      <!--opt, string, capability set of resource pool ID, range:[1,32], desc:resource pool ID-->test
    </resourceID>
    <transportProtocolType>
      <!--opt, enum, cloud storage transfer protocol type, subType:string, desc:HTTP; HTTPS;-->HTTP
    </transportProtocolType>
  </V2.0>
  <ImageAndVideoCloud>
    <!--opt, object, cloud storage protocol of picture and video library, dep:and,{$.CloudStorage.protocolType,eq,Image and Video Cloud}-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:ipaddress: IP address; hostname: host server name;-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host server name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--req, int, port No., range:[1,65535]-->1
    </port>
    <accessKey>
      <!--opt, string, access key, range:[1,64]-->test
    </accessKey>
    <secretKey>
      <!--opt, string, secret key, range:[1,64]-->test
    </secretKey>
    <postPoolID>
      <!--opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID>
      <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--opt, int, vehicle detection picture pool ID, range:[0,4294967295]-->1
    </vehicleDetectionID>
  </ImageAndVideoCloud>
  <bucketName>
    <!--opt, string, name of cloud storage bucket, desc:bucket name of cloud storage-->test
  </bucketName>
</CloudStorage>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

14.3.1.4 Search for test accessing cloud storage for test information

Request URL

POST /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage/test

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No. The default value is 1.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorageTestDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, test conditions, attr:version{req, string, protocolVersion}-->
  <addressingFormatType>
    <!--req, enum, address types, subType:string, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <portNo>
    <!--req, int, range:[1,65535]-->1
  </portNo>
  <username>
    <!--opt, string, user name, range:[1,64]-->test
  </username>
  <password>
    <!--opt, string, password, range:[1,32]-->test
  </password>
  <postPoolID>
    <!--opt, int, checkpoint picture pool ID, range:[0,4294967295]-->1
  </postPoolID>
  <illegalPoolID>
    <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
  </illegalPoolID>
  <vehicleDetectionID>
    <!--opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
  </vehicleDetectionID>
  <protocolType>
    <!--req, enum, cloud storage protocol type, subType:string, desc:cloud storage protocol type: "V1.0","V2.0","Image and Video Cloud"-->V1.0
  </protocolType>
  <cloudStorageType>
    <!--opt, enum, storage server type, subType:int, desc:0-cloud storage (default),1-CVR,2-pcnvr2.0; if this node is empty,it is the cloud storage by
default-->1
  </cloudStorageType>
  <V2.0>
    <!--opt, object, it is valid when <protocolType> is "V2.0", dep:and,{$.CloudStorage.protocolType,eq,V2.0}-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--req, int, range:[1,65535]-->1
    </port>
    <accessKey>
      <!--opt, string, password, range:[1,64], desc:password-->test
    </accessKey>
    <secretKey>
      <!--opt, string, secret key, range:[1,64], desc:secret key-->test
    </secretKey>
    <postPoolID>
      <!--opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID>
      <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
    </vehicleDetectionID>
    <resourceID>
      <!--opt, string, both resourceID and cloudPoolID can be used to configure the cloud storage poolID. But different from cloudPoolId, the value of
resourceID can be in string format. When both resourceID and cloudPoolId are configured, cloudPoolId has priority over resourceID. If this node is not
supported, you can configure the pool ID via cloudPoolId, range:[1,32], desc:both resourceID and cloudPoolID can be used to configure the cloud storage
poolID. But different from cloudPoolId, the value of resourceID can be in string format. When both resourceID and cloudPoolId are configured, cloudPoolId
has priority over resourceID. If this node is not supported, you can configure the pool ID via cloudPoolId-->test
    </resourceID>
    <transportProtocolType>
      <!--opt, enum, cloud storage server type, subType:string, desc:cloud storage server type-->HTTP
    </transportProtocolType>
  </V2.0>
  <ImageAndVideoCloud>
    <!--opt, object, dep:and,{$.CloudStorage.protocolType,eq,Image and Video Cloud}-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
```

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```
      <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--req, int, port No., range:[1,65535]-->1
    </port>
    <accessKey>
      <!--opt, string, password, range:[1,64], desc:password-->test
    </accessKey>
    <secretKey>
      <!--opt, string, secret key, range:[1,64], desc:secret key-->test
    </secretKey>
    <postPoolID>
      <!--opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID>
      <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
    </vehicleDetectionID>
  </ImageAndVideoCloud>
  <bucketName>
    <!--opt, string, bucket name of cloud storage, desc:bucket name of cloud storage-->test
  </bucketName>
</CloudStorageTestDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorageTestResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, test result, attr:version{req, string, protocolVersion}-->
  <errorDescription>
    <!--ro, req, enum, error description, subType:string, desc:error description-->ok
  </errorDescription>
</CloudStorageTestResult>
```

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14.3.1.5 Get the access parameters of cloud storage in a batch

Request URL

GET /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No. The default value is 1.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorageList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, cloud storage service list, subType:object, attr:version{req, string, protocolVersion}-->
  <CloudStorage>
    <!--ro, opt, object, cloud storage service-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
```

```
</ipv6Address>
<port>
  <!--ro, req, int, port No., range:[1,65535]-->1
</port>
<userName>
  <!--ro, opt, string, range:[1,64]-->test
</userName>
<postPoolID>
  <!--ro, opt, int, range:[0,4294967295]-->1
</postPoolID>
<illegalPoolID>
  <!--ro, opt, int, range:[0,4294967295]-->1
</illegalPoolID>
<vehicleDetectionID>
  <!--ro, opt, int, range:[0,4294967295]-->1
</vehicleDetectionID>
<protocolType>
  <!--ro, opt, enum, subType:string-->V1.0
</protocolType>
<cloudStorageType>
  <!--ro, opt, enum, storage server type, subType:int, desc:storage server type-->1
</cloudStorageType>
<V2.0>
  <!--ro, opt, object, dep:and,{$.CloudStorage.protocolType,eq,V2.0}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <addressingFormatType>
    <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <port>
    <!--ro, req, int, port No., range:[1,65535]-->1
  </port>
  <postPoolID>
    <!--ro, opt, int, range:[0,4294967295]-->1
  </postPoolID>
  <illegalPoolID>
    <!--ro, opt, int, range:[0,4294967295]-->1
  </illegalPoolID>
  <vehicleDetectionID>
    <!--ro, opt, int, range:[0,4294967295]-->1
  </vehicleDetectionID>
  <resourceID>
    <!--ro, opt, string, range:[1,32]-->test
  </resourceID>
  <transportProtocolType>
    <!--ro, opt, enum, subType:string-->HTTP
  </transportProtocolType>
</V2.0>
<ImageAndVideoCloud>
  <!--ro, opt, object, dep:and,{$.CloudStorage.protocolType,eq,Image and Video Cloud}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <addressingFormatType>
    <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <port>
    <!--ro, req, int, port No., range:[1,65535]-->1
  </port>
  <postPoolID>
    <!--ro, opt, int, range:[0,4294967295]-->1
  </postPoolID>
  <illegalPoolID>
    <!--ro, opt, int, range:[0,4294967295]-->1
  </illegalPoolID>
  <vehicleDetectionID>
    <!--ro, opt, int, range:[0,4294967295]-->1
  </vehicleDetectionID>
</ImageAndVideoCloud>
<bucketName>
  <!--ro, opt, string-->test
</bucketName>
<associateToCenter>
  <!--ro, opt, object-->
```

drackum@gmail.com
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```
      <centerType>
        <!--ro, req, enum, subType:string-->ISUP
      </centerType>
      <centerID>
        <!--ro, req, int, range:[1,4]-->0
      </centerID>
    </associateToCenter>
  </CloudStorage>
</CloudStorageList>
```

14.3.1.6 Set the access parameters of cloud storage in a batch

Request URL

PUT /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorageList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, cloud storage service list, subType:object, attr:version{req, string, protocolVersion}-->
  <CloudStorage>
    <!--opt, object, cloud storage service-->
    <id>
      <!--req, string, ID-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--req, int, port No., range:[1,65535]-->1
    </port>
    <userName>
      <!--opt, string, range:[1,64]-->test
    </userName>
    <password>
      <!--opt, string, password, range:[1,32]-->test
    </password>
    <postPoolID>
      <!--opt, int, range:[0,4294967295]-->1
    </postPoolID>
    <illegalPoolID>
      <!--opt, int, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--opt, int, range:[0,4294967295]-->1
    </vehicleDetectionID>
    <protocolType>
      <!--opt, enum, subType:string-->V1.0
    </protocolType>
    <cloudStorageType>
      <!--opt, enum, storage server type, subType:int-->1
    </cloudStorageType>
    <V2.0>
      <!--opt, object, dep:and,{$.CloudStorage.protocolType,eq,V2.0}-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
      </addressingFormatType>
      <hostName>
        <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
      </ipAddress>
      <ipv6Address>
```

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```
        <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
      </ipv6Address>
      <port>
        <!--req, int, port No., range:[1,65535]-->1
      </port>
      <accessKey>
        <!--opt, string, password, range:[1,64]-->test
      </accessKey>
      <secretKey>
        <!--opt, string, secret key, range:[1,64]-->test
      </secretKey>
      <postPoolID>
        <!--opt, int, range:[0,4294967295]-->1
      </postPoolID>
      <illegalPoolID>
        <!--opt, int, range:[0,4294967295]-->1
      </illegalPoolID>
      <vehicleDetectionID>
        <!--opt, int, range:[0,4294967295]-->1
      </vehicleDetectionID>
      <resourceID>
        <!--opt, string, range:[1,32]-->test
      </resourceID>
      <transportProtocolType>
        <!--opt, enum, subType:string-->HTTP
      </transportProtocolType>
    </V2.0>
    <ImageAndVideoCloud>
      <!--opt, object, dep:and,{$.CloudStorage.protocolType,eq,Image and Video Cloud}-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
      </addressingFormatType>
      <hostName>
        <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
      </ipv6Address>
      <port>
        <!--req, int, port No., range:[1,65535]-->1
      </port>
      <accessKey>
        <!--opt, string, password, range:[1,64]-->test
      </accessKey>
      <secretKey>
        <!--opt, string, secret key, range:[1,64]-->test
      </secretKey>
      <postPoolID>
        <!--opt, int, range:[0,4294967295]-->1
      </postPoolID>
      <illegalPoolID>
        <!--opt, int, range:[0,4294967295]-->1
      </illegalPoolID>
      <vehicleDetectionID>
        <!--opt, int, range:[0,4294967295]-->1
      </vehicleDetectionID>
    </ImageAndVideoCloud>
    <bucketName>
      <!--opt, string-->test
    </bucketName>
  </CloudStorage>
</CloudStorageList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.3.1.7 Set the third-party cloud access parameters

Request URL

PUT /ISAPI/ContentMgmt/Storage/cloud

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Cloud xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--req, bool-->true
  </enable>
  <type opt="OneDrive,GoogleDrive ,DropBox">
    <!--req, enum, subType:string, attr:opt{req, string}-->OneDrive
  </type>
  <status opt="Online,Offline">
    <!--req, enum, subType:string, attr:opt{req, string}-->Online
  </status>
  <authCode min="0" max="64">
    <!--req, string, attr:min{req, int},max{req, int}-->test
  </authCode>
  <alias min="0" max="32">
    <!--opt, string, attr:min{req, int},max{req, int}-->test
  </alias>
  <tolalCapacity>
    <!--opt, int, unit:MB-->1
  </tolalCapacity>
  <usedSpace>
    <!--opt, int, unit:MB-->1
  </usedSpace>
</Cloud>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.3.1.8 Get the third-party cloud access parameters

drackum@gmail.com
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Request URL

GET /ISAPI/ContentMgmt/Storage/cloud

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Cloud xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--ro, req, bool-->true
  </enable>
  <type opt="OneDrive,GoogleDrive ,DropBox">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->OneDrive
  </type>
  <status opt="Online,Offline">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->Online
  </status>
  <authCode min="0" max="64">
    <!--ro, req, string, attr:min{req, int},max{req, int}-->test
  </authCode>
  <alias min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </alias>
  <tolalCapacity>
    <!--ro, opt, int, unit:MB-->1
  </tolalCapacity>
  <usedSpace>
    <!--ro, opt, int, unit:MB-->1
  </usedSpace>
</Cloud>
```

14.3.1.9 Get the third-party cloud access capability

Request URL

GET /ISAPI/ContentMgmt/Storage/cloud/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Cloud xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--ro, req, bool-->true
  </enable>
  <type opt="OneDrive,GoogleDrive ,DropBox">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->OneDrive
  </type>
  <status opt="Online,Offline">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->Online
  </status>
  <authCode min="0" max="64">
    <!--ro, req, string, attr:min{req, int},max{req, int}-->test
  </authCode>
  <alias min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </alias>
  <tolalCapacity>
    <!--ro, opt, int, unit:MB-->1
  </tolalCapacity>
  <usedSpace>
    <!--ro, opt, int, unit:MB-->1
  </usedSpace>
</Cloud>
```

drackum@gmail.com
DETEKTOR

14.3.1.10 Get storage strategy parameters for the third-party cloud access

Request URL

GET /ISAPI/ContentMgmt/Storage/cloud/channels/<channelID>/uploadStrategy

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<UploadStrategy xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <strategyType opt="record,picture">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->record
  </strategyType>
  <Record>
    <!--ro, opt, object, dep:and,{$.UploadStrategy.strategyType,eq,record}-->
    <allEvent>
      <!--ro, opt, bool-->true
    </allEvent>
    <motion>
      <!--ro, opt, bool-->true
    </motion>
    <alarm>
      <!--ro, opt, bool-->true
    </alarm>
    <VCA>
      <!--ro, opt, bool-->true
    </VCA>
  </Record>
</UploadStrategy>
```

14.3.1.11 Set storage strategy parameters for the third-party cloud access

Request URL

PUT /ISAPI/ContentMgmt/Storage/cloud/channels/<channelID>/uploadStrategy

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<UploadStrategy xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <strategyType opt="record,picture">
    <!--opt, enum, subType:string, attr:opt{req, string}-->record
  </strategyType>
  <Record>
    <!--opt, object, dep:and,{$.UploadStrategy.strategyType,eq,record}-->
    <allEvent>
      <!--opt, bool-->true
    </allEvent>
    <motion>
      <!--opt, bool-->true
    </motion>
    <alarm>
      <!--opt, bool-->true
    </alarm>
    <VCA>
      <!--opt, bool-->true
    </VCA>
  </Record>
</UploadStrategy>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.3.1.12 Get the capability of storage strategy parameters for the third-party cloud access

Request URL

GET /ISAPI/ContentMgmt/Storage/cloud/channels/<channelID>/uploadStrategy/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<UploadStrategy xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <strategyType opt="record,picture">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->record
  </strategyType>
  <Record>
    <!--ro, opt, object, dep:and,{$.UploadStrategy.strategyType,eq,record}-->
    <allEvent>
      <!--ro, opt, bool-->true
    </allEvent>
    <motion>
      <!--ro, opt, bool-->true
    </motion>
    <alarm>
      <!--ro, opt, bool-->true
    </alarm>
    <VCA>
      <!--ro, opt, bool-->true
    </VCA>
  </Record>
</UploadStrategy>
```

14.3.1.13 Get the URL capability of the third-party cloud

Request URL

GET /ISAPI/ContentMgmt/Storage/cloud/URL/capabilities

Query Parameter

None

Request Message

None

Response Message

dr.ackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<CloudURL xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <url min="0" max="256">
    <!--ro, req, string, attr:min{req, int},max{req, int}-->test
  </url>
</CloudURL>
```

14.3.2 存储策略

14.3.2.1 Get AcuSearch capability

Request URL

GET /ISAPI/ContentMgmt/PicSearchByCoordinate/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "PicSearchByCoordinateCap": {
    /*opt, object, AcuSearch configuration capability, desc:corresponding API: GET or PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json*/
        "searchPicEnabled": {
        /*req, object, whether to enable AcuSearch, desc:true (only enable AcuSearch for the following configured channels), false (disable AcuSearch for
all channels). When enabled, the device will perform modeling and analysis for following configured channels and support searching pictures by target
afterwards.*/
            "@opt": [],
            /*req, array, AcuSearch enabling/disabling, subType:boolean, range:[1,2]*/
            "@def":  "true"
            /*opt, boolean, default value*/
        },
        "targetTypeList": {
        /*opt, object, list of target types, desc:support selecting specified target type for modeling and analysis*/
            "@size":  4,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  0,
            /*opt, int, the minimum number of array elements*/
            "targetType": {
            /*req, object, target type range*/
                "@opt": [],
                /*opt, array, enumeration list of target type, subType:string, range:[1,4]*/
                "@def":  "face"
                /*opt, string, default value*/
            },
            "humanEnabled": {
            /*opt, object, whether to enable human body as a target type*/
                "@opt": [],
                /*req, array, subType:boolean, range:[1,2]*/
                "@def":  "true"
                /*opt, boolean, default value*/
            },
            "vehicleEnabled": {
            /*opt, object, whether to enable vehicle as a target type*/
                "@opt": [],
                /*req, array, subType:boolean, range:[1,2]*/
                "@def":  "true"
                /*opt, boolean, default value*/
            },
            "faceEnabled": {
            /*opt, object, whether to enable face as a target type*/
                "@opt": [],
                /*req, array, subType:boolean, range:[1,2]*/
                "@def":  "true"
                /*opt, boolean, default value*/
            },
            "humanAttrEnabled": {
            /*opt, object, desc:when enabled, this function consumes system resources and may impact the target search efficiency. Attribute information
will only be available in AcuSearch if this is enabled*/
                "@opt": [],
                /*req, array, whether to enable human body attribute as a target type, subType:boolean, range:[1,2]*/
                "@def":  "false"
                /*opt, boolean, default value*/
            },
            "vehicleAttrEnabled": {
            /*opt, object, whether to enable vehicle attribute as a target type, desc:when enabled, this function consumes system resources and may impact
the target search efficiency. Attribute information will only be available in AcuSearch if this is enabled*/
                "@opt": [],
                /*req, array, subType:boolean, range:[1,2]*/
```

dracium@gmail.com DETEKTOR

```
                "@def":  "false"
                /*opt, boolean, default value*/
            },
            "animalEnabled": {
            /*opt, object, whether to enable animal as a target type*/
                "@opt": [],
                /*req, array, subType:boolean, range:[1,2]*/
                "@def":  "false"
                /*opt, boolean, default value*/
            }
        },
        "channelList": {
        /*opt, object, channel No. list with AI by camera enabled, desc:cameras connected to these channels support AcuSearch modeling and the analysis is conducted by the camera's AI function. If this field is absent, it indicates that all the channels use AI by camera. It corresponds to the "channelID" of the "dualVCAType" = "remote" returned by URL (/ISAPI/Streaming/channels/dualVCA?format=json)*/
            "@size":  256,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  0,
            /*opt, int, the minimum number of array elements*/
            "@min":  1,
            /*req, int, minimum value*/
            "@max":  256
            /*req, int, maximum value*/
        },
        "localChannelList": {
        /*opt, object, channel No. list with AI by NVR/DVR enabled, desc:cameras connected to these channels do not support AcuSearch modeling and the analysis is conducted by the AI by NVR/DVR. If this field is absent, it indicates that all the channels use AI by NVR/DVR*/
            "@size":  4,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  0,
            /*opt, int, the minimum number of array elements*/
            "@min":  1,
            /*req, int, minimum value*/
            "@max":  256
            /*req, int, maximum value*/
        }
    }
}
```

14.3.2.2 Get AcuSearch configurations

Request URL

GET /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
{
    "searchPicEnabled":  "true",
    /*req, boolean, whether to enable AcuSearch, desc:true (only enable AcuSearch for the following configured channels), false (disable AcuSearch for all channels). When enabled, the device will perform modeling and analysis for following configured channels and support searching pictures by target afterwards.*/
    "targetTypeList": [
    /*opt, array, modeling target list, subType:object, range:[0,4], desc:support selecting specified target type for modeling and analysis*/
        {
            "targetType":  "face",
            /*req, enum, target type, subType:string, desc:"face", "human" (human body), "vehicle", "animal"*/
            "humanEnabled":  "true",
            /*opt, boolean, whether to enable human body as a target type, dep:and,{targetType,eq,human}*/
            "vehicleEnabled":  "true",
            /*opt, boolean, whether to enable vehicle as a target type, dep:and,{targetType,eq,vehicle}*/
            "faceEnabled":  "true",
            /*opt, boolean, whether to enable face as a target type, dep:and,{targetType,eq,face}*/
            "humanAttrEnabled":  "false",
            /*opt, boolean, whether to enable human body attribute as a target type, dep:and,{targetType,eq,human}, desc:when enabled, this function consumes system resources and may impact the target search efficiency. Attribute information will only be available in AcuSearch if this is enabled*/
            "vehicleAttrEnabled":  "false",
            /*opt, boolean, whether to enable vehicle attribute as a target type, dep:and,{targetType,eq,vehicle}, desc:when enabled, this function consumes system resources and may impact the target search efficiency. Attribute information will only be available in AcuSearch if this is enabled*/
            "animalEnabled":  "false"
            /*opt, boolean, whether to enable animal as a target type, dep:and,{targetType,eq,animal}*/
        }
    ],
    "targetResourcesInfo": [
    /*opt, array, (read-only) information about occupied resource channels information after enabling modeling target, subType:object, range:[0,4], desc:used to inform users of the number of occupied resources channels when enabling modeling target*/
        {
            "targetType":  "face",
            /*req, enum, target type, subType:string, desc:"face", "human" (human body), "vehicle", "animal"*/
            "resourcesNum":  2
            /*req, int, number of consumed resource channels when enabling modeling target, range:[0,100], desc:for example, if the "targetType" = "face" and "resourcesNum" = 2, it indicates that enabling face as modeling target (setting "faceEnabled" to true) will reduce the device's resource analysis capacity by 2 channels*/
        }
    ],
    "channelResourcesInfo": [
    /*opt, array, (read-only) estimated resource consumption of each channel after enabling local AI by NVR/DVR, subType:object, range:[0,256], desc:used to inform users of the occupied channels when enabling local AI by NVR/DVR*/
        {
            "channelID":  1,
            /*req, int, channel No., range:[0,100]*/
            "localAnalysisNum":  "0.1"
            /*req, number, estimated resource consumption, desc:for example, if the "channel ID" = 1 and "localAnalysisNum" = 2, it indicates that when enabling local AI by NVR/DVR of channel 1, 2 channels will be occupied for analysis*/
        }
    ],
    "channelList": [],
    /*opt, array, channel No. list with AI by camera enabled, subType:integer, range:[0,256], desc:cameras connected to these channels support AcuSearch modeling and the analysis is conducted by the camera's AI function. If this field is absent, it indicates that all the channels use AI by camera. It corresponds to the "channelID" of the "dualVCAType" = "remote" returned by URL (/ISAPI/Streaming/channels/dualVCA?format=json)*/
    "localChannelList": [],
    /*opt, array, channel No. list with AI by NVR/DVR enabled, subType:integer, range:[0,4], desc:cameras connected to these channels do not support AcuSearch modeling and the analysis is conducted by the AI by NVR/DVR. If this field is absent, it indicates that all the channels use AI by NVR/DVR*/
    "preStreamIDList": [],
    /*opt, array, stream ID list of AI by camera, subType:string, range:[0,2048]*/
    "localStreamIDList": [],
    /*opt, array, stream ID list of AI by NVR/DVR, subType:string, range:[0,2048]*/
    "sourceList": [
    /*opt, array, modeling data source of AI by camera, subType:object, range:[0,256], desc:if this field is absent, the modeling data source of all AI by camera channels is the intelligent information overlaid on video streams*/
        {
            "channelID":  1,
            /*req, int, channel No., range:[1,256]*/
            "sourceType":  "dualVCA"
            /*req, enum, data source, subType:string, desc:"alarmImage" (target information on alarm picture); "dualVCA" (intelligent information overlaid on video streams); "originalAlarm" (original event information).*/
        }
    ]
}
```

dracium@gmail.com
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14.3.2.3 Set AcuSearch parameters

Request URL

PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json

Query Parameter

None

Request Message

```
{
    "searchPicEnabled":  "true",
    /*req, boolean, whether to enable AcuSearch, desc:true (only enable AcuSearch for the following configured channels), false (disable AcuSearch for all channels). When enabled, the device will perform modeling and analysis for following configured channels and support searching pictures by target afterwards.*/
    "targetTypeList": [
    /*opt, array, modeling target list, subType:object, range:[0,4], desc:support selecting specified target type for modeling and analysis*/
        {
            "targetType":  "face",
            /*req, enum, target type, subType:string, desc:"face", "human" (human body), "vehicle", "animal"*/
            "humanEnabled":  "true",
            /*opt, boolean, whether to enable human body as a target type, dep:and,{targetType,eq,human}*/
            "vehicleEnabled":  "true",
            /*opt, boolean, whether to enable vehicle as a target type, dep:and,{targetType,eq,vehicle}*/
            "faceEnabled":  "true",
            /*opt, boolean, whether to enable face as a target type, dep:and,{targetType,eq,face}*/
            "humanAttrEnabled":  "false",
            /*opt, boolean, whether to enable human body attribute as a target type, dep:and,{targetType,eq,human}, desc:when enabled, this function consumes system resources and may impact the target search efficiency. Attribute information will only be available in AcuSearch if this is enabled*/
            "vehicleAttrEnabled":  "false",
            /*opt, boolean, whether to enable vehicle attribute as a target type, dep:and,{targetType,eq,vehicle}, desc:when enabled, this function consumes system resources and may impact the target search efficiency. Attribute information will only be available in AcuSearch if this is enabled*/
            "animalEnabled":  "false"
            /*opt, boolean, whether to enable animal as a target type, dep:and,{targetType,eq,animal}*/
        }
    ],
    "channelList": [],
    /*opt, array, channel No. list with AI by camera enabled, subType:integer, range:[0,256], desc:cameras connected to these channels support AcuSearch modeling and the analysis is conducted by the camera's AI function. If this field is absent, it indicates that all the channels use AI by camera. It corresponds to the "channelID" of the "dualVCAType" = "remote" returned by URL (/ISAPI/Streaming/channels/dualVCA?format=json)*/
    "localChannelList": []
    /*opt, array, channel No. list with AI by NVR/DVR enabled, subType:integer, range:[0,4], desc:cameras connected to these channels do not support AcuSearch modeling and the analysis is conducted by the AI by NVR/DVR. If this field is absent, it indicates that all the channels use AI by NVR/DVR*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*opt, int, Status Code, desc:this field is required when the status code is not "1" (success)*/
    "statusString":  "OK",
    /*opt, string, status description, range:[1,64], desc:this field is required when the status code is not "OK" (success)*/
    "subStatusCode":  "ok",
    /*opt, string, sub status code, range:[1,64], desc:this field is required when the status description is not "ok"(success)*/
    "errorCode":  1,
    /*opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*opt, string, error message, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field*/
    "MErrDevSelfEx":  "0x00000000"
    /*opt, string, error codes categorized by functional modules, desc:N/A*/
}
```

dracium@gmail.com
DETEKTOR

14.3.2.4 Get parameters of all video storage nodes

Request URL

GET /ISAPI/ContentMgmt/record/storageMounts

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MountList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Mount>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->1
    </id>
    <path>
      <!--ro, req, string-->test
    </path>
    <dir>
      <!--ro, req, string-->test
    </dir>
    <size>
      <!--ro, req, string-->test
    </size>
    <descr>
      <!--ro, req, string-->test
    </descr>
  </Mount>
</MountList>
```

14.3.2.5 Get parameters of a specified video storage node

Request URL

GET /ISAPI/ContentMgmt/record/storageMounts/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Mount xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <path>
    <!--ro, req, string-->test
  </path>
  <dir>
    <!--ro, req, string-->test
  </dir>
  <size>
    <!--ro, req, string-->test
  </size>
  <descr>
    <!--ro, req, string-->test
  </descr>
</Mount>
```

14.3.2.6 Get capability of uploading files to cloud or downloading files to local storage

Request URL

GET /ISAPI/ContentMgmt/RedirectStorageUrl/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
{
    "RedirectStorageUrlCap": {
    /*opt, object, capability of uploading files to cloud or downloading files to local storage, desc:Corresponding URL: POST
/ISAPI/ContentMgmt/RedirectStorageUrl?format=json.*/
        "sourceUrlList": {
        /*req, object, array element value*/
            "@size":  32,
            /*opt, int, the maximum number of array elements*/
            "@minSize":  1,
            /*opt, int, the minimum number of array elements*/
            "sourceUrl": {
            /*req, object, source file storage information*/
                "filePathType": {
                /*req, object, range of file storage path type, desc:localPath#local storage of device: the file is stored on the device, and the value of
filepah is a path generated by device. Authentication is required when the URL is used to download files from device, and the method is the same with that
in device login. Direct request is allowed if the URL parameter contains token(?token=). For other ways, the authentication method is the same with that in
device login. If the returned URL contains the IP and port, the device is only accessible under a direct network connection, the platform should replace the
IP and port in the URL based on the device's network configuration. localStorageDir#local storage directory: the "filepath" content indicates the path,
which is generated by the device. It cannot be accessed via domain name or IP address directly. URL#cloud storage URL: The file storage address. It is
usually HTTP or HTTPS and can be directly accessed to get files. Authentication is required when you use the URL access files.*/
                    "@opt": [],
                    /*opt, array, enumeration list of file storage path types, subType:string, range:[1,3]*/
                    "@def":  "localPath"
                    /*opt, string, default value*/
                },
                "filePath": {
                /*req, object, range of file storage path type, desc:different filePathType values suggest different ways of file access*/
                    "@def":  "undefined",
                    /*opt, string, default value*/
                    "@min":  0,
                    /*req, int, minimum length*/
                    "@max":  1024
                    /*req, int, maximum length*/
                }
            },
            "redirectType": {
            /*opt, object, target storage type range*/
                "@opt": [],
                /*opt, array, target storage type enumeration, subType:string, range:[1,4]*/
                "@def":  "allocatedFilePath"
                /*opt, string, default value*/
            },
            "redirectFilePath": {
            /*opt, object, destination address of file storage, desc:the platform can directly specify the destination address for picture transfer storage.
The device will upload pictures to this destination address directly*/
                "@def":  "undefined",
                /*opt, string, default value*/
                "@min":  0,
                /*req, int, minimum length*/
                "@max":  1024
                /*req, int, maximum length*/
            }
        },
        "childDevID": {
        /*opt, object, sub-device ID range, desc:this field is used by the gateway to transmitting a URL to the sub device, specifying which sub device the
redirected URL is for. The NVR supports dual NICs, the corresponding redirected URL needs to be assigned based on the NIC to which the sub-device is
connected*/
            "@def":  "3434645611",
            /*opt, string, default value*/
            "@min":  9,
            /*req, int, minimum length*/
            "@max":  32
            /*req, int, maximum length*/
        },
        "taskID": {
        /*opt, object, range of task ID, desc:used for the platform to specify its own task ID. Applied by platform by default; if not, generated by
device*/
            "@def":  "fdb4465d7aa5495884845127a8bbbe3c",
            /*opt, string, default value*/
            "@min":  1,
            /*req, int, minimum length*/
            "@max":  32
            /*req, int, maximum length*/
        }
    }
}
```

dracku@gmail.com
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14.3.2.7 Get the parameters of current storage mode

Request URL

GET /ISAPI/ContentMgmt/Storage

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<storage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, storage information, attr:version{req, string, protocolVersion}-->
  <hddList size="10">
    <!--ro, opt, object, HDD information list, attr:size{req, int}-->
    <hdd>
      <!--ro, opt, object, HDD information-->
      <id>
        <!--ro, req, string, HDD No.-->test
      </id>
      <hddName>
        <!--ro, req, string, HDD name-->test
      </hddName>
      <hddPath>
        <!--ro, opt, string, HDD path-->test
      </hddPath>
      <hddType>
        <!--ro, req, string, HDD type, desc:"IDE", "SATA", "eSATA", "NFS", "iSCSI", "Virtual Disk", "NTFS"-->test
      </hddType>
      <status>
        <!--ro, req, enum, HDD status, subType:string, desc:"ok", "unformatted", "error", "idle", "mismatch", "offline", "smartFailed" (smart detection
failed), "reparing" (repairing), "formating" (HDD is formatting), "notexist" (HDD does not exist), "unRecordHostFormatted" (unformatted in education sharing
system), "maintaining" (HDD is in maintaining), "synching" (synchronizing), "syncError" (synchronization exception), "unloaded", "unSupported" (HDD is not
supported by the device), "waiting" (waiting for formatting)-->ok
      </status>
      <capacity>
        <!--ro, req, float, capacity, unit:MB-->50.000
      </capacity>
      <freeSpace>
        <!--ro, req, float, free space, unit:MB-->20.000
      </freeSpace>
      <property>
        <!--ro, req, string, HDD property, desc:"RW" (read and write), "RO" (read-only), "Redund" (redundant)-->test
      </property>
    </hdd>
  </hddList>
  <nasList size="10">
    <!--ro, opt, object, NAS information list, attr:size{req, int}-->
    <nas>
      <!--ro, opt, object, NAS information-->
      <id>
        <!--ro, req, string, index No.-->test
      </id>
      <addressingFormatType>
        <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
      </addressingFormatType>
      <ipAddress>
        <!--ro, opt, string, IPv4 address, dep:and,{$.nasList.nas.addressingFormatType,eq,ipaddress}, desc:it is valid when the value of
addressingFormatType is "ipaddress"-->192.168.230.1
      </ipAddress>
      <portNo>
        <!--ro, req, int, port No.-->1
      </portNo>
      <nasType>
        <!--ro, req, enum, NAS disk type, subType:string, desc:; "NFS", "iSCSI"-->NFS
      </nasType>
      <path>
        <!--ro, req, string, NAS disk path-->test
      </path>
      <status>
        <!--ro, req, enum, current status, subType:string, desc:"ok", "unformatted", "error", "idle", "mismatch", "offline", "smartFailed" (smart detection
failed), "reparing" (repairing), "formating" (in formatting), "notexist" (HDD does not exist), "waiting" (waiting for formatting)-->ok
      </status>
      <capacity>
        <!--ro, req, float, capacity, desc:unit: MB-->0.000
      </capacity>
      <freeSpace>
        <!--ro, req, float, free space, desc:unit: MB-->0.000
      </freeSpace>
      <property>
        <!--ro, req, enum, property, subType:string, desc:"RW" (read and write), "RO" (read-only), "redund" (redundant)-->RW
      </property>
    </nas>
  </nasList>
  <workMode opt="group,quota,extract,auto,age">
    <!--ro, opt, enum, working mode, subType:string, attr:opt{req, string}, desc:"group","quota","extract", "auto"-->group
  </workMode>
</storage>
```

dr'ackum@gmail.com
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14.3.2.8 Set the current storage mode parameters

Request URL

PUT /ISAPI/ContentMgmt/Storage

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<storage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, storage information, attr:version{req, string, protocolVersion}-->
  <hddList size="10">
    <!--opt, object, HDD information list, attr:size{req, int}-->
    <hdd>
      <!--opt, object, HDD information-->
      <id>
        <!--req, string, HDD No.-->test
      </id>
      <hddName>
        <!--req, string, HDD name-->test
      </hddName>
      <hddPath>
        <!--opt, string, HDD path-->test
      </hddPath>
      <hddType>
        <!--req, string, HDD type, desc:"IDE", "SATA", "eSATA", "NFS", "iSCSI", "Virtual Disk", "NTFS"-->test
      </hddType>
      <status>
        <!--req, enum, HDD status, subType:string, desc:"ok", "unformatted", "error", "idle", "mismatch", "offline", "smartFailed" (smart detection failed),
"reparing" (repairing), "formating" (HDD is formatting), "notexist" (HDD does not exist), "unRecordHostFormatted" (unformatted in education sharing system),
"maintaining" (HDD is in maintaining), "synching" (synchronizing), "syncError" (synchronization exception), "unloaded", "waiting" (waiting for formatting),
"initializing"-->ok
      </status>
      <capacity>
        <!--req, float, capacity, unit:MB-->50.000
      </capacity>
      <freeSpace>
        <!--req, float, free space, unit:MB-->20.000
      </freeSpace>
      <property>
        <!--req, string, HDD property, desc:"RW" (read and write), "RO" (read-only), "Redund" (redundant)-->test
      </property>
      <group>
        <!--opt, string, the No. of the disk group to which the HDD belongs-->test
      </group>
      <DataModeList>
        <!--opt, array, data mode list, subType:object-->
        <DataMode>
          <!--opt, object, data mode-->
          <type>
            <!--req, string, application mode type, desc:"recordStorage" (video storage), "pictureCloudStorage" (picture to be saved in cloud storage),
"fileStorage" (file storage), only one type can be selected-->test
          </type>
          <occupancyRate>
            <!--req, int, disk usage, range:[0,100], desc:the disk space occupied by the application mode-->0
          </occupancyRate>
        </DataMode>
      </DataModeList>
      <formatType>
        <!--opt, enum, disk file type, subType:string, desc:"FAT32", "EXT4"-->FAT32
      </formatType>
      <ipAddress>
        <!--opt, string, IPv4, desc:IPv4 address-->172.6.64.7
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6, desc:IPv6 address-->1080:0:0:0:8:800:200C:417A
      </ipv6Address>
      <encryptionStatus>
        <!--opt, enum, encryption status, subType:string, desc:"unencrypted", "encrypted", "verfyFailed" (verification failed)-->unencrypted
      </encryptionStatus>
      <logStorageEnable>
        <!--opt, bool, whether the current disk is a log storage disk-->true
      </logStorageEnable>
      <installationTime>
        <!--opt, time, memory card installation time-->12:05:00+08:00
      </installationTime>
      <hddModel>
        <!--opt, string, HDD model-->test
      </hddModel>
      <hddSerialNumber>
        <!--opt, string, HDD serial No.-->test
      </hddSerialNumber>
      <hddMode>
        <!--ro, opt, enum, subType:string-->record
      </hddMode>
    </hdd>
  </hddList>
  <nasList size="10">
    <!--opt, object, NAS information list, attr:size{req, int}-->
    <nas>
      <!--opt, object, NAS information-->
      <id>
        <!--req, string, index No.-->test
```

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```
        <!--req, string, index No.-->test
      </id>
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"ipaddress" (IP address), "hostname" (host server name)-->ipaddress
      </addressingFormatType>
      <hostName>
        <!--opt, string, host server name, dep:and,{$.nasList.nas.addressingFormatType,eq,hostname}, desc:it is valid when the value of addressingFormatType
is "hostname"-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, dep:and,{$.nasList.nas.addressingFormatType,eq,ipaddress}, desc:it is valid when the value of addressingFormatType is
"ipaddress"-->192.168.230.1
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6 address, dep:and,{$.nasList.nas.addressingFormatType,eq,ipaddress}, desc:it is valid when the value of addressingFormatType is
"ipaddress"-->CDCD:910A:2222:5498:8475:1111:3900:2020
      </ipv6Address>
      <portNo>
        <!--req, int, port No.-->1
      </portNo>
      <userName>
        <!--opt, string, user name-->test
      </userName>
      <password>
        <!--opt, string, password-->test
      </password>
      <nasType>
        <!--req, enum, NAS disk type, subType:string, desc:"NFS", "iSCSI"-->NFS
      </nasType>
      <path>
        <!--req, string, NAS disk path-->test
      </path>
      <status>
        <!--req, enum, current status, subType:string, desc:"ok", "unformatted", "error", "idle", "mismatch", "offline", "smartFailed" (smart detection
failed), "reparing" (repairing), "formating" (in formatting), "notexist" (HDD does not exist), "waiting" (waiting for formatting), “initializing”-->ok
      </status>
      <capacity>
        <!--req, float, capacity, desc:unit: MB-->0.000
      </capacity>
      <freeSpace>
        <!--req, float, free space, desc:unit: MB-->0.000
      </freeSpace>
      <property>
        <!--req, enum, property, subType:string, desc:; "RW" (read and write), "RO" (read-only), "redund" (redundant)-->RW
      </property>
      <group>
        <!--opt, string, disk group No.-->test
      </group>
    </nas>
  </nasList>
  <workMode opt="group,quota,extract,auto,age">
    <!--opt, enum, working mode, subType:string, attr:opt{req, string}, desc:; "group","quota","extract", "auto"-->group
  </workMode>
  <ssdList size="10">
    <!--opt, object, SSD information list, attr:size{req, int}-->
    <ssd>
      <!--opt, object, SSD information-->
      <id>
        <!--req, string, index No.-->test
      </id>
      <ssdName>
        <!--req, string, SSD name-->test
      </ssdName>
      <version>
        <!--opt, string, SSD firmware version-->test
      </version>
    </ssd>
  </ssdList>
  <posInfoStoragingEnabled>
    <!--opt, bool, whether to enable POS information storage-->true
  </posInfoStoragingEnabled>
</storage>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

14.3.2.9 Get the configuration capability of current storage working mode

Request URL

GET /ISAPI/ContentMgmt/Storage/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<storage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, storage information, attr:version{req, string, protocolVersion}-->
  <hddList size="10">
    <!--ro, opt, object, HDD information list, attr:size{req, int}-->
  </hddList>
  <nasList size="10">
    <!--ro, opt, object, NAS information list, attr:size{req, int}-->
  </nasList>
  <workMode opt="group,quota,extract">
    <!--ro, opt, enum, working mode, subType:string, attr:opt{req, string}, desc:"group", "quota", "extract", "auto"-->group
  </workMode>
  <supportDataReconstruction opt="all,record">
    <!--ro, opt, enum, the supported data type for reconstruction, subType:string, attr:opt{req, string}, desc:"all", "record" (recorded video)-->all
  </supportDataReconstruction>
</storage>
```

14.3.2.10 Get the disk data reconstruction progress

Request URL

GET /ISAPI/ContentMgmt/Storage/dataReconstruction/progress?format=json

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
{
    "ProgressLists": [
    /*ro, opt, array, subType:object*/
        {
            "diskID":  1,
            /*ro, req, int, disk ID*/
            "progress":  100,
            /*ro, req, int, disk data reconstruction progress*/
            "status":  1
            /*ro, req, enum, status, subType:int, desc:0 (reconstructing), 1 (device exception (applying for memory failed/opening files failed/deleting files failed)), 2 (abnormal disconnection), 3 (reconstructed)*/
        }
    ]
}
```

14.3.2.11 Get storage strategy parameters

Request URL

GET /ISAPI/ContentMgmt/Storage/extension

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<storageExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, storage strategy parameters, attr:version{req, string, protocolVersion}-->
  <LoopEnable>
    <!--ro, opt, bool, whether to enable HDD overwriting-->true
  </LoopEnable>
  <enableDormant>
    <!--ro, opt, bool, whether to enable HDD sleeping-->true
  </enableDormant>
  <packDuration>
    <!--ro, opt, int, packaging time of video file, unit:min-->1
  </packDuration>
  <logStorageMode>
    <!--ro, opt, enum, log storage mode, subType:string, desc:; "system" (system default mode), "custom" (custom mode)-->system
  </logStorageMode>
  <logStorageId>
    <!--ro, opt, int, disk or array No., dep:and,{$.storageExtension.logStorageMode,eq,custom}, desc:it refers to disk No. when the device uses a single disk for storage, or array No. when the device uses disk arrays for storage. It is valid when the value of logStorageMode is "custom"-->1
  </logStorageId>
  <logStorageCycle>
    <!--ro, opt, int, log retention period, dep:and,{$.storageExtension.logStorageMode,eq,custom}, desc:this node is valid only when logStorageMode is "custom"-->1
  </logStorageCycle>
  <printLogSaveEnable>
    <!--ro, opt, bool, whether to enable record printing-->true
  </printLogSaveEnable>
  <picPatition>
    <!--ro, opt, int, picture partition percentage-->1
  </picPatition>
  <recordCycle>
    <!--ro, opt, int, the retention period of videos, range:[0,90], unit:天, desc:the device estimates the video duration supported by the storage space according to the resolution, picture quality, frame rate, bit rate, and other information of current stream-->14
  </recordCycle>
  <dualVCAEnabled>
    <!--ro, opt, bool, whether to enable the intelligent information overlay-->true
  </dualVCAEnabled>
  <searchPicEnabled>
    <!--ro, opt, bool-->true
  </searchPicEnabled>
</storageExtension>
```

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14.3.2.12 Set storage strategy parameters

Request URL

PUT /ISAPI/ContentMgmt/Storage/extension

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<storageExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, storage strategy parameters, attr:version{req, string, protocolVersion}-->
  <LoopEnable>
    <!--opt, bool, whether to enable HDD overwriting-->true
  </LoopEnable>
  <enableDormant>
    <!--ro, opt, bool, whether to enable HDD sleeping-->true
  </enableDormant>
  <packDuration>
    <!--opt, int, packaging time of video file, unit:min-->1
  </packDuration>
  <logStorageMode>
    <!--opt, enum, log storage mode, subType:string, desc:"system" (system default mode), "custom" (custom mode)-->system
  </logStorageMode>
  <logStorageId>
    <!--opt, int, disk or array No., dep:and,{$.storageExtension.logStorageMode,eq,custom}, desc:it refers to disk No. when the device uses a single disk
for storage, or array No. when the device uses disk arrays for storage. It is valid when the value of logStorageMode is "custom"-->1
  </logStorageId>
  <logStorageCycle>
    <!--opt, int, log retention period, dep:and,{$.storageExtension.logStorageMode,eq,custom}, desc:this node is valid only when logStorageMode is "custom"-
->1
  </logStorageCycle>
  <printLogSaveEnable>
    <!--opt, bool, whether to enable record printing-->true
  </printLogSaveEnable>
  <picPatition>
    <!--opt, int, picture partition percentage-->1
  </picPatition>
  <dualVCAEnabled>
    <!--opt, bool, whether to enable dual-VCA, desc:"true" (the NVR will enable the intelligent information overlay and save the metadata information from
cameras for dual-VCA, which is used to mark targets in playback by color on the timebar)-->true
  </dualVCAEnabled>
  <searchPicEnabled>
    <!--opt, bool, whether to enable searching pictures by picture during playback, desc:call /ISAPI/Streaming/channels/dualVCA?format=json to configure the
channel for searching pictures by picture and enable the intelligent information overlay of cameras; "true" (the device will save the target frame
information from cameras for searching the specified face / human body pictures in playback); the dual-VCA and searching pictures by picture functions
depend on the intelligent information overlaid in the stream from cameras, but they are mutually independent; the DSP of NVR stores the two kinds of
intelligent data for search in different applications-->true
  </searchPicEnabled>
</storageExtension>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.3.2.13 Get capability of storage strategy parameters

Request URL

GET /ISAPI/ContentMgmt/Storage/extension/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<storageExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, storage strategy parameters, attr:version{req, string, protocolVersion}-->
  <LoopEnable opt="true,false">
    <!--ro, opt, bool, whether to enable overwriting, attr:opt{req, string}-->true
  </LoopEnable>
  <enableDormant opt="true,false">
    <!--ro, opt, bool, whether to enable HDD sleeping, attr:opt{req, string}-->true
  </enableDormant>
  <packDuration max="300" min="1">
    <!--ro, opt, int, packaging time of video file, unit:min, attr:max{req, int},min{req, int}-->0
  </packDuration>
  <logStorageMode opt="system,custom">
    <!--ro, opt, enum, log storage mode, subType:string, attr:opt{req, string}, desc:"system" (system default mode, in which, the capacity of each HDD is 64
MB, and the existing data will be overwritten if the HDD is full), "custom" (custom mode)-->system
  </logStorageMode>
  <logStorageId>
    <!--ro, opt, int, disk or array No., dep:and,{$.storageExtension.logStorageMode,eq,custom}, desc:it refers to disk No. when the device uses a single
disk for storage, or array No. when the device uses disk arrays for storage-->0
  </logStorageId>
  <logStorageCycle min="0" max="90" def="90">
    <!--ro, opt, int, log retention period, dep:and,{$.storageExtension.logStorageMode,eq,custom}, attr:min{req, int},max{req, int},def{req, int}-->0
  </logStorageCycle>
  <printLogSaveEnable opt="true,false">
    <!--ro, opt, bool, whether to enable record printing, attr:opt{req, string}-->true
  </printLogSaveEnable>
  <picPatition min="0" max="100">
    <!--ro, opt, int, picture partition percentage, attr:min{req, int},max{req, int}-->0
  </picPatition>
  <dualVCAEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable the intelligent information overlay, attr:opt{req, string}-->true
  </dualVCAEnabled>
  <searchPicEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </searchPicEnabled>
</storageExtension>
```

14.3.2.14 Get the parameters for storing additional information

Request URL

GET /ISAPI/ContentMgmt/Storage/ExtraInfo

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ExtraInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the parameters for storing additional information, attr:version{req, string, protocolVersion}-->
  <diskMode opt="singleDisk,IRAID">
    <!--ro, req, enum, disk mode, subType:string, attr:opt{req, string}, desc:"singleDisk" (single disk), "RAID". The device needs to reboot after changing
the disk mode-->singleDisk
  </diskMode>
  <applyMode opt="recordStorage,pictureCloudStorage,fileStorage,recordPictureStorage,recordPictureFileStorage">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->recordStorage
  </applyMode>
  <recordStorageRatio min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.ExtraInfo.applyMode,eq,recordStorage},{$.ExtraInfo.applyMode,eq,recordPictureStorage},
{$.ExtraInfo.applyMode,eq,recordPictureFileStorage}, attr:min{req, int},max{req, int}-->1
  </recordStorageRatio>
  <pictureCloudStorageRatio min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.ExtraInfo.applyMode,eq,pictureCloudStorage},{$.ExtraInfo.applyMode,eq,recordPictureStorage},
{$.ExtraInfo.applyMode,eq,recordPictureFileStorage}, attr:min{req, int},max{req, int}-->1
  </pictureCloudStorageRatio>
  <fileStorageRatio min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.ExtraInfo.applyMode,eq,fileStorage},{$.ExtraInfo.applyMode,eq,recordPictureFileStorage}, attr:min{req,
int},max{req, int}-->1
  </fileStorageRatio>
</ExtraInfo>
```

14.3.2.15 Set the parameters for storing additional information

Request URL

PUT /ISAPI/ContentMgmt/Storage/ExtraInfo

drackum@gmail.com
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Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ExtraInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, the parameters for storing additional information, attr:version{req, string, protocolVersion}-->
  <diskMode opt="singleDisk,IRAID">
    <!--req, enum, disk mode, subType:string, attr:opt{req, string}, desc:"singleDisk" (single disk), "IRAID". The device needs to reboot after changing the
disk mode-->singleDisk
  </diskMode>
  <applyMode opt="recordStorage,pictureCloudStorage,fileStorage,recordPictureStorage,recordPictureFileStorage">
    <!--opt, enum, subType:string, attr:opt{req, string}-->recordStorage
  </applyMode>
  <recordStorageRatio min="0" max="100">
    <!--opt, int, range:[0,100], dep:and,{$.ExtraInfo.applyMode,eq,recordStorage},{$.ExtraInfo.applyMode,eq,recordPictureStorage},
{$.ExtraInfo.applyMode,eq,recordPictureFileStorage}, attr:min{req, int},max{req, int}-->1
  </recordStorageRatio>
  <pictureCloudStorageRatio min="0" max="100">
    <!--opt, int, range:[0,100], dep:and,{$.ExtraInfo.applyMode,eq,pictureCloudStorage},{$.ExtraInfo.applyMode,eq,recordPictureStorage},
{$.ExtraInfo.applyMode,eq,recordPictureFileStorage}, attr:min{req, int},max{req, int}-->1
  </pictureCloudStorageRatio>
  <fileStorageRatio min="0" max="100">
    <!--opt, int, range:[0,100], dep:and,{$.ExtraInfo.applyMode,eq,fileStorage},{$.ExtraInfo.applyMode,eq,recordPictureFileStorage}, attr:min{req,
int},max{req, int}-->1
  </fileStorageRatio>
</ExtraInfo>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

dr4ckum@gmail.com
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14.3.2.16 Get the capability of storing additional information

Request URL

GET /ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ExtraInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <diskMode opt="singleDisk,IRAID">
    <!--ro, req, enum, disk mode, subType:string, attr:opt{req, string}, desc:"singleDisk" (single disk), "RAID". The device needs to reboot after changing the disk mode-->singleDisk
  </diskMode>
  <applyMode opt="recordStorage,pictureCloudStorage,fileStorage,recordPictureStorage,recordPictureFileStorage">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->recordStorage
  </applyMode>
  <recordStorageRatio min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.ExtraInfo.applyMode,eq,recordStorage},{$.ExtraInfo.applyMode,eq,recordPictureStorage},{$.ExtraInfo.applyMode,eq,recordPictureFileStorage}, attr:min{req, int},max{req, int}-->1
  </recordStorageRatio>
  <pictureCloudStorageRatio min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.ExtraInfo.applyMode,eq,pictureCloudStorage},{$.ExtraInfo.applyMode,eq,recordPictureStorage},{$.ExtraInfo.applyMode,eq,recordPictureFileStorage}, attr:min{req, int},max{req, int}-->1
  </pictureCloudStorageRatio>
  <fileStorageRatio min="0" max="100">
    <!--ro, opt, int, range:[0,100], dep:and,{$.ExtraInfo.applyMode,eq,fileStorage},{$.ExtraInfo.applyMode,eq,recordPictureFileStorage}, attr:min{req, int},max{req, int}-->1
  </fileStorageRatio>
</ExtraInfo>
```

14.3.3 录像查询

14.3.3.1 Refresh the video index manually before playback

Request URL

PUT /ISAPI/ContentMgmt/record/control/manualRefresh/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

14.3.3.2 Get the capability of manually refreshing video index

Request URL

GET /ISAPI/ContentMgmt/record/control/manualRefresh/channels/capabilities?format=json

Query Parameter

None

Request Message

drackum@gmail.com DETEKTOR

None

Response Message

```
{
    "isSupportManualRefresh":  true
    /*ro, opt, bool, whether the device supports limiting the frequency of refreshing video index, desc:device of new version will limit the frequency of refreshing video index to less than once every 10 seconds by default; device of old version will not limit the frequency of refreshing video index, but frequent refresh will cause device exception*/
}
```

14.3.3.3 Get video track type

Request URL

GET /ISAPI/ContentMgmt/record/profile

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CMRecordProfile xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <trackType>
    <!--ro, opt, enum, video track type, subType:string, desc:video track type-->standard
  </trackType>
</CMRecordProfile>
```

14.3.3.4 Search recorded video files by date

Request URL

POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution?type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	Channel No. * 100 + stream type (-1: main stream, -2: sub-stream).
type	enum	"channelID" (search for recordings by channel ID), "trackStreamID" (channel No.), "streamID" (search for recordings by stream ID), "trackStreamID" (stream ID). When there is no 'type" node or "type" is "channelID", it indicates searching by channel ID, and "trackStreamID" is the channel No.; when "type" is "streamID", it indicates searching by stream ID, and "trackStreamID" is the stream ID.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<trackDailyParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <year>
    <!--req, int, year-->1
  </year>
  <monthOfYear>
    <!--req, int, month of the year-->1
  </monthOfYear>
  <drawFrame>
    <!--opt, bool-->false
  </drawFrame>
</trackDailyParam>
```

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<trackDailyDistribution xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <dayList>
    <!--ro, opt, array, subType:object-->
    <day>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int, ID-->1
      </id>
      <dayOfMonth>
        <!--ro, req, int, day of the month,starts from 1st-->1
      </dayOfMonth>
      <record>
        <!--ro, req, bool, true-with video,false-no video-->true
      </record>
      <recordType>
        <!--ro, opt, enum, subType:string-->event
      </recordType>
    </day>
  </dayList>
</trackDailyDistribution>
```

14.3.3.5 Get video search conditions

Request URL

GET /ISAPI/ContentMgmt/search/profile

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CMSearchProfile xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <searchProfile>
    <!--ro, opt, enum, video search conditions, subType:string, desc:video search conditions-->basic
  </searchProfile>
  <textSearch>
    <!--ro, opt, enum, text type to be searched, such as recording information text and metadata text, subType:string, desc:text type to be searched,such as
recording information text and metadata text-->true
  </textSearch>
  <maxSearchTimespans>
    <!--ro, req, int, maximum time period of results returned in each search-->1
  </maxSearchTimespans>
  <maxSearchTracks>
    <!--ro, opt, int, maximum tracks can be searched-->1
  </maxSearchTracks>
  <maxSearchSources>
    <!--ro, opt, int, maximum recording channels can be searched-->1
  </maxSearchSources>
  <maxSearchMetadatas>
    <!--ro, opt, int, maximum number of metadata can be searched-->1
  </maxSearchMetadatas>
  <maxSearchMatchResults>
    <!--ro, req, int, maximum search results can be returned-->1
  </maxSearchMatchResults>
  <maxSearchTimeout>
    <!--ro, req, int, maximum search timeout-->1
  </maxSearchTimeout>
  <maxConcurrentSearches>
    <!--ro, req, int, maximum concurrent search operations-->1
  </maxConcurrentSearches>
</CMSearchProfile>
```

14.3.3.6 Get the capability of searching for videos by target type

Request URL

GET /ISAPI/ContentMgmt/SearchByTargetType/capabilities?format=json

Query Parameter

None

dra.kum@gmail.com DETEKTOR

Request Message

None

Response Message

```
{
    "SearchDescriptionCap": {
    /*ro, req, object, search description capability*/
        "@size":  1024,
        /*ro, req, int, supported maximum number of search conditions*/
        "channelID": {
        /*ro, req, object, the range of the channel No.*/
            "@min":  1,
            /*ro, req, int, the minimum value of channel No.*/
            "@max":  256
            /*ro, req, int, the maximum value of channel No.*/
        },
        "targetTypes": {
        /*ro, req, object, the search target type supported by the device*/
            "@opt": ["human", "vehicle"]
            /*ro, req, array, options, subType:string, desc:"human", "vehicle"*/
        }
    },
    "SearchResultCap": {
    /*ro, opt, object, the capability of results searching*/
        "@size":  1024,
        /*ro, req, int, supported maximum number of search results*/
        "channelID": {
        /*ro, req, object, the range of the channel No.*/
            "@min":  1,
            /*ro, req, int, the minimum value of channel No.*/
            "@max":  256
            /*ro, req, int, the maximum value of channel No.*/
        },
        "targetTypes": {
        /*ro, req, object, the search target type supported by the device*/
            "@opt": ["human", "vehicle"]
            /*ro, req, array, options, subType:string, desc:"human", "vehicle"*/
        }
    }
}
```

drackum@gmail.com
DETEKTOR

14.3.3.7 Search for videos by target type

Request URL

POST /ISAPI/ContentMgmt/SearchByTargetType?format=json

Query Parameter

None

Request Message

```
{
    "SearchDescription": {
    /*opt, object, search description*/
        "searchID":  "test",
        /*req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two
searches, the search history will be saved in the memory to speed up next search*/
        "searchResultPosition":  0,
        /*req, int, the start position of the search result in the result list*/
        "maxResults":  100,
        /*req, int, the maximum number of search results this time*/
        "SearchCondList": [
        /*req, array, list of search conditions, subType:object*/
            {
                "channelID":  1,
                /*req, int, channel No.*/
                "targetTypes": ["human", "vehicle"],
                /*req, array, search target type, subType:string, desc:"human", "vehicle"*/
                "searchTimeList": [
                /*opt, array, list of search period, subType:object, desc:if this node is not applied, it indicates that the search will not be performed by
time*/
                    {
                        "searchTime": {
                        /*req, object, search time period*/
                            "startTime":  "1970-01-01T08:00:00+08:00",
                            /*req, datetime, start time of the search, desc:ISO8601 time (TD format, local time and time difference)*/
                            "endTime":  "1970-01-01T00:09:00+08:00"
                            /*req, datetime, end time of the search, desc:ISO8601 time (TD format, local time and time difference)*/
                        }
                    }
                ]
            }
        ]
    }
}
```

Response Message

```
{
    "SearchResult": {
    /*ro, opt, object, search result*/
        "searchID":  "test",
        /*ro, req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during
two searching, the search history will be saved in the memory to speed up next searching*/
        "responseStatus":  true,
        /*ro, req, bool, status of search results*/
        "numOfMatches":  100,
        /*ro, opt, int, number of results returned this time*/
        "totalMatches":  200,
        /*ro, opt, int, total number of matched results*/
        "responseStatusStrg":  "OK",
        /*ro, opt, enum, search status, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data)*/
        "matchList": [
        /*ro, req, array, the information list of returned results, subType:object*/
            {
                "channelID":  1,
                /*ro, req, int, channel No.*/
                "targetTypes": ["human", "vehicle"],
                /*ro, req, array, search target type, subType:string, desc:"human", "vehicle"; multiple elements in the array indicates multi-type, e.g.,
human and vehicle*/
                "RecordInfoList": [
                /*ro, req, array, list of video information, subType:object*/
                    {
                        "RecordTime": {
                        /*ro, req, object, video time range*/
                            "startTime":  "1970-01-01T08:00:00+08:00",
                            /*ro, req, datetime, video start time, desc:ISO8601 time (TD format, local time and time difference)*/
                            "endTime":  "1970-01-01T09:00:00+08:00"
                            /*ro, req, datetime, video end time, desc:ISO8601 time (TD format, local time and time difference)*/
                        }
                    }
                ]
            }
        ]
    }
}
```

dracku@gmail.com
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14.3.3.8 Search for video files by VCA event

Request URL

POST /ISAPI/ContentMgmt/SmartSearch

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SmartSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, search condition, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of records for a single search (display on each page)-->1
  </maxResults>
  <trackID>
    <!--req, int, channel No. and stream type, desc:channel No. * 100 + stream type (-1: main stream, -2: sub-stream)-->1
  </trackID>
  <startTime>
    <!--req, datetime, start time, desc:ISO8601 time (TZ format: UTC time)-->1970-01-01T00:00:00Z
  </startTime>
  <endTime>
    <!--req, datetime, end time, desc:ISO8601 time (TZ format: UTC time)-->1970-01-01T01:00:00Z
  </endTime>
  <type opt="motionDetection,traversePlaneDetection,FieldDetection,faceDetection">
    <!--req, enum, type for search, subType:string, attr:opt{req, string}, desc:;"motionDetection" (motion detection); "traversePlaneDetection" (line crossing detection); "FieldDetection" (intrusion detection); "faceDetection" (facial detection)-->motionDetection
  </type>
  <MotionDetection>
    <!--opt, object, motion detection parameter, dep:and,{$.SmartSearchDescription.type,eq,motionDetection}-->
    <Grid>
      <!--opt, object, grid area-->
      <rowGranularity>
        <!--req, float, row-->0.000
      </rowGranularity>
      <columnGranularity>
        <!--req, float, column-->0.000
      </columnGranularity>
    </Grid>
    <MotionDetectionLayout>
      <!--opt, object, motion detection layout-->
      <layout>
        <!--opt, object, layout-->
        <gridMap>
          <!--opt, string, grid image, desc:hexadecimal, the image is divided in to N × N grids,each grid can be enabled (value: 1) or disabled (value 0) motion detection,each four horizontal grids consist of a hexadecimal number,e.g.,the corresponding hexadecimal number of "0011" is 3-->0x3
        </gridMap>
      </layout>
    </MotionDetectionLayout>
    <sensitivity min="1" max="100">
      <!--opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
  </MotionDetection>
  <TraversePlane>
    <!--opt, object, line crossing parameters, dep:and,{$.SmartSearchDescription.type,eq,traversePlaneDetection}-->
    <TraversePlaneParam>
      <!--opt, object, line crossing-->
      <LineList>
        <!--opt, array, line list, subType:object-->
        <Line>
          <!--opt, object, line-->
          <StartPoint>
            <!--opt, object, start point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </StartPoint>
          <EndPoint>
            <!--opt, object, end point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </EndPoint>
        </Line>
      </LineList>
      <crossDirection opt="bothDirection,leftToRight,rightToLeft">
        <!--opt, enum, line crossing direction, subType:string, attr:opt{req, string}, desc:; "bothDirection" (bidirectional); "leftToRight" (from left to right); "rightToLeft" (from right to left)-->bothDirection
      </crossDirection>
      <sensitivity min="1" max="100">
        <!--opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
      </sensitivity>
      <planeHeight>
        <!--opt, int, plane height-->1
      </planeHeight>
```

drackum@gmail.com
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```
      <detectionTarget opt="all,human,vehicle">
        <!--req, enum, detection target, subType:string, attr:opt{req, string}, desc:; "all" (humans and vehicles); "human"; "vehicle"-->all
      </detectionTarget>
    </TraversePlaneParam>
    <dwPreTime>
      <!--opt, int, pre-record-->1
    </dwPreTime>
    <dwDelayTime>
      <!--opt, int, post-record-->1
    </dwDelayTime>
    <PTZPos>
      <!--opt, object, PTZ coordinate-->
      <P>
        <!--opt, int, panning adjustment-->1
      </P>
      <T>
        <!--opt, int, tilting adjustment-->1
      </T>
      <Z>
        <!--opt, int, zooming adjustment-->1
      </Z>
    </PTZPos>
    <advancedType opt="humanFeature,objectFeature">
      <!--opt, enum, advanced parameter type, subType:string, attr:opt{req, string}, desc:; "humanFeature" (human body feature); "objectFeature" (object
feature)-->humanFeature
    </advancedType>
    <advance>
      <!--opt, object, advanced parameters-->
      <HumanFeature>
        <!--opt, object, human body feature, dep:and,{$.SmartSearchDescription.TraversePlane.advancedType,eq,humanFeature}-->
        <age min="0" max="100">
          <!--opt, int, age, attr:min{req, int},max{req, int}-->1
        </age>
        <sex opt="male,female">
          <!--opt, enum, name, subType:string, attr:opt{req, string}, desc:; "male"; "female"-->male
        </sex>
        <eyeGlass>
          <!--opt, bool, whether the person wears glasses-->true
        </eyeGlass>
        <ageGroup>
          <!--opt, enum, age group, subType:string, desc:refer to the dictionary for details; "any" (no limitation, and used for search)-->any
        </ageGroup>
        <ageDeviation>
          <!--opt, int, age tolerance, range:[0,100]-->0
        </ageDeviation>
      </HumanFeature>
      <ObjectFeature>
        <!--opt, object, target features, dep:and,{$.SmartSearchDescription.TraversePlane.advancedType,eq,objectFeature}-->
        <colorRatel>
          <!--opt, int, color ratio-->1
        </colorRatel>
        <R>
          <!--opt, int, red-->1
        </R>
        <G>
          <!--opt, int, green-->1
        </G>
        <B>
          <!--opt, int, blue-->1
        </B>
      </ObjectFeature>
    </advance>
  </TraversePlane>
  <FieldDetection>
    <!--opt, object, intrusion detection, dep:and,{$.SmartSearchDescription.type,eq,FieldDetection}-->
    <normalizedScreenSize>
      <!--opt, object, normalized coordinates-->
      <normalizedScreenWidth>
        <!--opt, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--opt, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <param>
      <!--opt, object, parameters-->
      <region>
        <!--opt, object, detection region-->
        <pointList>
          <!--opt, array, point list, subType:object, range:[0,4], desc:rectangle-->
          <point>
            <!--opt, object, point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </point>
        </pointList>
      </region>
    </param>
    <duration min="1" max="100">
```

drackum@gmail.com
DETEKTOR

```
      <!--opt, int, duration, attr:min{req, int},max{req, int}-->1
    </duration>
    <sensitivity min="1" max="100">
      <!--opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
    <rate min="1" max="100">
      <!--opt, int, speed, attr:min{req, int},max{req, int}-->1
    </rate>
    <detectionTarget opt="all,human,vehicle">
      <!--req, enum, detection target, subType:string, attr:opt{req, string}, desc:; "all" (humans and vehicles); "human"; "vehicle"-->all
    </detectionTarget>
    <dwPreTime>
      <!--opt, int, pre-record-->1
    </dwPreTime>
    <dwDelayTime>
      <!--opt, int, Post-record-->1
    </dwDelayTime>
    <PTZPos>
      <!--opt, object, PTZ coordinate-->
      <P>
        <!--opt, int, panning adjustment-->1
      </P>
      <T>
        <!--opt, int, tilting adjustment-->1
      </T>
      <Z>
        <!--opt, int, zooming adjustment-->1
      </Z>
    </PTZPos>
    <advancedType opt="humanFeature,objectFeature">
      <!--opt, enum, advanced parameter type, subType:string, attr:opt{req, string}, desc:; "humanFeature" (human body feature); "objectFeature" (object
feature)-->humanFeature
    </advancedType>
    <advance>
      <!--opt, object, advanced parameters-->
      <HumanFeature>
        <!--opt, object, human body feature, dep:and,{$.SmartSearchDescription.FieldDetection.advancedType,eq,humanFeature}-->
        <age min="0" max="100">
          <!--opt, int, age, attr:min{req, int},max{req, int}-->1
        </age>
        <sex opt="male,female">
          <!--opt, enum, name, subType:string, attr:opt{req, string}, desc:; "male"; "female"-->male
        </sex>
        <eyeGlass>
          <!--opt, bool, whether the person wears glasses-->true
        </eyeGlass>
        <ageGroup>
          <!--opt, enum, age group, subType:string, desc:refer to the dictionary for details; "any" (no limitation, and used for search)-->any
        </ageGroup>
        <ageDeviation>
          <!--opt, int, age tolerance, range:[0,100]-->0
        </ageDeviation>
      </HumanFeature>
      <ObjectFeature>
        <!--opt, object, target features, dep:and,{$.SmartSearchDescription.FieldDetection.advancedType,eq,objectFeature}-->
        <colorRatel>
          <!--opt, int, color ratio-->1
        </colorRatel>
        <R>
          <!--opt, int, red-->1
        </R>
        <G>
          <!--opt, int, green-->1
        </G>
        <B>
          <!--opt, int, blue-->1
        </B>
      </ObjectFeature>
    </advance>
  </FieldDetection>
  <FaceDetection>
    <!--opt, object, face detection, dep:and,{$.SmartSearchDescription.type,eq,faceDetection}-->
    <param>
      <!--opt, object, parameters-->
      <region>
        <!--opt, object, detection region-->
        <pointList>
          <!--opt, array, point list, subType:object, range:[0,4], desc:rectangle-->
          <point>
            <!--opt, object, point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </point>
        </pointList>
      </region>
    </param>
    <duration min="1" max="100">
      <!--opt, int, duration, attr:min{req, int},max{req, int}-->1
    </duration>
    <sensitivity min="1" max="100">
```

dracklm@gmail.com
DETEKTOR

```
      <!--opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
    <rate min="1" max="100">
      <!--opt, int, speed, attr:min{req, int},max{req, int}-->1
    </rate>
    <detectionTarget opt="all,human,vehicle">
      <!--req, enum, detection target, subType:string, attr:opt{req, string}, desc:; "all" (humans and vehicles); "human"; "vehicle"-->all
    </detectionTarget>
    <dwPreTime>
      <!--opt, int, pre-record-->1
    </dwPreTime>
    <dwDelayTime>
      <!--opt, int, post-record-->1
    </dwDelayTime>
    <PTZPos>
      <!--opt, object, PTZ coordinates-->
      <P>
        <!--opt, int, panning adjustment-->1
      </P>
      <T>
        <!--opt, int, tilting adjustment-->1
      </T>
      <Z>
        <!--opt, int, zooming adjustment-->1
      </Z>
    </PTZPos>
    <advancedType opt="humanFeature,objectFeature">
      <!--opt, enum, advanced parameter type, subType:string, attr:opt{req, string}, desc:; "humanFeature" (human body feature); "objectFeature" (object
feature)-->humanFeature
    </advancedType>
    <advance>
      <!--opt, object, advanced parameters-->
      <HumanFeature>
        <!--opt, object, human body feature, dep:and,{$.SmartSearchDescription.FaceDetection.advancedType,eq,humanFeature}-->
        <age min="0" max="100">
          <!--opt, int, age, attr:min{req, int},max{req, int}-->1
        </age>
        <sex opt="male,female">
          <!--opt, enum, name, subType:string, attr:opt{req, string}, desc:; "male"; "female"-->male
        </sex>
        <eyeGlass>
          <!--opt, bool, whether the person wears glasses-->true
        </eyeGlass>
        <ageGroup>
          <!--opt, enum, age group, subType:string, desc:refer to the dictionary for details; "any" (no limitation, and used for search)-->any
        </ageGroup>
        <ageDeviation>
          <!--opt, int, age tolerance, range:[0,100]-->0
        </ageDeviation>
      </HumanFeature>
      <ObjectFeature>
        <!--opt, object, target features, dep:and,{$.SmartSearchDescription.FaceDetection.advancedType,eq,objectFeature}-->
        <colorRatel>
          <!--opt, int, color ratio-->1
        </colorRatel>
        <R>
          <!--opt, int, red-->1
        </R>
        <G>
          <!--opt, int, green-->1
        </G>
        <B>
          <!--opt, int, blue-->1
        </B>
      </ObjectFeature>
    </advance>
  </FaceDetection>
</SmartSearchDescription>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SmartSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search result, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, search record ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one
during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search result, subType:string, desc:; “true” (succeeded); “false” (failed)-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, search status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched
data).-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <matchList>
    <!--ro, opt, array, matched result list, subType:object-->
    <searchMatchItem>
      <!--ro, opt, object, matched results-->
      <id>
        <!--ro, req, int, index No.-->1
      </id>
      <timeSpan>
        <!--ro, opt, object, time period-->
        <startTime>
          <!--ro, req, datetime, start time, desc:ISO8601 time (TZ format: UTC time)-->1970-01-01T00:10:00Z
        </startTime>
        <endTime>
          <!--ro, req, datetime, end time, desc:ISO8601 time (TZ format: UTC time)-->1970-01-01T01:20:00Z
        </endTime>
      </timeSpan>
    </searchMatchItem>
  </matchList>
</SmartSearchResult>
```

14.3.3.9 Get capability of searching videos by VCA event

Request URL

GET /ISAPI/ContentMgmt/SmartSearch/capabilities

Query Parameter

None

Request Message

None

Response Message

dr4ckun@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<SmartSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, search for videos by VCA event, attr:version{req, string, protocolVersion}-->
  <maxResults>
    <!--ro, req, int, the maximum number of search results this time-->1
  </maxResults>
  <type opt="motionDetection,traversePlaneDetection,FieldDetection,faceDetection">
    <!--ro, req, enum, type for search, subType:string, attr:opt{req, string}, desc:;"motionDetection" (motion detection); "traversePlaneDetection" (line
crossing detection); "FieldDetection" (intrusion detection); "faceDetection" (facial detection)-->motionDetection
  </type>
  <TraversePlane>
    <!--ro, opt, object, it is valid when type is "traversePlaneDetection", dep:and,{$.SmartSearchDescription.type,eq,traversePlaneDetection}-->
    <TraversePlaneParam>
      <!--ro, opt, object, line crossing-->
      <detectionTarget opt="all,human,vehicle">
        <!--ro, req, enum, detection target, subType:string, attr:opt{req, string}, desc:; "all" (humans and vehicles); "human"; "vehicle"-->all
      </detectionTarget>
    </TraversePlaneParam>
  </TraversePlane>
  <FieldDetection>
    <!--ro, opt, object, intrusion detection, dep:and,{$.SmartSearchDescription.type,eq,FieldDetection}-->
    <param>
      <!--ro, opt, object, parameters-->
      <region>
        <!--ro, opt, object, area-->
        <pointList>
          <!--ro, opt, array, point list, subType:object, range:[0,4], desc:rectangle-->
        </pointList>
      </region>
    </param>
    <detectionTarget opt="all,human,vehicle">
      <!--ro, req, enum, detection target, subType:string, attr:opt{req, string}, desc:; "all" (humans and vehicles); "human"; "vehicle"-->all
    </detectionTarget>
  </FieldDetection>
</SmartSearchDescription>
```

14.4 云台控制

14.4.1 预置点管理

14.4.1.1 Set parameters of all patrols for a specified channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

dnackum@gmail.com DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, list of patrol parameters, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPatrol>
    <!--opt, object, patrol parameters-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <id>
      <!--req, string, ID-->test
    </id>
    <patrolName>
      <!--req, string, patrol name-->test
    </patrolName>
    <resumeType>
      <!--opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
    </resumeType>
    <PatrolSequenceList>
      <!--req, array, list of patrol results, subType:object-->
      <PatrolSequence>
        <!--req, object, patrol result-->
        <presetID>
          <!--req, string, preset ID-->1
        </presetID>
        <seqSpeed>
          <!--req, string, patrol speed-->1
        </seqSpeed>
        <delay>
          <!--req, string, delay-->1
        </delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

dracku@gmail.com
DEFEKTOR

14.4.1.2 Get the parameters of all patrols of a specific channel

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPatrol>
    <!--ro, opt, object, patrol parameters-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <patrolName>
      <!--ro, req, string, patrol name-->test
    </patrolName>
    <resumeType>
      <!--ro, opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
    </resumeType>
    <PatrolSequenceList>
      <!--ro, req, array, subType:object-->
      <PatrolSequence>
        <!--ro, req, object, list-->
        <presetID>
          <!--ro, req, string, preset ID-->1
        </presetID>
        <seqSpeed>
          <!--ro, req, string, patrol speed-->1
        </seqSpeed>
        <delay>
          <!--ro, req, int, delay, unit:ms-->1
        </delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

14.4.1.3 Get patrol parameters of a specified channel

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

drackun@gmail.com DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <patrolName>
    <!--ro, req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--ro, opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--ro, req, array, patrol result list, subType:object-->
    <PatrolSequence>
      <!--ro, req, object, patrol result-->
      <presetID>
        <!--ro, req, string, preset ID-->1
      </presetID>
      <seqSpeed>
        <!--ro, req, string, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--ro, req, int, delay, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

14.4.1.4 Delete a patrol by channel

Request URL

DELETE /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

drackum@gmail.com
DETEKTOR

14.4.1.5 Set path parameters of a single patrol for a specified channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string, ID-->test
  </id>
  <patrolName>
    <!--req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--req, array, patrol result list, subType:object-->
    <PatrolSequence>
      <!--req, object, patrol result-->
      <presetID>
        <!--req, string, preset ID-->1
      </presetID>
      <seqSpeed>
        <!--req, string, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--req, int, delay, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.4.1.6 Call a preset

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/presets/<presetID>/goto

Query Parameter

drackum@gmail.com
DETEKTOR

Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

14.4.2 云台维护

14.4.2.1 Set a specified PTZ channel's parameters

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, PTZ channel's parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <serialNumber>
    <!--req, int, serial port No., desc:this field should be supported by both GET and PUT methods. If the value is 0, it indicates that no serial port is
linked-->1
  </serialNumber>
  <videoInputID>
    <!--req, int, video input ID-->1
  </videoInputID>
  <panMaxSpeed>
    <!--opt, int, the maximum panning speed, unit: degrees/sec-->1
  </panMaxSpeed>
  <tiltMaxSpeed>
    <!--opt, int, the maximum titling speed, unit: degrees/sec-->1
  </tiltMaxSpeed>
  <presetSpeed>
    <!--opt, int, preset speed-->1
  </presetSpeed>
  <autoPatrolSpeed>
    <!--opt, int, automatic patrol speed-->1
  </autoPatrolSpeed>
  <keyBoardControlSpeed>
    <!--opt, int, keypad control speed, range:[0,100]-->1
  </keyBoardControlSpeed>
  <controlProtocol>
```

dracknum@gmail.com
DETEKTOR

```
    <!--opt, enum, control protocol, subType:string, desc:"pelco-d", "modbus-RTU", "modbus-ASCII"-->pelco-d
  </controlProtocol>
  <controlAddress>
    <!--req, object, control address-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <Address>
      <!--opt, string, address, range:[1,255]-->test
    </Address>
  </controlAddress>
  <defaultPresetID>
    <!--opt, string, preset ID-->test
  </defaultPresetID>
  <PTZRs485Para>
    <!--opt, object-->
    <baudRate>
      <!--req, int, baud rate-->9600
    </baudRate>
    <dataBits>
      <!--req, int, data bit-->1
    </dataBits>
    <parityType>
      <!--req, enum, parity type, subType:string, desc:"none" (no parity bit), "even" (even parity), "odd" (odd parity), "mark" (total number of parity bits
is 1), "space" (total number of parity bits is 0)-->space
    </parityType>
    <stopBits>
      <!--req, enum, stop bit, subType:string, desc:"1", "1.5", "2"-->1
    </stopBits>
    <flowCtrl>
      <!--req, enum, flow control type, subType:string, desc:"none", "software", "hardware"-->hardware
    </flowCtrl>
  </PTZRs485Para>
  <manualControlSpeed>
    <!--opt, enum, manual control speed type, subType:string, desc:"pedestrian", "nonMotorVehicle" (non-motor vehicles), "motorVehicle" (vehicle),
"selfadaptive" (self-adaptive), "compatible"-->pedestrian
  </manualControlSpeed>
  <panSupport>
    <!--opt, bool, whether it supports panning-->true
  </panSupport>
  <tiltSupport>
    <!--opt, bool, whether it supports tilting-->true
  </tiltSupport>
  <zoomSupport>
    <!--opt, bool, whether it supports zooming-->true
  </zoomSupport>
  <PTPositiveDirection>
    <!--opt, object, panning and tilting positive direction, desc:panning and tilting positive direction-->
    <pan>
      <!--opt, enum, panning positive direction, subType:string, desc:"left", "right"-->left
    </pan>
    <tile>
      <!--opt, enum, tilting positive direction, subType:string, desc:"up", "down"-->up
    </tile>
  </PTPositiveDirection>
  <PTQueryFrequency>
    <!--opt, enum, subType:int-->30
  </PTQueryFrequency>
</PTZChannel>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.4.3 云台控制

14.4.3.1 Set the PTZ absolute positioning data of a specific channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/absolute

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, PTZ absolute positioning data, attr:version{req, string, protocolVersion}-->
  <AbsoluteHigh>
    <!--opt, object, absolute height-->
    <elevation>
      <!--opt, int, altitude, range:[-900,2700]-->1
    </elevation>
    <azimuth>
      <!--opt, int, panning angle, range:[0,3600]-->1
    </azimuth>
    <absoluteZoom>
      <!--opt, int, absolute zoom, range:[0,1000]-->1
    </absoluteZoom>
  </AbsoluteHigh>
</PTZData>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

dirackum@gmail.com
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14.5 异常行为检测

14.5.1 快速移动

14.5.1.1 Get the arming schedule of fast moving detection for all channels

Request URL

GET /ISAPI/Event/schedules/rapidMoves

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RapidMoveScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->rapidMove
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RapidMoveScheduleList>
```

drackum@gmail.com
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14.5.1.2 Set the arming schedule of fast moving detection for all channels

Request URL

PUT /ISAPI/Event/schedules/rapidMoves

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RapidMoveScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, enum, subType:string-->rapidMove
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RapidMoveScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.1.3 Set the linkage parameters of fast moving detection

Request URL

PUT /ISAPI/Event/triggers/rapidMove-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--req, enum, subType:string-->rapidMove
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.1.4 Get the linkage parameters of fast moving detection

Request URL

GET /ISAPI/Event/triggers/rapidMove-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->rapidMove
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

14.5.1.5 Fast moving

EventType:rapidMove

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, transmitted parameters, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
```

drackum@gmail.com
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```
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"rapidMove" (fast moving detection)-->rapidMove
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->0
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
```

drackum@gmail.com
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```
    <targetAttrs>
      <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
    </targetAttrs>
    <TaskInfo>
      <!--ro, opt, object, task information submitted by data center products-->
      <taskID>
        <!--ro, req, string, task ID, range:[0,64]-->test
      </taskID>
      <algorithmType>
        <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
      </algorithmType>
      <taskName>
        <!--ro, opt, string, task name-->test
      </taskName>
      <algorithmID>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmID>
      <monitorPointID>
        <!--ro, opt, string, camera No.-->test
      </monitorPointID>
      <monitorPointName>
        <!--ro, opt, string, camera name-->test
      </monitorPointName>
      <streamType>
        <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
      </streamType>
      <ruleID>
        <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
      </ruleID>
      <ruleCustomName>
        <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
      </ruleCustomName>
      <algorithmInfo>
        <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
        <algorithmId>
          <!--ro, opt, string, algorithm package ID-->test
        </algorithmId>
        <algoVendorId>
          <!--ro, opt, string, manufacturer ID-->test
        </algoVendorId>
        <algoVersionId>
          <!--ro, opt, string, algorithm version No.-->test
        </algoVersionId>
        <algoChipType>
          <!--ro, opt, string, chip type-->test
        </algoChipType>
        <algoModelVersion>
          <!--ro, opt, string, model version-->test
        </algoModelVersion>
      </algorithmInfo>
    </TaskInfo>
    <timeStamp>
      <!--ro, opt, datetime, timestamp of the frame, which triggers the alarm-->1970-01-01T00:00:00+08:00
    </timeStamp>
    <isDataRetransmission>
      <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
    </isDataRetransmission>
    <ruleID>
      <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
    </ruleID>
    <ruleName>
      <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
    </ruleName>
  </EventNotificationAlert>
```

drackum@gmail.com
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14.5.2 徘徊侦测

14.5.2.1 Set arming schedules of loitering detection of all channels

Request URL

PUT /ISAPI/Event/schedules/loiterings

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, arming schedule list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming schedule-->
    <id>
      <!--req, string, ID, desc:ID-->loitering-1
    </id>
    <eventType>
      <!--opt, string, event type-->loitering
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming schedule list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1: Monday; 2: Tuesday; 3: Wednesday; 4: Thursday; 5: Friday; 6: Saturday; 7: Sunday;-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming schedule, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LoiteringScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.2.2 Get arming schedules of the loitering detection of all channels

Request URL

GET /ISAPI/Event/schedules/loiterings

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, arming schedule list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule information-->
    <id>
      <!--ro, req, string, ID-->loitering-1
    </id>
    <eventType>
      <!--ro, opt, string, event type-->loitering
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, arming schedule list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1: Monday), 2: Tuesday, 3: Wednesday, 4: Thursday, 5: Friday, 6: Saturday, 7: Sunday-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time period-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time period-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LoiteringScheduleList>
```

drackum@gmail.com
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14.5.2.3 Get the linkage parameters of loitering detection

Request URL

GET /ISAPI/Event/triggers/loitering-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->loitering
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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14.5.2.4 Set the linkage parameters of loitering detection

Request URL

PUT /ISAPI/Event/triggers/loitering-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--req, enum, subType:string-->loitering
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.2.5 Get loitering detection parameters of a channel

Request URL

GET /ISAPI/Smart/loitering/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Loitering xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, loitering detection parameters configuration of a channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <LoiteringRegionList>
    <!--ro, opt, array, information list of loitering detection areas, subType:object-->
    <LoiteringRegion>
      <!--ro, opt, object, loitering detection area-->
      <id>
        <!--ro, req, string, area ID-->1
      </id>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
      </sensitivityLevel>
      <timeThreshold>
        <!--ro, opt, int, time threshold-->1
      </timeThreshold>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </LoiteringRegion>
  </LoiteringRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, ro,"PDC", attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool-->true
  </isSupportMultiScene>
</Loitering>
```

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14.5.2.6 Loitering detection

EventType:loitering

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
```

```
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"loitering" (loitering detection)-->loitering
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, detection area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human"-->human
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
```

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```
      <algorithmID>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmID>
      <monitorPointID>
        <!--ro, opt, string, camera No.-->test
      </monitorPointID>
      <monitorPointName>
        <!--ro, opt, string, camera name-->test
      </monitorPointName>
      <streamType>
        <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
      </streamType>
      <ruleID>
        <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
      </ruleID>
      <ruleCustomName>
        <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
      </ruleCustomName>
      <algorithmInfo>
        <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
        <algorithmId>
          <!--ro, opt, string, algorithm package ID-->test
        </algorithmId>
        <algoVendorId>
          <!--ro, opt, string, manufacturer ID-->test
        </algoVendorId>
        <algoVersionId>
          <!--ro, opt, string, algorithm version No.-->test
        </algoVersionId>
        <algoChipType>
          <!--ro, opt, string, chip type-->test
        </algoChipType>
        <algoModelVersion>
          <!--ro, opt, string, model version-->test
        </algoModelVersion>
      </algorithmInfo>
    </TaskInfo>
    <timeStamp>
      <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
    </timeStamp>
    <isDataRetransmission>
      <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
    </isDataRetransmission>
    <ruleID>
      <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
    </ruleID>
    <ruleName>
      <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
    </ruleName>
  </EventNotificationAlert>
```

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14.5.3 物品拿取

14.5.3.1 Get the arming schedule of object removal detection for all channels

Request URL

GET /ISAPI/Event/schedules/attendedBaggages

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<AttendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->attendedBaggage-1
    </id>
    <eventType>
      <!--ro, opt, string-->attendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</AttendBaggageScheduleList>
```

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14.5.3.2 Set the arming schedule of object removal detection for all channels

Request URL

PUT /ISAPI/Event/schedules/attendedBaggages

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AttendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->attendedBaggage-1
    </id>
    <eventType>
      <!--opt, string-->attendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</AttendBaggageScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.3.3 Get the linkage parameters of object removal detection

Request URL

GET /ISAPI/Event/triggers/attendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->attendedBaggage-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->attendedBaggage
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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14.5.3.4 Set the linkage parameters of object removal detection

Request URL

PUT /ISAPI/Event/triggers/attendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->attendedBaggage-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->attendedBaggage
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.3.5 Object removal

EventType:attendedBaggage

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, object removal alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, event triggering frequency, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"attendedBaggage" (object removal)-->attendedBaggage
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
```

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```
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"others"-->others
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
```

dinckum@gmail.com
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```
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

14.5.4 人员聚集

14.5.4.1 Get the arming time parameters of people gathering for all channels

Request URL

GET /ISAPI/Event/schedules/groups

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<GroupScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, people gathering arming time configuration list, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, people gathering arming time configuration-->
    <id>
      <!--ro, req, string, index, desc:the format is event type-channel No.-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string, desc:;group:group-->group
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-
->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming time-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start Time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</GroupScheduleList>
```

drackum@gmail.com
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14.5.4.2 Set the arming time parameters of people gathering for all channels

Request URL

PUT /ISAPI/Event/schedules/groups

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, people gathering arming time configuration list, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, people gathering arming time configuration-->
    <id>
      <!--req, string, index, desc:the format is event type-channel No.-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string, desc:;group:group-->group
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start Time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start Time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</GroupScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.4.3 Set the linkage parameters of people gathering

Request URL

PUT /ISAPI/Event/triggers/group-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, people gathering linkage configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, desc:the format is event type-channel No.-->test
  </id>
  <eventType>
    <!--req, enum, event type, subType:string, desc:;group:group-->group
  </eventType>
  <eventDescription>
    <!--opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string, video input channel ID, desc:if <eventType> is
"VMD,videoloss,tamperdetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage"-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, event capture interval, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string, wireless ID-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, linkage action list, subType:object-->
    <EventTriggerNotification>
      <!--opt, object, linkage action-->
      <id>
        <!--req, string, index-->test
      </id>
      <notificationMethod>
        <!--req, enum, linkage action, subType:string, desc:"email", "IM" (instant messaging), "IO" (I/O output), "syslog" (system log), "HTTP" (uploading
via HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "monitorAlarm" (alarm on monitor), "center" (uploading to the
center), "lightAudioAlarm" (light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio)-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, linkage cycle, subType:string, desc:"beginning", "beginningandend", "recurring"-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, linkage time interval, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string, video channel ID-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string, dynamic video input channel ID-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object, PTZ linkage-->
        <ptzChannelID>
          <!--req, string, PTZ linked channel No.-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
        </actionName>
        <actionNum>
          <!--opt, int, linked No., desc:depend on the content of <actionName>, it can be preset No., patrol No., and pattern No.-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object, linkage parameters of the white light, desc:it is valid when the value of notificationMethod is "whiteLight"-->
        <whiteLightDurationTime>
          <!--opt, int, light flashing duration, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

dracku@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.4.4 Get linkage parameters of people gathering

Request URL

GET /ISAPI/Event/triggers/group-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, people gathering linkage configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index, desc:the format is event type-channel No.-->test
  </id>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:group:group-->group
  </eventType>
  <eventDescription>
    <!--ro, opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID, desc:if <eventType> is
"VMD,videoloss,tamperdetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage"-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, event capture interval, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string, wireless ID-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, linkage action list, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object, linkage action-->
      <id>
        <!--ro, req, string, index-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, linkage action, subType:string, desc:"email", "IM" (instant messaging), "IO" (I/O output), "syslog" (system log), "HTTP"
(uploading via HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "monitorAlarm" (alarm on monitor), "center" (uploading
to the center), "LightAudioAlarm" (light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio)-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, linkage cycle, subType:string, desc:"beginning", "beginningandend", "recurring"-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, linkage time interval, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string, video channel ID-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string, dynamic video input channel ID-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object, PTZ linkage-->
        <ptzChannelID>
          <!--ro, req, string, PTZ linked channel No.-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int, linked No., desc:depend on the content of <actionName>, it can be preset No., patrol No., and pattern No.-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object, linkage parameters of the white light, desc:it is valid when the value of notificationMethod is "whiteLight"-->
        <whiteLightDurationTime>
          <!--ro, opt, int, light flashing duration, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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14.5.4.5 Get parameters of people gathering for a specified channel

Request URL

PUT /ISAPI/Smart/group/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Group xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters configuration of people gathering for a channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <GroupRegionList>
    <!--opt, array, people gathering area list, subType:object-->
    <GroupRegion>
      <!--opt, object, people gathering area-->
      <id>
        <!--req, string, area ID-->1
      </id>
      <objectOccupation>
        <!--opt, int, percentage-->1
      </objectOccupation>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
  </GroupRegionList>
  <mutexAbility opt="PDC">
    <!--opt, string, mutual exclusion ability, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--opt, bool, whether supports multiple scene detection-->true
  </isSupportMultiScene>
</Group>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.5.4.6 Get parameters of people gathering for a specified channel

Request URL

GET /ISAPI/Smart/group/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<Group xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters configuration of people gathering for a channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <GroupRegionList>
    <!--ro, opt, array, people gathering area list, subType:object-->
    <GroupRegion>
      <!--ro, opt, object, people gathering area-->
      <id>
        <!--ro, req, string, area ID-->1
      </id>
      <objectOccupation>
        <!--ro, opt, int, percentage-->1
      </objectOccupation>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
    <headCount>
      <!--ro, opt, int-->5
    </headCount>
    <duration>
      <!--ro, opt, int-->10
    </duration>
  </GroupRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, mutual exclusion ability, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool, whether supports multiple scene detection-->true
  </isSupportMultiScene>
  <shieldedRegionList>
    <!--ro, opt, array, subType:object-->
    <GroupRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
  </shieldedRegionList>
  <sensitivityLevel>
    <!--ro, opt, int, range:[0,100]-->1
  </sensitivityLevel>
</Group>
```

drackum@gmail.com
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14.5.4.7 People gathering

EventType:group

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
```

```
<!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
<ipAddress>
<!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
</ipAddress>
<ipv6Address>
<!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
</ipv6Address>
<portNo>
<!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
</portNo>
<protocol>
<!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
</protocol>
<macAddress>
<!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
</macAddress>
<dynChannelID>
<!--ro, opt, string, digital channel No.-->test
</dynChannelID>
<channelID>
<!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
</channelID>
<releatedChannelList>
<!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
</releatedChannelList>
<dateTime>
<!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
</dateTime>
<activePostCount>
<!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
</activePostCount>
<eventType>
<!--ro, req, enum, event type, subType:string, desc:"group" (people gathering)-->group
</eventType>
<eventState>
<!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
</eventState>
<eventDescription>
<!--ro, req, string, event description-->test
</eventDescription>
<deviceID>
<!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
</deviceID>
<DetectionRegionList>
<!--ro, opt, array, detection area list, subType:object-->
<DetectionRegionEntry>
<!--ro, opt, object, detection area-->
<regionID>
<!--ro, req, string, area ID-->test
</regionID>
<sensitivityLevel>
<!--ro, opt, int, sensitivity, range:[0,100]-->1
</sensitivityLevel>
<RegionCoordinatesList>
<!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
<RegionCoordinates>
<!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
<positionX>
<!--ro, req, int, X-coordinate, range:[0,1000]-->50
</positionX>
<positionY>
<!--ro, req, int, Y-coordinate, range:[0,1000]-->50
</positionY>
</RegionCoordinates>
</RegionCoordinatesList>
<objectOccupation>
<!--ro, opt, int, percentage, range:[1,100]-->1
</objectOccupation>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName>
<!--ro, opt, string, channel name, range:[1,64]-->test
</channelName>
<Extensions>
<!--ro, opt, object, extended information-->
<serialNumber>
<!--ro, opt, string, serial No.-->test
</serialNumber>
<eventPush>
<!--ro, opt, string, event notification-->test
</eventPush>
</Extensions>
<detectionPictureTransType>
<!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
</detectionPictureTransType>
<detectionPicturesNumber>
<!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
</detectionPicturesNumber>
<bkgUrl>
<!--ro, opt, string, background picture URL, desc:background picture URL-->test
```

drackun@gmail.com
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```
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

drackum@gmail.com
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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
group	[Message content]	application/xml	--	--	--
.value	[Binary picture data]	image/jpeg	随机值	pic.jpeg	--
video	[Video data]	video/mp4	随机值	video.mp4	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

14.5.5 物品遗留

14.5.5.1 Set the arming schedule of unattended baggage detection for all channels

Request URL

PUT /ISAPI/Event/schedules/unattendedBaggages

Query Parameter

None

Request Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<UnattendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->unattendedBaggage-1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->unattendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, range:[0,56], attr:size{req, int, range:[0,8]}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object, range:[0,8]-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</UnattendBaggageScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.5.2 Get the arming schedule of unattended baggage detection for all channels

Request URL

GET /ISAPI/Event/schedules/unattendedBaggages

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<UnattendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->unattendedBaggage-1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->unattendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, range:[0,56], attr:size{req, int, range:[0,8]}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object, range:[0,8]-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</UnattendBaggageScheduleList>
```

drackum@gmail.com
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14.5.5.3 Get the linkage parameters of unattended baggage detection

Request URL

GET /ISAPI/Event/triggers/unattendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->unattendedBaggage-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->unattendedBaggage
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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14.5.5.4 Set the linkage parameters of unattended baggage detection

Request URL

PUT /ISAPI/Event/triggers/unattendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->unattendedBaggage-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->unattendedBaggage
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.5.5.5 Unattended baggage detection

EventType:unattendedBaggage

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, unattended baggage alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, event triggering frequency, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"unattendedBaggage" (unattended baggage)-->unattendedBaggage
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
```

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```
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"others"-->others
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
```

dinckum@gmail.com
DEMEKTOR

```
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

14.6 人脸抓拍

14.6.1 人脸抓拍

14.6.1.1 Get the arming schedule of face capture

Request URL

GET /ISAPI/Event/schedules/faceSnap

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, arming schedule of face capture for all channels, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule of face capture for a specified channel-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <eventType>
      <!--ro, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpLoad",
"HFP", "IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fielddetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "LowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->ADAS
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel No.-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, arming time list, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--ro, opt, object, arming advanced configurations-->
          <vehicleDetectSceneID>
            <!--ro, opt, int, vehicle detection scene ID-->1
          </vehicleDetectSceneID>
          <sceneID>
            <!--ro, opt, int, scene ID-->1
          </sceneID>
          <SceneList>
            <!--ro, opt, array, switching rules of multiple scenes, subType:object, desc:mutually exclusive with "sceneID"-->
            <Scene>
              <!--ro, opt, object, multiple scene list-->
              <sceneID>
                <!--ro, req, string, scene ID-->1
              </sceneID>
              <durationTime>
                <!--ro, req, int, stayed time of a single scene-->1
              </durationTime>
            </Scene>
          </SceneList>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapScheduleList>
```

drackum@gmail.com
DETEKTOR

14.6.1.2 Set the arming schedule of face capture

Request URL

PUT /ISAPI/Event/schedules/faceSnap

Query Parameter

None

Request Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, req, array, arming schedule of face capture for all channels, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming schedule of face capture for a specified channel-->
    <id>
      <!--wo, req, string, channel No.-->test
    </id>
    <eventType>
      <!--wo, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpLoad",
"HFP", "IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fieldetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "LowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->faceSnap
    </eventType>
    <videoInputChannelID>
      <!--wo, opt, string, video input channel No.-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--wo, req, array, arming time list, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--wo, opt, object, arming slots-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:days of the week-->1
        </dayOfWeek>
        <TimeRange>
          <!--wo, req, object, time range-->
          <beginTime>
            <!--wo, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--wo, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--wo, opt, object, arming advanced configurations-->
          <vehicleDetectSceneID>
            <!--wo, opt, int, vehicle detection scene-->1
          </vehicleDetectSceneID>
          <sceneID>
            <!--wo, opt, int, scene ID-->1
          </sceneID>
          <SceneList>
            <!--wo, opt, array, switching rules of multiple scenes, subType:object, desc:mutually exclusive with "sceneID"-->
            <Scene>
              <!--wo, opt, object, multiple scene list-->
              <sceneID>
                <!--wo, req, string, scene ID-->1
              </sceneID>
              <durationTime>
                <!--wo, req, int, stayed time of a single scene-->1
              </durationTime>
            </Scene>
          </SceneList>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--wo, opt, array, holiday arming time list, subType:object-->
      <TimeBlock>
        <!--wo, opt, object, arming time-->
        <TimeRange>
          <!--wo, req, object, time range-->
          <beginTime>
            <!--wo, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--wo, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.1.3 Get the arming schedule of face capture for a specified channel

Request URL

GET /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, arming schedule of face capture for a specified channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:“ADAS”, “ADASAlarm”, “AID”, “ANPR”, “AccessControllerEvent”, “CDsStatus”, “DBD”, “GPSUpload”, “HFP”,
“IO”, “IOTD”, “LES”, ‘LFPD“, ”PALMismatch“, ”PIR“, ”PeopleCounting“, ”PeopleNumChange”, “Standup”, “TMA”, “TMPAM”, “VMD”, “abnormalAcceleration”,
“abnormalDriving”, “advReachHeight”, “alarmResult”, “attendance”, “attendedBaggage”, “audioAbnormal”, “audioexception”, “behaviorResult”,
“blindSpotDetection”, “cardMatch”, “changedStatus”, “collision”, “containerDetection”, “crowdSituationAnalysis”, “databaseException”, “defocus”,
“diskUnformat”, “diskerror”, “diskfull”, “driverConditionMonitor”, “emergencyAlarm”, “faceCapture”, “faceSnapModeling”, “facedetection”, “failDown”,
“faultAlarm”, “fielddetection”, “fireDetection”, “fireEscapeDetection”, “flowOverrun”, “framesPeopleCounting”, “getUp”, “group”, “hdBadBlock”, “hdImpact”,
“heatmap”, “highHDTemperature”, “highTempAlarm”, “hotSpare”, “illaccess”, “ipcTransferAbnorma”, “ipconflict”, “keyPersonGetUp”, “leavePosition”,
“linedetection”, “listSyncException”, “loitering”, “LowHDTemperature”, "mixedTargetDetection”, “modelError”, “nicbroken”, “nodeOffline”,
“nonPoliceIntrusio”, “overSpeed”, “overtimeTarry”, “parking”, “peopleNumChange”, “peopleNumCounting”, “personAbnormalAlarm”, “personDensityDetection”,
“personQueueCounting”, “personQueueDetection”, “personQueueRealTime”, “personQueueTime”, “playCellphone”, “pocException”, “poe”, “policeAbsent”,
“radarAlarm”, “radarFieldDetection”, “radarLineDetection”, “radarPerimeterRule”, “radarTargetDetection”, “radarVideoDetection”, “raidException”,
“rapidMove”, “reachHeight”, “recordCycleAbnormal”, “recordException”, “regionEntrance”, “regionExiting”, “retention”, “rollover”, “running”,
“safetyHelmetDetection”, “scenechangedetection”, “sensorAlarm”, “severeHDFailure”, “shelteralarm”, “shipsDetection”, “sitQuietly”, “smokeAndFireDetection”,
“smokeDetection”, “softIO”, “spacingChange”, “sysStorFull”, “takingElevatorDetection”, “targetCapture”, “temperature”, “thermometry”, “thirdPartyException”,
“toiletTarry”, “tollCodeInfo”, “tossing”, “unattendedBaggage”, “vehicleMatchResult”, “vehicleRcogResult”, “versionAbnormal”, “videoException”, “videoloss”,
“violationAlarm”, “violentMotion”, “yardTarry”, “faceSnap”-->faceSnap
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel No.-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00+08:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--ro, opt, object-->
        <vehicleDetectSceneID>
          <!--ro, opt, int-->1
        </vehicleDetectSceneID>
        <sceneID>
          <!--ro, opt, int-->1
        </sceneID>
        <SceneList>
          <!--ro, opt, array, subType:object-->
          <Scene>
            <!--ro, opt, object-->
            <sceneID>
              <!--ro, req, string-->1
            </sceneID>
            <durationTime>
              <!--ro, req, int-->1
            </durationTime>
          </Scene>
        </SceneList>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
  <backToSceneTime>
    <!--ro, opt, int, range:[5,100], step:1, unit:s-->15
  </backToSceneTime>
</Schedule>
```

drackum@gmail.com
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14.6.1.4 Set the arming schedule of face capture for a specified channel

Request URL

PUT /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, opt, object, arming schedule of face capture for a specified channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--wo, req, string, channel No.-->test
  </id>
  <eventType>
    <!--wo, req, enum, event type, subType:string, desc:“ADAS”, “ADASAlarm”, AID, ANPR, AccessControllerEvent, CDsStatus, DBD, GPSUpload, HFP, IO, IOTD,
LES, LFPD, PALMismatch, PIR, PeopleCounting, PeopleNumChange, Standup, TMA, TMPAM, VMD, abnormalAcceleration, abnormalDriving, advReachHeight, alarmResult,
attendance, attendedBaggage, audioAbnormal, audioexception, behaviorResult, blindSpotDetection, cardMatch, changedStatus, collision, containerDetection,
crowdSituationAnalysis, databaseException, defocus, diskUnformat, diskerror, diskfull, driverConditionMonitor, emergencyAlarm, faceCapture,
faceSnapModeling, facedetection, failDown, faultAlarm, fielddetection, fireDetection, fireEscapeDetection, flowOverrun, framesPeopleCounting, getUp, group,
hdBadBlock, hdImpact, heatmap, highHDTemperature, highTempAlarm, hotSpare, illaccess, ipcTransferAbnorma, ipconflict, keyPersonGetUp, leavePosition,
linedetection , listSyncException, loitering, lowHDTemperature, mixedTargetDetection, modelError, nicbroken, nodeOffline, nonPoliceIntrusio, overSpeed,
overtimeTarry, parking, peopleNumChange, peopleNumCounting, personAbnormalAlarm:, personDensityDetection, personQueueCounting, personQueueDetection,
personQueueRealTime, personQueueTime, playCellphone, pocException, poe, policeAbsent, radarAlarm, radarFieldDetection, radarlineDetection,
radarPerimeterRule, radarTargetDetection, radarVideoDetection, raidException, rapidMove, reachHeight, recordCycleAbnormal, recordException, regionEntrance,
regionExiting, retention, rollover, running, safetyHelmetDetection, scenechangedetection, sensorAlarm, severeHDFailure, shelteralarm, shipsDetection,
sitQuietly, smokeAndFireDetection, smokeDetection, softIO, spacingChange, sysStorFull, takingElevatorDetection, targetCapture, temperature, thermometry,
thirdPartyException, toiletTarry, tollCodeInfo, tossing, unattendedBaggage, vehicleMatchResult, vehicleRcogResult, versionAbnormal, videoException,
videoloss, violationAlarm, violentMotion, yardTarry, faceSnap-->faceSnap
  </eventType>
  <videoInputChannelID>
    <!--wo, opt, string, video input channel No.-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--wo, req, array, arming time list, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--wo, opt, object, arming time-->
      <dayOfWeek>
        <!--wo, opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)--
>1
      </dayOfWeek>
      <TimeRange>
        <!--wo, req, object, time range-->
        <beginTime>
          <!--wo, req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--wo, req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--wo, opt, object, arming advanced configurations-->
        <vehicleDetectSceneID>
          <!--wo, opt, int, vehicle detection scene-->1
        </vehicleDetectSceneID>
        <sceneID>
          <!--wo, opt, int, scene ID-->1
        </sceneID>
        <SceneList>
          <!--wo, opt, array, switching rules of multiple scenes, subType:object, desc:mutually exclusive with “sceneID”-->
          <Scene>
            <!--wo, opt, object, multiple scene list-->
            <sceneID>
              <!--wo, req, string, scene ID-->1
            </sceneID>
            <durationTime>
              <!--req, int, stayed time of a single scene-->1
            </durationTime>
          </Scene>
        </SceneList>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, holiday arming time list, subType:object-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <TimeRange>
        <!--req, object, time range-->
        <beginTime>
          <!--req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.1.5 Delete the linkage of face capture

Request URL

DELETE /ISAPI/Event/triggers/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.6.1.6 Set the linkage parameters of face capture

Request URL

PUT /ISAPI/Event/triggers/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, req, object, Configure linkage actions:, attr:version{req, string, protocolVersion}-->
  <id>
    <!--wo, req, string, channel No.-->test
  </id>
  <eventType>
    <!--wo, req, enum, event type, subType:string, desc:“ADAS”, “ADASAlarm”, “AID”, “ANPR”, “AccessControllerEvent”, “CDsStatus”, “DBD”, “GPSUpload”, “HFP”,
“IO”, “IOTD”, “LES”, ‘LFPD“, ”PALMismatch“, ”PIR“, ”PeopleCounting“, ”PeopleNumChange”, “Standup”, “TMA”, “TMPAM”, “VMD”, “abnormalAcceleration”,
“abnormalDriving”, “advReachHeight”, “alarmResult”, “attendance”, “attendedBaggage”, “audioAbnormal”, “audioexception”, “behaviorResult”,
“blindSpotDetection”, “cardMatch”, “changedStatus”, “collision”, “containerDetection”, “crowdSituationAnalysis”, “databaseException”, “defocus”,
“diskUnformat”, “diskerror”, “diskfull”, “driverConditionMonitor”, “emergencyAlarm”, “faceCapture”, “faceSnapModeling”, “facedetection”, “failDown”,
“faultAlarm”, “fielddetection”, “fireDetection”, “fireEscapeDetection”, “flowOverrun”, “framesPeopleCounting”, “getUp”, “group”, “hdBadBlock”, “hdImpact”,
“heatmap”, “highHDTemperature”, “highTempAlarm”, “hotSpare”, “illaccess”, “ipcTransferAbnorma”, “ipconflict”, “keyPersonGetUp”, “LeavePosition”,
“linedetection”, “ListSyncException”, “loitering”, “lowHDTemperature”, "mixedTargetDetection", “modelError”, “nicbroken”, “nodeOffline”,
“nonPoliceIntrusio”, “overSpeed”, “overtimeTarry”, “parking”, “peopleNumChange”, “peopleNumCounting”, “personAbnormalAlarm”, “personDensityDetection”,
“personQueueCounting”, “personQueueDetection”, “personQueueRealTime”, “personQueueTime”, “playCellphone”, “pocException”, “poe”, “policeAbsent”,
“radarAlarm”, “radarFieldDetection”, “radarLineDetection”, “radarPerimeterRule”, “radarTargetDetection”, “radarVideoDetection”, “raidException”,
“rapidMove”, “reachHeight”, “recordCycleAbnormal”, “recordException”, “regionEntrance”, “regionExiting”, “retention”, “rollover”, “running”,
“safetyHelmetDetection”, “scenechangedetection”, “sensorAlarm”, “severeHDFailure”, “shelteralarm”, “shipsDetection”, “sitQuietly”, “smokeAndFireDetection”,
“smokeDetection”, “softIO”, “spacingChange”, “sysStorFull”, “takingElevatorDetection”, “targetCapture”, “temperature”, “thermometry”, “thirdPartyException”,
“toiletTarry”, “tollCodeInfo”, “tossing”, “unattendedBaggage”, “vehicleMatchResult”, “vehicleRcogResult”, “versionAbnormal”, “videoException”, “videoloss”,
“violationAlarm”, “violentMotion”, “yardTarry”, “faceSnap”-->faceSnap
  </eventType>
  <eventDescription>
    <!--wo, opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--wo, opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--wo, opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--wo, opt, string, video input channel ID, desc:only valid when event types are VMD, videoloss, shelteralarm, regionEntrance, regionExiting,
loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, thermal, zoneTriggeredAlarm, criticalAlerts, businessCounselingLinkage-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--wo, opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--wo, opt, int, event capture interval, step:1, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--wo, opt, string, wireless ID-->test
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--wo, opt, array, configuration list of event linkage, subType:object-->
    <EventTriggerNotification>
      <!--wo, opt, object, list-->
      <id>
        <!--wo, req, string, linkage configuration ID-->test
      </id>
      <notificationMethod>
        <!--wo, req, enum, linkage action, subType:string, desc:"FTP" (uploading via FTP), "HTTP" (uploading via HTTP), "IM" (instant messaging), "IO" (I/O
output), "LightAudioAlarm" (light and audio alarm), "audio" (audio), "beep" (buzzer), "center" (uploading to the center), "cloud" (cloud storage), "email",
"focus" (focus), "monitorAlarm" (alarm on monitor), "ptz" (PTZ), "record" (recording), "syslog" (system log), "trace" (tracking), "whiteLight" (white
light)-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--wo, opt, enum, linkage cycle, subType:string, desc:“beginning”, “beginningandend”, “recurring”-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--wo, opt, int, linkage time interval, step:1, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--wo, opt, string, video input ID-->test
      </videoInputID>
      <dynVideoInputID>
        <!--wo, opt, string, dynamic video input ID-->test
      </dynVideoInputID>
      <ptzAction>
        <!--wo, opt, object, PTZ linkage action-->
        <ptzChannelID>
          <!--wo, req, string, PTZ linked channel No.-->test
        </ptzChannelID>
        <actionName>
          <!--wo, req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
        </actionName>
        <actionNum>
          <!--wo, opt, int, linkage value-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--wo, opt, object, white light linkage-->
        <whiteLightDurationTime>
          <!--wo, req, int, light flashing duration, range:[1,60], step:1, unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
```

drackum@gmail.com
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```
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.1.7 Get the linkage parameters of face capture

Request URL

GET /ISAPI/Event/triggers/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, linkage configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:“ADAS”, “ADASAlarm”, “AID”, “ANPR”, “AccessControllerEvent”, “CDsStatus”, “DBD”, “GPSUpload”, “HFP”,
“IO”, “IOTD”, “LES”, ‘LFPD“, ”PALMismatch“, ”PIR“, ”PeopleCounting“, ”PeopleNumChange”, “Standup”, “TMA”, “TMPAM”, “VMD”, “abnormalAcceleration”,
“abnormalDriving”, “advReachHeight”, “alarmResult”, “attendance”, “attendedBaggage”, “audioAbnormal”, “audioexception”, “behaviorResult”,
“blindSpotDetection”, “cardMatch”, “changedStatus”, “collision”, “containerDetection”, “crowdSituationAnalysis”, “databaseException”, “defocus”,
“diskUnformat”, “diskerror”, “diskfull”, “driverConditionMonitor”, “emergencyAlarm”, “faceCapture”, “faceSnapModeling”, “facedetection”, “failDown”,
“faultAlarm”, “fieldetection”, “fireDetection”, “fireEscapeDetection”, “flowOverrun”, “framesPeopleCounting”, “getUp”, “group”, “hdBadBlock”, “hdImpact”,
“heatmap”, “highHDTemperature”, “highTempAlarm”, “hotSpare”, “illaccess”, “ipcTransferAbnorma”, “ipconflict”, “keyPersonGetUp”, “leavePosition”,
“linedetection”, “listSyncException”, “loitering”, “lowHDTemperature”, "mixedTargetDetection”, “modelError”, “nicbroken”, “nodeOffline”,
“nonPoliceIntrusio”, “overSpeed”, “overtimeTarry”, “parking”, “peopleNumChange”, “peopleNumCounting”, “personAbnormalAlarm”, “personDensityDetection”,
“personQueueCounting”, “personQueueDetection”, “personQueueRealTime”, “personQueueTime”, “playCellphone”, “pocException”, “poe”, “policeAbsent”,
“radarAlarm”, “radarFieldDetection”, “radarLineDetection”, “radarPerimeterRule”, “radarTargetDetection”, “radarVideoDetection”, “raidException”,
“rapidMove”, “reachHeight”, “recordCycleAbnormal”, “recordException”, “regionEntrance”, “regionExiting”, “retention”, “rollover”, “running”,
“safetyHelmetDetection”, “scenechangedetection”, “sensorAlarm”, “severeHDFailure”, “shelteralarm”, “shipsDetection”, “sitQuietly”, “smokeAndFireDetection”,
“smokeDetection”, “softIO”, “spacingChange”, “sysStorFull”, “takingElevatorDetection”, “targetCapture”, “temperature”, “thermometry”, “thirdPartyException”,
“toiletTarry”, “tollCodeInfo”, “tossing”, “unattendedBaggage”, “vehicleMatchResult”, “vehicleRcogResult”, “versionAbnormal”, “videoException”, “videoloss”,
“violationAlarm”, “violentMotion”, “yardTarry”, “faceSnap”-->faceSnap
  </eventType>
  <eventDescription>
    <!--ro, opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID, desc:only valid when event types are VMD, videoloss, shelteralarm, regionEntrance, regionExiting,
loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, thermal, zoneTriggeredAlarm, criticalAlerts, businessCounselingLinkage-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
```

drackum@gmail.com
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```
    <intervalBetweenEvents>
      <!--ro, opt, int, event capture interval, step:1, unit:s-->1
    </intervalBetweenEvents>
    <WLSensorID>
      <!--ro, opt, string, wireless ID-->test
    </WLSensorID>
    <EventTriggerNotificationList>
      <!--ro, opt, array, subType:object-->
      <EventTriggerNotification>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->test
        </id>
        <notificationMethod>
          <!--ro, req, enum, subType:string-->FTP
        </notificationMethod>
        <notificationRecurrence>
          <!--ro, opt, enum, subType:string-->beginning
        </notificationRecurrence>
        <notificationInterval>
          <!--ro, opt, int, step:1, unit:ms-->1
        </notificationInterval>
        <videoInputID>
          <!--ro, opt, string-->test
        </videoInputID>
        <dynVideoInputID>
          <!--ro, opt, string-->test
        </dynVideoInputID>
        <ptzAction>
          <!--ro, opt, object-->
          <ptzChannelID>
            <!--ro, req, string-->test
          </ptzChannelID>
          <actionName>
            <!--ro, req, enum, subType:string-->preset
          </actionName>
          <actionNum>
            <!--ro, opt, int-->1
          </actionNum>
        </ptzAction>
        <WhiteLightAction>
          <!--ro, opt, object-->
          <whiteLightDurationTime>
            <!--ro, req, int, range:[1,60], step:1, unit:s-->1
          </whiteLightDurationTime>
        </WhiteLightAction>
        <childDevID>
          <!--ro, opt, string-->1
        </childDevID>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>
```

drackum@gmail.com
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14.6.1.8 Set the face capture rule parameters of a device

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceRule

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, face capture rule capability, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--req, enum, whether to enable the function, subType:string, desc:“true”, “false”-->true
  </enabled>
  <FaceDetectionRegion>
    <!--opt, object, detection area-->
    <RegionCoordinatesList>
      <!--opt, array, list of region coordinates, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--req, object, area point coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
```

```
    </RegionCoordinatesList>
  </FaceDetectionRegion>
  <InterPupileRegion>
    <!--opt, object, rectangle frame of minimum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--req, int, pupil distance, range:[0,1000]-->50
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->50
    </height>
  </InterPupileRegion>
  <MaxInterPupilRegion>
    <!--opt, object, the maximum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--req, int, pupil distance, range:[0,1000]-->50
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->50
    </height>
  </MaxInterPupilRegion>
  <SizeFilter>
    <!--opt, object, size filter-->
    <enabled>
      <!--req, bool, whether to enable size filter-->true
    </enabled>
    <ratio>
      <!--opt, float, ratio of head-shoulder size filter to minimum pupil distance, range:[4.0,6.0]-->5.0
    </ratio>
    <MaxTargetSize>
      <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000], desc:X-coordinate-->0
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </MaxTargetSize>
    <MinTargetSize>
      <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </MinTargetSize>
  </SizeFilter>
  <detectionType>
    <!--opt, string, face type for detection, desc:"abnormalFace" (abnormal face),"normalFace" (normal face),"multiFace" (multiple faces),"sunglassesFace"
(face,on which wearing sunglasses),"callFace" (face of a person who is making a phone call). Multiple values are separated by commas--
>abnormalFace,normalFace
  </detectionType>
  <autoROIEnable>
    <!--opt, bool, whether to enable face automatic ROI, desc:the value of this node is false by default-->false
  </autoROIEnable>
</FaceRule>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.1.9 Get the face capture rule parameters of a device

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule?defualt=<defualt>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
defualt	string	Get default parameters via FaceRule.defualt in /ISAPI/Intelligent/channels//faceRule/capabilities.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, face capture rule capability, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--ro, req, enum, whether to enable, subType:string, desc:“true”, “false”-->true
  </enabled>
  <FaceDetectionRegion>
    <!--ro, opt, object, detection area-->
    <RegionCoordinatesList>
      <!--ro, opt, array, list of region coordinates, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, req, object, area point coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </FaceDetectionRegion>
  <InterPupileRegion>
    <!--ro, opt, object, rectangle frame of minimum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--ro, req, int, pupillary distance, range:[0,1000]-->50
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->50
    </height>
  </InterPupileRegion>
  <MaxInterPupilRegion>
    <!--ro, opt, object, the maximum pupillary distance, desc:The origin is the lower-left corner of the screen.-->
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,1000]-->50
```

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```
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
      </positionY>
      <width>
        <!--ro, req, int, pupillary distance, range:[0,1000]-->50
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->50
      </height>
    </MaxInterPupilRegion>
    <SizeFilter>
      <!--ro, opt, object, size filter-->
      <enabled>
        <!--ro, req, bool, whether to enable size filter-->true
      </enabled>
      <ratio>
        <!--ro, opt, float, ratio of head-shoulder size filter to minimum pupil distance, range:[4.0,6.0]-->5.0
      </ratio>
      <MaxTargetSize>
        <!--ro, opt, object, target frame on the largest face, desc:The origin is the lower-left corner of the screen.-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000], desc:Value range: [0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
        <width>
          <!--ro, req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--ro, req, int, height, range:[0,1000]-->0
        </height>
      </MaxTargetSize>
      <MinTargetSize>
        <!--ro, opt, object, target frame on the smallest face, desc:The origin is the lower-left corner of the screen.-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
        <width>
          <!--ro, req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--ro, req, int, height, range:[0,1000]-->0
        </height>
      </MinTargetSize>
    </SizeFilter>
    <detectionType>
      <!--ro, opt, string, face type for detection, desc:abnormalFace (abnormal face), normalFace (normal face), multiFace (multiple faces), sunglassesFace
(face with sunglasses), callFace (face of a person who is making a phone call, which is used to detect faces in ATM mode). This node supports to be set to
multiple values and they should be separated by commas.-->abnormalFace,normalFace
    </detectionType>
    <autoROIEnable>
      <!--ro, opt, bool, whether to enable automatic face ROI, desc:By default it is false.-->false
    </autoROIEnable>
    <minInterPupileDistance>
      <!--ro, opt, int, range:[13,666], step:1, unit:px-->58
    </minInterPupileDistance>
  </FaceRule>
```

crackum@gmail.com
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14.6.1.10 Clear the face picture library

Request URL

DELETE /ISAPI/Intelligent/FDLib/<FDID>/picture?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
identityKey	string	--
FDType	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.1.11 Face capture and recognition event

EventType:faceCapture

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded*/
    "eventType":  "faceCapture",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)*/
    "eventDescription":  "faceCapture",
    /*ro, req, string, event description, desc:face capture data upload*/
    "channelName":  "ABC",
    /*ro, req, string, channel name (camera name)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID (PUID)*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished*/
    "faceCapture": [
    /*ro, opt, array, face capture and recognition results, one result can contain multiple pictures and one picture can contains multiple faces,
subType:object*/
        {
            "errorCode":  1,
            /*ro, opt, int, error code*/
            "errorMsg":  "ok",
            /*ro, opt, string, error code description*/
            "taskid":  "temp_dfasdffsadfdsadada",
            /*ro, opt, string, task ID*/
            "faceTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, capture time*/
            "traceUuid":  "sddddddddff",
            /*ro, opt, string, target ID, the same person in multiple capture results corresponds to same target ID*/
            "traceIdx":  1,
            /*ro, opt, int, No. of picture with highest comparison similarity*/
            "targetAttrs": {
            /*ro, opt, object, face attributes*/
                "deviceId":  "ee3c4000-a22d-11b4-8334-5803fb77894c",
                /*ro, opt, string, device UUID*/
                "deviceChannel":  1,
                /*ro, opt, int, device channel No.*/
                "deviceName":  "shebei",
                /*ro, opt, string, device name*/
                "faceTime":  "2004-05-03T17:30:08+08:00",
                /*ro, opt, datetime, capture time*/
                "bkgUrl":  "test",
                /*ro, opt, string, background picture URL*/
                "thermalURL":  "test",
                /*ro, opt, string, thermal picture URL*/
                "contentID":  "backgroundImage",
                /*ro, opt, string, background picture (binary)*/
```

drackum@gmail.com
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```
                "pId":  "F4F665D6A18E41308CE9934DCDDD1111"
                /*ro, opt, string*/
            },
            "faces": [
            /*ro, opt, array, face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on,
subType:object*/
                {
                    "faceId":  1,
                    /*ro, opt, int, face ID*/
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "age": {
                    /*ro, opt, object, age*/
                        "range":  5,
                        /*ro, opt, int, age deviation*/
                        "value":  23,
                        /*ro, opt, int, age*/
                        "ageGroup":  "old"
                        /*ro, opt, string, age group*/
                    },
                    "gender": {
                    /*ro, opt, object, gender*/
                        "confidence":  99.9999,
                        /*ro, opt, float, confidence*/
                        "value":  "male"
                        /*ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                    },
                    "glass": {
                    /*ro, opt, object, whether the person wears glasses*/
                        "confidence":  99.945,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes"*/
                    },
                    "mask": {
                    /*ro, opt, object, whether the person wears a mask*/
                        "confidence":  99.945,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "yes"*/
                    },
                    "smile": {
                    /*ro, opt, object, whether the person is smiling*/
                        "confidence":  99.945,
                        /*ro, opt, float, confidence*/
                        "value":  "no"
                        /*ro, opt, enum, whether the person is smiling, subType:string, desc:"no", "yes"*/
                    },
                    "faceExpression": {
                    /*ro, opt, object, facial expression*/
                        "value":  "unknown"
                        /*ro, opt, enum, facial expression, subType:string, desc:"unknown", "surprised", "panic", "disgusted", "happy", "sad", "angry",
"poker-faced"*/
                    },
                    "hat": {
                    /*ro, opt, object, whether the person wears a hat*/
                        "value":  "no"
                        /*ro, opt, enum, whether the person wears a hat, subType:string, desc:"no", "yes", "unknown"*/
                    },
                    "URL":  "xxxxxxx",
                    /*ro, opt, string, face picture thumbnail URL*/
                    "contentID":  "faceImage",
                    /*ro, opt, string, face picture thumbnail (binary)*/
                    "stayDuration":  20,
                    /*ro, opt, int, dwell time duration in the image, unit:ms*/
                    "faceScore":  30,
                    /*ro, req, int, face score (face picture quality), range:[0,100]*/
                    "captureEndMark":  true,
                    /*ro, req, bool, end flag of capture, desc:true (ended), false (not ended)*/
                    "targetID":  "xxxxxxx",
                    /*ro, opt, string, face and human body linkage ID*/
                    "FacePictureRect": {
                    /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, req, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, req, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "blockingState":  "noBlocking",
                    /*ro, opt, enum, target face shielding status, subType:string, desc:"noBlocking" (unshielded), "instantaneousInsignificantBlocking"
(lightly shielded), "continuousInsignificantBlocking" (keep shielding), "significantBlocking" (seriously shielded), "OSD", "unknown"*/
```


drackun@gmail.com
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```
(highly shielded), "continuousInsignificantBlocking" (keep shielding), "significantBlocking" (seriously shielded), "OSD", "unknown"*/
                    "swingAngle": -90,
                    /*ro, opt, int, panning angle, range:[-90,90]*/
                    "tiltAngle": -90,
                    /*ro, opt, int, tilting angle, range:[-90,90]*/
                    "pupilDistance": 10,
                    /*ro, opt, int, pupil distance of target face, desc:the maximum value is 10 bytes, and the maximum value is (resolution width/1.6),
unit: pixel*/
                    "livenessDetectionStatus": "liveFace",
                    /*ro, opt, enum, face anti-spoofing detection status, subType:string, desc:"liveFace" (live face), "notLiveFace" (the face is not real),
"detectionFailed" (detection failed), "notEnabled" (disabled).*/
                    "pId": "F4F665D6A18E41308CE9934DCDDD2222",
                    /*ro, opt, string*/
                    "enterTime": "2004-05-03T17:30:08+08:00",
                    /*ro, opt, datetime*/
                    "exitTime": "2004-05-03T17:30:08+08:00",
                    /*ro, opt, datetime*/
                    "perClassify": 0,
                    /*ro, opt, int, face data presorting*/
                    "faceSnapThermometryEnabled": true,
                    /*ro, opt, bool, whether to enable uploading captured face picture with temperature information*/
                    "currTemperature": 36.8,
                    /*ro, opt, float, face temperature*/
                    "isAbnomalTemperature": true,
                    /*ro, opt, bool, whether the temperature is abnormal or not: true, false*/
                    "thermometryUnit": "celsius",
                    /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                    "alarmTemperature": 36.7,
                    /*ro, opt, float, temperature threshold to trigger alarm*/
                    "interiorPosition": "driver"
                    /*ro, opt, enum, subType:string*/
                }
            ],
            "uid": "20190520103922026O0Kb4s3iQPzLignc1sRUbFT0hR039Y0YzuLt0VLfl99yZi",
            /*ro, opt, string, the unique ID of event uploading*/
            "OperatorInfo": {
            /*ro, opt, object, operator information, desc:operator information*/
                "personUUID": "test",
                /*ro, opt, string, range:[1,32]*/
                "personName": "test"
                /*ro, opt, string, range:[1,64]*/
            },
            "MEFData": [
            /*ro, opt, array, subType:object*/
                {
                    "time": "2004-05-03T17:30:08.000+08:00",
                    /*ro, req, datetime, it should be accurate to the millisecond, desc:it should be accurate to the millisecond*/
                    "IMSI": "460001357924680",
                    /*ro, req, string, range:[1,15]*/
                    "bandIdx": 11
                    /*ro, opt, enum, subType:int*/
                }
            ],
            "isNeedVerification": true
            /*ro, opt, bool*/
        }
    ],
    "GPS": {
    /*ro, opt, object*/
        "divisionEW": "E",
        /*ro, req, enum, "E" (Eastern Hemisphere), W" (West Hemisphere), subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)*/
        "longitude": 100,
        /*ro, req, int, longitude, desc:longitude*/
        "divisionNS": "N",
        /*ro, req, enum, "S" (Southern Hemisphere), "N" (North Hemisphere), subType:string, desc:"S" (Southern Hemisphere), "N" (North Hemisphere)*/
        "latitude": 100
        /*ro, req, int, latitude, desc:latitude*/
    },
    "URLCertificationType": "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (no authentication, it is used for the cloud protocol). "digest" (digest
authentication, it is used for local picture URL returned by NVR or DVR)*/
}
```


dracku m@gmail.com
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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceCapture	[Message content]	application/json	--	--	--
..	[Binary picture data]	image/jpeg	backgroundImage	backgroundImage.jpeg	--
..	[Binary picture data]	image/jpeg	faceImage	faceImage.jpeg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

14.6.2 人脸建模比对

14.6.2.1 Get the alarm information of captured pictures

Request URL

GET /ISAPI/Intelligent/FDLib/alarmInfo?format=json&listType=<listType>

Query Parameter

Parameter Name	Parameter Type	Description
listType	enum	--

Request Message

None

Response Message

```
{
    "alarmInfo": [
    /*ro, opt, array, supports alarm recognition results output of (multiple/single) person(s), subType:object, desc:supports alarm recognition results
output of (multiple/single) person(s); select the capture of maximum similarity*/
        {
            "errorCode":  1,
            /*ro, opt, int, error code*/
            "errorMsg":  "ok",
            /*ro, opt, string, error description*/
            "image":  "http",
            /*ro, opt, string, the picture URL when submitting the task*/
            "traceUuid":  "sddddddddff",
            /*ro, opt, string, trace*/
            "traceIdx":  1,
            /*ro, opt, int, picture index*/
            "targetAttrs": {
            /*ro, opt, object, target attributes*/
                "deviceId":  "1",
                /*ro, opt, string, device ID*/
                "monitorId":  "1",
                /*ro, opt, string, stream ID (encoding stream ID in CVR). This node will not be returned if the device does not support this function*/
                "deviceChannel":  1,
                /*ro, opt, int, device Channel No.*/
                "deviceName":  "shebei",
                /*ro, opt, string, device name*/
                "faceTime":  "2016-09-10 00:00:00.000",
                /*ro, opt, string, time of face picture comparison*/
                "rect": {
                /*ro, opt, object, background picture frame information of face, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "bkgUrl":  "http"
                /*ro, opt, string, picture URL*/
            },
            "faces": [
```

drackum@gmail.com DETEKTOR

```
"faces": [
/*ro, opt, array, face, subType:object*/
    {
        "faceId": 1,
        /*ro, opt, int, face ID*/
        "eyesPosition": {
        /*ro, opt, object, eyesPosition*/
            "leftEye": {
            /*ro, opt, object, left eye, desc:the origin is the upper-left corner of the screen*/
                "x": 0.000,
                /*ro, opt, float, x-coordinate of top-left corner, range:[0.000,1.000]*/
                "y": 0.000
                /*ro, opt, float, y-coordinate of top-left corner, range:[0.000,1.000]*/
            },
            "rightEye": {
            /*ro, opt, object, right eye, desc:the origin is the upper-left corner of the screen*/
                "x": 0.000,
                /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                "y": 0.000
                /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
            }
        },
        "faceRect": {
        /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
            "height": 1.000,
            /*ro, opt, float, height, range:[0.000,1.000]*/
            "width": 1.000,
            /*ro, opt, float, width, range:[0.000,1.000]*/
            "x": 0.000,
            /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
            "y": 0.000
            /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
        },
        "facePose": {
        /*ro, opt, object, submit the task*/
            "pitch": 0.9980,
            /*ro, opt, float, face pitch angle, face pitching up is regarded as positive*/
            "roll": 0.6230,
            /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive*/
            "yaw": 0.8760
            /*ro, opt, float, face yaw angle, face yawing left is regarded as positive*/
        },
        "age": {
        /*ro, opt, object, age*/
            "range": 5,
            /*ro, opt, int, age deviation*/
            "value": 23,
            /*ro, opt, int, age*/
            "ageGroup": "old"
            /*ro, opt, enum, child, subType:string, desc:"unknown", "child", "young", "middle", "old"*/
        },
        "gender": {
        /*ro, opt, object, gender*/
            "confidence": 99.99990,
            /*ro, opt, float, confidence*/
            "value": "male "
            /*ro, opt, string, gender*/
        },
        "glass": {
        /*ro, opt, object, whether the person wears glasses*/
            "confidence": 99.94499999999999,
            /*ro, opt, float, confidence*/
            "value": "yes"
            /*ro, opt, string, whether the person wears glasses*/
        },
        "smile": {
        /*ro, opt, object, whether the person is smiling*/
            "confidence": 99.99990,
            /*ro, opt, float, confidence*/
            "value": "yes"
            /*ro, opt, string, whether the person is smiling*/
        },
        "identify": [
        /*ro, opt, array, recognition result of the group, subType:object*/
            {
                "relationId": 9887,
                /*ro, opt, int, linkage relationship of arming*/
                "maxsimilarity": 0.780,
                /*ro, opt, float, maximum similarity*/
                "similarityRange": 1,
                /*ro, opt, int, similarity range*/
                "candidate": [
                /*ro, opt, array, candidate, subType:object*/
                    {
                        "alarmId": 1,
                        /*ro, opt, int, alarm ID*/
                        "blacklist_id": "2323",
                        /*ro, opt, string, alarm bloccklist library ID*/
                        "human_data": [
                        /*ro, opt, array, human body data, subType:object*/
                            {
                                "bkg_picurl": "http",
                                /*ro, opt, string, Background picture URL.*/
                                "face_id": "123445",
```


hirackum@gmail.com
DETEKTOR

```
                                        /*ro, opt, string, face ID*/
                                        "face_picurl":  "http",
                                        /*ro, opt, string, face thumbnail URL*/
                                        "face_rect": {
                                        /*ro, opt, object, front face, desc:the origin is the upper-left corner of the screen*/
                                            "height":  1.000,
                                            /*ro, opt, float, height, range:[0.000,1.000]*/
                                            "width":  1.000,
                                            /*ro, opt, float, width, range:[0.000,1.000]*/
                                            "x":  0.000,
                                            /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                                            "y":  0.000
                                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                                        },
                                        "similarity":  0.780
                                        /*ro, opt, float, person*/
                                    }
                                ],
                                "human_id":  "123",
                                /*ro, opt, string, blocklist person ID*/
                                "reserve_field": {
                                /*ro, opt, object, reserved area*/
                                    "name":  "张三",
                                    /*ro, opt, string, name*/
                                    "gender":  "male",
                                    /*ro, opt, string, gender*/
                                    "bornTime":  "2004-05-03",
                                    /*ro, opt, string, date of birth*/
                                    "city":  "130100",
                                    /*ro, opt, string, city*/
                                    "certificateType":  "officerID",
                                    /*ro, opt, string, certificate type*/
                                    "certificateNumber":  "test",
                                    /*ro, opt, string, certificate No.*/
                                    "caseInfo":  "0",
                                    /*ro, opt, string, case information*/
                                    "tag":  "aa,bb,cc"
                                    /*ro, opt, string, custom tag*/
                                },
                                "similarity":  0.780,
                                /*ro, opt, float, the maximum similarity of a same person in all pictures*/
                                "listType":  "blackList"
                                /*ro, opt, enum, list type, subType:string, desc:the maximum length is 20; "blackList" (blocklist), "whiteList"
(allowlist)*/
                            }
                        ]
                    }
                ]
            }
        ],
        "alarmRecordID":  "test",
        /*ro, opt, string, alarm record ID*/
        "alarmAckState":  "acknowledged",
        /*ro, opt, enum, alarm acknowledgment status, subType:string, desc:"acknowledged", "pending" (to be acknowledged. This value can be used when
the user cannot acknowledge the alarm when it is triggered but wants to prioritize acknowledging it later), "notAcknowledged" (not acknowledgment), "all"*/
        "alarmCustomInfo":  "test"
        /*ro, opt, string, custom information*/
    }
]
}
```

darackum@gmail.com DETEKTOR

14.6.3 人脸比对

14.6.3.1 Get the arming schedule of all face comparison channels

Request URL

GET /ISAPI/Event/schedules/faceContrast

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, arming schedule list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, req, object, arming schedule-->
    <id>
      <!--ro, req, string-->faceContrast-1
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->faceContrast
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, arming schedule list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedule-->
        <dayOfWeek>
          <!--ro, opt, enum, day of the week, subType:int, desc:1: Monday; 2: Tuesday; 3: Wednesday; 4: Thursday; 5: Friday; 6: Saturday; 7: Sunday;-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time period-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, holiday arming schedule, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming schedule-->
        <TimeRange>
          <!--ro, req, object, time period-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceContrastScheduleList>
```

drackum@gmail.com
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14.6.3.2 Set the arming schedule of multiple face picture comparison alarms

Request URL

PUT /ISAPI/Event/schedules/faceContrast

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, schedule information list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--req, object, schedule information-->
    <id>
      <!--req, string, index, desc:the format is event type-channel No.-->faceContrast-1
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->faceContrast
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming time list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceContrastScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.3.3 Get the arming schedule of face picture comparison for a specified channel

Request URL

GET /ISAPI/Event/schedules/faceContrast/faceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrast-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->faceContrast
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

drackum@gmail.com
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14.6.3.4 Set the arming schedule of face picture comparison for a specified channel

Request URL

PUT /ISAPI/Event/schedules/faceContrast/faceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceContrast-1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->faceContrast
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.6.3.5 Set the linkage parameters of face picture comparison

Request URL

PUT /ISAPI/Event/triggers/faceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceContrast-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->faceContrast
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.3.6 Get the linkage parameters of face picture comparison

Request URL

GET /ISAPI/Event/triggers/faceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrast-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->faceContrast
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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14.6.3.7 Set the linkage parameters of event when face picture comparison failed

Request URL

PUT /ISAPI/Event/triggers/faceContrastFailure-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceContrastFailure-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->faceContrastFailure
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.3.8 Get the linkage parameters of event when face picture comparison failed

Request URL

GET /ISAPI/Event/triggers/faceContrastFailure-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrastFailure-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->faceContrastFailure
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoLoss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
```

dracnum@gmail.com
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```
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <AudioAction>
        <!--ro, opt, object, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,audio}-->
        <audioID>
          <!--ro, opt, int-->1
        </audioID>
        <alarmTimes>
          <!--ro, opt, int, range:[0,255]-->1
        </alarmTimes>
        <childDevID>
          <!--ro, opt, string, range:[9,32]-->3434645611
        </childDevID>
        <audioContent>
          <!--ro, opt, string, range:[0,256]-->test
        </audioContent>
      </AudioAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

14.6.3.9 Get the linkage parameters of event when face picture comparison completed

Request URL

GET /ISAPI/Event/triggers/faceContrastSuccess-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrastSuccess-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->faceContrastSuccess
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
```

drackum@gmail.com
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```
    <dynVideoInputChannelID>
      <!--ro, opt, string-->1
    </dynVideoInputChannelID>
    <intervalBetweenEvents>
      <!--ro, opt, int, unit:s-->1
    </intervalBetweenEvents>
    <WLSensorID>
      <!--ro, opt, string-->1
    </WLSensorID>
    <EventTriggerNotificationList>
      <!--ro, opt, array, subType:object-->
      <EventTriggerNotification>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->test
        </id>
        <notificationMethod>
          <!--ro, req, enum, subType:string-->FTP
        </notificationMethod>
        <notificationRecurrence>
          <!--ro, opt, enum, subType:string-->beginning
        </notificationRecurrence>
        <notificationInterval>
          <!--ro, opt, int, unit:ms-->1
        </notificationInterval>
        <videoInputID>
          <!--ro, opt, string-->1
        </videoInputID>
        <dynVideoInputID>
          <!--ro, opt, string-->1
        </dynVideoInputID>
        <ptzAction>
          <!--ro, opt, object-->
          <ptzChannelID>
            <!--ro, req, string-->1
          </ptzChannelID>
          <actionName>
            <!--ro, req, enum, subType:string-->preset
          </actionName>
          <actionNum>
            <!--ro, opt, int-->1
          </actionNum>
        </ptzAction>
        <WhiteLightAction>
          <!--ro, opt, object-->
          <whiteLightDurationTime>
            <!--ro, opt, int, range:[1,60], unit:s-->1
          </whiteLightDurationTime>
        </WhiteLightAction>
        <AudioAction>
          <!--ro, opt, object, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,audio}-->
          <audioID>
            <!--ro, opt, int-->1
          </audioID>
          <alarmTimes>
            <!--ro, opt, int, range:[0,255]-->1
          </alarmTimes>
          <childDevID>
            <!--ro, opt, string, range:[9,32]-->3434645611
          </childDevID>
          <audioContent>
            <!--ro, opt, string, range:[0,256]-->test
          </audioContent>
        </AudioAction>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>
```

drackum@gmail.com
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14.6.3.10 Set the linkage parameters of event when face picture comparison completed

Request URL

PUT /ISAPI/Event/triggers/faceContrastSuccess-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceContrastSuccess-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->faceContrastSuccess
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.3.11 Set the face picture comparison parameters

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceContrast

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FaceContrast>
    <!--opt, object-->
    <id>
      <!--req, int-->1
    </id>
    <enable>
      <!--opt, bool-->true
    </enable>
    <faceContrastType>
      <!--req, enum, face picture comparison type, subType:string-->faceContrast
    </faceContrastType>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->1
    </thresholdValue>
    <FDLibList>
      <!--opt, array, subType:object-->
      <FDLib>
        <!--opt, object, face picture library-->
        <id>
          <!--req, int-->1
        </id>
        <FDID>
          <!--req, string, a unique ID of face picture library-->test
        </FDID>
        <thresholdValue>
          <!--opt, int, range:[0,100]-->1
        </thresholdValue>
        <customFaceLibID>
          <!--opt, string, custom face picture library ID-->test
        </customFaceLibID>
      </FDLib>
    </FDLibList>
    <faceSnapDataUpload>
      <!--opt, bool, upload captured face image, desc:true by default-->true
    </faceSnapDataUpload>
    <AttendanceSaveEnable>
      <!--opt, bool, desc:true by default-->true
    </AttendanceSaveEnable>
    <contrastFailureAlarmUpload>
      <!--opt, bool, desc:true by default-->true
    </contrastFailureAlarmUpload>
    <alarmOutType>
      <!--opt, enum, subType:string-->off
    </alarmOutType>
    <plusesTime>
      <!--opt, int, unit:ms-->1
    </plusesTime>
    <contrastSuccessAlarmUpload>
      <!--opt, bool-->true
    </contrastSuccessAlarmUpload>
```

drackum@gmail.com
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```
<LivenessDetection>
  <!--opt, object-->
  <enable>
    <!--req, bool-->true
  </enable>
</LivenessDetection>
<voiceEnabled>
  <!--opt, bool-->true
</voiceEnabled>
<faceSnapDataSave>
  <!--opt, bool-->true
</faceSnapDataSave>
<QuickContrast>
  <!--opt, object-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <snapTime>
    <!--req, float, unit:s-->0.000
  </snapTime>
  <threshold>
    <!--opt, int-->1
  </threshold>
  <quickConfigMode>
    <!--opt, enum, subType:string-->custom
  </quickConfigMode>
  <Custom>
    <!--opt, object-->
    <timeOutMode>
      <!--opt, enum, subType:string-->infinite
    </timeOutMode>
    <timeOut>
      <!--opt, int, range:[1,99], unit:s-->1
    </timeOut>
    <duplicateContrastMode>
      <!--opt, enum, subType:string-->success
    </duplicateContrastMode>
    <duplicateContrastTimes>
      <!--opt, int, range:[1,99]-->1
    </duplicateContrastTimes>
  </Custom>
</QuickContrast>
<mixedTargetDetectionWithFaceContrast>
  <!--opt, bool-->true
</mixedTargetDetectionWithFaceContrast>
<brokenNetHttp>
  <!--opt, bool-->true
</brokenNetHttp>
<relationId>
  <!--opt, string, arming ID-->test
</relationId>
<alarmStorageEnable>
  <!--opt, bool-->true
</alarmStorageEnable>
<realTimeContrast>
  <!--opt, bool-->true
</realTimeContrast>
<analysisTime>
  <!--opt, datetime-->1970-01-01T00:00:00+08:00
</analysisTime>
<contrastSuccessAlarmout>
  <!--opt, bool, desc:true by default-->true
</contrastSuccessAlarmout>
<contrastFailureAlarmout>
  <!--opt, bool, desc:true by default-->true
</contrastFailureAlarmout>
<removeDuplicateEnabled>
  <!--opt, bool-->true
</removeDuplicateEnabled>
<removeDuplicateTime>
  <!--opt, float, unit:s-->0.000
</removeDuplicateTime>
<repetitionTime>
  <!--opt, int-->1
</repetitionTime>
<topOverlayNameEnabled>
  <!--opt, bool-->true
</topOverlayNameEnabled>
<personListShowEnabled>
  <!--opt, bool-->true
</personListShowEnabled>
<refreshListTime>
  <!--opt, int, unit:s-->1
</refreshListTime>
<personCountShowEnabled>
  <!--opt, bool-->true
</personCountShowEnabled>
<FaceROI>
  <!--opt, object-->
  <enabled>
    <!--opt, bool-->true
  </enabled>
  <level>
```

drackum@gmail.com
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```
        <!--opt, int-->1
      </level>
    </FaceROI>
    <uploadResultCount>
      <!--opt, int-->1
    </uploadResultCount>
    <normalTemperatureAlarmout>
      <!--opt, bool-->true
    </normalTemperatureAlarmout>
    <abnormalTemperatureAlarmout>
      <!--opt, bool-->true
    </abnormalTemperatureAlarmout>
    <matchType>
      <!--opt, enum, subType:int-->0
    </matchType>
    <alarmSourceType>
      <!--opt, enum, subType:string-->timing
    </alarmSourceType>
    <timingInterval>
      <!--opt, enum, subType:int, dep:and,{$.FaceContrastList[*].FaceContrast.alarmSourceType,eq,timing}-->5
    </timingInterval>
  </FaceContrast>
</FaceContrastList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
DETEKTOR

14.6.3.12 Get the face picture comparison parameters.

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceContrast

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<FaceContrastList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FaceContrast>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int, ID-->1
    </id>
    <enable>
      <!--ro, opt, bool-->true
    </enable>
    <faceContrastType>
      <!--ro, req, enum, subType:string-->faceContrast
    </faceContrastType>
    <thresholdValue>
      <!--ro, opt, int, Detection Threshold, range:[0,100], desc:Detection Threshold-->1
    </thresholdValue>
    <FDLibList>
      <!--ro, opt, array, subType:object-->
      <FDLib>
        <!--ro, opt, object, Face Picture Library-->
        <id>
          <!--ro, req, int, ID-->1
        </id>
        <FDID>
          <!--ro, req, string-->test
        </FDID>
        <thresholdValue>
          <!--ro, opt, int, range:[0,100]-->1
        </thresholdValue>
      </FDLib>
    </FDLibList>
    <contrastFailureAlarmUpload>
      <!--ro, opt, bool, true by default, desc:true by default-->true
    </contrastFailureAlarmUpload>
  </FaceContrast>
</FaceContrastList>
```

14.6.3.13 Get the face picture comparison capability

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceContrast/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<FaceContrastCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enable>
  <faceContrastType opt="faceContrast,blacklistFaceContrast,whitelistFaceContrast">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->faceContrast
  </faceContrastType>
  <thresholdValue min="1" max="10">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
  </thresholdValue>
  <SupportFDLibList opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </SupportFDLibList>
  <plusesTime min="100" max="900">
    <!--ro, opt, int, unit:ms, attr:min{req, int},max{req, int}-->1
  </plusesTime>
  <isSupportSingleThresholdValue>
    <!--ro, opt, bool-->true
  </isSupportSingleThresholdValue>
  <isSupportFaceScore>
    <!--ro, opt, bool-->true
  </isSupportFaceScore>
  <isSupportCustomFaceLibID>
    <!--ro, opt, bool-->true
  </isSupportCustomFaceLibID>
  <FaceContrast>
    <!--ro, opt, object-->
    <linkageType opt="faceContrast,faceContrastSuccess,faceContrastFailure,faceContrastNonRealTime">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </linkageType>
  </FaceContrast>
  <supportFDLibNum min="1" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </supportFDLibNum>
  <supportFDLibType opt="ordinary,private,stranger">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->ordinary
  </supportFDLibType>
</FaceContrastCap>
```

14.6.3.14 Search for face picture comparison results of a single channel

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string-->test
  </searchID>
  <searchResultPosition>
    <!--req, int-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int-->1
  </maxResults>
  <snapStartTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </snapStartTime>
  <snapEndTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </snapEndTime>
  <faceMatchInfoEnable>
    <!--opt, bool-->true
  </faceMatchInfoEnable>
  <monitorName>
    <!--opt, string-->test
  </monitorName>
  <similarity>
    <!--opt, float, range:[0.0,100.0]-->0.0
  </similarity>
  <eventType>
    <!--opt, enum, subType:string-->unlimit
  </eventType>
```

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```
  <FaceModeList>
    <!--opt, array, subType:object-->
    <FaceMode>
      <!--opt, object-->
      <ModeInfo>
        <!--opt, object-->
        <similarity>
          <!--opt, float, range:[0.0,100.0]-->0.0
        </similarity>
        <modeData>
          <!--opt, string-->test
        </modeData>
      </ModeInfo>
    </FaceMode>
  </FaceModeList>
  <PictureInfoList>
    <!--opt, array, subType:object-->
    <PictureInfo>
      <!--opt, object-->
      <FDID>
        <!--req, string-->test
      </FDID>
      <PID>
        <!--req, string-->test
      </PID>
    </PictureInfo>
  </PictureInfoList>
  <totalSearchResults>
    <!--opt, int-->1
  </totalSearchResults>
  <FDIDList>
    <!--opt, array, subType:object-->
    <FDID>
      <!--opt, string-->test
    </FDID>
  </FDIDList>
  <sortord>
    <!--opt, enum, subType:string-->similarity
  </sortord>
  <sex>
    <!--opt, enum, subType:string-->male
  </sex>
  <glass>
    <!--opt, enum, subType:string-->yes
  </glass>
  <smile>
    <!--opt, enum, subType:string-->yes
  </smile>
  <high_risk_group>
    <!--opt, enum, subType:string-->yes
  </high_risk_group>
  <importLib>
    <!--opt, enum, subType:string-->yes
  </importLib>
  <CustomFaceLibIDList>
    <!--opt, array, subType:object-->
    <customFaceLibID>
      <!--opt, string-->test
    </customFaceLibID>
  </CustomFaceLibIDList>
</FCSearchDescription>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string-->test
  </searchID>
  <responseStatus>
    <!--ro, req, bool-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, subType:string-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int-->1
  </totalMatches>
  <progress>
    <!--ro, opt, int, range:[0,100]-->1
  </progress>
  <MatchList>
    <!--ro, opt, array, subType:object-->
    <MatchElement>
      <!--ro, opt, object-->
```

```
<snapPicURL>
  <!--ro, opt, string-->test
</snapPicURL>
<snapTime>
  <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
</snapTime>
<facePicURL>
  <!--ro, opt, string-->test
</facePicURL>
<FaceMatchInfoList>
  <!--ro, opt, array, subType:object-->
  <FaceMatchInfo>
    <!--ro, opt, object-->
    <FDID>
      <!--ro, req, string-->test
    </FDID>
    <FDname>
      <!--ro, opt, string-->test
    </FDname>
    <thresholdValue>
      <!--ro, opt, int, range:[0,100]-->1
    </thresholdValue>
    <bornTime>
      <!--ro, opt, date-->1970-01-01+08:00
    </bornTime>
    <name>
      <!--ro, opt, string-->test
    </name>
    <sex>
      <!--ro, opt, enum, subType:string-->male
    </sex>
    <province>
      <!--ro, req, string-->0
    </province>
    <city>
      <!--ro, opt, string-->11
    </city>
    <certificateType>
      <!--ro, req, enum, subType:string-->ID
    </certificateType>
    <certificateNumber>
      <!--ro, opt, string, range:[8,32]-->test
    </certificateNumber>
    <picURL>
      <!--ro, req, string-->test
    </picURL>
    <PID>
      <!--ro, req, string-->test
    </PID>
    <PersonInfoExtendList>
      <!--ro, opt, array, subType:object-->
      <PersonInfoExtend>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enable>
          <!--ro, req, bool-->true
        </enable>
        <name>
          <!--ro, opt, string-->test
        </name>
        <value>
          <!--ro, opt, string-->test
        </value>
      </PersonInfoExtend>
    </PersonInfoExtendList>
    <phoneNumber>
      <!--ro, opt, string-->test
    </phoneNumber>
    <caseInfo>
      <!--ro, opt, string-->test
    </caseInfo>
    <similarity>
      <!--ro, opt, float, range:[0.0,100.0]-->0.0
    </similarity>
    <modelStatus>
      <!--ro, opt, enum, subType:string-->modeling
    </modelStatus>
    <ModelingStatus>
      <!--ro, opt, object-->
      <FDID>
        <!--ro, opt, string-->test
      </FDID>
      <PID>
        <!--ro, opt, string-->test
      </PID>
      <name>
        <!--ro, req, string-->test
      </name>
      <status>
        <!--ro, req, enum, subType:string-->abnormal
      </status>
      <reason>
```

drackum@gmail.com
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```
            <reason>
              <!--ro, req, enum, subType:string-->analysisFailed
            </reason>
            <customHumanID>
              <!--ro, opt, string-->123456789012345678901234567890 12
            </customHumanID>
            <customFaceLibID>
              <!--ro, opt, string-->123456789012345678901234567890 12
            </customFaceLibID>
            <modeData>
              <!--ro, opt, string-->test
            </modeData>
          </ModelingStatus>
        </FaceMatchInfo>
      </FaceMatchInfoList>
      <monitorName>
        <!--ro, opt, string-->test
      </monitorName>
      <glass>
        <!--ro, req, enum, subType:string-->any
      </glass>
      <smile>
        <!--ro, req, enum, subType:string-->no
      </smile>
      <high_risk_group>
        <!--ro, opt, enum, subType:string-->yes
      </high_risk_group>
      <similarity>
        <!--ro, opt, float, range:[0.0,100.0]-->0.0
      </similarity>
      <Rect>
        <!--ro, opt, object-->
        <height>
          <!--ro, req, float, range:[0.000,1.000]-->0.000
        </height>
        <width>
          <!--ro, req, float, range:[0.000,1.000]-->0.000
        </width>
        <x>
          <!--ro, req, float, range:[0.000,1.000]-->0.000
        </x>
        <y>
          <!--ro, req, float, range:[0.000,1.000]-->0.000
        </y>
      </Rect>
      <channelID>
        <!--ro, opt, int-->1
      </channelID>
      <name>
        <!--ro, opt, string-->test
      </name>
      <sex>
        <!--ro, opt, enum, subType:string-->male
      </sex>
      <ageGroup>
        <!--ro, req, enum, subType:string-->any
      </ageGroup>
      <mask>
        <!--ro, req, enum, subType:string-->no
      </mask>
      <imageQuality>
        <!--ro, opt, int-->1
      </imageQuality>
      <faceExpression>
        <!--ro, req, enum, subType:string-->angry
      </faceExpression>
      <contrastStatus>
        <!--ro, opt, bool-->true
      </contrastStatus>
      <age>
        <!--ro, opt, int, range:[1,255]-->1
      </age>
      <targetID>
        <!--ro, opt, string, range:[1,48]-->test
      </targetID>
      <MEFData>
        <!--ro, opt, array, subType:object-->
        <MEF>
          <!--ro, opt, object-->
          <time>
            <!--ro, req, datetime-->2004-05-03T17:30:08+08:00.000
          </time>
          <IMSI>
            <!--ro, req, string, range:[1,15]-->test
          </IMSI>
        </MEF>
      </MEFData>
    </MatchElement>
  </MatchList>
</FCSearchResult>
```

drackum@gmail.com
DETEKTOR

14.6.3.15 Search for the face picture comparison alarm information

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same
one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time, desc:the maximum number of search results this time-->1
  </maxResults>
  <snapStartTime>
    <!--opt, datetime, capture start time with ISO8601 time format-->1970-01-01T00:00:00+08:00
  </snapStartTime>
  <snapEndTime>
    <!--opt, datetime, capture end time with ISO8601 time format-->1970-01-01T00:00:00+08:00
  </snapEndTime>
  <faceMatchInfoEnable>
    <!--opt, bool, whether to only upload data with matched information-->true
  </faceMatchInfoEnable>
  <monitorName>
    <!--opt, string, camera name-->test
  </monitorName>
  <similarity>
    <!--opt, float, similarity, range:[0.0,100.0]-->0.0
  </similarity>
  <eventType>
    <!--opt, enum, event type, subType:string, desc:event type-->unlimit
  </eventType>
  <FaceModeList>
    <!--opt, array, face picture information list, subType:object-->
    <FaceMode>
      <!--opt, object, face picture information-->
      <ModeInfo>
        <!--opt, object, model information-->
        <similarity>
          <!--opt, float, similarity, range:[0.0,100.0]-->0.0
        </similarity>
        <modeData>
          <!--opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method-
->test
        </modeData>
      </ModeInfo>
    </FaceMode>
  </FaceModeList>
  <PictureInfoList>
    <!--opt, array, list of the picture information, subType:object, desc:list of the picture information-->
    <PictureInfo>
      <!--opt, object, picture information-->
      <FDID>
        <!--req, string-->test
      </FDID>
      <PID>
        <!--req, string-->test
      </PID>
    </PictureInfo>
  </PictureInfoList>
  <ChannelList>
    <!--opt, array, search channel list, subType:object-->
    <Channel>
      <!--opt, object-->
      <channelID>
        <!--req, int, channel No.-->1
      </channelID>
    </Channel>
  </ChannelList>
  <totalSearchResults>
    <!--opt, int, the total number of results in a single search, desc:the total number of results in a single search-->1
  </totalSearchResults>
  <FDIDList>
    <!--opt, array, face picture list, subType:object-->
    <FDID>
      <!--opt, string, face picture library ID-->test
    </FDID>
  </FDIDList>
```

dr.ackum@gmail.com DETEKTOR

```
  <sortord>
    <!--opt, enum, subType:string-->similarity
  </sortord>
  <sex>
    <!--opt, enum, gender, subType:string, desc:gender-->male
  </sex>
  <glass>
    <!--opt, enum, whether the person wears glasses, subType:string, desc:whether the person wears glasses-->yes
  </glass>
  <smile>
    <!--opt, enum, whether the person wears a smile, subType:string, desc:whether the person wears a smile-->yes
  </smile>
  <high_risk_group>
    <!--opt, enum, subType:string-->yes
  </high_risk_group>
  <importLib>
    <!--opt, enum, subType:string-->yes
  </importLib>
  <CustomFaceLibIDList>
    <!--opt, array, custom list of face picture library ID, subType:object-->
    <customFaceLibID>
      <!--opt, string, custom face picture library ID-->test
    </customFaceLibID>
  </CustomFaceLibIDList>
  <thermometryUnit>
    <!--opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
  </thermometryUnit>
  <currTemperature>
    <!--opt, float, face temperature, range:[-20.0,105.0], unit:℃, desc:face temperature-->36.7
  </currTemperature>
  <temperatureCondition>
    <!--opt, enum, face temperature status, subType:string, desc:face temperature status-->normal
  </temperatureCondition>
  <temperatureValue>
    <!--opt, float, temperature value, range:[-20,105], unit:℃, dep:and,{$.FCSearchDescription.temperatureCondition,eq,above}-->37.2
  </temperatureValue>
  <humanIDList>
    <!--opt, array, subType:object-->
    <humanID>
      <!--req, string-->test
    </humanID>
  </humanIDList>
  <collectEnabled>
    <!--opt, enum, subType:string-->yes
  </collectEnabled>
</FCSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search results, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search result, subType:string, desc:search result-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data),"OK" (searching completed)-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <progress>
    <!--ro, opt, int, range:[0,100]-->1
  </progress>
  <MatchList>
    <!--ro, opt, array, matched results list, subType:object-->
    <MatchElement>
      <!--ro, opt, object, matched results-->
      <snapPicURL>
        <!--ro, opt, string, URL of captured background picture-->test
      </snapPicURL>
      <snapTime>
        <!--ro, opt, datetime, capture time-->1970-01-01T00:00:00+08:00
      </snapTime>
      <facePicURL>
        <!--ro, opt, string, face sub-picture URL-->test
      </facePicURL>
      <FaceMatchInfoList>
        <!--ro, opt, array, subType:object-->
        <FaceMatchInfo>
          <!--ro, opt, object-->
          <FPID>
```

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```
            <FDID>
              <!--ro, req, string-->test
            </FDID>
            <FDname>
              <!--ro, opt, string, face picture library name-->test
            </FDname>
            <thresholdValue>
              <!--ro, opt, int, detection threshold, range:[0,100], desc:detection threshold-->1
            </thresholdValue>
            <bornTime>
              <!--ro, opt, date, date of birth-->1970-01-01+08:00
            </bornTime>
            <name>
              <!--ro, opt, string, name-->test
            </name>
            <sex>
              <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
            </sex>
            <province>
              <!--ro, req, string-->0
            </province>
            <city>
              <!--ro, opt, string, city, desc:see the data dictionary for details-->11
            </city>
            <certificateType>
              <!--ro, req, enum, certificate type, subType:string, desc:certificate type-->ID
            </certificateType>
            <certificateNumber>
              <!--ro, opt, string, certificate No., range:[8,32], desc:certificate No.-->test
            </certificateNumber>
            <picURL>
              <!--ro, req, string-->test
            </picURL>
            <PID>
              <!--ro, req, string, picture ID-->test
            </PID>
            <PersonInfoExtendList>
              <!--ro, opt, array, person extended information list, subType:object-->
              <PersonInfoExtend>
                <!--ro, opt, object, person extended information-->
                <id>
                  <!--ro, req, int, serial No. of person custom extension information-->1
                </id>
                <enable>
                  <!--ro, req, bool, enable person extended information or not-->true
                </enable>
                <name>
                  <!--ro, opt, string, person tag extended information name-->test
                </name>
                <value>
                  <!--ro, opt, string, person tag extended information-->test
                </value>
              </PersonInfoExtend>
            </PersonInfoExtendList>
            <phoneNumber>
              <!--ro, opt, string, phone No.-->test
            </phoneNumber>
            <caseInfo>
              <!--ro, opt, string, case description-->test
            </caseInfo>
            <similarity>
              <!--ro, opt, float, similarity, range:[0.0,100.0]-->0.0
            </similarity>
            <modelStatus>
              <!--ro, opt, enum, subType:string-->modeling
            </modelStatus>
            <ModelingStatus>
              <!--ro, opt, object-->
              <FDID>
                <!--ro, opt, string, face picture library ID-->test
              </FDID>
              <PID>
                <!--ro, opt, string, picture ID-->test
              </PID>
              <name>
                <!--ro, req, string, picture name-->test
              </name>
              <status>
                <!--ro, req, enum, subType:string-->abnormal
              </status>
              <reason>
                <!--ro, req, enum, subType:string-->analysisFailed
              </reason>
              <customHumanID>
                <!--ro, opt, string-->123456789012345678901234567890l2
              </customHumanID>
              <customFaceLibID>
                <!--ro, opt, string, custom face picture library ID, desc:custom face picture library ID-->123456789012345678901234567890l2
              </customFaceLibID>
              <modeData>
                <!--ro, opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64
method-->test
              </modeData>
            </ModelingStatus>
```

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```
        </FaceMatchInfo>
      </FaceMatchInfoList>
      <monitorName>
        <!--ro, opt, string, camera information-->test
      </monitorName>
      <glass>
        <!--ro, req, enum, whether the person wears glasses, subType:string, desc:"any" (unlimited condition which is used in search), "no" (not wear
glasses), "sunglasses" (wears sunglasses), "unknown"-->any
      </glass>
      <smile>
        <!--ro, req, enum, whether the person wears a smile, subType:string, desc:whether the person wears a smile-->no
      </smile>
      <high_risk_group>
        <!--ro, opt, enum, subType:string-->yes
      </high_risk_group>
      <similarity>
        <!--ro, opt, float, similarity, range:[0.0,100.0], desc:similarity-->0.0
      </similarity>
      <Rect>
        <!--ro, opt, object-->
        <height>
          <!--ro, req, float, width, range:[0.000,1.000]-->0.000
        </height>
        <width>
          <!--ro, req, float, height, range:[0.000,1.000]-->0.000
        </width>
        <x>
          <!--ro, req, float, X-coordinate, range:[0.000,1.000], desc:the reference origin is the upper left corner of image-->0.000
        </x>
        <y>
          <!--ro, req, float, Y-coordinate, range:[0.000,1.000], desc:the reference origin is the upper left corner of image-->0.000
        </y>
      </Rect>
      <channelID>
        <!--ro, opt, int, channel No., desc:channel No.-->1
      </channelID>
      <name>
        <!--ro, opt, string-->test
      </name>
      <sex>
        <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
      </sex>
      <ageGroup>
        <!--ro, req, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime",
"teenager", "unknown", "young"-->any
      </ageGroup>
      <mask>
        <!--ro, req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"-->no
      </mask>
      <imageQuality>
        <!--ro, opt, int, picture quality-->1
      </imageQuality>
      <faceExpression>
        <!--ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-faced", "sad",
"surprised", "unknown"-->angry
      </faceExpression>
      <contrastStatus>
        <!--ro, opt, bool, comparison result-->true
      </contrastStatus>
      <age>
        <!--ro, opt, int, age, range:[1,255]-->1
      </age>
      <targetID>
        <!--ro, opt, string, target ID, range:[1,48]-->test
      </targetID>
      <thermometryUnit>
        <!--ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
      </thermometryUnit>
      <currTemperature>
        <!--ro, opt, float, face temperature, range:[-20.0,105.0], unit:℃, desc:face temperature-->36.7
      </currTemperature>
      <temperatureCondition>
        <!--ro, opt, enum, face temperature status, subType:string, desc:face temperature status-->normal
      </temperatureCondition>
      <temperatureValue>
        <!--ro, opt, float, temperature value, range:[-20,105], unit:℃, dep:and,{$.FCSearchResult.MatchList[*].MatchElement.temperatureCondition,eq,above}-
->37.2
      </temperatureValue>
      <MEFData>
        <!--ro, opt, array, subType:object-->
        <MEF>
          <!--ro, opt, object, electronic fence data-->
          <time>
            <!--ro, req, datetime-->2004-05-03T17:30:08+08:00.000
          </time>
          <IMSI>
            <!--ro, req, string, range:[1,15]-->test
          </IMSI>
        </MEF>
      </MEFData>
      <alarmTemperature>
        <!--ro, opt, float, alarm temperature threshold, range:[-20.0,60.0], unit:℃-->37.6
      </alarmTemperature>
      <alertEnabled>
```

drackum@gmail.com
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```
      <collectEnabled>
        <!--ro, opt, bool-->false
      </collectEnabled>
    </MatchElement>
  </MatchList>
</FCSearchResult>
```

14.6.3.16 Face picture comparison event

EventType:alarmResult

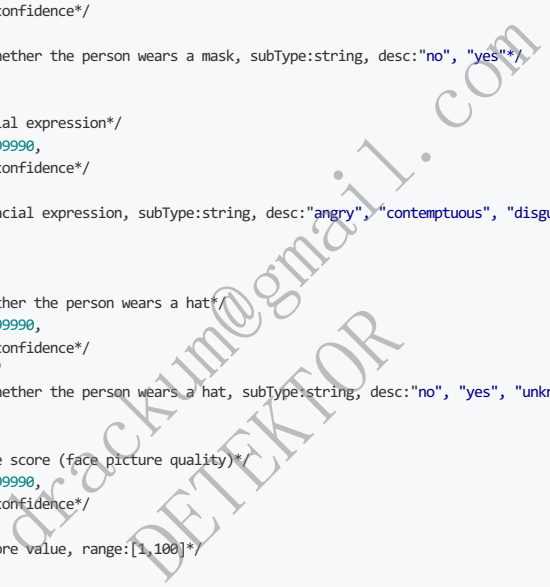
```
{
    "ipAddress": "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm, range:[1,32]*/
    "ipv6Address": "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm, range:[1,128]*/
    "portNo": 80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm, range:[1,65535]*/
    "protocol": "HTTP",
    /*ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"*/
    "macAddress": "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address, range:[1,32]*/
    "channelID": 1,
    /*ro, opt, int, channel No. of the device that triggers the alarm*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime": "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount": 1,
    /*ro, req, int, times that the same alarm has been uploaded*/
    "eventType": "alarmResult",
    /*ro, req, enum, event type, subType:string, desc:"alarmResult" (face picture comparison)*/
    "eventState": "active",
    /*ro, req, enum, continuous event status, subType:string, desc:active (valid event), inactive (invalid event)*/
    "eventDescription": "alarmResult",
    /*ro, req, string, event description, range:[1,128], desc:event description*/
    "channelName": "ABC",
    /*ro, req, string, channel name (camera name), range:[1,128]*/
    "deviceID": "test0123",
    /*ro, opt, string, device ID (PUID), range:[1,64], desc:this field must be returned when ISAPI event information is transmitted via ISUP*/
    "isDataRetransmission": true,
    /*ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be
uploaded again after the device recovers*/
    "alarmResult": [
    /*ro, opt, array, face picture comparison results, subType:object, desc:face picture comparison results*/
        {
            "errorCode": 1,
            /*ro, opt, int, when analyzing face picture failed, the corresponding error code will be returned*/
            "errorMsg": "ok",
            /*ro, opt, string, when analyzing face picture failed, the corresponding error information will be returned, desc:when analyzing face picture
failed, the corresponding error information will be returned*/
            "image": " http://10.7.52.174:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/011_163957_23006&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, opt, string, picture URL*/
            "modelData": "test",
            /*ro, opt, string, modeling data of background picture, which is encoded by Base64, range:[1,2048]*/
            "traceUuid": "sddddddddff",
            /*ro, opt, string, target ID, desc:one person corresponds to one target ID in multiple alarms (related URI:
/ISAPI/SDT/Management/Task/facePicture); the maximum size*/
            "traceIdx": 1,
            /*ro, opt, int, picture index, desc:one person can correspond to multiple pictures*/
            "targetAttrs": {
            /*ro, opt, object, target attributes*/
                "deviceId": "ee3c4000-a22d-11b4-8334-5803fb77894c",
                /*ro, opt, string, device UUID, range:[1,64]*/
                "deviceChannel": 1,
                /*ro, opt, int, device channel No.*/
                "deviceName": "shebei",
                /*ro, opt, string, device name*/
                "faceTime": "2016-09-10 00:00:00.000",
                /*ro, opt, string, capture time, range:[1,64]*/
                "rect": {
                /*ro, opt, object, background picture frame information of face, desc:the origin is the upper-left corner of the screen*/
                    "height": 1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width": 1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x": 0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y": 0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "contentID": "backgroundImage",
                /*ro, opt, string, contentID of the picture data binary data, range:[1,128], desc:the bkgURL cannot be returned when this field is
returned*/
                "bkgUrl": "http://absddsdfsdfsdk.jpg",
                /*ro, opt, string, background picture URL*/
                "pId": "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, captured background picture ID, range:[1,32], desc:captured background picture ID*/
```

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```
                "ThermalImage": {
                /*ro, opt, object, thermal imaging picture information*/
                    "resourcesContentType":  "binary",
                    /*ro, opt, string, resource transmission type, desc:"url" (transmitted via URL), "binary" (transmitted in binary format). When the
returned alarm message contains additional resource (e.g., pictures), this field should be returned*/
                    "resourcesContent":  "test"
                    /*ro, opt, string, resource ID, desc:when the returned alarm message contains additional resources (e.g., pictures), this field should
be returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of resourcesContentType
is "url", its value is the resource URL*/
                }
            },
            "faces": [
            /*ro, opt, array, face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on,
subType:object, desc:face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on*/
                {
                    "faceId":  1,
                    /*ro, opt, int, face ID, desc:the face ID cannot be duplicated in the same picture*/
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:it is used for modeling. The origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "faceMark": {
                    /*ro, opt, object, face features in the picture*/
                        "leftEye": {
                        /*ro, opt, object, left eye coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "rightEye": {
                        /*ro, opt, object, right eye's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "noseTip": {
                        /*ro, opt, object, nose tip's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "leftMouth": {
                        /*ro, opt, object, left mouth corner's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "rightMouth": {
                        /*ro, opt, object, right mouth corner's position, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        }
                    },
                    "recommendFaceRect": {
                    /*ro, opt, object, recommended face frame coordinates, desc:recommended face frame coordinates*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "facePose": {
                    /*ro, opt, object, face angle*/
                        "pitch":  0.998,
                        /*ro, opt, float, face pitch angle, face pitching up is regarded as positive, range:[-90.000,90.000], desc:this field does not
exists if it is invalid. It is accurate to three decimal places*/
                        "yaw":  0.876,
                        /*ro, opt, float, face yaw angle, face yawing left is regarded as positive, range:[-90.000,90.000], desc:this field does not exists
if it is invalid. It is accurate to three decimal places*/
                        "roll":  0.623
                        /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive, range:[-90.000,90.000], desc:this field does not
exists if it is invalid. It is accurate to three decimal places*/
                    },
                    "age": {
                    /*ro, opt, object, age*/
                        "range":  5,
                        /*ro, opt, int, age deviation; if the age deviation already exists, the age is required*/
                        "value":  23
```

drackum@gmail.com
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```
                        "value":  23,
                        /*ro, opt, int, age*/
                        "ageGroup":  "any"
                        /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old",
"prime", "teenager", "unknown", "young"*/
                    },
                    "gender": {
                    /*ro, opt, object, gender*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "male"
                        /*ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                    },
                    "glass": {
                    /*ro, opt, object, whether the person wears glasses*/
                        "confidence":  99.94499999999999,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes", "sunglasses"*/
                    },
                    "smile": {
                    /*ro, opt, object, whether the person is smiling*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person is smiling, subType:string, desc:"no", "yes"*/
                    },
                    "mask": {
                    /*ro, opt, object, whether the person wears a mask*/
                        "confidence":  99.94499999999999,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "yes"*/
                    },
                    "faceExpression": {
                    /*ro, opt, object, facial expression*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "angry"
                        /*ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-
faced", "sad", "surprised", "unknown"*/
                    },
                    "hat": {
                    /*ro, opt, object, whether the person wears a hat*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "unknown"
                        /*ro, opt, enum, whether the person wears a hat, subType:string, desc:"no", "yes", "unknown"*/
                    },
                    "score": {
                    /*ro, opt, object, face score (face picture quality)*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  65
                        /*ro, opt, int, score value, range:[1,100]*/
                    },
                    "modeldata":  "test",
                    /*ro, opt, string, face model data (encoded by base64)*/
                    "AbsoluteHigh": {
                    /*ro, opt, object, PTZ position*/
                        "elevation":  -900,
                        /*ro, opt, int, tilting angle, range:[-900,2700]*/
                        "azimuth":  22,
                        /*ro, opt, int, azimuth, range:[0,3600]*/
                        "absoluteZoom":  1
                        /*ro, opt, int, zooming ratio, range:[1,1000]*/
                    },
                    "identify": [
                    /*ro, opt, array, arming linkage information, subType:object*/
                        {
                            "relationId":  "9887",
                            /*ro, opt, string, arming linkage ID, range:[0,64], desc:arming linkage ID*/
                            "maxsimilarity":  0.780,
                            /*ro, opt, float, the maximum similarity of face comparison results, range:[0.000,1.000], step:0.001*/
                            "similarityRange":  1,
                            /*ro, opt, enum, similarity range, subType:int, desc:0 (smaller than the minimum value), 1 (between the minimum value and
maximum value), 2 (larger than the maximum value). If the similarity is smaller than the minimum value, the person will be marked*/
                            "candidate": [
                            /*ro, opt, array, matched face picture information, subType:object*/
                                {
                                    "alarmId":  1,
                                    /*ro, opt, int, alarm ID, desc:it cannot be duplicated of one arming linkage*/
                                    "blacklist_id":  "2323",
                                    /*ro, opt, string, face picture library ID, range:[1,64], desc:it corresponds to the FDID of face picture library. For
cameras, only the default FDID will be returned, the custom face picture library ID will be returned in customFaceLibID*/
                                    "human_data": [
                                    /*ro, opt, array, person information of matched face pictures, subType:object*/
                                        {
                                            "face_id":  "123445",
                                            /*ro, opt, string, face ID, range:[1,32]*/
                                            "face_picurl":  "http://127.0.0.1:6120/face_picurl",
                                            /*ro, opt, string, face picture URL*/
                                            "pId":  "F4F665D6A18E41308CE9934DCDDD766F",
                                            /*ro, opt, string, face thumbnail ID, range:[1,64], desc:face thumbnail ID*/
```

```
                                        "face_rect": {
                                        /*ro, opt, object, face thumbnail coordinates, desc:the origin is the upper-left corner of the screen*/
                                            "height":  1.000,
                                            /*ro, opt, float, height, range:[0.000,1.000]*/
                                            "width":  1.000,
                                            /*ro, opt, float, width, range:[0.000,1.000]*/
                                            "x":  0.000,
                                            /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                                            "y":  0.000
                                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                                        },
                                        "similarity":  0.997000,
                                        /*ro, opt, float, face picture similarity*/
                                        "isNoSaveFDPicture":  true,
                                        /*ro, opt, bool, whether to not save the picture of face picture library, desc:true (do not save); if this node
is not returned, it indicates that saving the picture*/
                                        "contentID":  "faceLibImage",
                                        /*ro, opt, string, picture in the face picture library, desc:picture in the face picture library*/
                                        "bkg_picurl":  "http://10639103:80/FDLib?
FDID=529FB9EAC2A04041A603E18F300CD3A5&pId=F4F665D6A18E41308CE9934DCDDD766F&size=22332"
                                        /*ro, opt, string, background picture URL*/
                                    }
                                ],
                                "human_id":  "123",
                                /*ro, opt, string, person ID, desc:person ID in the list, which corresponds to FPID in the face picture library*/
                                "reserve_field": {
                                /*ro, opt, object, reserved (transmitted data)*/
                                    "name":  "张三",
                                    /*ro, req, string, name, range:[1,256]*/
                                    "gender":  "any",
                                    /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male",
"female", "unknown"*/
                                    "bornTime":  "2004-05-03",
                                    /*ro, req, string, date of birth*/
                                    "city":  "130100",
                                    /*ro, opt, string, city code of person birth, range:[1,32], desc:see the data dictionary for details*/
                                    "certificateType":  "officerID",
                                    /*ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport"
(passport), "other" (the other certificate)*/
                                    "certificateNumber":  "test",
                                    /*ro, opt, string, certificate No., range:[0,32]*/
                                    "ageGroup":  "any",
                                    /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle
age), "old", "prime", "teenager", "unknown", "young"*/
                                    "phoneNumber":  "test",
                                    /*ro, opt, string, phone number, range:[1,64]*/
                                    "cityName":  "石家庄市",
                                    /*ro, opt, string, city (corresponding to the field "city"), range:[0,32], desc:city (corresponding to the field
"city")*/
                                    "provinceName":  "河北省"
                                    /*ro, opt, string, province, range:[0,32], desc:province*/
                                },
                                "similarity":  0.780,
                                /*ro, opt, float, range:[0.000,1.000]*/
                                "listType":  "blackList",
                                /*ro, opt, enum, list type, subType:string, desc:"blacklist" (blocklist), "whitelist" (allowlist)*/
                                "extendData": [
                                /*ro, opt, array, person extended information, subType:object*/
                                    {
                                        "extendID":  1,
                                        /*ro, opt, int, person extended information ID, desc:it starts from 1*/
                                        "enable":  true,
                                        /*ro, opt, bool, enable person extended information or not*/
                                        "name":  "test",
                                        /*ro, opt, string, name of person tag extended information, range:[1,96]*/
                                        "value":  "test"
                                        /*ro, opt, string, person tag extended information, range:[0,64]*/
                                    }
                                ],
                                "FDLibName":  "faceLib1",
                                /*ro, opt, string, face picture library name, desc:face picture library name*/
                                "FDLibThreshold":  50,
                                /*ro, opt, int, detection threshold of face picture library, range:[0,100]*/
                                "customFaceLibID":  "test",
                                /*ro, opt, string, custom face picture library ID*/
                                "customHumanID":  "test"
                                /*ro, opt, string, custom person ID*/
                            }
                        ],
                        "alarmRecordID":  "test",
                        /*ro, opt, string, alarm record ID*/
                        "libAttribute":  "general"
                        /*ro, opt, enum, library type, subType:string, desc:"general" (normal library), "blackList" (blocklist library), "VIP" (VIP
library), "passerby" (allowlist library)*/
                    }
                ],
                "contentID":  "faceImage",
                /*ro, opt, string, background picture ID, desc:background picture ID*/
                "URL":  "test",
                /*ro, opt, string, face picture thumbnail URL, range:[1,256], desc:face picture thumbnail URL*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, picture ID in face picture library, range:[1,32]*/
                "stayDuration":  20,
```


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```
                    /*ro, opt, int, dwell time duration in the image, unit:ms*/
                    "prompt": "test",
                    /*ro, opt, string, prompt contents of face picture comparison, range:[1,16]*/
                    "contrastStatus": true,
                    /*ro, opt, bool, comparison result*/
                    "realTimeContrast": true,
                    /*ro, opt, bool, whether to enable real-time face picture comparison*/
                    "remainReportNum": 0,
                    /*ro, opt, int, the remaining times of uploading the comparison alarm of the current target, desc:the remaining times of uploading the
comparison alarm of the current target*/
                    "UID": "test",
                    /*ro, opt, string*/
                    "UUPID": "test",
                    /*ro, opt, string, background picture ID of face picture library, range:[1,64], desc:it consists of device serial No., imported time,
and random digits*/
                    "frontMatchType": 1,
                    /*ro, opt, enum, face picture library type, subType:int, desc:1 (cluster library), 0 (normal library)*/
                    "isImportFD": true,
                    /*ro, opt, bool, whether to import the captured face data to the library, desc:true (yes), false (no). When face picture comparison
failed, the face picture quality meets the requirements, and is imported to the library, true will be returned. When face picture comparison failed, and the
face picture quality does not meet the requirements, false will be returned*/
                    "faceSnapThermometryEnabled": true,
                    /*ro, opt, bool, whether to enable temperature screening when capturing face pictures*/
                    "currTemperature": 36.7,
                    /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃, desc:it should be used with thermometryUnit together*/
                    "isAbnomalTemperature": true,
                    /*ro, opt, bool, whether the temperature is abnormal, desc:true (yes), false (no)*/
                    "thermometryUnit": "celsius",
                    /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                    "alarmTemperature": 36.7,
                    /*ro, opt, float, range:[-20.0,150.0], unit:℃*/
                    "facePicQuality": "low"
                    /*ro, opt, enum, subType:string*/
                }
            ],
            "alarmRecordID": "test",
            /*ro, opt, string, alarm record ID*/
            "uid": "test",
            /*ro, opt, string, the unique ID of event uploading, range:[1,64], desc:it can be composed of time (accurate to the millisecond) and random
digits*/
            "remainReportNum": 0,
            /*ro, opt, int, the remaining times of uploading the comparison alarm of the current target*/
            "algorithmInfo": {
            /*ro, opt, object, algorithm version information, desc:this field is required for secondary analysis*/
                "algorithmId": "test",
                /*ro, opt, string, algorithm package ID*/
                "algoVendorId": "test",
                /*ro, opt, string, manufacturer ID*/
                "algoVersionId": "test",
                /*ro, opt, string, algorithm version No.*/
                "algoChipType": "test",
                /*ro, opt, string, chip type*/
                "algoModelVersion": "test"
                /*ro, opt, string, model version*/
            }
        }
    ],
    "URLCertificationType": "digest",
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no"-no authentication, "digest"-digest authentication (it is used for local
picture URL returned by NVR or DVR)*/
    "GPS": {
    /*ro, opt, object, device GPS information*/
        "divisionEW": "E",
        /*ro, req, enum, "E" (Eastern Hemisphere), W" (West Hemisphere), subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)*/
        "longitude": 100,
        /*ro, req, int, longitude, desc:longitude*/
        "divisionNS": "N",
        /*ro, req, enum, subType:string*/
        "latitude": 100
        /*ro, req, int, latitude, desc:latitude*/
    }
}
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
alarmResult	[Message content]	application/json	--	--	--
..value	[Binary picture data]	image/jpeg	backgroundImage	backgroundImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
faceLibImage	[Binary picture data]	image/jpeg	faceLibImage	faceLibImage.jpg	--
...value	[Binary picture data]	image/jpeg			--
..value	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

dtackum@gmail.com
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14.6.4 人脸库管理

14.6.4.1 Delete all face picture libraries

Request URL

DELETE /ISAPI/Intelligent/FDLib

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.4.2 Import the face pictures to a specific face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/<FDID>/data?checkCode=<checkCode>&FDType=<FDType>&mode=<mode>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
checkCode	string	Verification code, and the maximum length is 16. When "security" is 1, it indicates that "checkCode" in the URL is encrypted in AES128 CBC mode; when "security" is 2, it indicates that "checkCode" in the URL is encrypted in AES256 CBC mode.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.
mode	string	When importing a face library, an error "faceLibraryNumError" occurs to prompt the user whether to overwrite. If the overwrite import mode is selected, and the face library has reached its maximum capacity, the subsequently imported faces will overwrite the earliest imported face information.

Request Message

Binary Data

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.6.4.3 Update the specified face picture of a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>/imageData?identityKey=<identityKey>&FDType=<FDType>&modeling=<modeling>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	face picture library ID
pictureID	string	picture ID
identityKey	string	The maximum length is 64. This parameter stands for the identity key of the operation platform. The device can know the operator's identity by checking this parameter in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.
modeling	enum	whether it is synchronous modeling true (synchronous modeling), false or not applied (asynchronous modeling).

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ImageData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, picture parameters, attr:version{req, string, protocolVersion}-->
  <picType>
    <!--req, enum, picture type, subType:string, desc:"binary" (binary picture in form format), "url" (picture transmitted in URL)-->binary
  </picType>
  <picUrl>
    <!--opt, picurl, picture URL-->http://127.0.0.1
  </picUrl>
  <Rect>
    <!--opt, object, target frame, desc:the origin is the upper-left corner of the screen. When there are multiple face targets in one picture, you need to
specify the modeling target-->
    <height>
      <!--req, float, height, range:[0.000,1.000]-->0.000
    </height>
    <width>
      <!--req, float, width, range:[0.000,1.000]-->0.000
    </width>
    <x>
      <!--req, float, X-coordinate, range:[0.000,1.000]-->0.000
    </x>
    <y>
      <!--req, float, Y-coordinate, range:[0.000,1.000]-->0.000
    </y>
  </Rect>
</ImageData>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
ImageData	[Message content]	application/xml	--	--	--
faceImage	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

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- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

14.6.4.4 Get the information of the person linked with the face picture.

Request URL

GET /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
pictureID	string	Picture ID
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FaceAppendData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <bornTime>
    <!--ro, opt, string-->test
  </bornTime>
  <name>
    <!--ro, req, string, range:[1,32]-->test
  </name>
  <sex>
    <!--ro, opt, enum, gender, subType:string, desc:male (male), female (female)-->male
  </sex>
  <province>
    <!--ro, opt, string-->11
  </province>
  <city>
    <!--ro, opt, string-->11
  </city>
  <certificateType>
    <!--ro, opt, enum, ID type, subType:string, desc:officerID (officer ID), ID (ID card), passportID (passport), other (other)-->officerID
  </certificateType>
  <certificateNumber>
    <!--ro, opt, string, ID No., range:[8,32], desc:When the value of certificateType is officerID, ID, or other, the size of the value of certificateNumber
can be between 8 bytes and 18 bytes. When the value of certificateType is passportID, the size of the value of certificateType can be between 8 bytes and 32
bytes.-->test
  </certificateNumber>
  <PersonInfoExtendList>
    <!--ro, opt, array, subType:object-->
    <PersonInfoExtend>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int, person extension information No., desc:The No. starts from 1 and increases by 1 subsequently.-->1
      </id>
      <enable>
        <!--ro, req, bool-->true
      </enable>
      <name>
        <!--ro, opt, string-->test
      </name>
      <value>
        <!--ro, opt, string-->test
      </value>
    </PersonInfoExtend>
  </PersonInfoExtendList>
  <phoneNumber>
    <!--ro, opt, string, range:[1,64]-->test
  </phoneNumber>
  <RegionCoordinatesList>
    <!--ro, opt, array, face target, subType:object, range:[0,4], desc:This node determines a rectangle frame to guide data modeling and processing in later
searching.-->
    <RegionCoordinates>
      <!--ro, opt, object, coordinates of the rectangle frame, desc:The origin is the upper-left corner of the screen.-->
      <positionX>
        <!--ro, req, int, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <customHumanId>
    <!--ro, opt, string, range:[1,64]-->test
  </customHumanId>
  <faceDescribe>
    <!--ro, opt, string-->test
  </faceDescribe>
  <IMSIList>
    <!--ro, opt, array, IMSI list, subType:object, desc:If the face picture library is an allowed library, it must be linked to at least one IMSI number and
up to 5 ones.-->
    <IMSI>
      <!--ro, opt, string, range:[1,15]-->123456789012345
    </IMSI>
  </IMSIList>
  <plateNo>
    <!--ro, opt, string, range:[0,32]-->test
  </plateNo>
</FaceAppendData>
```

drackum@gmail.com
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14.6.4.5 Set the face picture related information

Request URL

PUT /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
pictureID	string	--
identityKey	string	--
FDType	string	--

Request Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<FaceAppendData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, face picture related information, attr:version{req, string, protocolVersion}-->
  <bornTime>
    <!--opt, string, date of birth-->test
  </bornTime>
  <name>
    <!--req, string, name, range:[1,32]-->test
  </name>
  <sex>
    <!--opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
  </sex>
  <province>
    <!--opt, string, province, desc:see the data dictionary for details-->11
  </province>
  <city>
    <!--opt, string, city, desc:see the data dictionary for details-->11
  </city>
  <certificateType>
    <!--opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport" (passport), "other" (other
certificate)-->officerID
  </certificateType>
  <certificateNumber>
    <!--opt, string, certificate No., range:[8,32], desc:certificate No.-->test
  </certificateNumber>
  <PersonInfoExtendList>
    <!--opt, array, custom extension information of person, subType:object-->
    <PersonInfoExtend>
      <!--opt, object-->
      <id>
        <!--req, int, serial No. of person custom extension information, desc:serial No. of person custom extension information-->1
      </id>
      <enable>
        <!--req, bool-->true
      </enable>
      <name>
        <!--opt, string, name of person tag extended information-->test
      </name>
      <value>
        <!--opt, string, person tag extended information-->test
      </value>
    </PersonInfoExtend>
  </PersonInfoExtendList>
  <phoneNumber>
    <!--opt, string, range:[1,64]-->test
  </phoneNumber>
  <RegionCoordinatesList>
    <!--opt, array, face picture target, subType:object, range:[0,4], desc:face picture target-->
    <RegionCoordinates>
      <!--opt, object, coordinates, desc:the origin is the upper-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <customHumanId>
    <!--opt, string, custom person ID, range:[1,64]-->test
  </customHumanId>
  <faceDescribe>
    <!--opt, string-->test
  </faceDescribe>
  <IMSIList>
    <!--opt, array, subType:object-->
    <IMSI>
      <!--opt, string, range:[1,15]-->123456789012345
    </IMSI>
  </IMSIList>
  <plateNo>
    <!--opt, string, range:[0,32]-->test
  </plateNo>
</FaceAppendData>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.4.6 Delete the linked information of a picture in the face picture

library **Request URL**

DELETE /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
pictureID	string	Picture ID
identityKey	string	The maximum length is 64. This parameter stands for the identity key of the operation platform. The device can know the operator's identity by checking this parameter in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

crackun@gmail.com DEFENDER

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.6.4.7 Delete a face picture library

Request URL

DELETE /ISAPI/Intelligent/FDLib/<FDID>?FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

crackum@gmail.com DEFEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.6.4.8 Set the basic parameters of a face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/<FDID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID.
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for 8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, face picture library basic information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, index-->1
  </id>
  <FDID>
    <!--opt, string, face picture library ID-->test
  </FDID>
  <name>
    <!--opt, string, face picture library name-->test
  </name>
  <thresholdValue>
    <!--opt, int, detection threshold, range:[0,100]-->1
  </thresholdValue>
  <customInfo>
    <!--opt, string, custom information-->test
  </customInfo>
  <customFaceLibID>
```

dnackum@gmail.com
HIKTOR

```
    <!--opt, string, ID of custom human body picture library-->test
  </customFaceLibID>
  <faceLibType>
    <!--opt, enum, face picture library type, subType:string, desc:"ordinary" (normal library), "private" (private library), "stranger" (stranger library),
"allowable" (allowlist library), "privacyMask" (privacy mask library)-->ordinary
  </faceLibType>
  <autoUpdata>
    <!--opt, bool, whether to enable auto update of background picture in face picture library, desc:if autoUpdata in the capability set only supports
"true", the auto update feature is enabled by default; if not, it is disabled-->true
  </autoUpdata>
  <qualityThreshold>
    <!--opt, int, picture quality threshold, range:[0,100], desc:this node is valid when <autoUpdata> is "true"-->1
  </qualityThreshold>
  <importFDThreshold>
    <!--opt, int, put-in-storage similarity threshold, range:[0,100], desc:this node is valid when <autoUpdata> is "true"-->1
  </importFDThreshold>
  <Age>
    <!--opt, object, configuration capability of face picture library clustering aging parameters, desc:this node is valid when <autoUpdata> is "true"-->
    <judgementCycleType>
      <!--req, enum, evaluation cycle type of clustering data aging in face picture library, subType:string, desc:"day" (evaluate at 0 every day), "week"
(evaluate at 0 every Sunday), "month" (default; evaluate at 0 on the first day of every month), "duration" (evaluate by duration), "quarter" (evaluate at 0
on the first day of every quarter), "halfYear" (evaluate at 0 on the first day of every six months)-->day
    </judgementCycleType>
    <statisticalCycleType>
      <!--opt, enum, statistics cycle type of clustering matches in face picture library, subType:string, desc:"day" (daily), "week" (weekly), "month"
(monthly)-->day
    </statisticalCycleType>
    <statisticalCycle>
      <!--opt, int, statistics cycle range of clustering matches in face picture library, range:[0,99999], desc:this node is based on statisticalCycleType--
>1
    </statisticalCycle>
    <minFMCount>
      <!--opt, int, min. number of face picture matches within statistics cycle, range:[0,99999], desc:the default value is 30-->1
    </minFMCount>
    <FMCountEnabled>
      <!--opt, bool, whether to enable restriction on total number of clustering matches in face picture library-->true
    </FMCountEnabled>
    <FMCount>
      <!--opt, int, range of total face picture matches, range:[0,99999]-->100
    </FMCount>
    <backgroundPicThreshold>
      <!--opt, int, background picture replacement similarity threshold, range:[1,100]-->90
    </backgroundPicThreshold>
    <judgementCycle>
      <!--opt, int, evaluation cycle range of clustering data aging in face picture library, range:[1,365], unit:d, dep:and,
{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
    </judgementCycle>
    <agingExecDataTime>
      <!--opt, time, clustering library aging execution time-->00:00:00+08:00
    </agingExecDataTime>
    <automaticClearEnabled>
      <!--opt, bool, whether to enable auto clear-up when the clustering library is full-->true
    </automaticClearEnabled>
  </Age>
  <FaceCounting>
    <!--opt, object, face picture statistics-->
    <enabled>
      <!--req, bool, whether to enable face-picture people counting deduplication-->true
    </enabled>
    <uploadType>
      <!--req, enum, reporting mode after face-picture people counting deduplication is enabled, subType:string, desc:"out_repetition" (report outside
deduplication time), "noUpload" (do not report)-->out_repetition
    </uploadType>
    <channelList>
      <!--opt, object, video channel list, desc:this node is used on normal cameras accessed to NVR/CVR to specify which video channels should use the face
picture library to perform people counting-->
      <channelID>
        <!--req, int, video channel No., range:[1,256]-->true
      </channelID>
    </channelList>
  </FaceCounting>
  <faceMonitor>
    <!--opt, bool, whether it is the face picture library linked with intelligent face picture arming-->true
  </faceMonitor>
  <monitorFaceNum>
    <!--opt, bool, max. face pictures supported by face picture arming-->true
  </monitorFaceNum>
  <customNoConfig>
    <!--opt, bool, this node is invalid for Web client, desc:it is only valid in person arming mode-->true
  </customNoConfig>
</FDLibBaseCfg>
```

Response Message

dracku@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

14.6.4.9 Get the basic parameters of a face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/<FDID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID.
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, face picture library basic information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, index-->1
  </id>
  <FDID>
    <!--ro, opt, string, face picture library ID-->test
  </FDID>
  <name>
    <!--ro, opt, string, face picture library name-->test
  </name>
  <thresholdValue>
    <!--ro, opt, int, detection threshold, range:[0,100]-->1
  </thresholdValue>
  <customInfo>
    <!--ro, opt, string, custom information-->test
  </customInfo>
  <customFaceLibID>
    <!--ro, opt, string, custom face picture library ID-->test
```

dachun@gmail.com
HIKVISION

```
  </customFaceLibID>
  <faceLibType>
    <!--ro, opt, enum, face picture library type, subType:string, desc:"ordinary" (normal library), "private" (private library), "stranger" (stranger library (clustering library)), "allowable" (allowlist library), "privacyMask" (privacy mask library)-->ordinary
  </faceLibType>
  <autoUpdata>
    <!--ro, opt, bool, whether to enable auto update of background picture in face picture library, desc:if autoUpdata in the capability set only supports "true", the auto update feature is enabled by default; if not, it is disabled.-->true
  </autoUpdata>
  <qualityThreshold>
    <!--ro, opt, int, picture quality threshold, range:[0,100], desc:this node is valid when <autoUpdata> is "true"-->1
  </qualityThreshold>
  <importFDThreshold>
    <!--ro, opt, int, put-in-storage similarity threshold, range:[0,100], desc:this node is valid when <autoUpdata> is "true"-->1
  </importFDThreshold>
  <Age>
    <!--ro, opt, object, evaluation cycle, desc:this node is valid when <autoUpdata> is "true"-->
    <judgementCycleType>
      <!--ro, req, enum, evaluation cycle type, subType:string, desc:"day" (evaluate at 0 o'clock every day), "week" (evaluate at 0 o'clock every Sunday), "month" (default; evaluate at 0 o'clock on the first day of every month), "duration" (evaluate by duration), "quarter" (evaluate at 0 o'clock on the first day of every quarter), "halfYear" (evaluate at 0 o'clock on the first day of every six months)-->day
    </judgementCycleType>
    <statisticalCycleType>
      <!--ro, opt, enum, statistics cycle type of face picture matches, subType:string, desc:"day" (daily), "week" (weekly), "month" (monthly)-->day
    </statisticalCycleType>
    <statisticalCycle>
      <!--ro, opt, int, statistics cycle of face picture matches, range:[0,99999]-->1
    </statisticalCycle>
    <minFMCount>
      <!--ro, opt, int, min. number of face picture matches within statistics cycle, range:[0,99999], desc:the default value is 30-->1
    </minFMCount>
    <FMCountEnabled>
      <!--ro, opt, bool, whether to enable total number of face picture matches-->true
    </FMCountEnabled>
    <FMCount>
      <!--ro, opt, int, total number of face picture matches, range:[0,99999]-->100
    </FMCount>
    <backgroundPicThreshold>
      <!--ro, opt, int, background picture replacement similarity threshold, range:[1,100]-->90
    </backgroundPicThreshold>
    <judgementCycle>
      <!--ro, opt, int, evaluation cycle of clustering data aging in face picture library, range:[1,365], unit:d, dep:and, {$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
    </judgementCycle>
    <agingExecDataTime>
      <!--ro, opt, time, clustering library aging execution time-->00:00:00+08:00
    </agingExecDataTime>
    <automaticClearEnabled>
      <!--ro, opt, bool, whether to enable auto clear-up when the clustering library is full-->true
    </automaticClearEnabled>
  </Age>
  <FaceCounting>
    <!--ro, opt, object, face picture statistics-->
    <enabled>
      <!--ro, req, bool, whether to enable face-picture people counting deduplication-->true
    </enabled>
    <uploadType>
      <!--ro, req, enum, reporting mode after face-picture people counting deduplication is enabled, subType:string, desc:"out_repetition" (report outside deduplication time), "noUpload" (do not report)-->out_repetition
    </uploadType>
    <channelList>
      <!--ro, opt, object, video channel list, desc:this node is used on normal cameras accessed to NVR/CVR to specify which video channels should use the face picture library to perform people counting-->
      <channelID>
        <!--ro, req, int, video channel No., range:[1,256]-->true
      </channelID>
    </channelList>
  </FaceCounting>
  <faceMonitor>
    <!--ro, opt, bool, whether it is the face picture library linked with intelligent face picture arming-->true
  </faceMonitor>
  <monitorFaceNum>
    <!--ro, opt, bool, max. face pictures supported by face picture arming-->true
  </monitorFaceNum>
  <customNoConfig>
    <!--ro, opt, bool, this node is not valid for Web client, desc:it is only valid in person arming mode-->true
  </customNoConfig>
</FDLibBaseCfg>
```

drackum@gmail.com
DETEKTOR

14.6.4.10 Get face picture library capability

Request URL

GET /ISAPI/Intelligent/FDLib/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<FDLibCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, face picture library capability object, attr:version{opt, string, protocolVersion}-->
  <CreateFDLibList size="15">
    <!--ro, req, array, face picture library capability object array, subType:object, attr:size{opt, string}-->
    <CreateFDLib>
      <!--ro, req, object, face picture library capability object-->
      <id max="15" min="1">
        <!--ro, req, int, number range capability of face picture library, range:[1,15], attr:max{opt, string},min{opt, string}-->10
      </id>
      <name max="64" min="1">
        <!--ro, opt, string, name length of face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-->64
      </name>
      <faceDataMax max="150000" min="1">
        <!--ro, opt, int, the maximum number of human face base picture allowed in the face picture library, range:[1,150000], attr:max{opt,
string},min{opt, string}-->30000
      </faceDataMax>
      <customFaceLibID max="64" min="1">
        <!--ro, opt, string, the custom ID length in the face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-->64
      </customFaceLibID>
      <indentityKey max="64" min="0">
        <!--ro, opt, string, description length of the secret key in the private face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-
->64
      </indentityKey>
    </CreateFDLib>
  </CreateFDLibList>
  <FaceAppendData>
    <!--ro, opt, object, the object capability of background picture data extension of face picture library-->
    <name max="64" min="1">
      <!--ro, opt, string, the description length of person name, range:[1,64], attr:max{opt, string},min{opt, string}-->64
    </name>
    <isSupportPhoneNumber>
      <!--ro, opt, bool, whether the device supports telephone number capability-->true
    </isSupportPhoneNumber>
    <customHumanID max="128" min="1">
      <!--ro, opt, string, the capability of the person ID of custom face picture, range:[1,128], attr:max{opt, string},min{opt, string}-->64
    </customHumanID>
  </FaceAppendData>
  <FDSearchDescription>
    <!--ro, opt, object, the search capability of face picture library-->
    <searchID max="63" min="1">
      <!--ro, opt, string, search ID, attr:max{opt, string},min{opt, string}, desc:it is used to confirm the upper-level platform or system. If the platform
or the system is the same one during two searches, the search history will be saved in the memory to speed up next search; min/max will be returned
according to the device actual capability, if not returned: min=1 max=63 (default)-->63
    </searchID>
    <searchResultPosition>
      <!--ro, opt, int, start position, desc:if this node is not returned, confirm <isSupportStandardSearch> exists and the node "true" is returned; it will
start from 0 by default; if not returned, and it will start from 1 by default-->1
    </searchResultPosition>
    <maxResults>
      <!--ro, opt, int, the number of search results this time, desc:the number of search results in a single search is between 1 and 500; if exceeding the
max number of search results supported by the device, the device will return the max number of results in a single search by default; if this node is not
returned, the number of search results is between 1 and 500 by default-->30
    </maxResults>
    <FDID>
      <!--ro, opt, string, face picture library ID-->1
    </FDID>
    <name max="64" min="1">
      <!--ro, opt, string, person name, attr:max{opt, string},min{opt, string}-->64
    </name>
    <modelStatus opt="modeling,unmodeled,modelingFailed">
      <!--ro, opt, enum, the modeling status of the background picture in the face picture library, subType:string, attr:opt{opt, string}, desc:"modeling"
(modeled), "unmodeled" (not modeled), "modelingFailed" (modeling failed)-->modeling
    </modelStatus>
    <modelingStatus opt="success,failed,none">
      <!--ro, opt, enum, the modeling result of the background picture in the face picture library, subType:string, attr:opt{opt, string}, desc:"success"
(modeling succeeded), "failed" (modeling failed), "none" (not modeled)-->success
    </modelingStatus>
    <customFaceLibID max="64" min="0">
      <!--ro, opt, string, the custom ID length in the face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-->64
    </customFaceLibID>
    <indentityKey max="64" min="0">
      <!--ro, opt, string, the private key length capability of the private library, attr:max{opt, string},min{opt, string}-->64
    </indentityKey>
    <faceScore min="-1" max="100">
      <!--ro, opt, int, face score (face picture quality), range:[-1,100], attr:min{opt, string},max{opt, string}, desc:valid range: [0-100]; -1 (rating
unknown, i.e., the face picture is not rated)-->-1
    </faceScore>
  </FDSearchDescription>
  <isSuportAnalysisFace>
    <!--ro, opt, bool, whether the device supports face analysis-->true
  </isSuportAnalysisFace>
  <isSuportFCSearch opt="true,false">
    <!--ro, opt, bool, whether supports searching captured face picture library, attr:opt{opt, string}-->true
  </isSuportFCSearch>
  <isSupportFDLibFaceImport>
```

dreckum@gmail.com DEFEKTOR

```
<isSupportFDLibEachImport>
  <!--ro, opt, bool, the background picture import data of the face picture library-->true
</isSupportFDLibEachImport>
<FCSearchDescription>
  <!--ro, opt, object, the object capability of searching captured face picture library-->
  <searchID max="63" min="1">
    <!--ro, opt, string, search ID, attr:max{opt, string},min{opt, string}-->63
  </searchID>
  <searchResultPosition>
    <!--ro, opt, int, start position; this node is not returned-->1
  </searchResultPosition>
  <maxResults>
    <!--ro, opt, int, the number of search results this time-->30
  </maxResults>
  <snapStartTime>
    <!--ro, opt, datetime, ISO 8601 time-->1970-01-01T00:00:00+08:00
  </snapStartTime>
  <snapEndTime>
    <!--ro, opt, datetime, ISO 8601 time-->1970-01-01T00:00:00+08:00
  </snapEndTime>
  <eventType opt="unlimit,faceContrast,whiteFaceContrast,faceCapture,faceContrastFailure">
    <!--ro, opt, enum, type of face capture event, subType:string, attr:opt{opt, string}, desc:"unlimit", "faceContrast" (face picture comparison),
"whiteFaceContrast" (allowlist face picture comparison (strangers alarm)), "faceCapture" (face capture), "faceContrastFailure" (face picture comparison
failed)-->unlimit
  </eventType>
  <FaceModeCap>
    <!--ro, opt, object, the object capability of searching picture by picture-->
    <FaceMode>
      <!--ro, opt, object, face model-->
      <similarity max="100.0" min="0.0">
        <!--ro, opt, float, similarity, range:[0.0,100.0], attr:max{opt, string},min{opt, string}-->50.0
      </similarity>
    </FaceMode>
    <maxResultNum>
      <!--ro, opt, int, the maximum number of returned results, desc:due to its large storage capacity, the search time for face capture library can be
long; consequently, in order to improve search efficiency, the number of samples and results of the search is limited-->30
    </maxResultNum>
  </FaceModeCap>
  <isSupportMultiChannelSearch>
    <!--ro, opt, bool, whether the device supports multichannel search-->true
  </isSupportMultiChannelSearch>
  <isSupportTotalSearchResult>
    <!--ro, opt, bool, whether the device supports the device result limit for each search-->true
  </isSupportTotalSearchResult>
</FCSearchDescription>
<isSupportSurplusCapacity>
  <!--ro, opt, bool, whether to support searching for the number of remaining face pictures of specified face picture library-->true
</isSupportSurplusCapacity>
<isSupportStandardSearch>
  <!--ro, opt, bool, this node is used for the inconsistency between the implementation of some devices and documentation, desc:due to the inconsistency
between the implementation of some devices and documentation, the modification is as follows: the responseStatus returns true, capitalize the first letters
of matchList and matchElement, in order to keep consistent with the documentation; the face picture library search (/ISAPI/Intelligent/FDLib/FDSearch), the
face comparison search (/ISAPI/Intelligent/FDLib/FCSearch), the face modeling status search and searchResultPosition, all starts from 0-->true
</isSupportStandardSearch>
<isSupportFaceDataExport>
  <!--ro, opt, bool, it indicates whether to support exporting the data of face picture library-->true
</isSupportFaceDataExport>
<isSupportNewlyPictureUpload>
  <!--ro, opt, bool, whether to support importing pictures into the face picture library, desc:it is used for the device to be compatible with large-sized
picture import-->true
</isSupportNewlyPictureUpload>
<isSupportFCSearchNormal>
  <!--ro, opt, bool, whether to support searching for all face picture data, desc:it is used for searching for face picture result via ISAPI protocols; if
the search condition is not configured, by default, it will search all and get pictures via the URI: NET_DVR_SmartSearchPicture-->true
</isSupportFCSearchNormal>
<isSupportCustomHumanID>
  <!--ro, opt, bool, whether to support custom person ID for developer-->true
</isSupportCustomHumanID>
<isSupportCustomFaceLibID>
  <!--ro, opt, bool, whether to support custom face picture library ID for developer-->true
</isSupportCustomFaceLibID>
<isSupportFCSearchJsonFormat>
  <!--ro, opt, bool, whether to support searching for the face comparison alarm information in json format, desc:if the device supports, corresponding
URL: /ISAPI/Intelligent/FDLib/FCSearch/capabilities?format=json; the function supported by the device can be judged via the URL-->true
</isSupportFCSearchJsonFormat>
<SupportUploadPictureType opt="binary,url">
  <!--ro, opt, enum, the supported types of the pictures to be imported, subType:string, attr:opt{opt, string}, desc:"binary", "url"; if it is not
returned, it is binary by default-->binary
</SupportUploadPictureType>
<isSupportFaceLibFormat>
  <!--ro, opt, bool, whether to support formating face picture library, desc:corresponding URL: /ISAPI/Intelligent/FDLib/format?format=json-->true
</isSupportFaceLibFormat>
</FDLibCap>
```

14.6.4.11 Import face pictures and the additional information to a specific face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/pictureUpload?identityKey=<identityKey>&type=<type>&dataType=<dataType>&pictureSaveMode=<pictureSaveMode>

dr.ackum@gmail.com DEIKTOR

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	--
type	string	--
dataType	string	--
pictureSaveMode	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PictureUploadData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, picture information, attr:version{req, string, protocolVersion}-->
  <FDID>
    <!--req, string, a unique ID of face comparison picture library-->test
  </FDID>
  <customFaceLibID>
    <!--opt, string, custom face picture library ID, desc:custom face picture library ID-->test
  </customFaceLibID>
  <FaceAppendData>
    <!--opt, object, face picture information-->
    <bornTime>
      <!--opt, string, date of birth-->test
    </bornTime>
    <name>
      <!--req, string, name, range:[1,32]-->test
    </name>
    <sex>
      <!--opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
    </sex>
    <province>
      <!--opt, string, province, desc:see the data dictionary for details-->11
    </province>
    <city>
      <!--opt, string, city, desc:see the data dictionary for details-->11
    </city>
    <certificateType>
      <!--opt, enum, certificate type, subType:string, desc:"officerID"-officer ID,"ID"-identify card,"passportID"-passport,"other"-other certificate--
>officerID
    </certificateType>
    <certificateNumber>
      <!--opt, string, certificate No., range:[8,32], desc:certificate No.-->test
    </certificateNumber>
    <PersonInfoExtendList>
      <!--opt, array, custom extension information of person, subType:object-->
      <PersonInfoExtend>
        <!--opt, object, person information-->
        <id>
          <!--req, int, serial No. of person custom extension information, desc:serial No. of person custom extension information-->1
        </id>
        <enable>
          <!--req, bool, enable person extended information or not-->true
        </enable>
        <name>
          <!--opt, string, person tag extended information name-->test
        </name>
        <value>
          <!--opt, string, person tag extended information-->test
        </value>
      </PersonInfoExtend>
    </PersonInfoExtendList>
    <caseInfo>
      <!--opt, string, case information, range:[1,64]-->test
    </caseInfo>
    <customHumanID>
      <!--opt, string, custom person ID, range:[1,64]-->test
    </customHumanID>
    <phoneNumber>
      <!--opt, string, phone number, range:[1,64]-->test
    </phoneNumber>
    <RegionCoordinatesList>
      <!--opt, array, face picture target, subType:object, range:[0,4], desc:face picture target-->
      <RegionCoordinates>
        <!--opt, object, coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <modeldata>
```

drackun@gmail.com
DETEKTOR

```
      <!--opt, string, modeling data, desc:modeling data-->test
    </modeldata>
    <IMSIList>
      <!--opt, array, subType:object-->
      <IMSI>
        <!--opt, string, range:[1,15]-->123456789012345
      </IMSI>
    </IMSIList>
    <userType>
      <!--opt, enum, subType:string-->normal
    </userType>
    <associatedPlateType>
      <!--opt, enum, subType:string-->greenChannel
    </associatedPlateType>
  </FaceAppendData>
  <picURL>
    <!--opt, string, picture URL-->test
  </picURL>
</PictureUploadData>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
PictureUploadData	[Message content]	application/xml	--	--	--
importImage	[Binary picture data]	image/jpeg		file.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

dracku@gmail.com DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<MaskInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, information, attr:version{req, string, protocolVersion}, desc:information-->
  <PID>
    <!--ro, opt, string, picture ID, desc:picture ID-->test
  </PID>
  <picURL>
    <!--ro, opt, picurl, picture URL generated by device after importing pictures, desc:picture URL generated by device after importing pictures--
>http://127.0.0.1
  </picURL>
  <isNoSavePDPicture>
    <!--ro, opt, bool-->true
  </isNoSavePDPicture>
  <FaceGradingDetails>
    <!--ro, opt, object, face score (face picture quality), desc:face score (face picture quality)-->
    <featureConfidence>
      <!--ro, opt, float, [0,1], range:[0.000,1.000], desc:[0,1]-->0.000
    </featureConfidence>
    <faceRect>
      <!--ro, opt, enum, rectangle frame for human face, subType:int, desc:rectangle frame for human face-->1
    </faceRect>
    <pupillaryDistance>
      <!--ro, opt, int, pupil distance, desc:pupil distance-->1
    </pupillaryDistance>
    <colorConfidence>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </colorConfidence>
    <grayScale>
      <!--ro, opt, int, grayscale value, range:[0,255]-->1
    </grayScale>
    <grayMean>
      <!--ro, opt, int, range:[0,255]-->1
    </grayMean>
    <grayVariance>
      <!--ro, opt, int, range:[0,128]-->1
    </grayVariance>
    <clearityScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </clearityScore>
    <pitchAngle>
      <!--ro, opt, float, range:[-90.000,90.000]-->0.000
    </pitchAngle>
    <yawAngle>
      <!--ro, opt, float, range:[-90.000,90.000]-->0.000
    </yawAngle>
    <rollAngle>
      <!--ro, opt, float, face roll angle, range:[-90.000,90.000], desc:face roll angle-->0.000
    </rollAngle>
    <poseConfidence>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </poseConfidence>
    <frontalScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </frontalScore>
    <visibleScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </visibleScore>
    <totalScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </totalScore>
    <detectConfidence>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </detectConfidence>
    <invisibleClass>
      <!--ro, opt, array, subType:object, range:[1,8]-->
    </invisibleClass>
    <faceClass>
      <!--ro, opt, enum, face picture type, subType:int, desc:face picture type-->0
    </faceClass>
    <illuminationClass>
      <!--ro, opt, enum, subType:int-->0
    </illuminationClass>
    <leftEyeOpen>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </leftEyeOpen>
    <rightEyeOpen>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </rightEyeOpen>
    <mouthOpen>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </mouthOpen>
    <pictureStatus>
      <!--ro, opt, enum, subType:string-->normal
    </pictureStatus>
  </FaceGradingDetails>
</MaskInfo>
```

drackum@gmail.com
DETEKTOR

14.6.4.12 Set the face picture library parameters

Request URL

PUT /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	--

Request Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfgList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, face picture library information list, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibBaseCfg>
    <!--opt, object, face picture library information-->
    <id>
      <!--opt, int, range:[1,15]-->1
    </id>
    <FDID>
      <!--opt, string, unique ID of face picture library-->test
    </FDID>
    <name>
      <!--req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--opt, string, additional information of face picture library, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <faceLibType>
      <!--ro, opt, enum, face picture library type, subType:string, desc:"ordinary" (normal library), "private" (private library), "stranger" (stranger
library), "allowable" (allowlist library), "privacyMask" (privacy mask library)-->allowable
    </faceLibType>
    <autoUpdata>
      <!--opt, bool-->true
    </autoUpdata>
    <qualityThreshold>
      <!--opt, int, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--opt, int, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--opt, object-->
      <judgementCycleType>
        <!--req, enum, subType:string-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--ro, req, enum, subType:string-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--req, int, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--req, int, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--opt, bool-->true
      </FMCountEnabled>
      <FMCount>
        <!--opt, int, range:[0,99999]-->1
      </FMCount>
      <backgoundPicThreshold>
        <!--opt, int, range:[1,100]-->90
      </backgoundPicThreshold>
      <judgementCycle>
        <!--opt, int, range:[1,365], unit:d, dep:and,{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
      </judgementCycle>
    </Age>
    <FaceCounting>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <uploadType>
        <!--ro, req, enum, subType:string-->noUpload
      </uploadType>
    </FaceCounting>
    <faceMonitor>
      <!--ro, opt, bool-->true
    </faceMonitor>
    <monitorFaceNum>
      <!--ro, opt, int-->1
    </monitorFaceNum>
    <customNoConfig>
      <!--ro, opt, bool-->true
    </customNoConfig>
    <uploadID>
      <!--opt, string, range:[1,128], dep:and,{$.FDLibBaseCfglist.FDLibBaseCfg.autoUpdata,eq,true}-->test
    </uploadID>
  </FDLibBaseCfg>
</FDLibBaseCfgList>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.6.4.13 Create a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64 characters. This parameter stands for the identity key of the operation platform. The device can know the operator's identity by checking this parameter in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.

Request Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<CreateFDLibList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, create the list of face picture library, subType:object, attr:version{req, string, protocolVersion}-->
  <CreateFDLib>
    <!--opt, object, create a face picture library-->
    <id>
      <!--opt, int, sub item No. in the list; it increases from 1, range:[1,15]-->1
    </id>
    <name>
      <!--req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--opt, string, custom information, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <autoUpdata>
      <!--opt, bool, whether to enable auto update of background picture in face picture library-->true
    </autoUpdata>
    <qualityThreshold>
      <!--opt, int, picture quality threshold, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--opt, int, inbound similarity threshold, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--opt, object, the aggregate aging parameters configuration of face picture library; it is valid when autoUpdata is true-->
      <judgementCycleType>
        <!--req, enum, the time type for aging judgement cycle of the aggregate data, subType:string, desc:"day", "week", "month", "duration" (judge
according to actual duration)-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--req, enum, statistics cycle type of clustering matches in face picture library, subType:string, desc:"day", "week", "month"-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--req, int, statistics cycle of face picture matches, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--req, int, min. number of face picture matches within statistics cycle, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--opt, bool, whether to enable total number of face picture matches-->true
      </FMCountEnabled>
      <FMCount>
        <!--opt, int, total number of face picture matches, range:[0,99999]-->1
      </FMCount>
      <backgroundPicThreshold>
        <!--opt, int, range:[1,100]-->90
      </backgroundPicThreshold>
      <judgementCycle>
        <!--opt, int, evaluation cycle range of clustering data aging in face picture library, range:[1,365], unit:d, dep:and,
{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->60
      </judgementCycle>
      <agingExecDataTime>
        <!--opt, time-->00:00:00+08:00
      </agingExecDataTime>
      <automaticClearEnabled>
        <!--opt, bool-->true
      </automaticClearEnabled>
    </Age>
    <FaceCounting>
      <!--opt, object, the configuration for customer traffic (face) deduplication of face picture comparison-->
      <enabled>
        <!--req, bool, whether to enable customer traffic (face) deduplication-->true
      </enabled>
      <uploadType>
        <!--req, enum, uploading mode after the customer traffic (face) deduplication is enabled, subType:string, desc:"out_repetition" (uploading outside
the deduplication period); "noUpload" (not uploading outside the deduplication period)-->noUpload
      </uploadType>
    </FaceCounting>
    <faceLibType>
      <!--opt, enum, face picture library type, subType:string, desc:if this node is not returned, a normal library is created by default; "ordinary"
(normal library); "privacyMask" (privacy mask library)-->ordinary
    </faceLibType>
  </CreateFDLib>
</CreateFDLibList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, create the result list of face picture library, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibInfo>
    <!--ro, opt, object, face picture library information-->
    <id>
      <!--ro, req, int, sub item No. in the list; it increases from 1 and corresponds to the id of CreateFDLibList-->1
    </id>
    <name>
      <!--ro, req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <FDID>
      <!--ro, req, string, returned face picture library ID when it created, the library ID of the same type is unique, range:[1,64]-->test
    </FDID>
    <statusCode>
      <!--ro, opt, enum, status code; it is returned when creating a face picture library failed, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3
(Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->1
    </statusCode>
    <subStatusCode>
      <!--ro, opt, string, sub status code; it is returned when creating a face picture library failed-->existSameNameFacelib
    </subStatusCode>
  </FDLibInfo>
</FDLibInfoList>
```

14.6.4.14 Get the information, including library ID, library type, name, and custom information, of all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64 characters.

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfgList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibBaseCfg>
    <!--ro, opt, object, face picture library information-->
    <id>
      <!--ro, opt, int, range:[1,15]-->1
    </id>
    <FDID>
      <!--ro, opt, string, face picture library ID-->test
    </FDID>
    <name>
      <!--ro, req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--ro, opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--ro, opt, string, custom information, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--ro, opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <faceLibType>
      <!--ro, opt, enum, face picture library type, subType:string, desc:face picture library type-->ordinary
    </faceLibType>
    <autoUpdata>
      <!--ro, opt, bool-->true
    </autoUpdata>
    <qualityThreshold>
      <!--ro, opt, int, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--ro, opt, int, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--ro, opt, object-->
      <judgementCycleType>
        <!--ro, req, enum, subType:string-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--ro, req, enum, "day", "week", "month", subType:string, desc:"day", "week", "month"-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--ro, req, int, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--ro, req, int, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--ro, opt, bool-->true
      </FMCountEnabled>
      <FMCount>
        <!--ro, opt, int, range:[0,99999]-->1
      </FMCount>
      <backgroundPicThreshold>
        <!--ro, opt, int, range:[1,100]-->90
      </backgroundPicThreshold>
      <judgementCycle>
        <!--ro, opt, int, range:[1,365], unit:d, dep:and,{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
      </judgementCycle>
    </Age>
    <FaceCounting>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <uploadType>
        <!--ro, req, enum, subType:string-->noUpload
      </uploadType>
      <faceSimilarity>
        <!--ro, opt, int, range:[0,100]-->40
      </faceSimilarity>
    </FaceCounting>
    <faceMonitor>
      <!--ro, opt, bool-->true
    </faceMonitor>
    <monitorFaceNum>
      <!--ro, opt, int-->1
    </monitorFaceNum>
    <customNoConfig>
      <!--ro, opt, bool-->true
    </customNoConfig>
    <uploadID>
      <!--ro, opt, string, range:[1,128], dep:and,{$.FDLibBaseCfglist.FDLibBaseCfg.autoUpdata,eq,true}-->test
    </uploadID>
  </FDLibBaseCfg>
</FDLibBaseCfgList>
```

drackum@gmail.com
DETEKTOR

14.6.4.15 Export the package of face capture library attributes

Request URL

POST /ISAPI/SDT/Face/captureSearch/dataPackage

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int*/
    "maxResults":  100,
    /*req, int*/
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*opt, datetime*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*opt, datetime*/
    "choiceChannel": [
    /*opt, array, subType:object*/
        {
            "channelID":  "test"
            /*opt, string*/
        }
    ],
    "ageGroup":  "any",
    /*opt, enum, subType:string*/
    "gender":  "male",
    /*opt, enum, subType:string*/
    "glasses":  "yes",
    /*opt, enum, subType:string*/
    "smile":  "yes",
    /*opt, enum, subType:string*/
    "isInLibrary":  "yes",
    /*opt, enum, subType:string*/
    "thermometryUnit":  "celsius",
    /*opt, enum, subType:string*/
    "currTemperature":  38.0,
    /*opt, float, range:[-20.0,150.0]*/
    "temperatureCondition":  "normal",
    /*opt, enum, subType:string*/
    "FPID":  " "
    /*opt, string*/
}
```

Response Message

Binary Data

drackum@gmail.com
DETEKTOR

14.6.4.16 Delete the task according to task ID

Request URL

DELETE /ISAPI/SDT/Management/Task/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.6.4.17 Pause the task according to task ID

Request URL

GET /ISAPI/SDT/Management/Task/pause/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field must be returned when the value of statusCode is not 1*/
}
```

dracknm@gmail.com
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14.6.4.18 Obtain the task progress

Request URL

GET /ISAPI/SDT/Management/Task/progress/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "progress": {
    /*ro, opt, object, task progress*/
        "value":  22
        /*ro, req, int, value of task searching progress, range:[0,100]*/
    }
}
```

14.6.4.19 Resume the task according to task ID

Request URL

GET /ISAPI/SDT/Management/Task/resume/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:status code*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:status description*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:which corresponds to subStatusCode, this field is required when statusCode is not 1. The returned value is the
transformed decimal number*/
    "errorMsg":  "ok"
    /*ro, opt, string, error details, desc:this field is required when statusCode is not 1*/
}
```

14.6.5 人脸特征搜图

14.6.5.1 Get the capability set of searching by picture in captured picture library

Request URL

GET /ISAPI/SDT/Face/searchByPic/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, string, search start time*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, string, search end time*/
    "channelNum":  100,
    /*ro, req, int, number of cameras can be selected*/
    "resultMaxNum":  500,
    /*ro, req, int, maximum results can be searched*/
    "modelThreshold": [
    /*ro, opt, array, maximum and minimum threshold for comparison, subType:object*/
        {
            "min":  0,
            /*ro, opt, int, the minimum value*/
            "max":  100
            /*ro, req, int, the maximum value*/
        }
    ],
    "faceURL": {
    /*ro, opt, object, length of picture URL*/
        "min":  0,
        /*ro, opt, int, the minimum value*/
        "max":  100
        /*ro, opt, int, the maximum value*/
    },
    "targetModelData": {
    /*ro, opt, object, length of target model data*/
        "min":  0,
        /*ro, opt, int, the minimum value*/
        "max":  100
        /*ro, opt, int, the maximum value*/
    },
    "sortType":  "similarity,time",
    /*ro, opt, string, sorting type, desc:“similarity” (by similarity), “time” (by time)*/
    "URLType":  "cloudURL",
    /*ro, opt, string, URL type of the returned picture after search, desc:this node is available for devices supporting local URL (such as NVR) and by
default the device will return the local URL*/
    "maxResultsLocal": {
    /*ro, req, object, the maximum records of current search*/
        "min":  0,
        /*ro, opt, int, the minimum value*/
        "max":  1000
        /*ro, opt, int, the maximum value*/
    },
    "isSupportAsync":  true,
    /*ro, opt, bool, whether the device supports asynchronously searching face by picture*/
}
```

drackun@gmail.com
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14.6.5.2 Get task progress of using picture to search picture

Request URL

GET /ISAPI/SDT/Face/searchByPic/progress?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID": "ffc98d57-c156-40a8-ae64-84cb98dcee10",
    /*ro, req, string, task ID of using picture to search picture*/
    "progress": 100,
    /*ro, req, int, task progress of using picture to search picture*/
    "analysisModelNum": 100
    /*ro, req, int, total number of models to be compared*/
}
```

14.6.5.3 Search for the results of asynchronously using picture to search picture

Request URL

POST /ISAPI/SDT/Face/searchByPic/result?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

```
{
    "searchResultPosition": 0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults": 100
    /*req, int, the maximum number of search results this time*/
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "activeNumMax",
    /*ro, req, enum, sub status code, subType:string, desc:"ok" (succeeded). It is required when an error occurred. "activeNumMax" (the max. number of device allowed for License access is reached), "licenseExpired" (License expired)*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "taskID": "test",
    /*ro, req, string, task ID of using picture to search picture*/
    "responseStatusStrg": "OK",
    /*ro, req, string, searching status description*/
    "numOfMatches": 1,
    /*ro, req, int, number of matched results returned this time*/
    "totalMatches": 1,
    /*ro, req, int, total number of matched results by this search*/
    "targets": [
    /*ro, opt, array, list of matched records, subType:object*/
        {
            "id": 1,
            /*ro, req, int, No.*/
            "captureTime": "2004-05-03T17:30:08+08:00",
            /*ro, opt, string, capture time*/
            "isSummerTime": false,
            /*ro, opt, bool, whether the time returned currently is in DST (Daylight Saving Time) system*/
            "deviceType": "Blade",
            /*ro, req, enum, device type, subType:string, desc:"Blade" (vehicle analysis server), "BladePS" (security vehicle analysis server), "CVR","DM" (full analysis server), "DMbehavior" (abnormal event detection device), "DVR","DVS", "DockStation" (dock station), "FA" (real-time face picture comparison and alarm server), "FD" (capture detection server), "FS" (face picture static data server), "HAWK" (video structural analysis server), "HMCP" (model comparison server), "HybirdNVR" (hybrid DVR), "IPCamera" (network camera), "IPDome" (network speed dome), "IPZoom" (integrated camera), "InfoReleaseSys" (information release system), "InfoTerminal" (information release terminal), "MCU", "NVR", "PHA" (security control panel), "PURE" (pure analysis server), "RFID", "SipServer" (SIP server), "Switch", "lightFace" (lightweight facial recognition server)*/
            "deviceId": "test",
            /*ro, opt, string, camera ID*/
            "captureSite": "3#大门处",
            /*ro, opt, string, capture site (camera name)*/
```

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```
            "monitorId":  "1",
            /*ro, opt, string, stream ID (encoding stream ID in CVR). This node will not be returned if the device does not support this function*/
            "ageGroup":  "any",
            /*ro, req, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime",
"teenager", "unknown", "young"*/
            "gender":  "any",
            /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
            "glasses":  "no",
            /*ro, req, enum, whether the person wears glasses, subType:string, desc:"no" (not wear glasses), "sunglasses" (wears sunglasses), "unknown",
"yes"*/
            "smile":  "no",
            /*ro, req, enum, whether the person wears a smile, subType:string, desc:smile or not: "unknown", "yes", "no"*/
            "picUrl":  "xxx",
            /*ro, opt, string, background picture URL*/
            "subpicUrl":  "xxx",
            /*ro, req, string, face picture thumbnail URL*/
            "rect": {
            /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                "height":  1.000,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width":  1.000,
                /*ro, opt, float, width, range:[0.000,1.000]*/
                "x":  0.000,
                /*ro, opt, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, opt, float, y-coordinate, range:[0.000,1.000]*/
            },
            "face_pose": {
            /*ro, opt, object, face angle*/
                "pitch":  0.514,
                /*ro, opt, float, face pitch angle, face pitching up is regarded as positive*/
                "yaw":  0.2909,
                /*ro, opt, float, face yaw angle, face yawing left is regarded as positive*/
                "roll":  0.3669
                /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive*/
            },
            "face_rect": {
            /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                "height":  1.000,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width":  1.000,
                /*ro, opt, float, width, range:[0.000,1.000]*/
                "x":  0.000,
                /*ro, opt, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, opt, float, y-coordinate, range:[0.000,1.000]*/
            },
            "similarity":  1.00,
            /*ro, opt, float, similarity*/
            "errInfo": {
            /*ro, opt, object, error description, desc:object, error information. This node is valid when the value of the node subpicUrl is empty*/
                "errMsg":  "test",
                /*ro, req, string, error description, desc:the reason when the value of the node subpicUrl is empty*/
                "errType":  0
                /*ro, req, int, error type, desc:error type, which is the reason type when the value of the node subpicUrl is empty*/
            },
            "channelID":  1,
            /*ro, opt, int, channel No.*/
            "URLType":  "cloudURL",
            /*ro, opt, string, type of returned picture URL, desc:type of returned picture URL*/
            "faceExpression":  "angry",
            /*ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-faced",
"sad", "surprised", "unknown"*/
            "mask":  "no",
            /*ro, req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"*/
            "linkFaceBodyId":  "XXX",
            /*ro, opt, string, linkage ID between face and human body picture*/
            "monitorPointID":  "test",
            /*ro, opt, string, camera ID*/
            "thermometryUnit":  "celsius",
            /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
            "currTemperature":  38.0,
            /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
            "temperatureCondition":  "normal"
            /*ro, opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
        }
    ]
}
```

drackum@gmail.com
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14.6.5.4 Submit the task of asynchronously searching by picture

Request URL

POST /ISAPI/SDT/Face/searchByPic?format=json

Query Parameter

None

Request Message

```
{
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*opt, string, start time of search*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*opt, string, end time of search*/
    "choiceChannel": [
    /*opt, array, list of selected cameras, subType:object*/
        {
            "channelID":  "test",
            /*opt, string, camera ID*/
            "channelName":  "channel1"
            /*opt, string, camera name, range:[0,64]*/
        }
    ],
    "similarityMin":  0.3,
    /*req, float, minimum threshold for comparison, range:[0.00,1.00], desc:minimum threshold for comparison*/
    "similarityMax":  1.0,
    /*req, float, maximum threshold for comparison, range:[0.00,1.00], desc:maximum threshold for comparison*/
    "age":  23,
    /*opt, int, age*/
    "ageGroup":  "any",
    /*req, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime", "teenager",
"unknown", "young"*/
    "gender":  "any",
    /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male", "female", "unknown"*/
    "glasses":  "no",
    /*req, enum, whether the person wears glasses, subType:string, desc:"no" (not wear glasses), "sunglasses" (wears sunglasses), "unknown", "yes"*/
    "smile":  "no",
    /*req, enum, whether the person is smiling, subType:string, desc:"no", "unknown", "yes"*/
    "mask":  "no",
    /*req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"*/
    "modelMaxNum":  100,
    /*req, int, maximum number of model comparison results (the maximum number of returned results), desc:maximum number of model comparison results (the
maximum number of returned results)*/
    "targetModelData":  "test",
    /*req, string, target model data*/
    "sortType":  "similarity",
    /*opt, enum, sorting type, subType:string, desc:"similarity" (sort by similarity), "time" (sort by time)*/
    "URLType":  "cloudURL",
    /*opt, string, type of returned picture URL, desc:the URL type of searched picture*/
    "pictureMerge":  false,
    /*opt, bool, whether to merge multiple pictures of one person, desc:the value of this node is false by default*/
    "thermometryUnit":  "celsius",
    /*opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
    "currTemperature":  38.0,
    /*opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
    "temperatureCondition":  "normal"
    /*opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
}
```

dracku@gmail.com
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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceInfo	[Message content]	application/json	--	--	--
faceImage	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "ffc98d57-c156-40a8-ae64-84cb98dcee10"
    /*ro, req, string, task ID of searching by picture*/
}
```

14.6.5.5 Get the task information

Request URL

GET /ISAPI/SDT/Face/searchByPic?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, req, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskInfo": [
    /*ro, opt, array, task information, subType:object*/
        {
            "taskID":  "abc",
            /*ro, req, string, task ID of searching by picture, range:[0,64]*/
            "taskCreateTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, task creation time*/
            "timeUsed":  100,
            /*ro, opt, int, task execution duration, unit:s, desc:unit: second*/
            "progress":  100,
            /*ro, opt, int, task progress*/
            "TaskDetails": {
            /*ro, opt, object, task details*/
                "startTime":  "2004-05-03T17:30:08+08:00",
                /*ro, opt, string, start time of search*/
                "endTime":  "2004-05-03T17:30:08+08:00",
                /*ro, opt, string, end time of search*/
                "choiceChannel": [
                /*ro, opt, array, list of selected cameras, subType:object*/
                    {
                        "channelID":  "test",
                        /*ro, opt, string, camera ID*/
                        "channelName":  "channel1"
                        /*ro, opt, string, camera name*/
                    }
                ],
                "similarityMin":  0.3,
                /*ro, req, float, minimum threshold for comparison, range:[0.00,1.00], desc:it should be accurate to two decimal place, the default value is
0*/
                "similarityMax":  1.0,
                /*ro, opt, float, maximum threshold for comparison, range:[0.00,1.00], desc:it should be accurate to two decimal place, the default value is
0*/
                "age":  23,
                /*ro, opt, int, age*/
                "ageGroup":  "any",
                /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime",
"teenager", "unknown", "young"*/
                "gender":  "any",
                /*ro, opt, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
                "glasses":  "no",
                /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no" (not wear glasses), "sunglasses" (wears sunglasses),
"unknown"*/
                "smile":  "no",
                /*ro, opt, enum, whether the person wears a smile, subType:string, desc:"no", "unknown", "yes"*/
                "mask":  "no",
                /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"*/
                "modelMaxNum":  100,
                /*ro, opt, int, maximum number of model comparison results, desc:the maximum number of returned results*/
                "targetModelData":  "test",
                /*ro, opt, string, target model data*/
                "sortType":  "similarity",
                /*ro, opt, enum, sorting type, subType:string, desc:"similarity" (sort by similarity), "time" (sort by time)*/
                "URLType":  "cloudURL",
                /*ro, opt, string, type of returned picture URL, desc:the URL type of searched picture*/
                "pictureMerge":  false,
                /*ro, opt, bool, whether to merge multiple pictures of one person, desc:the value of this node is false by default*/
                "thermometryUnit":  "celsius",
                /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                "currTemperature":  38.0,
                /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
                "temperatureCondition":  "normal"
                /*ro, opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
            }
        }
    ]
}
```

drackun@gmail.com
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14.6.5.6 Delete the task of searching picture by picture asynchronously

Request URL

DELETE /ISAPI/SDT/Face/searchByPic?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

14.7 人体识别

14.7.1 以人搜人

14.7.1.1 Analyze and model human body pictures

Request URL

POST /ISAPI/Intelligent/humanModel?format=json

Query Parameter

None

Request Message

```
{
    "URL":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314",
    /*opt, string, picture URL,the picture data will be uploaded in form format if no value is assigned to this node, desc:picture URL,the picture data will
be uploaded in form format if no value is assigned to this node*/
    "picData":  "JXU4RkQ5JXU5MUNDJXU2NjJGJXU4OTgxJXU3RjE2JXU3ODAxJXU3Njg0JXU1MTg1JXU1QkI5JXVGRjAx",
    /*opt, string*/
    "algorithm": [
    /*opt, array, algorithm type, subType:object, desc:algorithm type*/
        {
            "classificationID":  "1",
            /*req, string, algorithm category ID, desc:algorithm category ID*/
            "targetType":  "1",
            /*req, string, target type, desc:target type*/
            "analysisType":  "1,2,3",
            /*opt, string, algorithm analysis type, desc:algorithm analysis type*/
            "analysisSourceType":  "video",
            /*req, string, VCA source type, desc:VCA source type*/
            "version":  "1.0",
            /*opt, string, algorithm version, desc:algorithm version*/
            "vendor":  ""
            /*opt, string, algorithm manufacturer, desc:algorithm manufacturer*/
        }
    ],
    "Rect": {
    /*opt, object, target frame, desc:target frame*/
        "height":  0.107407,
        /*req, float, height, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
        "width":  0.025,
        /*req, float, width, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
        "x":  0.645833,
        /*req, float, X-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
        "y":  0.651852
        /*req, float, Y-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
    }
}
```

drackum@gmail.com
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Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "confidence":  99.99,
    /*ro, opt, float, confidence, range:[0.00,100.00], desc:confidence*/
    "modelData":  "abc"
    /*ro, opt, string, target model data, desc:used for searching human body data*/
}
```

14.7.1.2 Get the capability of detecting human bodies in the picture

Request URL

GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL, range:[0,256]*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "URLLen": {
    /*ro, req, object, URL length*/
        "min":  0,
        /*ro, req, int, the minimum value*/
        "max":  256
        /*ro, req, int, the maximum value*/
    },
    "Rect": {
    /*ro, req, object, target frame, desc:the origin is the lower-left corner of the screen*/
        "height": {
        /*ro, req, object, height*/
            "min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "max":  1.000
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
        },
        "width": {
        /*ro, req, object, width*/
            "min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "max":  1.000
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
        },
        "x": {
        /*ro, req, object, x-coordinate*/
            "min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "max":  1.000
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, req, object, Y-coordinate*/
            "min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "max":  1.000
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
        }
    },
    "modelDataLen": {
```

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```
    /*ro, req, object, data length of the target model*/
        "min": 0,
        /*ro, req, int, the minimum value*/
        "max": 256
        /*ro, req, int, the maximum value*/
    },
    "maxResults": {
    /*ro, req, object, the maximum number of search results this time*/
        "min": 0,
        /*ro, req, int, the minimum value*/
        "max": 1000
        /*ro, req, int, the maximum value*/
    },
    "startTime": "1970-01-01T00:00:00+08:00",
    /*ro, req, datetime, start time*/
    "endTime": "1970-01-01T00:00:00+08:00",
    /*ro, req, datetime, end time*/
    "channelIDLen": {
    /*ro, req, object, camera ID length*/
        "min": 0,
        /*ro, req, int, the minimum value*/
        "max": 64
        /*ro, req, int, the maximum value*/
    },
    "similarity": {
    /*ro, opt, object, similarity*/
        "min": 0.0,
        /*ro, req, float, the minimum value*/
        "max": 1.00
        /*ro, req, float, the maximum value*/
    },
    "taskIDLen": {
    /*ro, req, object, task ID length*/
        "min": 0,
        /*ro, req, int, the minimum value*/
        "max": 64
        /*ro, req, int, the maximum value*/
    },
    "picData": {
    /*ro, opt, object, size of the picture encoded by Base64*/
        "@min": 0,
        /*ro, req, int, the minimum value*/
        "@max": 10000000000
        /*ro, req, float, the maximum value*/
    },
    "isSupportSync": true,
    /*ro, opt, bool, whether it supports search human body in synchronization mode, desc:whether it supports search human body in synchronization mode*/
    "isSupportExport": true,
    /*ro, opt, bool,
    desc:/ISAPI/Intelligent/humanRecognition/searchByPic/result/export?format=json
    /ISAPI/Intelligent/humanRecognition/searchByPic/result/export/progress?format=json
    /ISAPI/Intelligent/humanRecognition/searchByPic/result/export/stop?format=json*/
    "sortType": {
    /*ro, opt, object, sorting type*/
        "@opt": ["similarity", "time"]
        /*ro, req, array, range, subType:string, desc:similarity (by similarity(by default)), time (by time)*/
    },
    "URLType": {
    /*ro, opt, object, the URL type of searched picture*/
        "@opt": ["cloudURL"]
        /*ro, req, array, range, subType:string*/
    },
    "maxCloudURLResults": {
    /*ro, req, object, the maximum number of search results this time, desc:the maximum records of current search. It is valid when the value of URLType is
"cloudURL"*/
        "min": 0,
        /*ro, req, int, the minimum value*/
        "max": 1000
        /*ro, req, int, the maximum value*/
    },
    "isSupportAsync": true,
    /*ro, opt, bool, whether the device supports asynchronously searching human body by picture, desc:this node is used only for DeepinMind devices*/
    "isSupportTaskInfo": true
    /*ro, opt, bool, desc:/ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo/capabilities?format=json*/
}
```

dnckum@gmail.com
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14.7.1.3 Get task progress of searching for human body by picture

Request URL

GET /ISAPI/Intelligent/humanRecognition/searchByPic/progress?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	UUID format

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode, this field is required when statusCode is not 1*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "progress":  1,
    /*ro, req, int, progress of searching by picture task, range:[0,100], desc:progress of searching by picture task*/
    "matchePicNums":  100
    /*ro, opt, int, range:[0,10000]*/
}
```

14.7.1.4 Get progress of exporting results of searching by human body picture

Request URL

GET /ISAPI/Intelligent/humanRecognition/searchByPic/result/export/progress?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	UUID format

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode, this field is required when statusCode is not 1*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "progress":  0,
    /*ro, opt, int, when errcode is 1 and errMsg is "ok",value range: [0,100], range:[0,100], desc:when errcode is 1 and errMsg is "ok",value range:
[0,100]*/
    "URL":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314"
    /*ro, opt, string, URL for downloading the exported results, desc:this field is required when progress is 100*/
}
```

14.7.1.5 Stop exporting the results of human body search

Request URL

PUT /ISAPI/Intelligent/humanRecognition/searchByPic/result/export/stop?format=json

Query Parameter

None

Request Message

```
{
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008"
    /*req, string, task ID, desc:task ID*/
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

14.7.1.6 Export result of human body search

Request URL

POST /ISAPI/Intelligent/humanRecognition/searchByPic/result/export?format=json

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, desc:the start position of the search result in the result list*/
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*req, string, start time, desc:ISO 8601 format*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*req, string, end time, desc:ISO 8601 format*/
    "choiceChannel": [
    /*opt, array, camera list, subType:object, desc:which specifies cameras where the human body pictures are captured. If this node is not configured, it
indicates no camera limit*/
        {
            "channelID":  "abc"
            /*opt, string, camera ID, desc:the maximum string size is 32*/
        }
    ],
    "similarity":  1,
    /*opt, float, similarity, range:[0.00,1.00], desc:the larger the value is, the more accurate it will be*/
    "modelData":  "abc",
    /*req, string, target model data, desc:target model data*/
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008",
    /*opt, string, task ID of searching by picture, desc:task ID of searching by picture*/
    "URLType":  "cloudURL",
    /*opt, string, the URL type of searched picture, desc:this node is for devices supporting local URL, and the local URL will be returned by default*/
    "maxResults":  100
    /*opt, int, the maximum number of search results this time, desc:the search will stop when the number of results reaches the configured value*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008"
    /*ro, req, string, task ID, desc:task ID*/
}
```

14.7.1.7 Search the result of human body search

Request URL

POST /ISAPI/Intelligent/humanRecognition/searchByPic/result?format=json

Query Parameter

None

Request Message

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```
{
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008",
    /*req, string, task ID, desc:task ID*/
    "searchResultPosition":  0,
    /*req, int, The end position of search result in result list. In a single search,if you cannot get all the records in the result list,you can mark the end position and get the following records after the marked position in the next search., desc:The end position of search result in result list. In a single search,if you cannot get all the records in the result list,you can mark the end position and get the following records after the marked position in the next search.*/
    "maxResults":  100
    /*req, int, the maximum number of records that can be searched in this time, desc:the maximum number of records that can be searched in this time*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "activeNumMax",
    /*ro, opt, enum, sub status code, subType:string, desc:"ok" (succeeded). It is required when an error occurred. "activeNumMax" (the max. number of device allowed for License access is reached), "licenseExpired" (License expired)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008",
    /*ro, req, string, task ID, desc:task ID*/
    "responseStatusStrg":  "OK",
    /*ro, opt, enum, searching status, subType:string, desc:searching status: "OK"-searching ended,"NO MATCHES"-no data found,"MORE"-searching,the max. size is 32 bytes. It is valid only when errorCode is 1 and errorMsgis ok*/
    "numOfMatches":  1,
    /*ro, opt, int, returned number of results for current search, desc:returned number of results for current search*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results, desc:total number of matched results*/
    "targetInfo": [
    /*ro, req, array, target information, subType:object, desc:target information*/
        {
            "monitorPointIndexCode":  "abc",
            /*ro, req, string, camera No., desc:camera No.*/
            "monitorPointName":  "abc",
            /*ro, req, string, camera name, desc:camera name*/
            "captureTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, string, capture time, desc:capturing time,e.g.,2004-05-03T17:30:08+08:00*/
            "isSummerTime":  false,
            /*ro, opt, bool, whether the time returned currently is in DST (Daylight Saving Time) system*/
            "streamType":  "localvideo",
            /*ro, opt, enum, streaming type, subType:string, desc:streaming type: "realtime,historyvideo,localvideo"*/
            "targetSize":  "unknown",
            /*ro, opt, enum, target size, subType:string, desc:target size*/
            "direction":  "lowerLeft",
            /*ro, opt, enum, target direction, subType:string, desc:target direction*/
            "speed":  "unknown",
            /*ro, opt, enum, vehicle speed, subType:string, desc:target speed*/
            "targetType":  "human",
            /*ro, opt, enum, target type, subType:string, desc:"human", "twoWheelVehicle" (two wheeler), "threeWheelVehicle" (tricycle)*/
            "gender":  "unknown",
            /*ro, opt, enum, gender, subType:string, desc:gender*/
            "glass":  "unknown",
            /*ro, opt, enum, whether the person wears glasses, subType:string, desc:whether the person wears glasses*/
            "ageGroup":  "unknown",
            /*ro, opt, enum, age group, subType:string, desc:age group*/
            "ride":  "unknown",
            /*ro, opt, enum, whether the person is riding a bike, subType:string, desc:whether the person is riding a bike*/
            "bag":  "unknown",
            /*ro, opt, enum, whether the person is carrying a bag, subType:string, desc:whether the person is with a bag on his/her back*/
            "things":  "unknown",
            /*ro, opt, enum, whether the person is carrying things, subType:string, desc:whether the person is with things on his/her hands*/
            "hat":  "unknown",
            /*ro, opt, enum, whether the person wears a hat, subType:string, desc:whether the person is wearing hat*/
            "jacketType":  "unknown",
            /*ro, opt, enum, top's type, subType:string, desc:jacket type*/
            "trousersType":  "unknown",
            /*ro, opt, enum, trousers type, subType:string, desc:trousers type*/
            "jacketColor":  "unknown",
            /*ro, opt, enum, jacket color, subType:string, desc:"unknown", "white", "silver", "gray", "black", "red", "deepBlue", "blue", "yellow", "green", "brown", "pink", "purple", "deepGray", "cyan", "orange", "mixture"*/
            "trousersColor":  "unknown",
            /*ro, opt, enum, trousers color, subType:string, desc:"unknown", "white", "silver", "gray", "black", "red", "deepBlue", "blue", "yellow", "green", "brown", "pink", "purple", "deepGray", "cyan", "orange", "mixture"*/
            "hairStyle":  "shortHair",
            /*ro, opt, enum, hairstyle, subType:string, desc:hairstyle*/
            "mask":  "unknown",
            /*ro, opt, enum, whether the person wears a mask, subType:string, desc:whether the person is wearing mask or not*/
            "cyclingType":  "unknown",
            /*ro, opt, enum, riding type, subType:string, desc:cycling type*/
            "cyclingPersonNumber":  "unknown",
            /*ro, opt, enum, number of riders, subType:string, desc:number of clcling person*/
            "similarity":  0.53,
```

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```
            /*ro, opt, float, similarity, desc:float,similarity,value ranges from 0.0 to 1.00*/
            "targetRect": {
            /*ro, opt, object, target frame, desc:target frame*/
                "height":  0.107407,
                /*ro, req, float, height, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
                "width":  0.025,
                /*ro, req, float, width, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
                "x":  0.645833,
                /*ro, req, float, x-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
                "y":  0.651852
                /*ro, req, float, Y-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
            },
            "picUrl":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, req, string, picture URL, desc:picture URL*/
            "subPicUrl":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, opt, string, thumbnail URL, desc:thumbnail picture URL*/
            "URLType":  "cloudURL"
            /*ro, opt, string, the URL type of searched picture. This node is applicable to the device which supports local URL. To get the picture of could
storage,set the value to "cloudURL", desc:the URL type of searched picture. This node is applicable to the device which supports local URL. To get the
picture of could storage,set the value to "cloudURL"*/
        }
    ]
}
```

14.7.1.8 Search for human body via picture comparison in synchronous mode

Request URL

POST /ISAPI/Intelligent/humanRecognition/searchByPic/sync?format=json

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, desc:the start position of the search result in the result list*/
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*req, string, start time, desc:ISO 8601 format*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*req, string, end time, desc:ISO 8601 format*/
    "choiceChannel": [
    /*opt, array, camera list, subType:object, desc:which specifies cameras where the human body pictures are captured. If this node is not configured, it
indicates no camera limit*/
        {
            "channelID":  "abc"
            /*opt, string, camera ID, desc:the maximum string size is 32*/
        }
    ],
    "similarity":  0.89,
    /*opt, float, similarity, range:[0.00,1.00], desc:the larger the value is, the more accurate it will be*/
    "modelData":  "abc",
    /*req, string, target modeling data, desc:target modeling data*/
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008",
    /*opt, string, task ID of searching by picture, desc:the task ID returned from /ISAPI/SDT/Management/Task/Video?format=json*/
    "URLType":  "cloudURL",
    /*opt, string, the URL type of searched picture, desc:this node is for devices supporting local URL, and the local URL will be returned by default*/
    "maxResults":  100,
    /*opt, int, the maximum number of search results this time, desc:the search will stop when the number of results reaches the configured value*/
    "modelMaxNum":  100,
    /*opt, int, maximum number of returned results of model comparison, desc:maximum expected number of returned results from picture comparison*/
    "sortType":  "similarity"
    /*opt, enum, sorting type, desc:"similarity" (sort by similarity; default), "time" (sort by time)*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
human	[Message content]	application/json	--	--	--
humanImage	[Binary picture data]	image/jpeg	humanImage	humanImage.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "activeNumMax",
    /*ro, opt, enum, sub status code, subType:string, desc:"ok" (succeeded). It is required when an error occurred. "activeNumMax" (the max. number of
device allowed for License access is reached), "licenseExpired" (License expired)*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "responseStatusStrg": "OK",
    /*ro, opt, enum, searching status, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data).*/
    "numOfMatches": 1,
    /*ro, opt, int, returned number of results for current search, desc:no greater than <maxResults>*/
    "totalMatches": 1,
    /*ro, opt, int, total number of matched results, desc:for checking whether all results have been obtained and calculating the number of pages*/
    "targetInfo": [
    /*ro, req, array, target information, subType:object*/
        {
            "monitorPointIndexCode": "abc",
            /*ro, req, string, camera No., desc:information of the camera which captured the human body*/
            "monitorPointName": "abc",
            /*ro, req, string, camera name, desc:information of the camera which captured the human body*/
            "captureTime": "2004-05-03T17:30:08+08:00",
            /*ro, opt, string, capture time, desc:capture time in format as the shown example*/
            "isSummerTime": false,
            /*ro, opt, bool, whether the time returned currently is in DST (Daylight Saving Time) system*/
            "streamType": "localvideo",
            /*ro, opt, enum, streaming type, subType:string, desc:human body record resoure: "realtime" (real-time stream), "historyvideo" (historical
stream), "localvideo" (local stream)*/
            "targetSize": "unknown",
            /*ro, opt, enum, target size, subType:string, desc:"unknown", "big", "medium", "small"*/
            "direction": "lowerLeft",
            /*ro, opt, enum, target direction, subType:string, desc:"lowerLeft", "upperLeft", "lowerRight", "upperRight", "unknown", "forward", "backward",
"leftward", "rightward"*/
            "speed": "unknown",
            /*ro, opt, enum, vehicle speed, subType:string, desc:"unknown", "fast", "medium", "slow"*/
            "targetType": "human",
            /*ro, opt, enum, target type, subType:string, desc:"human", "twoWheelVehicle" (two wheeler), "threeWheelVehicle" (tricycle)*/
            "gender": "unknown",
            /*ro, opt, enum, gender, subType:string, desc:"unknown", "male", "female"*/
            "glass": "unknown",
            /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"unknown", "no", "yes", "sunglasses" (wears sunglasses)*/
            "ageGroup": "unknown",
            /*ro, opt, enum, age group, subType:string, desc:"unknown", "child", "young", "middle", "old", "infant", "kid", "teenager", "prime",
"middleAged" (the middle age)*/
            "ride": "unknown",
            /*ro, opt, enum, whether the person is riding a bike, subType:string, desc:"unknown", "no", "yes"*/
            "bag": "unknown",
            /*ro, opt, enum, whether the person is carrying a bag, subType:string, desc:"unknown", "no", "yes"*/
            "things": "unknown",
            /*ro, opt, enum, whether the person is carrying things, subType:string, desc:"unknown", "no", "yes"*/
            "hat": "unknown",
            /*ro, opt, enum, whether the person wears a hat, subType:string, desc:"unknown", "no", "yes"*/
            "jacketType": "unknown",
            /*ro, opt, enum, top type, subType:string, desc:"unknown", "shortSleeve", "longSleeve"*/
            "trousersType": "unknown",
            /*ro, opt, enum, bottom's type, subType:string, desc:"unknown", "shortTrousers" (shorts), "longTrousers" (trousers), "skirt"*/
            "jacketColor": "unknown",
            /*ro, opt, enum, jacket color, subType:string, desc:"unknown", "white", "silver", "gray", "black", "red", "deepBlue", "blue", "yellow", "green",
"brown", "pink", "purple", "deepGray", "cyan", "orange", "mixture"*/
            "trousersColor": "unknown",
            /*ro, opt, enum, bottom's color, subType:string, desc:"unknown", "white", "silver", "gray", "black", "red", "deepBlue", "blue", "yellow",
"green", "brown", "pink", "purple", "deepGray", "cyan", "orange", "mixture"*/
            "hairStyle": "shortHair",
            /*ro, opt, enum, hairstyle, subType:string, desc:"shortHair", "longHair", "unknown"*/
```

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```
            "mask":  "unknown",
            /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"unknown", "no", "yes"*/
            "cyclingType":  "unknown",
            /*ro, opt, enum, riding type, subType:string, desc:"unknown", "bicycle", "motorcycle" (motorcycle / electric moped), "threeWheelVehicle"
(tricycle)*/
            "cyclingPersonNumber":  "unknown",
            /*ro, opt, enum, number of riders, subType:string, desc:"unknown", "onePerson", "multiplePersons"*/
            "similarity":  0.53,
            /*ro, opt, float, similarity, desc:the larger the value, the higher the similarity*/
            "targetRect": {
            /*ro, opt, object, human body coordinate frame, desc:the origin is the upper-left corner of the screen, with X starting from left to right and Y
from top to bottom, both normalized to [0.000, 1.000]*/
                "height":  0.107407,
                /*ro, req, float, height, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
                "width":  0.025,
                /*ro, req, float, width, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
                "x":  0.645833,
                /*ro, req, float, x-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
                "y":  0.651852
                /*ro, req, float, y-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
            },
            "picUrl":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, req, string, picture URL, desc:human body picture download URL*/
            "subPicUrl":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, opt, string, thumbnail URL, desc:human body thumbnail download URL*/
            "linkFaceBodyId":  "XXX",
            /*ro, opt, string, linkage ID between face and human body picture, desc:linkage ID between face and human body picture*/
            "rowKey":  "test"
            /*ro, opt, string, main key of human body record library., range:[1,128], desc:search by <rowKey> to improve the efficiency, as every piece of
data in the big data system is identified by <rowKey>*/
        }
    ]
}
```

14.7.1.9 Get the capability of searching for the task of searching by human body picture

Request URL

GET /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "searchID": {
    /*ro, req, object, search ID, desc:this node is used to check whether searches are the same one; due to device performance limitation, you can apply the
search conditions multiple times (the search ID should be the same) to complete one search and get the search progress*/
        "@min":  0,
        /*ro, opt, int, the minimum value, range:[0,32]*/
        "@max":  32
        /*ro, opt, int, the maximum value, range:[0,32]*/
    },
    "maxResultNumber": {
    /*ro, opt, object, the maximum number of records that need to be searched*/
        "@min":  0,
        /*ro, opt, int, the minimum value, range:[0,4294967295]*/
        "@max":  2000,
        /*ro, opt, int, the maximum value, range:[0,4294967295]*/
        "@def":  0
        /*ro, opt, int, the default value, range:[0,4294967295]*/
    },
    "maxResults": {
    /*ro, req, object, the maximum number of records that can be displayed on each page by this search*/
        "@min":  0,
        /*ro, opt, int, the minimum value, range:[0,100]*/
        "@max":  100
        /*ro, opt, int, the maximum value, range:[1,100]*/
    }
}
```

14.7.1.10 Get the tasks of searching by human body picture

Request URL

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POST /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo?format=json

Query Parameter

None

Request Message

```
{
    "searchID":  "C93A238C-6AD0-0001-4CB8-1D961A901701",
    /*req, string, search ID, range:[0,64]*/
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, range:[0,4294967295], desc:in a single search, if you cannot get all the records
in the result list, you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  100
    /*opt, int, the maximum number of search results this time, range:[0,100]*/
}
```

Response Message

```
{
    "responseStatusStrg":  "MORE",
    /*ro, req, enum, searching status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data),"OK" (searching
completed)*/
    "numOfMatches":  1,
    /*ro, opt, int, number of results returned this time, range:[0,100]*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results, range:[0,4294967295]*/
    "TaskInfo": [
    /*ro, opt, array, task information, subType:object*/
        {
            "taskID":  "abc",
            /*ro, req, string, task ID of searching by picture, range:[0,64]*/
            "taskCreateTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, task creation time*/
            "taskFinishTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, task end time*/
            "timeUsed":  100,
            /*ro, opt, int, task execution duration, unit:s, desc:unit: second*/
            "progress":  100,
            /*ro, opt, int, task progress, range:[0,100]*/
            "TaskDetails": {
            /*ro, opt, object, task details*/
                "startTime":  "2004-05-03T17:30:08+08:00",
                /*ro, req, string, start time, desc:ISO 8601 format*/
                "endTime":  "2004-05-03T17:30:08+08:00",
                /*ro, req, string, end time, desc:ISO 8601 format*/
                "choiceChannel": [
                /*ro, opt, array, camera list, subType:object, desc:which specifies cameras where the human body pictures are captured. If this node is not
configured, it indicates no camera limit*/
                    {
                        "channelID":  "abc",
                        /*ro, opt, string, camera ID, desc:the maximum string size is 32*/
                        "channelName":  "channel1"
                        /*ro, opt, string, camera name, range:[0,64]*/
                    }
                ],
                "similarity":  1,
                /*ro, opt, int, similarity, range:[0.00,1.00], desc:the larger the value is, the more accurate it will be*/
                "modelData":  "abc",
                /*ro, opt, string, target model data, desc:target model data*/
                "taskID":  "5d811723-d165-11ea-9b07-005056c00008",
                /*ro, opt, string, task ID of searching by picture, desc:task ID of searching by picture*/
                "URLType":  "cloudURL",
                /*ro, opt, string, the URL type of searched picture, desc:this node is for devices supporting local URL, and the local URL will be returned
by default*/
                "maxResults":  100
                /*ro, opt, int, the maximum number of search results this time, desc:the search will stop when the number of results reaches the configured
value*/
            }
        }
    ]
}
```

dr.ekum@gmail.com
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14.7.1.11 Set conditions of searching for human body by modeled picture data and submit the search task

Request URL

POST /ISAPI/Intelligent/humanRecognition/searchByPic?format=json

Query Parameter

None

Request Message

```
{
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*req, string, start time, desc:ISO 8601 format*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*req, string, end time, desc:ISO 8601 format*/
    "choiceChannel": [
    /*opt, array, camera list, subType:object, desc:which specifies cameras where the human body pictures are captured. If this node is not configured, it
indicates no camera limit*/
        {
            "channelID":  "abc",
            /*opt, string, camera ID, desc:the maximum string size is 32*/
            "channelName":  "channel1"
            /*opt, string, camera name, range:[0,64]*/
        }
    ],
    "similarity":  0.99,
    /*opt, float, similarity, range:[0.00,1.00], desc:the larger the value is, the more accurate it will be*/
    "modelData":  "abc",
    /*req, string, target model data, desc:target model data*/
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008",
    /*opt, string, task ID of searching by picture, desc:task ID of searching by picture*/
    "URLType":  "cloudURL",
    /*opt, string, the URL type of searched picture, desc:this node is for devices supporting local URL, and the local URL will be returned by default*/
    "maxResults":  100
    /*opt, int, the maximum number of search results this time, desc:the search will stop when the number of results reaches the configured value*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
searchByPic	[Message content]	application/json	--	--	--
humanImage	[Binary picture data]	image/jpeg	humanImage	humanImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

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- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "5d811723-d165-11ea-9b07-005056c00008"
    /*ro, req, string, task ID, desc:task ID*/
}
```

14.7.1.12 Delete a task of searching by human body picture

Request URL

DELETE /ISAPI/Intelligent/humanRecognition/searchByPic?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID, which is returned by the URL (POST /ISAPI/Intelligent/humanRecognition/searchByPic?format=json).

Request Message

None

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, Status Code, desc:this field is required when the status code is not "1" (success)*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status code is not "OK" (success)*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status code is not "ok" (success)*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode": "0x00000000",
    /*ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field*/
    "MErrDevSelfEx": "0x00000000"
    /*ro, opt, string, error codes categorized by functional modules, desc:N/A*/
}
```

14.7.1.13 Detect human body

Request URL

POST /ISAPI/Intelligent/humanRecognition?format=json

Query Parameter

None

Request Message

```
{
    "URL": "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314",
    /*opt, string, picture URL,the picture data will be uploaded in form format if no value is assigned to this node, desc:picture URL,the picture data will
be uploaded in form format if no value is assigned to this node*/
    "picData": "JXU4RkQ5JXU5MUNDJXU2NjJGJXU4OTgxJXU3RjE2JXU3ODAxJXU3Njg0JXU1MTg1JXU1QkI5JXVGRjAx",
    /*opt, string*/
    "algorithm": [
    /*opt, array, algorithm type, subType:object, desc:algorithm type*/
        {
            "classificationID": "1",
            /*req, string, algorithm category ID, desc:algorithm category ID*/
            "targetType": "1",
            /*req, string, target type: 1-face,2-vehicle,3-active targets (human body and vehicle),4-human body, desc:target type: 1-face,2-vehicle,3-active
targets (human body and vehicle),4-human body*/
            "analysisType": "1,2,3",
            /*opt, string, algorithm analysis type, desc:algorithm analysis type*/
            "analysisSourceType": "video",
            /*req, string, VCA source type, desc:VCA source type*/
            "version": "1.0",
            /*opt, string, algorithm version, desc:algorithm version*/
            "vendor": ""
            /*opt, string, algorithm manufacturer, desc:algorithm manufacturer*/
        }
    ]
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
humanRecognition	[Message content]	application/json	--	--	--
human	[Binary picture data]	image/jpeg	human	human	--

drackum@gmail.com DETEKTOR

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "Rect": [
    /*ro, opt, array, target frame, subType:object, range:[0.000,1.000], desc:target frame*/
        {
            "height":  0.107407,
            /*ro, req, float, height, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
            "width":  0.025,
            /*ro, req, float, width, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
            "x":  0.645833,
            /*ro, req, float, X-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
            "y":  0.651852
            /*ro, req, float, Y-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
        }
    ]
}
```

drackum@gmail.com
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14.8 周界入侵检测

14.8.1 区域入侵

14.8.1.1 Set intrusion detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/fieldDetections

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->fieldDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FieldDetectionScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.1.2 Get arming schedules of all channels for intrusion detection

Request URL

GET /ISAPI/Event/schedules/fieldDetections

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, configuration of intrusion detection arming schedules-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->fieldDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, list of arming schedules, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming schedules-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FieldDetectionScheduleList>
```

drackum@gmail.com
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14.8.1.3 Set the linkage triggering parameters of intrusion detection event for a specified channel

Request URL

PUT /ISAPI/Event/triggers/FieldDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <notificationMethod>
      <!--req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object-->
      <ptzChannelID>
        <!--req, string-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object-->
      <whiteLightDurationTime>
        <!--opt, int, range:[1,60]-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <AudioAction>
      <!--opt, object-->
      <audioID>
        <!--req, int-->1
      </audioID>
      <alarmTimes>
        <!--req, int, range:[0,9]-->1
      </alarmTimes>
    </AudioAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.1.4 Get the linkage triggering parameters of intrusion detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/FieldDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60]-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <AudioAction>
      <!--ro, opt, object-->
      <audioID>
        <!--ro, req, int-->1
      </audioID>
      <alarmTimes>
        <!--ro, req, int, range:[0,9]-->1
      </alarmTimes>
    </AudioAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

drackum@gmail.com
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14.8.1.5 Set the linkage parameters of intrusion detection event for a specified channel

Request URL

PUT /ISAPI/Event/triggers/FieldDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->fielddetection
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->1
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->1
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <LightAudioByIPAction>
```

drackum@gmail.com
DETEKTOR

```
        <!--opt, object-->
        <notificationType>
          <!--req, enum, subType:string-->audio
        </notificationType>
        <notificationName>
          <!--opt, string-->test
        </notificationName>
      </LightAudioByIPAction>
      <SupplementLightAlarm>
        <!--opt, object-->
        <durationTime>
          <!--opt, int, range:[0,90], unit:s-->5
        </durationTime>
      </SupplementLightAlarm>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

dractum@gmail.com
DETEKTOR

14.8.1.6 Get the linkage parameters of intrusion detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/FieldDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->fielddetection
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

14.8.1.7 Set intrusion detection parameters of all channels

Request URL

PUT /ISAPI/Smart/FieldDetection

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FieldDetection>
    <!--opt, object-->
    <id>
      <!--req, string, channel No.-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <intelliBackSearch>
      <!--opt, bool, whether the device supports video dual-VCA-->true
    </intelliBackSearch>
    <startTriggerTime>
      <!--req, int, unit:ms-->1
    </startTriggerTime>
    <endTriggerTime>
      <!--req, int, unit:ms-->1
    </endTriggerTime>
    <normalizedScreenSize>
      <!--opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <minObjectSize>
      <!--opt, int, minimum pixels of each object-->1
    </minObjectSize>
    <maxObjectSize>
      <!--opt, int, maximum pixels of each object-->1
    </maxObjectSize>
    <FieldDetectionRegionList size="4">
      <!--opt, array, subType:object, attr:size{req, int}-->
      <FieldDetectionRegion>
        <!--opt, object, intrusion region-->
        <id>
          <!--req, string-->1
        </id>
        <enabled>
```

crackum@gmail.com
DETEKTOR

```
          <enabled>
            <!--req, bool, whether to enable the function-->true
          </enabled>
          <sensitivityLevel>
            <!--req, int, sensitivity, range:[0,100]-->1
          </sensitivityLevel>
          <timeThreshold>
            <!--req, int-->1
          </timeThreshold>
          <objectOccupation>
            <!--req, int, percentage-->1
          </objectOccupation>
          <detectionTarget>
            <!--opt, enum, detection target, subType:string, desc:detection target-->all
          </detectionTarget>
          <RegionCoordinatesList>
            <!--opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
            <RegionCoordinates>
              <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->1
              </positionX>
              <positionY>
                <!--req, int, y-coordinate, range:[0,1000]-->1
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
          <humanMisinfoFilterEnabled>
            <!--opt, bool, whether to enable false alarm filtering (human)-->true
          </humanMisinfoFilterEnabled>
          <vehicleMisinfoFilterEnabled>
            <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
          </vehicleMisinfoFilterEnabled>
          <priority>
            <!--opt, enum, priority, subType:string, desc:priority-->low
          </priority>
          <alarmConfidence>
            <!--opt, enum, subType:string-->low
          </alarmConfidence>
          <recordConfidence>
            <!--opt, enum, subType:string-->low
          </recordConfidence>
        </FieldDetectionRegion>
      </FieldDetectionRegionList>
      <humanMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering (human)-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
      </vehicleMisinfoFilterEnabled>
    </FieldDetection>
  </FieldDetectionList>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.1.8 Get intrusion detection parameters of all channels

Request URL

GET /ISAPI/Smart/FieldDetection

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, intrusion detection parameter list, subType:object, attr:version{req, string, protocolVersion}-->
  <FieldDetection>
    <!--ro, opt, object, intrusion detection parameters-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <intelliBackSearch>
      <!--ro, opt, bool, whether the device supports video dual-VCA-->true
    </intelliBackSearch>
    <startTriggerTime>
      <!--ro, req, int, unit:ms-->1
    </startTriggerTime>
    <endTriggerTime>
      <!--ro, req, int, unit:ms-->1
    </endTriggerTime>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <minObjectSize>
      <!--ro, opt, int, minimum pixels of each object-->1
    </minObjectSize>
    <maxObjectSize>
      <!--ro, opt, int, maximum pixels of each object-->1
    </maxObjectSize>
    <FieldDetectionRegionList size="4">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <FieldDetectionRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->1
        </id>
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
        <sensitivityLevel>
          <!--ro, req, int, sensitivity-->1
        </sensitivityLevel>
        <timeThreshold>
          <!--ro, req, int, time threshold-->1
        </timeThreshold>
        <objectOccupation>
          <!--ro, req, int, percentage-->1
        </objectOccupation>
        <detectionTarget>
          <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <RegionCoordinatesList>
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <humanMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
      </FieldDetectionRegion>
    </FieldDetectionRegionList>
    <humanMisinfoFilterEnabled>
```

drackum@gmail.com
DETEKTOR

```
      <!--ro, opt, bool-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </FieldDetection>
</FieldDetectionList>
```

14.8.1.9 Get intrusion detection parameters by video input channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, intrusion detection, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <intelliBackSearch>
    <!--ro, opt, bool, whether the device supports video dual-VCA-->true
  </intelliBackSearch>
  <startTriggerTime>
    <!--ro, req, int, unit: milliseconds, unit:ms-->1
  </startTriggerTime>
  <endTriggerTime>
    <!--ro, req, int, unit: milliseconds, unit:ms-->1
  </endTriggerTime>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <minObjectSize>
    <!--ro, opt, int, minimum pixels of each object-->1
  </minObjectSize>
  <maxObjectSize>
    <!--ro, opt, int, maximum pixels of each object-->1
  </maxObjectSize>
  <FieldDetectionRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <FieldDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, ID-->1
      </id>
      <enabled>
        <!--ro, opt, bool, whether to enable the function, desc:whether to enable the function-->true
      </enabled>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity-->1
      </sensitivityLevel>
      <timeThreshold>
        <!--ro, req, int, time threshold-->1
      </timeThreshold>
      <objectOccupation>
        <!--ro, req, int, percentage-->1
      </objectOccupation>
      <detectionTarget>
        <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
      </detectionTarget>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
```

drackun@gmail.com
DETEKTOR

```
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <humanMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
        <ruleName>
          <!--ro, opt, string, range:[1,64]-->test
        </ruleName>
        <detectionTargetProperty>
          <!--ro, opt, object-->
          <human>
            <!--ro, opt, object, dep:or,{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,all},
{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,human}-->
            <propertyList>
              <!--ro, opt, array, subType:object-->
              <property>
                <!--ro, opt, enum, subType:string-->nonMotor
              </property>
            </propertyList>
          </human>
          <animal>
            <!--ro, opt, object, dep:or,{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,all},
{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,animal}-->
            <propertyList>
              <!--ro, opt, array, subType:object-->
              <property>
                <!--ro, opt, enum, subType:string-->large
              </property>
            </propertyList>
          </animal>
        </detectionTargetProperty>
      </FieldDetectionRegion>
    </FieldDetectionRegionList>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <contAlarmForStaticTargetEnabled>
      <!--ro, opt, bool-->true
    </contAlarmForStaticTargetEnabled>
    <targetAlarmInterval>
      <!--ro, opt, int, range:[1,5], step:1, unit:s, dep:and,{$.FieldDetection.targetAlarmInterval,eq,true}-->1
    </targetAlarmInterval>
    <stackTarget>
      <!--ro, opt, bool-->true
    </stackTarget>
    <targetSensitivityLevel>
      <!--ro, opt, int, range:[0,100]-->50
    </targetSensitivityLevel>
  </FieldDetection>
```

drackum@gmail.com
DETEKTOR

14.8.1.10 Set intrusion detection parameters by channel

Request URL

PUT /ISAPI/Smart/FieldDetection/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
```

```
<FieldDetection xmlns= http://www.isapi.org/ver20/XMLSchema version= 2.0 >
  <!--opt, object, intrusion detection parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <intelliBackSearch>
    <!--opt, bool, whether the device supports video dual-VCA-->true
  </intelliBackSearch>
  <startTriggerTime>
    <!--req, int, trigger start time, unit:ms-->1
  </startTriggerTime>
  <endTriggerTime>
    <!--req, int, trigger end time, unit:ms-->1
  </endTriggerTime>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <minObjectSize>
    <!--opt, int, minimum pixels of each object-->1
  </minObjectSize>
  <maxObjectSize>
    <!--opt, int, maximum pixels of each object-->1
  </maxObjectSize>
  <FieldDetectionRegionList size="4">
    <!--opt, array, intrusion detection region list, subType:object, attr:size{req, int}-->
    <FieldDetectionRegion>
      <!--opt, object, intrusion detection region-->
      <id>
        <!--req, string, region ID-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <sensitivityLevel>
        <!--req, int, sensitivity-->1
      </sensitivityLevel>
      <timeThreshold>
        <!--req, int, time threshold-->1
      </timeThreshold>
      <objectOccupation>
        <!--req, int, percentage-->1
      </objectOccupation>
      <detectionTarget>
        <!--opt, enum, detection target, subType:string-->test
      </detectionTarget>
      <RegionCoordinatesList>
        <!--opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <humanMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
      </vehicleMisinfoFilterEnabled>
      <priority>
        <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
      </priority>
      <alarmConfidence>
        <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </alarmConfidence>
      <recordConfidence>
        <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </recordConfidence>
    </FieldDetectionRegion>
  </FieldDetectionRegionList>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
</FieldDetection>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.1.11 Get configuration capability of intrusion detection by channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<FieldDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <FieldDetectionRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <FieldDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, region ID-->1
      </id>
      <minRegionCoordinatesNum>
        <!--ro, opt, int-->1
      </minRegionCoordinatesNum>
      <maxRegionCoordinatesNum>
        <!--ro, opt, int-->1
      </maxRegionCoordinatesNum>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity-->1
      </sensitivityLevel>
      <timeThreshold min="0" max="60" def="0">
        <!--ro, req, int, time threshold, attr:min{req, int},max{req, int},def{req, int}-->1
      </timeThreshold>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <humanMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
      </vehicleMisinfoFilterEnabled>
    </FieldDetectionRegion>
  </FieldDetectionRegionList>
  <isSupportHumanMisinfoFilter>
    <!--ro, opt, bool, whether it supports false human body alarm prevention, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will not be returned-->true
  </isSupportHumanMisinfoFilter>
  <isSupportVehicleMisinfoFilter>
    <!--ro, opt, bool, whether it supports false vehicle alarm prevention, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned-->true
  </isSupportVehicleMisinfoFilter>
</FieldDetection>
```

drackum@gmail.com
DETEKTOR

14.8.1.12 Get parameters of all intrusion detection regions of a specific channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <channelEnabled>
    <!--ro, opt, bool-->true
  </channelEnabled>
  <FieldDetectionRegion>
    <!--ro, opt, object, intrusion detection-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--ro, req, int, time threshold-->1
    </timeThreshold>
    <objectOccupation>
      <!--ro, req, int-->1
    </objectOccupation>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
    </detectionTarget>
    <RegionCoordinatesList>
      <!--ro, opt, array, list of area coordinates, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:confidence of alarm notification-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, confidence of video recording, subType:string, desc:confidence of video recording-->low
    </recordConfidence>
  </FieldDetectionRegion>
</FieldDetectionRegionList>
```

drackum@gmail.com
DEFEKTOR

14.8.1.13 Set parameters of all intrusion detection regions of a specified channel

Request URL

PUT /ISAPI/Smart/FieldDetection/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameters of all intrusion detection regions, subType:object, attr:version{req, string, protocolVersion}-->
  <FieldDetectionRegion>
    <!--opt, object, intrusion detection regions-->
    <id>
      <!--req, string, No.-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, sensitivity-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--req, int, time threshold-->1
    </timeThreshold>
    <objectOccupation>
      <!--req, int, percentage-->1
    </objectOccupation>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <RegionCoordinatesList>
      <!--opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </FieldDetectionRegion>
</FieldDetectionRegionList>
```

dracku m@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.1.14 Get parameters of an intrusion detection region of a specific channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <sensitivityLevel>
    <!--ro, req, int, sensitivity, range:[0,100]-->1
  </sensitivityLevel>
  <timeThreshold>
    <!--ro, req, int, time threshold-->1
  </timeThreshold>
  <objectOccupation>
    <!--ro, req, int-->1
  </objectOccupation>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
  </detectionTarget>
  <RegionCoordinatesList>
    <!--ro, opt, object, list of area coordinates-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates-->
      <positionX>
        <!--ro, req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:confidence of alarm notification: "low","mediumLow","mediumHigh","high","low"-
->low
  </alarmConfidence>
  <recordConfidence>
    <!--ro, opt, enum, confidence of video recording, subType:string, desc:confidence of recording video: "low","mediumLow","mediumHigh","high","low"-->low
  </recordConfidence>
  <ruleName>
    <!--ro, opt, string, range:[1,64]-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object, dep:or,{$.FieldDetectionRegion.detectionTarget,eq,all},{$.FieldDetectionRegion.detectionTarget,eq,human}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object, dep:or,{$.FieldDetectionRegion.detectionTarget,eq,all},{$.FieldDetectionRegion.detectionTarget,eq,animal}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</FieldDetectionRegion>
```

drackum@gmail.com
DETEKTOR

14.8.1.15 Set parameters of an intrusion detection region of a specified channel

Request URL

PUT /ISAPI/Smart/FieldDetection/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of an intrusion detection region of a specified channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, No.-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <sensitivityLevel>
    <!--req, int, sensitivity-->1
  </sensitivityLevel>
  <timeThreshold>
    <!--req, int, time threshold-->1
  </timeThreshold>
  <objectOccupation>
    <!--req, int, percentage-->1
  </objectOccupation>
  <detectionTarget>
    <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all" is
mutually exclusive with "others"-->all
  </detectionTarget>
  <RegionCoordinatesList>
    <!--opt, object, region coordinate list-->
    <RegionCoordinates>
      <!--opt, object, region coordinates-->
      <positionX>
        <!--req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </alarmConfidence>
  <recordConfidence>
    <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </recordConfidence>
</FieldDetectionRegion>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.1.16 Get configuration capability of intrusion detection by channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/regions/<regionID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, detection area ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <sensitivityLevel min="1" max="100">
    <!--ro, req, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
  <timeThreshold min="1" max="100" def="1">
    <!--ro, req, int, range:[1,100], attr:min{req, int},max{req, int},def{req, int}-->1
  </timeThreshold>
  <objectOccupation min="1" max="100">
    <!--ro, req, int, range:[1,100], attr:min{req, int},max{req, int}-->1
  </objectOccupation>
  <maxRegionCoordinatesNum>
    <!--ro, req, int, the maximum number of coordinates-->1
  </maxRegionCoordinatesNum>
  <minRegionCoordinatesNum>
    <!--ro, req, int, the minimum number of coordinates-->1
  </minRegionCoordinatesNum>
  <detectionTarget opt="all,human,vehicle,animal,others">
    <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string}, desc:detection target-->all
  </detectionTarget>
  <RegionCoordinatesList>
    <!--ro, opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <humanMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, false alarm reduction of human body, attr:opt{req, string}-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, false alarm reduction of vehicle, attr:opt{req, string}-->true
  </vehicleMisinfoFilterEnabled>
  <priority opt="low,middle,high">
    <!--ro, opt, enum, priority, subType:string, attr:opt{req, string}, desc:priority: "low,middle,high"-->low
  </priority>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
  </isSupportAllTargetDetectionWebDisplay>
  <ruleName min="1" max="64">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object-->
      <propertyList size="1">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="nonMotor">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object-->
      <propertyList size="3">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="large,small,poultry">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</FieldDetectionRegion>
```

dracktm@gmail.com
DETEKTOR

14.8.1.17 Intrusion detection

EventType:fielddetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
```

```
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"fielddetection" (intrusion detection)-->fielddetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
        </height>
      </TargetRect>
      <sceneID>
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
```

drackum@gmail.com
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```
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <targetSpeed>
    <!--ro, opt, int-->0
  </targetSpeed>
```

drackum@gmail.com
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```
  <targetDistance>
    <!--ro, opt, int, unit:m-->2
  </targetDistance>
  <triggerType>
    <!--ro, opt, enum, subType:string-->video
  </triggerType>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits--
>test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <nonTargetRate>
    <!--ro, opt, int, range:[1,100], unit:%-->1
  </nonTargetRate>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </longitude>
    <latitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, range:[0,90]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </latitude>
  </deviceLocation>
  <laserRanging>
    <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
  </laserRanging>
</EventNotificationAlert>
```

drackum@gmail.com
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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
fielddetection	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

dracklin@gmail.com DETEKTOR

14.8.2 越界侦测

14.8.2.1 Get the arming schedules of line crossing detection of all channels

Request URL

GET /ISAPI/Event/schedules/lineDetections

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the arming schedules of line crossing detection of all channels, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule information-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->lineDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming time-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LineDetectionScheduleList>
```

drackum@gmail.com
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14.8.2.2 Set line crossing detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/lineDetections

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming information-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->lineDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LineDetectionScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.2.3 Set alarm linkage parameters of line crossing detection by channel

Request URL

PUT /ISAPI/Event/triggers/LineDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, linkage action list, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object, linkage action-->
    <id>
      <!--req, string, No.-->test
    </id>
    <notificationMethod>
      <!--req, enum, linkage action, subType:string, desc:"email", "IM" (instant messaging), "IO" (I/O output), "syslog" (system log), "HTTP" (uploading via
HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "monitorAlarm" (alarm on monitor), "center" (uploading to the center),
"LightAudioAlarm" (light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio), "supplementLight" (supplement
light)-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, linkage time interval, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string, video channel ID-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string, dynamic video input channel ID-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object, PTZ linkage-->
      <ptzChannelID>
        <!--req, string, PTZ linked channel No.-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object, linkage parameters of the white light, desc:it is valid when the value of notificationMethod is "whiteLight"-->
      <whiteLightDurationTime>
        <!--opt, int, light flashing duration, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, supplement light alarm duration, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

dr4ckum@gmail.com
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14.8.2.4 Get alarm linkage parameters of a line crossing detection channel

Request URL

GET /ISAPI/Event/triggers/LineDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object, linkage type-->
    <id>
      <!--ro, req, string-->test
    </id>
    <notificationMethod>
      <!--ro, req, enum, linkage type, subType:string, desc:linkage type-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

drackum@gmail.com
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14.8.2.5 Get the linkage parameters of line crossing detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/LineDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->linedetection
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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14.8.2.6 Set the linkage parameters of line crossing detection event for a specified channel

Request URL

PUT /ISAPI/Event/triggers/LineDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->LineDetection
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
```

```
    </inputIOPortID>
    <dynInputIOPortID>
      <!--opt, string-->1
    </dynInputIOPortID>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <dynVideoInputChannelID>
      <!--opt, string-->1
    </dynVideoInputChannelID>
    <intervalBetweenEvents>
      <!--opt, int, unit:s-->1
    </intervalBetweenEvents>
    <WLSensorID>
      <!--opt, string-->1
    </WLSensorID>
    <EventTriggerNotificationList>
      <!--opt, array, subType:object-->
      <EventTriggerNotification>
        <!--opt, object-->
        <id>
          <!--req, string-->test
        </id>
        <notificationMethod>
          <!--req, enum, subType:string-->email
        </notificationMethod>
        <notificationRecurrence>
          <!--opt, enum, subType:string-->beginning
        </notificationRecurrence>
        <notificationInterval>
          <!--opt, int, unit:ms-->1
        </notificationInterval>
        <outputIOPortID>
          <!--opt, string-->1
        </outputIOPortID>
        <dynOutputIOPortID>
          <!--opt, string-->1
        </dynOutputIOPortID>
        <videoInputID>
          <!--opt, string-->1
        </videoInputID>
        <dynVideoInputID>
          <!--opt, string-->1
        </dynVideoInputID>
        <ptzAction>
          <!--opt, object-->
          <ptzChannelID>
            <!--req, string-->1
          </ptzChannelID>
          <actionName>
            <!--req, enum, subType:string-->preset
          </actionName>
          <actionNum>
            <!--opt, int-->1
          </actionNum>
          <presetDurationTime>
            <!--opt, int, unit:s-->1
          </presetDurationTime>
        </ptzAction>
        <cellphoneNumber>
          <!--opt, string, range:[1,11]-->test
        </cellphoneNumber>
        <LightAudioByIPAction>
          <!--opt, object-->
          <notificationType>
            <!--req, enum, subType:string-->audio
          </notificationType>
          <notificationName>
            <!--opt, string-->test
          </notificationName>
        </LightAudioByIPAction>
        <SupplementLightAlarm>
          <!--opt, object-->
          <durationTime>
            <!--opt, int, range:[0,90], unit:s-->5
          </durationTime>
        </SupplementLightAlarm>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.2.7 Set line crossing detection parameters of all channels

Request URL

PUT /ISAPI/Smart/LineDetection

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineDetection>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <intelliBackSearch>
      <!--opt, bool-->true
    </intelliBackSearch>
    <duration>
      <!--opt, int-->1
    </duration>
    <startTriggerTime>
      <!--opt, int-->1
    </startTriggerTime>
    <endTriggerTime>
      <!--opt, int-->1
    </endTriggerTime>
    <normalizedScreenSize>
      <!--opt, object-->
      <normalizedScreenWidth>
        <!--req, int-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <minObjectSize>
      <!--opt, int-->1
    </minObjectSize>
    <maxObjectSize>
      <!--opt, int-->1
    </maxObjectSize>
    <LineItemList size="4">
      <!--opt, array, subType:object, attr:size{req, int}-->
      <LineItem>
        <!--opt, object-->
        <id>
          <!--req, string-->1
        </id>
        <enabled>
          <!--req, bool, whether to enable the function-->true
        </enabled>
        <sensitivityLevel>
          <!--req, int, detection sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <directionSensitivity>
          <!--opt, enum, detection direction, subType:string, desc:detection direction-->any
        </directionSensitivity>
        <CoordinatesList>
          <!--opt, array, subType:object, range:[0,2]-->
          <Coordinates>
```

drackum@gmail.com
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```
            <Coordinates>
              <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->1
              </positionX>
              <positionY>
                <!--req, int, y-coordinate, range:[0,1000]-->1
              </positionY>
            </Coordinates>
          </CoordinatesList>
          <detectionTarget>
            <!--opt, enum, detection target, subType:string, desc:detection target-->all
          </detectionTarget>
          <humanMisinfoFilterEnabled>
            <!--opt, bool, whether to enable false alarm filtering (human body)-->true
          </humanMisinfoFilterEnabled>
          <vehicleMisinfoFilterEnabled>
            <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
          </vehicleMisinfoFilterEnabled>
          <priority>
            <!--opt, enum, priority, subType:string, desc:priority-->low
          </priority>
          <alarmConfidence>
            <!--opt, enum, subType:string-->low
          </alarmConfidence>
          <recordConfidence>
            <!--opt, enum, subType:string-->low
          </recordConfidence>
        </LineItem>
      </LineItemList>
      <recogRuleType>
        <!--opt, enum, subType:string-->vectorMode
      </recogRuleType>
      <humanMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering (human body)-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
      </vehicleMisinfoFilterEnabled>
    </LineDetection>
  </LineDetectionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
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14.8.2.8 Get line crossing detection parameters of all channels

Request URL

GET /ISAPI/Smart/LineDetection

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineDetection>
    <!--ro, opt, object-->
```

```
      <id>
        <!--ro, req, string, channel No.-->test
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <intelliBackSearch>
        <!--ro, opt, bool-->true
      </intelliBackSearch>
      <duration>
        <!--ro, opt, int, duration-->1
      </duration>
      <startTriggerTime>
        <!--ro, opt, int-->1
      </startTriggerTime>
      <endTriggerTime>
        <!--ro, opt, int-->1
      </endTriggerTime>
      <normalizedScreenSize>
        <!--ro, opt, object-->
        <normalizedScreenWidth>
          <!--ro, req, int-->1
        </normalizedScreenWidth>
        <normalizedScreenHeight>
          <!--ro, req, int-->1
        </normalizedScreenHeight>
      </normalizedScreenSize>
      <minObjectSize>
        <!--ro, opt, int-->1
      </minObjectSize>
      <maxObjectSize>
        <!--ro, opt, int-->1
      </maxObjectSize>
      <LineItemList size="4">
        <!--ro, opt, array, subType:object, attr:size{req, int}-->
        <LineItem>
          <!--ro, opt, object-->
          <id>
            <!--ro, req, string, ID-->1
          </id>
          <enabled>
            <!--ro, req, bool, whether to enable the function-->true
          </enabled>
          <sensitivityLevel>
            <!--ro, req, int, detection sensitivity, range:[0,100]-->1
          </sensitivityLevel>
          <directionSensitivity>
            <!--ro, opt, enum, detection direction, subType:string, desc:detection direction-->any
          </directionSensitivity>
          <CoordinatesList>
            <!--ro, opt, array, rectangle, subType:object, range:[0,2], desc:rectangle-->
            <Coordinates>
              <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--ro, req, int, X-coordinate, range:[0,1000]-->1
              </positionX>
              <positionY>
                <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
              </positionY>
            </Coordinates>
          </CoordinatesList>
          <detectionTarget>
            <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
          </detectionTarget>
          <humanMisinfoFilterEnabled>
            <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
          </humanMisinfoFilterEnabled>
          <vehicleMisinfoFilterEnabled>
            <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
          </vehicleMisinfoFilterEnabled>
          <priority>
            <!--ro, opt, enum, priority, subType:string, desc:priority-->low
          </priority>
          <alarmConfidence>
            <!--ro, opt, enum, subType:string-->low
          </alarmConfidence>
          <recordConfidence>
            <!--ro, opt, enum, subType:string-->low
          </recordConfidence>
        </LineItem>
      </LineItemList>
      <recogRuleType>
        <!--ro, opt, enum, subType:string-->vectorMode
      </recogRuleType>
      <humanMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
      </vehicleMisinfoFilterEnabled>
    </LineDetection>
  </LineDetectionList>
```

dracknm@gmail.com
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14.8.2.9 Set line crossing detection parameters by channel

Request URL

PUT /ISAPI/Smart/LineDetection/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, line crossing detection parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <intelliBackSearch>
    <!--opt, bool, whether to enable dual-VCA-->true
  </intelliBackSearch>
  <duration>
    <!--opt, int, duration-->1
  </duration>
  <startTriggerTime>
    <!--opt, int, trigger start time-->1
  </startTriggerTime>
  <endTriggerTime>
    <!--opt, int, trigger end time-->1
  </endTriggerTime>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <minObjectSize>
    <!--opt, int, minimum number of pixels of each object-->1
  </minObjectSize>
  <maxObjectSize>
    <!--opt, int, maximum number of pixels of each object-->1
  </maxObjectSize>
  <LineItemList size="4">
    <!--opt, array, subType:object, attr:size{req, int}-->
    <id>
      <!--req, string, No.-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, detection sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <directionSensitivity>
      <!--opt, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any" (all directions)-
->any
    </directionSensitivity>
    <CoordinatesList>
      <!--opt, array, coordinates list, subType:object, range:[0,2], desc:line-->
      <Coordinates>
        <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </Coordinates>
    </CoordinatesList>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
```

drackum@gmail.com
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```
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </LineItemList>
  <recogRuleType>
    <!--opt, enum, direction recognition mode, subType:string, desc:"vectorMode" (vector mode), "slopeMode" (slope mode)-->vectorMode
  </recogRuleType>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
</LineDetection>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

dracknum@gmail.com
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14.8.2.10 Get configuration capability of line crossing detection by channel

Request URL

GET /ISAPI/Smart/LineDetection/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <LineItemList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <LineItem>
      <!--ro, opt, object-->
      <id>
```

```
        <!--ro, req, string-->test
      </id>
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
      </enabled>
      <sensitivityLevel min="1" max="100">
        <!--ro, req, int, detection sensitivity, attr:min{req, int},max{req, int}-->1
      </sensitivityLevel>
      <directionSensitivity opt="left-right,right-left,any">
        <!--ro, opt, enum, detection direction, subType:string, attr:opt{req, string}, desc:detection direction-->any
      </directionSensitivity>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,2], desc:line-->
        <RegionCoordinates>
          <!--ro, req, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget opt="all,human,vehicle,animal,others" def="human">
        <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string},def{req, string}, desc:detection target-->all
      </detectionTarget>
      <alarmConfidence opt="low,mediumLow,mediumHigh,high" def="low">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
      </alarmConfidence>
      <recordConfidence opt="low,mediumLow,mediumHigh,high" def="low">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
      </recordConfidence>
      <ruleName min="1" max="64">
        <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
      </ruleName>
      <detectionTargetProperty>
        <!--ro, opt, object-->
        <human>
          <!--ro, opt, object-->
          <propertyList size="1">
            <!--ro, opt, object, attr:size{req, int}-->
            <property opt="nonMotor">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
            </property>
          </propertyList>
        </human>
        <animal>
          <!--ro, opt, object-->
          <propertyList size="3">
            <!--ro, opt, object, attr:size{req, int}-->
            <property opt="large,small,poultry">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
            </property>
          </propertyList>
        </animal>
      </detectionTargetProperty>
    </LineItem>
  </LineItemList>
  <isSupportPerimeterDetection opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isSupportPerimeterDetection>
  <isSupportMultiScene opt="true,false">
    <!--ro, opt, bool, whether supports multiple scene detection, attr:opt{req, string}-->true
  </isSupportMultiScene>
  <recogRuleType>
    <!--ro, opt, enum, subType:string-->vectorMode
  </recogRuleType>
  <isSupportHumanMisinfoFilter>
    <!--ro, opt, bool, whether supports false human body alarm prevention, desc:If this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned.-->true
  </isSupportHumanMisinfoFilter>
  <isSupportVehicleMisinfoFilter>
    <!--ro, opt, bool, whether supports false vehicle alarm prevention, desc:If this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned.-->true
  </isSupportVehicleMisinfoFilter>
  <isSupportTargetMultiSelect>
    <!--ro, opt, bool, whether supports selecting multiple detection targets, desc:If this function is supported, this node will be returned and its value
is true; otherwise, this node will not be returned.-->true
  </isSupportTargetMultiSelect>
  <isSupportAllDayUpload>
    <!--ro, opt, bool, whether supports uploading all-day event-->true
  </isSupportAllDayUpload>
  <isSupportOverlapParams>
    <!--ro, opt, bool-->true
  </isSupportOverlapParams>
  <warningSystemEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </warningSystemEnabled>
  <isSupportStackTarget>
    <!--ro, opt, bool-->true
  </isSupportStackTarget>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
```

drackum@gmail.com
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```
    </isSupportAllTargetDetectionWebDisplay>
    <targetSensitivityLevel min="0" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
    </targetSensitivityLevel>
  </LineDetection>
```

14.8.2.11 Add detection lines for a specific channel

Request URL

POST /ISAPI/Smart/LineDetection/<channelID>/lineItem

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItemList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineItem>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, detection sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <directionSensitivity>
      <!--ro, opt, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any"-->any
    </directionSensitivity>
    <CoordinatesList>
      <!--ro, opt, array, list of coordinates, subType:object, range:[0,2], desc:line-->
      <Coordinates>
        <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </Coordinates>
    </CoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:"low", "middle" "high"-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, confidence of alarm notification, subType:string-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, confidence of video recording, subType:string-->low
    </recordConfidence>
  </LineItem>
</LineItemList>
```

drackum@gmail.com
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14.8.2.12 Set parameters of all line crossing detection lines by channel

Request URL

PUT /ISAPI/Smart/LineDetection/<channelID>/lineItem

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItemList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameters of all line crossing detection lines, subType:object, attr:version{req, string, protocolVersion}-->
  <LineItem>
    <!--opt, object, parameters of a line crossing detection line-->
    <id>
      <!--req, string, No.-->test
    </id>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, detection sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <directionSensitivity>
      <!--opt, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any" (all directions)-
->any
    </directionSensitivity>
    <CoordinatesList>
      <!--opt, array, coordinates list, subType:object, range:[0,2], desc:line-->
      <Coordinates>
        <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </Coordinates>
    </CoordinatesList>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </LineItem>
</LineItemList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

dracku m@gmail.com
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14.8.2.13 Get parameters of all detection lines of a specific channel

Request URL

GET /ISAPI/Smart/LineDetection/<channelID>/lineItem

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItemList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineItem>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, detection sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <directionSensitivity>
      <!--ro, opt, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any"-->any
    </directionSensitivity>
    <CoordinatesList>
      <!--ro, opt, array, list of coordinates, subType:object, range:[0,2], desc:line-->
      <Coordinates>
        <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </Coordinates>
    </CoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, subType:string-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, subType:string-->low
    </recordConfidence>
  </LineItem>
</LineItemList>
```

drackum@gmail.com
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14.8.2.14 Delete all detection lines of a specific channel

Request URL

DELETE /ISAPI/Smart/LineDetection/<channelID>/lineItem

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.2.15 Get parameters of a detection line of a specific channel

Request URL

GET /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
ruleID	string	Rule ID

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable the function, desc:whether to enable the function-->true
  </enabled>
  <sensitivityLevel>
    <!--ro, req, int, detection sensitivity-->1
  </sensitivityLevel>
  <directionSensitivity>
    <!--ro, opt, enum, detection direction, subType:string, desc:"left-right,right-left,any"-->any
  </directionSensitivity>
  <CoordinatesList>
    <!--ro, opt, array, line, subType:object, range:[0,2], desc:line-->
    <Coordinates>
      <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </Coordinates>
  </CoordinatesList>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all,human,vehicle,others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm reduction of vehicle-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--ro, opt, enum, priority, subType:string, desc:“low”, “middle”, “high”-->low
  </priority>
  <alarmConfidence>
    <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:"low", "mediumLow", "mediumHigh", "high", "low”-->low
  </alarmConfidence>
  <recordConfidence>
    <!--ro, opt, enum, confidence of recording video, subType:string, desc:"low", "mediumLow", "mediumHigh", "high", "low”-->low
  </recordConfidence>
  <ruleName>
    <!--ro, opt, string, range:[1,64]-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object, dep:or,{$.LineItem.detectionTarget,eq,all},{$.LineItem.detectionTarget,eq,human}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object, dep:or,{$.LineItem.detectionTarget,eq,all},{$.LineItem.detectionTarget,eq,animal}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</LineItem>
```

drackum@gmail.com
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14.8.2.16 Set parameters of a detection line of a specified channel

Request URL

PUT /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
ruleID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of a detection line, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, No.-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <sensitivityLevel>
    <!--req, int, detection sensitivity-->1
  </sensitivityLevel>
  <directionSensitivity>
    <!--opt, enum, detection direction, subType:string, desc:"Left-right" (from left to right), "right-left" (from right to left), "any" (all directions)--
>any
  </directionSensitivity>
  <CoordinatesList>
    <!--opt, array, coordinates list, subType:object, range:[0,2], desc:line-->
    <Coordinates>
      <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </Coordinates>
  </CoordinatesList>
  <detectionTarget>
    <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all" is
mutually exclusive with "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </alarmConfidence>
  <recordConfidence>
    <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </recordConfidence>
</LineItem>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.2.17 Delete a detection line of a specific channel

Request URL

DELETE /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--
ruleID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.2.18 Get configuration capability of line crossing detection by channel

Request URL

GET /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
ruleID	string	Rule ID

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <sensitivityLevel min="1" max="100">
    <!--ro, req, int, sensitivity, attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
  <directionSensitivity opt="left-right,right-left,any">
    <!--ro, opt, enum, direction, subType:string, attr:opt{req, string}, desc:direction-->any
  </directionSensitivity>
  <CoordinatesList>
    <!--ro, opt, array, line, subType:object, range:[0,2], desc:line-->
    <Coordinates>
      <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,1000]-->1
      </positionY>
    </Coordinates>
  </CoordinatesList>
  <detectionTarget opt="all,human,vehicle,animal,others">
    <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string}, desc:detection target-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable false alarm filtering (human body), attr:opt{req, string}-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle), attr:opt{req, string}-->true
  </vehicleMisinfoFilterEnabled>
  <priority opt="low,middle,high">
    <!--ro, opt, enum, priority, subType:string, attr:opt{req, string}, desc:priority-->low
  </priority>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
  </isSupportAllTargetDetectionWebDisplay>
  <ruleName min="1" max="64">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object-->
      <propertyList size="1">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="nonMotor">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object-->
      <propertyList size="3">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="large,small,poultry">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</LineItem>
```

14.8.2.19 Line crossing detection

EventType:linedetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
```

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```
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"linedetection" (Line Crossing Detection)-->linedetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
        </height>
      </TargetRect>
      <sceneID>
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
```

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```
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <targetSpeed>
    <!--ro, opt, datetime-->2019-12-30T15:29:23+08:00
  </targetSpeed>
  <targetDistance>
    <!--ro, opt, int, unit:m-->2
  </targetDistance>
  <triggerType>
    <!--ro, opt, enum, subType:string-->video
  </triggerType>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits-->test
```

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```
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <applicationScene>
    <!--ro, opt, enum, Application Scene, subType:string-->indoor
  </applicationScene>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </longitude>
    <latitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, range:[0,90]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </latitude>
  </deviceLocation>
  <laserRanging>
    <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
  </laserRanging>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
linedetection	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

dracklin@gmail.com DETEKTOR

14.8.3 进入区域

14.8.3.1 Set region entrance detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/regionEntrances

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, string, event type-->regionEntrance
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, day of a week, subType:int, desc:day of a week-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionEntranceScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.3.2 Get arming schedules of all channels for region entrance detection

Request URL

GET /ISAPI/Event/schedules/regionEntrances

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, configuration of region entrance detection arming schedules-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->1
    </id>
    <eventType>
      <!--ro, opt, string, event type-->regionEntrance
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, list of arming schedules, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming schedule-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionEntranceScheduleList>
```

drackum@gmail.com
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14.8.3.3 Get the linkage triggering parameters of region entrance for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionEntrance-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

drackum@gmail.com
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14.8.3.4 Set the linkage triggering parameters of region entrance for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionEntrance-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <notificationMethod>
      <!--req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object-->
      <ptzChannelID>
        <!--req, string-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object-->
      <whiteLightDurationTime>
        <!--opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.3.5 Get the linkage parameters of region entrance for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionEntrance-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->regionEntrance
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <triggerEnabled>
        <!--ro, opt, bool-->true
      </triggerEnabled>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->1
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->1
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
```

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```
            <!--ro, req, enum, subType:string-->preset
          </actionName>
          <actionNum>
            <!--ro, opt, int-->1
          </actionNum>
        </ptzAction>
        <WhiteLightAction>
          <!--ro, opt, object-->
          <whiteLightDurationTime>
            <!--ro, opt, int, range:[1,60], unit:s-->1
          </whiteLightDurationTime>
          <lightFlashFrequency>
            <!--ro, opt, enum, subType:string-->high
          </lightFlashFrequency>
          <brightness>
            <!--ro, opt, int, range:[0,100]-->1
          </brightness>
          <timeRangeList>
            <!--ro, opt, array, subType:object-->
            <timeRangeItem>
              <!--ro, opt, object-->
              <week>
                <!--ro, req, enum, subType:int-->1
              </week>
              <timeRange>
                <!--ro, req, array, subType:object-->
                <timeItem>
                  <!--ro, req, object-->
                  <timeRangeID>
                    <!--ro, req, int-->1
                  </timeRangeID>
                  <beginTime>
                    <!--ro, req, string-->08:00
                  </beginTime>
                  <endTime>
                    <!--ro, req, string-->24:00
                  </endTime>
                </timeItem>
              </timeRange>
            </timeRangeItem>
          </timeRangeList>
        </WhiteLightAction>
        <SupplementLightAlarm>
          <!--ro, opt, object-->
          <durationTime>
            <!--ro, opt, int, range:[0,90], unit:s-->5
          </durationTime>
        </SupplementLightAlarm>
        <customOverHTTPList>
          <!--ro, opt, array, subType:object, dep:and,
{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
          <customOverHTTP>
            <!--ro, opt, object-->
            <id>
              <!--ro, req, int-->1
            </id>
            <enabled>
              <!--ro, req, bool-->true
            </enabled>
            <hostID>
              <!--ro, opt, int-->1
            </hostID>
            <customInfo>
              <!--ro, opt, string, range:[1,512]-->test
            </customInfo>
          </customOverHTTP>
        </customOverHTTPList>
        <AudioAction>
          <!--ro, opt, object-->
          <audioID>
            <!--ro, opt, int-->1
          </audioID>
          <alarmTimes>
            <!--ro, opt, int, range:[0,9]-->1
          </alarmTimes>
          <childDevID>
            <!--ro, opt, string, range:[9,32]-->3434645611
          </childDevID>
          <childDevAudioEnabled>
            <!--ro, opt, bool-->true
          </childDevAudioEnabled>
          <childDevAudioRuleList>
            <!--ro, opt, array, subType:object, range:[1,3]-->
            <childDevAudioRuleItem>
              <!--ro, req, object-->
              <ruleIndexID>
                <!--ro, req, int, range:[1,3]-->1
              </ruleIndexID>
              <childDevAudioID>
                <!--ro, req, int, range:[1,64]-->1
              </childDevAudioID>
              <ruleName>
                <!--ro, opt, string, range:[1,128]-->test
```

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```
            </ruleName>
            <alarmTimes>
              <!--ro, opt, int, range:[1,50]-->1
            </alarmTimes>
          </childDevAudioRuleItem>
        </childDevAudioRuleList>
        <timeRangeList>
          <!--ro, opt, array, subType:object-->
          <timeRangeItem>
            <!--ro, opt, object-->
            <week>
              <!--ro, req, enum, subType:int-->1
            </week>
            <timeRange>
              <!--ro, req, array, subType:object-->
              <timeItem>
                <!--ro, req, object-->
                <timeRangeID>
                  <!--ro, req, int-->1
                </timeRangeID>
                <ruleIndexID>
                  <!--ro, opt, int, range:[1,3]-->1
                </ruleIndexID>
                <childDevAudioID>
                  <!--ro, opt, int, range:[1,64]-->1
                </childDevAudioID>
                <beginTime>
                  <!--ro, req, string-->08:00
                </beginTime>
                <endTime>
                  <!--ro, req, string-->24:00
                </endTime>
              </timeItem>
            </timeRange>
          </timeRangeItem>
        </timeRangeList>
      </AudioAction>
      <audioAlarmTriggerParam>
        <!--ro, opt, object, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,audio}-->
        <audioIndexList>
          <!--ro, opt, array, subType:object, range:[1,3]-->
          <audioIndexItem>
            <!--ro, req, object-->
            <indexID>
              <!--ro, req, int, range:[1,3]-->1
            </indexID>
            <indexName>
              <!--ro, opt, string, range:[1,128]-->test
            </indexName>
            <alarmTimes>
              <!--ro, opt, int, range:[1,50]-->1
            </alarmTimes>
            <audioID>
              <!--ro, opt, enum, subType:int-->1
            </audioID>
          </audioIndexItem>
        </audioIndexList>
        <timeRangeList>
          <!--ro, opt, array, subType:object-->
          <timeRangeItem>
            <!--ro, opt, object-->
            <week>
              <!--ro, req, enum, subType:int-->1
            </week>
            <timeRange>
              <!--ro, req, array, subType:object-->
              <timeItem>
                <!--ro, req, object-->
                <timeRangeID>
                  <!--ro, req, int-->1
                </timeRangeID>
                <indexID>
                  <!--ro, opt, int, range:[1,3]-->1
                </indexID>
                <audioID>
                  <!--ro, opt, enum, subType:int-->1
                </audioID>
                <beginTime>
                  <!--ro, req, string-->08:00
                </beginTime>
                <endTime>
                  <!--ro, req, string-->24:00
                </endTime>
              </timeItem>
            </timeRange>
          </timeRangeItem>
        </timeRangeList>
        <audioAlarmTriggerEnabled>
          <!--ro, opt, bool-->true
        </audioAlarmTriggerEnabled>
      </audioAlarmTriggerParam>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

drackum@gmail.com
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14.8.3.6 Set the linkage parameters of region entrance for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionEntrance-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->regionEntrance
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->1
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->1
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <SupplementLightAlarm>
        <!--opt, object-->
        <durationTime>
          <!--opt, int, range:[0,90], unit:s-->5
        </durationTime>
      </SupplementLightAlarm>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.3.7 Region entrance

EventType:regionEntrance

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"regionEntrance" (region entrance)-->regionEntrance
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area points coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
```

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```
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <detectionTarget>
          <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
        </detectionTarget>
        <TargetRect>
          <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
          <X>
            <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
          </X>
          <Y>
            <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
          </Y>
          <width>
            <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
          </width>
          <height>
            <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
          </height>
        </TargetRect>
        <sceneID>
          <!--ro, opt, int, scene ID-->1
        </sceneID>
      </DetectionRegionEntry>
    </DetectionRegionList>
    <channelName>
      <!--ro, opt, string, channel name, range:[1,64]-->test
    </channelName>
    <Extensions>
      <!--ro, opt, object, extended information-->
      <serialNumber>
        <!--ro, opt, string, serial No.-->test
      </serialNumber>
      <eventPush>
        <!--ro, opt, string, event notification-->test
      </eventPush>
    </Extensions>
    <detectionPictureTransType>
      <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
    </detectionPictureTransType>
    <detectionPicturesNumber>
      <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
    </detectionPicturesNumber>
    <bkgUrl>
      <!--ro, opt, string, background picture URL, desc:background picture URL-->test
    </bkgUrl>
    <URLCertificationType>
      <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication，"digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
    </URLCertificationType>
    <pId>
      <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
    </pId>
    <visibleLightURL>
      <!--ro, opt, string, visible light picture URL-->test
    </visibleLightURL>
    <thermalURL>
      <!--ro, opt, string, thermal imaging picture URL-->test
    </thermalURL>
    <targetAttrs>
      <!--ro, opt, object, transparent transmission field-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选，设备ID，string类型，最大长度为
64*/ "deviceChannel": 1, /*可选，设备通道号，integer32类型*/ "deviceName": "shebei", /*可选，设备名称，string类型，最大长度为128*/ }
    </targetAttrs>
    <TaskInfo>
      <!--ro, opt, object, task information submitted by data center products-->
      <taskID>
        <!--ro, req, string, task ID, range:[0,64]-->test
      </taskID>
      <algorithmType>
        <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
      </algorithmType>
      <taskName>
        <!--ro, opt, string, task name-->test
      </taskName>
      <algorithmID>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmID>
      <monitorPointID>
        <!--ro, opt, string, camera No.-->test
      </monitorPointID>
      <monitorPointName>
        <!--ro, opt, string, camera name-->test
      </monitorPointName>
      <streamType>
        <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "Localvideo" (local stream)--
>realtime
      </streamType>
```

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```
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits--
>test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:Rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:Rule Name-->test
  </ruleName>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
```

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```
      </longitude>
      <latitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, range:[0,90]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </latitude>
    </deviceLocation>
    <laserRanging>
      <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
    </laserRanging>
  </EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
regionEntrance	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

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- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

14.8.4 离开区域

14.8.4.1 Get arming schedules of all channels for region exiting detection

Request URL

GET /ISAPI/Event/schedules/regionExitings

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, list of arming schedules, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->test
    </id>
    <eventType>
      <!--ro, opt, string, event type-->regionExiting
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, list of arming schedules, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming schedule-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionExitingScheduleList>
```

drackum@gmail.com
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14.8.4.2 Set region exiting detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/regionExitings

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, string, event type-->regionExiting
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionExitingScheduleList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.4.3 Get the linkage triggering parameters of region exiting for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionExiting-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, enum, subType:string-->regionExiting
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

drackum@gmail.com
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14.8.4.4 Set the linkage triggering parameters of region exiting for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionExiting-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object-->
    <id>
      <!--req, enum, subType:string-->regionExiting
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object-->
      <ptzChannelID>
        <!--req, string-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object-->
      <whiteLightDurationTime>
        <!--opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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14.8.4.5 Set the linkage parameters of region exiting for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionExiting-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, string-->regionExiting
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <LightAudioByIPAction>
        <!--opt, object-->
        <notificationType>
          <!--req, enum, subType:string-->audio
        </notificationType>
        <notificationName>
          <!--opt, string-->10.16.99.98:8000:光
        </notificationName>
      </LightAudioByIPAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

14.8.4.6 Get the linkage parameters of region exiting for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionExiting-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, string-->regionEntrance
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
```

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```
        <triggerEnabled>
          <!--ro, opt, bool-->true
        </triggerEnabled>
        <notificationMethod>
          <!--ro, req, enum, subType:string-->email
        </notificationMethod>
        <notificationRecurrence>
          <!--ro, opt, enum, subType:string-->beginning
        </notificationRecurrence>
        <notificationInterval>
          <!--ro, opt, int, unit:ms-->1
        </notificationInterval>
        <outputIOPortID>
          <!--ro, opt, string--> l
        </outputIOPortID>
        <dynOutputIOPortID>
          <!--ro, opt, string--> l
        </dynOutputIOPortID>
        <videoInputID>
          <!--ro, opt, string-->1
        </videoInputID>
        <dynVideoInputID>
          <!--ro, opt, string-->1
        </dynVideoInputID>
        <ptzAction>
          <!--ro, opt, object-->
          <ptzChannelID>
            <!--ro, req, string-->1
          </ptzChannelID>
          <actionName>
            <!--ro, req, enum, subType:string-->preset
          </actionName>
          <actionNum>
            <!--ro, opt, int-->1
          </actionNum>
        </ptzAction>
        <WhiteLightAction>
          <!--ro, opt, object-->
          <whiteLightDurationTime>
            <!--ro, opt, int, range:[1,60], unit:s-->1
          </whiteLightDurationTime>
          <lightFlashFrequency>
            <!--ro, opt, enum, subType:string-->high
          </lightFlashFrequency>
          <brightness>
            <!--ro, opt, int, range:[0,100]-->1
          </brightness>
          <timeRangeList>
            <!--ro, opt, array, subType:object-->
            <timeRangeItem>
              <!--ro, opt, object-->
              <week>
                <!--ro, req, enum, subType:int-->1
              </week>
              <timeRange>
                <!--ro, req, array, subType:object-->
                <timeItem>
                  <!--ro, req, object-->
                  <timeRangeID>
                    <!--ro, req, int-->1
                  </timeRangeID>
                  <beginTime>
                    <!--ro, req, string-->08:00
                  </beginTime>
                  <endTime>
                    <!--ro, req, string-->24:00
                  </endTime>
                </timeItem>
              </timeRange>
            </timeRangeItem>
          </timeRangeList>
        </WhiteLightAction>
        <SupplementLightAlarm>
          <!--ro, opt, object-->
          <durationTime>
            <!--ro, opt, int, range:[0,90], unit:s-->5
          </durationTime>
        </SupplementLightAlarm>
        <customOverHTTPList>
          <!--ro, opt, array, subType:object, dep:and,
{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
          <customOverHTTP>
            <!--ro, opt, object-->
            <id>
              <!--ro, req, int-->1
            </id>
            <enabled>
              <!--ro, req, bool-->true
            </enabled>
            <hostID>
              <!--ro, opt, int-->1
            </hostID>
            <customInfo>
```

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```
        <!--ro, opt, string, range:[1,512]-->test
      </customInfo>
    </customOverHTTP>
  </customOverHTTPList>
  <AudioAction>
    <!--ro, opt, object-->
    <audioID>
      <!--ro, opt, int-->1
    </audioID>
    <alarmTimes>
      <!--ro, opt, int, range:[0,9]-->1
    </alarmTimes>
    <childDevID>
      <!--ro, opt, string, range:[9,32]-->3434645611
    </childDevID>
    <childDevAudioEnabled>
      <!--ro, opt, bool-->true
    </childDevAudioEnabled>
    <childDevAudioRuleList>
      <!--ro, opt, array, subType:object, range:[1,3]-->
      <childDevAudioRuleItem>
        <!--ro, req, object-->
        <ruleIndexID>
          <!--ro, req, int, range:[1,3]-->1
        </ruleIndexID>
        <childDevAudioID>
          <!--ro, req, int, range:[1,64]-->1
        </childDevAudioID>
        <ruleName>
          <!--ro, opt, string, range:[1,128]-->test
        </ruleName>
        <alarmTimes>
          <!--ro, opt, int, range:[1,50]-->1
        </alarmTimes>
      </childDevAudioRuleItem>
    </childDevAudioRuleList>
    <timeRangeList>
      <!--ro, opt, array, subType:object-->
      <timeRangeItem>
        <!--ro, opt, object-->
        <week>
          <!--ro, req, enum, subType:int-->1
        </week>
        <timeRange>
          <!--ro, req, array, subType:object-->
          <timeItem>
            <!--ro, req, object-->
            <timeRangeID>
              <!--ro, req, int-->1
            </timeRangeID>
            <ruleIndexID>
              <!--ro, opt, int, range:[1,3]-->1
            </ruleIndexID>
            <childDevAudioID>
              <!--ro, opt, int, range:[1,64]-->1
            </childDevAudioID>
            <beginTime>
              <!--ro, req, string-->08:00
            </beginTime>
            <endTime>
              <!--ro, req, string-->24:00
            </endTime>
          </timeItem>
        </timeRange>
      </timeRangeItem>
    </timeRangeList>
  </AudioAction>
  <audioAlarmTriggerParam>
    <!--ro, opt, object, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,audio}-->
    <audioIndexList>
      <!--ro, opt, array, subType:object, range:[1,3]-->
      <audioIndexItem>
        <!--ro, req, object-->
        <indexID>
          <!--ro, req, int, range:[1,3]-->1
        </indexID>
        <indexName>
          <!--ro, opt, string, range:[1,128]-->test
        </indexName>
        <alarmTimes>
          <!--ro, opt, int, range:[1,50]-->1
        </alarmTimes>
        <audioID>
          <!--ro, opt, enum, subType:int-->1
        </audioID>
      </audioIndexItem>
    </audioIndexList>
    <timeRangeList>
      <!--ro, opt, array, subType:object-->
      <timeRangeItem>
        <!--ro, opt, object-->
        <week>
          <!--ro, req, enum, subType:int-->1
        </week>
```

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```
            </week>
            <timeRange>
              <!--ro, req, array, subType:object-->
              <timeItem>
                <!--ro, req, object-->
                <timeRangeID>
                  <!--ro, req, int-->1
                </timeRangeID>
                <indexID>
                  <!--ro, opt, int, range:[1,3]-->1
                </indexID>
                <audioID>
                  <!--ro, opt, enum, subType:int-->1
                </audioID>
                <beginTime>
                  <!--ro, req, string-->08:00
                </beginTime>
                <endTime>
                  <!--ro, req, string-->24:00
                </endTime>
              </timeItem>
            </timeRange>
          </timeRangeItem>
        </timeRangeList>
        <audioAlarmTriggerEnabled>
          <!--ro, opt, bool-->true
        </audioAlarmTriggerEnabled>
      </audioAlarmTriggerParam>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

14.8.4.7 Region exiting

EventType:regionExiting

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome".
The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP--
>HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"regionExiting" (region exiting)-->regionExiting
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
```

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```
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height-->1.000
        </height>
      </TargetRect>
      <sceneID>
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transparent transmission field-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为
64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
```

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```
      <algorithmID>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmID>
      <monitorPointID>
        <!--ro, opt, string, camera No.-->test
      </monitorPointID>
      <monitorPointName>
        <!--ro, opt, string, camera name-->test
      </monitorPointName>
      <streamType>
        <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
      </streamType>
      <ruleID>
        <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
      </ruleID>
      <ruleCustomName>
        <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
      </ruleCustomName>
      <algorithmInfo>
        <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
        <algorithmId>
          <!--ro, opt, string, algorithm package ID-->test
        </algorithmId>
        <algoVendorId>
          <!--ro, opt, string, manufacturer ID-->test
        </algoVendorId>
        <algoVersionId>
          <!--ro, opt, string, algorithm version No.-->test
        </algoVersionId>
        <algoChipType>
          <!--ro, opt, string, chip type-->test
        </algoChipType>
        <algoModelVersion>
          <!--ro, opt, string, model version-->test
        </algoModelVersion>
      </algorithmInfo>
    </TaskInfo>
    <timeStamp>
      <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
    </timeStamp>
    <uid>
      <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits--
>test
    </uid>
    <isDataRetransmission>
      <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
    </isDataRetransmission>
    <ruleID>
      <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
    </ruleID>
    <ruleName>
      <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
    </ruleName>
    <visibleLightAbsoluteHigh>
      <!--ro, opt, object-->
      <elevation>
        <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
      </elevation>
      <azimuth>
        <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
      </absoluteZoom>
      <focus>
        <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
      </focus>
    </visibleLightAbsoluteHigh>
    <thermalAbsoluteHigh>
      <!--ro, opt, object-->
      <elevation>
        <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
      </elevation>
      <azimuth>
        <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
      </absoluteZoom>
      <focus>
        <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
      </focus>
    </thermalAbsoluteHigh>
    <deviceLocation>
      <!--ro, opt, object-->
      <longitudeType>
        <!--ro, req, enum, subType:string-->W
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
```

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```
      </latitudeType>
      <longitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, degree(s), range:[0,180]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </longitude>
      <latitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, range:[0,90]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </latitude>
    </deviceLocation>
    <laserRanging>
      <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
    </laserRanging>
  </EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
regionExiting	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

drackum@gmail.com DETEKTOR

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

14.9 混合目标检测

14.9.1 全局目标侦测

14.9.1.1 Get the global target detection capability

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/humanRecognition/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HumanRecognition xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1000
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <HumanRecognitionRegionList size="8">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <HumanRecognitionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <regionName min="1" max="10">
        <!--ro, req, string, attr:min{req, int},max{req, int}-->test
      </regionName>
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <LineCoordinatesList size="2">
        <!--ro, opt, array, subType:object, range:[0,2], attr:size{req, int}-->
        <LineCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </LineCoordinates>
      </LineCoordinatesList>
    </HumanRecognitionRegion>
  </HumanRecognitionRegionList>
  <isSupportHumanMisinfoFilter>
    <!--ro, opt, bool-->true
  </isSupportHumanMisinfoFilter>
  <isSupportPicRecognition>
    <!--ro, opt, bool-->true
  </isSupportPicRecognition>
  <HumanRecognitionModeSearchCap>
    <!--ro, opt, object-->
    <searchTargetsNumMax>
      <!--ro, req, int-->5
    </searchTargetsNumMax>
    <HumanMode>
      <!--ro, opt, object-->
      <age_group opt="test">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unknown
      </age_group>
      <gender opt="test">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unknown
```

drackum@gmail.com
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```
        </gender>
        <jacet_color opt="test">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unknown
        </jacet_color>
        <glass opt="test">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->no
        </glass>
        <bag opt="test">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->yes
        </bag>
        <ride opt="test">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unknown
        </ride>
        <similarity min="0" max="100">
          <!--ro, opt, float, range:[0.0,100.0], attr:min{req, int},max{req, int}-->0.000
        </similarity>
      </HumanMode>
      <eventType opt="unlimit,humanRecognition">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unlimit
      </eventType>
    </HumanRecognitionModeSearchCap>
    <analysisParamList>
      <!--ro, opt, array, subType:object-->
      <analysisParam>
        <!--ro, opt, object-->
        <analysisMode opt="normal,groupBehavior,individualBehavior">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->normal
        </analysisMode>
        <captureInterval min="5" max="3600" def="12">
          <!--ro, opt, int, range:[5,3600], unit:s, attr:min{req, int},max{req, int},def{req, int}-->12
        </captureInterval>
        <maxRegionNum>
          <!--ro, opt, int-->1
        </maxRegionNum>
        <cruiseLinkageCapture opt="true,false">
          <!--ro, opt, bool, attr:opt{req, string}-->true
        </cruiseLinkageCapture>
        <analysisControl opt="auto,non-auto">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
        </analysisControl>
        <timeInterval min="1" max="60" def="12">
          <!--ro, opt, int, range:[0,60], unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
        </timeInterval>
      </analysisParam>
    </analysisParamList>
    <isSupportConfidenceConfig>
      <!--ro, opt, bool-->true
    </isSupportConfidenceConfig>
    <InterPupileRegion>
      <!--ro, opt, object-->
      <positionX min="0" max="1000">
        <!--ro, req, int, range:[0,1000], attr:min{req, int},max{req, int}-->1
      </positionX>
      <positionY min="0" max="1000">
        <!--ro, req, int, range:[0,1000], attr:min{req, int},max{req, int}-->1
      </positionY>
      <width min="0" max="1000">
        <!--ro, req, int, range:[0,1000], attr:min{req, int},max{req, int}-->1
      </width>
      <height min="0" max="1000">
        <!--ro, req, int, range:[0,1000], attr:min{req, int},max{req, int}-->1
      </height>
    </InterPupileRegion>
    <sid min="1" max="8">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </sid>
    <sceneDurationTime min="1" max="86400" def="5">
      <!--ro, opt, int, range:[1,86400], unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
    </sceneDurationTime>
    <targetCutout>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, opt, bool-->true
      </enabled>
      <size>
        <!--ro, req, int-->1
      </size>
      <zoom>
        <!--ro, req, int-->1
      </zoom>
    </targetCutout>
    <isSupportpicOverlay>
      <!--ro, opt, bool-->true
    </isSupportpicOverlay>
    <isSupportRealTimeShow>
      <!--ro, opt, bool-->true
    </isSupportRealTimeShow>
    <enabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </enabled>
  </HumanRecognition>
```

drackum@gmail.com
DETEKTOR

14.10 视频识别

14.10.1 专业智能异常行为检测(S)

14.10.1.1 Get the intelligent capability

Request URL

GET /ISAPI/Intelligent/capabilities

Query Parameter

None

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>
<IntelliCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <isFaceSupport>
    <!--ro, opt, bool, whether it supports face detection, desc:whether it supports face detection-->true
  </isFaceSupport>
  <isBehaviorSupport>
    <!--ro, opt, bool, whether it supports abnormal event detection, desc:whether it supports abnormal event detection-->true
  </isBehaviorSupport>
  <isLineDetectionSupport>
    <!--ro, opt, bool, whether it supports line crossing detection, desc:whether it supports line crossing detection-->true
  </isLineDetectionSupport>
  <isFieldDetectionSupport>
    <!--ro, opt, bool, whether it supports intrusion detection, desc:whether it supports intrusion detection-->true
  </isFieldDetectionSupport>
  <isRegionEntranceSupport>
    <!--ro, opt, bool, whether it supports region entrance detection, desc:whether it supports region entrance detection-->true
  </isRegionEntranceSupport>
  <isRegionExitingSupport>
    <!--ro, opt, bool, whether it supports region exiting detection, desc:whether it supports region exiting detection-->true
  </isRegionExitingSupport>
  <isLoiteringSupport>
    <!--ro, opt, bool, whether it supports loitering detection, desc:whether it supports loitering detection-->true
  </isLoiteringSupport>
  <isGroupSupport>
    <!--ro, opt, bool, whether it supports people gathering detection, desc:whether it supports people gathering detection-->true
  </isGroupSupport>
  <isRapidMoveSupport>
    <!--ro, opt, bool, whether it supports fast moving detection, desc:whether it supports fast moving detection-->true
  </isRapidMoveSupport>
  <isParkingSupport>
    <!--ro, opt, bool, whether it supports parking detection, desc:whether it supports parking detection-->true
  </isParkingSupport>
  <isUnattendedBaggageSupport>
    <!--ro, opt, bool, whether it supports unattended baggage detection, desc:whether it supports unattended baggage detection-->true
  </isUnattendedBaggageSupport>
  <isAttendedBaggageSupport>
    <!--ro, opt, bool, whether it supports object removal detection, desc:whether it supports object removal detection-->true
  </isAttendedBaggageSupport>
  <isTeacherSupport>
    <!--ro, opt, bool, whether it supports teachers' behavior detection, desc:whether it supports teacher's behavior detection-->true
  </isTeacherSupport>
  <isStudentSupport>
    <!--ro, opt, bool, whether it supports students' behavior detection, desc:whether it supports student's behavior detection-->true
  </isStudentSupport>
  <HumanRecognitionModeSearchCap>
    <!--ro, opt, object, whether it is false human body recognition alarm: "true"-yes, "false"-no, desc:whether it is false human body recognition alarm:
"true"-yes,"false"-no-->
    <searchTargetsNumMax>
      <!--ro, opt, int, maximum number of sample pictures that can be imported for searching by picture-->1
    </searchTargetsNumMax>
    <HumanMode>
      <!--ro, opt, object, similarity,range: [0.0,100.0]-->
      <similarity min="0.0" max="100.0">
        <!--ro, opt, float, range:[0.0,100.0], attr:min{req, int},max{req, int}-->0.000
      </similarity>
    </HumanMode>
    <eventType opt="unlimit,humanRecognition">
      <!--ro, req, enum, event type, subType:string, attr:opt{req, string}, desc:"unlimit"-no limit, "humanRecognition"-human body detection alarm-->unlimit
    </eventType>
    <isSupportMultiChannelSearch>
      <!--ro, opt, bool, whether it supports multi-channel search-->true
    </isSupportMultiChannelSearch>
    <isSupportTotalSearchResult>
      <!--ro, opt, bool, whether it supports limiting number of results that can be obtained after a single search-->true
    </isSupportTotalSearchResult>
  </HumanRecognitionModeSearchCap>
  <VehicleRecognitionModeSearchCap>
    <!--ro, opt, object, capability of searching for vehicles, desc:event type: "unlimit"-no limit,"vehicleBlackList"-vehicle blocklist,"vehicleWhiteList"-
vehicle allowlist-->
    <VehicleMode>
      <!--ro, opt, object, similarity,range: [0.0,100.0]-->
    </VehicleMode>
    <isSupportMultiChannelSearch>
      <!--ro, opt, bool, whether it supports multi-channel search-->true
    </isSupportMultiChannelSearch>
    <isSupportTotalSearchResult>
      <!--ro, opt, bool, whether it supports limiting number of results that can be obtained after a single search-->true
    </isSupportTotalSearchResult>
  </VehicleRecognitionModeSearchCap>
  <isSupportUploadFacePictureByForm>
    <!--ro, opt, bool, whether it supports uploading face pictures by form, desc:whether it supports uploading face pictures by form-->true
  </isSupportUploadFacePictureByForm>
</IntelliCap>
```

drackun@gmail.com
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14.10.1.2 Get the auxiliary area parameters capability

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/Auxiliary/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AuxiliaryCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <maxAreaNum>
    <!--ro, req, int-->0
  </maxAreaNum>
  <areaType opt="bedLocation">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->bedLocation
  </areaType>
  <Region>
    <!--ro, opt, object-->
    <vertexNum min="0" max="10">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->0
    </vertexNum>
  </Region>
</AuxiliaryCap>
```

14.10.1.3 Get the intelligent capability of device

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>
<IntelliCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <isFaceSupport>
    <!--ro, req, bool, whether it supports face detection, desc:whether it supports face detection-->true
  </isFaceSupport>
  <isBehaviorSupport>
    <!--ro, req, bool, whether it supports abnormal event detection, desc:whether it supports abnormal event detection-->true
  </isBehaviorSupport>
  <isLineDetectionSupport>
    <!--ro, req, bool, whether it supports line crossing detection, desc:whether it supports line crossing detection-->true
  </isLineDetectionSupport>
  <isFieldDetectionSupport>
    <!--ro, req, bool, whether it supports intrusion detection, desc:whether it supports intrusion detection-->true
  </isFieldDetectionSupport>
  <isRegionEntranceSupport>
    <!--ro, req, bool, whether it supports region entrance detection, desc:whether it supports region entrance detection-->true
  </isRegionEntranceSupport>
  <isRegionExitingSupport>
    <!--ro, req, bool, whether it supports region exiting detection, desc:whether it supports region exiting detection-->true
  </isRegionExitingSupport>
  <isLoiteringSupport>
    <!--ro, req, bool, whether it supports loitering detection, desc:whether it supports loitering detection-->true
  </isLoiteringSupport>
  <isGroupSupport>
    <!--ro, opt, bool, whether it supports people gathering detection, desc:whether it supports people gathering detection-->true
  </isGroupSupport>
  <isRapidMoveSupport>
    <!--ro, opt, bool, whether it supports fast moving detection, desc:whether it supports fast moving detection-->true
  </isRapidMoveSupport>
  <isParkingSupport>
    <!--ro, opt, bool, whether it supports parking detection, desc:whether it supports parking detection-->true
  </isParkingSupport>
  <isUnattendedBaggageSupport>
    <!--ro, opt, bool, whether it supports unattended baggage detection, desc:whether it supports unattended baggage detection-->true
  </isUnattendedBaggageSupport>
  <isAttendedBaggageSupport>
    <!--ro, opt, bool, whether it supports object removal detection, desc:whether it supports object removal detection-->true
  </isAttendedBaggageSupport>
  <isTeacherSupport>
    <!--ro, opt, bool, whether it supports teacher's behavior detection, desc:whether it supports teacher's behavior detection-->true
  </isTeacherSupport>
  <isStudentSupport>
    <!--ro, opt, bool, whether it supports student's behavior detection, desc:whether it supports student's behavior detection-->true
  </isStudentSupport>
</IntelliCap>
```

14.10.1.4 Get the capability of smart tracking parameters

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/intelliTrace/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliTraceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <IntelliTraceBlockList size="10">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <IntelliTraceBlock>
      <!--ro, opt, object-->
      <sid min="1" max="10">
        <!--ro, req, string, range:[1,10], attr:min{req, int},max{req, int}-->test
      </sid>
      <sceneName min="1" max="10">
        <!--ro, req, string, attr:min{req, int},max{req, int}-->test
      </sceneName>
      <patrolId min="1" max="10">
        <!--ro, opt, int, range:[0,10], attr:min{req, int},max{req, int}-->1
      </patrolId>
      <dwelltime min="15" max="720">
        <!--ro, opt, int, range:[15,720], unit:s, attr:min{req, int},max{req, int}-->30
      </dwelltime>
```

drackun@gmail.com DETEKTOR

```
<dubiousConfirmationFrame min="1" max="300">
  <!--ro, opt, int, range:[1,300], attr:min{req, int},max{req, int}-->1
</dubiousConfirmationFrame>
<traceEnable opt="true,false">
  <!--ro, req, string, attr:opt{req, string}-->test
</traceEnable>
<trackDuration min="1" max="10">
  <!--ro, req, int, range:[0,300], unit:s, attr:min{req, int},max{req, int}-->1
</trackDuration>
<PTZLimitEnable opt="true,false">
  <!--ro, req, string, attr:opt{req, string}-->test
</PTZLimitEnable>
<SceneNameOverlay>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <positionX>
    <!--ro, req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--ro, req, int, range:[0,576]-->1
  </positionY>
</SceneNameOverlay>
<sceneRatio min="1" max="10">
  <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
</sceneRatio>
<sceneType opt="land,surface,skyDrone,electricFence">
  <!--ro, opt, enum, subType:string, attr:opt{req, string}-->land
</sceneType>
<skyDroneAlgorithmParams>
  <!--ro, opt, object-->
  <alertSensitivity min="1" max="100" def="50">
    <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int},def{req, int}-->50
  </alertSensitivity>
  <alertTime min="0" max="180" def="0">
    <!--ro, opt, int, range:[0,180], unit:s, attr:min{req, int},max{req, int},def{req, int}-->0
  </alertTime>
  <alertTimePercent min="0" max="100" def="30">
    <!--ro, opt, int, range:[0,100], unit:%, attr:min{req, int},max{req, int},def{req, int}-->30
  </alertTimePercent>
  <targetSize>
    <!--ro, opt, object-->
    <length min="1" max="1000" def="450">
      <!--ro, opt, int, range:[1,1000], unit:mm, attr:min{req, int},max{req, int},def{req, int}-->450
    </length>
    <width min="1" max="1000" def="100">
      <!--ro, opt, int, range:[1,1000], unit:mm, attr:min{req, int},max{req, int},def{req, int}-->100
    </width>
  </targetSize>
</skyDroneAlgorithmParams>
<targetAttributeOverlapEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</targetAttributeOverlapEnabled>
<randomFrameFilterEnable opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</randomFrameFilterEnable>
<backgroundModelOutputEnabled opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</backgroundModelOutputEnabled>
<movementFilter>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <humanMovement>
    <!--ro, opt, object-->
    <ratio min="0" max="1000" def="50">
      <!--ro, opt, int, range:[0,1000], attr:min{req, int},max{req, int},def{req, int}-->1
    </ratio>
  </humanMovement>
  <vehicleMovement>
    <!--ro, opt, object-->
    <ratio min="0" max="1000" def="50">
      <!--ro, opt, int, range:[0,1000], attr:min{req, int},max{req, int},def{req, int}-->1
    </ratio>
  </vehicleMovement>
  <backgroundModelObjectMovement>
    <!--ro, opt, object-->
    <ratio min="0" max="1000" def="50">
      <!--ro, opt, int, range:[0,1000], attr:min{req, int},max{req, int},def{req, int}-->1
    </ratio>
  </backgroundModelObjectMovement>
</movementFilter>
<BGSFilter>
```

crackun@gmail.com DETEKTOR

```
        <!--ro, opt, object-->
        <enabled opt="true,false">
          <!--ro, req, bool, attr:opt{req, string}-->true
        </enabled>
        <vehicleForeground>
          <!--ro, opt, object-->
          <percentage min="0" max="100" def="60">
            <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->1
          </percentage>
        </vehicleForeground>
        <humanForeground>
          <!--ro, opt, object-->
          <percentage min="0" max="100" def="20">
            <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->1
          </percentage>
        </humanForeground>
      </BGSFilter>
      <alarmDuration>
        <!--ro, opt, object-->
        <enable opt="true,false">
          <!--ro, req, bool, attr:opt{req, string}-->true
        </enable>
        <durationTime min="1" max="30">
          <!--ro, opt, int, range:[1,30], unit:s, attr:min{req, int},max{req, int}-->1
        </durationTime>
      </alarmDuration>
      <trackRatioMode opt="followConfig,followTarget">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->followConfig
      </trackRatioMode>
      <targetRatio min="5" max="200">
        <!--ro, opt, int, range:[5,200], unit:%, dep:and,{$.IntelliTraceBlockList[*].IntelliTraceBlock.trackRatioMode,eq,followTarget}, attr:min{req,
int},max{req, int}-->5
      </targetRatio>
    </IntelliTraceBlock>
  </IntelliTraceBlockList>
</IntelliTraceCap>
```

14.10.1.5 Get the scene parameters capability

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/scene/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SceneParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <sceneMode opt="commandingHeight,road,parkingLot,square,normal,panoramic">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->commandingHeight
  </sceneMode>
</SceneParam>
```

14.10.1.6 Get the configuration capability of zooming ratio in abnormal event detection scene

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/zoomRatio/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

drackuin@gmail.com DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ZoomRatioCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the configuration capability of zooming ratio in abnormal event detection scene, attr:version{req, string, protocolVersion}-->
  <isSupportZoomRatioInScene>
    <!--ro, opt, bool, whether it supports zooming ratio configuration of patrol tracking for multiple scenes-->true
  </isSupportZoomRatioInScene>
  <isSupportZoomRatioGoto>
    <!--ro, opt, bool, whether it supports optimizing zooming ratio of patrol tracking for multiple scenes-->true
  </isSupportZoomRatioGoto>
</ZoomRatioCap>
```

14.10.1.7 Get the intelligent channel list

Request URL

GET /ISAPI/Intelligent/intelliChannelList

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, intelligent channel list, subType:object, attr:version{req, string, protocolVersion}-->
  <IntelliChannel>
    <!--ro, opt, object, intelligent channel-->
    <id>
      <!--ro, req, string, channel ID-->test
    </id>
  </IntelliChannel>
</IntelliChannelList>
```

14.10.2 人脸库管理(S)

14.10.2.1 Get face picture library capability

Request URL

GET /ISAPI/Intelligent/FDLib/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, this field is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, detailed error description, this field is required when the value of statusCode is not 1*/
    "FDNameMaxLen":  64,
    /*ro, req, int, maximum length of face picture library name*/
    "customInfoMaxLen":  192,
    /*ro, req, int, maximum length of custom information*/
    "FDMaxNum":  3,
    /*ro, req, int, maximum number of face picture libraries*/
    "FDRecordDataMaxNum":  12345,
    /*ro, req, int, maximum face records supported by face picture library*/
```

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```
    "supportFDFunction":  "post,delete,put,get,setUp",
    /*ro, req, string, the supported operations on face picture library, desc:"post" (create), "delete" (delete), "put" (edit), "get" (search), "setUp"
(set)*/
    "isSuportFDSearch":  true,
    /*ro, req, bool, whether supports searching in face picture library, desc:whether supports searching in face picture library*/
    "isDisplayCaptureNum": {
    /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, opt, array, subType:bool*/
    },
    "isSupportFDSearchDataPackage":  true,
    /*ro, req, bool, whether supports packaging the found data in the face picture library*/
    "isSuportFSsearchByPic":  true,
    /*ro, req, bool, whether supports searching by picture in the face picture library, desc:whether supports searching by picture in the face picture
library*/
    "isSuportFSsearchByPicGenerate":  true,
    /*ro, req, bool, whether supports exporting results of using picture to search picture from the face picture library*/
    "isSuportFDSearchDuplicate":  true,
    /*ro, req, bool, whether supports duplication checking*/
    "isSuportFDSearchDuplicateGenerate":  true,
    /*ro, req, bool, whether supports exporting the duplication checking results*/
    "isSuportFCSearch":  true,
    /*ro, req, bool, whether supports searching face picture comparison alarms*/
    "isSupportFCSearchDataPackage":  true,
    /*ro, req, bool, whether supports packaging the search results of face picture comparison alarms*/
    "isSupportFDExecuteControl":  true,
    /*ro, req, bool, whether supports creating relation between face picture libraries and cameras*/
    "generateMaxNum":  1234,
    /*ro, opt, int, maximum face records can be exported from face picture library*/
    "faceLibType":  "blackFD,staticFD,infraredFD",
    /*ro, opt, string, face picture library type*/
    "modelMaxNum":  1000,
    /*ro, opt, int, the maximum number of search results*/
    "isSupportModelData":  true,
    /*ro, opt, bool, whether it supports applying model data*/
    "isSuportFDLibArmingType":  true,
    /*ro, opt, bool, whether it supports face picture library arming type*/
    "isSuportFDLibSearch":  true,
    /*ro, opt, bool, whether it supports searching face picture library*/
    "FDArmingRecordDataMaxNum":  12345,
    /*ro, opt, int, the supported maximum number of face records in the face picture arming library*/
    "isSupportControlPersonRecordByHumanId":  true,
    /*ro, opt, bool, whether it supports modifying and deleting the face record by humanId*/
    "isSupportControlPersonRecordByRowKey":  true,
    /*ro, opt, bool, whether it supports modifying and deleting the face record by rowKey*/
    "isSupportFaceLibRebuildCfg":  true,
    /*ro, opt, bool, whether it supports recreating face picture library information and configuration*/
    "isSupportFDMove":  true,
    /*ro, opt, bool, whether it supports moving face data in the face picture library in a batch, desc:the related URI is
/ISAPI/Intelligent/FDLib/FDMove/capabilities?format=json*/
    "faceURLLen":  1024,
    /*ro, opt, int, the maximum size of the face picture URL, desc:if this node is not returned, the default size of the face picture URL supported by the
device is 256 bytes; otherwise, the device should support that the value of this node is greater than or equal to 256*/
    "isSupportArmingLibCfg":  true,
    /*ro, opt, bool, whether it supports configuring parameters of the armed face picture library, desc:related URI:
/ISAPI/Intelligent/FDLib/armingLibCfg/capabilities?format=json*/
    "isSupportModelTransformation":  true,
    /*ro, opt, bool, whether it supports converting face picture models in the face picture list library, desc:related URI:
/ISAPI/Intelligent/FDLib/model/transformation/capabilities?format=json*/
    "featurePointTypeList": ["face", "leftEye", "rightEye", "leftMouthCorner", "rightMouthCorner", "nose"],
    /*ro, opt, array, feature point types of face pictures supported by the device, subType:string*/
    "isSupportPresort":  true,
    /*ro, opt, bool, whether it supports configuring pre-categorization, desc:related URI: /ISAPI/Intelligent/FDLib/presort/capabilities?format=json*/
    "libAttribute": {
    /*ro, opt, object, library attribute type, desc:"general" (normal library), "blockList" (blocklist library), "VIP" (VIP library), "passerby" (allowlist
library, which cannot be deleted)*/
        "@opt": ["general", "blackList", "VIP", "passerby"]
        /*ro, opt, array, options, subType:string*/
    },
    "faceType": {
    /*ro, opt, object, face picture type*/
        "@opt": ["normalFace", "patrolFace", "hijackFace", "superFace"]
        /*ro, opt, array, options, subType:string*/
    },
    "saveFacePic": {
    /*ro, opt, object, whether to save the registered face picture*/
        "@opt": [true, false]
        /*ro, opt, array, options, subType:bool*/
    },
    "leaderPermission": {
    /*ro, opt, object, first authentication permission*/
        "@size":  4,
        /*ro, opt, int, the maximum number of array members, range:[1,4]*/
        "@min":  1,
        /*ro, opt, int, member value range, range:[1,4]*/
        "@max":  4
        /*ro, opt, int, member value range, range:[1,4]*/
    },
    "facePicFormat": {
    /*ro, opt, object*/
        "@opt": ["jpg", "png", "bmp"]
        /*ro, req, array, subType:string*/
    }
}
```

drackum@gmail.com
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14.10.2.2 Search face picture library information by arming mode

Request URL

POST /ISAPI/Intelligent/FDLib/search?format=json

Query Parameter

None

Request Message

```
{
    "searchResultPosition": 0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults": 0,
    /*req, int, the maximum number of search results this time*/
    "libArmingType": "armingLib"
    /*opt, enum, arming type, subType:string, desc:"armingLib" (armed library (default)), "nonArmingLib" (not armed library)*/
}
```

Response Message

```
{
    "responseStatusStrg": "OK",
    /*ro, opt, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data).*/
    "numOfMatches": 1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches": 1,
    /*ro, opt, int, total number of matched results*/
    "FDLib": [
    /*ro, opt, array, face picture library information, subType:object*/
        {
            "FDID": "test",
            /*ro, opt, string, face picture library ID*/
            "faceLibType": "blackFD",
            /*ro, opt, enum, face picture library type, desc:"blackFD" (list library), "staticFD" (static library)*/
            "name": "test",
            /*ro, opt, string, name of the face picture library, range:[0,48]*/
            "customInfo": "test",
            /*ro, opt, string, custom information, range:[0,192]*/
            "libArmingType": "armingLib",
            /*ro, opt, enum, arming type, subType:string, desc:"armingLib" (armed library (default)), "nonArmingLib" (not armed library)*/
            "libAttribute": "blackList"
            /*ro, opt, enum, library attribute type, subType:string, desc:"blackList" (blocklist), "VIP" (VIP library), "passerby" (passer-by library, which cannot be deleted)*/
        }
    ]
}
```

drackum@gmail.com
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14.10.2.3 Delete all face picture libraries

Request URL

DELETE /ISAPI/Intelligent/FDLib?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

14.10.2.4 Create a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib?format=json

Query Parameter

None

Request Message

```
{
    "faceLibType":  "blackFD,staticFD",
    /*req, string, face picture library type, desc:face picture library type*/
    "name":  "test",
    /*req, string, name of the face picture library, range:[0,48], desc:name of the face picture library*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192], desc:custom information*/
    "libArmingType":  "armingLib",
    /*opt, enum, subType:string*/
    "libAttribute":  "blackList",
    /*opt, enum, subType:string*/
    "FDID":  "test"
    /*opt, string, face picture library ID, desc:face picture library ID*/
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, req, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "FDID":  "test"
    /*ro, opt, string, face picture library ID, desc:face picture library ID*/
}
```

drackum@gmail.com
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14.10.2.5 Delete a specified face picture library

Request URL

DELETE /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD" (list library), "staticFD" (static library)
terminalNo	string	Terminal No. It is an integer starting from 1.

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

14.10.2.6 Edit the information of a specific face picture library information, including name and custom information

Request URL

PUT /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	--
terminalNo	string	--

Request Message

```
{
    "name":  "test",
    /*opt, string, face picture library name, range:[0,48]*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192]*/
    "libArmingType":  "armingLib",
    /*opt, enum, arming type of the list library, subType:string, desc:“armingLib” (armed face picture library), “nonArmingLib” (not armed face picture library). The default value is "armingLib”*/
    "libAttribute":  "blackList"
    /*opt, enum, library type, subType:string, desc:“blackList” (blocklist library), “VIP” (VIP library), “passerby” (passerby library). The passerby library cannot be deleted*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode when statusCode is not 1*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

14.10.2.7 Get the information, including library ID, library type, name, and custom information, of all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

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Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD” (list library), "staticFD” (static library), "infraredFD" (IR face picture library)
terminalNo	string	Terminal No. It is an integer starting from 1.

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:status code*/
    "statusString":  "OK",
    /*ro, req, string, status description, range:[1,64], desc:status description*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:sub status code*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:error description*/
    "faceLibType":  "blackFD",
    /*ro, opt, enum, face picture library type, subType:string, desc:"blackFD” (list library), "staticFD” (static library)*/
    "name":  "test",
    /*ro, opt, string, face picture library name, range:[1,48]*/
    "customInfo":  "test",
    /*ro, opt, string, custom information, range:[0,192]*/
    "libArmingType":  "armingLib",
    /*ro, opt, enum, library arming type, subType:string, desc:library arming type*/
    "libAttribute":  "blackList",
    /*ro, opt, enum, subType:string*/
    "personnelFileEnabled":  true
    /*ro, opt, bool*/
}
```

drackum@gmail.com
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14.10.3 人脸库记录管理

14.10.3.1 Get the total number of face records in all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib/Count?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "requestURL":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, see the description of this node and above nodes in the message of JSON_ResponseStatus, desc:this node is required when the value of
statusCode is not 1*/
    "FDRecordDataInfo": [
    /*ro, opt, array, information of face records in face picture library, subType:object, desc:this node is valid when errorCode is 1 and errorMsg is
"ok"*/
        {
            "FDID":  "test",
            /*ro, opt, string, face picture library ID, desc:the maximum size is 63 bytes*/
            "faceLibType":  "blackFD",
            /*ro, opt, enum, face picture library type, subType:string, desc:face picture library type "blackFD" list library,"staticFD" static library, the
maximum size is 32 bytes*/
            "name":  "test",
            /*ro, opt, string, face picture library name, desc:the maximum size is 48 bytes*/
            "recordDataNumber":  123,
            /*ro, opt, int, number of records*/
            "libArmingType":  "armingLib",
            /*ro, opt, enum, arming type, subType:string, desc:arming type*/
            "libAttribute":  "general",
            /*ro, opt, enum, subType:string*/
            "personnelFileEnabled":  true
            /*ro, opt, bool*/
        }
    ],
    "totalRecordDataNumber":  500000,
    /*ro, opt, int, total number of records, desc:total number of records*/
    "FDCapacity": {
    /*ro, opt, object*/
        "total":  0.000,
        /*ro, req, float*/
        "use":  0.000,
        /*ro, req, float*/
        "remain":  0.000,
        /*ro, req, float, remaining space, desc:remaining space*/
        "maxRecordDataNumber":  0,
        /*ro, opt, int*/
        "useRecordDataNumber":  0,
        /*ro, opt, int*/
        "remainRecordDataNumber":  0
        /*ro, opt, int*/
    }
}
```

drackum@gmail.com
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14.10.3.2 Get the total number of face records in a face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/Count?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD" (list library), "staticFD" (static library), "infraredFD" (IR face picture library)
terminalNo	string	--

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "FDID":  "test",
    /*ro, opt, string, face picture library ID, range:[1,64]*/
    "faceLibType":  "blackFD",
    /*ro, opt, enum, face picture library type, subType:string, desc:face picture library type*/
    "name":  "test",
    /*ro, opt, string, name of the face picture library, desc:the maximum length is 48*/
    "recordDataNumber":  123
    /*ro, opt, int, total number of records*/
}
```

14.10.3.3 Add a face record to the face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/FaceDataRecord?format=json

Query Parameter

None

Request Message

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```
{
    "faceURL":  "test",
    /*opt, string, picture storage URL inputted when uploading the face picture by URL, desc:select this field or modelData*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture library type*/
    "FDID":  "test",
    /*req, string, face picture library ID, range:[1,64]*/
    "FPID":  " test",
    /*opt, string, face record ID, range:[1,1000], desc:face record ID,it can be generated by device or inputted. If it is inputted,it should be the unique ID with the combination of letters and digits,and the maximum length is 63 bytes; if it is generated by the device automatically,it is the same as the employee No. (person ID)*/
    "name":  "张三",
    /*req, string, name of person in the face picture, range:[1,1000], desc:the maximum size is 96 bytes*/
    "gender":  "male",
    /*opt, enum, gender of person in the face picture, subType:string, desc:“male”, “female”, “unknown”; the maximum size is 32 bytes*/
    "bornTime":  "2004-05-03",
    /*req, string, birthday of person in the face picture, desc:the maximum size is 20 bytes*/
    "city":  "130100",
    /*opt, string, city code of birth for the person in the face picture, desc:the maximum size is 32 bytes*/
    "certificateType":  "ID",
    /*opt, enum, certificate type, subType:string, desc:"ID" (ID card), "officerID" (officer ID), "other" (other type), "LawyerCertificate" (lawyer qualification), "birthCertificate", "householdRegister", "judgeCertificate", "studentIDCard", "workCard" (work permit),  "policeID", "foreignPermanentResidentIDCard", "passport", "pressCard", "prosecutorCard", "temporaryIDCard", "temporaryResidencePermit"*/
    "certificateNumber":  "test",
    /*opt, string, certificate No., range:[1,64], desc:the maximum size is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, case information, desc:the maximum size is 192 bytes; it is valid when faceLibType is "blackFD”*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag, desc:up to 4 tags, which are separated by commas. The maximum size is 195 bytes. It is valid when faceLibType is "blackFD”*/
    "address":  "test",
    /*opt, string, person address, desc:the maximum size is 192 bytes. It is valid when faceLibType is "staticFD”*/
    "customInfo":  "test",
    /*opt, string, custom information, desc:the maximum size is 192 bytes. It is valid when faceLibType is "staticFD"*/
    "modelData":  "test",
    /*opt, string, target modeling data, desc:target model data,non-modeled binary data needs to be encrypted by Base64 during transmission*/
    "transfer":  true,
    /*opt, bool, whether to enable transfer, desc:whether to enable transfer*/
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal” (operate by terminal)*/
    "terminalNoList": [1],
    /*opt, array, terminal ID list, subType:int, desc:this node is required when operation type is "byTerminal"; currently, only one terminal is supported*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied, subType:object*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, coordinates of the feature point, desc:object coordinates of the feature point*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:this node is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:this node is required when featurePointType is "face"*/
            }
        }
    ],
    "cutOutRect": {
    /*opt, object, picture cropping frame, desc:the origin is the upper-left corner of the screen, which is used for the target frame in matting*/
        "x":  0.000,
        /*req, float, x-coordinate, range:[0,1], desc:normalize coordinate*/
        "y":  0.000,
        /*req, float, y-Coordinate, range:[0,1], desc:normalize coordinate*/
        "width":  0.000,
        /*req, float, width, range:[0,1]*/
        "height":  0.000
        /*req, float, height, range:[0,1]*/
    },
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string, desc:"normalFace" (normal face, default value), "patrolFace" (patrol person face), "hijackFace" (duress person face), "superFace" (super person face)*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, first authentication permission, subType:int, range:[1,4], desc:if the array is empty, it indicates clearing first authentication permission of the person; if not, it indicates that the person have first authentication permission for the door 1, 2, 3, and 4*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceURL	[Message content]	application/json	--	--	--
img	[Binary picture data]	image/jpeg		facePic.jpg	--

dracKum@gmail.com
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Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:this field is required when the value of statusCode is not 1*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:see the description of this node and above nodes in the message of JSON_ResponseStatus*/
    "FPID":  "test",
    /*ro, opt, string, face record ID, range:[1,64], desc:face record ID returned when the face record is added,it is unique,and the maximum size is 63
bytes. This node is valid when errorCode is "1" and errorMsg is "ok"*/
    "rowKey":  "test"
    /*ro, opt, string, rowKey of face record, desc:rowKey of face record is returned after face data is added. It is unique and in string format. The
maximum length is 64 bytes.*/
}
```

dracku@gmail.com
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14.10.3.4 Edit face records in the face picture library in a batch

Request URL

PUT /ISAPI/Intelligent/FDLib/FDModify?format=json

Query Parameter

None

Request Message

```
{
    "faceURL":  "test",
    /*opt, string, picture storage URL inputted when uploading the face picture by URL, the maximum length is 256 bytes*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture library type: "blackFD"-list library,"staticFD"-static library,the maximum
length is 32 bytes*/
    "FDID":  "test",
    /*req, string, face picture library ID, desc:face picture library ID,the maximum length is 63 bytes,multiple face picture libraries should be separated
by commas*/
    "FPID":  " ",
    /*opt, string, face record ID*/
    "name":  "test",
    /*opt, string, name*/
    "gender":  "any",
    /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
    "bornTime":  "2004-05-03",
    /*req, time, date of birth of the person in the face picture in ISO8601 time format, the maximum length is 20 bytes, desc:date of birth of the person in
the face picture in ISO8601 time format,the maximum length is 20 bytes*/
    "city":  "130100",
    /*opt, string, code of the city of birth for the person in the face picture, the maximum length is 32 bytes, desc:code of the city of birth for the
person in the face picture,the maximum length is 32 bytes*/
    "certificateType":  "ID",
    /*req, enum, certificate type, subType:string*/
    "certificateNumber":  "test",
    /*opt, string, ID No., the maximum length is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, case information, it is valid when faceLibType is "blackFD", range:[0,192]*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag*/
    "address":  "test",
    /*opt, string, person address, it is valid when faceLibType is "staticFD"*/
    "customInfo":  "test",
    /*opt, string, custom information, it is valid when faceLibType is "staticFD"*/
    "modelData":  "test",
    /*opt, string, target model data, non-modeled binary data needs to be encrypted by base64 during transmission*/
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal"-by terminal*/
    "terminalNoList": [1],
    /*opt, array, terminal ID list, subType:int, desc:this node is required when operation type is "byTerminal"; currently, only one terminal is supported*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied. If the device only supports three types of feature points, when the platform applies more than three types
of feature points,the device will not return error information, subType:object*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of
mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, point coordinates*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:width which is between 0 and 1000. This node is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:height which is between 0 and 1000. This node is required when featurePointType is "face"*/
            }
        }
    ],
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, subType:int, range:[1,4]*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceURL	[Message content]	application/json	--	--	--
img	[Binary picture data]	image/jpeg		faceImage.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error details, desc:this field is required when statusCode is not 1*/
}
```

drackum@gmail.com
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14.10.3.5 Get the capability of moving face data in the face picture library in a batch

Request URL

GET /ISAPI/Intelligent/FDLib/FDMove/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FDMove": [
    /*ro, opt, array, subType:object*/
        {
            "id": {
            /*ro, opt, object, number of data that can be moved each time*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  0
                /*ro, opt, int, the maximum value*/
            },
            "FDSrc": {
            /*ro, opt, object, source face picture library from which the face record will be moved out*/
                "FDID": {
                /*ro, opt, object, face picture library ID*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int, the maximum value*/
                },
                "FPID": {
                /*ro, opt, object, face record ID*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int, the maximum value*/
                },
                "humanId": {
                /*ro, opt, object, person ID*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int, the maximum value*/
                }
            }
```

```
    },
    "FDDst": {
    /*ro, opt, object, destination face picture library which the face record will be moved to*/
        "FDID": {
        /*ro, opt, object, face picture library ID*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "FPID": {
        /*ro, opt, object, face record ID*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "humanId": {
        /*ro, opt, object, person ID*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "name": {
        /*ro, opt, object, name of the person in the face picture*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "gender": {
        /*ro, opt, object, gender of the person in the face picture*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "bornTime": {
        /*ro, opt, object, birth date of the person in the face picture*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "city": {
        /*ro, opt, object, city code of birth of the person in the face picture*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "certificateType": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int*/
        },
        "certificateNumber": {
        /*ro, opt, object, certificate No.*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "caseInfo": {
        /*ro, opt, object, remarks*/
            "@min":  0,
            /*ro, opt, int, the minimum value*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "tag": {
        /*ro, opt, object, custom tag*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "address": {
        /*ro, opt, object, person address*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
        "customInfo": {
        /*ro, opt, object, custom information*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  0
            /*ro, opt, int, the maximum value*/
        },
```

drackum@gmail.com
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```
                "rowKey": {
                /*ro, opt, object, main key of face picture library*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int*/
                },
                "transfer": {
                /*ro, opt, object, whether to transfer*/
                    "@opt": [true, false]
                    /*ro, opt, array, subType:bool*/
                }
            }
        }
    ]
}
```

14.10.3.6 Move face data in the face picture library in a batch

Request URL

POST /ISAPI/Intelligent/FDLib/FDMove?format=json

Query Parameter

None

Request Message

```
{
    "FDMove": [
    /*req, array, subType:object*/
        {
            "id":  1,
            /*req, int, serial No. of the element in the array,it starts from 1 and is used to match returned results*/
            "FDSrc": {
            /*opt, object, face records to be moved*/
                "FDID":  "test",
                /*req, string, face picture library ID*/
                "FPID":  "test",
                /*req, string, face record ID*/
                "humanId":  "humanIDXXX"
                /*opt, string, person ID*/
            },
            "FDDst": {
            /*opt, object, destination face picture library (face records will be moved to this face picture library)*/
                "FDID":  "test",
                /*req, string, face picture library ID, desc:the maximum length is 63 bytes. Multiple libraries are separated by commas*/
                "FPID":  " ",
                /*req, string, face record ID*/
                "humanId":  "humanIDXXX",
                /*opt, string, person ID*/
                "name":  "test",
                /*opt, string, name*/
                "gender":  "any",
                /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male", "female", "unknown"*/
                "bornTime":  "2004-05-03",
                /*req, string, birth date of the person in the face picture, range:[0,20]*/
                "city":  "130100",
                /*opt, string, city code, desc:city of birth*/
                "certificateType":  "ID",
                /*req, enum, certificate type, subType:string, desc:certificate type*/
                "certificateNumber":  "test",
                /*opt, string, certificate No.*/
                "caseInfo":  "test",
                /*opt, string, remarks*/
                "tag":  "aa,bb,cc,dd",
                /*opt, string, custom tag*/
                "address":  "test",
                /*opt, string, person address*/
                "customInfo":  "test",
                /*opt, string, custom information*/
                "rowKey ":  "test",
                /*opt, string, key of the face record database,the maximum size is 64 bytes. Searching by rowKey can be efficient*/
                "transfer":  true
                /*opt, bool, whether to enable transferring, desc:whether to enable transferring*/
            }
        }
    ]
}
```

dr4ckum@gmail.com
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Response Message

```
{
    "FDMoveResult": [
    /*ro, req, array, result of batch moving, subType:object*/
        {
            "id": 1,
            /*ro, req, int, serial No. of the element in the array,it starts from 1 and is used to match returned results, desc:serial No. of the element in the array,it starts from 1 and is used to match returned results*/
            "errorCode": 1,
            /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
            "errorMsg": "ok"
            /*ro, req, string, error description, desc:this node is required when the value of statusCode is not 1*/
        }
    ]
}
```

14.10.3.7 Set the face picture data in the face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/FDSetUp?format=json

Query Parameter

None

Request Message

```
{
    "faceURL": "http",
    /*opt, string, picture storage URL, which is inputted when uploading the face picture by URL*/
    "faceLibType": "blackFD",
    /*req, enum, face picture library type, subType:string, desc:"blackFD" (list library), "staticFD" (static library), the maximum length is 32 bytes*/
    "FDID": "test",
    /*req, string, face picture library ID, desc:the maximum length is 63 bytes. Multiple libraries are separated by commas*/
    "FPID": " ",
    /*opt, string, face record ID, desc:it can be generated by the device or inputted. If it is inputted, it should be the unique ID with the combination of letters and digits, and the maximum length is 63 bytes; if it is generated by the device automatically, it is the same as the employee No. (person ID)*/
    "deleteFP": true,
    /*opt, bool, whether to delete the face record, desc:true (yes). This field required when the face record needs to be deleted; for adding or editing the face record, it should not be configured*/
    "modelData": "test",
    /*opt, string, target model data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method*/
    "PicFeaturePoints": [
    /*opt, array, feature picture coordinates, subType:object, desc:if the device only supports three types of feature points, when the platform applies more than three types of feature points, the device will not return error information*/
        {
            "featurePointType": "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, coordinates of the feature point*/
                "x": 1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized coordinates*/
                "y": 1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized coordinates*/
                "width": 1,
                /*opt, int, width, range:[0,1000], desc:it is required when featurePointType is "face"*/
                "height": 1
                /*opt, int, height, range:[0,1000], desc:it is required when featurePointType is "face"*/
            }
        }
    ],
    "faceType": "normalFace",
    /*opt, enum, face picture type, subType:string, desc:"normalFace" (normal face, default value), "patrolFace" (patrol person face), "hijackFace" (duress person face), "superFace" (super person face)*/
    "saveFacePic": true,
    /*opt, bool, whether to save the registered face picture*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, first authentication permission, subType:int, range:[1,4], desc:first authentication permission*/
}
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceURL	[Message content]	application/json	--	--	--
img	[Binary picture data]	image/jpeg		faceImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

drackum@gmail.com
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14.11 语音对讲

14.11.1 语音对讲

14.11.1.1 Get the parameters of all two-way audio channels

Request URL

GET /ISAPI/System/TwoWayAudio/channels

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, parameters configuration of the two-way audio channel, subType:object, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--ro, opt, object, configuration of the two-way audio channel-->
    <id>
      <!--ro, req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable or not-->true
    </enabled>
    <audioCompressionType>
      <!--ro, req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM",
"MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType>
      <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM",
"MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume>
      <!--ro, opt, int, input volume, range:[1,100]-->100
    </speakerVolume>
    <microphoneVolume>
      <!--ro, opt, int, output volume, range:[1,100]-->100
    </microphoneVolume>
    <noisereduce>
      <!--ro, opt, bool, whether to enable the environmental noise filter or not-->true
    </noisereduce>
    <audioBitRate>
      <!--ro, opt, int, audio frame rate, unit:kbs-->100
    </audioBitRate>
    <audioInputType>
      <!--ro, opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-
adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--ro, opt, object, linked video channel-->
      <enabled>
        <!--ro, req, bool, whether to enable the linked video input channel or not-->true
      </enabled>
      <videoInputChannelList>
        <!--ro, req, array, list of linked video channels, subType:object-->
        <videoInputChannelID>
          <!--ro, opt, string, linked video channel-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate>
      <!--ro, opt, float, audio sampling rate, unit:kHz-->48.00
    </audioSamplingRate>
    <lineOutForbidden>
      <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
    </lineOutForbidden>
    <muteDuringPanTilt>
      <!--ro, opt, bool, whether to mute during motion-->true
    </muteDuringPanTilt>
    <audioOutputType>
      <!--ro, opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <matrixPickUp>
      <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it
is valid only when the audio input type is "Linein")-->true
    </matrixPickUp>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

drackum@gmail.com
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14.11.1.2 Set audio parameters for all two-way audio channels

Request URL

PUT /ISAPI/System/TwoWayAudio/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameters configuration of the two-way audio channel, subType:object, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--opt, object, configuration of the two-way audio channel-->
    <id>
      <!--req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function or not-->true
    </enabled>
    <audioCompressionType>
      <!--req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType>
      <!--req, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume>
      <!--opt, int, input volume-->100
    </speakerVolume>
    <microphoneVolume>
      <!--opt, int, output volume-->100
    </microphoneVolume>
    <noisereduce>
      <!--opt, bool, whether to enable the environmental noise filter or not-->true
    </noisereduce>
    <audioBitRate>
      <!--opt, int, audio frame rate, unit:kbs-->100
    </audioBitRate>
    <audioInputType>
      <!--req, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--opt, object, linked video channel-->
      <enabled>
        <!--req, bool, whether to enable the linked video input channel or not-->true
      </enabled>
      <videoInputChannelList>
        <!--req, array, list of linked video channels, subType:object-->
        <videoInputChannelID>
          <!--opt, string, linked video channel-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate>
      <!--opt, float, audio sampling rate, unit:kHz-->48.00
    </audioSamplingRate>
    <lineOutForbidden>
      <!--opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
    </lineOutForbidden>
    <muteDuringPanTilt>
      <!--opt, bool, whether to mute during motion-->true
    </muteDuringPanTilt>
    <audioOutputType>
      <!--opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <matrixPickUp>
      <!--opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
    </matrixPickUp>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.11.1.3 Receive two-way audio data

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData?sessionId=<audioSessionID>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--
audioSessionID	string	--

Request Message

Binary Data

Response Message

Binary Data

14.11.1.4 Send two-way audio data

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData?sessionId=<audioSessionID>&type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--
audioSessionID	string	Two-way audio session ID
type	enum	If this filed does not exist, it indicates two-way audio via channel, and the audioID is the channel ID. When the type is childDevID, it indicates two-way audio with a single sub-device, and the audioID is the gateway intercom resource index (obtained from /ISAPI/System/DynamicResource/SearchResources?format=json); when the type is childDevList, it indicates broadcasting to multiple sub-devices, with multiple childDevID values configured in /ISAPI/System/TwoWayAudio/channels//open.

Request Message

Binary Data

Response Message

drackum@gmail.com
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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, Custom Error Information Description, range:[0,1024], desc:Custom detailed error information returned by device application, used
for quick positioning and diagnosis.-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

14.11.1.5 Set audio parameters of a specified two-way audio channel

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, configuration of the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType>
    <!--req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume>
    <!--opt, int, input volume, range:[1,100]-->100
  </speakerVolume>
  <microphoneVolume>
    <!--opt, int, output volume, range:[1,100]-->100
  </microphoneVolume>
  <noisereduce>
    <!--opt, bool, whether to enable the environmental noise filter or not-->true
  </noisereduce>
  <audioBitRate>
    <!--opt, int, audio frame rate, unit:kbs-->100
  </audioBitRate>
  <audioInputType>
    <!--opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--opt, object, linked video channel-->
    <enabled>
      <!--req, bool, whether to enable the linked video input channel or not-->true
    </enabled>
    <videoInputChannelList>
      <!--req, array, list of linked video channels, subType:object-->
      <videoInputChannelID>
        <!--opt, string, linked video channel-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate>
    <!--opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <lineOutForbidden>
    <!--opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the value
is "true", audio output is not supported-->true
  </lineOutForbidden>
  <micInForbidden>
    <!--opt, bool, whether the audio input is not supported. If this node is not returned or its value is "false", audio input is supported;  if the value
is "true", audio input is not supported-->true
  </micInForbidden>
  <muteDuringPanTilt>
    <!--opt, bool, whether to mute during motion-->true
  </muteDuringPanTilt>
  <audioOutputType>
    <!--opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <timeOut>
    <!--opt, int, timeout during two-way audio, range:[10,600], unit:s, desc:when the Web Client initiates the two-way audio actively towards the radar
device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some noise.
When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client stops
speaking or not.-->10
  </timeOut>
  <matrixPickUp>
    <!--opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
  </matrixPickUp>
</TwoWayAudioChannel>
```

drackum@gmail.com
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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

14.11.1.6 Get the capability of configuring audio parameters for a specified two-way audio channel

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/capabilities?audioInputType=<audioInputType>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
audioInputType	string	Audio input type, corresponding to the "audioInputType" in the message. This parameter is mainly used to obtain the parameters corresponding to the specified audio input type.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability of configuring audios for the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,Opus,G.722.1">
    <!--ro, req, enum, encoding type of the audio output, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711uLaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,G.722.1">
    <!--ro, opt, enum, encoding type of the audio input, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume min="0" max="100">
    <!--ro, opt, int, input volume, attr:min{opt, int},max{opt, int}-->100
  </speakerVolume>
  <microphoneVolume min="0" max="100">
    <!--ro, opt, int, output volume, attr:min{opt, int},max{opt, int}-->100
  </microphoneVolume>
  <noisereduce opt="true,false">
    <!--ro, opt, bool, whether to enable the environmental noise filter or not, attr:opt{opt, string}-->true
  </noisereduce>
  <audioBitRate min="1" max="100">
    <!--ro, opt, int, audio frame rate, unit:kbs, attr:min{opt, int},max{opt, int}-->100
  </audioBitRate>
  <audioInputType opt="MicIn,LineIn,selfAdaptive,wirelessPickUp,doubleMic,networkPickUp">
    <!--ro, opt, enum, audio input type, subType:string, attr:opt{opt, string}, desc:"MicIn" (microphone-level input), "LineIn" (line-level input),
"selfAdaptive" (self-adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--ro, opt, object, linked video channel-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the linked video input channel or not, attr:opt{opt, string}-->true
    </enabled>
    <videoInputChannelList>
      <!--ro, req, array, list of linked video channels, subType:object-->
      <videoInputChannelID min="1" max="2048">
        <!--ro, opt, string, linked video channel, attr:min{opt, string},max{opt, string}-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate min="1.00" max="48.00">
    <!--ro, opt, float, audio sampling rate, unit:kHz, attr:min{opt, float},max{opt, float}-->48.00
  </audioSamplingRate>
  <lineOutForbidden opt="true,false">
    <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported, attr:opt{opt, string}-->true
  </lineOutForbidden>
  <muteDuringPanTilt opt="true,false">
    <!--ro, opt, bool, whether to mute during motion, attr:opt{opt, string}-->true
  </muteDuringPanTilt>
  <audioOutputType opt="Close,LineOut,Speaker,selfAdaptive,LineOut_Speaker,HDMI" def="LineOut">
    <!--ro, opt, enum, audio output type, subType:string, attr:opt{opt, string},def{opt, string}, desc:"Close" (output closed), "LineOut" (line-level
output), "Speaker" (speaker output), "selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <isSupportEchoCancellation>
    <!--ro, opt, bool, whether the device supports echo cancellation-->true
  </isSupportEchoCancellation>
  <timeOut min="10" max="600">
    <!--ro, opt, int, timeout during two-way audio, attr:min{req, int},max{req, int}, desc:when the Web Client initiates the two-way audio actively towards
the radar device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some
noise. When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client
stops speaking or not.-->300
  </timeOut>
  <matrixPickUp opt="true,false">
    <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it is
valid only when the audio input type is "Linein"), attr:opt{opt, string}-->false
  </matrixPickUp>
  <isSupportRestore>
    <!--ro, opt, bool, whether the device supports restoring audio parameters of the two-way audio channel, desc:corresponding API:
/ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json-->true
  </isSupportRestore>
  <networkPickUpStatus opt="online,offline">
    <!--ro, opt, enum, network pick-up status, subType:string, dep:and,{$.TwoWayAudioChannel.audioInputType,eq,networkPickUp}, attr:opt{opt, string},
desc:network status of the network pickup linked with the audio channel: "online", "offline"-->online
  </networkPickUpStatus>
  <noisereduceType opt="normal,smart">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->normal
  </noisereduceType>
</TwoWayAudioChannel>
```

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14.11.1.7 Stop two-way audio

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/close?sessionId=<audioSessionID>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-Way Audio Channel ID
audioSessionID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, Custom Error Information Description, range:[0,1024], desc:Custom detailed error information returned by device application, used
for quick positioning and diagnosis.-->badXmlFormat
  </description>
</ResponseStatus>
```

drackum@gmail.com
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14.11.1.8 Start two-way audio

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/open?type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
type	enum	If this filed does not exist, it indicates two-way audio via channel, and the audioID is the channel ID. When the type is childDevID, it indicates two-way audio with a single sub device, and the audioID is the gateway intercom resource index (obtained from /ISAPI/System/DynamicResource/SearchResources?format=json); when the type is childDevList, it indicates broadcasting to multiple sub devices, with multiple childDevID values specified in the message, and the audioID in the URL is invalid.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudio xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters for this two-way audio, attr:version{req, string, protocolVersion}-->
  <audioLevel>
    <!--opt, int, audio level, range:[0,15], desc:range: [0,15]-->0
  </audioLevel>
  <microphoneVolume>
    <!--opt, int, output volume-->100
  </microphoneVolume>
  <audioCompressionType>
    <!--opt, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioSamplingRate>
    <!--opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <isBroadcast>
    <!--opt, bool-->true
  </isBroadcast>
  <broadcastType>
    <!--opt, enum, subType:string-->alarm
  </broadcastType>
  <childList>
    <!--opt, array, subType:object, range:[0,8], dep:and,{$.TwoWayAudio.isBroadCast,eq,true}-->
    <childDevID>
      <!--opt, string, range:[9,32]-->test
    </childDevID>
    <voiceTalkIndex>
      <!--opt, int, range:[1,256]-->test
    </voiceTalkIndex>
  </childList>
</TwoWayAudio>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioSession xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, result returned when starting two-way audio succeeds, attr:version{opt, string, protocolVersion}-->
  <sessionId>
    <!--ro, req, string, two-way audio session ID-->1
  </sessionId>
</TwoWayAudioSession>
```

drackum@gmail.com
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14.11.1.9 Get the audio parameters of a specified two-way audio channel

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>?audioInputType=<audioInputType>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
audioInputType	string	Audio input type, corresponding to the "audioInputType" in the message. This parameter is mainly used to obtain the parameters corresponding to the specified audio input type.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, configuration of the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType>
    <!--ro, req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume>
    <!--ro, opt, int, input volume, range:[1,100]-->100
  </speakerVolume>
  <microphoneVolume>
    <!--ro, opt, int, output volume, range:[1,100]-->100
  </microphoneVolume>
  <noisereduce>
    <!--ro, opt, bool, whether to enable the environmental noise filter or not-->true
  </noisereduce>
  <audioBitRate>
    <!--ro, opt, int, audio frame rate, unit:kbs-->100
  </audioBitRate>
  <audioInputType>
    <!--ro, opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--ro, opt, object, linked video channel-->
    <enabled>
      <!--ro, req, bool, whether to enable the linked video input channel or not-->true
    </enabled>
    <videoInputChannelList>
      <!--ro, req, array, list of linked video channels, subType:object-->
      <videoInputChannelID>
        <!--ro, opt, string, linked video channel-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate>
    <!--ro, opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <lineOutForbidden>
    <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
  </lineOutForbidden>
  <micInForbidden>
    <!--ro, opt, bool, whether the audio input is not supported. If this node is not returned or its value is "false", audio input is supported;  if the
value is "true", audio input is not supported-->true
  </micInForbidden>
  <muteDuringPanTilt>
    <!--ro, opt, bool, whether to mute during motion-->true
  </muteDuringPanTilt>
  <audioOutputType>
    <!--ro, opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <timeOut>
    <!--ro, opt, int, timeout during two-way audio, range:[10,600], unit:s, desc:when the Web Client initiates the two-way audio actively towards the radar
device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some noise.
When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client stops
speaking or not.-->10
  </timeOut>
  <matrixPickUp>
    <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
  </matrixPickUp>
  <networkPickUpStatus>
    <!--ro, opt, enum, subType:string, dep:and,{$.TwoWayAudioChannel.audioInputType,eq,networkPickUp}-->online
  </networkPickUpStatus>
  <noisereduceType>
    <!--ro, opt, enum, subType:string-->normal
  </noisereduceType>
</TwoWayAudioChannel>
```

drackum@gmail.com
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14.11.1.10 Get the capability of configuring parameters for all two-way audio channels

Request URL

GET /ISAPI/System/TwoWayAudio/channels/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capability of configuring audios for all two-way audio channels, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--ro, opt, object, capability of configuring audios for the two-way audio channel-->
    <id>
      <!--ro, req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable or not-->true
    </enabled>
    <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,Opus">
      <!--ro, req, enum, encoding type of the audio output, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729",
"G.729a", "G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM">
      <!--ro, opt, enum, encoding type of the audio input, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume min="0" max="100">
      <!--ro, opt, int, input volume, attr:min{opt, int},max{opt, int}-->100
    </speakerVolume>
    <microphoneVolume min="0" max="100">
      <!--ro, opt, int, output volume, attr:min{opt, int},max{opt, int}-->100
    </microphoneVolume>
    <noisereduce opt="true,false">
      <!--ro, opt, bool, whether to enable the environmental noise filter or not, attr:opt{opt, string}-->true
    </noisereduce>
    <audioBitRate min="1" max="100">
      <!--ro, opt, int, audio frame rate, unit:kbs, attr:min{opt, string},max{opt, string}-->100
    </audioBitRate>
    <audioInputType opt="MicIn,LineIn,selfAdaptive,wirelessPickUp">
      <!--ro, opt, enum, audio input type, subType:string, attr:opt{opt, string}, desc:"MicIn" (microphone-level input), "LineIn" (line-level input),
"selfAdaptive" (self-adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--ro, opt, object, linked video channel-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the linked video input channel or not, attr:opt{opt, string}-->true
      </enabled>
      <videoInputChannelList>
        <!--ro, req, array, list of linked video channels, subType:object-->
        <videoInputChannelID min="1" max="2048">
          <!--ro, opt, string, linked video channel, attr:min{opt, string},max{opt, string}-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate min="1.00" max="48.00">
      <!--ro, opt, float, audio sampling rate, unit:kHz, attr:min{opt, float},max{opt, float}-->48.00
    </audioSamplingRate>
    <lineOutForbidden opt="true,false">
      <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported, attr:opt{opt, string}-->true
    </lineOutForbidden>
    <muteDuringPanTilt opt="true,false">
      <!--ro, opt, bool, whether to mute during motion, attr:opt{opt, string}-->true
    </muteDuringPanTilt>
    <audioOutputType opt="Close,LineOut,Speaker,selfAdaptive,LineOut_Speaker" def="LineOut">
      <!--ro, opt, enum, audio output type, subType:string, attr:opt{opt, string},def{opt, string}, desc:"Close" (output closed), "LineOut" (line-level
output), "Speaker" (speaker output), "selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <isSupportEchoCancellation>
      <!--ro, opt, bool, whether the device supports echo cancellation-->true
    </isSupportEchoCancellation>
    <matrixPickUp opt="true,false">
      <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it
is valid only when the audio input type is "Linein"), attr:opt{opt, string}-->false
    </matrixPickUp>
    <isSupportRestore>
      <!--ro, opt, bool, whether the device supports restoring audio parameters of the two-way audio channel, desc:corresponding API:
/ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json-->true
    </isSupportRestore>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

dracku...@gmail.com
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15 How-To Video Guidance

If you need access to corresponding video guidance for device integration, please register on https://tpp.hikvision.com and visit our Training Center: https://tpp.hikvision.com/tpp/Training. The Training Center is specifically designed to provide technical training and guidance resources for our partners. On this platform, you can find integration video tutorials for various devices, enabling better understanding and learning of the integration process. To offer more personalized service, our Training Center also supports filtering by integration protocols, devices, and applications.

drackum@gmail.com
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